JTEKT

Koyo® Bearing specification tables

Koyo Bearing specification tables Contents

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The bearing numbers listed in the bearing specification table include those that are no longer in production. Please contact JTEKT for production availability and stock status.

Deep groove ball bearings

Deep groove ball bearings are available in a variety of sizes, and are the most popular of all rolling bearings. This type of bearing supports radial load and a certain degree of axial load in both directions simultaneously.

■ Shielded / sealed type
- Simplifies sealing structure of applications.
- Greasing is not necessary because bearings are pre-lubricated.
- Table 1 on the next page lists major shielded and sealed bearing types and compares their performance.

■ With locating snap ring
- Bearings with a locating snap ring can be fit to the housing easily, as the locating snap ring facilitates axial positioning.

■ Extra-small ball bearings and miniature ball bearings
- The open type is widely used as well as shielded/sealed type.

Single-row deep groove ball bearings



Open type

Bore diameter **10** – **500** mm



Shielded/sealed type

Bore diameter **10** – **220** mm



With snap ring groove | With locating snap ring

Bore diameter **10** – **130** mm

Extra-small ball bearings and miniature ball bearings



Bore diameter **1** – **9** mm

Double-row deep groove ball bearings

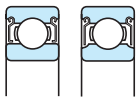
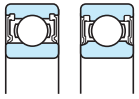
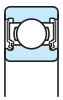


(with filling slot)

Bore diameter **10** – **75** mm



Table 1 Comparison of shielded and sealed bearing performance

Type / Characteristics	Shielded	Sealed			
	Non-contact type	Non-contact type	Contact type		Extremely light contact type
	ZZ type	2RU type	2RS type	2RK type	2RD type
	 (a) [1] (b)	 (c)	 (d) [2] (e)	 (f)	 (g)
Friction torque	Small	Small	Large	Large	Small
High speed performance	Good	Good	Limited because of contact		Good
Grease sealing property	Good	Better than ZZ type	Better than 2RU type for low-speed applications	Excellent	Excellent
Dirt resistance	Good	Better than ZZ type	Better than 2RU type	Excellent	Excellent
Water resistance	Economical	Better than ZZ type but inferior to 2RS, 2RK and 2RD types	Good	Excellent	Better than ZZ and 2RU types
Operating temperature [3]	– 30 to +110°C		– 30 to +100°C		– 30 to +110°C

[Notes] 1) Illustration (a) of the ZZ type shows the relatively small size bearing.
2) Illustration (d) of the 2RS type shows the relatively small size bearing.
3) The operating temperature range listed is for the standard type. It can be widened by using a different type of grease or sealing material. Consult with JTEKT for details.

■ Handling instructions

1) The shielded/sealed type deep groove ball bearing and the deep groove ball bearing with a locating snap ring are designed for use with the inner ring rotating.
Consult with JTEKT on use with the outer ring rotating.
2) When the axial load is large, make the shaft shoulder and housing shoulder larger than usual.
(Referring to the specification table, make the mounting dimension d_a larger and make D_a smaller.)

Boundary dimensions	The dimensions of standard series are as specified in JIS B 1512. For extra-small and miniature ball bearings, special series (ML) are specified together with those described above.
Tolerances	As specified in JIS B 1514-1. (refer to Table 7-3 on pp. A 60 – A 63.)
Radial internal clearance	■ Deep groove ball bearings (except extra-small ball bearings and miniature ball bearings) ..... as specified in JIS B 1520 (refer to Table 10-2 on p. A 102.) ■ Extra-small ball bearings and miniature ball bearings ..... (refer to Table 10-3 on p. A 102.) ■ Deep groove ball bearings for motors ..... (refer to Table 10-6 on p. A 105.)
Recommended fits	■ Bearings of classes 0 and 6 ..... (refer to Table 9-4 on pp. A 91, 92.) ■ Precision extra-small ball bearings and miniature ball bearings ..... (refer to Table 9-5 on p. A 93.)
Standard cages	• Synthetic resin molded cage (supplementary code : FG, MG) • Pressed cage (supplementary code : //) • Copper alloy machined cage (supplementary code : FY) [Remark] For certain applications, stainless steel sheet pressed cages (YS) may also be used.

Application of standard cages

Bearing series	Molded cage	Pressed cage	Machined cage
68	683 – 689	–	–
69	693 – 699	–	–
60	603 – 609	–	–
62	623 – 629	–	–
63	633 – 639	–	–
67	–	6700 – 6706	–
68	–	6800 – 6838	6840 – 68/600
69	–	6900 – 6918	6920 – 6980
160	–	16001 – 16028	16030 – 16072
60	6000 – 6009	6010 – 6034	6036 – 6084
62	6200 – 6208, 6210	6209, 6211 – 6230	6232 – 6248
63	6300 – 6309	6310 – 6328	6330 – 6340
64	–	6403 – 6418	–
42	–	4200 – 4215	–
43	–	4302 – 4315	–

Allowable misalignment	0.002 3 – 0.003 4 rad (8' – 12')

Equivalent radial load (Single/double-row)

Dynamic equivalent radial load
$P_r = XF_r + YF_a$
(refer to the table on the right for values X and Y.)

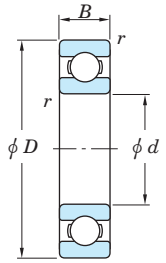
Static equivalent radial load
$P_{0r} = 0.6F_r + 0.5F_a$
(when the value of $P_{0r} < F_r$, $P_{0r} = F_r$)

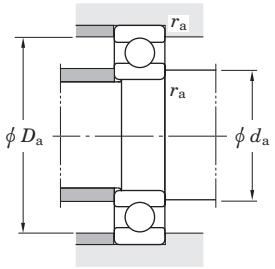
$\frac{if_0F_a}{C_{0r}}$	e	$\frac{F_a}{F_r} \leq e$		$\frac{F_a}{F_r} > e$	
		X	Y	X	Y
0.172	0.19	1	0	0.56	2.30
0.345	0.22				1.99
0.689	0.26				1.71
1.03	0.28				1.55
1.38	0.30				1.45
2.07	0.34				1.31
3.45	0.38				1.15
5.17	0.42				1.04
6.89	0.44				1.00

Factor f_0 is shown in the bearing dimension table.

open type

d 10 ~ (17) mm





Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
10	15	3	0.1	1.05	0.430	0.020	15.7	39 000	47 000	**6700**	10.8	14.2	0.1	0.002
	19	5	0.3	2.15	0.840	0.030	14.8	37 000	43 000	**6800**	12	17	0.3	0.005
	22	6	0.3	3.35	1.25	0.070	14.0	34 000	41 000	**6900**	12	20	0.3	0.010
	26	8	0.3	5.70	1.95	0.100	12.3	31 000	36 000	**6000**	12	24	0.3	0.019
	30	9	0.6	6.40	2.40	0.120	13.2	24 000	29 000	**6200**	14	26	0.6	0.032
	35	11	0.6	10.1	3.45	0.270	11.2	22 000	27 000	**6300**	14	31	0.6	0.053
12	18	4	0.2	1.15	0.530	0.023	16.2	34 000	41 000	**6701**	13.6	16.4	0.2	0.003
	21	5	0.3	2.40	1.05	0.040	15.3	33 000	39 000	**6801**	14	19	0.3	0.006
	24	6	0.3	3.60	1.45	0.080	14.5	31 000	36 000	**6901**	14	22	0.3	0.011
	28	7	0.3	6.40	2.40	0.120	13.2	27 000	32 000	**16001**	14	26	0.3	0.024
	28	8	0.3	6.40	2.40	0.120	13.2	27 000	32 000	**6001**	14	26	0.3	0.022
	32	10	0.6	8.50	3.05	0.240	12.3	22 000	27 000	**6201**	16	28	0.6	0.037
	37	12	1	12.1	4.20	0.420	11.1	20 000	25 000	**6301**	17	32	1	0.060
15	21	4	0.2	1.15	0.580	0.024	16.7	29 000	35 000	**6702**	16.6	19.4	0.2	0.004
	24	5	0.3	2.60	1.25	0.050	15.8	28 000	33 000	**6802**	17	22	0.3	0.007
	28	7	0.3	5.40	2.25	0.120	14.3	26 000	30 000	**6902**	17	26	0.3	0.017
	32	8	0.3	7.00	2.85	0.150	13.9	23 000	28 000	**16002**	17	30	0.3	0.025
	32	9	0.3	7.00	2.85	0.150	13.9	23 000	27 000	**6002**	17	30	0.3	0.030
	35	11	0.6	9.55	3.75	0.290	13.2	20 000	24 000	**6202**	19	31	0.6	0.045
	42	13	1	14.3	5.45	0.460	12.3	17 000	20 000	**6302**	20	37	1	0.082
17	23	4	0.2	1.25	0.660	0.027	16.9	27 000	32 000	**6703**	18.6	21.4	0.2	0.005
	26	5	0.3	3.30	1.55	0.060	15.7	26 000	30 000	**6803**	19	24	0.3	0.008
	30	7	0.3	5.75	2.55	0.130	14.7	23 000	28 000	**6903**	19	28	0.3	0.018
	35	8	0.3	7.50	3.25	0.170	14.4	21 000	25 000	**16003**	19	33	0.3	0.032
	35	10	0.3	7.50	3.25	0.170	14.4	21 000	25 000	**6003**	19	33	0.3	0.039
	40	12	0.6	12.0	4.80	0.370	13.2	17 000	21 000	**6203**	21	36	0.6	0.065
	47	14	1	17.0	6.65	0.550	12.4	15 000	18 000	**6303**	22	42	1	0.115
	47	14	1	19.6	7.60	0.680	12.0	15 000	18 000	**6303R**	22	42	1	0.121

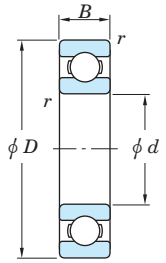
[Remark] Standard cage types used for the above bearings are described earlier in this section.

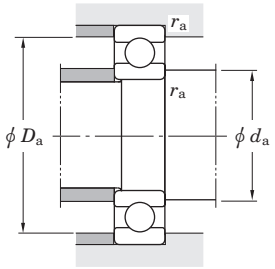
d (17) ~ 28 mm

Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
17	62	17	1.1	25.9	9.85	0.920	11.6	13 000	15 000	**6403**	23.5	55.5	1	0.270
20	27	4	0.2	1.30	0.730	0.030	16.1	23 000	27 000	**6704**	21.6	25.4	0.2	0.006
	32	7	0.3	5.00	2.45	0.100	15.5	21 000	25 000	**6804**	22	30	0.3	0.018
	37	9	0.3	7.95	3.70	0.190	14.7	19 000	23 000	**6904**	22	35	0.3	0.036
	42	8	0.3	9.90	4.50	0.290	14.4	17 000	21 000	**16004**	22	40	0.3	0.050
	42	12	0.6	11.7	5.05	0.350	13.9	17 000	21 000	**6004**	24	38	0.6	0.069
	42	12	0.6	14.4	5.85	0.460	13.0	18 000	21 000	**6004R**	24	38	0.6	0.073
	47	14	1	16.0	6.65	0.510	13.2	15 000	17 000	**6204**	25	42	1	0.106
	47	14	1	19.6	7.60	0.680	12.0	15 000	18 000	**6204R**	25	42	1	0.114
	52	15	1.1	19.9	7.85	0.660	12.3	14 000	17 000	**6304**	26.5	45.5	1	0.144
	52	15	1.1	22.6	8.95	0.790	12.0	14 000	16 000	**6304R**	26.5	45.5	1	0.151
	72	19	1.1	38.7	15.2	1.50	11.1	11 000	13 000	**6404**	26.5	65.5	1	0.400
22	44	12	0.6	11.7	5.15	0.350	14.1	17 000	20 000	**60/22**	26	40	0.6	0.073
	50	14	1	16.0	6.65	0.510	13.2	15 000	17 000	**62/22**	27	45	1	0.118
	56	16	1.1	23.1	9.40	0.770	12.6	13 000	15 000	**63/22**	28.5	49.5	1	0.201
25	32	4	0.2	1.35	0.840	0.035	15.8	19 000	22 000	**6705**	26.6	30.4	0.2	0.006
	37	7	0.3	5.40	2.95	0.120	16.0	18 000	21 000	**6805**	27	35	0.3	0.022
	42	9	0.3	8.75	4.55	0.230	15.4	16 000	19 000	**6905**	27	40	0.3	0.041
	47	8	0.3	11.1	5.60	0.340	15.1	15 000	18 000	**16005**	27	45	0.3	0.060
	47	12	0.6	12.6	5.85	0.380	14.5	15 000	18 000	**6005**	29	43	0.6	0.080
	52	15	1	17.5	7.85	0.550	13.9	13 000	15 000	**6205**	30	47	1	0.128
	52	15	1	22.1	9.30	0.740	12.8	13 000	16 000	**6205R**	30	47	1	0.138
	62	17	1.1	25.7	11.3	0.860	13.2	11 000	13 000	**6305**	31.5	55.5	1	0.232
	62	17	1.1	32.7	13.4	1.20	11.9	11 000	14 000	**6305R**	31.5	55.5	1	0.255
	80	21	1.5	45.2	19.4	1.65	12.2	9 100	11 000	**6405**	33	72	1.5	0.530
28	52	12	0.6	15.6	7.40	0.480	14.5	14 000	16 000	**60/28**	32	48	0.6	0.097
	58	16	1	22.4	9.75	0.720	13.4	12 000	14 000	**62/28**	33	53	1	0.173
	68	18	1.1	29.4	13.1	0.990	13.3	10 000	12 000	**63/28**	34.5	61.5	1	0.328

open type

d 30 ~ (40) mm





Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
30	37	4	0.2	1.45	0.950	0.040	15.7	16 000	19 000	**6706**	31.6	35.4	0.2	0.008
	42	7	0.3	5.65	3.40	0.140	16.4	15 000	18 000	**6806**	32	40	0.3	0.026
	47	9	0.3	9.05	5.00	0.260	15.8	14 000	17 000	**6906**	32	45	0.3	0.045
	55	9	0.3	14.1	7.35	0.440	15.2	13 000	15 000	**16006**	32	53	0.3	0.085
	55	13	1	16.5	8.25	0.530	14.7	13 000	15 000	**6006**	35	50	1	0.116
	62	16	1	24.3	11.3	0.800	13.9	11 000	13 000	**6206**	35	57	1	0.199
	62	16	1	29.2	12.8	1.00	13.0	11 000	13 000	**6206R**	35	57	1	0.212
	72	19	1.1	33.3	15.0	1.15	13.3	9 600	12 000	**6306**	36.5	65.5	1	0.346
	72	19	1.1	41.7	17.7	1.55	12.0	9 800	12 000	**6306R**	36.5	65.5	1	0.379
	90	23	1.5	54.2	23.9	2.05	12.3	8 100	9 700	**6406**	38	82	1.5	0.735
32	58	13	1	18.8	9.15	0.600	14.5	12 000	14 000	**60/32**	37	53	1	0.127
	65	17	1	29.4	13.1	0.990	13.3	10 000	12 000	**62/32**	37	60	1	0.228
	75	20	1.1	37.6	16.2	1.30	12.7	9 300	11 000	**63/32**	38.5	68.5	1	0.437
35	47	7	0.3	5.95	3.85	0.160	16.5	13 000	16 000	**6807**	37	45	0.3	0.030
	55	10	0.6	13.6	7.75	0.440	15.7	12 000	14 000	**6907**	39	51	0.6	0.073
	62	9	0.3	15.3	8.85	0.500	15.7	11 000	13 000	**16007**	37	60	0.3	0.110
	62	14	1	19.9	10.3	0.640	14.9	11 000	13 000	**6007**	40	58	1	0.155
	72	17	1.1	32.1	15.4	1.10	13.9	9 200	11 000	**6207**	41.5	65.5	1	0.288
	72	17	1.1	38.7	17.5	1.40	12.9	9 300	11 000	**6207R**	41.5	65.5	1	0.309
	80	21	1.5	41.7	19.3	1.45	13.2	8 500	10 000	**6307**	43	72	1.5	0.457
	80	21	1.5	50.0	21.7	1.90	12.1	8 700	10 000	**6307R**	43	72	1.5	0.494
	100	25	1.5	68.8	31.0	2.65	12.2	7 200	8 600	**6407**	43	92	1.5	0.952
40	52	7	0.3	6.15	4.20	0.180	16.3	12 000	14 000	**6808**	42	50	0.3	0.033
	62	12	0.6	17.1	9.95	0.570	15.6	11 000	13 000	**6908**	44	58	0.6	0.112
	68	9	0.3	15.8	9.65	0.530	16.0	9 800	12 000	**16008**	42	66	0.3	0.125
	68	15	1	20.9	11.5	0.690	15.2	10 000	12 000	**6008**	45	63	1	0.192
	80	18	1.1	36.4	17.8	1.25	14.0	8 300	10 000	**6208**	46.5	73.5	1	0.366
	90	23	1.5	50.9	24.0	1.85	13.2	7 700	9 200	**6308**	48	82	1.5	0.633

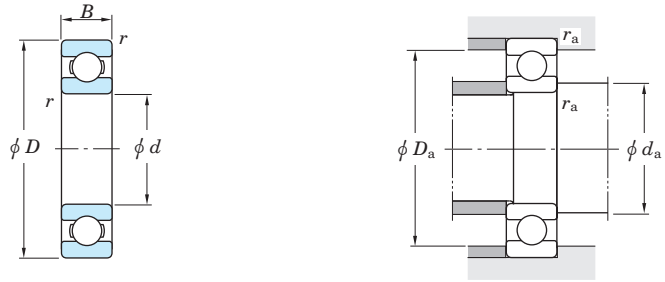
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (40) ~ (60) mm

Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
40	110	27	2	79.6	36.6	3.15	12.3	6 600	7 900	**6408**	49	101	2	1.23
45	58	7	0.3	7.75	5.40	0.230	16.3	11 000	13 000	**6809**	47	56	0.3	0.040
	68	12	0.6	17.7	10.9	0.600	15.9	9 700	11 000	**6909**	49	64	0.6	0.132
	75	10	0.6	19.4	12.3	0.670	16.0	8 900	10 000	**16009**	49	71	0.6	0.170
	75	16	1	26.2	15.1	0.900	15.3	9 200	11 000	**6009**	50	70	1	0.245
	85	19	1.1	40.9	20.3	1.40	14.0	7 700	9 200	**6209**	51.5	78.5	1	0.407
	100	25	1.5	61.1	29.5	2.25	13.3	6 800	8 100	**6309**	53	92	1.5	0.833
	120	29	2	96.5	45.1	3.90	12.2	6 000	7 200	**6409**	54	111	2	1.53
50	65	7	0.3	8.20	6.10	0.260	16.1	9 600	11 000	**6810**	52	63	0.3	0.052
	72	12	0.6	18.2	11.7	0.640	16.1	9 000	11 000	**6910**	54	68	0.6	0.133
	80	10	0.6	20.0	13.3	0.710	16.2	8 200	9 700	**16010**	54	76	0.6	0.180
	80	16	1	27.3	16.6	0.960	15.6	8 400	9 900	**6010**	55	75	1	0.261
	90	20	1.1	43.9	23.3	1.55	14.4	7 100	8 500	**6210**	56.5	83.5	1	0.463
	90	20	1.1	50.5	25.5	1.80	13.9	7 100	8 600	**6210R**	56.5	83.5	1	0.487
	110	27	2	77.5	38.3	2.90	13.2	6 100	7 300	**6310**	59	101	2	1.07
	130	31	2.1	104	49.5	4.10	12.5	5 500	6 600	**6410**	61	119	2	1.88
55	72	9	0.3	11.0	8.10	0.420	16.2	8 700	10 000	**6811**	57	70	0.3	0.083
	80	13	1	20.8	14.1	0.760	16.2	8 100	9 600	**6911**	60	75	1	0.185
	90	11	0.6	24.2	16.3	0.880	16.2	7 400	8 800	**16011**	59	86	0.6	0.260
	90	18	1.1	35.3	21.2	1.25	15.3	7 600	8 900	**6011**	61.5	83.5	1	0.385
	100	21	1.5	54.2	29.4	1.95	14.4	6 300	7 600	**6211**	63	92	1.5	0.607
	120	29	2	89.5	45.0	3.45	13.2	5 600	6 700	**6311**	64	111	2	1.37
	140	33	2.1	126	62.3	5.35	12.2	5 000	6 000	**6411**	66	129	2	2.29
60	78	10	0.3	14.3	10.6	0.550	16.3	8 000	9 400	**6812**	62	76	0.3	0.104
	85	13	1	25.2	17.3	0.940	16.2	7 500	8 900	**6912**	65	80	1	0.192
	95	11	0.6	24.8	17.6	0.930	16.4	6 900	8 100	**16012**	64	91	0.6	0.280
	95	18	1.1	36.8	23.2	1.35	15.6	7 100	8 400	**6012**	66.5	88.5	1	0.415

open type

d (60) ~ (80) mm



Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
60	110	22	1.5	65.6	36.2	2.40	14.4	5 700	6 900	**6212**	68	102	1.5	0.783
	130	31	2.1	102	52.2	3.95	13.2	5 200	6 200	**6312**	71	119	2	1.70
	150	35	2.1	137	70.8	5.75	12.4	4 600	5 500	**6412**	71	139	2	2.77
65	85	10	0.6	14.9	11.5	0.590	16.2	7 300	8 600	**6813**	69	81	0.6	0.126
	90	13	1	21.7	16.1	0.830	16.6	7 100	8 400	**6913**	70	85	1	0.211
	100	11	0.6	21.4	16.0	0.830	16.5	6 600	7 800	**16013**	69	96	0.6	0.300
	100	18	1.1	38.1	25.2	1.40	15.8	6 600	7 800	**6013**	71.5	93.5	1	0.435
	120	23	1.5	71.5	40.1	2.65	14.4	5 400	6 400	**6213**	73	112	1.5	0.990
	140	33	2.1	116	59.9	4.50	13.2	4 800	5 800	**6313**	76	129	2	2.08
	160	37	2.1	148	79.2	6.20	12.4	4 300	5 200	**6413**	76	149	2	3.30
70	90	10	0.6	15.1	11.9	0.620	16.1	6 800	8 100	**6814**	74	86	0.6	0.134
	100	16	1	29.7	21.2	1.10	16.3	6 400	7 600	**6914**	75	95	1	0.342
	110	13	0.6	37.6	25.6	1.40	16.0	6 100	7 200	**16014**	74	106	0.6	0.433
	110	20	1.1	47.6	30.9	1.80	15.6	6 100	7 200	**6014**	76.5	103.5	1	0.602
	125	24	1.5	77.8	44.1	2.90	14.5	5 100	6 100	**6214**	78	117	1.5	1.07
	150	35	2.1	130	68.2	4.95	13.2	4 500	5 400	**6314**	81	139	2	2.52
	180	42	3	181	104	10.2	12.2	3 900	4 600	**6414**	83	167	2.5	4.83
75	95	10	0.6	15.7	12.9	0.660	16.0	6 400	7 600	**6815**	79	91	0.6	0.142
	105	16	1	30.5	22.6	1.20	16.5	6 100	7 200	**6915**	80	100	1	0.363
	115	13	0.6	34.4	25.3	1.35	16.4	5 700	6 700	**16015**	79	111	0.6	0.457
	115	20	1.1	49.4	33.5	1.90	15.8	5 700	6 800	**6015**	81.5	108.5	1	0.638
	130	25	1.5	84.3	48.3	3.10	14.5	4 800	5 800	**6215**	83	122	1.5	1.18
	160	37	2.1	142	77.2	5.40	13.2	4 200	5 000	**6315**	86	149	2	3.02
	190	45	3	192	115	10.9	12.3	3 600	4 400	**6415**	88	177	2.5	5.87
80	100	10	0.6	15.9	13.3	0.690	16.0	6 100	7 200	**6816**	84	96	0.6	0.150
	110	16	1	31.2	24.0	1.25	16.6	5 700	6 800	**6916**	85	105	1	0.382
	125	14	0.6	39.7	29.7	1.50	16.4	5 200	6 100	**16016**	84	121	0.6	0.597

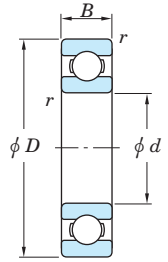
[Remark] Standard cage types used for the above bearings are described earlier in this section.

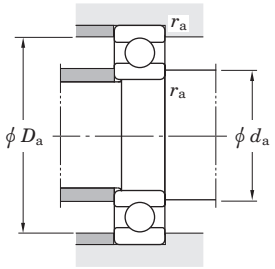
d (80) ~ (100) mm

Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
80	125	22	1.1	59.5	39.8	2.25	15.6	5 300	6 300	**6016**	86.5	118.5	1	0.850
	140	26	2	90.9	53.0	3.25	14.6	4 500	5 400	**6216**	89	131	2	1.40
	170	39	2.1	154	86.7	5.85	13.3	3 900	4 700	**6316**	91	159	2	3.59
	200	48	3	205	125	11.5	12.3	3 400	4 100	**6416**	93	187	2.5	6.84
85	110	13	1	23.4	19.0	0.980	16.2	5 600	6 600	**6817**	90	105	1	0.266
	120	18	1.1	39.9	29.6	1.55	16.4	5 300	6 300	**6917**	91.5	113.5	1	0.535
	130	14	0.6	40.8	31.7	1.55	16.5	4 900	5 800	**16017**	89	126	0.6	0.626
	130	22	1.1	61.8	43.1	2.35	15.8	5 000	5 900	**6017**	91.5	123.5	1	0.890
	150	28	2	105	61.9	3.70	14.5	4 200	5 000	**6217**	94	141	2	1.79
	180	41	3	166	96.8	6.35	13.3	3 700	4 400	**6317**	98	167	2.5	4.23
	210	52	4	217	136	12.2	12.3	3 300	3 900	**6417**	101	194	3	8.07
90	115	13	1	23.8	19.7	1.00	16.1	5 300	6 300	**6818**	95	110	1	0.279
	125	18	1.1	41.0	31.6	1.60	16.5	5 100	6 000	**6918**	96.5	118.5	1	0.565
	140	16	1	49.9	37.0	1.85	16.3	4 700	5 600	**16018**	95	135	1	0.848
	140	24	1.5	72.8	49.7	2.65	15.6	4 700	5 600	**6018**	98	132	1.5	1.16
	160	30	2	120	71.5	4.20	14.5	3 900	4 700	**6218**	99	151	2	2.15
	190	43	3	178	107	8.80	13.3	3 500	4 200	**6318**	103	177	2.5	4.91
	225	54	4	230	149	12.7	12.5	3 100	3 700	**6418**	106	209	3	9.78
95	130	18	1.1	42.1	33.5	1.65	16.6	4 800	5 700	**6919**	101.5	123.5	1	0.705
	145	16	1	51.5	39.6	1.90	16.4	4 500	5 300	**16019**	100	140	1	0.885
	145	24	1.5	75.5	53.9	2.75	15.8	4 400	5 200	**6019**	103	137	1.5	1.21
	170	32	2.1	136	81.9	4.65	14.4	3 700	4 400	**6219**	106	159	2	2.62
	200	45	3	191	119	9.45	13.3	3 300	4 000	**6319**	108	187	2.5	5.67
100	125	13	1	24.5	21.2	1.05	16.0	4 800	5 700	**6820**	105	120	1	0.309
	140	20	1.1	56.2	41.9	2.05	16.2	4 500	5 300	**6920**	106.5	133.5	1	0.960
	150	16	1	53.0	42.1	1.95	16.5	4 300	5 100	**16020**	105	145	1	0.910
	150	24	1.5	75.2	54.2	2.70	15.9	4 300	5 100	**6020**	108	142	1.5	1.25

open type

d (100) ~ (140) mm





Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
100	180	34	2.1	153	93.1	5.15	14.4	3 500	4 200	**6220**	111	169	2	3.14
	215	47	3	216	141	10.9	13.2	3 000	3 600	**6320**	113	202	2.5	7.00
105	145	20	1.1	58.1	44.8	2.10	16.4	4 300	5 100	**6921**	111.5	138.5	1	1.00
	160	18	1	52.3	42.2	1.90	16.5	4 100	4 800	**16021**	110	155	1	1.20
	160	26	2	90.4	65.8	3.20	15.8	4 000	4 700	**6021**	114	151	2	1.59
	190	36	2.1	166	105	5.70	14.4	3 300	3 900	**6221**	116	179	2	3.70
	225	49	3	230	153	11.7	13.2	2 900	3 500	**6321**	118	212	2.5	8.05
110	140	16	1	35.1	30.7	1.40	16.1	4 300	5 100	**6822**	115	135	1	0.506
	150	20	1.1	59.9	47.8	2.20	16.4	4 100	4 900	**6922**	116.5	143.5	1	1.04
	170	19	1	71.8	56.7	2.55	16.3	3 800	4 500	**16022**	115	165	1	1.46
	170	28	2	103	73.0	3.55	15.6	3 800	4 500	**6022**	119	161	2	1.96
	200	38	2.1	180	117	6.20	14.4	3 100	3 700	**6222**	121	189	2	4.36
	240	50	3	257	180	13.3	13.2	2 700	3 200	**6322**	123	227	2.5	9.54
120	150	16	1	36.2	33.0	1.45	16.0	4 000	4 700	**6824**	125	145	1	0.655
	165	22	1.1	71.6	56.9	2.50	16.4	3 800	4 400	**6924**	126.5	158.5	1	1.41
	180	19	1	79.0	63.3	2.75	16.4	3 600	4 200	**16024**	125	175	1	1.80
	180	28	2	106	79.3	3.60	15.9	3 600	4 200	**6024**	129	171	2	2.07
	215	40	2.1	194	131	6.65	14.4	2 900	3 400	**6224**	131	204	2	5.15
	260	55	3	258	185	12.6	13.5	2 500	3 000	**6324**	133	247	2.5	12.5
130	165	18	1.1	46.1	41.2	1.75	16.1	3 600	4 300	**6826**	136.5	158.5	1	0.939
	180	24	1.5	86.9	67.4	3.00	16.3	3 400	4 100	**6926**	138	172	1.5	1.86
	200	22	1.1	89.1	74.8	3.05	11.2	3 000	3 600	**16026**	136.5	193.5	1	2.69
	200	33	2	133	101	4.45	15.8	3 200	3 800	**6026**	139	191	2	3.16
	230	40	3	209	146	9.15	14.5	2 700	3 200	**6226**	143	217	2.5	5.82
	280	58	4	287	214	14.1	13.6	2 300	2 700	**6326**	146	264	3	15.1
140	175	18	1.1	47.8	44.4	1.85	16.0	3 400	4 000	**6828**	146.5	168.5	1	0.837
	190	24	1.5	89.1	74.8	3.05	16.5	3 200	3 800	**6928**	148	182	1.5	1.98
	210	22	1.1	82.2	71.1	2.80	16.5	2 900	3 400	**16028**	146.5	203.5	1	2.86

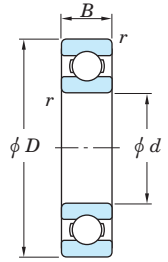
[Remark] Standard cage types used for the above bearings are described earlier in this section.

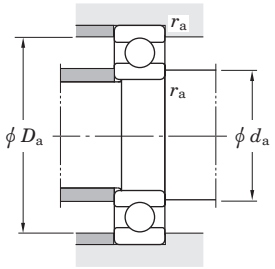
d (140) ~ (190) mm

Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
140	210	33	2	137	109	4.55	15.9	3 000	3 600	**6028**	149	201	2	3.55
	250	42	3	208	150	8.65	14.8	2 400	2 900	**6228**	153	237	2.5	7.45
	300	62	4	316	246	15.6	13.6	2 100	2 500	**6328**	156	284	3	19.4
150	190	20	1.1	59.7	54.9	2.20	16.1	3 100	3 700	**6830**	156.5	183.5	1	1.40
	210	28	2	117	94.3	3.75	16.2	2 900	3 400	**6930**	159	201	2	3.05
	225	24	1.1	114	99.3	3.70	16.6	2 700	3 100	**16030**	156.5	218.5	1	3.58
	225	35	2.1	157	126	5.10	16.0	2 800	3 300	**6030**	161	214	2	4.22
	270	45	3	220	168	9.05	15.1	2 200	2 700	**6230**	163	257	2.5	9.41
	320	65	4	343	284	16.6	13.9	1 900	2 300	**6330**	166	304	3	26.2
160	200	20	1.1	60.5	56.9	2.20	16.1	2 900	3 400	**6832**	166.5	193.5	1	1.45
	220	28	2	120	101	3.85	16.4	2 700	3 200	**6932**	169	211	2	3.20
	240	25	1.5	124	108	3.95	16.5	2 600	3 100	**16032**	168	232	1.5	4.25
	240	38	2.1	171	135	5.30	15.9	2 600	3 000	**6032**	171	229	2	5.22
	290	48	3	231	186	9.45	15.4	2 100	2 500	**6232**	173	277	2.5	14.3
	340	68	4	347	286	16.4	13.9	1 800	2 200	**6332**	176	324	3	29.0
170	215	22	1.1	74.8	70.5	2.60	16.1	2 700	3 200	**6834**	176.5	208.5	1	1.90
	230	28	2	124	108	3.95	16.5	2 600	3 100	**6934**	179	221	2	3.35
	260	28	1.5	142	127	4.45	16.5	2 300	2 700	**16034**	178	252	1.5	5.75
	260	42	2.1	201	161	6.20	15.8	2 400	2 800	**6034**	181	249	2	6.80
	310	52	4	265	223	11.1	15.3	1 900	2 300	**6234**	186	294	3	17.5
	360	72	4	408	355	20.5	13.6	1 700	2 000	**6334**	186	344	3	38.6
180	225	22	1.1	75.8	73.1	2.65	16.1	2 600	3 000	**6836**	186.5	218.5	1	2.00
	250	33	2	153	129	4.70	16.3	2 400	2 800	**6936**	189	241	2	4.90
	280	31	2	169	148	5.15	16.4	2 100	2 500	**16036**	189	271	2	7.55
	280	46	2.1	227	194	7.15	15.8	2 200	2 600	**6036**	191	269	2	10.3
	320	52	4	284	241	12.0	15.1	1 800	2 200	**6236**	196	304	3	18.3
	380	75	4	443	407	22.1	13.9	1 600	1 900	**6336**	196	364	3	44.7
190	240	24	1.5	91.4	88.1	3.10	16.1	2 400	2 800	**6838**	198	232	1.5	2.60

open type

d (190) ~ (260) mm





Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	(kg)
190	260	33	2	158	138	4.85	16.4	2 300	2 700	**6938**	199	251	2	5.20
	290	31	2	173	158	5.20	16.6	2 000	2 400	**16038**	199	281	2	7.85
	290	46	2.1	235	201	7.35	15.8	2 100	2 500	**6038**	201	279	2	10.8
	340	55	4	319	281	13.7	15.0	1 700	2 000	**6238**	206	324	3	23.0
	400	78	5	443	415	21.3	14.1	1 500	1 800	**6338**	210	380	4	51.5
200	250	24	1.5	97.6	93.6	3.20	16.1	2 300	2 700	**6840**	208	242	1.5	2.70
	280	38	2.1	196	168	5.80	16.2	2 100	2 500	**6940**	211	269	2	7.30
	310	34	2	201	180	5.95	16.4	1 900	2 300	**16040**	209	301	2	10.1
	310	51	2.1	272	243	11.3	15.6	1 900	2 300	**6040**	211	299	2	14.0
	360	58	4	336	311	14.4	15.2	1 600	1 900	**6240**	216	344	3	28.2
	420	80	5	513	506	25.5	14.0	1 300	1 600	**6340**	220	400	4	58.0
220	270	24	1.5	101	101	3.35	16.0	2 000	2 400	**6844**	228	262	1.5	3.00
	300	38	2.1	201	180	5.85	16.4	1 900	2 200	**6944**	231	289	2	7.90
	340	37	2.1	225	217	6.65	16.5	1 700	2 000	**16044**	231	329	2	13.2
	340	56	3	294	271	12.0	15.6	1 700	2 000	**6044**	233	327	2.5	18.3
	400	65	4	389	376	16.8	15.1	1 400	1 700	**6244**	236	384	3	37.0
	460	88	5	542	539	26.7	13.8	1 200	1 500	**6344**	240	440	4	71.6
240	300	28	2	135	135	4.25	16.1	1 800	2 100	**6848**	249	291	2	4.50
	320	38	2.1	205	192	5.95	16.5	1 700	2 000	**6948**	251	309	2	8.50
	360	37	2.1	230	228	6.75	16.5	1 600	1 800	**16048**	251	349	2	14.1
	360	56	3	305	296	12.3	15.9	1 600	1 900	**6048**	253	347	2.5	19.7
	440	72	4	424	431	18.2	15.2	1 200	1 500	**6248**	256	424	3	51.0
	500	95	5	587	624	28.2	14.2	1 100	1 300	**6348**	260	480	4	93.3
260	320	28	2	141	146	4.40	16.0	1 700	2 000	**6852**	269	311	2	4.80
	360	46	2.1	266	263	10.2	16.3	1 500	1 800	**6952**	271	349	2	14.4
	400	44	3	295	310	11.5	16.4	1 400	1 600	**16052**	273	387	2.5	21.6
	400	65	4	364	377	15.0	15.8	1 400	1 700	**6052**	276	384	3	29.3
	480	80	5	502	541	22.2	15.1	1 100	1 300	**6252**	280	460	4	68.2

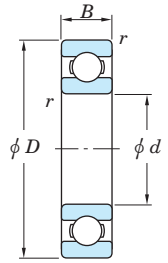
[Remark] Standard cage types used for the above bearings are described earlier in this section.

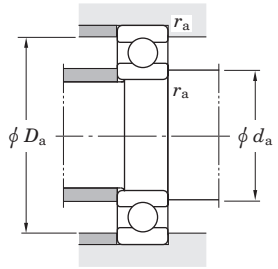
d (260) ~ (360) mm

Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	(kg)
260	540	102	6	663	741	32.4	14.2	990	1 200	**6352**	284	516	5	116
280	350	33	2	179	183	5.35	16.1	1 500	1 800	**6856**	289	341	2	7.40
	380	46	2.1	273	283	10.5	16.5	1 400	1 700	**6956**	291	369	2	15.1
	420	44	3	302	331	11.7	14.7	1 300	1 500	**16056**	293	407	2.5	22.9
	420	65	4	377	408	15.5	16.0	1 300	1 500	**6056**	296	404	3	31.0
	500	80	5	529	599	23.2	15.3	1 000	1 200	**6256**	300	480	4	71.8
	580	108	6	711	845	33.9	14.5	880	1 100	**6356**	304	556	5	145
300	380	38	2.1	224	230	6.45	16.2	1 400	1 600	**6860**	311	369	2	10.5
	420	56	3	345	377	13.7	16.2	1 300	1 500	**6960**	313	407	2.5	24.1
	460	50	4	355	405	14.0	16.4	1 100	1 400	**16060**	316	447	3	32.2
	460	74	4	444	482	18.4	15.6	1 200	1 400	**6060**	316	444	3	44.0
	540	85	5	551	663	23.5	15.6	880	1 100	**6260**	320	520	4	89.5
	620	109	7.5	741	886	35.0	14.4	810	970	**6360**	332	588	6	169
320	400	38	2.1	227	239	6.50	16.1	1 300	1 500	**6864**	331	389	2	11.0
	440	56	3	356	404	14.1	16.4	1 200	1 400	**6964**	333	427	2.5	25.5
	480	50	4	364	432	14.3	16.5	1 100	1 300	**16064**	336	467	3	33.9
	480	74	4	441	487	17.8	15.7	1 100	1 300	**6064**	336	464	3	46.0
	580	92	5	612	745	26.7	15.4	840	1 000	**6264**	340	560	4	113
	670	112	7.5	793	1 010	36.9	14.8	720	870	**6364**	352	638	6	207
340	420	38	2.1	231	249	6.60	16.1	1 200	1 400	**6868**	351	409	2	11.5
	460	56	3	352	407	13.7	16.5	1 100	1 300	**6968**	353	447	2.5	26.8
	520	57	4	419	512	16.8	16.4	980	1 200	**16068**	356	507	3	46.8
	520	82	5	552	661	23.7	15.6	980	1 200	**6068**	360	500	4	61.8
	620	92	6	639	817	27.7	15.6	760	910	**6268**	364	596	5	131
	710	118	7.5	880	1 160	41.7	14.7	660	790	**6368**	372	678	6	238
360	440	38	2.1	240	268	6.95	16.0	1 100	1 300	**6872**	371	429	2	12.0
	480	56	3	362	432	14.0	16.5	1 000	1 200	**6972**	373	467	2.5	28.2
	540	57	4	431	546	17.2	16.5	900	1 100	**16072**	376	527	3	49.0

open type

d (360) ~ (500) mm





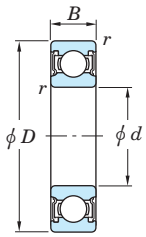
Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
360	540	82	5	548	668	23.0	15.7	920	1 100	**6072**	380	520	4	64.7
	650	95	6	696	904	30.4	15.4	700	840	**6272**	384	626	5	144
380	480	46	2.1	305	359	8.95	16.2	980	1 200	**6876**	391	469	2	20.0
	520	65	4	440	552	17.6	16.4	920	1 100	**6976**	396	504	3	40.8
	560	82	5	572	725	24.1	15.9	860	1 000	**6076**	400	540	4	67.6
	680	95	6	730	990	31.9	15.6	650	780	**6276**	404	656	5	162
400	500	46	2.1	311	374	9.10	16.1	920	1 100	**6880**	411	489	2	20.5
	540	65	4	453	588	18.1	16.5	860	1 000	**6980**	416	524	3	42.7
	600	63	5	447	587	17.5	16.5	780	920	**16080**	420	580	4	65.0
	600	90	5	635	824	27.0	15.7	780	920	**6080**	420	580	4	87.7
	720	103	6	785	1 080	34.2	15.5	590	710	**6280**	424	696	5	197
420	520	46	2.1	316	389	9.25	16.1	860	1 000	**6884**	431	509	2	21.5
	560	65	4	449	588	17.7	16.5	810	950	**6984**	436	544	3	43.5
	620	63	5	459	617	18.0	16.4	740	870	**16084**	440	600	4	69.9
	620	90	5	663	894	28.3	15.8	740	870	**6084**	440	600	4	91.2
440	540	46	2.1	321	404	9.40	16.0	810	950	**6888**	451	529	2	22.5
	600	74	4	529	676	21.4	16.4	740	870	**6988**	456	584	3	61.3
	650	67	5	508	710	20.2	16.5	680	810	**16088**	460	630	4	81.7
460	580	56	3	393	517	11.7	16.2	740	870	**6892**	473	567	2.5	35.0
	620	74	4	509	711	20.3	16.5	690	820	**6992**	476	604	3	61.7
	680	71	5	539	767	21.4	16.5	630	750	**16092**	480	660	4	91.2
480	600	56	3	401	539	12.0	16.1	690	820	**6896**	493	587	2.5	36.5
	650	78	5	540	768	21.5	16.5	640	760	**6996**	500	630	4	72.5
	700	71	5	554	807	22.1	16.5	600	710	**16096**	500	680	4	98.5
500	620	56	3	409	561	12.2	16.1	650	770	**68/500**	513	607	2.5	37.5
	670	78	5	556	807	22.2	16.5	610	720	**69/500**	520	650	4	75.2

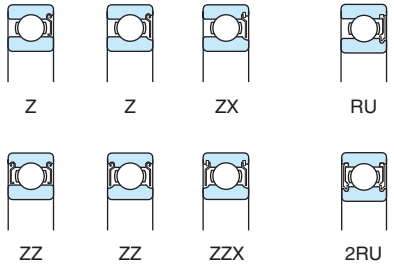
[Remark] Standard cage types used for the above bearings are described earlier in this section.

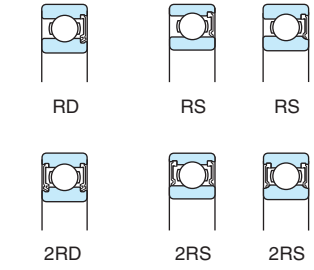
d (500) mm

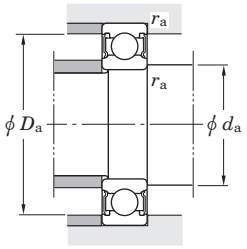
Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
500	720	71	5	568	846	22.7	16.4	560	660	**160/500**	520	700	4	102
	720	100	6	749	1 100	31.3	16.0	570	670	**60/500**	524	696	5	128

shielded type
sealed type
d 10 ~ (20) mm





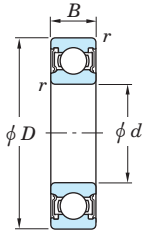


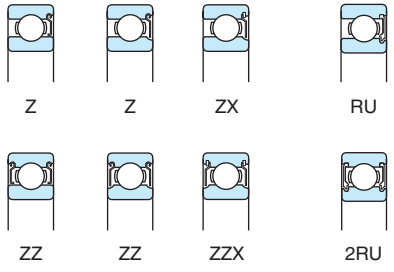


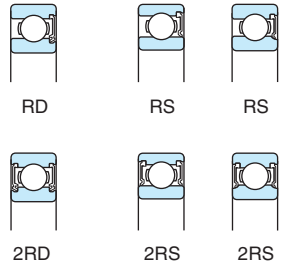
Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1}) Grease lub.			Oil lub.	Bearing No.				Mounting dimensions. (mm)				(Refer.) Mass
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Z, ZZ RU, 2RU	(RD, 2RD)	(RS, 2RS)	(Z)	Shielded	Non-contact sealed	Extremely light contact sealed	Contact sealed	d_a min.	d_a max.	D_a max.	r_a max.	Open type (kg)
10	19	5	0.3	2.15	0.840	0.030	14.8	37 000	—	22 000	43 000	**6800 ZZ**	**6800 2RU**	—	**6800 2RS**	12	12	17	0.3	0.005
	22	6	0.3	3.35	1.25	0.070	14.0	34 000	—	21 000	41 000	**6900 ZZ**	**6900 2RU**	—	**6900 2RS**	12	12.5	20	0.3	0.010
	26	8	0.3	5.70	1.95	0.100	12.3	31 000	28 000	19 000	36 000	**6000 ZZ**	**6000 2RU**	**6000 2RD**	**6000 2RS**	12	13	24	0.3	0.019
	30	9	0.6	6.40	2.40	0.120	13.2	24 000	22 000	16 000	29 000	**6200 ZZ**	**6200 2RU**	**6200 2RD**	**6200 2RS**	14	15	26	0.6	0.032
	35	11	0.6	10.1	3.45	0.270	11.2	22 000	20 000	16 000	27 000	**6300 ZZ**	**6300 2RU**	**6300 2RD**	**6300 2RS**	14	16	31	0.6	0.053
12	18	4	0.2	1.15	0.530	0.023	16.2	34 000	—	20 000	41 000	**6701 ZZX**	**6701 2RU**	—	**6701 2RS**	13.6	—	16.4	0.2	0.003
	21	5	0.3	2.40	1.05	0.040	15.3	33 000	30 000	20 000	39 000	**6801 ZZ**	**6801 2RU**	**6801 2RD**	**6801 2RS**	14	14	19	0.3	0.006
	24	6	0.3	3.60	1.45	0.080	14.5	31 000	28 000	18 000	36 000	**6901 ZZ**	**6901 2RU**	**6901 2RD**	**6901 2RS**	14	14	22	0.3	0.011
	28	8	0.3	6.40	2.40	0.120	13.2	27 000	24 000	17 000	32 000	**6001 ZZ**	**6001 2RU**	**6001 2RD**	**6001 2RS**	14	15	26	0.3	0.022
	32	10	0.6	8.50	3.05	0.240	12.3	22 000	20 000	15 000	27 000	**6201 ZZ**	**6201 2RU**	**6201 2RD**	**6201 2RS**	16	16.5	28	0.6	0.037
	37	12	1	12.1	4.20	0.420	11.1	20 000	18 000	15 000	25 000	**6301 ZZ**	**6301 2RU**	**6301 2RD**	**6301 2RS**	17	17.5	32	1	0.060
15	21	4	0.2	1.15	0.580	0.024	16.7	29 000	—	16 000	35 000	**6702 ZZX**	**6702 2RU**	—	**6702 2RS**	16.6	—	19.4	0.2	0.004
	24	5	0.3	2.60	1.25	0.050	15.8	28 000	—	16 000	33 000	**6802 ZZ**	**6802 2RU**	—	**6802 2RS**	17	17	22	0.3	0.007
	28	7	0.3	5.40	2.25	0.120	14.3	26 000	23 000	15 000	30 000	**6902 ZZ**	**6902 2RU**	**6902 2RD**	**6902 2RS**	17	18	26	0.3	0.017
	32	9	0.3	7.00	2.85	0.150	13.9	23 000	21 000	14 000	27 000	**6002 ZZ**	**6002 2RU**	**6002 2RD**	**6002 2RS**	17	18.5	30	0.3	0.030
	35	11	0.6	9.55	3.75	0.290	13.2	20 000	18 000	13 000	24 000	**6202 ZZ**	**6202 2RU**	**6202 2RD**	**6202 2RS**	19	19.5	31	0.6	0.045
	42	13	1	14.3	5.45	0.460	12.3	17 000	15 000	12 000	20 000	**6302 ZZ**	**6302 2RU**	**6302 2RD**	**6302 2RS**	20	21.5	37	1	0.082
17	23	4	0.2	1.25	0.660	0.027	16.9	27 000	—	15 000	32 000	**6703 ZZ**	**6703 2RU**	—	**6703 2RS**	18.6	—	21.4	0.2	0.005
	26	5	0.3	3.30	1.55	0.060	15.7	26 000	—	14 000	30 000	**6803 ZZ**	**6803 2RU**	—	**6803 2RS**	19	19	24	0.3	0.008
	30	7	0.3	5.75	2.55	0.130	14.7	23 000	21 000	13 000	28 000	**6903 ZZ**	**6903 2RU**	**6903 2RD**	**6903 2RS**	19	19.5	28	0.3	0.018
	35	10	0.3	7.50	3.25	0.170	14.4	21 000	19 000	12 000	25 000	**6003 ZZ**	**6003 2RU**	**6003 2RD**	**6003 2RS**	19	21	33	0.3	0.039
	40	12	0.6	12.0	4.80	0.370	13.2	17 000	15 000	12 000	21 000	**6203 ZZ**	**6203 2RU**	**6203 2RD**	**6203 2RS**	21	22	36	0.6	0.065
	47	14	1	17.0	6.65	0.550	12.4	15 000	14 000	10 000	18 000	**6303 ZZ**	**6303 2RU**	**6303 2RD**	**6303 2RS**	22	24.3	42	1	0.115
20	27	4	0.2	1.30	0.730	0.030	16.1	23 000	—	12 000	27 000	**6704 ZZ**	**6704 2RU**	—	**6704 2RS**	21.6	—	25.4	0.2	0.006
	32	7	0.3	5.00	2.45	0.100	15.5	21 000	—	12 000	25 000	**6804 ZZ**	**6804 2RU**	—	**6804 2RS**	22	22.5	30	0.3	0.018
	37	9	0.3	7.95	3.70	0.190	14.7	19 000	17 000	11 000	23 000	**6904 ZZ**	**6904 2RU**	**6904 2RD**	**6904 2RS**	22	23.5	35	0.3	0.036
	42	12	0.6	11.7	5.05	0.350	13.9	17 000	15 000	10 000	21 000	**6004 ZZ**	**6004 2RU**	**6004 2RD**	**6004 2RS**	24	25	38	0.6	0.069
	47	14	1	16.0	6.65	0.510	13.2	15 000	14 000	9 700	17 000	**6204 ZZ**	**6204 2RU**	**6204 2RD**	**6204 2RS**	25	26.5	42	1	0.106

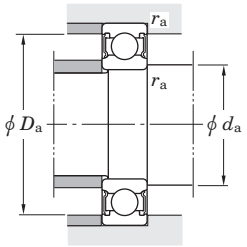
[Remark] Standard cage types used for the above bearings are described earlier in this section.

shielded type
sealed type
d (20) ~ 35 mm





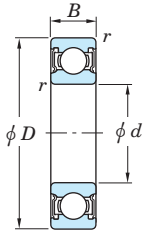


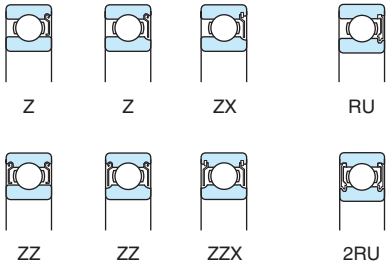


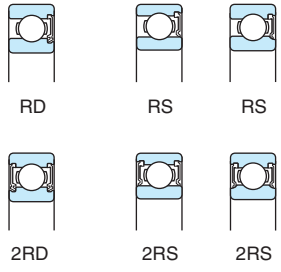
Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1}) Grease lub.			Oil lub.	Bearing No.				Mounting dimensions. (mm)				(Refer.) Mass
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Z, ZZ RU, 2RU	(RD, 2RD)	(RS, 2RS)	(Z)	Shielded	Non-contact sealed	Extremely light contact sealed	Contact sealed	d_a min.	d_a max.	D_a max.	r_a max.	Open type (kg)
20	52	15	1.1	19.9	7.85	0.660	12.3	14 000	13 000	9 500	17 000	**6304 ZZ**	**6304 2RU**	**6304 2RD**	**6304 2RS**	26.5	27	45.5	1	0.144
22	44	12	0.6	11.7	5.15	0.350	14.1	17 000	15 000	9 900	20 000	**60/22 ZZ**	**60/22 2RU**	**60/22 2RD**	**60/22 2RS**	26	26.5	40	0.6	0.073
	50	14	1	16.0	6.65	0.510	13.2	15 000	14 000	9 700	17 000	**62/22 ZZ**	**62/22 2RU**	**62/22 2RD**	**62/22 2RS**	27	27	45	1	0.118
	56	16	1.1	23.1	9.40	0.770	12.6	13 000	12 000	8 600	15 000	**63/22 ZZ**	**63/22 2RU**	**63/22 2RD**	**63/22 2RS**	28.5	29	49.5	1	0.201
25	32	4	0.2	1.35	0.840	0.035	15.8	19 000	—	10 000	22 000	**6705 ZZ**	**6705 2RU**	—	**6705 2RS**	26.6	—	30.4	0.2	0.006
	37	7	0.3	5.40	2.95	0.120	16.0	18 000	—	10 000	21 000	**6805 ZZ**	**6805 2RU**	—	**6805 2RS**	27	27.5	35	0.3	0.022
	42	9	0.3	8.75	4.55	0.230	15.4	16 000	14 000	9 300	19 000	**6905 ZZ**	**6905 2RU**	**6905 2RD**	**6905 2RS**	27	29	40	0.3	0.041
	47	12	0.6	12.6	5.85	0.380	14.5	15 000	14 000	9 000	18 000	**6005 ZZ**	**6005 2RU**	**6005 2RD**	**6005 2RS**	29	29.5	43	0.6	0.080
	52	15	1	17.5	7.85	0.550	13.9	13 000	12 000	8 400	15 000	**6205 ZZ**	**6205 2RU**	**6205 2RD**	**6205 2RS**	30	31.5	47	1	0.128
	62	17	1.1	25.7	11.3	0.860	13.2	11 000	9 900	7 500	13 000	**6305 ZZ**	**6305 2RU**	**6305 2RD**	**6305 2RS**	31.5	34	55.5	1	0.232
28	52	12	0.6	15.6	7.40	0.480	14.5	14 000	13 000	8 100	16 000	**60/28 ZZ**	**60/28 2RU**	**60/28 2RD**	**60/28 2RS1**	32	33	48	0.6	0.097
	58	16	1	22.4	9.75	0.720	13.4	12 000	11 000	7 600	14 000	**62/28 ZZ**	**62/28 2RU**	**62/28 2RD**	**62/28 2RS**	33	35	53	1	0.173
	68	18	1.1	29.4	13.1	0.990	13.3	10 000	9 000	6 900	12 000	**63/28 ZZ**	**63/28 2RU**	**63/28 2RD**	**63/28 2RS**	34.5	37.5	61.5	1	0.328
30	37	4	0.2	1.45	0.950	0.040	15.7	16 000	—	8 800	19 000	**6706 ZZ**	**6706 2RU**	—	**6706 2RS**	31.6	—	35.4	0.2	0.008
	42	7	0.3	5.65	3.40	0.140	16.4	15 000	—	8 600	18 000	**6806 ZZ**	**6806 2RU**	—	**6806 2RS**	32	32.5	40	0.3	0.026
	47	9	0.3	9.05	5.00	0.260	15.8	14 000	13 000	8 200	17 000	**6906 ZZ**	**6906 2RU**	**6906 2RD**	**6906 2RS**	32	33	45	0.3	0.045
	55	13	1	16.5	8.25	0.530	14.7	13 000	12 000	7 500	15 000	**6006 ZZ**	**6006 2RU**	**6006 2RD**	**6006 2RS**	35	36	50	1	0.116
	62	16	1	24.3	11.3	0.800	13.9	11 000	9 900	7 000	13 000	**6206 ZZ**	**6206 2RU**	**6206 2RD**	**6206 2RS**	35	37.5	57	1	0.199
	72	19	1.1	33.3	15.0	1.15	13.3	9 600	8 600	6 400	12 000	**6306 ZZ**	**6306 2RU**	**6306 2RD**	**6306 2RS**	36.5	40	65.5	1	0.346
32	58	13	1	18.8	9.15	0.600	14.5	12 000	11 000	7 200	14 000	**60/32 ZZ**	**60/32 2RU**	**60/32 2RD**	**60/32 2RS**	37	38	53	1	0.127
	65	17	1	29.4	13.1	0.990	13.3	10 000	9 000	6 900	12 000	**62/32 ZZ**	**62/32 2RU**	**62/32 2RD**	**62/32 2RS**	37	38.5	60	1	0.228
	75	20	1.1	37.6	16.2	1.30	12.7	9 300	8 400	6 400	11 000	**63/32 ZZ**	**63/32 2RU**	**63/32 2RD**	**63/32 2RS**	38.5	41	68.5	1	0.437
35	47	7	0.3	5.95	3.85	0.160	16.5	13 000	—	7 400	16 000	**6807 ZZ**	**6807 2RU**	—	**6807 2RS**	37	37.5	45	0.3	0.030
	55	10	0.6	13.6	7.75	0.440	15.7	12 000	11 000	6 800	14 000	**6907 ZZ**	**6907 2RU**	**6907 2RD**	**6907 2RS**	39	40	51	0.6	0.073
	62	14	1	19.9	10.3	0.640	14.9	11 000	9 900	6 500	13 000	**6007 ZZ**	**6007 2RU**	**6007 2RD**	**6007 2RS**	40	42	58	1	0.155
	72	17	1.1	32.1	15.4	1.10	13.9	9 200	8 300	6 000	11 000	**6207 ZZ**	**6207 2RU**	**6207 2RD**	**6207 2RS**	41.5	43.5	65.5	1	0.288
	80	21	1.5	41.7	19.3	1.45	13.2	8 500	7 700	5 700	10 000	**6307 ZZ**	**6307 2RU**	**6307 2RD**	**6307 2RS**	43	46	72	1.5	0.457

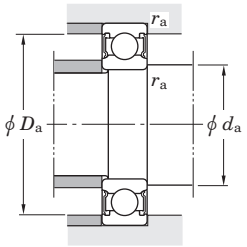
[Remark] Standard cage types used for the above bearings are described earlier in this section.

shielded type
sealed type
d 40 ~ (65) mm







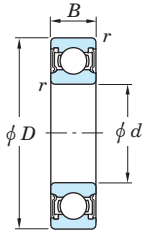


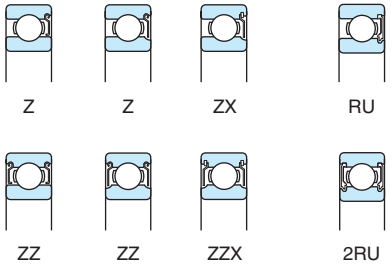
Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1}) Grease lub.			Oil lub.	Bearing No.				Mounting dimensions. (mm)				(Refer.) Mass
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Z, ZZ RU, 2RU	(RD, 2RD)	(RS, 2RS)	(Z)	Shielded	Non-contact sealed	Extremely light contact sealed	Contact sealed	d_a min.	d_a max.	D_a max.	r_a max.	Open type (kg)
40	52	7	0.3	6.15	4.20	0.180	16.3	12 000	11 000	6 700	14 000	**6808 ZZ**	**6808 2RU**	**6808 2RD**	**6808 2RS**	42	42	50	0.3	0.033
	62	12	0.6	17.1	9.95	0.570	15.6	11 000	9 900	6 100	13 000	**6908 ZZ**	**6908 2RU**	**6908 2RD**	**6908 2RS**	44	44.5	58	0.6	0.112
	68	15	1	20.9	11.5	0.690	15.2	10 000	9 000	5 800	12 000	**6008 ZZ**	**6008 2RU**	**6008 2RD**	**6008 2RS**	45	46.5	63	1	0.192
	80	18	1.1	36.4	17.8	1.25	14.0	8 300	7 500	5 400	10 000	**6208 ZZ**	**6208 2RU**	**6208 2RD**	**6208 2RS**	46.5	49	73.5	1	0.366
	90	23	1.5	50.9	24.0	1.85	13.2	7 700	6 900	5 100	9 200	**6308 ZZ**	**6308 2RU**	**6308 2RD**	**6308 2RS**	48	51.5	82	1.5	0.633
45	58	7	0.3	7.75	5.40	0.230	16.3	11 000	9 900	5 900	13 000	**6809 ZZ**	**6809 2RU**	**6809 2RD**	**6809 2RS**	47	47	56	0.3	0.040
	68	12	0.6	17.7	10.9	0.600	15.9	9 700	8 700	5 500	11 000	**6909 ZZ**	**6909 2RU**	**6909 2RD**	**6909 2RS**	49	50	64	0.6	0.132
	75	16	1	26.2	15.1	0.900	15.3	9 200	8 300	5 300	11 000	**6009 ZZ**	**6009 2RU**	**6009 2RD**	**6009 2RS**	50	51.5	70	1	0.245
	85	19	1.1	40.9	20.3	1.40	14.0	7 700	6 900	5 100	9 200	**6209 ZZ**	**6209 2RU**	**6209 2RD**	**6209 2RS**	51.5	53.5	78.5	1	0.407
	100	25	1.5	61.1	29.5	2.25	13.3	6 800	6 100	4 500	8 100	**6309 ZZ**	**6309 2RU**	**6309 2RD**	**6309 2RS**	53	59.5	92	1.5	0.833
50	65	7	0.3	8.20	6.10	0.260	16.1	9 600	8 600	5 200	11 000	**6810 ZZ**	**6810 2RU**	**6810 2RD**	**6810 2RS**	52	53	63	0.3	0.052
	72	12	0.6	18.2	11.7	0.640	16.1	9 000	—	5 000	11 000	**6910 ZZ**	**6910 2RU**	—	—	54	55.5	68	0.6	0.133
	80	16	1	27.3	16.6	0.960	15.6	8 400	7 600	4 800	9 900	**6010 ZZ**	**6010 2RU**	**6010 2RD**	**6010 2RS**	55	57	75	1	0.261
	90	20	1.1	43.9	23.3	1.55	14.4	7 100	6 400	4 600	8 500	**6210 ZZ**	**6210 2RU**	**6210 2RD**	**6210 2RS**	56.5	59	83.5	1	0.463
	110	27	2	77.5	38.3	2.90	13.2	6 100	5 500	4 100	7 300	**6310 ZZ**	**6310 2RU**	**6310 2RD**	**6310 2RS**	59	66.5	101	2	1.07
55	72	9	0.3	11.0	8.10	0.420	16.2	8 700	7 800	—	10 000	**6811 ZZ**	**6811 2RU**	**6811 2RD**	—	57	58.5	70	0.3	0.083
	80	13	1	20.8	14.1	0.760	16.2	8 100	7 300	4 500	9 600	**6911 ZZ**	**6911 2RU**	**6911 2RD**	**6911 2RS**	60	60.5	75	1	0.185
	90	18	1.1	35.3	21.2	1.25	15.3	7 600	6 800	4 300	8 900	**6011 ZZ**	**6011 2RU**	**6011 2RD**	**6011 2RS**	61.5	62	83.5	1	0.385
	100	21	1.5	54.2	29.4	1.95	14.4	6 300	5 700	4 100	7 600	**6211 ZZ**	**6211 2RU**	**6211 2RD**	**6211 2RS**	63	66	92	1.5	0.607
	120	29	2	89.5	45.0	3.45	13.2	5 600	—	3 700	6 700	**6311 ZZ**	**6311 2RU**	—	**6311 2RS**	64	74.5	111	2	1.37
60	78	10	0.3	14.3	10.6	0.550	16.3	8 000	7 200	—	9 400	**6812 ZZ**	**6812 2RU**	**6812 2RD**	—	62	63	76	0.3	0.104
	85	13	1	25.2	17.3	0.940	16.2	7 500	—	—	8 900	**6912 ZZ**	**6912 2RU**	—	—	65	66	80	1	0.192
	95	18	1.1	36.8	23.2	1.35	15.6	7 100	—	4 000	8 400	**6012 ZZ**	**6012 2RU**	—	**6012 2RS**	66.5	68.5	88.5	1	0.415
	110	22	1.5	65.6	36.2	2.40	14.4	5 700	5 100	3 700	6 900	**6212 ZZ**	**6212 2RU**	**6212 2RD**	**6212 2RS**	68	72.5	102	1.5	0.783
	130	31	2.1	102	52.2	3.95	13.2	5 200	—	3 500	6 200	**6312 ZZ**	**6312 2RU**	—	**6312 2RS**	71	80	119	2	1.70
65	85	10	0.6	14.9	11.5	0.590	16.2	7 300	6 600	—	8 600	**6813 ZZ**	**6813 2RU**	**6813 2RD**	—	69	69	81	0.6	0.126
	90	13	1	21.7	16.1	0.830	16.6	7 100	6 400	3 900	8 400	**6913 ZZ**	**6913 2RU**	**6913 2RD**	**6913 2RS**	70	71	85	1	0.211

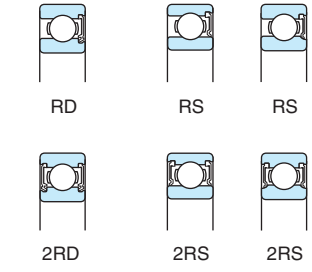
[Remark] Standard cage types used for the above bearings are described earlier in this section.

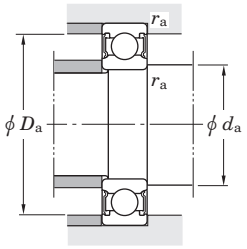
shielded type
sealed type

d (65) ~ (90) mm







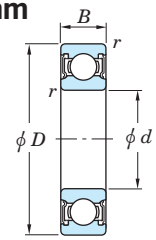


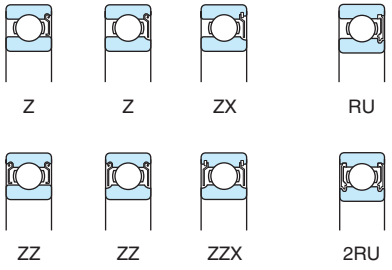
Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1}) Grease lub.			Oil lub.	Bearing No.				Mounting dimensions. (mm)				(Refer.) Mass
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Z, ZZ RU, 2RU	(RD, 2RD)	(RS, 2RS)	(Z)	Shielded	Non-contact sealed	Extremely light contact sealed	Contact sealed	d_a min.	d_a max.	D_a max.	r_a max.	Open type (kg)
65	100	18	1.1	38.1	25.2	1.40	15.8	6 600	—	3 700	7 800	**6013 ZZ**	**6013 2RU**	—	**6013 2RS**	71.5	74.5	93.5	1	0.435
	120	23	1.5	71.5	40.1	2.65	14.4	5 400	—	3 500	6 400	**6213 ZZ**	**6213 2RU**	—	**6213 2RS**	73	79	112	1.5	0.990
	140	33	2.1	116	59.9	4.50	13.2	4 800	—	3 200	5 800	**6313 ZZ**	**6313 2RU**	—	**6313 2RS**	76	86	129	2	2.08
70	90	10	0.6	15.1	11.9	0.620	16.1	6 800	6 100	—	8 100	**6814 ZZ**	**6814 2RU**	**6814 2RD**	—	74	74	86	0.6	0.134
	100	16	1	29.7	21.2	1.10	16.3	6 400	5 800	3 600	7 600	**6914 ZZ**	**6914 2RU**	**6914 2RD**	**6914 2RS**	75	76.5	95	1	0.342
	110	20	1.1	47.6	30.9	1.80	15.6	6 100	—	3 500	7 200	**6014 ZZ**	**6014 2RU**	—	**6014 2RS**	76.5	79.5	103.5	1	0.602
	125	24	1.5	77.8	44.1	2.90	14.5	5 100	—	3 300	6 100	**6214 ZZ**	**6214 2RU**	—	**6214 2RS**	78	84	117	1.5	1.07
	150	35	2.1	130	68.2	4.95	13.2	4 500	—	3 000	5 400	**6314 ZZ**	**6314 2RU**	—	**6314 2RS**	81	92	139	2	2.52
75	95	10	0.6	15.7	12.9	0.660	16.0	6 400	5 800	—	7 600	**6815 ZZ**	**6815 2RU**	**6815 2RD**	—	79	79	91	0.6	0.142
	105	16	1	30.5	22.6	1.20	16.5	6 100	—	—	7 200	**6915 ZZ**	**6915 2RU**	—	—	80	82.5	100	1	0.363
	115	20	1.1	49.4	33.5	1.90	15.8	5 700	—	3 300	6 800	**6015 ZZ**	**6015 2RU**	—	**6015 2RS**	81.5	84.5	108.5	1	0.638
	130	25	1.5	84.3	48.3	3.10	14.5	4 800	—	3 100	5 800	**6215 ZZ**	**6215 2RU**	—	**6215 2RS**	83	88.5	122	1.5	1.18
	160	37	2.1	142	77.2	5.40	13.2	4 200	—	2 800	5 000	**6315 ZZ**	**6315 2RU**	—	**6315 2RS**	86	97.5	149	2	3.02
80	100	10	0.6	15.9	13.3	0.690	16.0	6 100	5 500	—	7 200	**6816 ZZ**	**6816 2RU**	**6816 2RD**	—	84	84	96	0.6	0.150
	110	16	1	31.2	24.0	1.25	16.6	5 700	5 100	3 200	6 800	**6916 ZZ**	**6916 2RU**	**6916 2RD**	**6916 2RS**	85	86.5	105	1	0.382
	125	22	1.1	59.5	39.8	2.25	15.6	5 300	—	3 100	6 300	**6016 ZZ**	**6016 2RU**	—	**6016 2RS**	86.5	90	118.5	1	0.850
	140	26	2	90.9	53.0	3.25	14.6	4 500	—	2 900	5 400	**6216 ZZ**	**6216 2RU**	—	**6216 2RS**	89	93	131	2	1.40
	170	39	2.1	154	86.7	5.85	13.3	3 900	—	2 700	4 700	**6316 ZZ**	**6316 2RU**	—	**6316 2RS**	91	105	159	2	3.59
85	110	13	1	23.4	19.0	0.980	16.2	5 600	5 000	—	6 600	**6817 ZZ**	**6817 2RU**	**6817 2RD**	—	90	90.5	105	1	0.266
	120	18	1.1	39.9	29.6	1.55	16.4	5 300	4 800	3 000	6 300	**6917 ZZ**	**6917 2RU**	**6917 2RD**	**6917 2RS**	91.5	92.5	113.5	1	0.535
	130	22	1.1	61.8	43.1	2.35	15.8	5 000	—	2 900	5 900	**6017 ZZ**	**6017 2RU**	—	**6017 2RS**	91.5	96.5	123.5	1	0.890
	150	28	2	105	61.9	3.70	14.5	4 200	—	2 700	5 000	**6217 ZZ**	**6217 2RU**	—	**6217 2RS**	94	102	141	2	1.79
	180	41	3	166	96.8	6.35	13.3	3 700	—	2 500	4 400	**6317 ZZX**	**6317 2RU**	—	**6317 2RS**	98	111	167	2.5	4.23
90	115	13	1	23.8	19.7	1.00	16.1	5 300	4 800	—	6 300	**6818 ZZ**	**6818 2RU**	**6818 2RD**	—	95	95.5	110	1	0.279
	125	18	1.1	41.0	31.6	1.60	16.5	5 100	4 600	2 800	6 000	**6918 ZZ**	**6918 2RU**	**6918 2RD**	**6918 2RS**	96.5	97.5	118.5	1	0.565
	140	24	1.5	72.8	49.7	2.65	15.6	4 700	—	2 700	5 600	**6018 ZZ**	**6018 2RU**	—	**6018 2RS**	98	100.5	132	1.5	1.16
	160	30	2	120	71.5	4.20	14.5	3 900	—	2 600	4 700	**6218 ZZ**	**6218 2RU**	—	**6218 2RS**	99	108.5	151	2	2.15

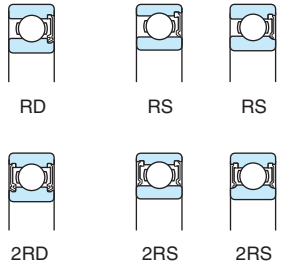
[Remark] Standard cage types used for the above bearings are described earlier in this section.

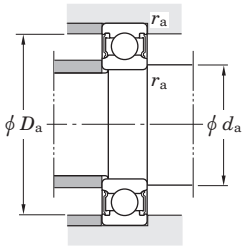
shielded type
sealed type

d (90) ~ (130) mm





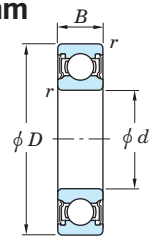


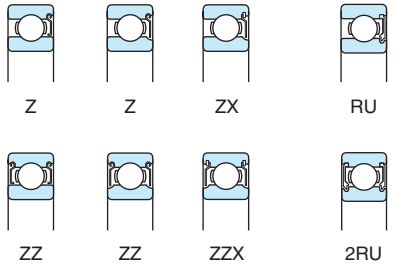


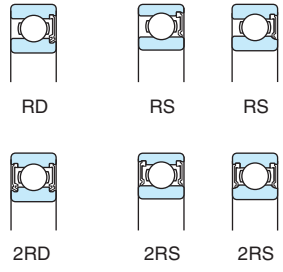
Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1}) Grease lub.			Oil lub.	Bearing No.				Mounting dimensions. (mm)				(Refer.) Mass
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Z, ZZ RU, 2RU	(RD, 2RD)	(RS, 2RS)	(Z)	Shielded	Non-contact sealed	Extremely light contact sealed	Contact sealed	d_a min.	d_a max.	D_a max.	r_a max.	Open type (kg)
90	190	43	3	178	107	8.80	13.3	3 500	—	2 400	4 200	**6318 ZZX**	**6318 2RU**	—	**6318 2RS**	103	117	177	2.5	4.91
95	130	18	1.1	42.1	33.5	1.65	16.6	4 800	4 300	2 700	5 700	**6919 ZZ**	**6919 2RU**	**6919 2RD**	**6919 2RS**	101.5	102	123.5	1	0.705
	145	24	1.5	75.5	53.9	2.75	15.8	4 400	—	2 500	5 200	**6019 ZZX**	**6019 2RU**	—	**6019 2RS**	103	107.5	137	1.5	1.21
	170	32	2.1	136	81.9	4.65	14.4	3 700	—	2 400	4 400	**6219 ZZX**	**6219 2RU**	—	**6219 2RS**	106	113	159	2	2.62
	200	45	3	191	119	9.45	13.3	3 300	—	2 200	4 000	**6319 ZZX**	**6319 2RU**	—	**6319 2RS**	108	122	187	2.5	5.67
100	125	13	1	24.5	21.2	1.05	16.0	4 800	4 300	—	5 700	**6820 ZZ**	**6820 2RU**	**6820 2RD**	—	105	105.5	120	1	0.309
	140	20	1.1	51.5	39.6	1.90	16.2	4 500	—	—	5 300	**6920-1 ZZ**	**6920-1 2RU**	—	—	106.5	110.5	133.5	1	0.960
	150	24	1.5	75.2	54.2	2.70	15.9	4 300	—	2 500	5 100	**6020 ZZ**	**6020 2RU**	—	**6020 2RS**	108	112	142	1.5	1.25
	180	34	2.1	153	93.1	5.15	14.4	3 500	—	2 300	4 200	**6220 ZZX**	**6220 2RU**	—	**6220 2RS**	111	122	169	2	3.14
	215	47	3	216	141	10.9	13.2	3 000	—	2 100	3 600	**6320 ZZX**	**6320 2RU**	—	**6320 2RS**	113	131	202	2.5	7.00
105	145	20	1.1	53.0	42.1	1.95	16.4	4 300	—	2 400	5 100	**6921-1 ZZ**	**6921-1 2RU**	—	**6921-1 2RS**	111.5	115	138.5	1	1.00
	160	26	2	90.4	65.8	3.20	15.8	4 000	—	2 300	4 700	**6021 ZZX**	**6021 2RU**	—	**6021 2RS**	114	119	151	2	1.59
	190	36	2.1	166	105	5.70	14.4	3 300	—	2 200	3 900	**6221 ZZX**	**6221 2RU**	—	**6221 2RS**	116	127	179	2	3.70
	225	49	3	230	153	11.7	13.2	2 900	—	2 000	3 500	**6321 ZZX**	**6321 2RU**	—	**6321 2RS**	118	136	212	2.5	8.05
110	140	16	1	35.1	30.7	1.40	16.1	4 300	3 900	—	5 100	**6822 ZZ**	**6822 2RU**	**6822 2RD**	—	115	116.5	135	1	0.606
	150	20	1.1	59.9	47.8	2.20	16.4	4 100	—	—	4 900	**6922 ZZ**	**6922 2RU**	—	—	116.5	119.5	143.5	1	1.04
	170	28	2	103	73.0	3.55	15.6	3 800	—	2 200	4 500	**6022 ZZX**	**6022 2RU**	—	**6022 2RS**	119	123	161	2	1.96
	200	38	2.1	180	117	6.20	14.4	3 100	—	2 000	3 700	**6222 ZZX**	**6222 2RU**	—	**6222 2RS**	121	136.5	189	2	4.36
	240	50	3	257	180	13.3	13.2	2 700	—	1 900	3 200	**6322 ZZX**	**6322 2RU**	—	**6322 2RS**	123	146.5	227	2.5	9.54
120	150	16	1	36.2	33.0	1.45	16.0	4 000	—	—	4 700	**6824 ZZ**	**6824 2RU**	—	—	125	128.5	145	1	0.655
	165	22	1.1	71.6	56.9	2.50	16.4	3 800	—	—	4 400	**6924 ZZ**	**6924 2RU**	—	—	126.5	131.5	158.5	1	1.41
	180	28	2	106	79.3	3.60	15.9	3 600	—	2 100	4 200	**6024 ZZX**	**6024 2RU**	—	**6024 2RS**	129	136	171	2	2.07
	215	40	2.1	194	131	6.65	14.4	2 900	—	1 900	3 400	**6224 ZZX**	**6224 2RU**	—	**6224 2RS**	131	144	204	2	5.15
	260	55	3	258	185	12.6	13.5	2 500	—	—	3 000	**6324 ZZX**	—	—	—	133	158	247	2.5	12.5
130	165	18	1.1	46.1	41.2	1.75	16.1	3 600	—	—	4 300	**6826 ZZ**	**6826 2RU**	—	—	136.5	139.5	158.5	1	0.939
	180	24	1.5	81.5	67.4	2.85	16.3	3 400	—	—	4 100	**6926-1 ZZ**	**6926-1 2RU**	—	—	138	144	172	1.5	1.86

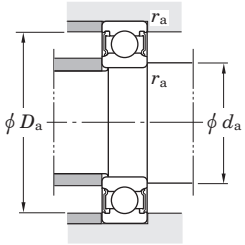
[Remark] Standard cage types used for the above bearings are described earlier in this section.

shielded type
sealed type
d (130) ~ 220 mm





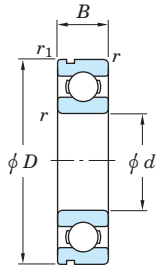




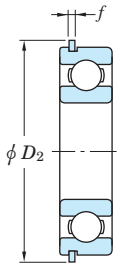
Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1}) Grease lub.			Oil lub.	Bearing No.				Mounting dimensions. (mm)				(Refer.) Mass
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Z, ZZ RU, 2RU	(RD, 2RD)	(RS, 2RS)	(Z)	Shielded	Non-contact sealed	Extremely light contact sealed	Contact sealed	d_a min.	d_a max.	D_a max.	r_a max.	Open type (kg)
130	200	33	2	133	101	4.45	15.8	3 200	—	1 900	3 800	**6026 ZZX**	**6026 2RU**	—	**6026 2RS**	139	146.5	191	2	3.16
	230	40	3	209	146	9.15	14.5	2 700	—	1 800	3 200	**6226 ZZX**	**6226 2RU**	—	**6226 2RS**	143	157	217	2.5	5.82
	280	58	4	287	214	14.1	13.6	2 300	—	—	2 700	**6326 ZZX**	—	—	—	146	171	264	3	15.1
140	175	18	1.1	47.8	44.4	1.85	16.0	3 400	3 100	—	4 000	**6828 ZZ**	—	**6828 2RD**	—	146.5	148	168.5	1	1.00
	190	24	1.5	83.3	71.6	2.90	16.5	3 200	—	—	3 800	**6928-1 ZZ**	**6928-1 2RU**	—	—	148	153	182	1.5	1.98
	210	33	2	137	109	4.55	15.9	3 000	—	1 800	3 600	**6028 ZZX**	**6028 2RU**	—	**6028 2RS**	149	158.5	201	2	3.55
	250	42	3	208	150	8.65	14.8	2 400	—	1 600	2 900	**6228 ZZX**	**6228 2RU**	—	**6228 2RS**	153	169	237	2.5	7.45
	300	62	4	316	246	15.6	13.6	2 100	—	—	2 500	**6328 ZZX**	—	—	—	156	184	284	3	19.4
150	210	28	2	117	94.3	3.75	16.2	2 900	—	1 700	3 400	**6930 ZZ**	**6930 2RU**	—	**6930 2RS**	159	165.5	201	2	3.05
	225	35	2.1	157	126	5.10	16.0	2 800	—	1 600	3 300	**6030 ZZX**	**6030 2RU**	—	**6030 2RS**	161	168.5	214	2	4.22
	270	45	3	220	168	9.05	15.1	2 200	—	—	2 700	**6230 ZZX**	—	—	—	163	183.5	257	2.5	9.41
160	200	20	1.1	60.5	56.9	2.20	16.1	2 900	2 600	—	3 400	**6832 ZZ**	—	**6832 2RD**	—	166.5	168.5	193.5	1	1.45
	240	38	2.1	171	135	5.30	15.9	2 600	—	1 500	3 000	**6032 ZZX**	**6032 2RU**	—	**6032 2RS**	171	178.5	229	2	5.22
	290	48	3	231	186	9.45	15.4	2 100	—	—	2 500	**6232 ZZX**	—	—	—	173	198	277	2.5	14.3
170	215	22	1.1	74.8	70.5	2.60	16.1	2 700	—	—	3 200	**6834 ZZ**	—	—	—	176.5	182.5	208.5	1	1.90
	260	42	2.1	201	161	6.20	15.8	2 400	—	—	2 800	**6034 ZZX**	**6034 2RU**	—	—	181	194	249	2	6.80
	310	52	4	265	223	11.1	15.3	1 900	—	—	2 300	**6234 ZZX**	—	—	—	186	210.5	294	3	17.5
180	225	22	1.1	75.8	73.1	2.65	16.1	2 600	2 300	—	3 000	**6836 ZZ**	—	**6836 2RD**	—	186.5	189.5	218.5	1	2.00
	280	46	2.1	227	194	7.15	15.8	2 200	—	—	2 600	**6036 ZZX**	**6036 2RU**	—	—	191	209.5	269	2	10.3
	320	52	4	264	226	10.8	15.1	1 800	—	—	2 200	**6236-1 ZZX**	—	—	—	196	220.5	304	3	18.3
190	240	24	1.5	91.4	88.1	3.10	16.1	2 400	—	—	2 800	**6838 ZZ**	—	—	—	198	202	232	1.5	2.60
	290	46	2.1	235	201	7.35	15.8	2 100	—	—	2 500	**6038 ZZX**	—	—	—	201	215	279	2	10.8
200	310	51	2.1	272	243	11.3	15.6	1 900	—	—	2 300	**6040 ZZX**	—	—	—	211	228	299	2	14.0
	360	58	4	314	293	13.1	15.2	1 600	—	—	1 900	**6240-1 ZZX**	—	—	—	216	250	344	3	28.2
220	340	56	3	294	271	12.0	15.6	1 700	—	—	2 000	**6044 ZZX**	—	—	—	233	251	327	2.5	18.3

[Remark] Standard cage types used for the above bearings are described earlier in this section.

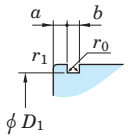
snap ring groove type
locating snap ring type
d 10 ~ (28) mm



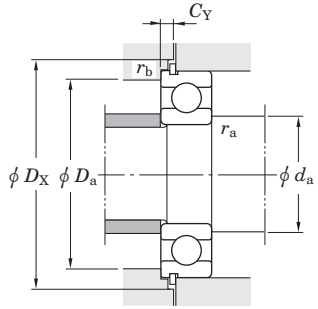
N
With snap ring groove



NR
With locating snap ring



Snap ring groove details



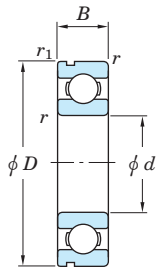

With locating snap ring and one shield

Boundary dimensions (mm)					Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.		Dimensions of snap ring groove (mm)				Dimensions of locating snap ring (mm)		Mounting dimensions (mm)							(Refer.) Mass	(Refer.) Bearing No.
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.	With snap ring groove	With locating snap ring	D_1 max.	a max.	b ±0.15	r_0 max.	D_2 max.	f ±0.05	d_a min.	D_a max.	D_X min.	C_Y max.	r_a max.	r_b max.	(kg)	
10	22	6	0.3	0.3	3.35	1.25	0.070	14.0	34 000	41 000	**6900N**	**6900NR**	20.8	1.05	0.925 [1]	0.2	24.8	0.65	12	20	25.5	1.5	0.3	0.3	0.010	**6900N**
	30	9	0.6	0.3	6.40	2.40	0.120	13.2	24 000	29 000	**6200N**	**6200NR**	28.17	2.06	1.5	0.4	34.7	1.07	14	26	35.5	2.92	0.6	0.3	0.032	**6200N**
	35	11	0.6	0.5	10.1	3.45	0.270	11.2	22 000	27 000	**6300N**	**6300NR**	33.17	2.06	1.5	0.4	39.7	1.07	14	31	40.5	2.92	0.6	0.5	0.053	**6300N**
12	24	6	0.3	0.3	3.60	1.45	0.080	14.5	31 000	36 000	**6901N**	**6901NR**	22.8	1.05	0.925 [1]	0.2	26.8	0.65	14	22	27.5	1.5	0.3	0.3	0.011	**6901N**
	32	10	0.6	0.3	8.50	3.05	0.240	12.3	22 000	27 000	**6201N**	**6201NR**	30.15	2.06	1.5	0.4	36.7	1.07	16	28	37.5	2.92	0.6	0.3	0.037	**6201N**
	37	12	1	0.5	12.1	4.20	0.420	11.1	20 000	25 000	**6301N**	**6301NR**	34.77	2.06	1.5	0.4	41.3	1.07	17	32	42	2.92	1	0.5	0.060	**6301N**
15	28	7	0.3	0.3	5.40	2.25	0.120	14.3	26 000	30 000	**6902N**	**6902NR**	26.7	1.3	1.075 [1]	0.25	30.8	0.8	17	26	31.5	1.9	0.3	0.3	0.017	**6902N**
	35	11	0.6	0.5	9.55	3.75	0.290	13.2	20 000	24 000	**6202N**	**6202NR**	33.17	2.06	1.5	0.4	39.7	1.07	19	31	40.5	2.92	0.6	0.5	0.045	**6202N**
	42	13	1	0.5	14.3	5.45	0.460	12.3	17 000	20 000	**6302N**	**6302NR**	39.75	2.06	1.5	0.4	46.3	1.07	20	37	47	2.92	1	0.5	0.082	**6302N**
17	30	7	0.3	0.3	5.75	2.55	0.130	14.7	23 000	28 000	**6903N**	**6903NR**	28.7	1.3	1.075 [1]	0.25	32.8	0.8	19	28	33.5	1.9	0.3	0.3	0.018	**6903N**
	40	12	0.6	0.5	12.0	4.80	0.370	13.2	17 000	21 000	**6203N**	**6203NR**	38.1	2.06	1.5	0.4	44.6	1.07	21	36	45.5	2.92	0.6	0.5	0.065	**6203N**
	47	14	1	0.5	17.0	6.65	0.550	12.4	15 000	18 000	**6303N**	**6303NR**	44.6	2.46	1.5	0.4	52.7	1.07	22	42	53.5	3.33	1	0.5	0.115	**6303N**
20	32	7	0.3	0.3	5.00	2.45	0.100	15.5	21 000	25 000	**6804N**	**6804NR**	30.7	1.3	1.075 [1]	0.25	34.8	0.8	22	30	35.5	1.9	0.3	0.3	0.018	**6804N**
	37	9	0.3	0.3	7.95	3.70	0.190	14.7	19 000	23 000	**6904N**	**6904NR**	35.7	1.7	1.075 [1]	0.25	39.8	0.8	22	35	40.5	2.3	0.3	0.3	0.036	**6904N**
	42	12	0.6	0.5	11.7	5.05	0.350	13.9	17 000	21 000	**6004N**	**6004NR**	39.75	2.06	1.5	0.4	46.3	1.07	24	38	47	2.92	0.6	0.5	0.069	**6004N**
	47	14	1	0.5	16.0	6.65	0.510	13.2	15 000	17 000	**6204N**	**6204NR**	44.6	2.46	1.5	0.4	52.7	1.07	25	42	53.5	3.33	1	0.5	0.106	**6204N**
	52	15	1.1	0.5	19.9	7.85	0.660	12.3	14 000	17 000	**6304N**	**6304NR**	49.73	2.46	1.5	0.4	57.9	1.07	26.5	45.5	58.5	3.33	1	0.5	0.144	**6304N**
22	44	12	0.6	0.5	11.7	5.15	0.350	14.1	17 000	20 000	**60/22N**	**60/22NR**	41.75	2.06	1.5	0.4	48.3	1.07	26	40	49	2.92	0.6	0.5	0.073	**60/22N**
	50	14	1	0.5	16.0	6.65	0.510	13.2	15 000	17 000	**62/22N**	**62/22NR**	47.6	2.46	1.5	0.4	55.7	1.07	27	45	56.5	3.33	1	0.5	0.118	**62/22N**
	56	16	1.1	0.5	23.1	9.40	0.770	12.6	13 000	15 000	**63/22N**	**63/22NR**	53.6	2.46	1.5	0.4	61.7	1.07	28.5	49.5	62.5	3.33	1	0.5	0.201	**63/22N**
25	37	7	0.3	0.3	5.40	2.95	0.120	16.0	18 000	21 000	**6805N**	**6805NR**	35.7	1.3	1.075 [1]	0.25	39.8	0.8	27	35	40.5	1.9	0.3	0.3	0.022	**6805N**
	42	9	0.3	0.3	8.75	4.55	0.230	15.4	16 000	19 000	**6905N**	**6905NR**	40.7	1.7	1.075 [1]	0.25	44.8	0.8	27	40	45.5	2.3	0.3	0.3	0.041	**6905N**
	47	12	0.6	0.5	12.6	5.85	0.380	14.5	15 000	18 000	**6005N**	**6005NR**	44.6	2.06	1.5	0.4	52.7	1.07	29	43	53.5	2.92	0.6	0.5	0.080	**6005N**
	52	15	1	0.5	17.5	7.85	0.550	13.9	13 000	15 000	**6205N**	**6205NR**	49.73	2.46	1.5	0.4	57.9	1.07	30	47	58.5	3.33	1	0.5	0.128	**6205N**
	62	17	1.1	0.5	25.7	11.3	0.860	13.2	11 000	13 000	**6305N**	**6305NR**	59.61	3.28	2.05	0.6	67.7	1.65	31.5	55.5	68.5	4.67	1	0.5	0.232	**6305N**
28	52	12	0.6	0.5	15.6	7.40	0.480	14.5	14 000	16 000	**60/28N**	**60/28NR**	49.73	2.06	1.5	0.4	57.9	1.07	32	48	58.5	2.92	0.6	0.5	0.097	**60/28N**
	58	16	1	0.5	22.4	9.75	0.720	13.4	12 000	14 000	**62/28N**	**62/28NR**	55.6	2.46	1.5	0.4	63.7	1.07	33	53	64.5	3.33	1	0.5	0.173	**62/28N**

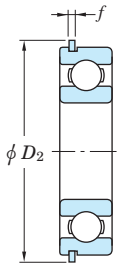
[Note] 1) The tolerance of the ring groove width is ±0.125.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

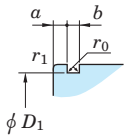
snap ring groove type
locating snap ring type
d (28) ~ (50) mm



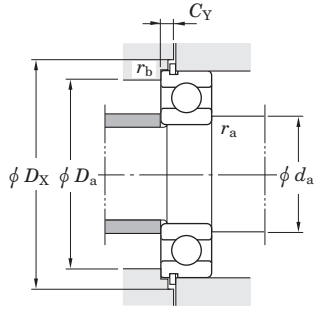
N
With snap ring groove



NR
With locating snap ring



Snap ring groove details



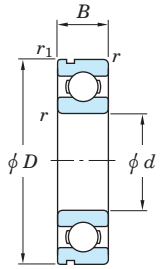

With locating snap ring and one shield

Boundary dimensions (mm)					Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.		Dimensions of snap ring groove (mm)				Dimensions of locating snap ring (mm)		Mounting dimensions (mm)						(Refer.) Mass	(Refer.) Bearing No.
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.	With snap ring groove	With locating snap ring	D_1 max.	a max.	b ±0.15	r_0 max.	D_2 max.	f ±0.05	d_a min.	D_a max.	D_X min.	C_Y max.	r_a max.	r_b max.	(kg)	
28	68	18	1.1	0.5	29.4	13.1	0.990	13.3	10 000	12 000	**63/28N**	**63/28NR**	64.82	3.28	2.05	0.6	74.6	1.65	34.5	61.5	76	4.67	1	0.5	0.328	**63/28N**
30	42	7	0.3	0.3	5.65	3.40	0.140	16.4	15 000	18 000	**6806N**	**6806NR**	40.7	1.3	1.075 [1]	0.25	44.8	0.8	32	40	45.5	1.9	0.3	0.3	0.026	**6806N**
	47	9	0.3	0.3	9.05	5.00	0.260	15.8	14 000	17 000	**6906N**	**6906NR**	45.7	1.7	1.075 [1]	0.25	49.8	0.8	32	45	50.5	2.3	0.3	0.3	0.045	**6906N**
	55	13	1	0.5	16.5	8.25	0.530	14.7	13 000	15 000	**6006N**	**6006NR**	52.6	2.08	1.5	0.4	60.7	1.07	35	50	61.5	2.9	1	0.5	0.116	**6006N**
	62	16	1	0.5	24.3	11.3	0.800	13.9	11 000	13 000	**6206N**	**6206NR**	59.61	3.28	2.05	0.6	67.7	1.65	35	57	68.5	4.67	1	0.5	0.199	**6206N**
	72	19	1.1	0.5	33.3	15.0	1.15	13.3	9 600	12 000	**6306N**	**6306NR**	68.81	3.28	2.05	0.6	78.6	1.65	36.5	65.5	80	4.67	1	0.5	0.346	**6306N**
32	58	13	1	0.5	18.8	9.15	0.600	14.5	12 000	14 000	**60/32N**	**60/32NR**	55.6	2.08	1.5	0.4	63.7	1.07	37	53	64.5	2.9	1	0.5	0.127	**60/32N**
	65	17	1	0.5	29.4	13.1	0.990	13.3	10 000	12 000	**62/32N**	**62/32NR**	62.6	3.28	2.05	0.6	70.7	1.65	37	60	71.5	4.67	1	0.5	0.228	**62/32N**
	75	20	1.1	0.5	37.6	16.2	1.30	12.7	9 300	11 000	**63/32N**	**63/32NR**	71.83	3.28	2.05	0.6	81.6	1.65	38.5	68.5	83	4.67	1	0.5	0.437	**63/32N**
35	47	7	0.3	0.3	5.95	3.85	0.160	16.5	13 000	16 000	**6807N**	**6807NR**	45.7	1.3	1.075 [1]	0.25	49.8	0.8	37	45	50.5	1.9	0.3	0.3	0.030	**6807N**
	55	10	0.6	0.6	13.6	7.75	0.440	15.7	12 000	14 000	**6907N**	**6907NR**	53.7	1.7	1.075 [1]	0.25	57.8	0.8	39	51	58.5	2.3	0.6	0.6	0.073	**6907N**
	62	14	1	0.5	19.9	10.3	0.640	14.9	11 000	13 000	**6007N**	**6007NR**	59.61	2.08	2.05	0.6	67.7	1.65	40	58	68.5	3.48	1	0.5	0.155	**6007N**
	72	17	1.1	0.5	32.1	15.4	1.10	13.9	9 200	11 000	**6207N**	**6207NR**	68.81	3.28	2.05	0.6	78.6	1.65	41.5	65.5	80	4.67	1	0.5	0.288	**6207N**
	80	21	1.5	0.5	41.7	19.3	1.45	13.2	8 500	10 000	**6307N**	**6307NR**	76.81	3.28	2.05	0.6	86.6	1.65	43	72	88	4.67	1.5	0.5	0.457	**6307N**
40	52	7	0.3	0.3	6.15	4.20	0.180	16.3	12 000	14 000	**6808N**	**6808NR**	50.7	1.3	1.075 [1]	0.25	54.8	0.8	42	50	55.5	1.9	0.3	0.3	0.033	**6808N**
	62	12	0.6	0.6	17.1	9.95	0.570	15.6	11 000	13 000	**6908N**	**6908NR**	60.7	1.7	1.075 [1]	0.25	64.8	0.8	44	58	65.5	2.3	0.6	0.6	0.112	**6908N**
	68	15	1	0.5	20.9	11.5	0.690	15.2	10 000	12 000	**6008N**	**6008NR**	64.82	2.49	2.05	0.6	74.6	1.65	45	63	76	3.89	1	0.5	0.192	**6008N**
	80	18	1.1	0.5	36.4	17.8	1.25	14.0	8 300	10 000	**6208N**	**6208NR**	76.81	3.28	2.05	0.6	86.6	1.65	46.5	73.5	88	4.67	1	0.5	0.366	**6208N**
	90	23	1.5	0.5	50.9	24.0	1.85	13.2	7 700	9 200	**6308N**	**6308NR**	86.79	3.28	2.85	0.6	96.5	2.41	48	82	98	5.43	1.5	0.5	0.633	**6308N**
45	58	7	0.3	0.3	7.75	5.40	0.230	16.3	11 000	13 000	**6809N**	**6809NR**	56.7	1.3	1.075 [1]	0.25	60.8	0.8	47	56	61.5	1.9	0.3	0.3	0.040	**6809N**
	68	12	0.6	0.6	17.7	10.9	0.600	15.9	9 700	11 000	**6909N**	**6909NR**	66.7	1.7	1.075 [1]	0.25	70.8	0.8	49	64	72	2.3	0.6	0.6	0.132	**6909N**
	75	16	1	0.5	26.2	15.1	0.900	15.3	9 200	11 000	**6009N**	**6009NR**	71.83	2.49	2.05	0.6	81.6	1.65	50	70	83	3.89	1	0.5	0.245	**6009N**
	85	19	1.1	0.5	40.9	20.3	1.40	14.0	7 700	9 200	**6209N**	**6209NR**	81.81	3.28	2.05	0.6	91.6	1.65	51.5	78.5	93	4.67	1	0.5	0.407	**6209N**
	100	25	1.5	0.5	61.1	29.5	2.25	13.3	6 800	8 100	**6309N**	**6309NR**	96.8	3.28	2.85	0.6	106.5	2.41	53	92	108	5.43	1.5	0.5	0.833	**6309N**
50	65	7	0.3	0.3	8.20	6.10	0.260	16.1	9 600	11 000	**6810N**	**6810NR**	63.7	1.3	1.075 [1]	0.25	67.8	0.8	52	63	68.5	1.9	0.3	0.3	0.052	**6810N**
	72	12	0.6	0.6	18.2	11.7	0.640	16.1	9 000	11 000	**6910N**	**6910NR**	70.7	1.7	1.075 [1]	0.25	74.8	0.8	54	68	76	2.3	0.6	0.6	0.133	**6910N**
	80	16	1	0.5	27.3	16.6	0.960	15.6	8 400	9 900	**6010N**	**6010NR**	76.81	2.49	2.05	0.6	86.6	1.65	55	75	88	3.89	1	0.5	0.261	**6010N**

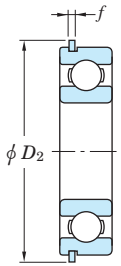
[Note] 1) The tolerance of the ring groove width is ±0.125.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

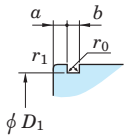
snap ring groove type
locating snap ring type
d (50) ~ 90 mm



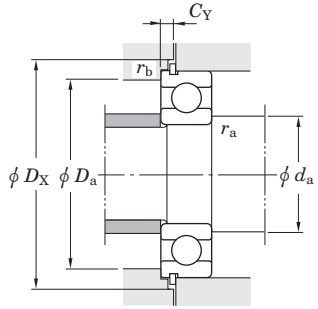
N
With snap ring groove



NR
With locating snap ring



Snap ring groove details



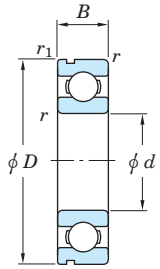


With locating snap ring and one shield

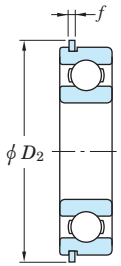
Boundary dimensions (mm)					Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.		Dimensions of snap ring groove (mm)				Dimensions of locating snap ring (mm)		Mounting dimensions (mm)						(Refer.) Mass	(Refer.) Bearing No.
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.	With snap ring groove	With locating snap ring	D_1 max.	a max.	b ±0.15	r_0 max.	D_2 max.	f ±0.05	d_a min.	D_a max.	D_X min.	C_Y max.	r_a max.	r_b max.	(kg)	
50	90	20	1.1	0.5	43.9	23.3	1.55	14.4	7 100	8 500	**6210N**	**6210NR**	86.79	3.28	2.85	0.6	96.5	2.41	56.5	83.5	98	5.43	1	0.5	0.463	**6210N**
	110	27	2	0.5	77.5	38.3	2.90	13.2	6 100	7 300	**6310N**	**6310NR**	106.81	3.28	2.85	0.6	116.6	2.41	59	101	118	5.43	2	0.5	1.07	**6310N**
55	90	18	1.1	0.5	35.3	21.2	1.25	15.3	7 600	8 900	**6011N**	**6011NR**	86.79	2.87	2.85	0.6	96.5	2.41	61.5	83.5	98	5.03	1	0.5	0.385	**6011N**
	100	21	1.5	0.5	54.2	29.4	1.95	14.4	6 300	7 600	**6211N**	**6211NR**	96.8	3.28	2.85	0.6	106.5	2.41	63	92	108	5.43	1.5	0.5	0.607	**6211N**
	120	29	2	0.5	89.5	45.0	3.45	13.2	5 600	6 700	**6311N**	**6311NR**	115.21	4.06	3.25	0.6	129.7	2.77	64	111	131.5	6.58	2	0.5	1.37	**6311N**
60	95	18	1.1	0.5	36.8	23.2	1.35	15.6	7 100	8 400	**6012N**	**6012NR**	91.82	2.87	2.85	0.6	101.6	2.41	66.5	88.5	103	5.03	1	0.5	0.415	**6012N**
	110	22	1.5	0.5	65.6	36.2	2.40	14.4	5 700	6 900	**6212N**	**6212NR**	106.81	3.28	2.85	0.6	116.6	2.41	68	102	118	5.43	1.5	0.5	0.783	**6212N**
	130	31	2.1	0.5	102	52.2	3.95	13.2	5 200	6 200	**6312N**	**6312NR**	125.22	4.06	3.25	0.6	139.7	2.77	71	119	141.5	6.58	2	0.5	1.70	**6312N**
65	100	18	1.1	0.5	38.1	25.2	1.40	15.8	6 600	7 800	**6013N**	**6013NR**	96.8	2.87	2.85	0.6	106.5	2.41	71.5	93.5	108	5.03	1	0.5	0.435	**6013N**
	120	23	1.5	0.5	71.5	40.1	2.65	14.4	5 400	6 400	**6213N**	**6213NR**	115.21	4.06	3.25	0.6	129.7	2.77	73	112	131.5	6.58	1.5	0.5	0.990	**6213N**
	140	33	2.1	0.5	116	59.9	4.50	13.2	4 800	5 800	**6313N**	**6313NR**	135.23	4.9	3.25	0.6	149.7	2.77	76	129	152	7.37	2	0.5	2.08	**6313N**
70	110	20	1.1	0.5	47.6	30.9	1.80	15.6	6 100	7 200	**6014N**	**6014NR**	106.81	2.87	2.85	0.6	116.6	2.41	76.5	103.5	118	5.03	1	0.5	0.602	**6014N**
	125	24	1.5	0.5	77.8	44.1	2.90	14.5	5 100	6 100	**6214N**	**6214NR**	120.22	4.06	3.25	0.6	134.7	2.77	78	117	136.5	6.58	1.5	0.5	1.07	**6214N**
	150	35	2.1	0.5	130	68.2	4.95	13.2	4 500	5 400	**6314N**	**6314NR**	145.24	4.9	3.25	0.6	159.7	2.77	81	139	162	7.37	2	0.5	2.52	**6314N**
75	115	20	1.1	0.5	49.4	33.5	1.90	15.8	5 700	6 800	**6015N**	**6015NR**	111.81	2.87	2.85	0.6	121.6	2.41	81.5	108.5	123	5.03	1	0.5	0.638	**6015N**
	130	25	1.5	0.5	84.3	48.3	3.10	14.5	4 800	5 800	**6215N**	**6215NR**	125.22	4.06	3.25	0.6	139.7	2.77	83	122	141.5	6.58	1.5	0.5	1.18	**6215N**
	160	37	2.1	0.5	142	77.2	5.40	13.2	4 200	5 000	**6315N**	**6315NR**	155.22	4.9	3.25	0.6	169.7	2.77	86	149	172	7.37	2	0.5	3.02	**6315N**
80	125	22	1.1	0.5	59.5	39.8	2.25	15.6	5 300	6 300	**6016N**	**6016NR**	120.22	2.87	3.25	0.6	134.7	2.77	86.5	118.5	136.5	5.39	1	0.5	0.850	**6016N**
	140	26	2	0.5	90.9	53.0	3.25	14.6	4 500	5 400	**6216N**	**6216NR**	135.23	4.9	3.25	0.6	149.7	2.77	89	131	152	7.37	2	0.5	1.40	**6216N**
	170	39	2.1	0.5	154	86.7	5.85	13.3	3 900	4 700	**6316N**	**6316NR**	163.65	5.69	3.65	0.6	182.9	3.05	91	159	185	8.44	2	0.5	3.59	**6316N**
85	130	22	1.1	0.5	61.8	43.1	2.35	15.8	5 000	5 900	**6017N**	**6017NR**	125.22	2.87	3.25	0.6	139.7	2.77	91.5	123.5	141.5	5.39	1	0.5	0.890	**6017N**
	150	28	2	0.5	105	61.9	3.70	14.5	4 200	5 000	**6217N**	**6217NR**	145.24	4.9	3.25	0.6	159.7	2.77	94	141	162	7.37	2	0.5	1.79	**6217N**
	180	41	3	0.5	166	96.8	6.35	13.3	3 700	4 400	**6317N**	**6317NR**	173.66	5.69	3.65	0.6	192.9	3.05	98	167	195	8.44	2.5	0.5	4.23	**6317N**
90	140	24	1.5	0.5	72.8	49.7	2.65	15.6	4 700	5 600	**6018N**	**6018NR**	135.23	3.71	3.25	0.6	149.7	2.77	98	132	152	6.17	1.5	0.5	1.16	**6018N**
	160	30	2	0.5	120	71.5	4.20	14.5	3 900	4 700	**6218N**	**6218NR**	155.22	4.9	3.25	0.6	169.7	2.77	99	151	172	7.37	2	0.5	2.15	**6218N**
	190	43	3	0.5	178	107	8.80	13.3	3 500	4 200	**6318N**	**6318NR**	183.64	5.69	3.65	0.6	202.9	3.05	103	177	205	8.44	2.5	0.5	4.91	**6318N**

[Remark] Standard cage types used for the above bearings are described earlier in this section.

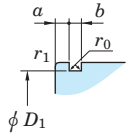
snap ring groove type
locating snap ring type
d 95 ~ 130 mm



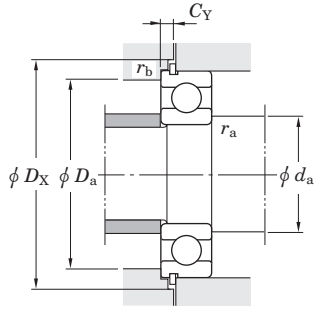
N
With snap ring groove



NR
With locating snap ring



Snap ring groove details



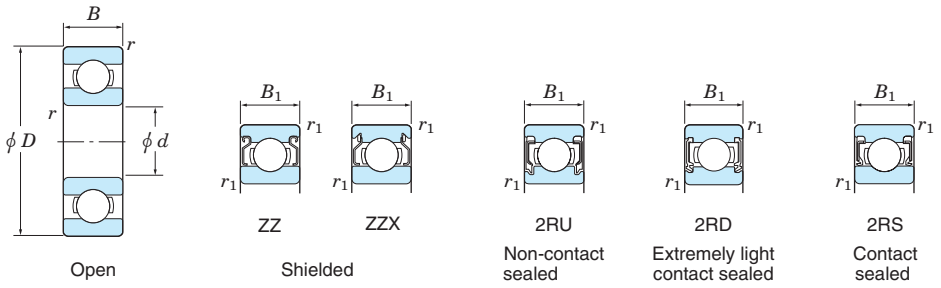


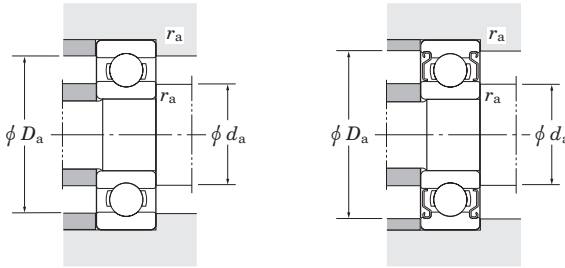
With locating snap ring and one shield

Boundary dimensions (mm)					Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.		Dimensions of snap ring groove (mm)				Dimensions of locating snap ring (mm)		Mounting dimensions (mm)						(Refer.) Mass	(Refer.) Bearing No.
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.	With snap ring groove	With locating snap ring	D_1 max.	a max.	b ±0.15	r_0 max.	D_2 max.	f ±0.05	d_a min.	D_a max.	D_X min.	C_Y max.	r_a max.	r_b max.	(kg)	
95	145	24	1.5	0.5	75.5	53.9	2.75	15.8	4 400	5 200	**6019N**	**6019NR**	140.23	3.71	3.25	0.6	154.7	2.77	103	137	157	6.17	1.5	0.5	1.21	**6019N**
	170	32	2.1	0.5	136	81.9	4.65	14.4	3 700	4 400	**6219N**	**6219NR**	163.65	5.69	3.65	0.6	182.9	3.05	106	159	185	8.44	2	0.5	2.62	**6219N**
	200	45	3	0.5	191	119	9.45	13.3	3 300	4 000	**6319N**	**6319NR**	193.65	5.69	3.65	0.6	212.9	3.05	108	187	215	8.44	2.5	0.5	5.67	**6319N**
100	150	24	1.5	0.5	75.2	54.2	2.70	15.9	4 300	5 100	**6020N**	**6020NR**	145.24	3.71	3.25	0.6	159.7	2.77	108	142	162	6.17	1.5	0.5	1.25	**6020N**
	180	34	2.1	0.5	153	93.1	5.15	14.4	3 500	4 200	**6220N**	**6220NR**	173.66	5.69	3.65	0.6	192.9	3.05	111	169	195	8.44	2	0.5	3.14	**6220N**
105	160	26	2	0.5	90.4	65.8	3.20	15.8	4 000	4 700	**6021N**	**6021NR**	155.22	3.71	3.25	0.6	169.7	2.77	114	151	172	6.17	2	0.5	1.59	**6021N**
	190	36	2.1	0.5	166	105	5.70	14.4	3 300	3 900	**6221N**	**6221NR**	183.64	5.69	3.65	0.6	202.9	3.05	116	179	205	8.44	2	0.5	3.70	**6221N**
110	170	28	2	0.5	103	73.0	3.55	15.6	3 800	4 500	**6022N**	**6022NR**	163.65	3.71	3.65	0.6	182.9	3.05	119	161	185	6.45	2	0.5	1.96	**6022N**
	200	38	2.1	0.5	180	117	6.20	14.4	3 100	3 700	**6222N**	**6222NR**	193.65	5.69	3.65	0.6	212.9	3.05	121	189	215	8.44	2	0.5	4.36	**6222N**
120	180	28	2	0.5	106	79.3	3.60	15.9	3 600	4 200	**6024N**	**6024NR**	173.66	3.71	3.65	0.6	192.9	3.05	129	171	195	6.45	2	0.5	2.07	**6024N**
130	200	33	2	0.5	133	101	4.45	15.8	3 200	3 800	**6026N**	**6026NR**	193.65	5.69	3.65	0.6	212.9	3.05	139	191	215	8.44	2	0.5	3.16	**6026N**

[Remark] Standard cage types used for the above bearings are described earlier in this section.

d 1 ~ (4) mm

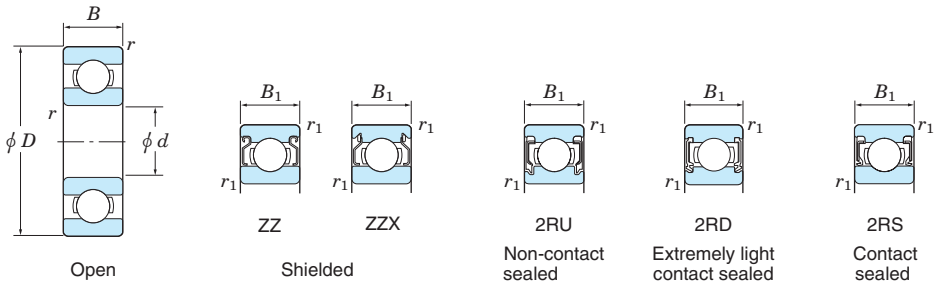


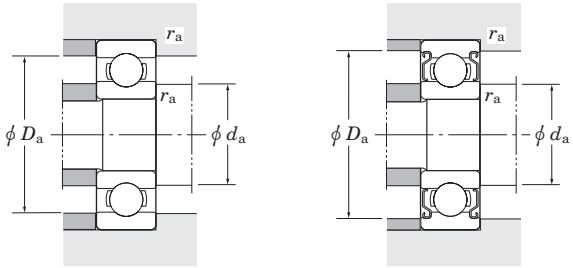


Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1}) Grease lub.			Oil lub.	Bearing No.					Mounting dimensions (mm)			(Refer.)
d	D	B	B_1	r 1) min.	r_1 1) min.	C_r	C_{0r}	C_u	f_0	Open ZZ, 2RU	(2RD)	(2RS)	Open Z	Open	Shielded	Non-contact sealed	Extremely light shielded	Contact sealed	d_a min.	D_a max.	r_a max.	Mass (g)
1	3	1	—	0.07	—	0.120	0.03	0.0007	11.6	130 000	—	—	150 000	**681**	—	—	—	—	1.6	2.4	0.05	0.03
	3	1.5	—	0.08	—	0.100	0.02	0.0006	12.8	130 000	—	—	150 000	**ML1003**	—	—	—	—	1.6	2.4	0.07	0.05
	4	1.6	—	0.1	—	0.170	0.04	0.001	11.4	120 000	—	—	140 000	**691**	—	—	—	—	1.8	3.2	0.1	0.1
1.2	4	1.8	—	0.08	—	0.140	0.03	0.0009	11.4	120 000	—	—	140 000	**ML1204**	—	—	—	—	1.8	3.4	0.07	0.1
1.5	4	1.2	2	0.1	0.1	0.140	0.03	0.0009	13.2	120 000	—	—	140 000	**68/1.5**	**W68/1.5 ZZ**	—	—	—	2.3	3.2	0.1	0.1
	5	2	2.6	0.15	0.15	0.300	0.07	0.002	13.3	110 000	—	—	130 000	**69/1.5**	**W69/1.5 ZZX**	—	—	—	2.7	3.8	0.15	0.1
	6	2.5	3	0.1	0.1	0.410	0.10	0.003	11.4	86 000	—	—	100 000	**ML1506**	**WML1506 ZZX**	—	—	—	2.3	5.2	0.1	0.3
2	5	1.5	2.3	0.1	0.1	0.210	0.05	0.001	13.3	98 000	—	—	110 000	**682**	**W682 ZZX**	—	—	—	2.8	4.4	0.1	0.1
	5	2	2.5	0.1	0.08	0.210	0.05	0.001	13.3	98 000	—	—	110 000	**ML2005**	**WML2005 ZZ**	—	—	—	2.6	4.2	0.07	0.1
	6	2.3	3	0.15	0.1	0.410	0.10	0.003	11.4	86 000	—	—	100 000	**692**	**W692 ZZ**	—	—	—	3.2	4.8	0.1	0.2
	6	2.5	3	0.1	0.1	0.410	0.10	0.003	11.4	86 000	—	—	100 000	**ML2006**	**WML2006 ZZX**	—	—	—	2.8	5.2	0.1	0.3
	7	2.5	3	0.15	0.15	0.480	0.13	0.003	12.6	67 000	—	—	79 000	**ML2007**	**WML2007 ZZX**	—	—	—	3.2	5.8	0.15	0.4
	7	2.8	3.5	0.15	0.15	0.480	0.13	0.003	12.6	67 000	—	—	79 000	**602**	**W602 ZZX**	—	—	—	3.2	5.8	0.15	0.5
2.5	6	1.8	2.6	0.1	0.1	0.240	0.06	0.002	14.3	75 000	—	—	89 000	**68/2.5**	**W68/2.5 ZZ**	—	—	—	3.3	5.2	0.1	0.2
	7	2.5	3.5	0.15	0.15	0.390	0.11	0.003	13.7	66 000	—	—	79 000	**69/2.5**	**W69/2.5 ZZ**	—	—	—	3.7	5.8	0.15	0.4
	8	2.5	—	0.1	—	0.540	0.15	0.004	13.4	63 000	—	—	75 000	**ML2508/1B**	—	—	—	—	3.3	7.2	0.1	0.6
	8	2.8	4	0.15	0.1	0.680	0.17	0.005	11.5	64 000	—	—	76 000	**ML2508**	**WML2508 ZZX**	—	—	—	3.7	6.8	0.1	0.6
3	6	2	2.5	0.08	0.05	0.240	0.06	0.002	14.3	75 000	—	—	89 000	**ML3006**	**WML3006 ZZ**	—	—	—	3.6	5.4	0.05	0.2
	7	2	3	(0.15)	(0.15)	0.390	0.11	0.003	13.7	66 000	—	—	79 000	**683**	**W683 ZZ**	—	—	—	4.2	5.8	0.1	0.3
	8	2.5	—	0.1	—	0.490	0.14	0.004	13.4	63 000	—	—	75 000	**ML3008**	—	—	—	—	3.8	7.2	0.1	0.5
	8	3	4	0.15	0.15	0.680	0.17	0.005	11.5	64 000	—	—	76 000	**693**	**W693 ZZ**	—	—	—	4.2	6.8	0.15	0.6
	9	3	5	0.15	0.15	0.540	0.16	0.004	14.0	60 000	—	—	72 000	**603**	**W603 ZZX**	—	—	—	4.2	7.8	0.15	0.9
	10	4	4	0.15	0.15	0.800	0.22	0.006	12.8	52 000	—	44 000	63 000	**623**	**623 ZZ**	—	—	**623 2RS**	4.2	8.8	0.15	1.6
	13	5	5	0.2	0.2	1.65	0.49	0.01	12.3	44 000	—	—	54 000	**633**	**633 ZZ**	—	—	—	4.6	11.4	0.2	3.0
4	7	2	2.5	0.08	0.05	0.320	0.11	0.003	15.1	64 000	—	—	76 000	**ML4007**	**WML4007 ZZ**	—	—	—	4.6	6.4	0.05	0.2
	8	2	3	0.1	0.08	0.490	0.14	0.004	14.6	61 000	—	—	73 000	**ML4008**	**WML4008 ZZ**	—	—	—	4.8	7.2	0.08	0.4
	9	2.5	4	(0.15)	(0.15)	0.800	0.23	0.006	12.8	59 000	—	—	70 000	**684**	**W684 ZZ**	—	—	—	5.2	7.8	0.1	0.6

[Note] 1) Numerical values in ( ) do not conform to JIS B 1521.

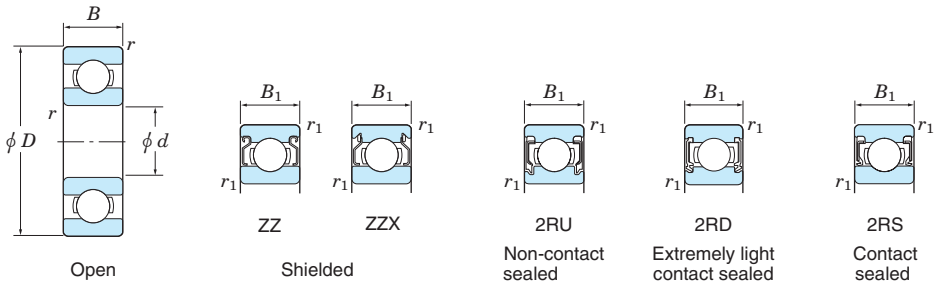
d (4) ~ (8) mm

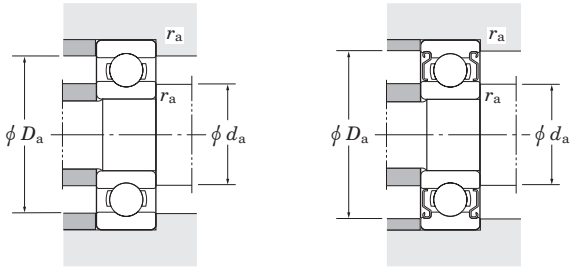




Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1}) Grease lub.			Oil lub.	Bearing No.					Mounting dimensions (mm)			(Refer.) Mass
d	D	B	B_1	r min.	r_1 min.	C_r	C_{0r}	C_u	f_0	Open ZZ, 2RU	(2RD)	(2RS)	Open Z	Open	Shielded	Non-contact sealed	Extremely light shielded	Contact sealed	d_a min.	D_a max.	r_a max.	(g)
4	10	3	4	0.15	0.1	0.810	0.23	0.006	13.3	56 000	—	—	67 000	**ML4010**	**WML4010 ZZ**	—	—	—	5.2	8.8	0.1	1.0
	11	4	4	0.15	0.15	1.20	0.35	0.009	12.4	54 000	—	44 000	65 000	**694**	**694 ZZ**	**694 2RU**	—	**694 2RS**	5.2	9.8	0.15	1.8
	12	4	4	0.2	0.2	1.20	0.35	0.009	12.4	53 000	—	—	63 000	**604**	**604 ZZ**	—	—	—	5.6	10.4	0.2	2.1
	13	5	5	0.2	0.2	1.65	0.48	0.010	12.3	44 000	—	39 000	54 000	**624**	**624 ZZ**	**624 2RU**	—	**624 2RS**	5.6	11.4	0.2	2.9
	16	5	5	0.3	0.3	1.70	0.52	0.010	12.4	40 000	—	—	49 000	**634**	**634 ZZ**	—	—	—	6	14	0.3	5.3
5	8	2	2.5	0.08	0.05	0.270	0.09	0.002	15.7	59 000	—	—	70 000	**ML5008**	**WML5008 ZZ**	—	—	—	5.6	7.4	0.05	0.3
	9	2.5	3	0.1	0.08	0.540	0.17	0.004	15.3	56 000	—	—	67 000	**ML5009**	**WML5009 ZZ**	—	—	—	5.8	8.2	0.08	0.5
	10	3	4	0.1	0.1	0.540	0.17	0.005	14.8	55 000	—	—	65 000	**ML5010**	**WML5010 ZZ**	—	—	—	5.8	9	0.1	0.9
	11	3	5	0.15	0.15	0.890	0.28	0.007	12.8	53 000	—	—	63 000	**685**	**W685 ZZ**	—	—	—	6.2	9.8	0.15	1.0
	13	4	4	0.2	0.2	1.35	0.43	0.010	12.3	50 000	45 000	42 000	60 000	**695**	**695 ZZ**	**695 2RU**	**695 2RD**	**695 2RS**	6.6	11.4	0.2	2.2
	14	5	5	0.2	0.2	1.65	0.49	0.010	12.3	50 000	—	—	60 000	**605**	**605 ZZ**	—	—	—	6.6	12.4	0.2	3.5
	16	5	5	0.3	0.3	2.15	0.67	0.030	12.4	40 000	36 000	33 000	49 000	**625**	**625 ZZ**	**625 2RU**	—	**625 2RS**	7	14	0.3	5.0
	19	6	6	0.3	0.3	2.90	0.89	0.040	12.3	35 000	32 000	27 000	43 000	**635**	**635 ZZ**	**635 2RU**	—	**635 2RS**	7	17	0.3	8.5
6	10	2.5	3	0.1	0.08	0.620	0.22	0.006	15.7	53 000	—	—	63 000	**ML6010**	**WML6010 ZZ**	—	—	—	6.8	9.2	0.08	0.6
	12	3	4	0.15	0.1	0.890	0.29	0.008	14.5	49 000	—	37 000	59 000	**ML6012**	**WML6012 ZZ**	—	—	**WML6012 2RS**	7.2	10.8	0.1	1.3
	13	3.5	5	0.15	0.15	1.35	0.44	0.010	13.7	48 000	43 000	36 000	57 000	**686**	**W686 ZZ**	—	—	**W686 2RS**	7.2	11.8	0.15	1.8
	15	5	5	0.2	0.2	1.70	0.52	0.010	12.4	45 000	41 000	32 000	54 000	**696**	**696 ZZ**	**696 2RU**	**696 2RD**	**696 2RS**	7.6	13.4	0.2	3.9
	17	6	6	0.3	0.3	2.45	0.74	0.030	12.2	43 000	39 000	—	51 000	**606**	**606 ZZ**	**606 2RU**	**606 2RD**	—	8	15	0.3	5.8
	19	6	6	0.3	0.3	3.25	1.05	0.040	12.3	35 000	32 000	27 000	43 000	**626**	**626 ZZ**	**626 2RU**	**626 2RD**	**626 2RS**	8	17	0.3	8.1
	22	7	7	0.3	0.3	4.10	1.35	0.060	12.4	31 000	—	23 000	37 000	**636**	**636 ZZ**	—	—	**636 2RS**	8	20	0.3	13
7	11	2.5	3	0.1	0.08	0.540	0.23	0.006	16.1	49 000	—	—	59 000	**ML7011**	**WML7011 ZZX**	—	—	—	7.8	10.2	0.08	0.7
	13	3	4	0.15	0.15	0.680	0.28	0.007	14.9	47 000	—	—	55 000	**ML7013**	**WML7013 ZZ**	—	—	—	8.2	11.8	0.15	1.4
	14	3.5	5	0.15	0.15	1.45	0.51	0.010	14.2	45 000	—	—	54 000	**687**	**W687 ZZ**	—	—	—	8.2	12.8	0.15	2.0
	17	5	5	0.3	0.3	2.00	0.71	0.02	14.0	42 000	—	28 000	50 000	**697**	**697 ZZ**	—	—	**697 2RS**	9	15	0.3	5.3
	19	6	6	0.3	0.3	2.95	0.89	0.040	12.3	40 000	36 000	27 000	47 000	**607**	**607 ZZ**	**607 2RU**	**607 2RD**	**607 2RS**	9	17	0.3	7.6
	22	7	7	0.3	0.3	4.10	1.35	0.060	12.4	31 000	28 000	23 000	37 000	**627**	**627 ZZ**	**627 2RU**	**627 2RD**	**627 2RS**	9	20	0.3	13
	26	9	9	0.3	0.3	5.65	1.95	0.100	12.3	26 000	—	—	32 000	**637**	**637 ZZ**	—	—	—	9	24	0.3	24
8	12	2.5	3.5	0.1	0.08	0.680	0.27	0.007	16.4	47 000	—	—	55 000	**ML8012**	**WML8012 ZZ**	—	—	—	8.8	11.2	0.08	0.8

d (8) ~ 9 mm

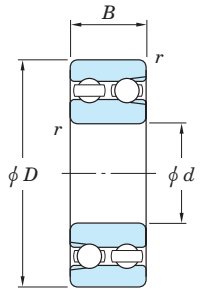




Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1}) Grease lub.			Oil lub.	Bearing No.					Mounting dimensions (mm)			(Refer.) Mass
d	D	B	B_1	r 1) min.	r_1 1) min.	C_r	C_{0r}	C_u	f_0	Open ZZ, 2RU	(2RD)	(2RS)	Open Z	Open	Shielded	Non-contact sealed	Extremely light shielded	Contact sealed	d_a min.	D_a max.	r_a max.	(g)
8	14	3.5	4	0.15	0.15	1.00	0.39	0.010	15.3	44 000	—	—	52 000	**ML8014**	**WML8014 ZZ**	—	—	—	9.2	12.8	0.15	1.8
	16	4	5	0.2	0.2	1.55	0.59	0.020	14.0	42 000	38 000	28 000	50 000	**688**	**W688 ZZ**	**W688 2RU**	**W688 2RD**	**W688 2RS**	9.6	14.4	0.2	3.2
	19	6	6	0.3	0.3	2.80	0.91	0.040	12.9	39 000	35 000	27 000	46 000	**698**	**698 ZZ**	—	**698 2RD**	**698 2RS**	10	17	0.3	7.2
	22	7	7	0.3	0.3	4.10	1.35	0.060	12.4	34 000	31 000	23 000	41 000	**608**	**608 ZZ**	**608 2RU**	**608 2RD**	**608 2RS**	10	20	0.3	12
	24	8	8	0.3	0.3	4.15	1.40	0.060	12.8	28 000	—	22 000	35 000	**628**	**628 ZZ**	**628 2RU**	—	**628 2RS**	10	22	0.3	18
	28	9	9	0.3	0.3	5.65	1.95	0.100	12.3	26 000	23 000	—	32 000	**638**	**638 ZZ**	—	**638 2RD**	—	10	26	0.3	29
9	17	4	5	0.2	0.2	1.65	0.66	0.020	14.9	39 000	35 000	—	46 000	**689**	**W689 ZZ**	**W689 2RU**	**W689 2RD**	—	10.6	15.4	0.2	3.5
	20	6	6	0.3	0.3	3.10	1.05	0.040	13.3	35 000	32 000	25 000	42 000	**699**	**699 ZZ**	—	**699 2RD**	**699 2RS**	11	18	0.3	7.5
	24	7	7	0.3	0.3	4.15	1.40	0.060	12.8	33 000	30 000	22 000	40 000	**609**	**609 ZZ**	**609 2RU**	**609 2RD**	**609 2RS**	11	22	0.3	15
	26	8	8	(0.6)	(0.6)	5.70	1.95	0.100	12.4	27 000	24 000	19 000	33 000	**629**	**629 ZZ**	**629 2RU**	**629 2RD**	**629 2RS**	12.1	22	0.3	20
	30	10	10	0.6	0.6	7.50	2.65	0.210	12.3	24 000	—	—	29 000	**639**	**639 ZZ**	—	—	—	13	26	0.6	35

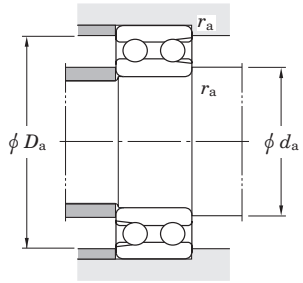
[Note] 1) Numerical values in ( ) do not conform to JIS B 1521.

d 10 ~ (60) mm



Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
10	30	14	0.6	8.30	4.65	0.239	10.2	15 000	20 000	**4200**	14	26	0.6	0.057
12	32	14	0.6	8.40	4.80	0.249	10.6	14 000	18 000	**4201**	16	28	0.6	0.062
15	35	14	0.6	10.6	7.05	0.364	11.1	12 000	16 000	**4202**	19	31	0.6	0.071
	42	17	1	14.2	9.15	0.472	10.7	11 000	14 000	**4302**	20	37	1	0.123
17	40	16	0.6	12.6	8.15	0.421	11.0	11 000	14 000	**4203**	21	36	0.6	0.106
	47	19	1	17.8	11.7	0.604	10.7	9 400	13 000	**4303**	22	42	1	0.171
20	47	18	1	17.8	12.5	0.646	11.1	9 000	12 000	**4204**	25	42	1	0.165
	52	21	1.1	16.3	10.0	0.519	10.9	8 300	11 000	**4304**	26.5	45.5	1	0.227
25	52	18	1	17.7	13.2	0.681	11.7	7 500	9 900	**4205**	30	47	1	0.189
	62	24	1.1	28.4	20.1	1.04	11.0	6 700	9 000	**4305**	31.5	55.5	1	0.365
30	62	20	1	23.9	19.3	1.00	11.8	6 400	8 500	**4206**	35	57	1	0.298
	72	27	1.1	38.4	28.0	1.45	11.0	5 700	7 600	**4306**	36.5	65.5	1	0.542
35	72	23	1.1	28.9	24.0	1.24	11.9	5 600	7 400	**4207**	41.5	65.5	1	0.460
	80	31	1.5	43.9	32.7	1.69	11.0	5 200	7 000	**4307**	43	72	1.5	0.752
40	80	23	1.1	36.8	33.1	1.71	12.1	4 700	6 300	**4208**	46.5	73.5	1	0.558
	90	33	1.5	49.8	38.2	1.97	11.2	4 600	6 100	**4308**	48	82	1.5	1.01
45	85	23	1.1	35.0	34.3	1.77	12.4	4 600	6 100	**4209**	51.5	78.5	1	0.605
	100	36	1.5	62.3	48.8	2.52	11.2	4 100	5 500	**4309**	53	92	1.5	1.35
50	90	23	1.1	34.6	34.8	1.80	12.6	4 200	5 600	**4210**	56.5	83.5	1	0.651
	110	40	2	76.1	60.7	3.14	11.1	3 700	5 000	**4310**	59	101	2	1.80
55	100	25	1.5	40.9	42.2	2.18	12.6	3 800	5 000	**4211**	63	92	1.5	0.882
	120	43	2	91.1	73.8	3.81	11.1	3 400	4 600	**4311**	64	111	2	2.29
60	110	28	1.5	52.6	52.7	2.72	12.4	3 500	4 700	**4212**	68	102	1.5	1.20

d (60) ~ 75 mm



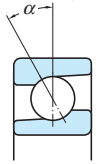
Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Factor	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_u	f_0	Grease lub.	Oil lub.		d_a min.	D_a max.	r_a max.	
60	130	46	2.1	107	88.4	4.57	11.1	3 100	4 200	**4312**	71	119	2	2.87
65	120	31	1.5	60.0	61.3	3.16	12.4	3 200	4 300	**4213**	73	112	1.5	1.59
	140	48	2.1	116	97.3	4.96	11.2	2 900	3 900	**4313**	76	129	2	3.46
70	125	31	1.5	68.2	70.1	3.63	12.4	3 100	4 100	**4214**	78	117	1.5	1.68
	150	51	2.1	125	106	5.25	11.3	2 700	3 600	**4314**	81	139	2	4.21
75	130	31	1.5	67.7	70.8	3.61	12.5	2 900	3 900	**4215**	83	122	1.5	1.77
	160	55	2.1	143	124	5.90	11.2	2 500	3 400	**4315**	86	149	2	5.15

Angular contact ball bearings

Angular contact ball bearings are suitable for applications which require high accuracy and good high-speed performance. This type of bearing is designed to carry a combined load.

■ Single-row angular contact ball bearings and matched pair angular contact ball bearings

Contact angle



- The standard contact angles are 15°, 30° and 40°. They are identified, respectively, by the supplementary codes "C", "A" (omitted) and "B". Bearings with a smaller contact angle are more suitable for applications involving high-speed rotation. Those with a larger contact angle feature superior axial load resistance.
- Angular contact ball bearings are often preloaded to enhance their rigidity and rotating performance. (refer to p. A 112.)
  For high-precision matched pair angular contact ball bearings of class 5 or higher, which are used in machine tools and other precision equipment, the standard preload is specified in four levels: slight (S), light (L), medium (M) and heavy (H). (refer to Table 11-2 on p. A 114.)
- When this type of bearing is loaded radially, an axial component of force is produced. In this case, two bearings are used together facing one another, or two or more bearings are matched and used. (refer to p. A 38.)
- Tables 1 and 2 list the different types of single-row and matched pair/stack angular contact ball bearings and describe their characteristics.

■ Double-row angular contact ball bearings

Consist of two single-row angular contact ball bearings matched back-to-back, with inner and outer rings integrated.
Table 3 shows major types and their characteristics.

■ Four-point contact ball bearings

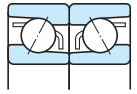
- Have a contact angle of 35° and an inner ring divided into two annular pieces. They are suitable for applications that involve either axial loading or combined loading, where the axial load makes up the major part of the load.
- Able to support both axial load and a certain degree of radial load. Each rolling element is in contact with each of the inner and outer rings at a single point, and both contact points lie on the contact angle line. The line runs to either the right or left depending on the direction of the axial load.

Single-row angular contact ball bearings



Bore diameter **10** – **380** mm

Matched pair angular contact ball bearings



Bore diameter **10** – **380** mm

Double-row angular contact ball bearings

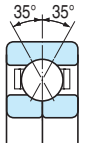


Bore diameter **10** – **110** mm

Four-point contact ball bearings



Bore diameter **20** – **110** mm



- Consult with JTEKT when using the four-point contact ball bearing because application conditions such as load magnitude should be examined carefully.

Table 1 Single-row angular contact ball bearings

Standard type



(with pressed cage)



(with machined cage)

- Single-row angular contact ball bearings accommodate radial load and axial load in one direction.
- Bearings with a machined cage are suitable for high-speed applications.

Reference **G-type bearing**

"G-type" bearings have a stand-out between the inner ring and outer ring on both sides that are equal in size. This arrangement is called "flush ground processing." These bearings can be matched in a variety of ways.

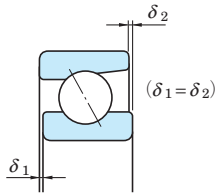
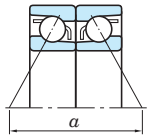
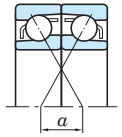
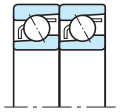
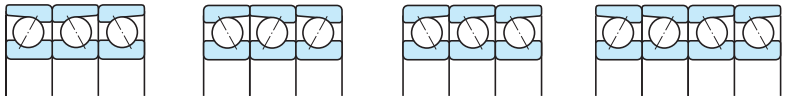


Table 2 Matched pair and stack angular contact ball bearings

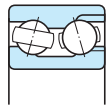
Back-to-back arrangement (DB)		• Carries radial load and axial load in both directions. • Suitable for applications involving moment loading because the distance between the load centers (a) is long. • As for the preloaded type, the clearance is pre-adjusted so that bearings will be preloaded the proper amount when the inner ring is fixed with a nut.
Face-to-face arrangement (DF)		• Carries radial load and axial load in both directions. • Has a smaller moment load accommodating capacity than the back-to-back arrangement, because the distance between the load centers (a) is shorter. • As for the preloaded type, the clearance is pre-adjusted so that bearings will be preloaded the proper amount when the outer rings are pressed together.
Tandem arrangement (DT)		• Carries radial load and axial load in one direction. • Suitable for applications which involve a high degree of axial loading.

Matched stack

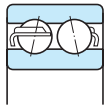


(DBD) (DFD) (DTD) (DBB)

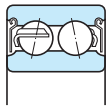
Table 3 Double-row angular contact ball bearings



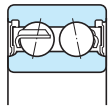
(with filling slot) 32, 33



(without filling slot) 52, 53



Shielded 52...ZZ, 53...ZZ



Sealed 52...2RS, 53...2RS

- Accommodates radial load and axial load in both directions. Also able to accommodate moment load. When installing bearings with filling slot (32 and 33 series), the raceway side without filling slot must accommodate main load.
- The 32 and 33 series are provided with a filling slot, while the 52 and 53 series are not.

  32 and 33 series : contact angle 32°
  52 and 53 series : contact angle 24°
- Inferior to single-row and matched pair angular contact ball bearings in terms of high-speed and high accuracy performance.
- Shielded or sealed 52 and 53 series bearings are also available.

Boundary dimensions	The dimensions of standard series are as specified in JIS B 1512.
Tolerances	· As specified in JIS B 1514-1. (refer to Table 7-3 on pp. A 60 – A 63.) · JTEKT has established "special tolerances" for bore diameter and outside diameter, as listed in the table to the right, to make it easy to produce high-precision matched stack bearings. Bearings which are produced based on these tolerances are identified by the supplementary code "K5."
Internal clearance	■ Matched pair bearing axial internal clearance.....(refer to Table 10-4 on p. A 103.) ■ Double-row bearing radial internal clearance.....(refer to Table 10-5 on p. A 104.)
Recommended fits	■ Classes 0 and 6 bearings.....(refer to Table 9-4 on pp. A 91, 92.) ■ Classes 5 and 4 bearings.....as listed in the table below. ■ Refer to Table 11-3 on page A 115 for the recommended fits of high-precision matched pair bearings (class 5 and class 4), which are used with light preload (L) or middle preload (M).

Special tolerances (K5) Unit : μm

Nominal bore diameter d (mm) over	up to	Single plane mean bore diameter (Δ_{dmp}) or single plane mean outside diameter deviation (Δ_{Dmp}) class 5 upper	class 5 lower	class 4 upper	class 4 lower
–	50	– 1	– 4	– 1	– 3
50	80	– 1	– 5	– 1	– 4
80	120	– 1	– 5	– 1	– 4

Fit		class 5 Tolerance class	class 4 Tolerance class
With shaft	Inner ring rotation	js 5	js 4
	Outer ring rotation	h 5	h 4
With housing	Fixed side	JS 6	JS 5
	Free side	H 6	H 5
	Outer ring rotation	M 5	M 4

Standard cages

- Pressed cage (supplementary code : //)
- Copper alloy machined cage (supplementary code : FY)

[Note] Machine tools are generally equipped with bearings that have a phenolic resin machined cage (FT). Bearings with a polyamide molded cage can also be used depending on the applications. Four-point contact ball bearings usually use a copper alloy machined cage.

Application of standard cages

Bearing series	Pressed cage	Machined cage
79C	—	7900C – 7932C
79CPA	—	7900CPA – 7932CPA
70	—	7000 – 7040
70B	—	7000B – 7040B
70C	—	7000C – 7040C
70CPA	—	7000CPA – 7034CPA
72	7200 – 7220	7200 – 7240
72B	7200B – 7220B	7200B – 7240B
72C	7200C – 7220C	7200C – 7240C
72CPA	—	7200CPA – 7230CPA
73	7300 – 7320	7300 – 7340
73B	7303B – 7320B	7303B – 7340B
73C	7303C – 7320C	7303C – 7334C
74	7405 – 7409	7404 – 7418
74B	7405B – 7409B	7404B – 7418B
32	3200 – 3215	3216 – 3222
33	3302 – 3313	3314 – 3322
52	5203 – 5214	—
53	5304 – 5315	—

Allowable misalignment

Single-row.....0.000 6 rad (2') : Matched pair, double-row.....misalignment not allowed

Equivalent radial load

[Single-row and matched pair angular contact ball bearings]

Dynamic equivalent radial load

$$P_r = XF_r + YF_a$$

Contact angle	$if_0F_a^*/C_{0r}$	e	Single-row and tandem arrangement				Back-to-back and face-to-face arrangement			
			$F_a/F_r \leqq e$		$F_a/F_r > e$		$F_a/F_r \leqq e$		$F_a/F_r > e$	
			X	Y	X	Y	X	Y	X	Y
15°	0.178	0.38	1	0	0.44	1.47	1	1.65	0.72	2.39
	0.357	0.40				1.40		1.57		2.28
	0.714	0.43				1.30		1.46		2.11
	1.07	0.46				1.23		1.38		2.00
	1.43	0.47				1.19		1.34		1.93
	2.14	0.50				1.12		1.26		1.82
	3.57	0.55				1.02		1.14		1.66
	5.35	0.56				1.00		1.12		1.63
	7.14	0.56				1.00		1.12		1.63
30°	–	0.80	1	0	0.39	0.76	1	0.78	0.63	1.24
40°	–	1.14	1	0	0.35	0.57	1	0.55	0.57	0.93

* For i, use 2 for DB&DF and 1 for single&DT.
Factor f_0 is shown in the bearing dimention table.

[Note] When two single-row angular contact ball bearings are used facing one another, an axial component of force is produced under radial load. In this case, refer to page A 38 for calculation of the dynamic equivalent radial load.

Static equivalent radial load

$$P_{0r} = X_0 F_r + Y_0 F_a$$

In reference to single-row and tandem arrangement bearings,

when $P_{0r} < F_r$, $P_{0r} = F_r$

Contact angle	Single-row and tandem arrangement		Back-to-back and face-to-face arrangement	
	X_0	Y_0	X_0	Y_0
15°	0.5	0.46	1	0.92
30°	0.5	0.33	1	0.66
40°	0.5	0.26	1	0.52

Equivalent radial load

[Double-row angular contact ball bearings]

Dynamic equivalent radial load

$$P_r = XF_r + YF_a$$

Contact angle	e	$F_a/F_r \leqq e$		$F_a/F_r > e$		(reference)
		X	Y	X	Y	
24°	0.66	1	0.95	0.68	1.45	52, 53 series
32°	0.86	1	0.73	0.62	1.17	32, 33 series

Static equivalent radial load

$$P_{0r} = X_0 F_r + Y_0 F_a$$

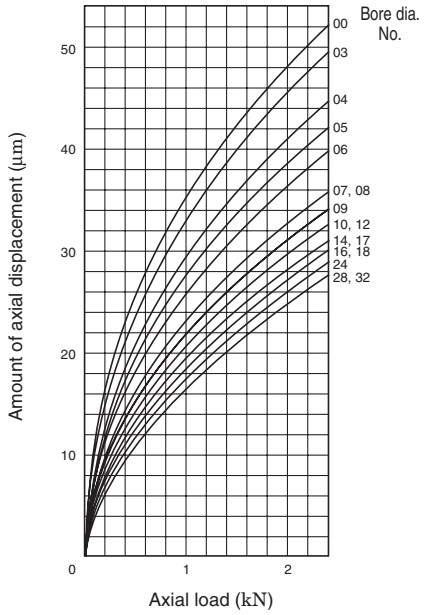
Contact angle	X_0	Y_0	(reference)
24°	1	0.78	52, 53 series
32°	1	0.63	32, 33 series

[Note] In angular contact ball bearings, slippage occurs between the balls and raceways under too small a load, causing smearing to develop.
Matched pair bearings may develop smearing when the ratio of the axial load to the radial load exceeds the value of e ($F_a / F_r > e$), as listed in the specification table. Consult with JTEKT when these bearings are used under the above conditions.

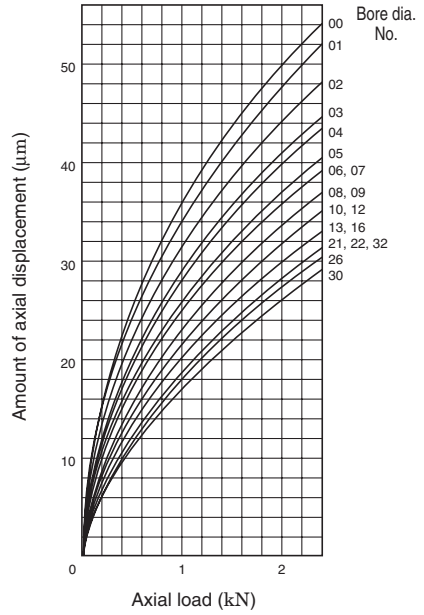
[Reference] Relationship between axial load and axial displacement

Diagrams (1) to (9) illustrate the relationship between axial load and axial displacement.

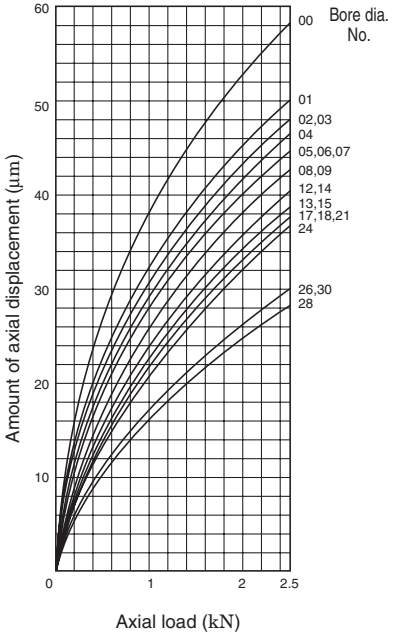
(1) 7900C (contact angle 15°)



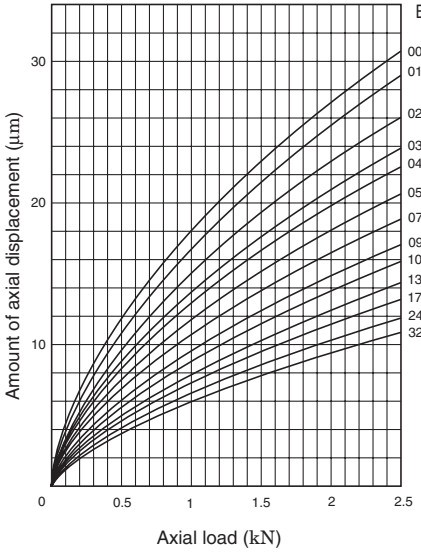
(2) 7000C (contact angle 15°)



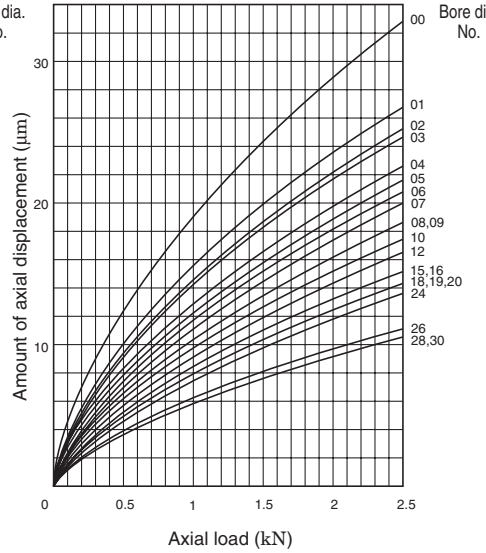
(3) 7200C (contact angle 15°)



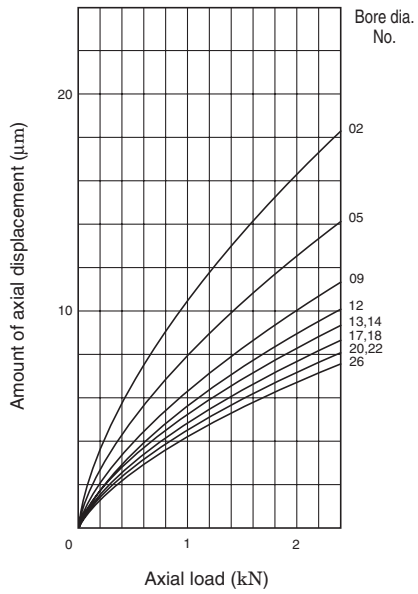
(4) 7000 (contact angle 30°)



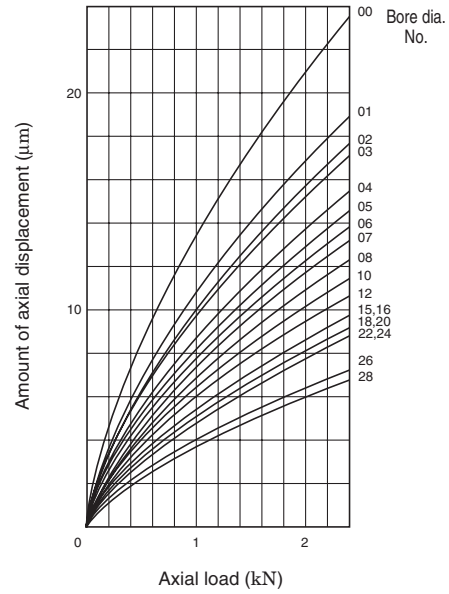
(5) 7200 (contact angle 30°)



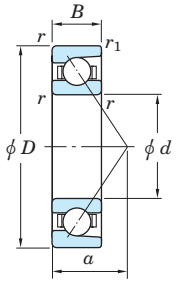
(6) 7000B (contact angle 40°)



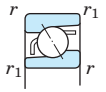
(7) 7200B (contact angle 40°)



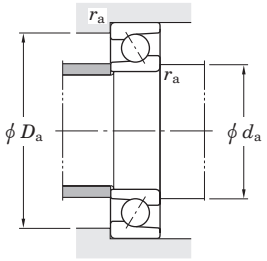
d 10 ~ (17) mm



With machined cage



With pressed cage



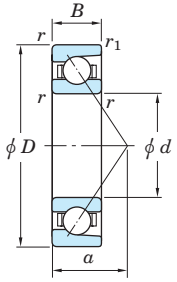
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN)		Factor		Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage		C_u											
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0		Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
10	22	6	0.3	0.15	3.75	1.50	—	—	0.060	—	14.2		52 000	69 000	**7900C**	5.1	12.5	19.5	0.3	0.008
	26	8	0.3	0.15	6.25	2.35	—	—	0.120	—	—		34 000	42 000	**7000**	9.1	12.5	23.5	0.3	0.021
	26	8	0.3	0.15	5.80	2.15	—	—	0.110	—	—		25 000	33 000	**7000B**	11.6	12.5	23.5	0.3	0.021
	26	8	0.3	0.15	6.60	2.45	—	—	0.130	—	12.5		47 000	62 000	**7000C**	6.4	12.5	23.5	0.3	0.021
	30	9	0.6	0.3	5.85	2.20	6.75	2.75	0.110	0.140	—		29 000	37 000	**7200**	10.4	14.5	25.5	0.6	0.031
	30	9	0.6	0.3	5.35	2.00	6.20	2.50	0.100	0.130	—		22 000	29 000	**7200B**	13.1	14.5	25.5	0.6	0.031
	30	9	0.6	0.3	6.25	2.35	7.25	2.95	0.120	0.150	13.4		40 000	54 000	**7200C**	7.2	14.5	25.5	0.6	0.031
	35	11	0.6	0.3	10.6	3.75	11.6	4.30	0.300	0.340	—		27 000	33 000	**7300**	12.0	14.5	30.5	0.6	0.054
12	24	6	0.3	0.15	4.00	1.70	—	—	0.070	—	14.7		48 000	62 000	**7901C**	5.4	14.5	21.5	0.3	0.010
	28	8	0.3	0.15	6.75	2.75	—	—	0.140	—	—		29 000	37 000	**7001**	9.9	14.5	25.5	0.3	0.024
	28	8	0.3	0.15	6.20	2.50	—	—	0.130	—	—		22 000	29 000	**7001B**	12.6	14.5	25.5	0.3	0.024
	28	8	0.3	0.15	7.25	2.95	—	—	0.150	—	13.4		40 000	54 000	**7001C**	6.7	14.5	25.5	0.3	0.024
	32	10	0.6	0.3	9.30	3.65	10.0	4.05	0.280	0.310	—		27 000	34 000	**7201**	11.4	16.5	27.5	0.6	0.038
	32	10	0.6	0.3	8.65	3.40	9.30	3.75	0.240	0.270	—		20 000	27 000	**7201B**	14.2	16.5	27.5	0.6	0.038
	32	10	0.6	0.3	9.90	3.85	10.6	4.30	0.300	0.330	12.5		38 000	50 000	**7201C**	7.9	16.5	27.5	0.6	0.038
	37	12	1	0.6	12.8	4.60	14.0	5.25	0.360	0.410	—		24 000	31 000	**7301**	13.1	17.5	31.5	1	0.065
15	28	7	0.3	0.15	5.95	2.65	—	—	0.110	—	14.5		39 000	52 000	**7902C**	6.4	17.5	25.5	0.3	0.015
	32	9	0.3	0.15	7.65	3.45	—	—	0.180	—	—		26 000	32 000	**7002**	11.3	17.5	29.5	0.3	0.035
	32	9	0.3	0.15	6.95	3.15	—	—	0.160	—	—		19 000	25 000	**7002B**	14.6	17.5	29.5	0.3	0.035
	32	9	0.3	0.15	8.25	3.70	—	—	0.190	—	14.1		35 000	47 000	**7002C**	7.6	17.5	29.5	0.3	0.035
	35	11	0.6	0.3	10.1	4.25	10.1	4.25	0.300	0.300	—		24 000	29 000	**7202**	12.9	19.5	30.5	0.6	0.048
	35	11	0.6	0.3	9.30	3.95	9.30	3.95	0.260	0.260	—		18 000	24 000	**7202B**	16.2	19.5	30.5	0.6	0.048
	35	11	0.6	0.3	10.8	4.55	10.8	4.55	0.340	0.340	13.3		33 000	43 000	**7202C**	8.9	19.5	30.5	0.6	0.048
	42	13	1	0.6	15.7	6.45	16.8	7.20	0.490	0.550	—		20 000	25 000	**7302**	15.0	20.5	36.5	1	0.088
17	30	7	0.3	0.15	6.25	2.95	—	—	0.120	—	14.9		36 000	47 000	**7903C**	6.7	19.5	27.5	0.3	0.016
	35	10	0.3	0.15	8.40	4.15	—	—	0.210	—	—		23 000	28 000	**7003**	12.7	19.5	32.5	0.3	0.045

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

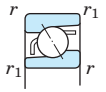
2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

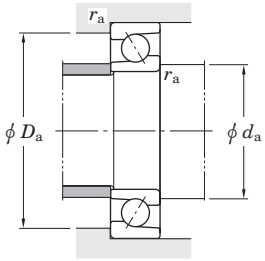
d (17) ~ (25) mm



With machined cage



With pressed cage



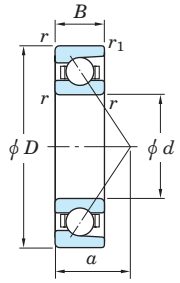
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage												
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0	Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
17	35	10	0.3	0.15	7.60	3.75	—	—	0.190	—	—	17 000	23 000	**7003B**	16.1	19.5	32.5	0.3	0.045
	35	10	0.3	0.15	9.15	4.45	—	—	0.230	—	14.6	31 000	41 000	**7003C**	8.6	19.5	32.5	0.3	0.045
	40	12	0.6	0.3	12.7	5.50	12.7	5.50	0.380	0.380	—	21 000	26 000	**7203**	14.4	21.5	35.5	0.6	0.070
	40	12	0.6	0.3	11.7	5.05	11.7	5.05	0.330	0.330	—	16 000	21 000	**7203B**	18.2	21.5	35.5	0.6	0.070
	40	12	0.6	0.3	13.6	5.90	13.6	5.90	0.440	0.440	13.4	29 000	38 000	**7203C**	9.9	21.5	35.5	0.6	0.070
	47	14	1	0.6	18.7	7.90	20.0	8.75	0.590	0.660	—	18 000	23 000	**7303**	16.5	22.5	41.5	1	0.120
	47	14	1	0.6	17.3	7.30	18.5	8.10	0.510	0.570	—	14 000	18 000	**7303B**	20.8	22.5	41.5	1	0.120
	47	14	1	0.6	19.8	8.40	19.8	8.40	0.650	0.650	12.6	25 000	33 000	**7303C**	11.4	22.5	41.5	1	0.120
20	37	9	0.3	0.15	9.10	4.55	—	—	0.240	—	14.9	30 000	39 000	**7904C**	8.3	22.5	34.5	0.3	0.035
	42	12	0.6	0.3	12.9	6.10	—	—	0.390	—	—	19 000	24 000	**7004**	15.1	24.5	37.5	0.6	0.079
	42	12	0.6	0.3	11.7	5.55	—	—	0.340	—	—	14 000	19 000	**7004B**	19.2	24.5	37.5	0.6	0.079
	42	12	0.6	0.3	13.9	6.60	—	—	0.450	—	14.1	26 000	35 000	**7004C**	10.2	24.5	37.5	0.6	0.079
	47	14	1	0.6	18.1	8.40	19.2	9.15	0.580	0.640	—	17 000	22 000	**7204**	17.0	25.5	41.5	1	0.112
	47	14	1	0.6	16.6	7.70	17.6	8.40	0.500	0.550	—	13 000	17 000	**7204B**	21.5	25.5	41.5	1	0.112
	47	14	1	0.6	19.4	9.00	20.6	9.80	0.670	0.730	13.4	24 000	32 000	**7204C**	11.6	25.5	41.5	1	0.112
	52	15	1.1	0.6	21.8	9.40	23.4	10.4	0.710	0.790	—	17 000	21 000	**7304**	17.9	27	45	1	0.150
	52	15	1.1	0.6	20.2	8.70	21.7	9.65	0.610	0.680	—	13 000	17 000	**7304B**	22.6	27	45	1	0.150
	52	15	1.1	0.6	23.1	9.95	24.8	11.1	0.780	0.860	12.6	23 000	31 000	**7304C**	12.3	27	45	1	0.150
	72	19	1.1	0.6	44.5	19.1	—	—	1.50	—	—	9 600	13 000	**7404**	23.1	27	65	1	0.395
	72	19	1.1	0.6	41.9	17.9	—	—	1.40	—	—	8 500	12 000	**7404B**	29.2	27	65	1	0.395
25	42	9	0.3	0.15	10.2	5.45	—	—	0.300	—	15.5	25 000	33 000	**7905C**	9.1	27.5	39.5	0.3	0.041
	47	12	0.6	0.3	14.1	7.40	—	—	0.450	—	—	17 000	21 000	**7005**	16.4	29.5	42.5	0.6	0.091
	47	12	0.6	0.3	12.8	6.70	—	—	0.390	—	—	12 000	17 000	**7005B**	21.1	29.5	42.5	0.6	0.091
	47	12	0.6	0.3	15.4	8.00	—	—	0.510	—	14.7	23 000	30 000	**7005C**	10.8	29.5	42.5	0.6	0.091
	52	15	1	0.6	19.2	9.50	20.2	10.3	0.620	0.670	—	15 000	19 000	**7205**	18.8	30.5	46.5	1	0.135
	52	15	1	0.6	17.5	8.70	18.4	9.40	0.530	0.580	—	12 000	15 000	**7205B**	23.9	30.5	46.5	1	0.135

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

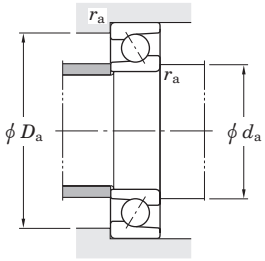
d (25) ~ (35) mm



With machined cage



With pressed cage



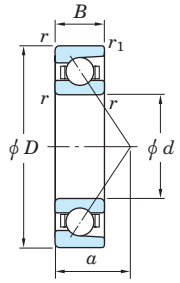
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN)		Factor		Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage		C_u											
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	(With machined cage)	(With pressed cage)	f_0		Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
25	52	15	1	0.6	20.7	10.2	21.9	11.1	0.710	0.770	14.0		21 000	28 000	**7205C**	12.7	30.5	46.5	1	0.135
	62	17	1.1	0.6	31.0	14.4	33.0	15.8	1.05	1.15	—		14 000	17 000	**7305**	21.1	32	55	1	0.243
	62	17	1.1	0.6	28.6	13.3	30.5	14.6	0.910	1.00	—		10 000	14 000	**7305B**	26.8	32	55	1	0.243
	62	17	1.1	0.6	33.0	15.3	35.1	16.8	1.20	1.30	12.8		19 000	25 000	**7305C**	14.3	32	55	1	0.243
	80	21	1.5	1	49.7	23.2	53.3	25.7	1.80	2.00	—		8 200	11 000	**7405**	26.4	33.5	71.5	1.5	0.527
	80	21	1.5	1	46.1	21.5	49.5	23.9	1.55	1.70	—		7 300	10 000	**7405B**	33.6	33.5	71.5	1.5	0.527
30	47	9	0.3	0.15	10.4	6.25	—	—	0.320	—	15.9		22 000	29 000	**7906C**	9.7	32.5	44.5	0.3	0.046
	55	13	1	0.6	18.2	10.1	—	—	0.610	—	—		14 000	18 000	**7006**	18.8	35.5	49.5	1	0.133
	55	13	1	0.6	16.4	9.20	—	—	0.530	—	—		11 000	14 000	**7006B**	24.3	35.5	49.5	1	0.133
	55	13	1	0.6	19.8	11.0	—	—	0.690	—	14.9		20 000	26 000	**7006C**	12.2	35.5	49.5	1	0.133
	62	16	1	0.6	26.7	13.7	28.1	14.8	0.890	0.970	—		13 000	16 000	**7206**	21.5	35.5	56.5	1	0.208
	62	16	1	0.6	24.3	12.5	25.6	13.6	0.770	0.840	—		9 600	13 000	**7206B**	27.6	35.5	56.5	1	0.208
	62	16	1	0.6	28.8	14.7	30.4	16.0	1.00	1.10	14.0		18 000	24 000	**7206C**	14.3	35.5	56.5	1	0.208
	72	19	1.1	0.6	37.6	18.9	39.9	20.6	1.30	1.45	—		12 000	14 000	**7306**	24.5	37	65	1	0.362
	72	19	1.1	0.6	34.5	17.4	36.6	19.0	1.15	1.25	—		8 700	12 000	**7306B**	31.3	37	65	1	0.362
	72	19	1.1	0.6	40.4	20.3	42.8	22.1	1.50	1.65	13.4		16 000	21 000	**7306C**	16.5	37	65	1	0.362
	90	23	1.5	1	59.5	28.4	63.9	31.6	2.20	2.45	—		7 300	9 700	**7406**	29.3	38.5	81.5	1.5	0.686
	90	23	1.5	1	55.2	26.4	59.3	29.3	1.90	2.10	—		6 500	8 900	**7406B**	37.3	38.5	81.5	1.5	0.686
35	55	10	0.6	0.3	15.7	9.70	—	—	0.550	—	15.7		19 000	25 000	**7907C**	11.0	39.5	50.5	0.6	0.074
	62	14	1	0.6	21.9	12.6	—	—	0.740	—	—		12 000	15 000	**7007**	21.2	40.5	56.5	1	0.170
	62	14	1	0.6	19.7	11.4	—	—	0.640	—	—		9 200	12 000	**7007B**	27.6	40.5	56.5	1	0.170
	62	14	1	0.6	23.9	13.7	—	—	0.840	—	15.0		17 000	22 000	**7007C**	13.5	40.5	56.5	1	0.170
	72	17	1.1	0.6	35.2	18.6	37.1	20.2	1.20	1.30	—		11 000	14 000	**7207**	24.2	42	65	1	0.295
	72	17	1.1	0.6	32.0	17.0	33.8	18.5	1.05	1.15	—		8 300	11 000	**7207B**	31.4	42	65	1	0.295
	72	17	1.1	0.6	38.0	20.1	40.1	21.7	1.40	1.50	14.0		15 000	20 000	**7207C**	15.8	42	65	1	0.295
	80	21	1.5	1	44.2	22.0	49.9	26.4	1.55	1.85	—		10 000	13 000	**7307**	27.4	43.5	71.5	1.5	0.475
	80	21	1.5	1	40.6	20.2	45.8	24.3	1.30	1.60	—		7 700	10 000	**7307B**	35.0	43.5	71.5	1.5	0.475

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

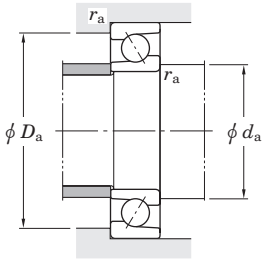
d (35) ~ 45 mm



With machined cage



With pressed cage



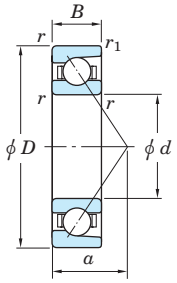
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor		Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage													
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0		Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
35	80	21	1.5	1	47.4	23.6	53.5	28.3	1.75	2.10	13.4		14 000	19 000	**7307C**	18.3	43.5	71.5	1.5	0.475
	100	25	1.5	1	75.6	37.0	81.1	41.1	2.85	3.20	—		6 500	8 600	**7407**	32.6	43.5	91.5	1.5	0.950
	100	25	1.5	1	70.2	34.3	75.3	38.1	2.45	2.75	—		5 700	7 900	**7407B**	41.7	43.5	91.5	1.5	0.950
40	62	12	0.6	0.3	19.7	12.4	—	—	0.710	—	15.7		17 000	22 000	**7908C**	12.8	44.5	57.5	0.6	0.107
	68	15	1	0.6	23.4	14.6	—	—	0.830	—	—		11 000	14 000	**7008**	23.2	45.5	62.5	1	0.210
	68	15	1	0.6	21.1	13.2	—	—	0.720	—	—		8 300	11 000	**7008B**	30.2	45.5	62.5	1	0.210
	68	15	1	0.6	25.7	15.9	—	—	0.940	—	15.4		15 000	20 000	**7008C**	14.8	45.5	62.5	1	0.210
	80	18	1.1	0.6	42.0	23.3	44.1	25.1	1.50	1.60	—		10 000	12 000	**7208**	26.3	47	73	1	0.382
	80	18	1.1	0.6	38.2	21.3	40.2	23.0	1.30	1.40	—		7 500	10 000	**7208B**	34.2	47	73	1	0.382
	80	18	1.1	0.6	45.4	25.2	47.7	27.1	1.70	1.85	14.2		14 000	18 000	**7208C**	17.0	47	73	1	0.382
	90	23	1.5	1	54.0	27.4	61.0	32.9	1.90	2.30	—		9 200	12 000	**7308**	30.3	48.5	81.5	1.5	0.657
	90	23	1.5	1	49.6	25.2	56.0	30.3	1.65	2.00	—		6 900	9 200	**7308B**	38.8	48.5	81.5	1.5	0.657
	90	23	1.5	1	57.9	29.4	65.4	35.3	2.20	2.65	13.4		13 000	17 000	**7308C**	20.2	48.5	81.5	1.5	0.657
	110	27	2	1	87.4	43.5	93.8	48.4	3.35	3.70	—		5 900	7 900	**7408**	35.5	50	100	2	1.23
	110	27	2	1	81.1	40.4	87.0	44.9	2.90	3.20	—		5 200	7 200	**7408B**	45.4	50	100	2	1.23
45	68	12	0.6	0.3	20.8	14.1	—	—	0.770	—	16.0		15 000	20 000	**7909C**	13.6	49.5	63.5	0.6	0.127
	75	16	1	0.6	27.8	17.7	—	—	1.00	—	—		10 000	12 000	**7009**	25.3	50.5	69.5	1	0.260
	75	16	1	0.6	25.0	16.0	—	—	0.870	—	—		7 500	10 000	**7009B**	33.2	50.5	69.5	1	0.260
	75	16	1	0.6	30.5	19.3	—	—	1.15	—	15.4		14 000	18 000	**7009C**	16.0	50.5	69.5	1	0.260
	85	19	1.1	0.6	47.2	26.6	49.6	28.6	1.70	1.85	—		9 400	12 000	**7209**	28.0	52	78	1	0.430
	85	19	1.1	0.6	42.9	24.3	45.1	26.1	1.50	1.60	—		7 000	9 400	**7209B**	36.4	52	78	1	0.430
	85	19	1.1	0.6	51.0	28.7	53.6	30.9	1.95	2.10	14.2		13 000	17 000	**7209C**	18.1	52	78	1	0.430
	100	25	1.5	1	68.9	37.1	73.1	40.4	2.55	2.80	—		8 200	10 000	**7309**	33.6	53.5	91.5	1.5	0.875
	100	25	1.5	1	63.2	34.1	67.0	37.2	2.20	2.40	—		6 200	8 200	**7309B**	43.1	53.5	91.5	1.5	0.875
	100	25	1.5	1	74.0	39.7	78.4	43.4	2.95	3.20	13.5		11 000	15 000	**7309C**	22.3	53.5	91.5	1.5	0.875
	120	29	2	1	106	53.8	114	59.8	4.20	4.65	—		5 400	7 100	**7409**	38.6	55	110	2	1.55
	120	29	2	1	98.7	50.0	106	55.5	3.60	4.00	—		4 800	6 600	**7409B**	49.5	55	110	2	1.55

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

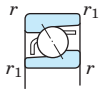
2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

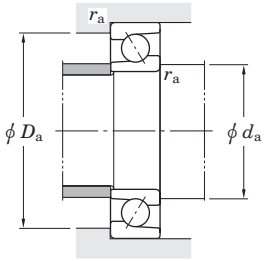
d 50 ~ (60) mm



With machined cage



With pressed cage



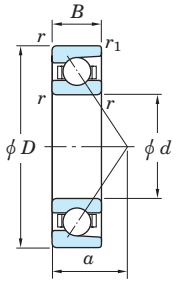
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN)		Factor		Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage		C_u											
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0		Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
50	72	12	0.6	0.3	21.8	15.7	—	—	0.840	—	16.2		14 000	18 000	**7910C**	14.2	54.5	67.5	0.6	0.128
	80	16	1	0.6	29.5	20.1	—	—	1.10	—	—		9 200	11 000	**7010**	26.9	55.5	74.5	1	0.290
	80	16	1	0.6	26.5	18.1	—	—	0.960	—	—		6 900	9 200	**7010B**	35.3	55.5	74.5	1	0.290
	80	16	1	0.6	32.5	21.9	—	—	1.25	—	15.7		13 000	17 000	**7010C**	16.8	55.5	74.5	1	0.290
	90	20	1.1	0.6	49.2	29.4	51.6	31.5	1.80	1.95	—		8 500	11 000	**7210**	30.4	57	83	1	0.485
	90	20	1.1	0.6	44.6	26.7	46.7	28.6	1.55	1.70	—		6 400	8 500	**7210B**	39.6	57	83	1	0.485
	90	20	1.1	0.6	53.5	31.8	56.0	34.1	2.05	2.20	14.6		12 000	16 000	**7210C**	19.4	57	83	1	0.485
	110	27	2	1	87.6	48.1	92.9	52.5	3.35	3.65	—		7 300	9 100	**7310**	37.2	60	100	2	1.14
	110	27	2	1	80.5	44.3	85.3	48.3	2.90	3.15	—		5 500	7 300	**7310B**	47.9	60	100	2	1.14
	110	27	2	1	93.9	51.6	99.5	56.2	3.85	4.20	13.4		10 000	13 000	**7310C**	24.5	60	100	2	1.14
	130	31	2.1	1.1	122	65.3	—	—	4.90	—	—		4 900	6 600	**7410**	41.6	62	118	2	1.92
	130	31	2.1	1.1	113	60.4	—	—	4.20	—	—		4 400	6 000	**7410B**	53.5	62	118	2	1.92
55	80	13	1	0.6	24.6	18.5	—	—	0.980	—	16.3		13 000	17 000	**7911C**	15.5	60.5	74.5	1	0.178
	90	18	1.1	0.6	38.9	26.3	—	—	1.50	—	—		8 300	10 000	**7011**	29.9	62	83	1	0.420
	90	18	1.1	0.6	34.9	23.7	—	—	1.30	—	—		6 200	8 300	**7011B**	39.4	62	83	1	0.420
	90	18	1.1	0.6	42.6	28.6	—	—	1.65	—	15.5		11 000	15 000	**7011C**	18.7	62	83	1	0.420
	100	21	1.5	1	60.9	37.1	63.7	39.8	2.30	2.45	—		7 600	9 500	**7211**	33.3	63.5	91.5	1.5	0.635
	100	21	1.5	1	55.1	33.8	57.7	36.2	2.00	2.15	—		5 700	7 600	**7211B**	43.6	63.5	91.5	1.5	0.635
	100	21	1.5	1	66.1	40.2	69.2	43.1	2.60	2.80	14.6		11 000	14 000	**7211C**	21.1	63.5	91.5	1.5	0.635
	120	29	2	1	101	56.5	107	61.7	3.95	4.30	—		6 700	8 400	**7311**	40.2	65	110	2	1.45
	120	29	2	1	92.9	52.0	98.4	56.7	3.40	3.70	—		5 000	6 700	**7311B**	51.8	65	110	2	1.45
	120	29	2	1	108	60.6	115	66.1	4.50	4.90	13.4		9 300	12 000	**7311C**	26.4	65	110	2	1.45
	140	33	2.1	1.1	148	82.4	—	—	6.40	—	—		4 500	6 000	**7411**	45.0	67	128	2	2.36
	140	33	2.1	1.1	138	76.5	—	—	5.50	—	—		4 000	5 500	**7411B**	57.8	67	128	2	2.36
60	85	13	1	0.6	29.0	21.8	—	—	1.15	—	16.3		12 000	16 000	**7912C**	16.3	65.5	79.5	1	0.187
	95	18	1.1	0.6	39.9	28.1	—	—	1.55	—	—		7 700	9 700	**7012**	31.4	67	88	1	0.450
	95	18	1.1	0.6	35.7	25.3	—	—	1.35	—	—		5 800	7 700	**7012B**	41.5	67	88	1	0.450

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

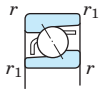
2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

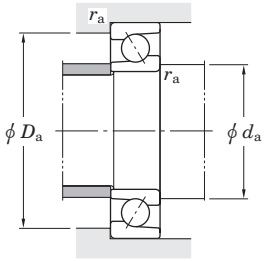
d (60) ~ (70) mm



With machined cage



With pressed cage



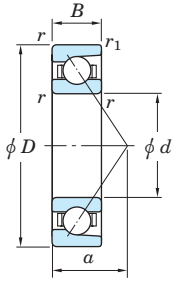
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN)		Factor		Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage		C_u											
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0		Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
60	95	18	1.1	0.6	43.8	30.6	—	—	1.75	—	15.7		11 000	14 000	**7012C**	19.4	67	88	1	0.450
	110	22	1.5	1	73.7	45.7	77.1	49.0	2.85	3.05	—		6 900	8 600	**7212**	36.1	68.5	101.5	1.5	0.820
	110	22	1.5	1	66.8	41.6	69.9	44.6	2.45	2.60	—		5 100	6 900	**7212B**	47.5	68.5	101.5	1.5	0.820
	110	22	1.5	1	80.0	49.5	83.8	53.0	3.20	3.45	14.5		9 500	13 000	**7212C**	22.7	68.5	101.5	1.5	0.820
	130	31	2.1	1.1	116	65.6	123	71.6	4.55	5.00	—		6 200	7 700	**7312**	43.2	72	118	2	1.81
	130	31	2.1	1.1	106	60.3	113	65.8	3.95	4.30	—		4 600	6 200	**7312B**	55.8	72	118	2	1.81
	130	31	2.1	1.1	124	70.3	131	76.7	5.25	5.70	13.4		8 600	11 000	**7312C**	28.4	72	118	2	1.81
	150	35	2.1	1.1	161	93.6	—	—	6.85	—	—		4 100	5 500	**7412**	48.5	72	138	2	2.85
	150	35	2.1	1.1	149	86.7	—	—	5.90	—	—		3 700	5 100	**7412B**	62.6	72	138	2	2.85
65	90	13	1	0.6	25.9	21.2	—	—	1.10	—	16.5		11 000	15 000	**7913C**	16.9	70.5	84.5	1	0.205
	100	18	1.1	0.6	42.1	31.4	—	—	1.70	—	—		7 200	9 000	**7013**	33.0	72	93	1	0.470
	100	18	1.1	0.6	37.7	28.3	—	—	1.45	—	—		5 400	7 200	**7013B**	43.8	72	93	1	0.470
	100	18	1.1	0.6	46.3	34.3	—	—	1.90	—	15.9		10 000	13 000	**7013C**	20.1	72	93	1	0.470
	120	23	1.5	1	84.1	54.2	87.8	57.8	3.35	3.55	—		6 400	8 000	**7213**	38.2	73.5	111.5	1.5	1.02
	120	23	1.5	1	76.2	49.3	79.5	52.6	2.90	3.10	—		4 800	6 400	**7213B**	50.3	73.5	111.5	1.5	1.02
	120	23	1.5	1	91.4	58.7	95.4	62.6	3.80	4.05	14.6		8 900	12 000	**7213C**	23.9	73.5	111.5	1.5	1.02
	140	33	2.1	1.1	131	75.3	139	82.2	5.15	5.65	—		5 800	7 200	**7313**	46.3	77	128	2	2.22
	140	33	2.1	1.1	120	69.3	127	75.6	4.45	4.85	—		4 300	5 800	**7313B**	59.7	77	128	2	2.22
	140	33	2.1	1.1	140	80.7	149	88.1	5.90	6.45	13.4		8 000	11 000	**7313C**	30.3	77	128	2	2.22
	160	37	2.1	1.1	174	104	—	—	7.40	—	—		3 900	5 200	**7413**	51.4	77	148	2	3.41
	160	37	2.1	1.1	161	96.8	—	—	6.35	—	—		3 500	4 800	**7413B**	66.3	77	148	2	3.41
70	100	16	1	0.6	36.2	29.0	—	—	1.55	—	16.4		10 000	12 000	**7914C**	19.4	75.5	94.5	1	0.332
	110	20	1.1	0.6	53.3	39.4	—	—	2.15	—	—		6 600	8 300	**7014**	36.0	77	103	1	0.660
	110	20	1.1	0.6	47.8	35.5	—	—	1.90	—	—		5 000	6 600	**7014B**	47.8	77	103	1	0.660
	110	20	1.1	0.6	58.6	43.0	—	—	2.45	—	15.7		9 200	12 000	**7014C**	22.1	77	103	1	0.660
	125	24	1.5	1	87.3	55.6	95.4	63.5	3.40	3.90	—		6 100	7 600	**7214**	40.2	78.5	116.5	1.5	1.12
	125	24	1.5	1	79.0	50.6	86.4	57.8	2.95	3.40	—		4 600	6 100	**7214B**	52.9	78.5	116.5	1.5	1.12

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

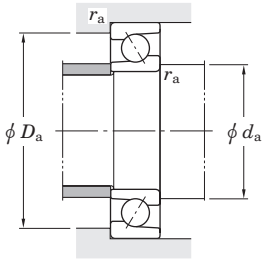
d (70) ~ (80) mm



With machined cage



With pressed cage



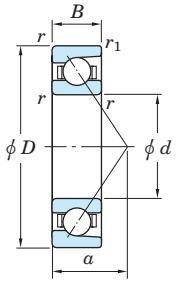
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor		Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage													
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0		Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
70	125	24	1.5	1	94.9	60.2	104	68.8	3.90	4.45	14.6		8 400	11 000	**7214C**	25.1	78.5	116.5	1.5	1.12
	150	35	2.1	1.1	147	85.8	156	93.6	5.70	6.20	—		5 400	6 700	**7314**	49.3	82	138	2	2.70
	150	35	2.1	1.1	135	78.9	143	86.0	4.90	5.35	—		4 000	5 400	**7314B**	63.7	82	138	2	2.70
	150	35	2.1	1.1	158	91.9	167	100	6.50	7.10	13.4		7 500	9 900	**7314C**	32.2	82	138	2	2.70
	180	42	3	1.1	187	115	—	—	5.30	—	—		3 500	4 600	**7414**	57.6	84	166	2.5	4.99
	180	42	3	1.1	185	119	—	—	5.45	—	—		3 100	4 300	**7414B**	74.2	84	166	2.5	4.99
75	105	16	1	0.6	36.7	30.5	—	—	1.60	—	16.5		9 300	12 000	**7915C**	20.1	80.5	99.5	1	0.350
	115	20	1.1	0.6	54.6	41.7	—	—	2.25	—	—		6 300	7 800	**7015**	37.4	82	108	1	0.690
	115	20	1.1	0.6	48.8	37.6	—	—	1.95	—	—		4 700	6 300	**7015B**	49.9	82	108	1	0.690
	115	20	1.1	0.6	60.1	45.6	—	—	2.55	—	15.9		8 700	11 000	**7015C**	22.7	82	108	1	0.690
	130	25	1.5	1	99.0	65.2	103	69.5	3.95	4.20	—		5 800	7 200	**7215**	42.1	83.5	121.5	1.5	1.23
	130	25	1.5	1	89.6	59.3	93.6	63.3	3.40	3.65	—		4 300	5 800	**7215B**	55.5	83.5	121.5	1.5	1.23
	130	25	1.5	1	108	70.6	112	75.3	4.50	4.80	14.6		8 000	11 000	**7215C**	26.2	83.5	121.5	1.5	1.23
	160	37	2.1	1.1	160	97.0	170	106	6.20	6.75	—		5 000	6 300	**7315**	52.4	87	148	2	3.15
	160	37	2.1	1.1	147	89.2	156	97.3	5.35	5.85	—		3 800	5 000	**7315B**	67.8	87	148	2	3.15
	160	37	2.1	1.1	172	104	182	113	7.10	7.75	13.4		7 000	9 200	**7315C**	34.2	87	148	2	3.15
	190	45	3	1.1	214	141	—	—	6.30	—	—		3 300	4 400	**7415**	61.3	89	176	2.5	5.90
	190	45	3	1.1	198	131	—	—	5.80	—	—		2 900	4 000	**7415B**	78.9	89	176	2.5	5.90
80	110	16	1	0.6	37.3	31.6	—	—	1.65	—	16.5		8 800	11 000	**7916C**	20.7	85.5	104.5	1	0.368
	125	22	1.1	0.6	66.7	50.6	—	—	2.75	—	—		5 800	7 200	**7016**	40.6	87	118	1	0.930
	125	22	1.1	0.6	59.8	45.7	—	—	2.40	—	—		4 300	5 800	**7016B**	54.0	87	118	1	0.930
	125	22	1.1	0.6	73.3	55.3	—	—	3.10	—	15.7		8 000	11 000	**7016C**	24.7	87	118	1	0.930
	140	26	2	1	107	71.5	111	76.2	4.10	4.40	—		5 400	6 700	**7216**	44.8	90	130	2	1.50
	140	26	2	1	96.4	65.0	101	69.3	3.55	3.80	—		4 000	5 400	**7216B**	59.2	90	130	2	1.50
	140	26	2	1	116	77.5	121	82.7	4.70	5.00	14.7		7 500	9 900	**7216C**	27.7	90	130	2	1.50
	170	39	2.1	1.1	174	109	184	119	6.75	7.35	—		4 700	5 900	**7316**	55.6	92	158	2	3.85
	170	39	2.1	1.1	159	100	169	109	5.80	6.35	—		3 500	4 700	**7316B**	71.9	92	158	2	3.85

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

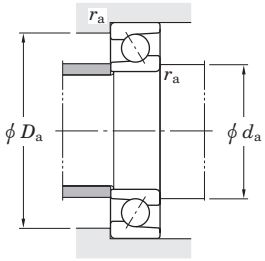
d (80) ~ 90 mm



With machined cage



With pressed cage



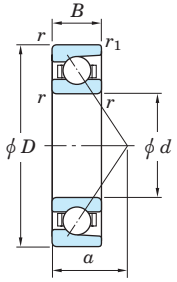
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage												
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0	Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
80	170	39	2.1	1.1	186	117	197	127	7.70	8.40	13.5	6 500	8 600	**7316C**	36.2	92	158	2	3.85
	200	48	3	1.1	241	166	—	—	7.20	—	—	3 100	4 100	**7416**	65.0	94	186	2.5	6.00
	200	48	3	1.1	223	154	—	—	6.65	—	—	2 700	3 800	**7416B**	83.6	94	186	2.5	6.00
85	120	18	1.1	0.6	48.6	40.6	—	—	2.10	—	16.5	8 100	11 000	**7917C**	22.7	92	113	1	0.523
	130	22	1.1	0.6	68.2	53.7	—	—	2.75	—	—	5 500	6 800	**7017**	42.3	92	123	1	0.970
	130	22	1.1	0.6	61.0	48.4	—	—	2.40	—	—	4 100	5 500	**7017B**	56.5	92	123	1	0.970
	130	22	1.1	0.6	75.1	58.7	—	—	3.15	—	15.9	7 600	10 000	**7017C**	25.5	92	123	1	0.970
	150	28	2	1	123	83.6	129	89.2	4.70	5.00	—	5 000	6 300	**7217**	47.9	95	140	2	1.87
	150	28	2	1	111	76.0	116	81.1	4.05	4.35	—	3 800	5 000	**7217B**	63.3	95	140	2	1.87
	150	28	2	1	134	90.6	140	96.6	5.35	5.70	14.7	7 000	9 200	**7217C**	29.7	95	140	2	1.87
	180	41	3	1.1	187	122	198	133	7.30	7.95	—	4 400	5 500	**7317**	58.8	99	166	2.5	4.53
	180	41	3	1.1	172	112	182	122	6.30	6.85	—	3 300	4 400	**7317B**	76.1	99	166	2.5	4.53
	180	41	3	1.1	201	130	213	142	8.35	9.10	13.5	6 100	8 100	**7317C**	38.3	99	166	2.5	4.53
	210	52	4	1.5	255	180	—	—	7.65	—	—	3 000	3 900	**7417**	68.7	103	192	3	8.54
	210	52	4	1.5	236	167	—	—	7.10	—	—	2 600	3 600	**7417B**	88.1	103	192	3	8.54
90	125	18	1.1	0.6	49.5	42.6	—	—	2.15	—	16.6	7 800	10 000	**7918C**	23.4	97	118	1	0.551
	140	24	1.5	1	81.5	63.3	—	—	3.25	—	—	5 100	6 400	**7018**	45.2	98.5	131.5	1.5	1.26
	140	24	1.5	1	73.0	57.1	—	—	2.80	—	—	3 900	5 100	**7018B**	60.2	98.5	131.5	1.5	1.26
	140	24	1.5	1	89.6	69.1	—	—	3.65	—	15.7	7 100	9 400	**7018C**	27.4	98.5	131.5	1.5	1.26
	160	30	2	1	141	96.7	147	103	5.30	5.65	—	4 700	5 900	**7218**	51.1	100	150	2	2.30
	160	30	2	1	128	88.0	133	93.8	4.60	4.90	—	3 500	4 700	**7218B**	67.4	100	150	2	2.30
	160	30	2	1	153	105	160	112	6.00	6.40	14.6	6 500	8 600	**7218C**	31.7	100	150	2	2.30
	190	43	3	1.1	201	135	213	147	5.90	6.40	—	4 200	5 200	**7318**	61.9	104	176	2.5	5.30
	190	43	3	1.1	184	124	195	135	5.40	5.90	—	3 100	4 200	**7318B**	80.2	104	176	2.5	5.30
	190	43	3	1.1	216	145	229	158	6.30	6.90	13.5	5 800	7 700	**7318C**	40.3	104	176	2.5	5.30
	225	54	4	1.5	270	196	—	—	8.10	—	—	2 800	3 700	**7418**	72.5	108	207	3	11.4
	225	54	4	1.5	250	182	—	—	7.50	—	—	2 500	3 400	**7418B**	93.1	108	207	3	11.4

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

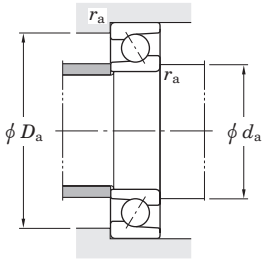
d 95 ~ (105) mm



With machined cage



With pressed cage



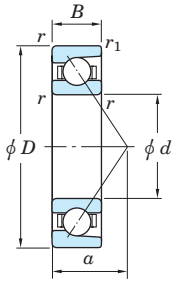
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN)		Factor		Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage		C_u											
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0		Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
95	130	18	1.1	0.6	50.3	44.1	—	—	2.15	—	16.5		7 400	9 800	**7919C**	24.1	102	123	1	0.574
	145	24	1.5	1	83.3	67.1	—	—	3.25	—	—		4 800	6 000	**7019**	47.2	103.5	136.5	1.5	1.32
	145	24	1.5	1	74.5	60.5	—	—	2.85	—	—		3 600	4 800	**7019B**	63.2	103.5	136.5	1.5	1.32
	145	24	1.5	1	91.7	73.4	—	—	3.70	—	15.9		6 700	8 900	**7019C**	28.3	103.5	136.5	1.5	1.32
	170	32	2.1	1.1	153	103	160	111	5.50	5.90	—		4 400	5 500	**7219**	54.3	107	158	2	2.78
	170	32	2.1	1.1	138	94.0	145	101	4.80	5.10	—		3 300	4 400	**7219B**	71.6	107	158	2	2.78
	170	32	2.1	1.1	166	112	174	120	6.30	6.75	14.6		6 100	8 100	**7219C**	33.8	107	158	2	2.78
	200	45	3	1.1	215	149	228	162	6.35	6.90	—		4 000	4 900	**7319**	65.1	109	186	2.5	6.12
	200	45	3	1.1	197	137	209	149	5.80	6.35	—		3 000	4 000	**7319B**	84.4	109	186	2.5	6.12
	200	45	3	1.1	231	160	245	174	6.80	7.40	13.5		5 500	7 300	**7319C**	42.3	109	186	2.5	6.12
100	140	20	1.1	0.6	69.4	58.5	—	—	2.85	—	16.3		7 000	9 200	**7920C**	26.1	107	133	1	0.773
	150	24	1.5	1	85.5	70.6	—	—	3.35	—	—		4 700	5 900	**7020**	48.1	108.5	141.5	1.5	1.37
	150	24	1.5	1	76.5	63.6	—	—	2.95	—	—		3 500	4 700	**7020B**	64.4	108.5	141.5	1.5	1.37
	150	24	1.5	1	94.2	77.2	—	—	3.80	—	16.0		6 500	8 600	**7020C**	28.7	108.5	141.5	1.5	1.37
	180	34	2.1	1.1	171	117	180	126	6.10	6.50	—		4 100	5 200	**7220**	57.7	112	168	2	3.32
	180	34	2.1	1.1	155	107	163	115	5.25	5.65	—		3 100	4 200	**7220B**	76.2	112	168	2	3.32
	180	34	2.1	1.1	186	127	195	136	6.95	7.40	14.6		5 700	7 600	**7220C**	35.9	112	168	2	3.32
	215	47	3	1.1	229	161	259	194	6.60	7.95	—		3 600	4 600	**7320**	69.4	114	201	2.5	7.53
	215	47	3	1.1	210	148	238	178	6.10	7.30	—		2 700	3 600	**7320B**	90.2	114	201	2.5	7.53
	215	47	3	1.1	246	173	278	208	7.10	8.50	13.4		5 000	6 700	**7320C**	44.8	114	201	2.5	7.53
105	145	20	1.1	0.6	70.8	61.5	—	—	2.90	—	16.4		6 700	8 800	**7921C**	26.7	112	138	1	0.810
	160	26	2	1	99.7	81.9	—	—	3.80	—	—		4 400	5 500	**7021**	51.8	115	150	2	1.73
	160	26	2	1	89.2	73.8	—	—	3.30	—	—		3 300	4 400	**7021B**	68.6	115	150	2	1.73
	160	26	2	1	110	89.6	—	—	4.30	—	15.9		6 000	8 000	**7021C**	31.0	115	150	2	1.73
	190	36	2.1	1.1	187	132	—	—	6.70	—	—		3 900	4 900	**7221**	61.0	117	178	2	3.95
	190	36	2.1	1.1	169	121	—	—	5.80	—	—		2 900	3 900	**7221B**	80.5	117	178	2	3.95

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

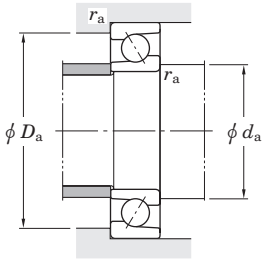
d (105) ~ (130) mm



With machined cage



With pressed cage



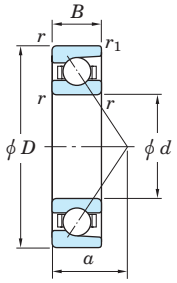
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN)		Factor		Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage		C_u											
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0		Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
105	190	36	2.1	1.1	203	143	—	—	7.60	—	14.6		5 400	7 200	**7221C**	38.0	117	178	2	3.95
	225	49	3	1.1	260	193	—	—	7.75	—	—		3 500	4 400	**7321**	72.1	119	211	2.5	8.62
	225	49	3	1.1	238	177	—	—	7.15	—	—		2 600	3 500	**7321B**	93.7	119	211	2.5	8.62
	225	49	3	1.1	278	207	—	—	8.30	—	13.4		4 800	6 400	**7321C**	46.6	119	211	2.5	8.62
110	150	20	1.1	0.6	72.2	64.4	—	—	2.95	—	16.5		6 400	8 500	**7922C**	27.4	117	143	1	0.840
	170	28	2	1	115	92.8	—	—	4.30	—	—		4 200	5 200	**7022**	54.4	120	160	2	2.14
	170	28	2	1	103	83.7	—	—	3.75	—	—		3 100	4 200	**7022B**	72.7	120	160	2	2.14
	170	28	2	1	126	101	—	—	4.85	—	15.7		5 800	7 700	**7022C**	32.8	120	160	2	2.14
	200	38	2.1	1.1	202	148	—	—	7.30	—	—		3 700	4 600	**7222**	64.3	122	188	2	4.65
	200	38	2.1	1.1	183	135	—	—	6.35	—	—		2 800	3 700	**7222B**	84.9	122	188	2	4.65
	200	38	2.1	1.1	220	160	—	—	8.35	—	14.5		5 100	6 800	**7222C**	40.0	122	188	2	4.65
	240	50	3	1.1	290	226	—	—	8.75	—	—		3 200	4 000	**7322**	76.4	124	226	2.5	10.1
	240	50	3	1.1	266	208	—	—	8.05	—	—		2 400	3 200	**7322B**	99.6	124	226	2.5	10.1
	240	50	3	1.1	311	242	—	—	9.40	—	13.4		4 500	5 900	**7322C**	48.8	124	226	2.5	10.1
120	165	22	1.1	0.6	89.7	81.2	—	—	3.55	—	16.5		5 900	7 800	**7924C**	30.1	127	158	1	1.15
	180	28	2	1	121	103	—	—	4.50	—	—		3 900	4 900	**7024**	57.3	130	170	2	2.27
	180	28	2	1	108	93.0	—	—	3.95	—	—		2 900	3 900	**7024B**	76.9	130	170	2	2.27
	180	28	2	1	133	113	—	—	5.10	—	16.0		5 400	7 100	**7024C**	34.1	130	170	2	2.27
	215	40	2.1	1.1	218	166	—	—	7.85	—	—		3 400	4 300	**7224**	68.5	132	203	2	5.49
	215	40	2.1	1.1	197	151	—	—	6.80	—	—		2 600	3 400	**7224B**	90.3	132	203	2	5.49
	215	40	2.1	1.1	237	180	—	—	8.95	—	14.6		4 800	6 300	**7224C**	42.5	132	203	2	5.49
	260	55	3	1.1	308	252	—	—	9.45	—	—		3 000	3 700	**7324**	82.3	134	246	2.5	12.6
	260	55	3	1.1	282	231	—	—	8.65	—	—		2 200	3 000	**7324B**	107.2	134	246	2.5	12.6
	260	55	3	1.1	331	271	—	—	10.2	—	13.7		4 100	5 500	**7324C**	53.0	134	246	2.5	12.6
130	180	24	1.5	1	109	99.9	—	—	4.20	—	16.4		5 400	7 100	**7926C**	32.8	138.5	171.5	1.5	1.50
	200	33	2	1	147	125	—	—	5.25	—	—		3 500	4 400	**7026**	64.1	140	190	2	3.43

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

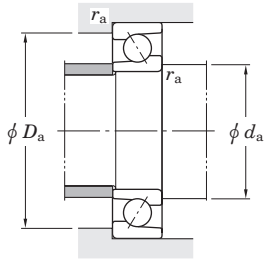
d (130) ~ (150) mm



With machined cage



With pressed cage



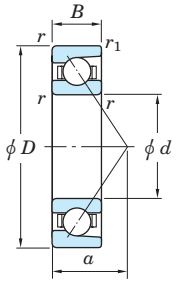
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN)		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage		C_u										
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	(With machined cage)	(With pressed cage)	f_0	Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
130	200	33	2	1	131	113	—	—	4.60	—	—	2 600	3 500	**7026B**	85.7	140	190	2	3.43
	200	33	2	1	161	137	—	—	5.95	—	15.9	4 800	6 400	**7026C**	38.6	140	190	2	3.43
	230	40	3	1.1	245	198	—	—	7.60	—	—	3 200	4 000	**7226**	72.0	144	216	2.5	6.21
	230	40	3	1.1	222	180	—	—	6.95	—	—	2 400	3 200	**7226B**	95.5	144	216	2.5	6.21
	230	40	3	1.1	266	214	—	—	8.25	—	14.7	4 400	5 800	**7226C**	44.1	144	216	2.5	6.21
	280	58	4	1.5	376	329	—	—	11.8	—	—	2 700	3 400	**7326**	88.8	148	262	3	15.4
	280	58	4	1.5	312	268	—	—	9.70	—	—	2 100	2 700	**7326B**	115.0	148	262	3	15.4
	280	58	4	1.5	368	314	—	—	11.3	—	13.7	3 800	5 000	**7326C**	56.5	148	262	3	15.4
140	190	24	1.5	1	110	105	—	—	4.20	—	16.6	5 100	6 700	**7928C**	34.1	148.5	181.5	1.5	1.59
	210	33	2	1	150	133	—	—	5.30	—	—	3 300	4 100	**7028**	67.0	150	200	2	3.64
	210	33	2	1	134	119	—	—	4.65	—	—	2 500	3 300	**7028B**	89.9	150	200	2	3.64
	210	33	2	1	165	145	—	—	6.00	—	16.0	4 500	6 000	**7028C**	39.9	150	200	2	3.64
	250	42	3	1.1	273	234	—	—	8.65	—	—	2 900	3 600	**7228**	77.3	154	236	2.5	7.76
	250	42	3	1.1	247	213	—	—	7.85	—	—	2 200	2 900	**7228B**	102.8	154	236	2.5	7.76
	250	42	3	1.1	297	254	—	—	9.40	—	14.8	4 000	5 300	**7228C**	47.1	154	236	2.5	7.76
	300	62	4	1.5	411	374	—	—	13.0	—	—	2 500	3 200	**7328**	94.5	158	282	3	18.8
	300	62	4	1.5	378	344	—	—	12.0	—	—	1 900	2 500	**7328B**	123.3	158	282	3	18.8
	300	62	4	1.5	441	401	—	—	14.0	—	13.4	3 500	4 600	**7328C**	60.5	158	282	3	18.8
150	210	28	2	1	148	132	—	—	5.45	—	16.3	4 700	6 200	**7930C**	38.1	160	200	2	2.47
	225	35	2.1	1.1	171	154	—	—	5.95	—	—	3 000	3 800	**7030**	72.1	162	213	2	4.43
	225	35	2.1	1.1	153	138	—	—	5.20	—	—	2 300	3 000	**7030B**	96.2	162	213	2	4.43
	225	35	2.1	1.1	188	169	—	—	6.70	—	16.1	4 200	5 500	**7030C**	42.8	162	213	2	4.43
	270	45	3	1.1	310	280	—	—	9.95	—	—	2 700	3 300	**7230**	83.1	164	256	2.5	9.75
	270	45	3	1.1	281	254	—	—	9.05	—	—	2 000	2 700	**7230B**	110.6	164	256	2.5	9.75
	270	45	3	1.1	338	303	—	—	10.8	—	14.7	3 700	4 900	**7230C**	50.6	164	256	2.5	9.75
	320	65	4	1.5	434	414	—	—	14.0	—	—	2 300	2 900	**7330**	100.3	168	302	3	22.4
	320	65	4	1.5	397	380	—	—	12.8	—	—	1 800	2 300	**7330B**	131.1	168	302	3	22.4

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

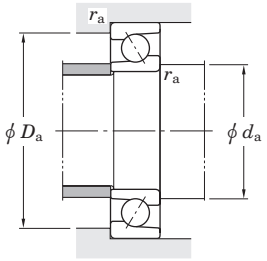
d (150) ~ (180) mm



With machined cage



With pressed cage



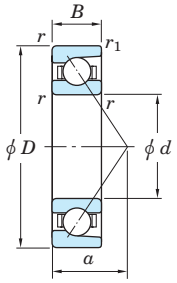
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN)		Factor		Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage		C_u											
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0		Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
150	320	65	4	1.5	468	445	—	—	15.0	—	13.7		3 200	4 300	**7330C**	64.0	168	302	3	22.4
160	220	28	2	1	151	144	—	—	5.45	—	16.5		4 400	5 800	**7932C**	39.5	170	210	2	2.60
	240	38	2.1	1.1	194	176	—	—	6.65	—	—		2 800	3 500	**7032**	76.8	172	228	2	5.45
	240	38	2.1	1.1	173	158	—	—	5.80	—	—		2 100	2 800	**7032B**	102.9	172	228	2	5.45
	240	38	2.1	1.1	214	193	—	—	7.50	—	16.0		3 900	5 200	**7032C**	45.8	172	228	2	5.45
	290	48	3	1.1	288	263	—	—	9.05	—	—		2 500	3 100	**7232**	89.0	174	276	2.5	12.1
	290	48	3	1.1	297	279	—	—	9.60	—	—		1 800	2 500	**7232B**	118.4	174	276	2.5	12.1
	290	48	3	1.1	315	333	—	—	9.85	—	15.2		3 400	4 500	**7232C**	54.1	174	276	2.5	12.1
	340	68	4	1.5	456	455	—	—	14.9	—	—		2 200	2 700	**7332**	106.2	178	322	3	26.4
	340	68	4	1.5	415	416	—	—	13.6	—	—		1 600	2 200	**7332B**	138.9	178	322	3	26.4
	340	68	4	1.5	492	490	—	—	16.0	—	14.0		3 000	4 000	**7332C**	67.5	168.5	322	3	26.4
170	230	28	2	1	157	151	—	—	5.75	—	16.6		3 900	5 100	**7934C**	40.8	180	220	2	3.21
	260	42	2.1	1.1	232	214	—	—	7.90	—	—		2 600	3 200	**7034**	83.1	182	248	2	7.58
	260	42	2.1	1.1	208	193	—	—	6.90	—	—		1 900	2 600	**7034B**	111.2	182	248	2	7.77
	260	42	2.1	1.1	256	234	—	—	8.95	—	15.9		3 600	4 800	**7034C**	49.8	182	248	2	7.57
	310	52	4	1.5	340	331	—	—	11.0	—	—		2 300	2 800	**7234**	95.3	188	292	3	15.1
	310	52	4	1.5	306	300	—	—	10.0	—	—		1 700	2 300	**7234B**	126.7	188	292	3	15.1
	310	52	4	1.5	371	359	—	—	12.0	—	15.1		3 100	4 200	**7234C**	58.2	188	292	3	15.1
	360	72	4	1.5	486	485	—	—	15.4	—	—		2 000	2 500	**7334**	112.5	188	342	3	31.2
	360	72	4	1.5	444	444	—	—	14.1	—	—		1 500	2 000	**7334B**	147.2	188	342	3	31.2
	360	72	4	1.5	523	521	—	—	16.5	—	13.8		2 800	3 700	**7334C**	71.5	188	342	3	31.2
180	250	33	2	1	200	188	—	—	7.05	—	16.4		3 600	4 700	**7936C**	45.3	190	240	2	4.68
	280	46	2.1	1.1	265	253	—	—	9.15	—	—		2 400	3 000	**7036**	89.4	192	268	2	10.1
	280	46	2.1	1.1	237	228	—	—	7.95	—	—		1 800	2 400	**7036B**	119.5	192	268	2	10.2
	280	46	2.1	1.1	291	276	—	—	10.4	—	15.7		3 300	4 400	**7036C**	53.8	192	268	2	9.96
	320	52	4	1.5	367	362	—	—	11.8	—	—		2 200	2 700	**7236**	98.2	198	302	3	15.7

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

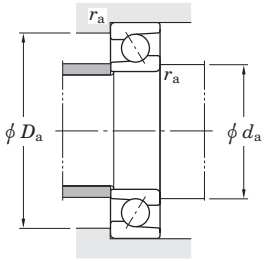
d (180) ~ (240) mm



With machined cage



With pressed cage



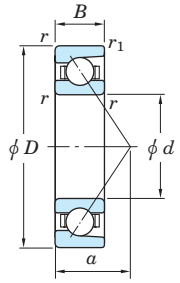
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN)		Factor		Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage		C_u											
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0		Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
180	320	52	4	1.5	331	329	—	—	10.7	—	—		1 600	2 200	**7236B**	130.9	198	302	3	15.7
	320	52	4	1.5	400	393	—	—	12.8	—	14.9		3 000	4 000	**7236C**	59.5	198	302	3	15.7
	380	75	4	1.5	512	534	—	—	16.5	—	—		1 900	2 400	**7336**	118.3	198	362	3	40.0
	380	75	4	1.5	466	488	—	—	15.1	—	—		1 400	1 900	**7336B**	155.0	198	362	3	40.0
190	260	33	2	1	198	197	—	—	6.85	—	16.5		3 300	4 500	**7938C**	46.6	200	250	2	4.83
	290	46	2.1	1.1	271	268	—	—	9.35	—	—		2 300	2 800	**7038**	92.3	202	278	2	10.8
	290	46	2.1	1.1	243	241	—	—	8.15	—	—		1 700	2 300	**7038B**	123.7	202	278	2	10.8
	290	46	2.1	1.1	299	293	—	—	10.6	—	15.9		3 100	4 200	**7038C**	55.2	202	278	2	10.8
	340	55	4	1.5	379	390	—	—	12.4	—	—		2 000	2 500	**7238**	104.0	208	322	3	18.8
	340	55	4	1.5	341	353	—	—	11.2	—	—		1 500	2 000	**7238B**	138.7	208	322	3	18.8
	340	55	4	1.5	414	424	—	—	13.5	—	15.1		2 800	3 700	**7238C**	63.0	208	322	3	18.8
	400	78	5	2	563	598	—	—	18.0	—	—		1 800	2 200	**7338**	124.2	212	378	4	45.5
	400	78	5	2	514	548	—	—	16.5	—	—		1 300	1 800	**7338B**	162.8	212	378	4	45.5
200	280	38	2.1	1.1	256	255	—	—	8.70	—	16.3		3 100	4 100	**7940C**	51.2	212	268	2	6.85
	310	51	2.1	1.1	304	309	—	—	10.0	—	—		2 100	2 600	**7040**	99.1	212	298	2	12.7
	310	51	2.1	1.1	273	279	—	—	9.05	—	—		1 600	2 100	**7040B**	132.5	212	298	2	12.7
	310	51	2.1	1.1	335	338	—	—	10.9	—	15.7		2 900	3 900	**7040C**	59.7	212	298	2	12.7
	360	58	4	1.5	405	423	—	—	13.1	—	—		1 900	2 400	**7240**	109.8	218	342	3	22.4
	360	58	4	1.5	365	384	—	—	11.9	—	—		1 400	1 900	**7240B**	146.5	218	342	3	22.4
	360	58	4	1.5	442	460	—	—	14.2	—	15.1		2 600	3 500	**7240C**	66.5	218	342	3	22.4
	420	80	5	2	593	658	—	—	19.3	—	—		1 700	2 100	**7340**	129.5	222	398	4	52.0
	420	80	5	2	541	602	—	—	17.7	—	—		1 200	1 700	**7340B**	170.1	222	398	4	52.0
220	340	56	3	1.1	334	353	—	—	10.9	—	—		1 900	2 400	**7044**	108.9	234	326	2.5	18.5
	340	56	3	1.1	299	318	—	—	9.80	—	—		1 400	1 900	**7044B**	145.5	234	326	2.5	18.9
240	360	56	3	1.1	364	375	—	—	12.3	—	—		1 700	2 200	**7048**	114.6	254	346	2.5	19.7
	360	56	3	1.1	325	338	—	—	11.1	—	—		1 300	1 700	**7048B**	153.9	254	346	2.5	20.1

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

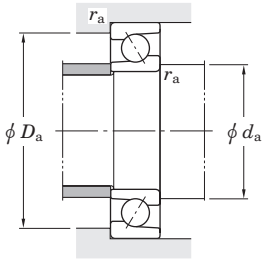
d (240) ~ 380 mm



With machined cage



With pressed cage



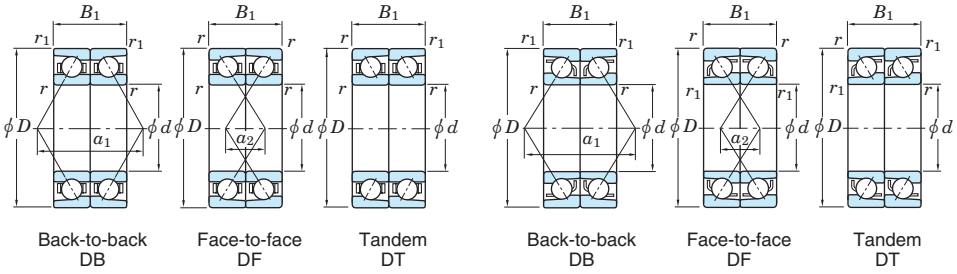
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor		Limiting speeds [1] (min^{-1})		Bearing No. [2]	Load center (mm)	Mounting dimensions (mm)			(Refer.) Mass (kg)
					With machined cage		With pressed cage													
d	D	B	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cage	With pressed cage	f_0		Grease lub.	Oil lub.		a	d_a min.	D_a max.	r_a max.	
240	440	72	4	1.5	504	595	—	—	16.7	—	—		1 500	1 800	**7248**	134.2	258	422	3	51.8
	440	72	4	1.5	453	539	—	—	15.1	—	—		1 100	1 500	**7248B**	178.6	258	422	3	52.8
260	400	65	4	1.5	407	478	—	—	13.6	—	—		1 500	1 900	**7052**	128.4	278	382	3	28.7
	400	65	4	1.5	364	431	—	—	12.2	—	—		1 100	1 500	**7052B**	171.0	278	382	3	29.3
280	420	65	4	1.5	415	507	—	—	14.0	—	—		1 400	1 800	**7056**	133.5	298	402	3	30.4
	420	65	4	1.5	384	453	—	—	13.1	—	—		1 100	1 400	**7056B**	179.3	298	402	3	31.0
300	460	74	4	1.5	533	680	—	—	18.0	—	—		1 300	1 600	**7060**	146.7	318	442	3	43.7
	460	74	4	1.5	478	613	—	—	16.3	—	—		960	1 300	**7060B**	196.4	318	442	3	44.9
320	480	74	4	1.5	546	722	—	—	18.6	—	—		1 200	1 500	**7064**	152.5	338	462	3	46.0
	480	74	4	1.5	489	651	—	—	16.8	—	—		890	1 200	**7064B**	204.8	338	462	3	47.2
340	520	82	5	2	628	861	—	—	21.4	—	—		1 100	1 300	**7068**	165.1	362	498	4	61.8
	520	82	5	2	563	777	—	—	19.4	—	—		800	1 100	**7068B**	221.4	362	498	4	63.3
360	540	82	5	2	644	913	—	—	22.2	—	—		1 000	1 300	**7072**	170.9	382	518	4	64.6
	540	82	5	2	577	824	—	—	20.1	—	—		750	1 000	**7072B**	229.8	382	518	4	66.2
380	560	82	5	2	660	966	—	—	23.0	—	—		940	1 200	**7076**	176.7	402	538	4	67.2
	560	82	5	2	590	870	—	—	20.7	—	—		700	940	**7076B**	238.2	402	538	4	69.1

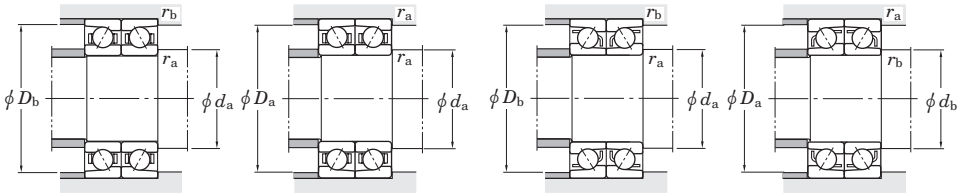
[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cage or molded cage.

2) B or no indication after the bearing number indicates nominal contact angle of 15° and 30° respectively.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

d 10 ~ (17) mm



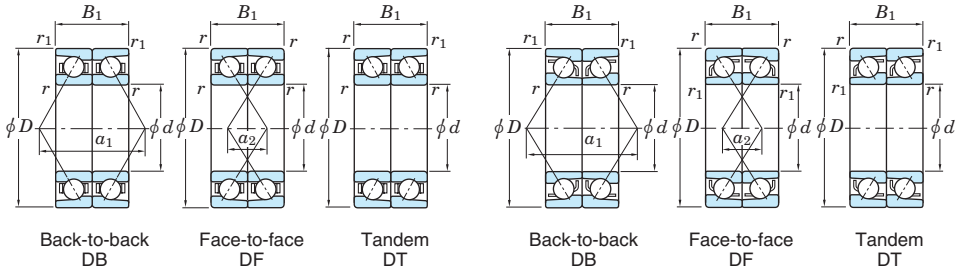


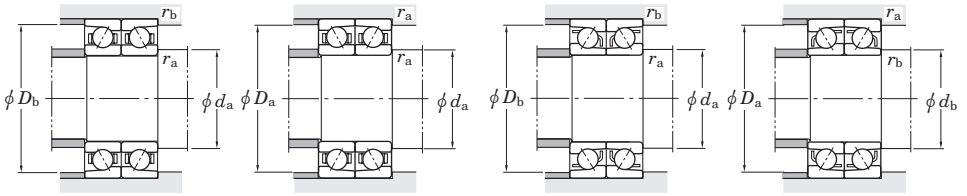
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)						(Refer.) Mass (kg)
					With machined cages		With pressed cages																		
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
10	22	12	0.3	0.15	6.10	3.05	—	—	0.120	—	14.2	42 000	55 000	**7900CDB**	**7900CDF**	**7900CDT**	10.3	1.7	12.5	—	19.5	20.8	0.3	0.15	0.016
	26	16	0.3	0.15	10.1	4.65	—	—	0.240	—	—	27 000	34 000	**7000DB**	**7000DF**	**7000DT**	18.2	2.2	12.5	—	23.5	24.8	0.3	0.15	0.042
	26	16	0.3	0.15	9.40	4.35	—	—	0.220	—	—	20 000	27 000	**7000BDB**	**7000BDF**	**7000BDT**	23.1	7.1	12.5	—	23.5	24.8	0.3	0.15	0.042
	26	16	0.3	0.15	10.7	4.95	—	—	0.250	—	12.5	37 000	50 000	**7000CDB**	**7000CDF**	**7000CDT**	12.7	3.3	12.5	—	23.5	24.8	0.3	0.15	0.042
	30	18	0.6	0.3	9.50	4.40	11.0	5.45	0.230	0.280	—	23 000	29 000	**7200DB**	**7200DF**	**7200DT**	20.8	2.8	14.5	12.5	25.5	27.5	0.6	0.3	0.062
	30	18	0.6	0.3	8.70	4.05	10.1	5.05	0.210	0.260	—	18 000	23 000	**7200BDB**	**7200BDF**	**7200BDT**	26.2	8.2	14.5	12.5	25.5	27.5	0.6	0.3	0.062
	30	18	0.6	0.3	10.2	4.70	11.8	5.85	0.240	0.300	13.4	32 000	43 000	**7200CDB**	**7200CDF**	**7200CDT**	14.5	3.5	14.5	12.5	25.5	27.5	0.6	0.3	0.062
	35	22	0.6	0.3	17.3	7.55	18.9	8.60	0.590	0.680	—	21 000	27 000	**7300DB**	**7300DF**	**7300DT**	24.0	2.0	14.5	12.5	30.5	32.5	0.6	0.3	0.108
12	24	12	0.3	0.15	6.45	3.45	—	—	0.140	—	14.7	37 000	49 000	**7901CDB**	**7901CDF**	**7901CDT**	10.8	1.2	14.5	—	21.5	22.8	0.3	0.15	0.020
	28	16	0.3	0.15	11.0	5.45	—	—	0.280	—	—	23 000	29 000	**7001DB**	**7001DF**	**7001DT**	19.9	3.9	14.5	—	25.5	26.8	0.3	0.15	0.048
	28	16	0.3	0.15	10.1	5.05	—	—	0.260	—	—	18 000	23 000	**7001BDB**	**7001BDF**	**7001BDT**	25.2	9.2	14.5	—	25.5	26.8	0.3	0.15	0.048
	28	16	0.3	0.15	11.8	5.85	—	—	0.300	—	13.4	32 000	43 000	**7001CDB**	**7001CDF**	**7001CDT**	13.5	2.5	14.5	—	25.5	26.8	0.3	0.15	0.048
	32	20	0.6	0.3	15.1	7.25	16.2	8.05	0.560	0.620	—	22 000	27 000	**7201DB**	**7201DF**	**7201DT**	22.7	2.7	16.5	14.5	27.5	29.5	0.6	0.3	0.076
	32	20	0.6	0.3	14.0	6.80	15.1	7.50	0.480	0.530	—	16 000	22 000	**7201BDB**	**7201BDF**	**7201BDT**	28.5	8.5	16.5	14.5	27.5	29.5	0.6	0.3	0.076
	32	20	0.6	0.3	16.0	7.70	17.2	8.55	0.600	0.670	12.5	30 000	40 000	**7201CDB**	**7201CDF**	**7201CDT**	15.9	4.1	16.5	14.5	27.5	29.5	0.6	0.3	0.076
	37	24	1	0.6	20.7	9.20	22.7	10.5	0.720	0.820	—	20 000	24 000	**7301DB**	**7301DF**	**7301DT**	26.2	2.2	17.5	16.5	31.5	32.5	1	0.6	0.130
15	28	14	0.3	0.15	9.65	5.30	—	—	0.210	—	14.5	31 000	41 000	**7902CDB**	**7902CDF**	**7902CDT**	12.8	1.2	17.5	—	25.5	26.8	0.3	0.15	0.030
	32	18	0.3	0.15	12.4	6.85	—	—	0.350	—	—	20 000	26 000	**7002DB**	**7002DF**	**7002DT**	22.6	4.6	17.5	—	29.5	30.8	0.3	0.15	0.070
	32	18	0.3	0.15	11.3	6.30	—	—	0.320	—	—	15 000	20 000	**7002BDB**	**7002BDF**	**7002BDT**	29.1	11.1	17.5	—	29.5	30.8	0.3	0.15	0.070
	32	18	0.3	0.15	13.4	7.40	—	—	0.380	—	14.1	28 000	37 000	**7002CDB**	**7002CDF**	**7002CDT**	15.3	2.7	17.5	—	29.5	30.8	0.3	0.15	0.070
	35	22	0.6	0.3	16.4	8.55	16.4	8.55	0.600	0.600	—	19 000	24 000	**7202DB**	**7202DF**	**7202DT**	25.7	3.7	19.5	17.5	30.5	32.5	0.6	0.3	0.096
	35	22	0.6	0.3	15.1	7.85	15.1	7.85	0.520	0.520	—	14 000	19 000	**7202BDB**	**7202BDF**	**7202BDT**	32.4	10.4	19.5	17.5	30.5	32.5	0.6	0.3	0.096
	35	22	0.6	0.3	17.6	9.15	17.6	9.15	0.680	0.680	13.3	26 000	35 000	**7202CDB**	**7202CDF**	**7202CDT**	17.8	4.2	19.5	17.5	30.5	32.5	0.6	0.3	0.096
	42	26	1	0.6	25.4	12.9	27.3	14.4	0.990	1.10	—	16 000	20 000	**7302DB**	**7302DF**	**7302DT**	30.0	4.0	20.5	19.5	36.5	37.5	1	0.6	0.176
17	30	14	0.3	0.15	10.1	5.90	—	—	0.240	—	14.9	28 000	38 000	**7903CDB**	**7903CDF**	**7903CDT**	13.4	0.6	19.5	—	27.5	28.8	0.3	0.15	0.032
	35	20	0.3	0.15	13.7	8.25	—	—	0.430	—	—	18 000	23 000	**7003DB**	**7003DF**	**7003DT**	25.3	5.3	19.5	—	32.5	33.8	0.3	0.15	0.090
	35	20	0.3	0.15	12.4	7.50	—	—	0.390	—	—	14 000	18 000	**7003BDB**	**7003BDF**	**7003BDT**	32.2	12.2	19.5	—	32.5	33.8	0.3	0.15	0.090

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (17) ~ (25) mm



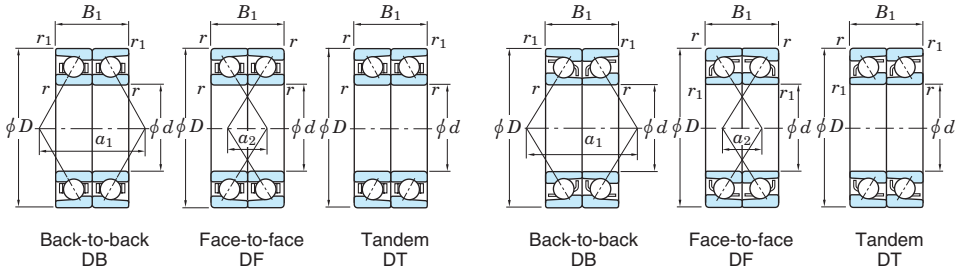


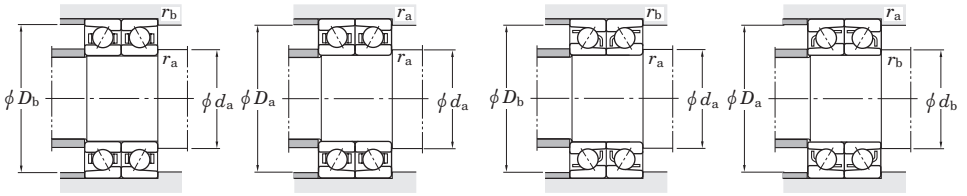
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)					(Refer.) Mass (kg)
					With machined cages		With pressed cages																	
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
17	35	20	0.3	0.15	14.8	8.95	—	—	0.460	—	14.6	25 000	33 000	**7003CDB**	**7003CDF**	**7003CDT**	17.1	2.9	19.5	—	32.5	33.8	0.3	0.15	0.090
	40	24	0.6	0.3	20.6	11.0	20.6	11.0	0.770	0.770	—	17 000	21 000	**7203DB**	**7203DF**	**7203DT**	28.8	4.8	21.5	19.5	35.5	37.5	0.6	0.3	0.140
	40	24	0.6	0.3	19.0	10.1	19.0	10.1	0.660	0.660	—	12 000	17 000	**7203BDB**	**7203BDF**	**7203BDT**	36.3	12.3	21.5	19.5	35.5	37.5	0.6	0.3	0.140
	40	24	0.6	0.3	22.1	11.8	22.1	11.8	0.880	0.880	13.4	23 000	30 000	**7203CDB**	**7203CDF**	**7203CDT**	19.8	4.2	21.5	19.5	35.5	37.5	0.6	0.3	0.140
	47	28	1	0.6	30.3	15.8	32.5	17.5	1.20	1.30	—	15 000	18 000	**7303DB**	**7303DF**	**7303DT**	33.1	5.1	22.5	21.5	41.5	42.5	1	0.6	0.240
	47	28	1	0.6	28.1	14.6	30.1	16.2	1.00	1.15	—	11 000	15 000	**7303BDB**	**7303BDF**	**7303BDT**	41.7	13.7	22.5	21.5	41.5	42.5	1	0.6	0.240
	47	28	1	0.6	32.2	16.8	32.2	16.8	1.30	1.30	12.6	20 000	27 000	**7303CDB**	**7303CDF**	**7303CDT**	22.8	5.2	22.5	21.5	41.5	42.5	1	0.6	0.240
20	37	18	0.3	0.15	14.8	9.15	—	—	0.470	—	14.9	24 000	31 000	**7904CDB**	**7904CDF**	**7904CDT**	16.6	1.4	22.5	—	34.5	35.8	0.3	0.15	0.070
	42	24	0.6	0.3	20.9	12.2	—	—	0.790	—	—	15 000	19 000	**7004DB**	**7004DF**	**7004DT**	30.2	6.2	24.5	—	37.5	39.5	0.6	0.3	0.158
	42	24	0.6	0.3	19.0	11.1	—	—	0.680	—	—	11 000	15 000	**7004BDB**	**7004BDF**	**7004BDT**	38.4	14.4	24.5	—	37.5	39.5	0.6	0.3	0.158
	42	24	0.6	0.3	22.6	13.2	—	—	0.900	—	14.1	21 000	28 000	**7004CDB**	**7004CDF**	**7004CDT**	20.4	3.6	24.5	—	37.5	39.5	0.6	0.3	0.158
	47	28	1	0.6	29.4	16.8	31.2	18.3	1.15	1.25	—	14 000	17 000	**7204DB**	**7204DF**	**7204DT**	33.9	5.9	25.5	24.5	41.5	42.5	1	0.6	0.224
	47	28	1	0.6	27.0	15.4	28.6	16.8	1.00	1.10	—	10 000	14 000	**7204BDB**	**7204BDF**	**7204BDT**	42.9	14.9	25.5	24.5	41.5	42.5	1	0.6	0.224
	47	28	1	0.6	31.5	18.0	33.4	19.6	1.35	1.45	13.4	19 000	26 000	**7204CDB**	**7204CDF**	**7204CDT**	23.2	4.8	25.5	24.5	41.5	42.5	1	0.6	0.224
	52	30	1.1	0.6	35.4	18.8	38.0	20.8	1.40	1.60	—	13 000	17 000	**7304DB**	**7304DF**	**7304DT**	35.8	5.8	27	24.5	45	47.5	1	0.6	0.300
	52	30	1.1	0.6	32.8	17.4	35.2	19.3	1.20	1.35	—	10 000	13 000	**7304BDB**	**7304BDF**	**7304BDT**	45.2	15.2	27	24.5	45	47.5	1	0.6	0.300
	52	30	1.1	0.6	37.6	19.9	40.3	22.2	1.55	1.75	12.6	18 000	24 000	**7304CDB**	**7304CDF**	**7304CDT**	24.6	5.4	27	24.5	45	47.5	1	0.6	0.300
	72	38	1.1	0.6	72.3	38.2	—	—	3.00	—	—	7 400	11 000	**7404DB**	**7404DF**	**7404DT**	46.1	8.1	27	—	65	67.5	1	0.6	0.790
	72	38	1.1	0.6	68.1	35.9	—	—	2.80	—	—	6 400	9 600	**7404BDB**	**7404BDF**	**7404BDT**	58.4	20.4	27	—	65	67.5	1	0.6	0.790
25	42	18	0.3	0.15	16.5	10.9	—	—	0.600	—	15.5	20 000	27 000	**7905CDB**	**7905CDF**	**7905CDT**	18.2	0.2	27.5	—	39.5	40.8	0.3	0.15	0.082
	47	24	0.6	0.3	22.9	14.8	—	—	0.900	—	—	13 000	17 000	**7005DB**	**7005DF**	**7005DT**	32.9	8.9	29.5	—	42.5	44.5	0.6	0.3	0.182
	47	24	0.6	0.3	20.7	13.4	—	—	0.780	—	—	10 000	13 000	**7005BDB**	**7005BDF**	**7005BDT**	42.3	18.3	29.5	—	42.5	44.5	0.6	0.3	0.182
	47	24	0.6	0.3	24.9	16.0	—	—	1.00	—	14.7	18 000	24 000	**7005CDB**	**7005CDF**	**7005CDT**	21.7	2.3	29.5	—	42.5	44.5	0.6	0.3	0.182
	52	30	1	0.6	31.2	19.0	32.9	20.6	1.25	1.35	—	12 000	15 000	**7205DB**	**7205DF**	**7205DT**	37.5	7.5	30.5	29.5	46.5	47.5	1	0.6	0.270
	52	30	1	0.6	28.4	17.4	29.9	18.8	1.05	1.15	—	9 200	12 000	**7205BDB**	**7205BDF**	**7205BDT**	47.7	17.7	30.5	29.5	46.5	47.5	1	0.6	0.270
	52	30	1	0.6	33.7	20.5	35.5	22.2	1.40	1.55	14.0	17 000	23 000	**7205CDB**	**7205CDF**	**7205CDT**	25.5	4.5	30.5	29.5	46.5	47.5	1	0.6	0.270
	62	34	1.1	0.6	50.3	28.8	53.6	31.6	2.10	2.35	—	11 000	14 000	**7305DB**	**7305DF**	**7305DT**	42.1	8.1	32	29.5	55	57.5	1	0.6	0.486

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (25) ~ (35) mm



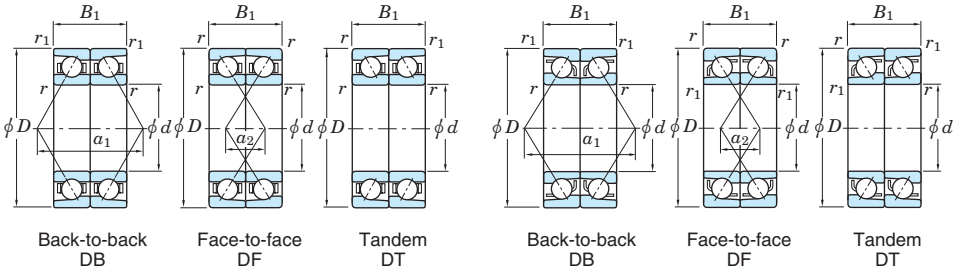


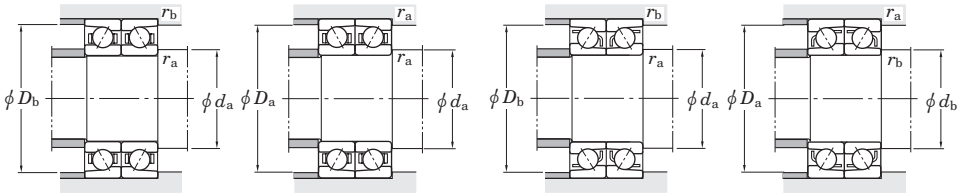
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)						(Refer.) Mass (kg)
					With machined cages		With pressed cages																		
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
25	62	34	1.1	0.6	46.5	26.6	49.5	29.2	1.85	2.00	—	8 300	11 000	**7305BDB**	**7305BDF**	**7305BDT**	53.5	19.5	32	29.5	55	57.5	1	0.6	0.486
	62	34	1.1	0.6	53.5	30.6	57.0	33.7	2.40	2.65	12.8	15 000	20 000	**7305CDB**	**7305CDF**	**7305CDT**	28.7	5.3	32	29.5	55	57.5	1	0.6	0.486
	80	42	1.5	1	80.7	46.3	86.6	51.5	3.60	4.00	—	6 400	9 100	**7405DB**	**7405DF**	**7405DT**	52.8	10.8	33.5	30.5	71.5	74.5	1.5	1	1.05
	80	42	1.5	1	74.9	43.0	80.4	47.8	3.10	3.40	—	5 500	8 200	**7405BDB**	**7405BDF**	**7405BDT**	67.2	25.2	33.5	30.5	71.5	74.5	1.5	1	1.05
30	47	18	0.3	0.15	16.8	12.5	—	—	0.650	—	15.9	18 000	23 000	**7906CDB**	**7906CDF**	**7906CDT**	19.3	1.3	32.5	—	44.5	45.8	0.3	0.15	0.092
	55	26	1	0.6	29.5	20.2	—	—	1.20	—	—	11 000	14 000	**7006DB**	**7006DF**	**7006DT**	37.5	11.5	35.5	—	49.5	50.5	1	0.6	0.266
	55	26	1	0.6	26.7	18.4	—	—	1.05	—	—	8 500	11 000	**7006BDB**	**7006BDF**	**7006BDT**	48.7	22.7	35.5	—	49.5	50.5	1	0.6	0.266
	55	26	1	0.6	32.2	22.0	—	—	1.40	—	14.9	16 000	21 000	**7006CDB**	**7006CDF**	**7006CDT**	24.4	1.6	35.5	—	49.5	50.5	1	0.6	0.266
	62	32	1	0.6	43.3	27.4	45.7	29.7	1.80	1.95	—	10 000	13 000	**7206DB**	**7206DF**	**7206DT**	43.0	11.0	35.5	34.5	56.5	57.5	1	0.6	0.416
	62	32	1	0.6	39.5	25.0	41.6	27.1	1.55	1.65	—	7 700	10 000	**7206BDB**	**7206BDF**	**7206BDT**	55.2	23.2	35.5	34.5	56.5	57.5	1	0.6	0.416
	62	32	1	0.6	46.8	29.5	49.4	32.0	2.05	2.20	14.0	14 000	19 000	**7206CDB**	**7206CDF**	**7206CDT**	28.5	3.5	35.5	34.5	56.5	57.5	1	0.6	0.416
	72	38	1.1	0.6	61.1	37.8	64.8	41.2	2.60	2.85	—	9 200	12 000	**7306DB**	**7306DF**	**7306DT**	49.0	11.0	37	34.5	65	67.5	1	0.6	0.724
	72	38	1.1	0.6	56.1	34.7	59.4	37.9	2.25	2.45	—	6 900	9 200	**7306BDB**	**7306BDF**	**7306BDT**	62.6	24.6	37	34.5	65	67.5	1	0.6	0.724
	72	38	1.1	0.6	65.6	40.5	69.5	44.2	3.00	3.25	13.4	13 000	17 000	**7306CDB**	**7306CDF**	**7306CDT**	32.9	5.1	37	34.5	65	67.5	1	0.6	0.724
	90	46	1.5	1	96.7	56.9	104	63.2	4.35	4.85	—	5 700	8 100	**7406DB**	**7406DF**	**7406DT**	58.5	12.5	38.5	35.5	81.5	84.5	1.5	1	1.37
	90	46	1.5	1	89.7	52.8	96.3	58.6	3.75	4.15	—	4 900	7 300	**7406BDB**	**7406BDF**	**7406BDT**	74.6	28.6	38.5	35.5	81.5	84.5	1.5	1	1.37
35	55	20	0.6	0.3	25.5	19.4	—	—	1.10	—	15.7	15 000	20 000	**7907CDB**	**7907CDF**	**7907CDT**	22.1	2.1	39.5	—	50.5	52.5	0.6	0.3	0.148
	62	28	1	0.6	35.5	25.2	—	—	1.50	—	—	9 800	12 000	**7007DB**	**7007DF**	**7007DT**	42.3	14.3	40.5	—	56.5	57.5	1	0.6	0.340
	62	28	1	0.6	32.0	22.8	—	—	1.30	—	—	7 300	9 800	**7007BDB**	**7007BDF**	**7007BDT**	55.1	27.1	40.5	—	56.5	57.5	1	0.6	0.340
	62	28	1	0.6	38.8	27.4	—	—	1.70	—	15.0	13 000	18 000	**7007CDB**	**7007CDF**	**7007CDT**	27.0	1.0	40.5	—	56.5	57.5	1	0.6	0.340
	72	34	1.1	0.6	57.1	37.3	60.3	40.4	2.45	2.65	—	8 800	11 000	**7207DB**	**7207DF**	**7207DT**	48.5	14.5	42	39.5	65	67.5	1	0.6	0.590
	72	34	1.1	0.6	52.1	34.1	54.9	36.9	2.10	2.25	—	6 600	8 800	**7207BDB**	**7207BDF**	**7207BDT**	62.7	28.7	42	39.5	65	67.5	1	0.6	0.590
	72	34	1.1	0.6	61.7	40.2	65.1	43.5	2.75	3.00	14.0	12 000	16 000	**7207CDB**	**7207CDF**	**7207CDT**	31.6	2.4	42	39.5	65	67.5	1	0.6	0.590
	80	42	1.5	1	71.8	44.0	81.1	52.8	3.05	3.65	—	8 200	10 000	**7307DB**	**7307DF**	**7307DT**	54.8	12.8	43.5	40.5	71.5	74.5	1.5	1	0.950
	80	42	1.5	1	65.9	40.5	74.4	48.6	2.65	3.15	—	6 200	8 200	**7307BDB**	**7307BDF**	**7307BDT**	70.1	28.1	43.5	40.5	71.5	74.5	1.5	1	0.950
	80	42	1.5	1	77.0	47.2	86.9	56.6	3.50	4.20	13.4	11 000	15 000	**7307CDB**	**7307CDF**	**7307CDT**	36.7	5.3	43.5	40.5	71.5	74.5	1.5	1	0.950
	100	50	1.5	1	123	73.9	132	82.1	5.70	6.35	—	5 000	7 200	**7407DB**	**7407DF**	**7407DT**	65.2	15.2	43.5	40.5	91.5	94.5	1.5	1	1.90

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (35) ~ (50) mm



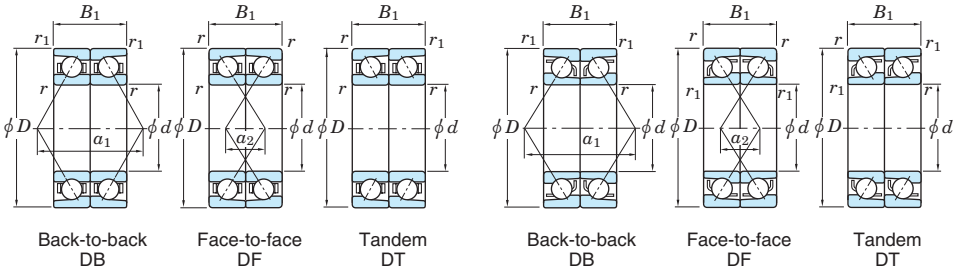


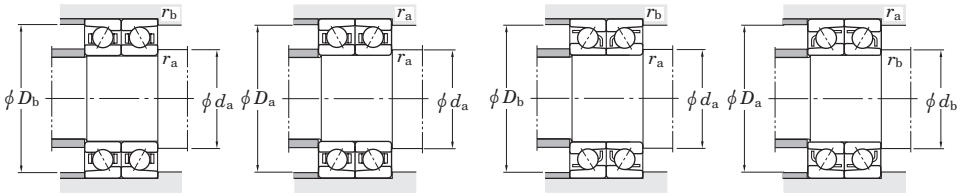
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)					(Refer.) Mass (kg)
					With machined cages		With pressed cages																	
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
35	100	50	1.5	1	114	68.6	122	76.2	4.90	5.45	—	4 300	6 500	**7407BDB**	**7407BDF**	**7407BDT**	83.3	33.3	43.5	40.5	91.5	94.5	1.5	1	1.90
40	62	24	0.6	0.3	32.0	24.9	—	—	1.40	—	15.7	13 000	18 000	**7908CDB**	**7908CDF**	**7908CDT**	25.7	1.7	44.5	—	57.5	59.5	0.6	0.3	0.214
	68	30	1	0.6	38.1	29.2	—	—	1.65	—	—	8 900	11 000	**7008DB**	**7008DF**	**7008DT**	46.3	16.3	45.5	—	62.5	63.5	1	0.6	0.420
	68	30	1	0.6	34.2	26.4	—	—	1.45	—	—	6 600	8 900	**7008BDB**	**7008BDF**	**7008BDT**	60.5	30.5	45.5	—	62.5	63.5	1	0.6	0.420
	68	30	1	0.6	41.7	31.8	—	—	1.90	—	15.4	12 000	16 000	**7008CDB**	**7008CDF**	**7008CDT**	29.5	0.5	45.5	—	62.5	63.5	1	0.6	0.420
	80	36	1.1	0.6	68.2	46.7	71.7	50.3	3.00	3.25	—	8 000	10 000	**7208DB**	**7208DF**	**7208DT**	52.7	16.7	47	44.5	73	75.5	1	0.6	0.764
	80	36	1.1	0.6	62.1	42.7	65.2	45.9	2.60	2.80	—	6 000	8 000	**7208BDB**	**7208BDF**	**7208BDT**	68.3	32.3	47	44.5	73	75.5	1	0.6	0.764
	80	36	1.1	0.6	73.8	50.4	77.6	54.3	3.45	3.70	14.2	11 000	15 000	**7208CDB**	**7208CDF**	**7208CDT**	34.1	1.9	47	44.5	73	75.5	1	0.6	0.764
	90	46	1.5	1	87.8	54.9	99.1	65.9	3.85	4.60	—	7 400	9 200	**7308DB**	**7308DF**	**7308DT**	60.5	14.5	48.5	45.5	81.5	84.5	1.5	1	1.31
	90	46	1.5	1	80.6	50.5	91.0	60.6	3.30	3.95	—	5 500	7 400	**7308BDB**	**7308BDF**	**7308BDT**	77.5	31.5	48.5	45.5	81.5	84.5	1.5	1	1.31
	90	46	1.5	1	94.1	58.8	106	70.5	4.40	5.25	13.4	10 000	14 000	**7308CDB**	**7308CDF**	**7308CDT**	40.4	5.6	48.5	45.5	81.5	84.5	1.5	1	1.31
	110	54	2	1	142	87.1	152	96.8	6.70	7.45	—	4 600	6 600	**7408DB**	**7408DF**	**7408DT**	70.9	16.9	50	45.5	100	104.5	2	1	2.46
	110	54	2	1	132	80.8	141	89.8	5.75	6.40	—	3 900	5 900	**7408BDB**	**7408BDF**	**7408BDT**	90.8	36.8	50	45.5	100	104.5	2	1	2.46
45	68	24	0.6	0.3	33.7	28.2	—	—	1.55	—	16.0	12 000	16 000	**7909CDB**	**7909CDF**	**7909CDT**	27.1	3.1	49.5	—	63.5	65.5	0.6	0.3	0.254
	75	32	1	0.6	45.2	35.4	—	—	2.00	—	—	8 000	10 000	**7009DB**	**7009DF**	**7009DT**	50.7	18.7	50.5	—	69.5	70.5	1	0.6	0.520
	75	32	1	0.6	40.6	32.0	—	—	1.75	—	—	6 000	8 000	**7009BDB**	**7009BDF**	**7009BDT**	66.3	34.3	50.5	—	69.5	70.5	1	0.6	0.520
	75	32	1	0.6	49.6	38.5	—	—	2.25	—	15.4	11 000	15 000	**7009CDB**	**7009CDF**	**7009CDT**	32.1	0.1	50.5	—	69.5	70.5	1	0.6	0.520
	85	38	1.1	0.6	76.6	53.2	80.5	57.2	3.40	3.70	—	7 500	9 400	**7209DB**	**7209DF**	**7209DT**	56.0	18.0	52	49.5	78	80.5	1	0.6	0.860
	85	38	1.1	0.6	69.7	48.6	73.2	52.3	2.95	3.20	—	5 600	7 500	**7209BDB**	**7209BDF**	**7209BDT**	72.8	34.8	52	49.5	78	80.5	1	0.6	0.860
	85	38	1.1	0.6	82.9	57.4	87.1	61.8	3.90	4.20	14.2	10 000	14 000	**7209CDB**	**7209CDF**	**7209CDT**	36.2	1.8	52	49.5	78	80.5	1	0.6	0.860
	100	50	1.5	1	112	74.2	119	80.9	5.15	5.60	—	6 600	8 200	**7309DB**	**7309DF**	**7309DT**	67.2	17.2	53.5	50.5	91.5	94.5	1.5	1	1.75
	100	50	1.5	1	103	68.2	109	74.3	4.40	4.85	—	4 900	6 600	**7309BDB**	**7309BDF**	**7309BDT**	86.3	36.3	53.5	50.5	91.5	94.5	1.5	1	1.75
	100	50	1.5	1	120	79.5	127	86.7	5.85	6.40	13.5	9 000	12 000	**7309CDB**	**7309CDF**	**7309CDT**	44.6	5.4	53.5	50.5	91.5	94.5	1.5	1	1.75
	120	58	2	1	173	108	185	120	8.35	9.30	—	4 200	6 000	**7409DB**	**7409DF**	**7409DT**	77.2	19.2	55	50.5	110	114.5	2	1	3.10
	120	58	2	1	160	100	172	111	7.20	8.00	—	3 600	5 400	**7409BDB**	**7409BDF**	**7409BDT**	99.1	41.1	55	50.5	110	114.5	2	1	3.10
50	72	24	0.6	0.3	35.4	31.4	—	—	1.70	—	16.2	11 000	15 000	**7910CDB**	**7910CDF**	**7910CDT**	28.3	4.3	54.5	—	67.5	69.5	0.6	0.3	0.256
	80	32	1	0.6	48.0	40.2	—	—	2.20	—	—	7 300	9 200	**7010DB**	**7010DF**	**7010DT**	53.8	21.8	55.5	—	74.5	75.5	1	0.6	0.580

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (50) ~ (60) mm



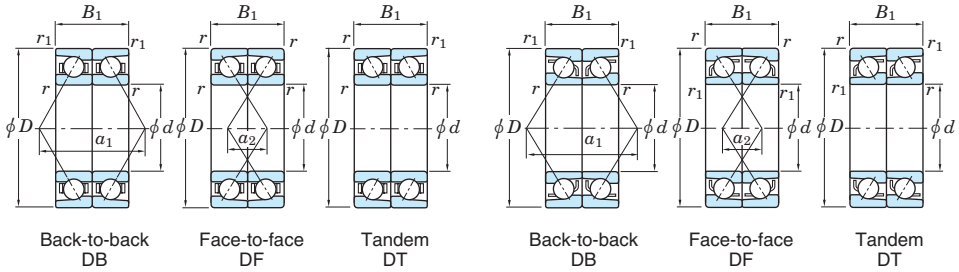


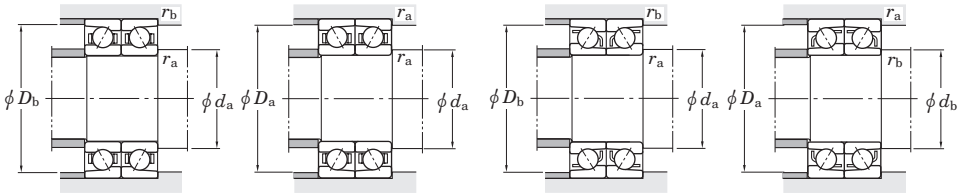
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN)		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)						(Refer.) Mass (kg)
					With machined cages		With pressed cages		C_u																
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
50	80	32	1	0.6	43.1	36.2	—	—	1.90	—	—	5 500	7 400	**7010BDB**	**7010BDF**	**7010BDT**	70.5	38.5	55.5	—	74.5	75.5	1	0.6	0.580
	80	32	1	0.6	52.7	43.9	—	—	2.50	—	15.7	10 000	13 000	**7010CDB**	**7010CDF**	**7010CDT**	33.6	1.6	55.5	—	74.5	75.5	1	0.6	0.580
	90	40	1.1	0.6	80.0	58.7	83.8	62.9	3.60	3.85	—	6 800	8 500	**7210DB**	**7210DF**	**7210DT**	60.7	20.7	57	54.5	83	85.5	1	0.6	0.970
	90	40	1.1	0.6	72.5	53.5	75.9	57.3	3.15	3.35	—	5 100	6 800	**7210BDB**	**7210BDF**	**7210BDT**	79.2	39.2	57	54.5	83	85.5	1	0.6	0.970
	90	40	1.1	0.6	86.9	63.6	91.0	68.1	4.10	4.40	14.6	9 400	12 000	**7210CDB**	**7210CDF**	**7210CDT**	38.9	1.1	57	54.5	83	85.5	1	0.6	0.970
	110	54	2	1	142	96.3	151	105	6.70	7.35	—	5 800	7 300	**7310DB**	**7310DF**	**7310DT**	74.4	20.4	60	55.5	100	104.5	2	1	2.28
	110	54	2	1	131	88.6	138	96.6	5.80	6.30	—	4 400	5 800	**7310BDB**	**7310BDF**	**7310BDT**	95.8	41.8	60	55.5	100	104.5	2	1	2.28
	110	54	2	1	153	103	162	112	7.70	8.40	13.4	8 000	11 000	**7310CDB**	**7310CDF**	**7310CDT**	49.0	5.0	60	55.5	100	104.5	2	1	2.28
	130	62	2.1	1.1	198	131	—	—	9.85	—	—	3 800	5 500	**7410DB**	**7410DF**	**7410DT**	83.3	21.3	62	—	118	123	2	1	3.84
	130	62	2.1	1.1	183	121	—	—	8.45	—	—	3 300	4 900	**7410BDB**	**7410BDF**	**7410BDT**	106.9	44.9	62	—	118	123	2	1	3.84
55	80	26	1	0.6	40.0	37.0	—	—	1.95	—	16.3	10 000	14 000	**7911CDB**	**7911CDF**	**7911CDT**	31.1	5.1	60.5	—	74.5	75.5	1	0.6	0.356
	90	36	1.1	0.6	63.2	52.5	—	—	2.95	—	—	6 600	8 300	**7011DB**	**7011DF**	**7011DT**	59.9	23.9	62	—	83	85.5	1	0.6	0.840
	90	36	1.1	0.6	56.7	47.5	—	—	2.55	—	—	5 000	6 600	**7011BDB**	**7011BDF**	**7011BDT**	78.8	42.8	62	—	83	85.5	1	0.6	0.840
	90	36	1.1	0.6	69.3	57.3	—	—	3.35	—	15.5	9 100	12 000	**7011CDB**	**7011CDF**	**7011CDT**	37.4	1.4	62	—	83	85.5	1	0.6	0.840
	100	42	1.5	1	98.9	74.2	104	79.6	4.60	4.90	—	6 100	7 600	**7211DB**	**7211DF**	**7211DT**	66.6	24.6	63.5	60.5	91.5	94.5	1.5	1	1.27
	100	42	1.5	1	89.6	67.6	93.8	72.4	3.95	4.25	—	4 600	6 100	**7211BDB**	**7211BDF**	**7211BDT**	87.3	45.3	63.5	60.5	91.5	94.5	1.5	1	1.27
	100	42	1.5	1	107	80.4	112	86.1	5.20	5.60	14.6	8 400	11 000	**7211CDB**	**7211CDF**	**7211CDT**	42.2	0.2	63.5	60.5	91.5	94.5	1.5	1	1.27
	120	58	2	1	164	113	174	123	7.90	8.60	—	5 400	6 700	**7311DB**	**7311DF**	**7311DT**	80.4	22.4	65	60.5	110	114.5	2	1	2.90
	120	58	2	1	151	104	160	113	6.80	7.40	—	4 000	5 400	**7311BDB**	**7311BDF**	**7311BDT**	103.7	45.7	65	60.5	110	114.5	2	1	2.90
	120	58	2	1	176	121	187	132	9.00	9.85	13.4	7 400	9 800	**7311CDB**	**7311CDF**	**7311CDT**	52.9	5.1	65	60.5	110	114.5	2	1	2.90
	140	66	2.1	1.1	241	165	—	—	12.8	—	—	3 500	5 000	**7411DB**	**7411DF**	**7411DT**	89.9	23.9	67	—	128	133	2	1	4.72
	140	66	2.1	1.1	224	153	—	—	11.0	—	—	3 000	4 500	**7411BDB**	**7411BDF**	**7411BDT**	115.7	49.7	67	—	128	133	2	1	4.72
60	85	26	1	0.6	47.2	43.6	—	—	2.35	—	16.3	9 100	13 000	**7912CDB**	**7912CDF**	**7912CDT**	32.6	6.6	65.5	—	79.5	80.5	1	0.6	0.374
	95	36	1.1	0.6	64.8	56.1	—	—	3.10	—	—	6 200	7 700	**7012DB**	**7012DF**	**7012DT**	62.8	26.8	67	—	88	90.5	1	0.6	0.900
	95	36	1.1	0.6	58.1	50.7	—	—	2.70	—	—	4 600	6 200	**7012BDB**	**7012BDF**	**7012BDT**	83.0	47.0	67	—	88	90.5	1	0.6	0.900
	95	36	1.1	0.6	71.2	61.3	—	—	3.50	—	15.7	8 500	11 000	**7012CDB**	**7012CDF**	**7012CDT**	38.8	2.8	67	—	88	90.5	1	0.6	0.900
	110	44	1.5	1	120	91.5	125	98.0	5.65	6.05	—	5 500	6 900	**7212DB**	**7212DF**	**7212DT**	72.3	28.3	68.5	65.5	101.5	104.5	1.5	1	1.64

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (60) ~ (70) mm



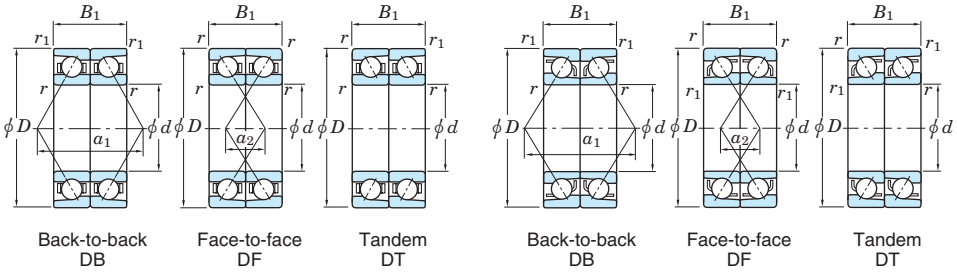


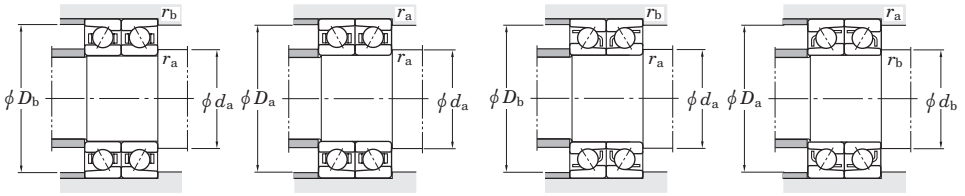
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)					(Refer.) Mass (kg)
					With machined cages		With pressed cages																	
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
60	110	44	1.5	1	108	83.3	114	89.2	4.90	5.25	—	4 100	5 500	**7212BDB**	**7212BDF**	**7212BDT**	95.0	51.0	68.5	65.5	101.5	104.5	1.5	1	1.64
	110	44	1.5	1	130	99.0	136	106	6.45	6.90	14.5	7 500	10 000	**7212CDB**	**7212CDF**	**7212CDT**	45.3	1.3	68.5	65.5	101.5	104.5	1.5	1	1.64
	130	62	2.1	1.1	188	131	199	143	9.15	10.0	—	5 000	6 200	**7312DB**	**7312DF**	**7312DT**	86.5	24.5	72	67	118	123	2	1	3.62
	130	62	2.1	1.1	172	121	183	132	7.90	8.60	—	3 700	5 000	**7312BDB**	**7312BDF**	**7312BDT**	111.6	49.6	72	67	118	123	2	1	3.62
	130	62	2.1	1.1	201	141	213	153	10.5	11.4	13.4	6 800	9 100	**7312CDB**	**7312CDF**	**7312CDT**	56.7	5.3	72	67	118	123	2	1	3.62
	150	70	2.1	1.1	262	187	—	—	13.7	—	—	3 200	4 600	**7412DB**	**7412DF**	**7412DT**	97.0	27.0	72	—	138	143	2	1	5.70
	150	70	2.1	1.1	243	173	—	—	11.8	—	—	2 800	4 100	**7412BDB**	**7412BDF**	**7412BDT**	125.1	55.1	72	—	138	143	2	1	5.70
65	90	26	1	0.6	42.2	42.3	—	—	2.20	—	16.5	8 600	12 000	**7913CDB**	**7913CDF**	**7913CDT**	33.8	7.8	70.5	—	84.5	85.5	1	0.6	0.410
	100	36	1.1	0.6	68.3	62.8	—	—	3.40	—	—	5 800	7 200	**7013DB**	**7013DF**	**7013DT**	65.9	29.9	72	—	93	95.5	1	0.6	0.940
	100	36	1.1	0.6	61.2	56.6	—	—	2.95	—	—	4 300	5 800	**7013BDB**	**7013BDF**	**7013BDT**	87.6	51.6	72	—	93	95.5	1	0.6	0.940
	100	36	1.1	0.6	75.2	68.7	—	—	3.85	—	15.9	7 900	11 000	**7013CDB**	**7013CDF**	**7013CDT**	40.2	4.2	72	—	93	95.5	1	0.6	0.940
	120	46	1.5	1	137	108	143	116	6.65	7.10	—	5 200	6 400	**7213DB**	**7213DF**	**7213DT**	76.4	30.4	73.5	70.5	111.5	114.5	1.5	1	2.04
	120	46	1.5	1	124	98.7	129	105	5.80	6.15	—	3 900	5 200	**7213BDB**	**7213BDF**	**7213BDT**	100.6	54.6	73.5	70.5	111.5	114.5	1.5	1	2.04
	120	46	1.5	1	148	117	155	125	7.60	8.10	14.6	7 100	9 400	**7213CDB**	**7213CDF**	**7213CDT**	47.8	1.8	73.5	70.5	111.5	114.5	1.5	1	2.04
	140	66	2.1	1.1	213	151	225	164	10.3	11.3	—	4 600	5 800	**7313DB**	**7313DF**	**7313DT**	92.5	26.5	77	72	128	133	2	1	4.44
	140	66	2.1	1.1	195	139	207	151	8.90	9.70	—	3 500	4 600	**7313BDB**	**7313BDF**	**7313BDT**	119.4	53.4	77	72	128	133	2	1	4.44
	140	66	2.1	1.1	228	161	242	176	11.8	12.9	13.4	6 300	8 500	**7313CDB**	**7313CDF**	**7313CDT**	60.6	5.4	77	72	128	133	2	1	4.44
	160	74	2.1	1.1	282	209	—	—	14.8	—	—	3 000	4 300	**7413DB**	**7413DF**	**7413DT**	102.9	28.9	77	—	148	153	2	1	6.82
	160	74	2.1	1.1	262	194	—	—	12.7	—	—	2 600	3 900	**7413BDB**	**7413BDF**	**7413BDT**	132.7	58.7	77	—	148	153	2	1	6.82
70	100	32	1	0.6	58.8	58.0	—	—	3.05	—	16.4	7 800	11 000	**7914CDB**	**7914CDF**	**7914CDT**	38.8	6.8	75.5	—	94.5	95.5	1	0.6	0.664
	110	40	1.1	0.6	86.7	78.7	—	—	4.30	—	—	5 300	6 600	**7014DB**	**7014DF**	**7014DT**	72.0	32.0	77	—	103	105.5	1	0.6	1.32
	110	40	1.1	0.6	77.7	71.1	—	—	3.75	—	—	4 000	5 300	**7014BDB**	**7014BDF**	**7014BDT**	95.5	55.5	77	—	103	105.5	1	0.6	1.32
	110	40	1.1	0.6	95.3	86.0	—	—	4.90	—	15.7	7 300	9 700	**7014CDB**	**7014CDF**	**7014CDT**	44.1	4.1	77	—	103	105.5	1	0.6	1.32
	125	48	1.5	1	142	111	155	127	6.85	7.80	—	4 900	6 100	**7214DB**	**7214DF**	**7214DT**	80.3	32.3	78.5	75.5	116.5	119.5	1.5	1	2.24
	125	48	1.5	1	128	101	140	116	5.90	6.75	—	3 700	4 900	**7214BDB**	**7214BDF**	**7214BDT**	105.8	57.8	78.5	75.5	116.5	119.5	1.5	1	2.24
	125	48	1.5	1	154	120	168	138	7.75	8.90	14.6	6 700	8 900	**7214CDB**	**7214CDF**	**7214CDT**	50.1	2.1	78.5	75.5	116.5	119.5	1.5	1	2.24
	150	70	2.1	1.1	239	172	253	187	11.4	12.4	—	4 300	5 400	**7314DB**	**7314DF**	**7314DT**	98.5	28.5	82	77	138	143	2	1	5.40

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (70) ~ (80) mm



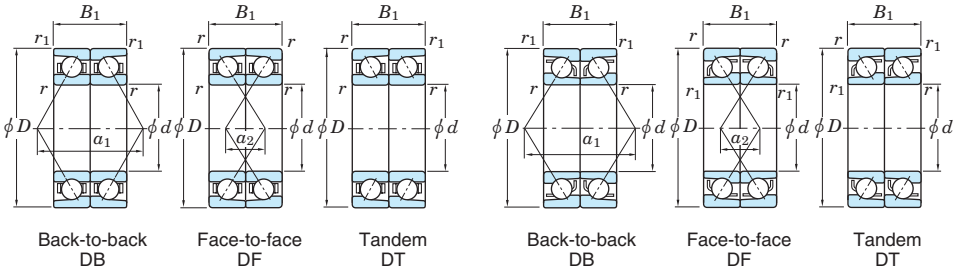


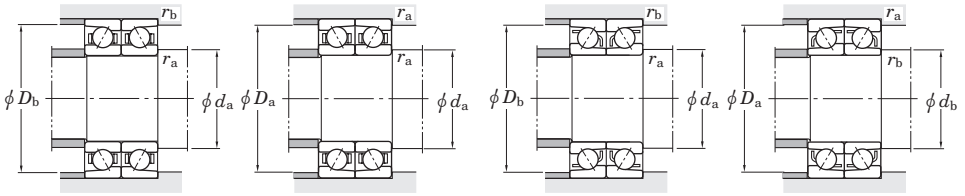
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)					(Refer.) Mass (kg)
					With machined cages		With pressed cages																	
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
70	150	70	2.1	1.1	219	158	232	172	9.80	10.7	—	3 200	4 300	**7314BDB**	**7314BDF**	**7314BDT**	127.3	57.3	82	77	138	143	2	1	5.40
	150	70	2.1	1.1	256	184	272	200	13.0	14.2	13.4	5 900	7 900	**7314CDB**	**7314CDF**	**7314CDT**	64.5	5.5	82	77	138	143	2	1	5.40
	180	84	3	1.1	303	230	—	—	10.6	—	—	2 700	3 900	**7414DB**	**7414DF**	**7414DT**	115.3	31.3	84	—	166	173	2.5	1	9.98
	180	84	3	1.1	301	237	—	—	10.9	—	—	2 300	3 500	**7414BDB**	**7414BDF**	**7414BDT**	148.4	64.4	84	—	166	173	2.5	1	9.98
75	105	32	1	0.6	59.7	60.9	—	—	3.15	—	16.5	7 400	9 800	**7915CDB**	**7915CDF**	**7915CDT**	40.1	8.1	80.5	—	99.5	100.5	1	0.6	0.700
	115	40	1.1	0.6	88.6	83.4	—	—	4.50	—	—	5 000	6 300	**7015DB**	**7015DF**	**7015DT**	74.9	34.9	82	—	108	110.5	1	0.6	1.38
	115	40	1.1	0.6	79.3	75.2	—	—	3.95	—	—	3 800	5 000	**7015BDB**	**7015BDF**	**7015BDT**	99.7	59.7	82	—	108	110.5	1	0.6	1.38
	115	40	1.1	0.6	97.6	91.3	—	—	5.10	—	15.9	6 900	9 200	**7015CDB**	**7015CDF**	**7015CDT**	45.5	5.5	82	—	108	110.5	1	0.6	1.38
	130	50	1.5	1	161	130	168	139	7.90	8.40	—	4 600	5 800	**7215DB**	**7215DF**	**7215DT**	84.2	34.2	83.5	80.5	121.5	124.5	1.5	1	2.46
	130	50	1.5	1	146	119	152	127	6.85	7.30	—	3 500	4 600	**7215BDB**	**7215BDF**	**7215BDT**	111.0	61.0	83.5	80.5	121.5	124.5	1.5	1	2.46
	130	50	1.5	1	175	141	183	151	8.95	9.55	14.6	6 400	8 500	**7215CDB**	**7215CDF**	**7215CDT**	52.5	2.5	83.5	80.5	121.5	124.5	1.5	1	2.46
	160	74	2.1	1.1	260	194	276	212	12.4	13.5	—	4 000	5 000	**7315DB**	**7315DF**	**7315DT**	104.9	30.9	87	82	148	153	2	1	6.30
	160	74	2.1	1.1	239	178	253	195	10.7	11.7	—	3 000	4 000	**7315BDB**	**7315BDF**	**7315BDT**	135.6	61.6	87	82	148	153	2	1	6.30
	160	74	2.1	1.1	279	208	296	227	14.2	15.5	13.4	5 500	7 400	**7315CDB**	**7315CDF**	**7315CDT**	68.5	5.5	87	82	148	153	2	1	6.30
	190	90	3	1.1	348	282	—	—	12.6	—	—	2 500	3 600	**7415DB**	**7415DF**	**7415DT**	122.7	32.7	89	—	176	183	2.5	1	11.8
	190	90	3	1.1	322	261	—	—	11.6	—	—	2 200	3 300	**7415BDB**	**7415BDF**	**7415BDT**	157.9	67.9	89	—	176	183	2.5	1	11.8
80	110	32	1	0.6	60.5	63.2	—	—	3.25	—	16.5	7 000	9 300	**7916CDB**	**7916CDF**	**7916CDT**	41.5	9.5	85.5	—	104.5	105.5	1	0.6	0.736
	125	44	1.1	0.6	108	101	—	—	5.50	—	—	4 600	5 800	**7016DB**	**7016DF**	**7016DT**	81.2	37.2	87	—	118	120.5	1	0.6	1.86
	125	44	1.1	0.6	97.1	91.3	—	—	4.75	—	—	3 500	4 600	**7016BDB**	**7016BDF**	**7016BDT**	108.0	64.0	87	—	118	120.5	1	0.6	1.86
	125	44	1.1	0.6	119	111	—	—	6.20	—	15.7	6 400	8 500	**7016CDB**	**7016CDF**	**7016CDT**	49.5	5.5	87	—	118	120.5	1	0.6	1.86
	140	52	2	1	173	143	181	152	8.25	8.80	—	4 300	5 400	**7216DB**	**7216DF**	**7216DT**	89.5	37.5	90	85.5	130	134.5	2	1	3.00
	140	52	2	1	157	130	163	139	7.15	7.60	—	3 200	4 300	**7216BDB**	**7216BDF**	**7216BDT**	118.3	66.3	90	85.5	130	134.5	2	1	3.00
	140	52	2	1	189	155	197	165	9.40	10.0	14.7	5 900	7 900	**7216CDB**	**7216CDF**	**7216CDT**	55.5	3.5	90	85.5	130	134.5	2	1	3.00
	170	78	2.1	1.1	282	218	299	238	13.5	14.7	—	3 800	4 700	**7316DB**	**7316DF**	**7316DT**	111.2	33.2	92	87	158	163	2	1	7.70
	170	78	2.1	1.1	259	200	274	218	11.6	12.7	—	2 800	3 800	**7316BDB**	**7316BDF**	**7316BDT**	143.9	65.9	92	87	158	163	2	1	7.70
	170	78	2.1	1.1	302	233	321	255	15.4	16.8	13.5	5 200	6 900	**7316CDB**	**7316CDF**	**7316CDT**	72.5	5.5	92	87	158	163	2	1	7.70
	200	96	3	1.1	391	332	—	—	14.4	—	—	2 400	3 400	**7416DB**	**7416DF**	**7416DT**	130.0	34.0	94	—	186	193	2.5	1	12.0

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (80) ~ (95) mm



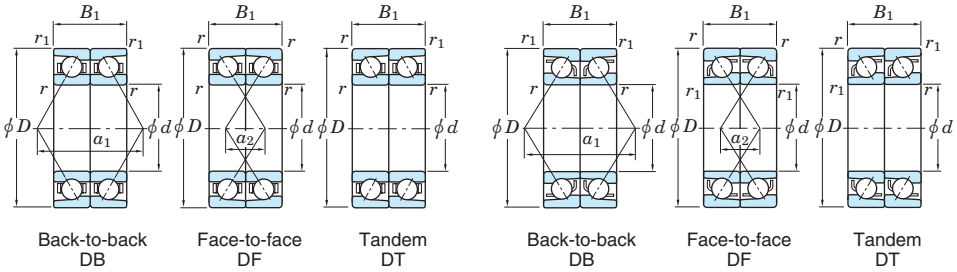


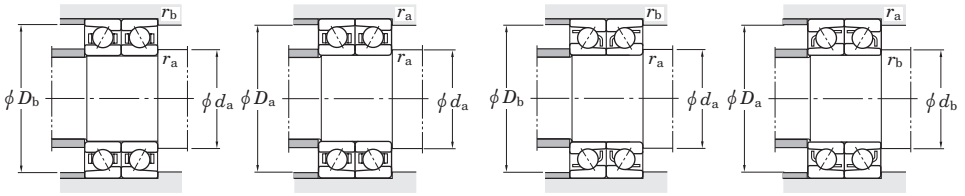
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)						(Refer.) Mass (kg)
					With machined cages		With pressed cages																		
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
80	200	96	3	1.1	363	307	—	—	13.3	—	—	2 100	3 100	**7416BDB**	**7416BDF**	**7416BDT**	167.2	71.2	94	—	186	193	2.5	1	12.0
85	120	36	1.1	0.6	79.0	81.3	—	—	4.20	—	16.5	6 500	8 600	**7917CDB**	**7917CDF**	**7917CDT**	45.5	9.5	92	—	113	115.5	1	0.6	1.05
	130	44	1.1	0.6	111	107	—	—	5.55	—	—	4 400	5 500	**7017DB**	**7017DF**	**7017DT**	84.7	40.7	92	—	123	125.5	1	0.6	1.94
	130	44	1.1	0.6	99.2	96.7	—	—	4.85	—	—	3 300	4 400	**7017BDB**	**7017BDF**	**7017BDT**	113.0	69.0	92	—	123	125.5	1	0.6	1.94
	130	44	1.1	0.6	122	117	—	—	6.30	—	15.9	6 000	8 000	**7017CDB**	**7017CDF**	**7017CDT**	51.1	7.1	92	—	123	125.5	1	0.6	1.94
	150	56	2	1	200	167	209	178	9.40	10.0	—	4 000	5 000	**7217DB**	**7217DF**	**7217DT**	95.9	39.9	95	90.5	140	144.5	2	1	3.74
	150	56	2	1	181	152	189	162	8.15	8.70	—	3 000	4 000	**7217BDB**	**7217BDF**	**7217BDT**	126.6	70.6	95	90.5	140	144.5	2	1	3.74
	150	56	2	1	218	181	227	193	10.7	11.4	14.7	5 500	7 400	**7217CDB**	**7217CDF**	**7217CDT**	59.5	3.5	95	90.5	140	144.5	2	1	3.74
	180	82	3	1.1	304	243	322	265	14.6	15.9	—	3 500	4 400	**7317DB**	**7317DF**	**7317DT**	117.5	35.5	99	92	166	173	2.5	1	9.06
	180	82	3	1.1	279	223	295	244	12.6	13.7	—	2 700	3 500	**7317BDB**	**7317BDF**	**7317BDT**	152.2	70.2	99	92	166	173	2.5	1	9.06
	180	82	3	1.1	326	261	346	284	16.7	18.2	13.5	4 900	6 500	**7317CDB**	**7317CDF**	**7317CDT**	76.5	5.5	99	92	166	173	2.5	1	9.06
	210	104	4	1.5	414	360	—	—	15.3	—	—	2 300	3 300	**7417DB**	**7417DF**	**7417DT**	137.5	33.5	103	—	192	201.5	3	1.5	17.1
	210	104	4	1.5	384	334	—	—	14.2	—	—	2 000	3 000	**7417BDB**	**7417BDF**	**7417BDT**	176.2	72.2	103	—	192	201.5	3	1.5	17.1
90	125	36	1.1	0.6	80.3	85.2	—	—	4.25	—	16.6	6 200	8 200	**7918CDB**	**7918CDF**	**7918CDT**	46.8	10.8	97	—	118	120.5	1	0.6	1.10
	140	48	1.5	1	132	127	—	—	6.45	—	—	4 100	5 100	**7018DB**	**7018DF**	**7018DT**	90.4	42.4	98.5	—	131.5	134.5	1.5	1	2.52
	140	48	1.5	1	119	114	—	—	5.60	—	—	3 100	4 100	**7018BDB**	**7018BDF**	**7018BDT**	120.5	72.5	98.5	—	131.5	134.5	1.5	1	2.52
	140	48	1.5	1	146	138	—	—	7.30	—	15.7	5 700	7 500	**7018CDB**	**7018CDF**	**7018CDT**	54.8	6.8	98.5	—	131.5	134.5	1.5	1	2.52
	160	60	2	1	229	193	239	206	10.6	11.3	—	3 800	4 700	**7218DB**	**7218DF**	**7218DT**	102.2	42.2	100	95.5	150	154.5	2	1	4.60
	160	60	2	1	207	176	217	188	9.15	9.80	—	2 800	3 800	**7218BDB**	**7218BDF**	**7218BDT**	134.9	74.9	100	95.5	150	154.5	2	1	4.60
	160	60	2	1	249	209	260	223	12.0	12.8	14.6	5 200	6 900	**7218CDB**	**7218CDF**	**7218CDT**	63.5	3.5	100	95.5	150	154.5	2	1	4.60
	190	86	3	1.1	327	270	346	294	11.8	12.8	—	3 300	4 200	**7318DB**	**7318DF**	**7318DT**	123.9	37.9	104	97	176	183	2.5	1	10.6
	190	86	3	1.1	300	248	317	270	10.8	11.8	—	2 500	3 300	**7318BDB**	**7318BDF**	**7318BDT**	160.5	74.5	104	97	176	183	2.5	1	10.6
	190	86	3	1.1	351	289	372	315	12.6	13.8	13.5	4 600	6 100	**7318CDB**	**7318CDF**	**7318CDT**	80.5	5.5	104	97	176	183	2.5	1	10.6
	225	108	4	1.5	439	393	—	—	16.2	—	—	2 100	3 100	**7418DB**	**7418DF**	**7418DT**	145.0	37.0	108	—	207	216.5	3	1.5	22.8
	225	108	4	1.5	406	364	—	—	15.0	—	—	1 800	2 800	**7418BDB**	**7418BDF**	**7418BDT**	186.2	78.2	108	—	207	216.5	3	1.5	22.8
95	130	36	1.1	0.6	81.6	88.3	—	—	4.30	—	16.5	5 900	7 900	**7919CDB**	**7919CDF**	**7919CDT**	48.1	12.1	102	—	123	125.5	1	0.6	1.15
	145	48	1.5	1	135	134	—	—	6.55	—	—	3 900	4 800	**7019DB**	**7019DF**	**7019DT**	94.5	46.5	103.5	—	136.5	139.5	1.5	1	2.64

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (95) ~ (105) mm



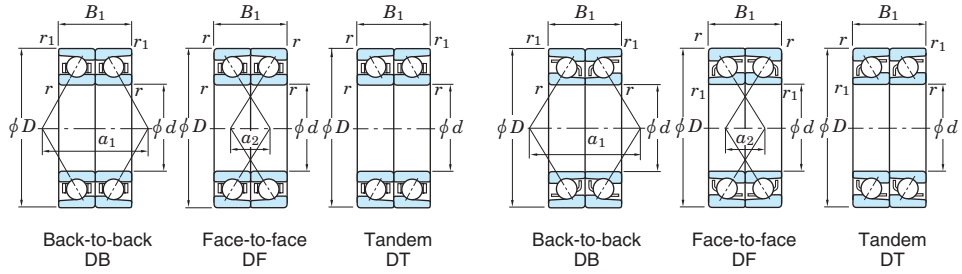


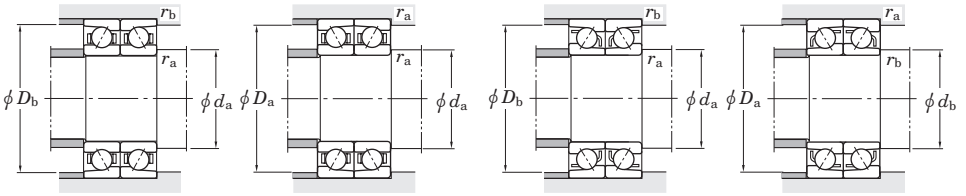
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)						(Refer.) Mass (kg)
					With machined cages		With pressed cages																		
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
95	145	48	1.5	1	121	121	—	—	5.70	—	—	2 900	3 900	**7019BDB**	**7019BDF**	**7019BDT**	126.4	78.4	103.5	—	136.5	139.5	1.5	1	2.64
	145	48	1.5	1	149	147	—	—	7.40	—	15.9	5 300	7 100	**7019CDB**	**7019CDF**	**7019CDT**	56.7	8.7	103.5	—	136.5	139.5	1.5	1	2.64
	170	64	2.1	1.1	248	207	260	221	11.0	11.8	—	3 500	4 400	**7219DB**	**7219DF**	**7219DT**	108.5	44.5	107	102	158	163	2	1	5.56
	170	64	2.1	1.1	224	188	235	201	9.55	10.2	—	2 700	3 500	**7219BDB**	**7219BDF**	**7219BDT**	143.2	79.2	107	102	158	163	2	1	5.56
	170	64	2.1	1.1	269	224	282	240	12.6	13.5	14.6	4 900	6 500	**7219CDB**	**7219CDF**	**7219CDT**	67.5	3.5	107	102	158	163	2	1	5.56
	200	90	3	1.1	350	298	371	325	12.7	13.8	—	3 200	4 000	**7319DB**	**7319DF**	**7319DT**	130.2	40.2	109	102	186	193	2.5	1	12.2
	200	90	3	1.1	321	273	340	298	11.6	12.7	—	2 400	3 200	**7319BDB**	**7319BDF**	**7319BDT**	168.8	78.8	109	102	186	193	2.5	1	12.2
	200	90	3	1.1	376	319	398	348	13.6	14.8	13.5	4 400	5 800	**7319CDB**	**7319CDF**	**7319CDT**	84.5	5.5	109	102	186	193	2.5	1	12.2
100	140	40	1.1	0.6	113	117	—	—	5.65	—	16.3	5 500	7 400	**7920CDB**	**7920CDF**	**7920CDT**	52.1	12.1	107	—	133	135.5	1	0.6	1.55
	150	48	1.5	1	139	141	—	—	6.75	—	—	3 800	4 700	**7020DB**	**7020DF**	**7020DT**	96.2	48.2	108.5	—	141.5	144.5	1.5	1	2.74
	150	48	1.5	1	124	127	—	—	5.90	—	—	2 800	3 800	**7020BDB**	**7020BDF**	**7020BDT**	128.9	80.9	108.5	—	141.5	144.5	1.5	1	2.74
	150	48	1.5	1	153	154	—	—	7.65	—	16.0	5 200	6 900	**7020CDB**	**7020CDF**	**7020CDT**	57.5	9.5	108.5	—	141.5	144.5	1.5	1	2.74
	180	68	2.1	1.1	279	235	292	252	12.2	13.0	—	3 300	4 100	**7220DB**	**7220DF**	**7220DT**	115.4	47.4	112	—	168	173	2	1	6.64
	180	68	2.1	1.1	252	214	264	229	10.5	11.3	—	2 500	3 300	**7220BDB**	**7220BDF**	**7220BDT**	152.3	84.3	112	—	168	173	2	1	6.64
	180	68	2.1	1.1	303	254	317	273	13.9	14.8	14.6	4 600	6 100	**7220CDB**	**7220CDF**	**7220CDT**	71.8	3.8	112	107	168	173	2	1	6.64
	215	94	3	1.1	373	323	421	387	13.2	15.9	—	2 900	3 600	**7320DB**	**7320DF**	**7320DT**	138.8	44.8	114	—	201	208	2.5	1	15.1
	215	94	3	1.1	342	297	386	356	12.2	14.6	—	2 200	2 900	**7320BDB**	**7320BDF**	**7320BDT**	180.4	86.4	114	—	201	208	2.5	1	15.1
	215	94	3	1.1	400	346	451	415	14.2	17.0	13.4	4 000	5 300	**7320CDB**	**7320CDF**	**7320CDT**	89.6	4.4	114	107	201	208	2.5	1	15.1
105	145	40	1.1	0.6	115	123	—	—	5.75	—	16.4	5 300	7 100	**7921CDB**	**7921CDF**	**7921CDT**	53.5	13.5	112	—	138	140.5	1	0.6	1.62
	160	52	2	1	162	164	—	—	7.60	—	—	3 500	4 400	**7021DB**	**7021DF**	**7021DT**	103.7	51.7	115	—	150	154.5	2	1	3.46
	160	52	2	1	145	148	—	—	6.65	—	—	2 600	3 500	**7021BDB**	**7021BDF**	**7021BDT**	137.2	85.2	115	—	150	154.5	2	1	3.46
	160	52	2	1	178	179	—	—	8.60	—	15.9	4 800	6 400	**7021CDB**	**7021CDF**	**7021CDT**	62.0	10.0	115	—	150	154.5	2	1	3.46
	190	72	2.1	1.1	303	265	—	—	13.4	—	—	3 100	3 900	**7221DB**	**7221DF**	**7221DT**	122.1	50.1	117	—	178	183	2	1	7.90
	190	72	2.1	1.1	275	241	—	—	11.6	—	—	2 300	3 100	**7221BDB**	**7221BDF**	**7221BDT**	161.0	89.0	117	—	178	183	2	1	7.90
	190	72	2.1	1.1	330	287	—	—	15.2	—	14.6	4 300	5 700	**7221CDB**	**7221CDF**	**7221CDT**	75.9	3.9	117	—	178	183	2	1	7.90
	225	98	3	1.1	422	386	—	—	15.5	—	—	2 800	3 500	**7321DB**	**7321DF**	**7321DT**	144.3	46.3	119	—	211	218	2.5	1	17.2
	225	98	3	1.1	387	355	—	—	14.3	—	—	2 100	2 800	**7321BDB**	**7321BDF**	**7321BDT**	187.5	89.5	119	—	211	218	2.5	1	17.2

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (105) ~ (130) mm



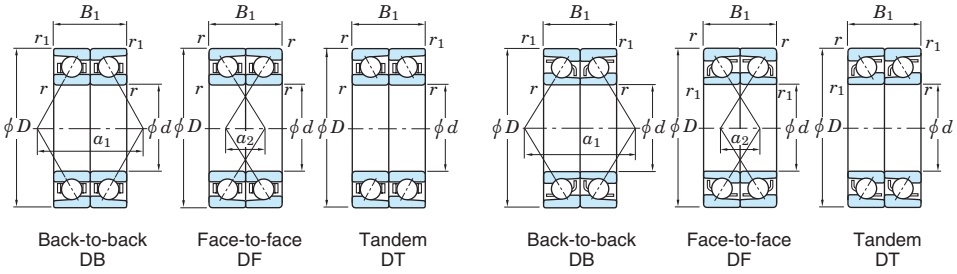


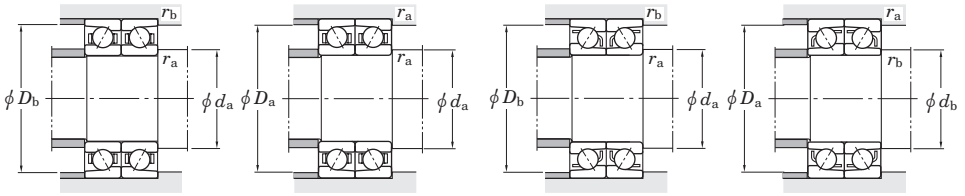
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)						(Refer.) Mass (kg)
					With machined cages		With pressed cages																		
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
105	225	98	3	1.1	452	413	—	—	16.6	—	13.4	3 900	5 100	**7321CDB**	**7321CDF**	**7321CDT**	93.2	4.8	119	—	211	218	2.5	1	17.2
110	150	40	1.1	0.6	117	129	—	—	5.85	—	16.5	5 100	6 800	**7922CDB**	**7922CDF**	**7922CDT**	54.8	14.8	117	—	143	145.5	1	0.6	1.68
	170	56	2	1	187	186	—	—	8.55	—	—	3 300	4 200	**7022DB**	**7022DF**	**7022DT**	108.9	52.9	120	—	160	164.5	2	1	4.28
	170	56	2	1	167	167	—	—	7.45	—	—	2 500	3 300	**7022BDB**	**7022BDF**	**7022BDT**	145.5	89.5	120	—	160	164.5	2	1	4.28
	170	56	2	1	205	203	—	—	9.70	—	15.7	4 600	6 100	**7022CDB**	**7022CDF**	**7022CDT**	65.5	9.5	120	—	160	164.5	2	1	4.28
	200	76	2.1	1.1	329	297	—	—	14.6	—	—	3 000	3 700	**7222DB**	**7222DF**	**7222DT**	128.7	52.7	122	—	188	193	2	1	9.30
	200	76	2.1	1.1	298	270	—	—	12.7	—	—	2 200	3 000	**7222BDB**	**7222BDF**	**7222BDT**	169.7	93.7	122	—	188	193	2	1	9.30
	200	76	2.1	1.1	357	321	—	—	16.7	—	14.5	4 100	5 400	**7222CDB**	**7222CDF**	**7222CDT**	80.1	4.1	122	—	188	193	2	1	9.30
	240	100	3	1.1	472	452	—	—	17.5	—	—	2 600	3 200	**7322DB**	**7322DF**	**7322DT**	152.7	52.7	124	—	226	233	2.5	1	20.2
	240	100	3	1.1	433	416	—	—	16.1	—	—	1 900	2 600	**7322BDB**	**7322BDF**	**7322BDT**	199.3	99.3	124	—	226	233	2.5	1	20.2
	240	100	3	1.1	505	484	—	—	18.8	—	13.4	3 500	4 700	**7322CDB**	**7322CDF**	**7322CDT**	97.7	2.3	124	—	226	233	2.5	1	20.2
120	165	44	1.1	0.6	146	162	—	—	7.10	—	16.5	4 700	6 200	**7924CDB**	**7924CDF**	**7924CDT**	60.2	16.2	127	—	158	160.5	1	0.6	2.30
	180	56	2	1	196	206	—	—	9.00	—	—	3 100	3 900	**7024DB**	**7024DF**	**7024DT**	114.6	58.6	130	—	170	174.5	2	1	4.54
	180	56	2	1	176	186	—	—	7.85	—	—	2 300	3 100	**7024BDB**	**7024BDF**	**7024BDT**	153.9	97.9	130	—	170	174.5	2	1	4.54
	180	56	2	1	216	226	—	—	10.2	—	16.0	4 300	5 700	**7024CDB**	**7024CDF**	**7024CDT**	68.2	12.2	130	—	170	174.5	2	1	4.54
	215	80	2.1	1.1	354	332	—	—	15.7	—	—	2 700	3 400	**7224DB**	**7224DF**	**7224DT**	137.0	57.0	132	—	203	208	2	1	11.0
	215	80	2.1	1.1	321	302	—	—	13.6	—	—	2 100	2 800	**7224BDB**	**7224BDF**	**7224BDT**	180.5	100.5	132	—	203	208	2	1	11.0
	215	80	2.1	1.1	385	359	—	—	17.9	—	14.6	3 800	5 000	**7224CDB**	**7224CDF**	**7224CDT**	85.0	5.0	132	—	203	208	2	1	11.0
	260	110	3	1.1	500	504	—	—	18.9	—	—	2 400	3 000	**7324DB**	**7324DF**	**7324DT**	164.7	54.7	134	—	246	253	2.5	1	25.2
	260	110	3	1.1	457	462	—	—	17.3	—	—	1 800	2 400	**7324BDB**	**7324BDF**	**7324BDT**	214.4	104.4	134	—	246	253	2.5	1	25.2
	260	110	3	1.1	538	542	—	—	20.3	—	13.7	3 300	4 400	**7324CDB**	**7324CDF**	**7324CDT**	105.9	4.1	134	—	246	253	2.5	1	25.2
130	180	48	1.5	1	177	200	—	—	8.45	—	16.4	4 300	5 700	**7926CDB**	**7926CDF**	**7926CDT**	65.5	17.5	138.5	—	171.5	174.5	1.5	1	3.00
	200	66	2	1	238	251	—	—	10.5	—	—	2 800	3 500	**7026DB**	**7026DF**	**7026DT**	128.3	62.3	140	—	190	194.5	2	1	6.86
	200	66	2	1	213	226	—	—	9.20	—	—	2 100	2 800	**7026BDB**	**7026BDF**	**7026BDT**	171.5	105.5	140	—	190	194.5	2	1	6.86
	200	66	2	1	262	274	—	—	11.9	—	15.9	3 900	5 100	**7026CDB**	**7026CDF**	**7026CDT**	77.2	11.2	140	—	190	194.5	2	1	6.86
	230	80	3	1.1	398	395	—	—	15.2	—	—	2 500	3 200	**7226DB**	**7226DF**	**7226DT**	143.9	63.9	144	—	216	223	2.5	1	12.4

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (130) ~ (160) mm



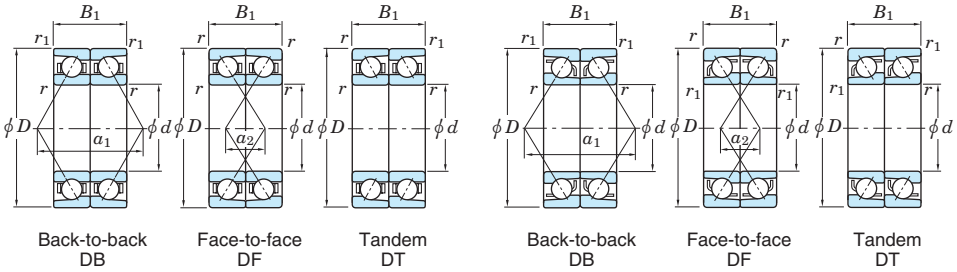


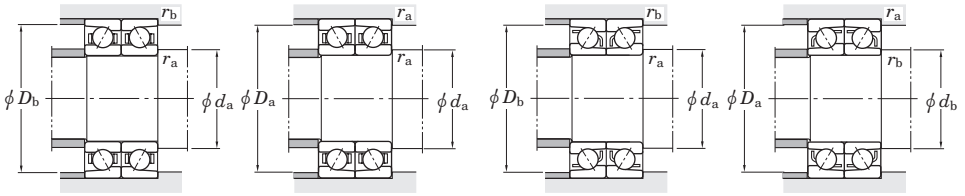
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)						(Refer.) Mass (kg)
					With machined cages		With pressed cages																		
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
130	230	80	3	1.1	360	360	—	—	13.9	—	—	1 900	2 500	**7226BDB**	**7226BDF**	**7226BDT**	191.0	111.0	144	—	216	223	2.5	1	12.4
	230	80	3	1.1	433	428	—	—	16.5	—	14.7	3 500	4 700	**7226CDB**	**7226CDF**	**7226CDT**	88.2	8.2	144	—	216	223	2.5	1	12.4
	280	116	4	1.5	611	659	—	—	23.7	—	—	2 200	2 700	**7326DB**	**7326DF**	**7326DT**	177.5	61.5	148	—	262	271.5	3	1.5	30.8
	280	116	4	1.5	507	536	—	—	19.4	—	—	1 600	2 200	**7326BDB**	**7326BDF**	**7326BDT**	230.0	114.0	148	—	262	271.5	3	1.5	30.8
	280	116	4	1.5	597	629	—	—	22.7	—	13.7	3 000	4 000	**7326CDB**	**7326CDF**	**7326CDT**	112.9	3.1	148	—	262	271.5	3	1.5	30.8
140	190	48	1.5	1	179	210	—	—	8.45	—	16.6	4 000	5 400	**7928CDB**	**7928CDF**	**7928CDT**	68.2	20.2	148.5	—	181.5	184.5	1.5	1	3.18
	210	66	2	1	243	265	—	—	10.6	—	—	2 600	3 300	**7028DB**	**7028DF**	**7028DT**	134.1	68.1	150	—	200	204.5	2	1	7.28
	210	66	2	1	217	237	—	—	9.25	—	—	2 000	2 600	**7028BDB**	**7028BDF**	**7028BDT**	179.8	113.8	150	—	200	204.5	2	1	7.28
	210	66	2	1	268	290	—	—	12.0	—	16.0	3 600	4 800	**7028CDB**	**7028CDF**	**7028CDT**	79.9	13.9	150	—	200	204.5	2	1	7.28
	250	84	3	1.1	443	468	—	—	17.3	—	—	2 300	2 900	**7228DB**	**7228DF**	**7228DT**	154.6	70.6	154	—	236	243	2.5	1	15.5
	250	84	3	1.1	401	426	—	—	15.7	—	—	1 700	2 300	**7228BDB**	**7228BDF**	**7228BDT**	205.6	121.6	154	—	236	243	2.5	1	15.5
	250	84	3	1.1	483	508	—	—	18.8	—	14.8	3 200	4 300	**7228CDB**	**7228CDF**	**7228CDT**	94.2	10.2	154	—	236	243	2.5	1	15.5
	300	124	4	1.5	668	748	—	—	26.1	—	—	2 000	2 500	**7328DB**	**7328DF**	**7328DT**	189.0	65.0	158	—	282	291.5	3	1.5	37.6
	300	124	4	1.5	613	688	—	—	24.0	—	—	1 500	2 000	**7328BDB**	**7328BDF**	**7328BDT**	246.6	122.6	158	—	282	291.5	3	1.5	37.6
	300	124	4	1.5	717	802	—	—	27.9	—	13.4	2 800	3 700	**7328CDB**	**7328CDF**	**7328CDT**	120.9	3.1	158	—	282	291.5	3	1.5	37.6
150	210	56	2	1	241	263	—	—	10.9	—	16.3	3 700	4 900	**7930CDB**	**7930CDF**	**7930CDT**	76.2	20.2	160	—	200	204.5	2	1	4.94
	225	70	2.1	1.1	278	308	—	—	11.9	—	—	2 400	3 000	**7030DB**	**7030DF**	**7030DT**	144.2	74.2	162	—	213	218	2	1	8.86
	225	70	2.1	1.1	249	275	—	—	10.4	—	—	1 800	2 400	**7030BDB**	**7030BDF**	**7030BDT**	192.3	122.3	162	—	213	218	2	1	8.86
	225	70	2.1	1.1	306	337	—	—	13.4	—	16.1	3 300	4 400	**7030CDB**	**7030CDF**	**7030CDT**	85.6	15.6	162	—	213	218	2	1	8.86
	270	90	3	1.1	504	560	—	—	19.9	—	—	2 100	2 700	**7230DB**	**7230DF**	**7230DT**	166.3	76.3	164	—	256	263	2.5	1	19.5
	270	90	3	1.1	456	509	—	—	18.1	—	—	1 600	2 100	**7230BDB**	**7230BDF**	**7230BDT**	221.2	131.2	164	—	256	263	2.5	1	19.5
	270	90	3	1.1	549	607	—	—	21.6	—	14.7	2 900	3 900	**7230CDB**	**7230CDF**	**7230CDT**	101.3	11.3	164	—	256	263	2.5	1	19.5
	320	130	4	1.5	706	829	—	—	27.9	—	—	1 900	2 300	**7330DB**	**7330DF**	**7330DT**	200.7	70.7	168	—	302	311.5	3	1.5	44.8
	320	130	4	1.5	645	760	—	—	25.6	—	—	1 400	1 900	**7330BDB**	**7330BDF**	**7330BDT**	262.2	132.2	168	—	302	311.5	3	1.5	44.8
	320	130	4	1.5	760	891	—	—	30.0	—	13.7	2 600	3 400	**7330CDB**	**7330CDF**	**7330CDT**	128.0	2.0	168	—	302	311.5	3	1.5	44.8
160	220	56	2	1	245	289	—	—	10.9	—	16.5	3 500	4 700	**7932CDB**	**7932CDF**	**7932CDT**	78.9	22.9	170	—	210	214.5	2	1	5.20

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (160) ~ (180) mm



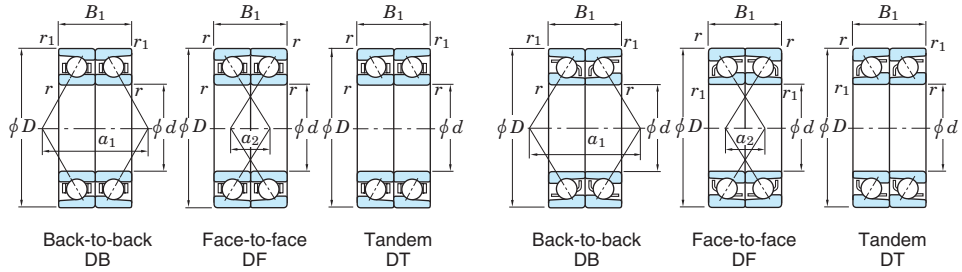


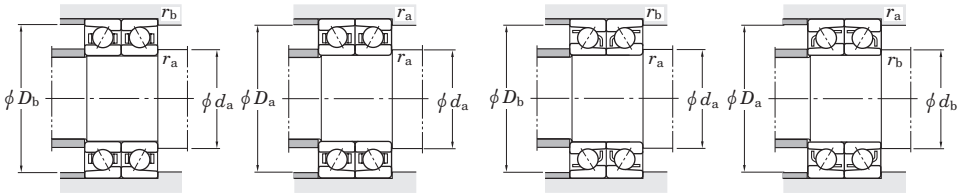
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)					(Refer.) Mass (kg)
					With machined cages		With pressed cages																	
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
160	240	76	2.1	1.1	315	353	—	—	13.3	—	—	2 300	2 800	**7032DB**	**7032DF**	**7032DT**	153.5	77.5	172	—	228	233	2	1	10.9
	240	76	2.1	1.1	282	316	—	—	11.6	—	—	1 700	2 300	**7032BDB**	**7032BDF**	**7032BDT**	205.8	129.8	172	—	228	233	2	1	10.9
	240	76	2.1	1.1	347	386	—	—	15.0	—	16.0	3 100	4 100	**7032CDB**	**7032CDF**	**7032CDT**	91.6	15.6	172	—	228	233	2	1	10.9
	290	96	3	1.1	468	525	—	—	18.1	—	—	2 000	2 500	**7232DB**	**7232DF**	**7232DT**	177.9	81.9	174	—	276	283	2.5	1	24.2
	290	96	3	1.1	482	557	—	—	19.2	—	—	1 500	2 000	**7232BDB**	**7232BDF**	**7232BDT**	236.8	140.8	174	—	276	283	2.5	1	24.2
	290	96	3	1.1	511	665	—	—	19.7	—	15.2	2 700	3 600	**7232CDB**	**7232CDF**	**7232CDT**	108.3	12.3	174	—	276	283	2.5	1	24.2
	340	136	4	1.5	741	909	—	—	29.7	—	—	1 700	2 200	**7332DB**	**7332DF**	**7332DT**	212.3	76.3	178	—	322	331.5	3	1.5	52.8
	340	136	4	1.5	675	831	—	—	27.2	—	—	1 300	1 700	**7332BDB**	**7332BDF**	**7332BDT**	277.8	141.8	178	—	322	331.5	3	1.5	52.8
	340	136	4	1.5	800	980	—	—	32.0	—	14.0	2 400	3 200	**7332CDB**	**7332CDF**	**7332CDT**	135.0	1.0	168.5	—	322	331.5	3	1.5	52.8
170	230	56	2	1	255	302	—	—	11.5	—	16.6	3 100	4 100	**7934CDB**	**7934CDF**	**7934CDT**	81.6	25.6	180	—	220	224.5	2	1	6.42
	260	84	2.1	1.1	377	429	—	—	15.8	—	—	2 100	2 600	**7034DB**	**7034DF**	**7034DT**	166.2	82.2	182	—	248	253	2	1	15.2
	260	84	2.1	1.1	338	386	—	—	13.8	—	—	1 600	2 100	**7034BDB**	**7034BDF**	**7034BDT**	222.4	138.4	182	—	248	253	2	1	15.5
	260	84	2.1	1.1	415	469	—	—	17.9	—	15.9	2 900	3 800	**7034CDB**	**7034CDF**	**7034CDT**	99.6	15.6	182	—	248	253	2	1	15.1
	310	104	4	1.5	552	661	—	—	22.0	—	—	1 800	2 300	**7234DB**	**7234DF**	**7234DT**	190.6	86.6	188	—	292	301.5	3	1.5	30.2
	310	104	4	1.5	497	600	—	—	20.0	—	—	1 400	1 800	**7234BDB**	**7234BDF**	**7234BDT**	253.4	149.4	188	—	292	301.5	3	1.5	30.2
	310	104	4	1.5	603	719	—	—	24.0	—	15.1	2 500	3 300	**7234CDB**	**7234CDF**	**7234CDT**	116.3	12.3	188	—	292	301.5	3	1.5	30.2
	360	144	4	1.5	789	969	—	—	30.7	—	—	1 600	2 000	**7334DB**	**7334DF**	**7334DT**	225.0	81.0	188	—	342	351.5	3	1.5	62.4
	360	144	4	1.5	721	888	—	—	28.2	—	—	1 200	1 600	**7334BDB**	**7334BDF**	**7334BDT**	294.4	150.4	188	—	342	351.5	3	1.5	62.4
	360	144	4	1.5	849	1 040	—	—	33.1	—	13.8	2 200	3 000	**7334CDB**	**7334CDF**	**7334CDT**	143.0	1.0	188	—	342	351.5	3	1.5	62.4
180	250	66	2	1	325	375	—	—	14.1	—	16.4	2 800	3 700	**7936CDB**	**7936CDF**	**7936CDT**	90.6	24.6	190	—	240	244.5	2	1	9.36
	280	92	2.1	1.1	430	506	—	—	18.3	—	—	1 900	2 400	**7036DB**	**7036DF**	**7036DT**	178.8	86.8	192	—	268	273	2	1	20.2
	280	92	2.1	1.1	385	457	—	—	15.9	—	—	1 400	1 900	**7036BDB**	**7036BDF**	**7036BDT**	239.0	147.0	192	—	268	273	2	1	20.4
	280	92	2.1	1.1	473	553	—	—	20.7	—	15.7	2 600	3 500	**7036CDB**	**7036CDF**	**7036CDT**	107.6	15.6	192	—	268	273	2	1	19.9
	320	104	4	1.5	596	724	—	—	23.7	—	—	1 700	2 200	**7236DB**	**7236DF**	**7236DT**	196.3	92.3	198	—	302	311.5	3	1.5	31.4
	320	104	4	1.5	538	657	—	—	21.5	—	—	1 300	1 700	**7236BDB**	**7236BDF**	**7236BDT**	261.8	157.8	198	—	302	311.5	3	1.5	31.4
	320	104	4	1.5	650	786	—	—	25.7	—	14.9	2 400	3 200	**7236CDB**	**7236CDF**	**7236CDT**	119.0	15.0	198	—	302	311.5	3	1.5	31.4
	380	150	4	1.5	831	1 070	—	—	33.0	—	—	1 500	1 900	**7336DB**	**7336DF**	**7336DT**	236.7	86.7	198	—	362	371.5	3	1.5	80.0

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d (180) ~ 240 mm



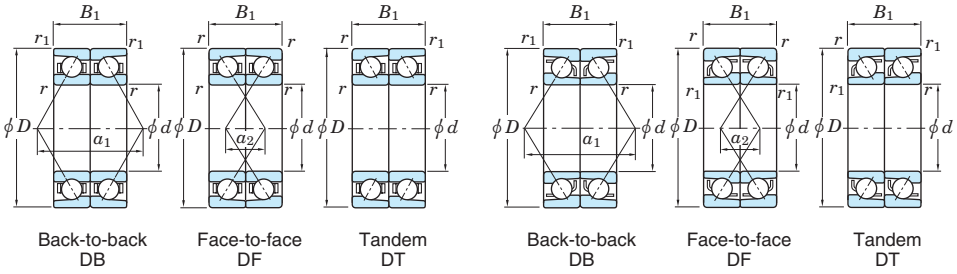


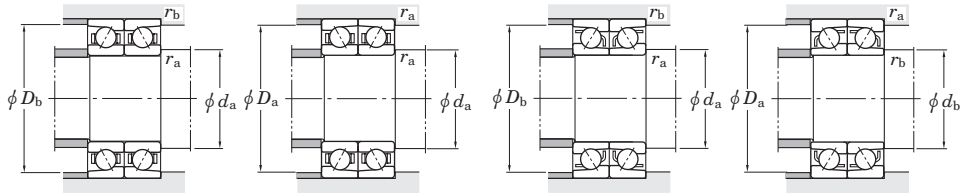
Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)						(Refer.) Mass (kg)
					With machined cages		With pressed cages																		
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
180	380	150	4	1.5	757	976	—	—	30.1	—	—	1 100	1 500	**7336BDB**	**7336BDF**	**7336BDT**	309.9	159.9	198	—	362	371.5	3	1.5	80.0
190	260	66	2	1	322	394	—	—	13.7	—	16.5	2 700	3 600	**7938CDB**	**7938CDF**	**7938CDT**	93.3	27.3	200	—	250	254.5	2	1	9.66
	290	92	2.1	1.1	441	535	—	—	18.7	—	—	1 800	2 300	**7038DB**	**7038DF**	**7038DT**	184.6	92.6	202	—	278	283	2	1	21.6
	290	92	2.1	1.1	395	483	—	—	16.3	—	—	1 400	1 800	**7038BDB**	**7038BDF**	**7038BDT**	247.4	155.4	202	—	278	283	2	1	21.6
	290	92	2.1	1.1	485	585	—	—	21.1	—	15.9	2 500	3 300	**7038CDB**	**7038CDF**	**7038CDT**	110.3	18.3	202	—	278	283	2	1	21.6
	340	110	4	1.5	616	779	—	—	24.7	—	—	1 600	2 000	**7238DB**	**7238DF**	**7238DT**	208.0	98.0	208	—	322	331.5	3	1.5	37.6
	340	110	4	1.5	555	706	—	—	22.4	—	—	1 200	1 600	**7238BDB**	**7238BDF**	**7238BDT**	277.4	167.4	208	—	322	331.5	3	1.5	37.6
	340	110	4	1.5	673	848	—	—	26.9	—	15.1	2 200	3 000	**7238CDB**	**7238CDF**	**7238CDT**	126.0	16.0	208	—	322	331.5	3	1.5	37.6
	400	156	5	2	914	1 200	—	—	36.0	—	—	1 400	1 800	**7338DB**	**7338DF**	**7338DT**	248.3	92.3	212	—	378	390	4	2	91.0
	400	156	5	2	835	1 100	—	—	33.0	—	—	1 100	1 400	**7338BDB**	**7338BDF**	**7338BDT**	325.5	169.5	212	—	378	390	4	2	91.0
200	280	76	2.1	1.1	415	509	—	—	17.4	—	16.3	2 500	3 300	**7940CDB**	**7940CDF**	**7940CDT**	102.3	26.3	212	—	268	273	2	1	13.7
	310	102	2.1	1.1	495	618	—	—	20.0	—	—	1 700	2 100	**7040DB**	**7040DF**	**7040DT**	198.3	96.3	212	—	298	303	2	1	25.4
	310	102	2.1	1.1	443	558	—	—	18.1	—	—	1 300	1 700	**7040BDB**	**7040BDF**	**7040BDT**	265.0	163.0	212	—	298	303	2	1	25.4
	310	102	2.1	1.1	544	676	—	—	21.9	—	15.7	2 300	3 100	**7040CDB**	**7040CDF**	**7040CDT**	119.3	17.3	212	—	298	303	2	1	25.4
	360	116	4	1.5	658	847	—	—	26.2	—	—	1 500	1 900	**7240DB**	**7240DF**	**7240DT**	219.7	103.7	218	—	342	351.5	3	1.5	44.8
	360	116	4	1.5	593	768	—	—	23.7	—	—	1 100	1 500	**7240BDB**	**7240BDF**	**7240BDT**	292.9	176.9	218	—	342	351.5	3	1.5	44.8
	360	116	4	1.5	718	921	—	—	28.4	—	15.1	2 100	2 800	**7240CDB**	**7240CDF**	**7240CDT**	133.0	17.0	218	—	342	351.5	3	1.5	44.8
	420	160	5	2	964	1 320	—	—	38.6	—	—	1 300	1 700	**7340DB**	**7340DF**	**7340DT**	259.0	99.0	222	—	398	410	4	2	104
	420	160	5	2	878	1 200	—	—	35.3	—	—	1 000	1 300	**7340BDB**	**7340BDF**	**7340BDT**	340.1	180.1	222	—	398	410	4	2	104
220	340	112	3	1.1	543	705	—	—	21.8	—	—	1 500	1 900	**7044DB**	**7044DF**	—	217.8	105.8	234	—	326	333	2.5	1	37.0
	340	112	3	1.1	486	636	—	—	19.6	—	—	1 100	1 500	**7044BDB**	**7044BDF**	—	290.9	178.9	234	—	326	333	2.5	1	37.8
240	360	112	3	1.1	591	751	—	—	24.6	—	—	1 400	1 700	**7048DB**	**7048DF**	—	229.2	117.2	254	—	346	353	2.5	1	39.4
	360	112	3	1.1	528	677	—	—	22.2	—	—	1 000	1 400	**7048BDB**	**7048BDF**	—	307.7	195.7	254	—	346	353	2.5	1	40.2
	440	144	4	1.5	819	1 190	—	—	33.4	—	—	1 200	1 500	**7248DB**	**7248DF**	—	268.3	124.3	258	—	422	431.5	3	1.5	104
	440	144	4	1.5	736	1 080	—	—	30.2	—	—	890	1 200	**7248BDB**	**7248BDF**	—	357.3	213.3	258	—	422	431.5	3	1.5	106

[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B, C or no indication after the bearing number indicates nominal contact angle of 40°, 15° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

d 260 ~ 380 mm



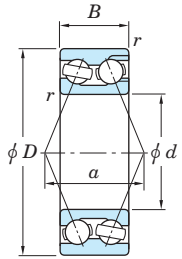


Boundary dimensions (mm)					Basic load ratings (kN)				Fatigue load limits (kN) C_u		Factor	Limiting speeds [1] (min^{-1})		Bearing No. [2]			Load center spread (mm)		Mounting dimensions (mm)						(Refer.) Mass (kg)
					With machined cages		With pressed cages																		
d	D	B_1	r min.	r_1 min.	C_r	C_{0r}	C_r	C_{0r}	With machined cages	With pressed cages	f_0	Grease lub.	Oil lub.	Back-to-back DB	Face-to-face DF	Tandem DT	a_1	a_2	d_a min.	d_b min.	D_a max.	D_b max.	r_a max.	r_b max.	
260	400	130	4	1.5	661	956	—	—	27.1	—	—	1 200	1 500	**7052DB**	**7052DF**	—	256.7	126.7	278	—	382	391.5	3	1.5	57.4
	400	130	4	1.5	592	862	—	—	24.4	—	—	910	1 200	**7052BDB**	**7052BDF**	—	341.9	211.9	278	—	382	391.5	3	1.5	58.6
280	420	130	4	1.5	675	1 010	—	—	27.9	—	—	1 100	1 400	**7056DB**	**7056DF**	—	267.1	137.1	298	—	402	411.5	3	1.5	60.8
	420	130	4	1.5	623	906	—	—	26.2	—	—	850	1 100	**7056BDB**	**7056BDF**	—	358.7	228.7	298	—	402	411.5	3	1.5	62.0
300	460	148	4	1.5	866	1 360	—	—	36.0	—	—	1 000	1 300	**7060DB**	**7060DF**	—	293.4	145.4	318	—	442	451.5	3	1.5	87.4
	460	148	4	1.5	776	1 230	—	—	32.5	—	—	770	1 000	**7060BDB**	**7060BDF**	—	392.9	244.9	318	—	442	451.5	3	1.5	89.8
320	480	148	4	1.5	887	1 440	—	—	37.3	—	—	950	1 200	**7064DB**	**7064DF**	—	304.9	156.9	338	—	462	471.5	3	1.5	92.0
	480	148	4	1.5	795	1 300	—	—	33.6	—	—	710	950	**7064BDB**	**7064BDF**	—	409.6	261.6	338	—	462	471.5	3	1.5	94.4
340	520	164	5	2	1 020	1 720	—	—	42.9	—	—	860	1 100	**7068DB**	**7068DF**	—	330.3	166.3	362	—	498	510	4	2	124
	520	164	5	2	914	1 550	—	—	38.7	—	—	640	860	**7068BDB**	**7068BDF**	—	442.8	278.8	362	—	498	510	4	2	127
360	540	164	5	2	1 050	1 830	—	—	44.5	—	—	800	1 000	**7072DB**	**7072DF**	—	341.8	177.8	382	—	518	530	4	2	129
	540	164	5	2	937	1 650	—	—	40.1	—	—	600	800	**7072BDB**	**7072BDF**	—	459.6	295.6	382	—	518	530	4	2	132
380	560	164	5	2	1 070	1 930	—	—	46.0	—	—	750	940	**7076DB**	**7076DF**	—	353.4	189.4	402	—	538	550	4	2	134
	560	164	5	2	959	1 740	—	—	41.5	—	—	560	750	**7076BDB**	**7076BDF**	—	476.4	312.4	402	—	538	550	4	2	138

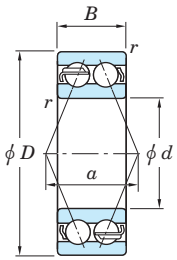
[Notes] 1) Limiting speeds shown above are applicable to machined cage bearings.
Limiting speeds of pressed cage bearings should be kept to under 80% of this value.
For bearings with 15° contact angle, this figure is applied to the high precision bearings ranked higher than class 5, used with machined cages or molded cages.

2) B or no indication after the bearing number indicates nominal contact angle of 40° and 30° respectively.
[Remark] Standard cage types used for the above bearings are described earlier in this section.

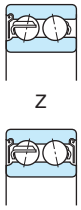
d 10 ~ (40) mm



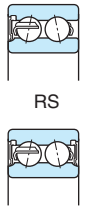
32, 33 series
(With filling slot)



Open

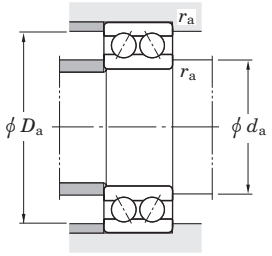


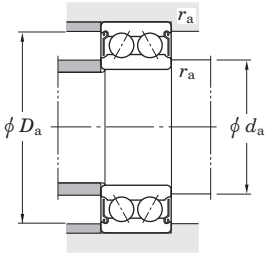
Shielded



Contact sealed

52, 53 series
(Without filling slot)



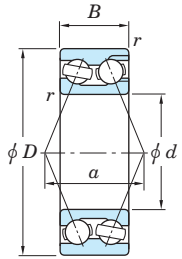


Boundary dimensions (mm)				Basic load ratings (kN) Open		Shielded/sealed		Fatigue load limits (kN) C_u		Limiting speeds (min^{-1}) Grease lub.		Oil lub.	Bearing No.			Load center spread (mm) Open	Mounting dimensions [1] (mm)				(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_r	C_{0r}	(Open)	(Shielded/sealed)	Open Z, ZZ	(RS, 2RS)	Open Z	Open	Shielded	Sealed	a	d_a min.	d_a max.	D_a max.	r_a max.	
10	30	14.3	0.6	9.15	5.35	—	—	0.280	—	15 000	—	20 000	**3200**	—	—	19.5	14.5	—	25.5	0.6	0.052
12	32	15.9	0.6	12.1	7.15	—	—	0.370	—	14 000	—	18 000	**3201**	—	—	21.7	16.5	—	27.5	0.6	0.063
15	35	15.9	0.6	12.1	7.45	—	—	0.390	—	12 000	—	16 000	**3202**	—	—	23.6	19.5	—	30.5	0.6	0.072
	42	19	1	19.0	11.9	—	—	0.610	—	10 000	—	14 000	**3302**	—	—	27.6	20.5	—	36.5	1	0.132
17	40	17.5	0.6	17.2	10.8	—	—	0.560	—	11 000	—	14 000	**3203**	—	—	26.6	21.5	—	35.5	0.6	0.100
	40	17.5	0.6	16.5	8.15	15.9	8.35	0.420	0.430	11 000	11 000	14 000	**5203**	**5203 ZZ**	**5203 2RS**	20.0	21.5	23.5	35.5	0.6	0.091
	47	22.2	1	23.0	17.1	—	—	0.760	—	9 400	—	13 000	**3303**	—	—	31.0	22.5	—	41.5	1	0.192
20	47	20.6	1	21.5	15.0	—	—	0.770	—	9 000	—	12 000	**3204**	—	—	31.5	25.5	—	41.5	1	0.170
	47	20.6	1	24.6	12.5	20.0	10.8	0.640	0.560	8 800	8 800	12 000	**5204**	**5204 ZZ**	**5204 2RS**	23.5	25.5	26.6	41.5	1	0.158
	52	22.2	1.1	26.0	18.4	—	—	0.950	—	8 200	—	11 000	**3304**	—	—	33.8	27	—	45	1	0.230
	52	22.2	1.1	30.9	15.0	24.7	12.8	0.780	0.660	8 300	8 300	11 000	**5304**	**5304 ZZ**	**5304 2RS**	25.9	27	28.3	45	1	0.230
25	52	20.6	1	23.7	18.2	—	—	0.940	—	7 800	—	10 000	**3205**	—	—	34.4	30.5	—	46.5	1	0.190
	52	20.6	1	26.7	14.8	23.6	13.8	0.760	0.710	7 700	7 700	10 000	**5205**	**5205 ZZ**	**5205 2RS**	26.1	30.5	32.3	46.5	1	0.190
	62	25.4	1.1	36.2	26.5	—	—	1.35	—	6 800	—	9 100	**3305**	—	—	40.5	32	—	55	1	0.369
	62	25.4	1.1	40.9	20.8	34.3	18.5	1.05	0.960	6 900	6 900	9 200	**5305**	**5305 ZZ**	**5305 2RS**	31.1	32	33.4	55	1	0.340
30	62	23.8	1	34.1	27.0	—	—	1.40	—	6 500	—	8 700	**3206**	—	—	40.7	35.5	—	56.5	1	0.320
	62	23.8	1	37.2	21.3	31.7	18.3	1.10	0.950	6 400	6 400	8 600	**5206**	**5206 ZZ**	**5206 2RS**	30.8	35.5	38.6	56.5	1	0.290
	72	30.2	1.1	47.7	36.1	—	—	1.85	—	5 800	—	7 800	**3306**	—	—	47.2	37	—	65	1	0.585
	72	30.2	1.1	51.2	28.5	42.9	25.2	1.45	1.30	5 800	5 800	7 700	**5306**	**5306 ZZ**	**5306 2RS**	36.2	37	41.3	65	1	0.510
35	72	27	1.1	46.0	37.5	—	—	1.95	—	5 600	—	7 500	**3207**	—	—	46.9	42	—	65	1	0.480
	72	27	1.1	49.0	29.0	39.7	24.6	1.50	1.25	5 500	5 500	7 300	**5207**	**5207 ZZ**	**5207 2RS**	36.1	42	43.9	65	1	0.430
	80	34.9	1.5	60.7	46.8	—	—	2.40	—	5 200	—	7 000	**3307**	—	—	53.4	43.5	—	71.5	1.5	0.816
	80	34.9	1.5	64.0	36.2	57.6	32.8	1.85	1.70	5 100	5 100	6 800	**5307**	**5307 ZZ**	**5307 2RS**	41.0	43.5	45.5	71.5	1.5	0.790
40	80	30.2	1.1	52.4	43.9	—	—	2.25	—	5 000	—	6 700	**3208**	—	—	52.6	47	—	73	1	0.650
	80	30.2	1.1	55.5	33.6	45.7	29.1	1.75	1.50	5 000	5 000	6 700	**5208**	**5208 ZZ**	**5208 2RS**	39.2	47	49.5	73	1	0.570

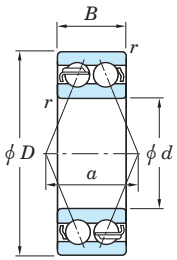
[Note] 1) The maximum value of d_a is applied to shielded and sealed type bearings.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

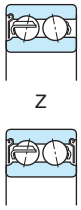
d (40) ~ 70 mm



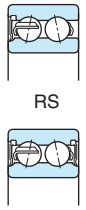
32, 33 series
(With filling slot)



Open

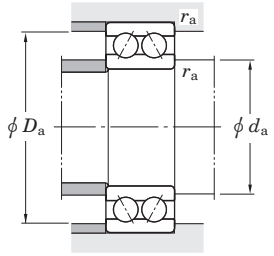


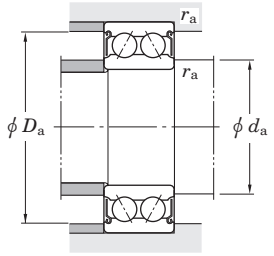
Shielded



Contact sealed

52, 53 series
(Without filling slot)



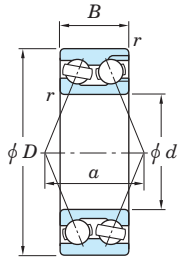


Boundary dimensions (mm)				Basic load ratings (kN) Open		Shielded/sealed		Fatigue load limits (kN) C_u		Limiting speeds (min^{-1}) Grease lub.		Oil lub.	Bearing No.			Load center spread (mm) Open	Mounting dimensions [1] (mm)				(Refer.) Mass
d	D	B	r min.	C_r	C_{0r}	C_r	C_{0r}	(Open)	(Shielded/sealed)	Open Z, ZZ	(RS, 2RS)	Open Z	Open	Shielded	Sealed	a	d_a min.	d_a max.	D_a max.	r_a max.	(kg)
40	90	36.5	1.5	67.6	53.8	—	—	2.80	—	4 600	—	6 100	**3308**	—	—	58.9	48.5	—	81.5	1.5	1.07
	90	36.5	1.5	78.3	45.4	64.3	37.8	2.35	1.95	4 600	4 600	6 100	**5308**	**5308 ZZ**	**5308 2RS**	44.9	48.5	52.1	81.5	1.5	1.05
45	85	30.2	1.1	56.8	51.4	—	—	2.65	—	4 600	—	6 100	**3209**	—	—	56.3	52	—	78	1	0.710
	85	30.2	1.1	62.3	38.4	52.1	33.9	2.00	1.75	4 600	4 600	6 100	**5209**	**5209 ZZ**	**5209 2RS**	42.2	52	55.3	78	1	0.620
	100	39.7	1.5	82.6	67.3	—	—	3.50	—	4 100	—	5 500	**3309**	—	—	65.6	53.5	—	91.5	1.5	1.42
	100	39.7	1.5	93.8	55.7	86.1	51.4	2.90	2.65	4 100	4 100	5 500	**5309**	**5309 ZZ**	**5309 2RS**	51.0	53.5	58.2	91.5	1.5	1.42
50	90	30.2	1.1	56.4	52.1	—	—	2.70	—	4 300	—	5 700	**3210**	—	—	58.8	57	—	83	1	0.760
	90	30.2	1.1	66.7	43.6	55.2	37.9	2.25	1.95	4 300	4 300	5 600	**5210**	**5210 ZZ**	**5210 2RS**	44.5	57	58.9	83	1	0.670
	110	44.4	2	108	88.6	—	—	4.60	—	3 800	—	5 000	**3310**	—	—	71.7	60	—	100	2	1.95
	110	44.4	2	111	67.0	102	62.2	3.45	3.20	3 600	3 600	4 800	**5310**	**5310 ZZ**	**5310 2RS**	56.6	60	64.4	100	2	1.93
55	100	33.3	1.5	63.6	60.2	—	—	3.10	—	3 900	—	5 100	**3211**	—	—	65.0	63.5	—	91.5	1.5	1.05
	100	33.3	1.5	82.3	55.2	66.1	44.7	2.85	2.30	3 800	3 800	5 100	**5211**	**5211 ZZ**	**5211 2RS**	50.2	63.5	66.2	91.5	1.5	0.960
	120	49.2	2	126	106	—	—	5.45	—	3 400	—	4 500	**3311**	—	—	79.3	65	—	110	2	2.53
	120	49.2	2	138	85.1	120	74.3	4.40	3.85	3 300	3 300	4 500	**5311**	**5311 ZZ**	**5311 2RS**	61.6	65	71.8	110	2	2.30
60	110	36.5	1.5	80.0	76.8	—	—	3.95	—	3 500	—	4 700	**3212**	—	—	71.3	68.5	—	101.5	1.5	1.40
	110	36.5	1.5	93.0	60.8	78.3	55.9	3.15	2.90	3 500	3 500	4 700	**5212**	**5212 ZZ**	**5212 2RS**	53.8	68.5	74.1	101.5	1.5	1.36
	130	54	2.1	156	132	—	—	6.85	—	3 100	—	4 200	**3312**	—	—	87.4	72	—	118	2	3.24
	130	54	2.1	157	98.7	138	87.1	5.10	4.50	3 100	3 100	4 100	**5312**	**5312 ZZ**	**5312 2RS**	67.2	72	79.2	118	2	3.16
65	120	38.1	1.5	95.5	97.4	—	—	5.05	—	3 200	—	4 300	**3213**	—	—	76.8	73.5	—	111.5	1.5	1.75
	120	38.1	1.5	109	75.3	86.5	63.1	3.90	3.25	3 200	3 200	4 300	**5213**	**5213 ZZ**	**5213 2RS**	58.8	73.5	79.0	111.5	1.5	1.66
	140	58.7	2.1	177	153	—	—	7.80	—	2 900	—	3 900	**3313**	—	—	92.7	77	—	128	2	4.08
	140	58.7	2.1	178	113	178	113	5.75	5.75	2 900	2 900	3 900	**5313**	**5313 ZZ**	**5313 2RS**	70.9	77	85.9	128	2	3.91
70	125	39.7	1.5	97.4	96.4	—	—	5.00	—	3 100	—	4 100	**3214**	—	—	80.7	78.5	—	116.5	1.5	1.92
	125	39.7	1.5	118	82.6	95.4	70.3	4.25	3.65	3 100	3 100	4 100	**5214**	**5214 ZZ**	**5214 2RS**	61.4	78.5	83.5	116.5	1.5	1.81
	150	63.5	2.1	188	160	—	—	7.90	—	2 700	—	3 600	**3314**	—	—	99.7	82	—	138	2	5.04
	150	63.5	2.1	200	129	200	129	6.35	6.35	2 700	2 700	3 600	**5314**	**5314 ZZ**	**5314 2RS**	76.0	82	92.9	138	2	4.89

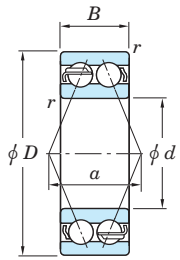
[Note] 1) The maximum value of d_a is applied to shielded and sealed type bearings.

[Remark] Standard cage types used for the above bearings are described earlier in this section.

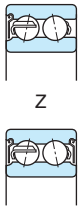
d 75 ~ 110 mm



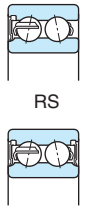
32, 33 series
(With filling slot)



Open

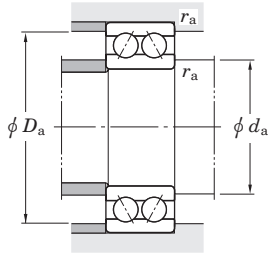


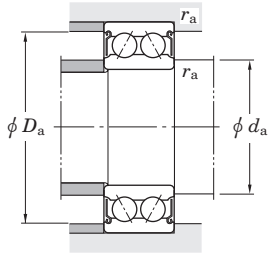
Shielded



Contact sealed

52, 53 series
(Without filling slot)





Boundary dimensions (mm)				Basic load ratings (kN) Open		Shielded/sealed		Fatigue load limits (kN) C_u		Limiting speeds (min^{-1}) Grease lub.		Oil lub.	Bearing No.			Load center spread (mm) Open	Mounting dimensions [1] (mm)				(Refer.) Mass (kg)
d	D	B	r min.	C_r	C_{0r}	C_r	C_{0r}	(Open)	(Shielded/sealed)	Open Z, ZZ	(RS, 2RS)	Open Z	Open	Shielded	Sealed	a	d_a min.	d_a max.	D_a max.	r_a max.	
75	130	41.3	1.5	116	120	—	—	6.15	—	2 900	—	3 900	**3215**	—	—	84.7	83.5	—	121.5	1.5	2.10
	160	68.3	2.1	211	189	—	—	9.00	—	2 500	—	3 300	**3315**	—	—	108.7	87	—	148	2	6.16
	160	68.3	2.1	218	147	218	147	6.95	6.95	2 500	2 500	3 300	**5315**	**5315 ZZ**	**5315 2RS**	81.5	87	99.6	148	2	5.97
80	140	44.4	2	122	121	—	—	5.95	—	2 700	—	3 600	**3216**	—	—	90.7	90	—	130	2	2.64
	170	68.3	2.1	230	213	—	—	9.85	—	2 400	—	3 100	**3316**	—	—	113.1	92	—	158	2	6.93
85	150	49.2	2	143	143	—	—	6.80	—	2 500	—	3 400	**3217**	—	—	98.4	95	—	140	2	3.39
	180	73	3	235	219	—	—	9.80	—	2 200	—	3 000	**3317**	—	—	118.8	99	—	166	2.5	8.30
90	160	52.4	2	165	167	—	—	7.70	—	2 400	—	3 100	**3218**	—	—	104.1	100	—	150	2	4.14
	190	73	3	256	242	—	—	10.6	—	2 100	—	2 800	**3318**	—	—	125.5	104	—	176	2.5	9.23
95	170	55.6	2.1	189	193	—	—	8.65	—	2 200	—	3 000	**3219**	—	—	110.6	107	—	158	2	5.00
	200	77.8	3	273	270	—	—	14.9	—	2 000	—	2 600	**3319**	—	—	132.2	109	—	186	2.5	10.9
100	180	60.3	2.1	215	221	—	—	9.65	—	2 100	—	2 800	**3220**	—	—	116.8	112	—	168	2	6.10
	215	82.6	3	312	324	—	—	17.4	—	1 800	—	2 500	**3320**	—	—	140.4	114	—	201	2.5	13.5
105	190	65.1	2.1	227	237	—	—	11.5	—	2 000	—	2 600	**3221**	—	—	124.2	117	—	178	2	7.37
	225	87.3	3	331	354	—	—	18.5	—	1 800	—	2 300	**3321**	—	—	148.1	119	—	211	2.5	15.6
110	200	69.8	2.1	251	263	—	—	10.9	—	1 900	—	2 500	**3222**	—	—	131.4	122	—	188	2	8.80
	240	92.1	3	352	388	—	—	15.1	—	1 600	—	2 200	**3322**	—	—	156.4	124	—	226	2.5	18.9

[Note] 1) The maximum value of d_a is applied to shielded and sealed type bearings.

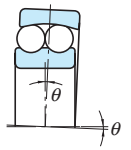
[Remark] Standard cage types used for the above bearings are described earlier in this section.

Self-aligning ball bearings

Self-aligning ball bearings have a spherical outer ring raceway, the center of whose curvature meets that of the bearing itself, so that the inner ring, balls and cage continue to rotate, aligning themselves if they have become misaligned within design limits.

This type of bearing is suitable when the displacement of the centers around which the shaft and housing rotate and shaft deflection are likely to occur.

Bearings with a tapered bore can easily be fit to the shaft with an adapter assembly.



θ : Allowable aligning angle

Self-aligning ball bearings



Cylindrical bore Tapered bore

Bore diameter **10 – 90** mm



Sealed type

Bore diameter **10 – 55** mm

Adapter assemblies



Bore diameter **17 – 80** mm

Boundary dimensions	The dimensions of standard series are as specified in JIS B 1512.
Tolerances	As specified in JIS B 1514-1, class 0. (refer to Table 7-3 on pp. A 60 – A 63.)
Radial internal clearance	As specified in JIS B 1520. (refer to Table 10-6 on p. A 105.)
Recommended fits	Refer to Table 9-4 on pp. A 91, 92.
Standard cages	• Staggered type pressed steel cage (application : all dimensional range of 12, 13, 22...2RS and 23...2RS series) • Snap type pressed steel cage (application : all dimensional range of 22 series and those of No. 2300 thru 2316.)
Allowable aligning angle	· 12 and 22 series.....0.044 rad (2.5°) · 13 and 23 series.....0.052 rad (3°) · 22...2RS and 23...2RS series.....0.026 rad (1.5°)

Dynamic equivalent radial load

$P_r = XF_r + YF_a$

$F_a/F_r \leqq e$		$F_a/F_r > e$	
X	Y	X	Y
1	Y_1	0.65	Y_2

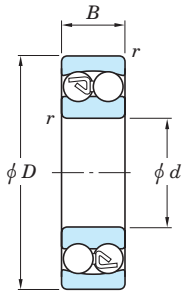
Static equivalent radial load

$P_{0r} = F_r + Y_0F_a$

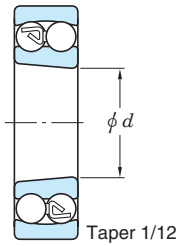
Refer to the bearing specification table for values of e, Y_1, Y_2 and Y_0.

open type

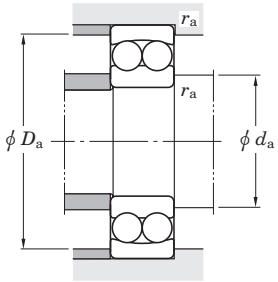
d 10 ~ (35) mm



Cylindrical bore



Tapered bore

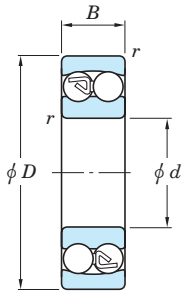


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)			Constant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	D_a max.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
10	30	9	0.6	6.85	1.20	0.08	23 000	28 000	**1200**	—	14	26	0.6	0.33	1.92	2.97	2.01	0.034	—
	30	14	0.6	9.20	1.60	0.10	23 000	29 000	**2200**	—	14	26	0.6	0.59	1.07	1.65	1.12	0.047	—
12	32	10	0.6	7.00	1.25	0.08	21 000	26 000	**1201**	—	16	28	0.6	0.33	1.89	2.93	1.98	0.040	—
	32	14	0.6	9.55	1.75	0.11	21 000	26 000	**2201**	—	16	28	0.6	0.53	1.18	1.83	1.24	0.053	—
15	35	11	0.6	9.35	1.75	0.11	18 000	22 000	**1202**	—	19	31	0.6	0.33	1.90	2.95	2.00	0.049	—
	35	14	0.6	9.60	1.85	0.12	18 000	22 000	**2202**	—	19	31	0.6	0.50	1.27	1.97	1.33	0.060	—
	42	13	1	12.0	2.30	0.14	16 000	20 000	**1302**	—	20	37	1	0.34	1.86	2.88	1.95	0.094	—
	42	17	1	15.1	2.90	0.18	14 000	20 000	**2302**	—	20	37	1	0.50	1.27	1.96	1.33	0.114	—
17	40	12	0.6	9.85	2.05	0.13	16 000	20 000	**1203**	—	21	36	0.6	0.31	2.03	3.14	2.12	0.073	—
	40	16	0.6	12.3	2.40	0.15	16 000	20 000	**2203**	—	21	36	0.6	0.50	1.27	1.96	1.33	0.088	—
	47	14	1	15.7	3.20	0.20	14 000	17 000	**1303**	—	22	42	1	0.33	1.92	2.97	2.01	0.130	—
	47	19	1	18.1	3.60	0.23	13 000	18 000	**2303**	—	22	42	1	0.49	1.28	1.98	1.34	0.158	—
20	47	14	1	12.4	2.65	0.16	14 000	17 000	**1204**	**1204K**	25	42	1	0.29	2.16	3.35	2.27	0.120	0.118
	47	18	1	15.8	3.25	0.21	14 000	17 000	**2204**	**2204K**	25	42	1	0.48	1.31	2.02	1.37	0.140	0.136
	52	15	1.1	15.5	3.35	0.21	13 000	15 000	**1304**	**1304K**	26.5	45.5	1	0.30	2.12	3.28	2.22	0.163	0.161
	52	21	1.1	22.5	4.65	0.30	11 000	15 000	**2304**	**2304K**	26.5	45.5	1	0.49	1.29	2.00	1.35	0.209	0.205
25	52	15	1	15.1	3.30	0.21	12 000	14 000	**1205**	**1205K**	30	47	1	0.28	2.28	3.52	2.39	0.141	0.138
	52	18	1	15.7	3.50	0.22	12 000	15 000	**2205**	**2205K**	30	47	1	0.40	1.58	2.45	1.66	0.163	0.158
	62	17	1.1	22.5	5.05	0.32	9 900	12 000	**1305**	**1305K**	31.5	55.5	1	0.27	2.31	3.57	2.42	0.257	0.252
	62	24	1.1	30.6	6.55	0.42	9 400	13 000	**2305**	**2305K**	31.5	55.5	1	0.46	1.36	2.10	1.42	0.335	0.327
30	62	16	1	19.6	4.70	0.29	9 900	12 000	**1206**	**1206K**	35	57	1	0.25	2.55	3.94	2.67	0.220	0.216
	62	20	1	19.3	4.65	0.29	10 000	12 000	**2206**	**2206K**	35	57	1	0.35	1.79	2.77	1.87	0.260	0.254
	72	19	1.1	26.6	6.30	0.40	8 700	11 000	**1306**	**1306K**	36.5	65.5	1	0.26	2.40	3.72	2.52	0.387	0.381
	72	27	1.1	39.3	8.70	0.55	8 000	11 000	**2306**	**2306K**	36.5	65.5	1	0.44	1.44	2.23	1.51	0.500	0.489
35	72	17	1.1	19.7	5.15	0.32	8 500	10 000	**1207**	**1207K**	41.5	65.5	1	0.23	2.71	4.20	2.84	0.323	0.317

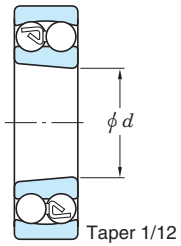
[Remark] Standard cage types used for the above bearings are described earlier in this section.

open type

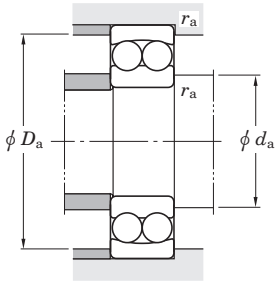
d (35) ~ 65 mm



Cylindrical bore



Tapered bore

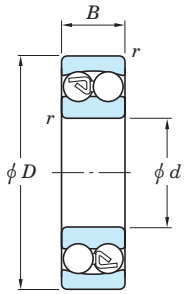


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)			Con-stant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	D_a max.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
35	72	23	1.1	27.1	6.65	0.42	8 500	10 000	**2207**	**2207K**	41.5	65.5	1	0.37	1.71	2.65	1.79	0.403	0.396
	80	21	1.5	31.4	7.95	0.49	7 600	9 300	**1307**	**1307K**	43	72	1.5	0.25	2.48	3.84	2.60	0.510	0.502
	80	31	1.5	49.4	11.1	0.71	7 100	9 800	**2307**	**2307K**	43	72	1.5	0.45	1.39	2.15	1.46	0.675	0.657
40	80	18	1.1	24.1	6.50	0.41	7 500	9 200	**1208**	**1208K**	46.5	73.5	1	0.22	2.83	4.38	2.97	0.417	0.411
	80	23	1.1	28.0	7.35	0.46	7 600	9 300	**2208**	**2208K**	46.5	73.5	1	0.33	1.92	2.96	2.01	0.505	0.494
	90	23	1.5	36.9	9.80	0.61	6 900	8 400	**1308**	**1308K**	48	82	1.5	0.25	2.57	3.98	2.69	0.715	0.704
	90	33	1.5	56.1	13.4	0.85	6 200	8 600	**2308**	**2308K**	48	82	1.5	0.43	1.47	2.27	1.54	0.925	0.903
45	85	19	1.1	27.2	7.35	0.46	7 000	8 500	**1209**	**1209K**	51.5	78.5	1	0.21	2.94	4.56	3.09	0.465	0.459
	85	23	1.1	29.1	8.15	0.51	7 000	8 500	**2209**	**2209K**	51.5	78.5	1	0.30	2.09	3.23	2.19	0.545	0.533
	100	25	1.5	47.6	12.9	0.80	6 100	7 500	**1309**	**1309K**	53	92	1.5	0.25	2.56	3.95	2.68	0.957	0.942
	100	36	1.5	68.0	16.6	1.05	5 600	7 700	**2309**	**2309K**	53	92	1.5	0.42	1.51	2.33	1.58	1.23	1.20
50	90	20	1.1	28.4	8.10	0.51	6 500	7 900	**1210**	**1210K**	56.5	83.5	1	0.21	3.07	4.76	3.22	0.525	0.515
	90	23	1.1	29.2	8.50	0.53	6 500	7 900	**2210**	**2210K**	56.5	83.5	1	0.27	2.33	3.61	2.45	0.590	0.577
	110	27	2	54.3	14.2	0.89	5 600	6 800	**1310**	**1310K**	59	101	2	0.23	2.70	4.17	2.83	1.21	1.19
	110	40	2	80.7	20.1	1.25	5 100	7 000	**2310**	**2310K**	59	101	2	0.40	1.56	2.41	1.63	1.64	1.60
55	100	21	1.5	33.5	10.0	0.63	5 800	7 100	**1211**	**1211K**	63	92	1.5	0.20	3.19	4.94	3.34	0.705	0.693
	100	25	1.5	33.2	9.95	0.62	5 800	7 100	**2211**	**2211K**	63	92	1.5	0.27	2.35	3.64	2.47	0.810	0.792
	120	29	2	64.2	18.1	1.10	5 000	6 200	**1311**	**1311K**	64	111	2	0.23	2.70	4.18	2.83	1.58	1.56
	120	43	2	94.2	23.8	1.50	4 600	6 400	**2311**	**2311K**	64	111	2	0.41	1.53	2.37	1.60	2.10	2.05
60	110	22	1.5	37.7	11.6	0.73	5 200	6 400	**1212**	**1212K**	68	102	1.5	0.19	3.37	5.22	3.53	0.900	0.885
	110	28	1.5	42.7	12.5	0.80	5 300	6 500	**2212**	**2212K**	68	102	1.5	0.28	2.26	3.49	2.36	1.09	1.07
	130	31	2.1	71.4	20.8	1.30	4 500	5 500	**1312**	**1312K**	71	119	2	0.22	2.91	4.50	3.05	1.96	1.93
	130	46	2.1	109	28.1	1.80	4 200	5 800	**2312**	**2312K**	71	119	2	0.39	1.62	2.51	1.70	2.60	2.53
65	120	23	1.5	38.7	12.4	0.79	4 800	5 800	**1213**	**1213K**	73	112	1.5	0.17	3.67	5.68	3.84	1.15	1.13
	120	31	1.5	54.5	16.4	1.05	4 900	5 900	**2213**	**2213K**	73	112	1.5	0.28	2.24	3.47	2.35	1.46	1.43
	140	33	2.1	77.2	22.9	1.40	4 300	5 200	**1313**	**1313K**	76	129	2	0.23	2.73	4.23	2.86	2.45	2.41

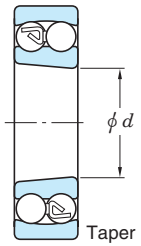
[Remark] Standard cage types used for the above bearings are described earlier in this section.

open type

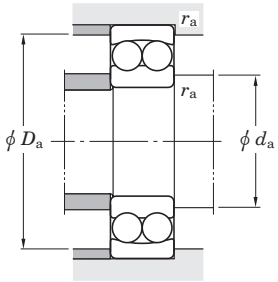
d 70 ~ 90 mm



Cylindrical bore



Tapered bore

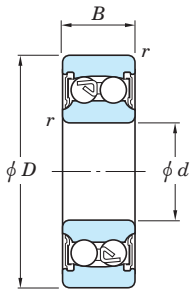


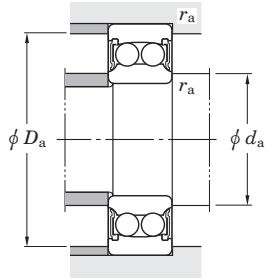
Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.			Mounting dimensions (mm)			Con-stant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore		d_a min.	D_a max.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
70	125	24	1.5	43.3	13.7	0.87	4 600	5 700	**1214**	—		78	117	1.5	0.18	3.48	5.38	3.64	1.26	—
	150	35	2.1	92.5	27.6	1.65	4 000	4 900	**1314**	—		81	139	2	0.22	2.84	4.40	2.98	2.99	—
75	130	25	1.5	48.5	15.5	0.97	4 300	5 300	**1215**	**1215K**		83	122	1.5	0.17	3.60	5.58	3.77	1.36	1.34
	160	37	2.1	98.6	29.9	1.70	4 000	4 900	**1315**	**1315K**		86	149	2	0.23	2.80	4.33	2.93	3.56	3.51
80	140	26	2	49.6	16.9	1.00	4 000	4 900	**1216**	**1216K**		89	131	2	0.16	3.90	6.03	4.08	1.67	1.64
	170	39	2.1	110	32.9	1.85	3 500	4 300	**1316**	**1316K**		91	159	2	0.22	2.90	4.49	3.04	4.18	4.12
85	150	28	2	61.5	20.5	1.20	3 800	4 600	**1217**	**1217K**		94	141	2	0.17	3.61	5.59	3.78	2.07	2.04
	180	41	3	122	37.8	2.05	3 300	4 000	**1317**	**1317K**		98	167	2.5	0.22	2.93	4.53	3.07	4.98	4.91
90	160	30	2	71.0	23.4	1.30	3 500	4 300	**1218**	**1218K**		99	151	2	0.17	3.69	5.70	3.86	2.52	2.48
	190	43	3	145	44.7	2.35	3 100	3 800	**1318**	**1318K**		103	177	2.5	0.22	2.81	4.35	2.94	5.80	5.71

[Remark] Standard cage types used for the above bearings are described earlier in this section.

sealed type

d 10 ~ 55 mm

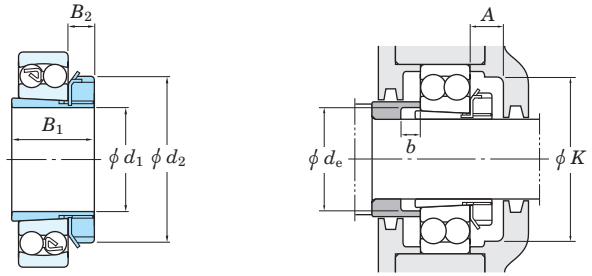




Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speed (min^{-1})	Bearing No.		Mounting dimensions (mm)				Con-stant	Axial load factors			(Refer.) Mass
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.			d_a min.	d_a max.	D_a max.	r_a max.	e	Y_1	Y_2	Y_0	(kg)
10	30	14	0.6	6.85	1.20	0.08	15 000	**2200 2RS**		13.7	13.7	25	0.6	0.33	1.92	2.97	2.01	0.047
12	32	14	0.6	7.00	1.25	0.08	14 000	**2201 2RS**		15.2	15.2	27	0.6	0.33	1.89	2.93	1.98	0.053
15	35	14	0.6	9.35	1.75	0.11	12 000	**2202 2RS**		18.0	18.0	30	0.6	0.33	1.90	2.95	2.00	0.060
	42	17	1	12.0	2.30	0.14	11 000	**2302 2RS**		20.0	20.0	36	1	0.34	1.86	2.88	1.95	0.114
17	40	16	0.6	9.85	2.05	0.13	11 000	**2203 2RS**		20.2	20.2	35	0.6	0.31	2.03	3.14	2.12	0.088
	47	19	1	15.7	3.20	0.20	9 400	**2303 2RS**		22.1	22.1	41	1	0.33	1.92	2.97	2.01	0.158
20	47	18	1	12.4	2.65	0.16	9 100	**2204 2RS**		24.1	24.1	41	1	0.29	2.16	3.35	2.27	0.140
	52	21	1.1	15.5	3.35	0.21	8 300	**2304 2RS**		26.2	26.2	45	1	0.30	2.12	3.28	2.22	0.209
25	52	18	1	15.1	3.30	0.21	7 900	**2205 2RS**		29.4	29.4	46	1	0.28	2.28	3.52	2.39	0.163
	62	24	1.1	22.5	5.05	0.32	6 600	**2305 2RS**		32	33.9	55	1	0.27	2.31	3.57	2.42	0.335
30	62	20	1	19.6	4.70	0.29	6 600	**2206 2RS**		35.5	35.5	56	1	0.25	2.55	3.94	2.67	0.260
	72	27	1.1	26.6	6.30	0.40	5 800	**2306 2RS**		37	37.8	65	1	0.26	2.40	3.72	2.52	0.500
35	72	23	1.1	19.7	5.15	0.32	5 700	**2207 2RS**		40.9	40.9	65	1	0.23	2.71	4.20	2.84	0.403
	80	31	1.5	31.4	7.95	0.49	5 100	**2307 2RS**		43.5	45.0	71.5	1.5	0.25	2.48	3.84	2.60	0.675
40	80	23	1.1	24.1	6.50	0.41	5 000	**2208 2RS**		47	48.1	73	1	0.22	2.83	4.38	2.97	0.505
	90	33	1.5	36.9	9.80	0.61	4 600	**2308 2RS**		48.5	49.6	81.5	1.5	0.25	2.57	3.98	2.69	0.925
45	85	23	1.1	27.2	7.35	0.46	4 600	**2209 2RS**		52	52.4	78	1	0.21	2.94	4.56	3.09	0.545
	100	36	1.5	47.6	12.9	0.80	4 100	**2309 2RS**		53.5	56.6	91.5	1.5	0.25	2.56	3.95	2.68	1.23
50	90	23	1.1	28.4	8.10	0.51	4 300	**2210 2RS**		56.5	56.5	83	1	0.21	3.07	4.76	3.22	0.590
	110	40	2	54.3	14.2	0.89	3 700	**2310 2RS**		60	62.5	100	2	0.23	2.70	4.17	2.83	1.64
55	100	25	1.5	33.5	10.0	0.63	3 900	**2211 2RS**		63.5	63.5	91.5	1.5	0.20	3.19	4.94	3.34	0.810

[Remark] Standard cage types used for the above bearings are described earlier in this section.

d_1 17 ~ (45) mm



Boundary dimensions (mm)				Brg. bore	Designations	Mounting dimensions (mm)				Mass	(Refer.)	
d_1	B_1	d_2	B_2	d (mm)	Bearing + **adapter ass'y**	A min.	K min.	d_e min.	b min.	Brg.+adapter ass'y (kg)	Adapter sleeve No.	Locknut No.
17	24	32	7	20	1204K+**H204X**	—	—	23	5	0.162	A204X	AN04
	28	32	7	20	2204K+**H304X**	—	—	24	5	0.185	A304X	AN04
	28	32	7	20	1304K+**H304X**	—	—	24	8	0.210	A304X	AN04
	31	32	7	20	2304K+**H2304X**	—	—	24	5	0.257	A2304X	AN04
20	26	38	8	25	1205K+**H205X**	15	45	28	5	0.218	A205X	AN05
	29	38	8	25	2205K+**H305X**	15	45	29	5	0.243	A305X	AN05
	29	38	8	25	1305K+**H305X**	15	45	29	6	0.337	A305X	AN05
	35	38	8	25	2305K+**H2305X**	15	45	29	5	0.424	A2305X	AN05
25	27	45	8	30	1206K+**H206X**	15	50	33	5	0.320	A206X	AN06
	31	45	8	30	2206K+**H306X**	15	50	34	5	0.368	A306X	AN06
	31	45	8	30	1306K+**H306X**	15	50	34	6	0.495	A306X	AN06
	38	45	8	30	2306K+**H2306X**	15	50	35	5	0.620	A2306X	AN06
30	29	52	9	35	1207K+**H207X**	17	58	38	5	0.462	A207X	AN07
	35	52	9	35	2207K+**H307X**	17	58	39	5	0.557	A307X	AN07
	35	52	9	35	1307K+**H307X**	17	58	39	7	0.663	A307X	AN07
	43	52	9	35	2307K+**H2307X**	17	58	40	5	0.843	A2307X	AN07
35	31	58	10	40	1208K+**H208X**	17	65	44	5	0.597	A208X	AN08
	36	58	10	40	2208K+**H308X**	17	65	44	5	0.696	A308X	AN08
	36	58	10	40	1308K+**H308X**	17	65	44	5	0.906	A308X	AN08
	46	58	10	40	2308K+**H2308X**	17	65	45	5	1.14	A2308X	AN08
40	33	65	11	45	1209K+**H209X**	17	72	49	5	0.701	A209X	AN09
	39	65	11	45	2209K+**H309X**	17	72	49	8	0.798	A309X	AN09
	39	65	11	45	1309K+**H309X**	17	72	49	5	1.21	A309X	AN09
	50	65	11	45	2309K+**H2309X**	17	72	50	5	1.51	A2309X	AN09
45	35	70	12	50	1210K+**H210X**	19	76	53	5	0.804	A210X	AN10
	42	70	12	50	2210K+**H310X**	19	76	54	10	0.896	A310X	AN10

d_1 (45) ~ 80 mm

Boundary dimensions (mm)				Brg. bore	Designations	Mounting dimensions (mm)				Mass	(Refer.)	
d_1	B_1	d_2	B_2	d (mm)	Bearing + **adapter ass'y**	A min.	K min.	d_e min.	b min.	Brg.+adapter ass'y (kg)	Adapter sleeve No.	Locknut No.
45	42	70	12	50	1310K+**H310X**	19	76	54	5	1.51	A310X	AN10
	55	70	12	50	2310K+**H2310X**	19	76	56	5	1.98	A2310X	AN10
50	37	75	12	55	1211K+**H211X**	19	85	60	6	1.02	A211X	AN11
	45	75	12	55	2211K+**H311X**	19	85	60	11	1.16	A311X	AN11
	45	75	12	55	1311K+**H311X**	19	85	60	6	1.93	A311X	AN11
	59	75	12	55	2311K+**H2311X**	19	85	61	6	2.50	A2311X	AN11
55	38	80	13	60	1212K+**H212X**	20	90	61	5	1.25	A212X	AN12
	47	80	13	60	2212K+**H312X**	20	90	65	9	1.49	A312X	AN12
	47	80	13	60	1312K+**H312X**	20	90	65	5	2.35	A312X	AN12
	62	80	13	60	2312K+**H2312X**	20	90	66	5	3.04	A2312X	AN12
60	40	85	14	65	1213K+**H213X**	21	96	70	5	1.56	A213X	AN13
	50	85	14	65	2213K+**H313X**	21	96	70	8	1.92	A313X	AN13
	50	85	14	65	1313K+**H313X**	21	96	70	5	2.90	A313X	AN13
65	43	98	15	75	1215K+**H215X**	23	110	80	5	2.09	A215X	AN15
	55	98	15	75	1315K+**H315X**	23	110	80	5	4.40	A315X	AN15
70	46	105	17	80	1216K+**H216X**	25	120	85	5	2.57	A216X	AN16
	59	105	17	80	1316K+**H316X**	25	120	86	5	5.21	A316X	AN16
75	50	110	18	85	1217K+**H217X**	27	128	90	6	3.11	A217X	AN17
	63	110	18	85	1317K+**H317X**	27	128	91	6	6.15	A317X	AN17
80	52	120	18	90	1218K+**H218X**	28	139	95	6	3.75	A218X	AN18
	65	120	18	90	1318K+**H318X**	28	139	96	6	7.16	A318X	AN18

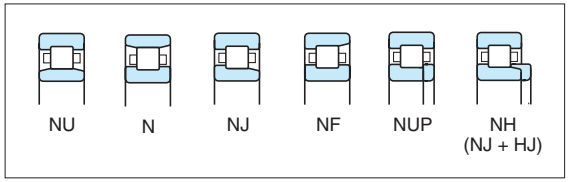
Cylindrical roller bearings

Cylindrical roller bearings feature high radial load capacity because the rollers and raceway are in linear contact. These bearings are suitable for applications that involve heavy radial and impact loading.

They are also appropriate for high-speed applications in that they can be machined very accurately due to their structure.

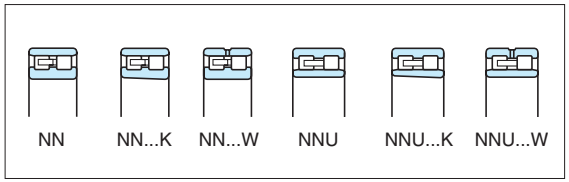
Having a separable inner ring or outer ring, these bearings can be mounted and dismounted easily.

Single-row cylindrical roller bearings



- The NU and N types exhibit their best performance when used as free side bearings since they adjust to the shaft's axial movement, to a certain extent, relative to the housing position.
- The NJ and NF types carry axial load in one direction, while the NUP and NH types can carry a certain degree of axial load in both directions.
- Type R cylindrical roller bearings feature enhanced load rating compared with standard series, though both have equal dimensions.
  This is because type R bearings have different internal design. They are identified by supplementary code "R".

Double-row cylindrical roller bearings



- Double-row cylindrical roller bearings come in two types : with a cylindrical bore, and with a tapered bore. As for those with a tapered bore, the specified amount of clearance can be obtained by adjusting the press-in distance. Some bearings have lubrication holes and lubrication grooves on the outer ring. They are identified by supplementary code "W".
- These bearings can accommodate high radial loads, and are often used on machine tool spindles.

Single-row cylindrical roller bearings



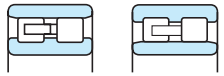
Bore diameter **20** – **460** mm



Thrust collar

Bore diameter **20** – **320** mm

Double-row cylindrical roller bearings



Bore diameter **25** – **480** mm

Boundary dimensions

The dimensions of standard series are as specified in JIS B 1512.

Tolerances

As specified in JIS B 1514-1 (refer to Table 7-3 on pp. A 60 – A 63).

Tolerances of roller set bore diameter F_w and roller set outside diameter E_w of interchangeable bearings are as follows :

Unit : μm

Nominal bore diameter d (mm)		Roller set bore diameter deviation Δ_{Fw}		Roller set outside diameter deviation Δ_{Ew}	
over	up to	upper	lower	upper	lower
–	**20**	+ 10	0	0	– 10
20	**50**	+ 15	0	0	– 15
50	**120**	+ 20	0	0	– 20
120	**200**	+ 25	0	0	– 25
200	**250**	+ 30	0	0	– 30
250	**315**	+ 35	0	0	– 35
315	**400**	+ 40	0	0	– 40
400	**500**	+ 45	0	–	–

[Remark] Interchangeable bearings have an inner ring with rollers that can be matched with the outer ring, or an outer ring with rollers that can be matched with the inner ring, without affecting performance in the bearing that has the same bearing number in one category.

Tapered bore tolerance and allowable values of high precision double-row cylindrical roller bearings (classes 5 and 4) are provided in JTEKT standards (refer to Table 7-11 on p. A 76).

Radial internal clearance

· Cylindrical bore and tapered bore bearings
.....(refer to Table 10-8 on pp. A 106, 107.)
· Motor bearings.....(refer to Table 10-7 on p. A 105.)

Recommended fits

Refer to Table 9-4 on pp. A 91, 92.

Standard cages

■ For single-row cylindrical roller bearings :
· Pressed cage (supplementary code : //)
· Synthetic resin molded cage (supplementary code : FG)
· Copper alloy machined cage (supplementary code : FY)
→ For application range, refer to Table 1.

Copper alloy machined cages without rivets (LY) are also used for some special purposes.

■ For double-row cylindrical roller bearings :
· Prong type copper alloy machined cage (supplementary code : FY)
· Separable prong type copper alloy machined cage (supplementary code : FW)
.....for class 5 or higher precision bearings

Allowable misalignment

Allowable misalignment of single-row cylindrical roller bearings depends on bearing type and specification.
General values are as follows :
1) When P_r / C_r is approx. 8% under load of normal use
.....0.000 6 rad (2') – 0.000 9 rad (3')
2) When P_r / C_r is approx. 5% under load lighter than 1)
.....0.001 2 rad (4')
When very large allowable misalignment is required, consult with JTEKT.

Equivalent radial load

Dynamic equivalent radial load $P_r = F_r$
Static equivalent radial load $P_{0r} = F_r$

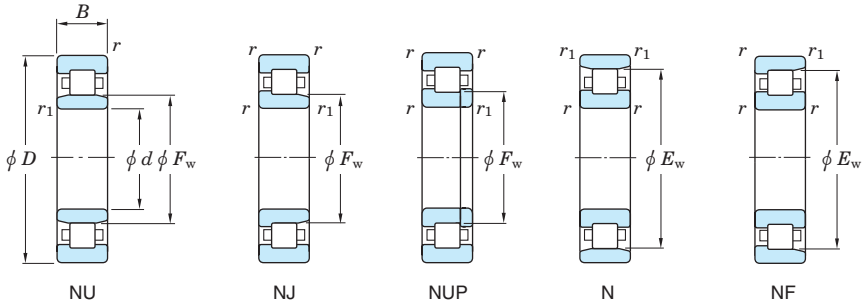
Allowable axial load

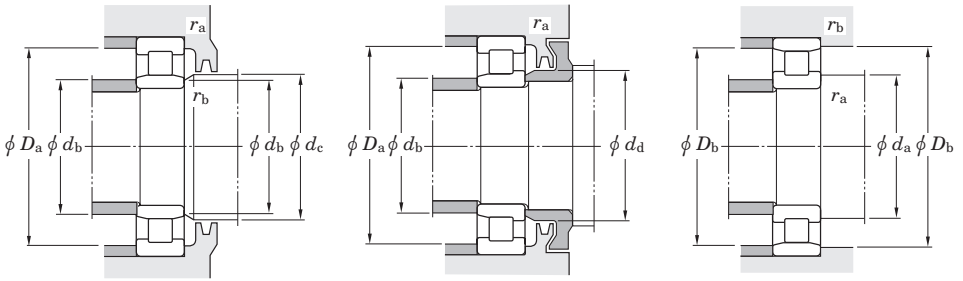
Cylindrical roller bearings with ribs, including loose rib and thrust collar, on both inner and outer rings accommodate axial load to a certain extent.
(NJ and NF types accommodate load applied in one direction : NUP and NH in both directions.)
For calculation of allowable axial load, refer to p. A 44.

Table 1 Application of standard cages

Bearing series	Pressed cage	Synthetic resin molded cage	Machined cage
NU, NUP 10	—	—	1005 – 1092
N, NF 2	204 – 220	—	204 – 264
NU, NJ, NUP 2	—	—	244 – 264
NU, NJ, NUP 2 R	—	204R – 213R	214R – 240R
NU, NJ, NUP 22	2204 – 2220	—	2204 – 2252
NU, NJ, NUP 22 R	2204R – 2220R	—	2204R – 2240R
NU 32	—	—	3206 – 3252
N, NF 3	304 – 320	—	304 – 348
NU, NJ, NUP 3	—	—	334 – 348
NU, NJ, NUP 3 R	—	304R – 314R	315R – 332R
NU, NJ, NUP 23	2304 – 2320	—	2304 – 2340
NU, NJ, NUP 23 R	2304R – 2320R	—	2304R – 2332R
NU 33	—	—	3306 – 3352
NU, NJ, NUP, NF 4	406 – 420	—	406 – 430

d 20 ~ (30) mm



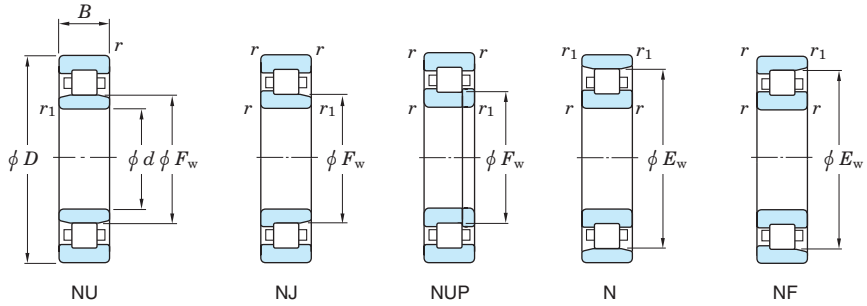


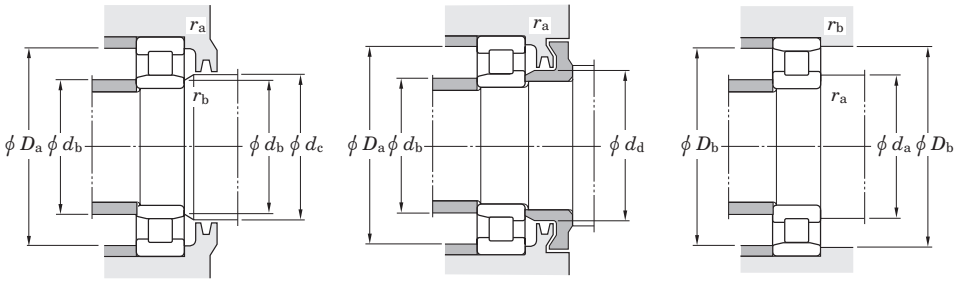
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
20	47	14	1	0.6	—	40	19.3	12.7	1.45	15 000	18 000	—	—	—	**N204**	**NF204**	25	—	—	—	32	42	43	42	1	0.6	(0.108)
	47	14	1	0.6	26.5	—	32.2	22.6	3.05	15 000	18 000	**NU204R**	**NJ204R**	**NUP204R**	—	—	25	24	26	29	32	42	—	—	1	0.6	0.112
	47	18	1	0.6	27	—	27.8	18.4	2.70	13 000	18 000	**NU2204**	**NJ2204**	**NUP2204**	—	—	25	24	26	29	32	42	—	—	1	0.6	0.146
	47	18	1	0.6	26.5	—	38.3	28.3	3.60	13 000	18 000	**NU2204R**	**NJ2204R**	**NUP2204R**	—	—	25	24	26	29	32	42	—	—	1	0.6	0.146
	52	15	1.1	0.6	—	44.5	28.9	19.2	2.50	12 000	16 000	—	—	—	**N304**	**NF304**	26.5	—	—	—	33	45.5	48	45.5	1	0.6	(0.147)
	52	15	1.1	0.6	27.5	—	39.4	26.9	3.75	12 000	16 000	**NU304R**	**NJ304R**	**NUP304R**	—	—	26.5	24	27	30	33	45.5	—	—	1	0.6	0.153
	52	21	1.1	0.6	28.5	—	38.0	30.2	3.60	11 000	16 000	**NU2304**	**NJ2304**	**NUP2304**	—	—	26.5	24	27	30	33	45.5	—	—	1	0.6	0.212
	52	21	1.1	0.6	27.5	—	52.5	38.8	5.40	11 000	16 000	**NU2304R**	**NJ2304R**	**NUP2304R**	—	—	26.5	24	27	30	33	45.5	—	—	1	1	0.215
25	47	12	0.6	0.3	30.5	—	17.8	13.1	2.25	15 000	18 000	**NU1005**	—	**NUP1005**	—	—	29	27	30	32	—	43	—	—	0.6	0.3	0.084
	52	15	1	0.6	—	45	22.1	15.7	1.80	13 000	16 000	—	—	—	**N205**	**NF205**	30	—	—	—	37	47	48	47	1	0.6	(0.132)
	52	15	1	0.6	31.5	—	36.7	27.7	3.75	13 000	15 000	**NU205R**	**NJ205R**	**NUP205R**	—	—	30	29	31	34	37	47	—	—	1	0.6	0.138
	52	18	1	0.6	32	—	29.6	22.8	3.05	12 000	16 000	**NU2205**	**NJ2205**	**NUP2205**	—	—	30	29	31	34	37	47	—	—	1	0.6	0.163
	52	18	1	0.6	31.5	—	43.6	34.6	4.40	12 000	15 000	**NU2205R**	**NJ2205R**	**NUP2205R**	—	—	30	29	31	34	37	47	—	—	1	0.6	0.166
	62	17	1.1	1.1	—	53	36.6	25.2	3.45	10 000	14 000	—	—	—	**N305**	**NF305**	31.5	—	—	—	40	55.5	55.5	55	1	1	(0.235)
	62	17	1.1	1.1	34	—	51.9	37.4	4.85	10 000	14 000	**NU305R**	**NJ305R**	**NUP305R**	—	—	31.5	31.5	33	37	40	55.5	—	—	1	1	0.243
	62	24	1.1	1.1	35	—	53.4	40.9	5.70	9 100	14 000	**NU2305**	**NJ2305**	**NUP2305**	—	—	31.5	31.5	33	37	40	55.5	—	—	1	1	0.340
	62	24	1.1	1.1	34	—	71.2	56.1	7.50	9 100	14 000	**NU2305R**	**NJ2305R**	**NUP2305R**	—	—	31.5	31.5	33	37	40	55.5	—	—	1	1	0.350
30	55	13	1	0.6	36.5	—	23.4	18.4	2.05	13 000	15 000	**NU1006**	—	**NUP1006**	—	—	35	34	35	38	—	50	—	—	1	0.6	0.121
	62	16	1	0.6	—	53.5	31.1	21.5	2.95	11 000	13 000	—	—	—	**N206**	**NF206**	35	—	—	—	44	57	58	56	1	0.6	(0.206)
	62	16	1	0.6	37.5	—	48.9	37.4	5.25	11 000	13 000	**NU206R**	**NJ206R**	**NUP206R**	—	—	35	34	37	40	44	57	—	—	1	0.6	0.209
	62	20	1	0.6	38.5	—	41.0	33.1	4.20	9 800	13 000	**NU2206**	**NJ2206**	**NUP2206**	—	—	35	34	37	40	44	57	—	—	1	0.6	0.262
	62	20	1	0.6	37.5	—	61.2	49.8	6.80	9 700	13 000	**NU2206R**	**NJ2206R**	**NUP2206R**	—	—	35	34	37	40	44	57	—	—	1	0.6	0.262
	62	23.8	1	1	38.5	—	53.3	46.4	5.95	8 700	13 000	**NU3206**	—	—	—	—	35	35	37	40	—	57	—	—	1	0.6	0.343
	72	19	1.1	1.1	—	62	48.3	35.2	5.00	8 700	12 000	—	—	—	**N306**	**NF306**	36.5	—	—	—	48	65.5	65.5	64	1	1	(0.353)
	72	19	1.1	1.1	40.5	—	66.5	50.2	6.80	8 700	12 000	**NU306R**	**NJ306R**	**NUP306R**	—	—	36.5	36.5	40	44	48	65.5	—	—	1	1	0.361
	72	27	1.1	1.1	42	—	64.3	50.8	7.15	7 700	12 000	**NU2306**	**NJ2306**	**NUP2306**	—	—	36.5	36.5	40	44	48	65.5	—	—	1	1	0.500
	72	27	1.1	1.1	40.5	—	93.3	77.6	10.1	7 800	12 000	**NU2306R**	**NJ2306R**	**NUP2306R**	—	—	36.5	36.5	40	44	48	65.5	—	—	1	1	0.534

[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d (30) ~ (45) mm



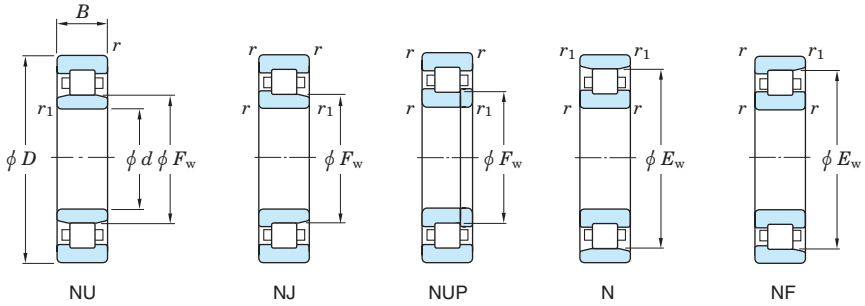


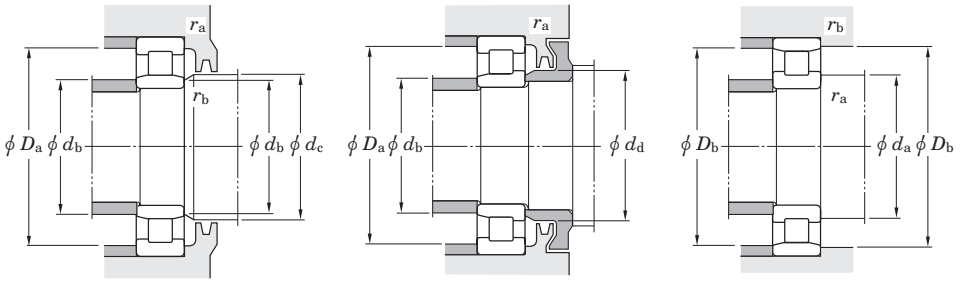
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min⁻¹)		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
30	72	30.2	1.1	1.1	42	—	86.4	74.3	9.95	7 700	12 000	**NU3306**	—	—	—	—	36.5	36.5	40	44	—	65.5	—	—	1	1	0.650
	90	23	1.5	1.5	45	73	78.3	55.0	7.95	7 600	10 000	**NU406**	**NJ406**	**NUP406**	**N406**	**NF406**	38	38	44	47	52	82	82	74	1.5	1.5	0.753
35	62	14	1	0.6	42	—	28.3	23.2	2.65	11 000	13 000	**NU1007**	—	**NUP1007**	—	—	40	39	41	44	—	57	—	—	1	0.5	0.182
	72	17	1.1	0.6	—	61.8	44.6	31.5	4.70	9 500	11 000	—	—	—	**N207**	**NF207**	41.5	—	—	—	50	65.5	68	64	1	0.6	(0.293)
	72	17	1.1	0.6	44	—	62.9	50.2	6.55	9 300	11 000	**NU207R**	**NJ207R**	**NUP207R**	—	—	41.5	39	43	46	50	65.5	—	—	1	0.6	0.306
	72	23	1.1	0.6	43.8	—	61.3	51.2	7.15	8 500	11 000	**NU2207**	**NJ2207**	**NUP2207**	—	—	41.5	39	43	46	50	65.5	—	—	1	0.6	0.402
	72	23	1.1	0.6	44	—	77.1	65.3	9.20	8 300	11 000	**NU2207R**	**NJ2207R**	**NUP2207R**	—	—	41.5	39	43	46	50	65.5	—	—	1	0.6	0.404
	72	27	1.1	1.1	43.8	—	68.5	59.1	7.90	7 600	11 000	**NU3207**	—	—	—	—	41.5	41.5	43	46	—	65.5	—	—	1	0.6	0.524
	80	21	1.5	1.1	—	68.2	62.0	46.9	6.20	7 900	10 000	—	—	—	**N307**	**NF307**	43	—	—	—	53	72	73.5	71	1.5	1	(0.477)
	80	21	1.5	1.1	46.2	—	83.3	65.4	9.35	7 700	10 000	**NU307R**	**NJ307R**	**NUP307R**	—	—	43	41.5	45	48	53	72	—	—	1.5	1	0.482
	80	31	1.5	1.1	46.2	—	75.5	65.7	7.95	7 000	10 000	**NU2307**	**NJ2307**	**NUP2307**	—	—	43	41.5	45	48	53	72	—	—	1.5	1	0.696
	80	31	1.5	1.1	46.2	—	116	101	15.0	6 900	10 000	**NU2307R**	**NJ2307R**	**NUP2307R**	—	—	43	41.5	45	48	53	72	—	—	1.5	1	0.729
	80	34.9	1.5	1.5	46.2	—	102	89.1	12.0	7 000	10 000	**NU3307**	—	—	—	—	43	43	45	48	—	72	—	—	1.5	1	0.908
	100	25	1.5	1.5	53	83	94.1	68.9	9.25	6 600	8 800	**NU407**	**NJ407**	**NUP407**	**N407**	**NF407**	43	43	52	55	61	92	92	84	1.5	1.5	1.02
40	68	15	1	0.6	47	—	31.2	25.7	3.10	10 000	12 000	**NU1008**	—	**NUP1008**	—	—	45	44	46	49	—	63	—	—	1	0.6	0.223
	80	18	1.1	1.1	—	70	54.7	42.9	6.15	8 300	10 000	—	—	—	**N208**	**NF208**	46.5	—	—	—	56	73.5	73.5	72	1	1	(0.374)
	80	18	1.1	1.1	49.5	—	69.6	55.4	7.35	8 300	9 900	**NU208R**	**NJ208R**	**NUP208R**	—	—	46.5	46.5	49	52	56	73.5	—	—	1	1	0.384
	80	23	1.1	1.1	50	—	72.8	62.0	8.75	7 500	10 000	**NU2208**	**NJ2208**	**NUP2208**	—	—	46.5	46.5	49	52	56	73.5	—	—	1	1	0.490
	80	23	1.1	1.1	49.5	—	90.5	77.6	10.3	7 400	9 900	**NU2208R**	**NJ2208R**	**NUP2208R**	—	—	46.5	46.5	49	52	56	73.5	—	—	1	1	0.490
	80	30.2	1.1	1.1	50	—	97.8	90.6	12.2	6 700	10 000	**NU3208**	—	—	—	—	46.5	46.5	49	52	—	73.5	—	—	1	1	0.711
	90	23	1.5	1.5	—	77.5	73.4	56.9	7.85	6 900	9 100	—	—	—	**N308**	**NF308**	48	—	—	—	60	82	82	80	1.5	1.5	(0.646)
	90	23	1.5	1.5	52	—	104	81.5	11.0	6 800	9 100	**NU308R**	**NJ308R**	**NUP308R**	—	—	48	48	51	55	60	82	—	—	1.5	1.5	0.664
	90	33	1.5	1.5	53.5	—	103	88.0	11.6	6 100	9 100	**NU2308**	**NJ2308**	**NUP2308**	—	—	48	48	51	55	60	82	—	—	1.5	1.5	0.956
	90	33	1.5	1.5	52	—	143	122	18.4	6 100	9 100	**NU2308R**	**NJ2308R**	**NUP2308R**	—	—	48	48	51	55	60	82	—	—	1.5	1.5	0.962
	90	36.5	1.5	1.5	53.5	—	130	119	17.6	6 100	9 100	**NU3308**	—	—	—	—	48	48	51	55	—	82	—	—	1.5	1.5	1.19
	110	27	2	2	58	92	120	89.1	12.6	6 000	8 000	**NU408**	**NJ408**	**NUP408**	**N408**	**NF408**	49	49	57	60	67	101	101	93	2	2	1.30
45	75	16	1	0.6	52.5	—	38.9	33.8	4.30	9 200	11 000	**NU1009**	—	**NUP1009**	—	—	50	49	52	54	—	70	—	—	1	0.6	0.289

[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d (45) ~ (55) mm



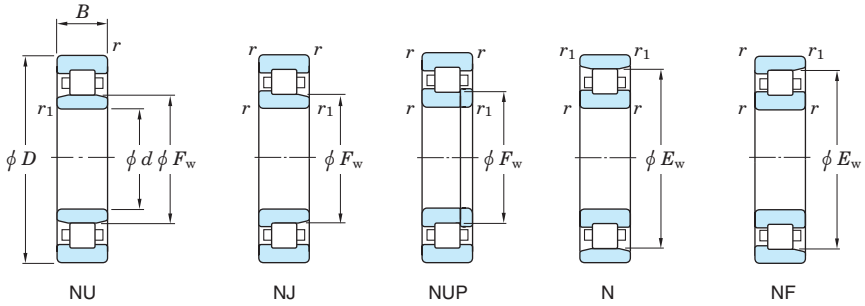


Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
45	85	19	1.1	1.1	—	75	57.6	46.9	6.70	7 700	9 200	—	—	—	**N209**	**NF209**	51.5	—	—	—	61	78.5	78.5	77	1	1	(0.427)
	85	19	1.1	1.1	54.5	—	78.9	66.4	9.05	7 600	9 200	**NU209R**	**NJ209R**	**NUP209R**	—	—	51.5	51.5	54	57	61	78.5	—	—	1	1	0.439
	85	23	1.1	1.1	55	—	76.6	67.8	9.60	6 900	9 200	**NU2209**	**NJ2209**	**NUP2209**	—	—	51.5	51.5	54	57	61	78.5	—	—	1	1	0.536
	85	23	1.1	1.1	54.5	—	95.1	84.6	11.2	6 900	9 200	**NU2209R**	**NJ2209R**	**NUP2209R**	—	—	51.5	51.5	54	57	61	78.5	—	—	1	1	0.536
	85	30.2	1.1	1.1	55	—	103	99.0	13.3	6 100	9 200	**NU3209**	—	—	—	—	51.5	51.5	54	57	—	78.5	—	—	1	1	0.770
	100	25	1.5	1.5	—	86.5	98.5	77.5	11.3	6 200	8 300	—	—	—	**N309**	**NF309**	53	—	—	—	66	92	92	89	1.5	1.5	(0.865)
	100	25	1.5	1.5	58.5	—	122	98.3	13.5	6 100	8 200	**NU309R**	**NJ309R**	**NUP309R**	—	—	53	53	57	60	66	92	—	—	1.5	1.5	0.909
	100	36	1.5	1.5	58.5	—	124	113	14.3	5 500	8 300	**NU2309**	**NJ2309**	**NUP2309**	—	—	53	53	57	60	66	92	—	—	1.5	1.5	1.25
	100	36	1.5	1.5	58.5	—	172	153	23.0	5 400	8 200	**NU2309R**	**NJ2309R**	**NUP2309R**	—	—	53	53	57	60	66	92	—	—	1.5	1.5	1.32
	100	39.7	1.5	1.5	58.5	—	164	149	22.6	5 500	8 300	**NU3309**	—	—	—	—	53	53	57	60	—	92	—	—	1.5	1.5	1.59
	120	29	2	2	64.5	100.5	134	112	13.8	5 400	7 200	**NU409**	**NJ409**	**NUP409**	**N409**	**NF409**	54	54	63	66	74	111	111	102	2	2	1.64
50	80	16	1	0.6	57.5	—	42.2	36.8	4.80	8 400	9 900	**NU1010**	—	**NUP1010**	—	—	55	54	57	59	—	75	—	—	1	0.6	0.306
	90	20	1.1	1.1	—	80.4	60.3	51.0	7.30	7 100	8 500	—	—	—	**N210**	**NF210**	56.5	—	—	—	67	83.5	83.5	82	1	1	(0.479)
	90	20	1.1	1.1	59.5	—	82.5	71.9	9.85	7 100	8 500	**NU210R**	**NJ210R**	**NUP210R**	—	—	56.5	56.5	58	62	67	83.5	—	—	1	1	0.497
	90	23	1.1	1.1	60.4	—	80.3	73.6	10.4	6 400	8 500	**NU2210**	**NJ2210**	**NUP2210**	—	—	56.5	56.5	58	62	67	83.5	—	—	1	1	0.580
	90	23	1.1	1.1	59.5	—	99.5	91.5	12.1	6 400	8 500	**NU2210R**	**NJ2210R**	**NUP2210R**	—	—	56.5	56.5	58	62	67	83.5	—	—	1	1	0.580
	90	30.2	1.1	1.1	60.4	—	108	108	14.5	5 700	8 500	**NU3210**	—	—	—	—	56.5	56.5	58	62	—	83.5	—	—	1	1	0.829
	110	27	2	2	—	95	109	93.4	11.7	5 600	7 500	—	—	—	**N310**	**NF310**	59	—	—	—	73	101	101	98	2	2	(1.15)
	110	27	2	2	65	—	138	113	16.0	5 500	7 400	**NU310R**	**NJ310R**	**NUP310R**	—	—	59	59	63	67	73	101	—	—	2	2	1.15
	110	40	2	2	65	—	151	142	20.1	5 000	7 500	**NU2310**	**NJ2310**	**NUP2310**	—	—	59	59	63	67	73	101	—	—	2	2	1.69
	110	40	2	2	65	—	203	187	28.6	4 900	7 400	**NU2310R**	**NJ2310R**	**NUP2310R**	—	—	59	59	63	67	73	101	—	—	2	2	1.76
	110	44.4	2	2	65	—	195	183	27.3	5 000	7 500	**NU3310**	—	—	—	—	59	59	63	67	—	101	—	—	2	2	2.14
	130	31	2.1	2.1	70.8	110.8	161	136	17.4	4 900	6 600	**NU410**	**NJ410**	**NUP410**	**N410**	**NF410**	61	61	69	73	81	119	119	112	2	2	2.01
55	90	18	1.1	1	64.5	—	47.1	43.8	5.75	7 600	8 900	**NU1011**	—	**NUP1011**	—	—	61.5	60	63	66	—	83.5	—	—	1	1	0.445
	100	21	1.5	1.1	—	88.5	72.5	62.3	8.30	6 400	7 700	—	—	—	**N211**	**NF211**	63	—	—	—	73	92	93.5	91	1.5	1	(0.633)
	100	21	1.5	1.1	66	—	108	98.7	14.2	6 400	7 700	**NU211R**	**NJ211R**	**NUP211R**	—	—	63	61.5	65	68	73	92	—	—	1.5	1	0.650
	100	25	1.5	1.1	66.5	—	94.2	87.2	11.6	5 800	7 700	**NU2211**	**NJ2211**	**NUP2211**	—	—	63	61.5	65	68	73	92	—	—	1.5	1	0.780

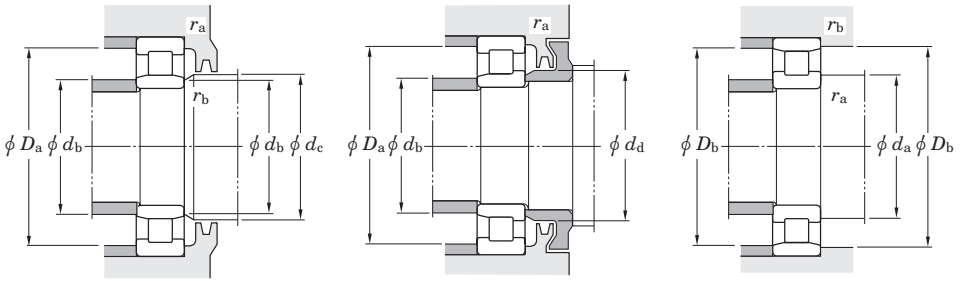
[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d (55) ~ (65) mm



NU NJ NUP N NF

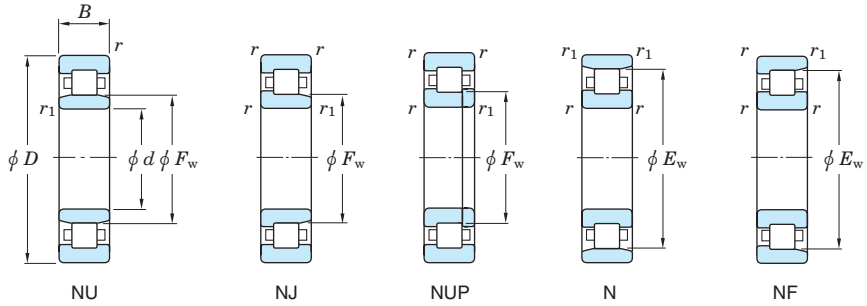


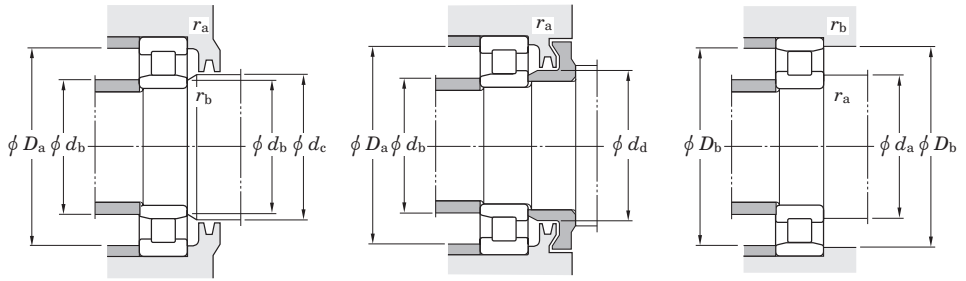
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
55	100	25	1.5	1.1	66	—	127	122	16.9	5 800	7 700	**NU2211R**	**NJ2211R**	**NUP2211R**	—	—	63	61.5	65	68	73	92	—	—	1.5	1	0.806
	100	33.3	1.5	1.5	66.5	—	119	118	16.1	5 100	7 700	**NU3211**	—	—	—	—	63	63	65	68	—	92	—	—	1.5	1	1.14
	120	29	2	2	—	104.5	138	111	15.8	5 100	6 800	—	—	—	**N311**	**NF311**	64	—	—	—	80	111	111	107	2	2	(1.44)
	120	29	2	2	70.5	—	172	143	19.8	5 100	6 700	**NU311R**	**NJ311R**	**NUP311R**	—	—	64	64	69	72	80	111	—	—	2	2	1.50
	120	43	2	2	70.5	—	185	162	24.6	4 500	6 800	**NU2311**	**NJ2311**	**NUP2311**	—	—	64	64	69	72	80	111	—	—	2	2	2.10
	120	43	2	2	70.5	—	251	233	35.3	4 500	6 700	**NU2311R**	**NJ2311R**	**NUP2311R**	—	—	64	64	69	72	80	111	—	—	2	2	2.25
	120	49.2	2	2	70.5	—	235	220	32.8	4 500	6 800	**NU3311**	—	—	—	—	64	64	69	72	—	111	—	—	2	2	2.81
	140	33	2.1	2.1	77.2	117.2	174	138	19.6	4 600	6 100	**NU411**	**NJ411**	**NUP411**	**N411**	**NF411**	66	66	76	79	87	129	129	119	2	2	2.51
60	95	18	1.1	1	69.5	—	53.0	50.0	6.75	7 000	8 300	**NU1012**	—	**NUP1012**	—	—	66.5	65	68	71	—	88.5	—	—	1	1	0.477
	110	22	1.5	1.5	—	97.5	85.7	79.9	10.4	5 800	7 000	—	—	—	**N212**	**NF212**	68	—	—	—	80	102	102	100	1.5	1.5	(0.823)
	110	22	1.5	1.5	72	—	122	107	15.7	5 800	6 900	**NU212R**	**NJ212R**	**NUP212R**	—	—	68	68	71	75	80	102	—	—	1.5	1.5	0.830
	110	28	1.5	1.5	73.5	—	120	123	15.3	5 200	7 000	**NU2212**	**NJ2212**	**NUP2212**	—	—	68	68	71	75	80	102	—	—	1.5	1.5	1.07
	110	28	1.5	1.5	72	—	164	157	21.7	5 200	6 900	**NU2212R**	**NJ2212R**	**NUP2212R**	—	—	68	68	71	75	80	102	—	—	1.5	1.5	1.09
	110	36.5	1.5	1.5	73.5	—	160	167	24.7	4 700	7 000	**NU3212**	—	—	—	—	68	68	71	75	—	102	—	—	1.5	1.5	1.52
	130	31	2.1	2.1	—	113	155	126	17.3	4 700	6 300	—	—	—	**N312**	**NF312**	71	—	—	—	86	119	119	116	2	2	(1.83)
	130	31	2.1	2.1	77	—	187	157	22.1	4 600	6 200	**NU312R**	**NJ312R**	**NUP312R**	—	—	71	71	75	79	86	119	—	—	2	2	1.87
	130	46	2.1	2.1	77	—	211	188	29.4	4 200	6 300	**NU2312**	**NJ2312**	**NUP2312**	—	—	71	71	75	79	86	119	—	—	2	2	2.69
	130	46	2.1	2.1	77	—	278	262	39.6	4 100	6 200	**NU2312R**	**NJ2312R**	**NUP2312R**	—	—	71	71	75	79	86	119	—	—	2	2	2.81
	130	54	2.1	2.1	77	—	275	265	39.9	4 200	6 300	**NU3312**	—	—	—	—	71	71	75	79	—	119	—	—	2	2	3.61
	150	35	2.1	2.1	83	127	209	184	26.1	4 200	5 700	**NU412**	**NJ412**	**NUP412**	**N412**	**NF412**	71	71	82	85	94	139	139	128	2	2	3.02
65	100	18	1.1	1	74.5	—	54.4	52.9	7.15	6 600	7 800	**NU1013**	—	**NUP1013**	—	—	71.5	70	73	76	—	93.5	—	—	1	1	0.506
	120	23	1.5	1.5	—	105.6	105	94.4	13.5	5 400	6 400	—	—	—	**N213**	**NF213**	73	—	—	—	87	112	112	108	1.5	1.5	(1.05)
	120	23	1.5	1.5	78.5	—	134	119	16.1	5 300	6 400	**NU213R**	**NJ213R**	**NUP213R**	—	—	73	73	77	81	87	112	—	—	1.5	1.5	1.05
	120	31	1.5	1.5	79.6	—	150	149	20.6	4 800	6 400	**NU2213**	**NJ2213**	**NUP2213**	—	—	73	73	77	81	87	112	—	—	1.5	1.5	1.43
	120	31	1.5	1.5	78.5	—	186	181	27.7	4 800	6 400	**NU2213R**	**NJ2213R**	**NUP2213R**	—	—	73	73	77	81	87	112	—	—	1.5	1.5	1.45
	120	38.1	1.5	1.5	79.6	—	186	197	29.7	4 300	6 400	**NU3213**	—	—	—	—	73	73	77	81	—	112	—	—	1.5	1.5	1.90
	140	33	2.1	2.1	—	121.5	169	139	19.2	4 300	5 800	—	—	—	**N313**	**NF313**	76	—	—	—	93	129	129	125	2	2	(2.19)

[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d (65) ~ (75) mm



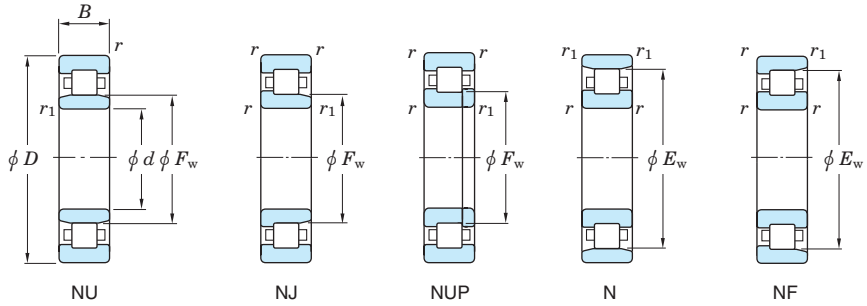


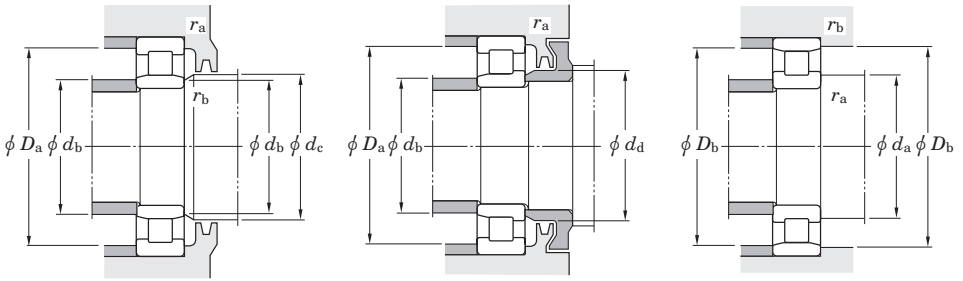
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
65	140	33	2.1	2.1	82.5	—	226	191	29.8	4 300	5 700	**NU313R**	**NJ313R**	**NUP313R**	—	—	76	76	81	85	93	129	—	—	2	2	2.31
	140	48	2.1	2.1	83.5	—	235	212	32.9	3 900	5 800	**NU2313**	**NJ2313**	**NUP2313**	—	—	76	76	81	85	93	129	—	—	2	2	3.25
	140	48	2.1	2.1	82.5	—	310	287	43.3	3 800	5 700	**NU2313R**	**NJ2313R**	**NUP2313R**	—	—	76	76	81	85	93	129	—	—	2	2	3.36
	140	58.7	2.1	2.1	83.5	—	302	294	43.9	3 900	5 800	**NU3313**	—	—	—	—	76	76	81	85	—	129	—	—	2	2	4.53
	160	37	2.1	2.1	89.3	135.3	228	203	28.2	4 000	5 300	**NU413**	**NJ413**	**NUP413**	**N413**	**NF413**	76	76	88	91	100	149	149	137	2	2	3.58
70	110	20	1.1	1	80	—	72.9	70.4	10.1	6 100	7 200	**NU1014**	—	**NUP1014**	—	—	76.5	75	78	82	—	103.5	—	—	1	1	0.702
	125	24	1.5	1.5	—	110.5	104	95.2	13.6	5 100	6 100	—	—	—	**N214**	**NF214**	78	—	—	—	92	117	117	114	1.5	1.5	(1.15)
	125	24	1.5	1.5	83.5	—	148	137	19.0	5 000	6 000	**NU214R**	**NJ214R**	**NUP214R**	—	—	78	78	82	86	92	117	—	—	1.5	1.5	1.16
	125	31	1.5	1.5	84.5	—	149	151	20.8	4 600	6 100	**NU2214**	**NJ2214**	**NUP2214**	—	—	78	78	82	86	92	117	—	—	1.5	1.5	1.52
	125	31	1.5	1.5	83.5	—	194	194	29.8	4 500	6 000	**NU2214R**	**NJ2214R**	**NUP2214R**	—	—	78	78	82	86	92	117	—	—	1.5	1.5	1.53
	125	39.7	1.5	1.5	84.5	—	185	198	30.0	4 100	6 100	**NU3214**	—	—	—	—	78	78	82	86	—	117	—	—	1.5	1.5	2.09
	150	35	2.1	2.1	—	130	198	168	23.3	4 000	5 400	—	—	—	**N314**	**NF314**	81	—	—	—	100	139	139	134	2	2	(2.73)
	150	35	2.1	2.1	89	—	256	222	33.4	4 000	5 300	**NU314R**	**NJ314R**	**NUP314R**	—	—	81	81	87	92	100	139	—	—	2	2	2.81
	150	51	2.1	2.1	90	—	279	262	39.3	3 600	5 400	**NU2314**	**NJ2314**	**NUP2314**	—	—	81	81	87	92	100	139	—	—	2	2	3.97
	150	51	2.1	2.1	89	—	342	323	47.1	3 600	5 300	**NU2314R**	**NJ2314R**	**NUP2314R**	—	—	81	81	87	92	100	139	—	—	2	2	4.08
	150	63.5	2.1	2.1	90	—	354	356	51.5	3 600	5 400	**NU3314**	—	—	—	—	81	81	87	92	—	139	—	—	2	2	5.62
	180	42	3	3	100	152	285	257	35.2	3 500	4 700	**NU414**	**NJ414**	**NUP414**	**N414**	**NF414**	83	83	99	102	112	167	167	153	2.5	2.5	5.26
75	115	20	1.1	1	85	—	80.0	78.1	10.2	5 700	6 800	**NU1015**	—	**NUP1015**	—	—	81.5	80	83	87	—	108.5	—	—	1	1	0.735
	130	25	1.5	1.5	—	116.5	121	118	16.1	4 800	5 800	—	—	—	**N215**	**NF215**	83	—	—	—	96	122	122	120	1.5	1.5	(1.24)
	130	25	1.5	1.5	88.5	—	163	156	21.9	4 800	5 700	**NU215R**	**NJ215R**	**NUP215R**	—	—	83	83	87	90	96	122	—	—	1.5	1.5	1.29
	130	31	1.5	1.5	88.5	—	162	172	22.3	4 300	5 800	**NU2215**	**NJ2215**	**NUP2215**	—	—	83	83	87	90	96	122	—	—	1.5	1.5	1.57
	130	31	1.5	1.5	88.5	—	202	207	31.5	4 300	5 700	**NU2215R**	**NJ2215R**	**NUP2215R**	—	—	83	83	87	90	96	122	—	—	1.5	1.5	1.61
	130	41.3	1.5	1.5	88.5	—	210	226	34.1	3 900	5 800	**NU3215**	—	—	—	—	83	83	87	90	—	122	—	—	1.5	1.5	2.28
	160	37	2.1	2.1	—	139.5	224	205	28.4	3 800	5 000	—	—	—	**N315**	**NF315**	86	—	—	—	106	149	149	143	2	2	(3.19)
	160	37	2.1	2.1	95	—	300	263	39.9	3 700	5 000	**NU315R**	**NJ315R**	**NUP315R**	—	—	86	86	93	97	106	149	—	—	2	2	3.37
	160	55	2.1	2.1	95.5	—	323	327	43.4	3 400	5 000	**NU2315**	**NJ2315**	**NUP2315**	—	—	86	86	93	97	106	149	—	—	2	2	4.84
	160	55	2.1	2.1	95	—	412	395	57.3	3 300	5 000	**NU2315R**	**NJ2315R**	**NUP2315R**	—	—	86	86	93	97	106	149	—	—	2	2	5.00

[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d (75) ~ (90) mm



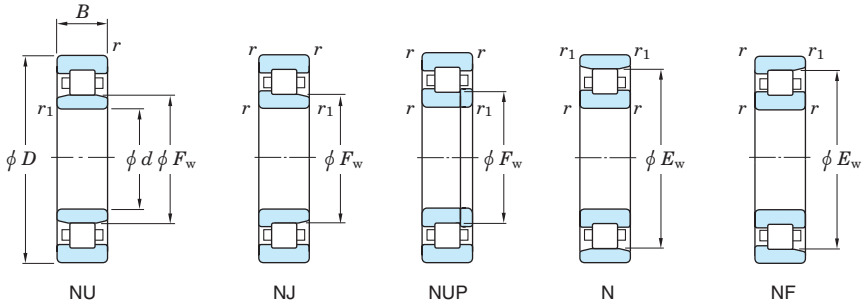


Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min⁻¹)		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
75	160	68.3	2.1	2.1	95.5	—	423	430	62.1	3 400	5 000	**NU3315**	—	—	—	—	86	86	93	97	—	149	—	—	2	2	6.86
	190	45	3	3	104.5	160.5	328	274	40.2	3 300	4 400	**NU415**	**NJ415**	**NUP415**	**N415**	**NF415**	88	88	103	107	118	177	177	162	2.5	2.5	6.25
80	125	22	1.1	1	91.5	—	87.2	86.4	11.5	5 300	6 300	**NU1016**	—	**NUP1016**	—	—	86.5	85	90	94	—	118.5	—	—	1	1	0.994
	140	26	2	2	—	125.3	133	122	16.2	4 500	5 400	—	—	—	**N216**	**NF216**	89	—	—	—	104	131	131	128	2	2	(1.51)
	140	26	2	2	95.3	—	174	167	23.0	4 400	5 300	**NU216R**	**NJ216R**	**NUP216R**	—	—	89	89	94	97	104	131	—	—	2	2	1.56
	140	33	2	2	95.3	—	184	186	27.8	4 000	5 400	**NU2216**	**NJ2216**	**NUP2216**	—	—	89	89	94	97	104	131	—	—	2	2	1.96
	140	33	2	2	95.3	—	233	243	35.8	4 000	5 300	**NU2216R**	**NJ2216R**	**NUP2216R**	—	—	89	89	94	97	104	131	—	—	2	2	2.03
	140	44.4	2	2	95.3	—	238	259	37.8	3 600	5 400	**NU3216**	—	—	—	—	89	89	94	97	—	131	—	—	2	2	2.87
	170	39	2.1	2.1	—	147	238	207	30.7	3 500	4 700	—	—	—	**N316**	**NF316**	91	—	—	—	114	159	159	151	2	2	(3.83)
	170	39	2.1	2.1	101	—	320	282	42.1	3 500	4 700	**NU316R**	**NJ316R**	**NUP316R**	—	—	91	91	99	105	114	159	—	—	2	2	4.00
	170	58	2.1	2.1	103	—	343	332	46.9	3 100	4 700	**NU2316**	**NJ2316**	**NUP2316**	—	—	91	91	99	105	114	159	—	—	2	2	5.83
	170	58	2.1	2.1	101	—	445	431	61.1	3 100	4 700	**NU2316R**	**NJ2316R**	**NUP2316R**	—	—	91	91	99	105	114	159	—	—	2	2	5.95
	170	68.3	2.1	2.1	103	—	423	436	61.9	3 100	4 700	**NU3316**	—	—	—	—	91	91	99	105	—	159	—	—	2	2	7.72
	200	48	3	3	110	170	374	315	45.2	3 100	4 200	**NU416**	**NJ416**	**NUP416**	**N416**	**NF416**	93	93	109	112	124	187	187	172	2.5	2.5	7.28
85	130	22	1.1	1	96.5	—	89.8	91.2	12.0	5 100	6 000	**NU1017**	—	**NUP1017**	—	—	91.5	90	95	99	—	123.5	—	—	1	1	1.04
	150	28	2	2	—	133.8	151	140	18.7	4 200	5 000	—	—	—	**N217**	**NF217**	94	—	—	—	110	141	141	137	2	2	(1.90)
	150	28	2	2	100.5	—	209	199	26.3	4 200	5 000	**NU217R**	**NJ217R**	**NUP217R**	—	—	94	94	99	104	110	141	—	—	2	2	1.94
	150	36	2	2	101.8	—	212	218	31.6	3 800	5 000	**NU2217**	**NJ2217**	**NUP2217**	—	—	94	94	99	104	110	141	—	—	2	2	2.50
	150	36	2	2	100.5	—	272	279	41.6	3 700	5 000	**NU2217R**	**NJ2217R**	**NUP2217R**	—	—	94	94	99	104	110	141	—	—	2	2	2.53
	150	49.2	2	2	101.8	—	269	296	42.1	3 300	5 000	**NU3217**	—	—	—	—	94	94	99	104	—	141	—	—	2	2	3.67
	180	41	3	3	—	156	281	247	35.6	3 300	4 500	—	—	—	**N317**	**NF317**	98	—	—	—	119	167	167	160	2.5	2.5	(4.52)
	180	41	3	3	108	—	364	330	48.3	3 300	4 400	**NU317R**	**NJ317R**	**NUP317R**	—	—	98	98	106	110	119	167	—	—	2.5	2.5	4.80
	180	60	3	3	108	—	394	382	54.2	3 000	4 500	**NU2317**	**NJ2317**	**NUP2317**	—	—	98	98	106	110	119	167	—	—	2.5	2.5	6.62
	180	60	3	3	108	—	491	485	67.7	2 900	4 400	**NU2317R**	**NJ2317R**	**NUP2317R**	—	—	98	98	106	110	119	167	—	—	2.5	2.5	6.98
	180	73	3	3	108	—	499	517	71.5	3 000	4 500	**NU3317**	—	—	—	—	98	98	106	110	—	167	—	—	2.5	2.5	9.23
	210	52	4	4	113	177	416	350	49.7	3 000	4 000	**NU417**	**NJ417**	**NUP417**	**N417**	**NF417**	101	101	111	115	128	194	194	179	3	3	8.68
90	140	24	1.5	1.1	103	—	106	109	14.6	4 700	5 600	**NU1018**	—	**NUP1018**	—	—	98	96.5	101	106	—	132	—	—	1.5	1	1.34

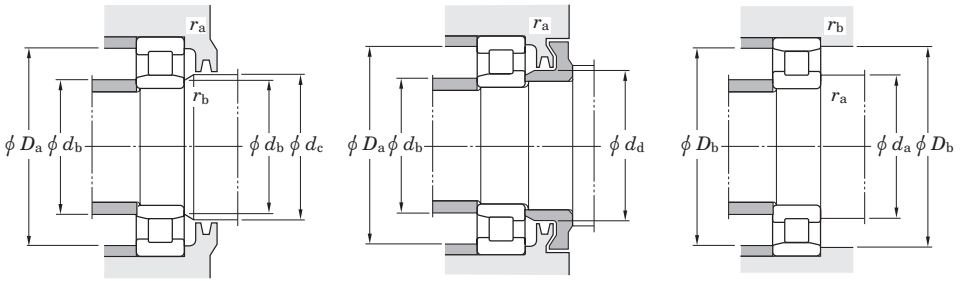
[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d (90) ~ (100) mm



NU NJ NUP N NF

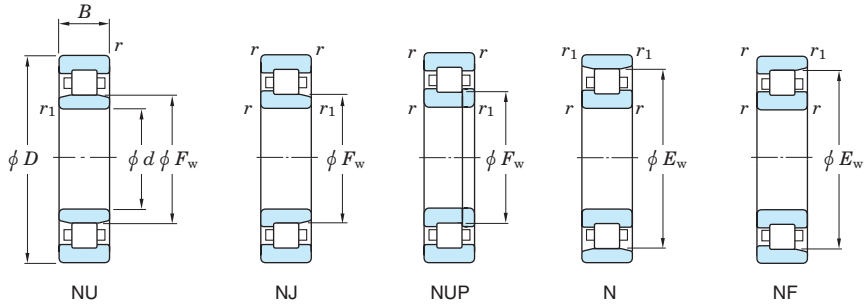


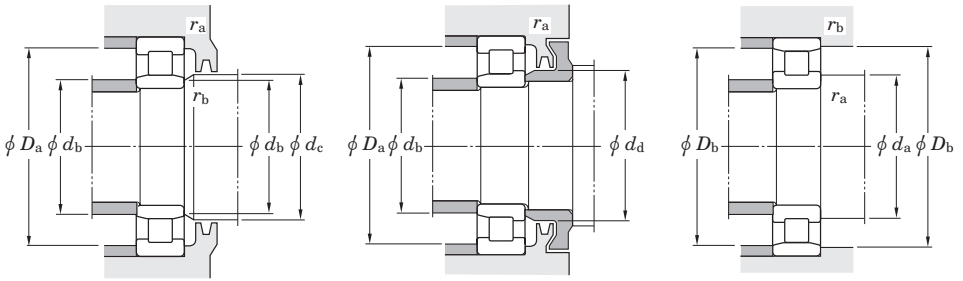
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
90	160	30	2	2	—	143	190	178	22.9	3 900	4 700	—	—	—	**N218**	**NF218**	99	—	—	—	116	151	151	146	2	2	(2.32)
	160	30	2	2	107	—	227	217	28.7	3 900	4 700	**NU218R**	**NJ218R**	**NUP218R**	—	—	99	99	105	109	116	151	—	—	2	2	2.38
	160	40	2	2	107	—	259	265	38.9	3 500	4 700	**NU2218**	**NJ2218**	**NUP2218**	—	—	99	99	105	109	116	151	—	—	2	2	3.10
	160	40	2	2	107	—	302	314	45.8	3 500	4 700	**NU2218R**	**NJ2218R**	**NUP2218R**	—	—	99	99	105	109	116	151	—	—	2	2	3.21
	160	52.4	2	2	107	—	338	373	52.8	3 100	4 700	**NU3218**	—	—	—	—	99	99	105	109	—	151	—	—	2	2	4.49
	190	43	3	3	—	165	300	265	38.7	3 100	4 200	—	—	—	**N318**	**NF318**	103	—	—	—	127	177	177	169	2.5	2.5	(5.27)
	190	43	3	3	113.5	—	395	355	50.6	3 100	4 100	**NU318R**	**NJ318R**	**NUP318R**	—	—	103	103	111	117	127	177	—	—	2.5	2.5	5.47
	190	64	3	3	115	—	408	395	55.5	2 800	4 200	**NU2318**	**NJ2318**	**NUP2318**	—	—	103	103	111	117	127	177	—	—	2.5	2.5	7.90
	190	64	3	3	113.5	—	544	534	74.5	2 800	4 100	**NU2318R**	**NJ2318R**	**NUP2318R**	—	—	103	103	111	117	127	177	—	—	2.5	2.5	8.12
	190	73	3	3	115	—	535	559	75.6	2 800	4 200	**NU3318**	—	—	—	—	103	103	111	117	—	177	—	—	2.5	2.5	10.3
	225	54	4	4	123.5	191.5	468	400	55.1	2 800	3 700	**NU418**	**NJ418**	**NUP418**	**N418**	**NF418**	106	106	122	125	139	209	209	194	3	3	10.3
95	145	24	1.5	1.1	108	—	110	115	15.2	4 500	5 300	**NU1019**	—	**NUP1019**	—	—	103	101.5	106	111	—	137	—	—	1.5	1	1.40
	170	32	2.1	2.1	—	151.5	207	195	25.1	3 700	4 400	—	—	—	**N219**	**NF219**	106	—	—	—	123	159	159	155	2	2	(2.80)
	170	32	2.1	2.1	112.5	—	275	265	38.3	3 700	4 400	**NU219R**	**NJ219R**	**NUP219R**	—	—	106	106	111	116	123	159	—	—	2	2	2.92
	170	43	2.1	2.1	113.5	—	288	298	42.9	3 300	4 400	**NU2219**	**NJ2219**	**NUP2219**	—	—	106	106	111	116	123	159	—	—	2	2	3.85
	170	43	2.1	2.1	112.5	—	358	371	52.8	3 300	4 400	**NU2219R**	**NJ2219R**	**NUP2219R**	—	—	106	106	111	116	123	159	—	—	2	2	3.93
	170	55.6	2.1	2.1	113.5	—	371	412	57.2	3 000	4 400	**NU3219**	—	—	—	—	106	106	111	116	—	159	—	—	2	2	5.42
	200	45	3	3	—	173.5	323	311	41.3	3 000	4 000	—	—	—	**N319**	**NF319**	108	—	—	—	134	187	187	178	2.5	2.5	(6.10)
	200	45	3	3	121.5	—	418	387	54.3	2 900	3 900	**NU319R**	**NJ319R**	**NUP319R**	—	—	108	108	119	124	134	187	—	—	2.5	2.5	6.42
	200	67	3	3	121.5	—	465	496	62.6	2 600	4 000	**NU2319**	**NJ2319**	**NUP2319**	—	—	108	108	119	124	134	187	—	—	2.5	2.5	9.39
	200	77.8	3	3	121.5	—	609	654	86.8	2 600	4 000	**NU3319**	—	—	—	—	108	108	119	124	—	187	—	—	2.5	2.5	12.1
	240	55	4	4	133.5	201.5	502	444	60.1	2 600	3 400	**NU419**	**NJ419**	**NUP419**	**N419**	**NF419**	111	111	132	136	149	224	224	204	3	3	13.6
100	150	24	1.5	1.1	113	—	114	120	15.8	4 300	5 100	**NU1020**	—	**NUP1020**	—	—	108	106.5	111	116	—	142	—	—	1.5	1	1.46
	180	34	2.1	2.1	—	160	229	217	28.1	3 500	4 200	—	—	—	**N220**	**NF220**	111	—	—	—	130	169	169	164	2	2	(3.38)
	180	34	2.1	2.1	119	—	312	306	43.0	3 500	4 200	**NU220R**	**NJ220R**	**NUP220R**	—	—	111	111	117	122	130	169	—	—	2	2	3.52
	180	46	2.1	2.1	120	—	322	338	47.3	3 100	4 200	**NU2220**	**NJ2220**	**NUP2220**	—	—	111	111	117	122	130	169	—	—	2	2	4.67
	180	46	2.1	2.1	119	—	417	444	60.7	3 100	4 200	**NU2220R**	**NJ2220R**	**NUP2220R**	—	—	111	111	117	122	130	169	—	—	2	2	4.82

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2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d (100) ~ (120) mm



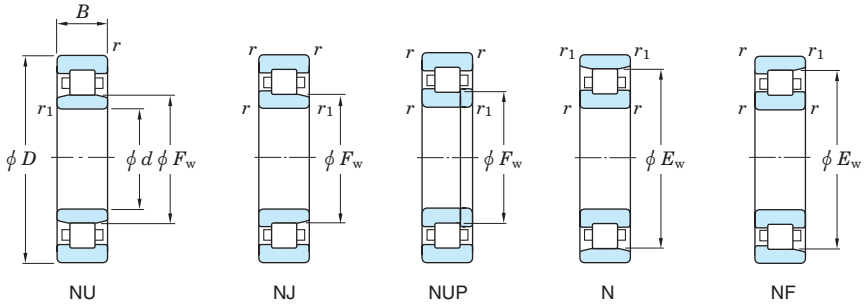


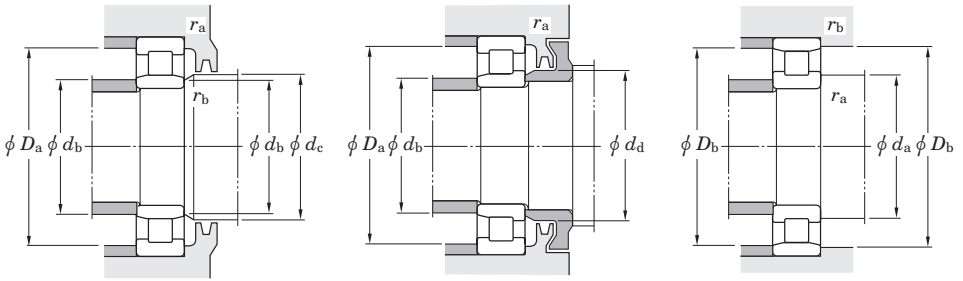
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
100	180	60.3	2.1	2.1	120	—	409	459	61.9	2 800	4 200	**NU3220**	—	—	—	—	111	111	117	122	—	169	—	—	2	2	6.62
	215	47	3	3	—	185.5	373	337	47.2	2 800	3 700	—	—	—	**N320**	**NF320**	113	—	—	—	143	202	202	190	2.5	2.5	(7.59)
	215	47	3	3	127.5	—	474	424	58.7	2 700	3 600	**NU320R**	**NJ320R**	**NUP320R**	—	—	113	113	125	132	143	202	—	—	2.5	2.5	7.75
	215	73	3	3	129.5	—	544	548	74.1	2 500	3 700	**NU2320**	**NJ2320**	**NUP2320**	—	—	113	113	125	132	143	202	—	—	2.5	2.5	11.9
	215	73	3	3	127.5	—	713	717	94.7	2 400	3 600	**NU2320R**	**NJ2320R**	**NUP2320R**	—	—	113	113	125	132	143	202	—	—	2.5	2.5	12.1
	215	82.6	3	3	129.5	—	663	706	93.2	2 500	3 700	**NU3320**	—	—	—	—	113	113	125	132	—	202	—	—	2.5	2.5	15.0
	250	58	4	4	139	211	560	498	67.3	2 500	3 300	**NU420**	**NJ420**	**NUP420**	**N420**	**NF420**	116	116	137	141	156	234	234	213	3	3	14.0
105	160	26	2	1.1	119.5	—	136	149	19.6	4 100	4 800	**NU1021**	—	**NUP1021**	—	—	114	111.5	118	122	—	151	—	—	2	1	1.85
	190	36	2.1	2.1	—	168.8	251	241	34.1	3 300	3 900	—	—	—	**N221**	**NF221**	116	—	—	—	137	179	179	173	2	2	(4.44)
	190	65.1	2.1	2.1	126.8	—	431	482	64.3	2 600	3 900	**NU3221**	—	—	—	—	116	116	124	129	—	179	—	—	2	2	8.00
	225	49	3	3	—	195	426	417	53.1	2 600	3 500	—	—	—	**N321**	**NF321**	118	—	—	—	149	212	212	199	2.5	2.5	(8.68)
	225	77	3	3	135	—	711	750	97.3	2 300	3 500	**NU2321**	—	**NUP2321**	—	—	118	118	131	138	—	212	—	—	2.5	2.5	15.6
	225	87.3	3	3	135	—	799	871	113	2 300	3 500	**NU3321**	—	—	—	—	118	118	132	137	—	212	—	—	2.5	2.5	17.4
	260	60	4	4	144.5	220.5	581	510	67.6	2 400	3 100	**NU421**	**NJ421**	**NUP421**	**N421**	**NF421**	121	121	143	147	162	244	244	223	3	3	19.1
110	170	28	2	1.1	125	—	168	171	21.7	3 800	4 500	**NU1022**	—	**NUP1022**	—	—	119	116.5	124	128	—	161	—	—	2	1	2.31
	200	38	2.1	2.1	—	178.5	300	290	40.1	3 100	3 700	—	—	—	**N222**	**NF222**	121	—	—	—	144	189	189	182	2	2	(5.24)
	200	38	2.1	2.1	132.5	—	366	365	51.1	3 100	3 700	**NU222R**	**NJ222R**	**NUP222R**	—	—	121	121	130	135	144	189	—	—	2	2	4.90
	200	53	2.1	2.1	132.5	—	397	442	55.1	2 800	3 700	**NU2222**	**NJ2222**	**NUP2222**	—	—	121	121	130	135	144	189	—	—	2	2	6.93
	200	53	2.1	2.1	132.5	—	479	517	69.9	2 800	3 700	**NU2222R**	**NJ2222R**	**NUP2222R**	—	—	121	121	130	135	144	189	—	—	2	2	6.93
	200	69.8	2.1	2.1	132.5	—	533	607	80.6	2 500	3 700	**NU3222**	—	—	—	—	121	121	130	135	—	189	—	—	2	2	9.55
	240	50	3	3	—	207	502	467	62.9	2 500	3 300	—	—	—	**N322**	**NF322**	123	—	—	—	158	227	227	211	2.5	2.5	(10.4)
	240	50	3	3	143	—	564	525	70.0	2 400	3 200	**NU322R**	**NJ322R**	**NUP322R**	—	—	123	123	140	145	158	227	—	—	2.5	2.5	10.7
	240	80	3	3	143	—	755	789	102	2 200	3 300	**NU2322**	**NJ2322**	**NUP2322**	—	—	123	123	140	145	158	227	—	—	2.5	2.5	18.8
	240	80	3	3	143	—	843	880	112	2 200	3 200	**NU2322R**	**NJ2322R**	**NUP2322R**	—	—	123	123	140	145	158	227	—	—	2.5	2.5	18.8
	240	92.1	3	3	143	—	849	918	118	2 200	3 300	**NU3322**	—	—	—	—	123	123	140	145	—	227	—	—	2.5	2.5	21.1
	280	65	4	4	155	235	685	621	80.8	2 200	2 900	**NU422**	**NJ422**	**NUP422**	**N422**	**NF422**	126	126	153	157	173	264	264	237	3	3	19.9
120	180	28	2	1.1	135	—	173	181	22.6	3 500	4 200	**NU1024**	—	**NUP1024**	—	—	129	126.5	134	138	—	171	—	—	2	1	2.47

[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d (120) ~ (140) mm



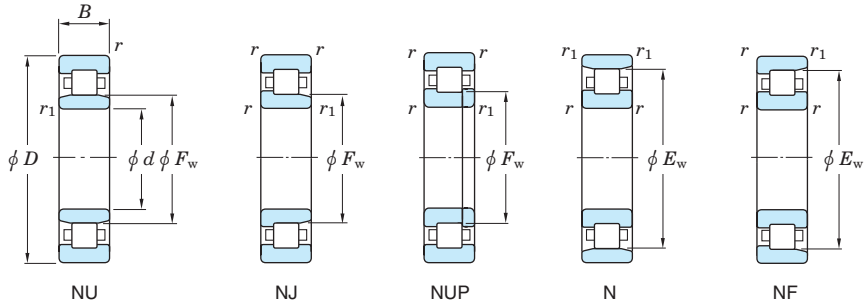


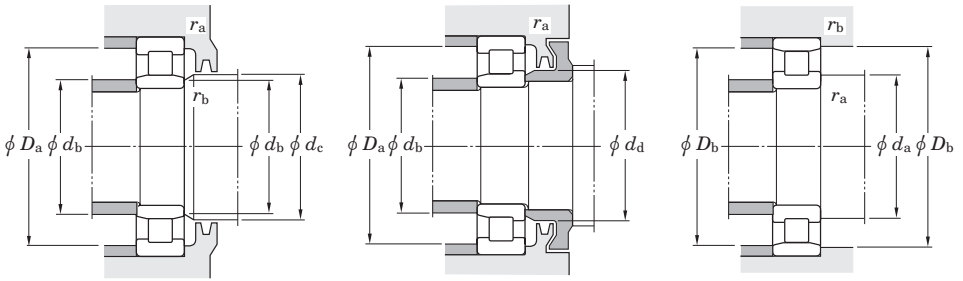
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
120	215	40	2.1	2.1	—	191.5	325	318	42.9	2 900	3 400	—	—	—	**N224**	**NF224**	131	—	—	—	156	204	204	196	2	2	(6.31)
	215	40	2.1	2.1	143.5	—	419	421	57.6	2 800	3 400	**NU224R**	**NJ224R**	**NUP224R**	—	—	131	131	141	146	156	204	—	—	2	2	5.85
	215	58	2.1	2.1	143.5	—	434	492	61.2	2 600	3 400	**NU2224**	**NJ2224**	**NUP2224**	—	—	131	131	141	146	156	204	—	—	2	2	8.56
	215	58	2.1	2.1	143.5	—	565	619	80.9	2 600	3 400	**NU2224R**	**NJ2224R**	**NUP2224R**	—	—	131	131	141	146	156	204	—	—	2	2	8.56
	215	76	2.1	2.1	143.5	—	596	695	89.2	2 300	3 400	**NU3224**	—	—	—	—	131	131	141	146	—	204	—	—	2	2	11.9
	260	55	3	3	—	226	596	551	72.7	2 200	3 000	—	—	—	**N324**	**NF324**	133	—	—	—	171	247	247	230	2.5	2.5	(13.1)
	260	55	3	3	154	—	660	610	79.8	2 200	3 000	**NU324R**	**NJ324R**	**NUP324R**	—	—	133	133	151	156	171	247	—	—	2.5	2.5	13.4
	260	86	3	3	154	—	886	918	116	2 000	3 000	**NU2324**	**NJ2324**	**NUP2324**	—	—	133	133	151	156	171	247	—	—	2.5	2.5	23.1
	260	86	3	3	154	—	991	1 030	129	2 000	3 000	**NU2324R**	**NJ2324R**	**NUP2324R**	—	—	133	133	151	156	172	247	—	—	2.5	2.5	23.1
	260	106	3	3	154	—	1 030	1 120	139	2 000	3 000	**NU3324**	—	—	—	—	133	133	151	156	—	247	—	—	2.5	2.5	28.3
	310	72	5	5	170	260	841	770	98.7	1 900	2 600	**NU424**	**NJ424**	**NUP424**	**N424**	**NF424**	140	140	168	172	190	290	290	262	4	4	28.0
130	200	33	2	1.1	148	—	215	238	29.5	3 200	3 800	**NU1026**	—	**NUP1026**	—	—	139	136.5	146	151	—	191	—	—	2	1	3.77
	230	40	3	3	—	204	338	362	45.2	2 700	3 200	—	—	—	**N226**	**NF226**	143	—	—	—	168	217	217	208	2.5	2.5	(7.21)
	230	40	3	3	153.5	—	454	453	61.0	2 600	3 200	**NU226R**	**NJ226R**	**NUP226R**	—	—	143	143	151	158	168	217	—	—	2.5	2.5	6.60
	230	64	3	3	156	—	474	560	68.7	2 400	3 200	**NU2226**	**NJ2226**	**NUP2226**	—	—	143	143	151	158	168	217	—	—	2.5	2.5	11.2
	230	64	3	3	153.5	—	662	737	95.8	2 400	3 200	**NU2226R**	**NJ2226R**	**NUP2226R**	—	—	143	143	151	158	168	217	—	—	2.5	2.5	11.2
	230	80	3	3	156	—	689	857	107	2 100	3 200	**NU3226**	—	—	—	—	143	143	151	158	—	217	—	—	2.5	2.5	14.1
	280	58	4	4	—	243	699	667	85.7	2 100	2 700	—	—	—	**N326**	**NF326**	146	—	—	—	184	264	264	247	3	3	(16.4)
	280	58	4	4	167	—	771	736	94.1	2 000	2 700	**NU326R**	**NJ326R**	**NUP326R**	—	—	146	146	164	169	184	264	—	—	3	3	16.7
	280	93	4	4	167	—	1 050	1 130	138	1 800	2 700	**NU2326**	**NJ2326**	**NUP2326**	—	—	146	146	164	169	184	264	—	—	3	3	29.1
	280	93	4	4	167	—	1 150	1 230	150	1 800	2 700	**NU2326R**	**NJ2326R**	**NUP2326R**	—	—	146	146	164	169	186	264	—	—	3	3	29.1
	280	112	4	4	167	—	1 170	1 290	158	1 800	2 700	**NU3326**	—	—	—	—	146	146	164	169	—	264	—	—	3	3	34.6
	340	78	5	5	185	285	964	876	108	1 800	2 300	**NU426**	**NJ426**	**NUP426**	**N426**	**NF426**	150	150	183	187	208	320	320	287	4	4	36.1
140	210	33	2	1.1	158	—	220	250	30.5	3 000	3 600	**NU1028**	—	**NUP1028**	—	—	149	146.5	156	161	—	201	—	—	2	1	4.00
	250	42	3	3	—	221	406	421	55.5	2 400	2 900	—	—	—	**N228**	**NF228**	153	—	—	—	182	237	237	228	2.5	2.5	(8.78)
	250	42	3	3	169	—	491	514	67.5	2 400	2 900	**NU228R**	**NJ228R**	**NUP228R**	—	—	153	153	166	171	182	237	—	—	2.5	2.5	8.50
	250	68	3	3	169	—	583	671	84.3	2 200	2 900	**NU2228**	**NJ2228**	**NUP2228**	—	—	153	153	166	171	182	237	—	—	2.5	2.5	14.3
	250	68	3	3	169	—	716	835	106	2 200	2 900	**NU2228R**	**NJ2228R**	**NUP2228R**	—	—	153	153	166	171	182	237	—	—	2.5	2.5	14.3

[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d (140) ~ (160) mm



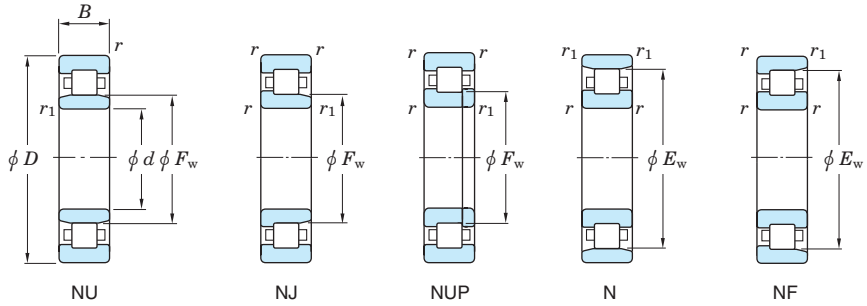


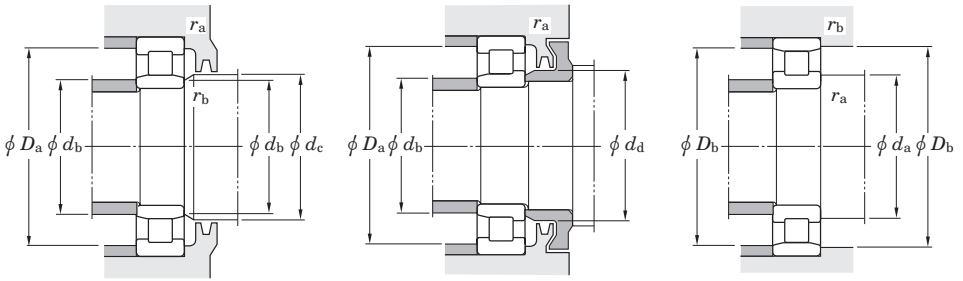
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
140	250	88	3	3	169	—	757	939	114	1 900	2 900	**NU3228**	—	—	—	—	153	153	166	171	—	237	—	—	2.5	2.5	18.5
	300	62	4	4	—	260	771	746	93.8	1 900	2 500	—	—	—	**N328**	**NF328**	156	—	—	—	198	284	284	264	3	3	(21.8)
	300	62	4	4	180	—	829	797	99.4	1 900	2 500	**NU328R**	**NJ328R**	**NUP328R**	—	—	156	156	176	182	198	284	—	—	3	3	21.8
	300	102	4	4	180	—	1 150	1 250	150	1 700	2 500	**NU2328**	**NJ2328**	**NUP2328**	—	—	156	156	176	182	198	284	—	—	3	3	36.8
	300	102	4	4	180	—	1 270	1 380	167	1 700	2 500	**NU2328R**	**NJ2328R**	**NUP2328R**	—	—	156	156	176	182	200	284	—	—	3	3	36.8
	300	118	4	4	180	—	1 360	1 550	185	1 700	2 500	**NU3328**	—	—	—	—	156	156	176	182	—	284	—	—	3	3	41.5
	360	82	5	5	198	302	1 090	1 020	124	1 600	2 200	**NU428**	**NJ428**	**NUP428**	**N428**	**NF428**	160	160	195	200	222	340	340	304	4	4	46.8
150	225	35	2.1	1.5	169.5	—	252	281	32.8	2 800	3 300	**NU1030**	—	**NUP1030**	—	—	161	158	167	173	—	214	—	—	2	1.5	4.83
	270	45	3	3	—	238	468	492	63.4	2 200	2 700	—	—	—	**N230**	**NF230**	163	—	—	—	196	257	257	245	2.5	2.5	(11.1)
	270	45	3	3	182	—	560	594	75.8	2 200	2 600	**NU230R**	**NJ230R**	**NUP230R**	—	—	163	163	179	184	196	257	—	—	2.5	2.5	10.7
	270	73	3	3	182	—	683	800	99.7	2 000	2 700	**NU2230**	**NJ2230**	**NUP2230**	—	—	163	163	179	184	196	257	—	—	2.5	2.5	18.7
	270	73	3	3	182	—	828	982	120	2 000	2 600	**NU2230R**	**NJ2230R**	**NUP2230R**	—	—	163	163	179	184	196	257	—	—	2.5	2.5	18.7
	270	96	3	3	182	—	939	1 200	143	1 800	2 700	**NU3230**	—	—	—	—	163	163	179	184	—	257	—	—	2.5	2.5	23.7
	320	65	4	4	—	277	829	807	99.1	1 800	2 300	—	—	—	**N330**	**NF330**	166	—	—	—	213	304	304	281	3	3	(25.6)
	320	65	4	4	193	—	948	922	115	1 700	2 300	**NU330R**	**NJ330R**	**NUP330R**	—	—	166	166	190	195	213	304	—	—	3	3	27.0
	320	108	4	4	193	—	1 270	1 400	167	1 600	2 300	**NU2330**	**NJ2330**	**NUP2330**	—	—	166	166	190	195	213	304	—	—	3	3	44.7
	320	108	4	4	193	—	1 450	1 600	187	1 500	2 300	**NU2330R**	**NJ2330R**	**NUP2330R**	—	—	166	166	190	195	213	304	—	—	3	3	44.7
	320	128	4	4	193	—	1 610	1 890	217	1 600	2 300	**NU3330**	—	—	—	—	166	166	190	195	—	304	—	—	3	3	51.4
	380	85	5	5	213	317	1 160	1 120	134	1 500	2 000	**NU430**	**NJ430**	**NUP430**	**N430**	**NF430**	170	170	210	216	237	360	360	319	4	4	53.3
160	240	38	2.1	1.5	180	—	297	330	42.8	2 600	3 000	**NU1032**	—	**NUP1032**	—	—	171	168	178	184	—	229	—	—	2	1.5	5.93
	290	48	3	3	—	255	535	568	71.3	2 100	2 500	—	—	—	**N232**	**NF232**	173	—	—	—	210	277	277	262	2.5	2.5	(13.9)
	290	48	3	3	195	—	624	666	83.3	2 000	2 400	**NU232R**	**NJ232R**	**NUP232R**	—	—	173	173	192	197	210	277	—	—	2.5	2.5	14.8
	290	80	3	3	195	—	790	939	113	1 800	2 500	**NU2232**	**NJ2232**	**NUP2232**	—	—	173	173	192	197	210	277	—	—	2.5	2.5	23.6
	290	80	3	3	193	—	1 010	1 190	141	1 800	2 400	**NU2232R**	**NJ2232R**	**NUP2232R**	—	—	173	173	192	197	210	277	—	—	2.5	2.5	23.6
	290	104	3	3	195	—	1 070	1 390	163	1 600	2 500	**NU3232**	—	—	—	—	173	173	192	197	—	277	—	—	2.5	2.5	29.8
	340	68	4	4	—	292	872	876	106	1 600	2 200	—	—	—	**N332**	**NF332**	176	—	—	—	228	324	324	296	3	3	(30.2)
	340	68	4	4	204	—	1 070	1 050	128	1 600	2 100	**NU332R**	**NJ332R**	**NUP332R**	—	—	176	176	200	211	228	324	—	—	3	3	32.0

[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d (160) ~ (190) mm



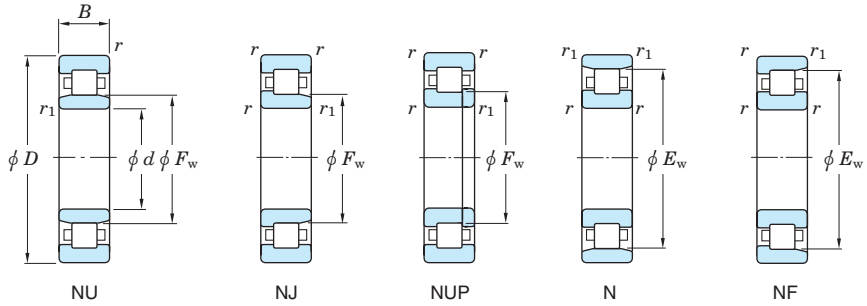


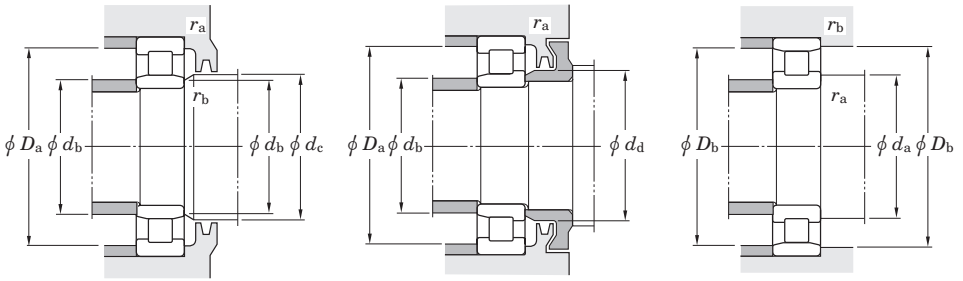
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.					Mounting dimensions (mm)											(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
160	340	114	4	4	208	—	1 340	1 520	178	1 400	2 200	**NU2332**	**NJ2332**	**NUP2332**	—	—	176	176	200	211	228	324	—	—	3	3	53.1
	340	114	4	4	204	—	1 640	1 820	212	1 400	2 100	**NU2332R**	**NJ2332R**	**NUP2332R**	—	—	176	176	200	211	228	324	—	—	3	3	53.1
	340	136	4	4	208	—	1 590	1 890	216	1 400	2 200	**NU3332**	—	—	—	—	176	176	200	211	—	324	—	—	3	3	61.5
170	260	42	2.1	2.1	193	—	347	400	50.5	2 400	2 800	**NU1034**	—	**NUP1034**	—	—	181	181	190	197	—	249	—	—	2	2	7.90
	310	52	4	4	—	272	596	637	78.4	1 900	2 300	—	—	—	**N234**	**NF234**	186	—	—	—	223	294	294	280	3	3	(17.2)
	310	52	4	4	207	—	754	802	98.7	1 900	2 200	**NU234R**	**NJ234R**	**NUP234R**	—	—	186	186	204	211	223	294	—	—	3	3	18.6
	310	86	4	4	208	—	896	1 080	127	1 700	2 300	**NU2234**	**NJ2234**	**NUP2234**	—	—	186	186	204	211	223	294	—	—	3	3	29.2
	310	86	4	4	205	—	1 210	1 410	166	1 700	2 200	**NU2234R**	**NJ2234R**	**NUP2234R**	—	—	186	186	204	211	223	294	—	—	3	3	29.2
	310	110	4	4	208	—	1 210	1 580	181	1 500	2 300	**NU3234**	—	—	—	—	186	186	204	211	—	294	—	—	3	3	36.2
	360	72	4	4	220	310	997	1 010	122	1 500	2 000	**NU334**	**NJ334**	**NUP334**	**N334**	**NF334**	186	186	216	223	241	344	344	314	3	3	38.6
	360	120	4	4	220	—	1 530	1 750	199	1 300	2 000	**NU2334**	**NJ2334**	**NUP2334**	—	—	186	186	216	223	241	344	—	—	3	3	62.6
	360	140	4	4	220	—	1 770	2 120	240	1 300	2 000	**NU3334**	—	—	—	—	186	186	216	223	—	344	—	—	3	3	70.8
180	280	46	2.1	2.1	205	—	447	503	63.2	2 200	2 600	**NU1036**	—	**NUP1036**	—	—	191	191	203	209	—	269	—	—	2	2	10.5
	320	52	4	4	—	282	618	677	82.2	1 800	2 200	—	—	—	**N236**	**NF236**	196	—	—	—	233	304	304	290	3	3	(18.0)
	320	52	4	4	217	—	783	852	104	1 800	2 100	**NU236R**	**NJ236R**	**NUP236R**	—	—	196	196	214	221	233	304	—	—	3	3	19.3
	320	86	4	4	218	—	929	1 140	133	1 600	2 200	**NU2236**	**NJ2236**	**NUP2236**	—	—	196	196	214	221	233	304	—	—	3	3	30.4
	320	86	4	4	215	—	1 260	1 510	175	1 600	2 100	**NU2236R**	**NJ2236R**	**NUP2236R**	—	—	196	196	214	221	233	304	—	—	3	3	30.4
	320	112	4	4	218	—	1 250	1 680	190	1 400	2 200	**NU3236**	—	—	—	—	196	196	214	221	—	304	—	—	3	3	38.4
	380	75	4	4	232	328	1 130	1 150	136	1 400	1 900	**NU336**	**NJ336**	**NUP336**	**N336**	**NF336**	196	196	227	235	255	364	364	332	3	3	42.6
	380	126	4	4	232	—	1 690	1 940	220	1 300	1 900	**NU2336**	**NJ2336**	**NUP2336**	—	—	196	196	227	235	255	364	—	—	3	3	73.0
	380	150	4	4	232	—	2 070	2 520	276	1 300	1 900	**NU3336**	—	—	—	—	196	196	227	235	—	364	—	—	3	3	84.4
190	290	46	2.1	2.1	215	—	460	530	65.7	2 100	2 500	**NU1038**	—	**NUP1038**	—	—	201	201	213	219	—	279	—	—	2	2	10.9
	340	55	4	4	—	299	694	768	91.3	1 700	2 000	—	—	—	**N238**	**NF238**	206	—	—	—	247	324	324	310	3	3	(21.5)
	340	55	4	4	230	—	869	954	114	1 700	2 000	**NU238R**	**NJ238R**	**NUP238R**	—	—	206	206	227	234	247	324	—	—	3	3	23.3
	340	92	4	4	231	—	1 040	1 290	146	1 500	2 000	**NU2238**	**NJ2238**	**NUP2238**	—	—	206	206	227	234	247	324	—	—	3	3	37.0
	340	92	4	4	228	—	1 380	1 670	189	1 500	2 000	**NU2238R**	**NJ2238R**	**NUP2238R**	—	—	206	206	227	234	247	324	—	—	3	3	37.0
	340	120	4	4	231	—	1 420	1 930	216	1 300	2 000	**NU3238**	—	—	—	—	206	206	227	234	—	324	—	—	3	3	46.8

[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d (190) ~ 240 mm



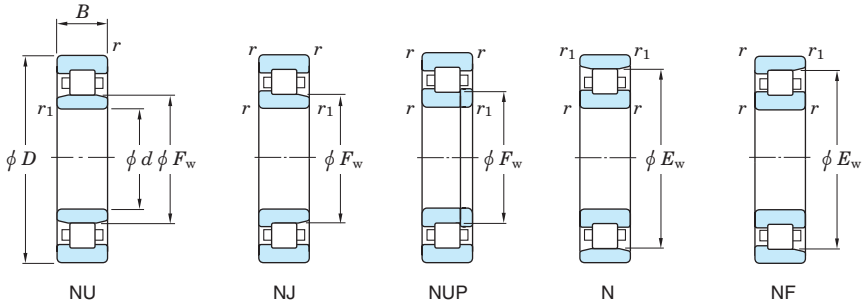


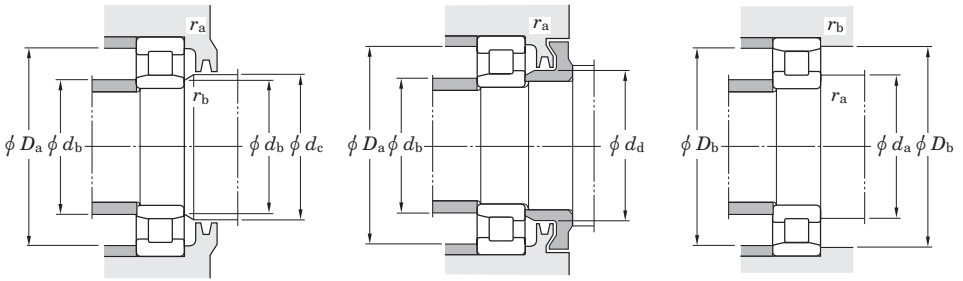
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
190	400	78	5	5	245	345	1 220	1 260	146	1 300	1 800	**NU338**	**NJ338**	**NUP338**	**N338**	**NF338**	210	210	240	248	268	380	380	349	4	4	49.9
	400	132	5	5	245	—	1 900	2 220	245	1 200	1 800	**NU2338**	**NJ2338**	**NUP2338**	—	—	210	210	240	248	268	380	—	—	4	4	84.7
	400	155	5	5	245	—	2 340	2 910	316	1 200	1 800	**NU3338**	—	—	—	—	210	210	240	248	—	380	—	—	4	4	96.5
200	310	51	2.1	2.1	229	—	487	582	71.0	1 900	2 300	**NU1040**	—	**NUP1040**	—	—	211	211	226	233	—	299	—	—	2	2	14.1
	360	58	4	4	—	316	775	865	102	1 600	1 900	—	—	—	**N240**	**NF240**	216	—	—	—	261	344	344	328	3	3	(25.7)
	360	58	4	4	243	—	958	1 060	124	1 600	1 900	**NU240R**	**NJ240R**	**NUP240R**	—	—	216	216	240	247	261	344	—	—	3	3	27.2
	360	98	4	4	244	—	1 190	1 490	169	1 400	1 900	**NU2240**	**NJ2240**	**NUP2240**	—	—	216	216	240	247	261	344	—	—	3	3	44.4
	360	98	4	4	241	—	1 530	1 870	211	1 400	1 900	**NU2240R**	**NJ2240R**	**NUP2240R**	—	—	216	216	240	247	261	344	—	—	3	3	44.4
	360	128	4	4	244	—	1 500	2 020	223	1 300	1 900	**NU3240**	—	—	—	—	216	216	240	247	—	344	—	—	3	3	56.2
	420	80	5	5	260	360	1 220	1 270	145	1 200	1 700	**NU340**	**NJ340**	**NUP340**	**N340**	**NF340**	220	220	254	263	283	400	400	364	4	4	56.2
	420	138	5	5	260	—	1 890	2 240	244	1 100	1 700	**NU2340**	**NJ2340**	**NUP2340**	—	—	220	220	254	263	283	400	—	—	4	4	97.4
	420	165	5	5	260	—	2 330	2 930	314	1 100	1 700	**NU3340**	—	—	—	—	220	220	250	258	—	400	—	—	4	4	113
220	340	56	3	3	250	—	637	748	88.1	1 700	2 000	**NU1044**	—	**NUP1044**	—	—	233	233	248	254	—	327	—	—	2.5	2.5	18.5
	400	65	4	4	270	350	949	1 080	123	1 400	1 700	**NU244**	**NJ244**	**NUP244**	**N244**	**NF244**	236	236	266	273	289	384	384	362	3	3	38.5
	400	108	4	4	270	—	1 420	1 810	196	1 200	1 700	**NU2244**	**NJ2244**	—	—	—	236	236	266	273	289	384	—	—	3	3	60.9
	400	144	4	4	270	—	2 040	2 880	308	1 100	1 700	**NU3244**	—	—	—	—	236	236	266	273	—	384	—	—	3	3	78.8
	460	88	5	5	284	396	1 490	1 570	176	1 100	1 500	**NU344**	**NJ344**	**NUP344**	**N344**	**NF344**	240	240	279	287	309	440	440	400	4	4	74.4
	460	145	5	5	284	—	2 260	2 690	287	990	1 500	**NU2344**	—	**NUP2344**	—	—	240	240	276	287	—	440	—	—	4	4	119
	460	180	5	5	284	—	2 660	3 300	347	990	1 500	**NU3344**	—	—	—	—	240	240	279	287	—	440	—	—	4	4	148
240	360	56	3	3	270	—	673	822	95.0	1 600	1 900	**NU1048**	—	**NUP1048**	—	—	253	253	268	275	—	347	—	—	2.5	2.5	20.1
	440	72	4	4	295	385	1 170	1 340	150	1 200	1 500	**NU248**	**NJ248**	**NUP248**	**N248**	**NF248**	256	256	293	298	316	424	424	397	3	3	52.1
	440	120	4	4	295	—	1 790	2 320	246	1 100	1 500	**NU2248**	**NJ2248**	—	—	—	256	256	293	298	316	424	—	—	3	3	82.5
	440	160	4	4	295	—	2 450	3 460	358	990	1 500	**NU3248**	—	—	—	—	256	256	293	298	—	424	—	—	3	3	107
	500	95	5	5	310	430	1 790	1 950	211	990	1 300	**NU348**	**NJ348**	**NUP348**	**N348**	**NF348**	260	260	305	313	337	480	480	434	4	4	94.6
	500	155	5	5	310	—	2 710	3 320	346	880	1 300	**NU2348**	—	**NUP2348**	—	—	260	260	303	313	—	480	—	—	4	4	152
	500	195	5	5	310	—	3 170	4 070	414	880	1 300	**NU3348**	—	—	—	—	260	260	305	313	—	480	—	—	4	4	189

[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d 260 ~ 460 mm



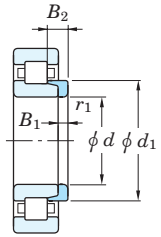


Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.					Mounting dimensions (mm)										(Refer.) Mass
d	D	B	r min.	r_1 min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	NU	NJ	NUP	N	NF	d_a min.	d_b min.	d_b max.	d_c min.	d_d min.	D_a max.	D_b max.	D_b min.	r_a max.	r_b max.	NU (N) (kg)
260	400	65	4	4	296	—	819	979	110	1 400	1 700	**NU1052**	—	**NUP1052**	—	—	276	276	292	300	—	384	—	—	3	3	29.2
	480	80	5	5	320	420	1 380	1 580	171	1 100	1 300	**NU252**	**NJ252**	**NUP252**	**N252**	**NF252**	280	280	318	323	343	460	460	432	4	4	69.0
	480	130	5	5	320	—	2 240	2 950	305	990	1 300	**NU2252**	**NJ2252**	—	—	—	280	280	318	323	343	460	—	—	4	4	107
	480	174	5	5	320	—	2 680	3 680	373	880	1 300	**NU3252**	—	—	—	—	280	280	318	323	—	460	—	—	4	4	139
	540	165	6	6	336	—	3 030	3 750	385	790	1 200	**NU2352**	—	**NUP2352**	—	—	284	284	327	339	—	516	—	—	5	5	185
	540	206	6	6	336	—	3 670	4 790	473	790	1 200	**NU3352**	—	—	—	—	284	284	330	339	—	516	—	—	5	5	232
280	420	65	4	4	316	—	841	1 030	114	1 300	1 500	**NU1056**	—	**NUP1056**	—	—	296	296	313	320	—	404	—	—	3	3	35.2
	500	80	5	5	340	440	1 430	1 680	179	1 000	1 200	**NU256**	**NJ256**	**NUP256**	**N256**	**NF256**	300	300	336	343	365	480	480	452	4	4	72.7
300	460	74	4	4	340	—	1 120	1 380	147	1 200	1 400	**NU1060**	—	**NUP1060**	—	—	316	316	337	344	—	444	—	—	3	3	44.1
	540	85	5	5	364	476	1 690	1 960	206	920	1 100	**NU260**	**NJ260**	**NUP260**	**N260**	**NF260**	320	320	361	368	392	520	520	487	4	4	90.7
320	480	74	4	4	360	—	1 150	1 450	152	1 100	1 300	**NU1064**	—	**NUP1064**	—	—	336	336	356	365	—	464	—	—	3	3	48.4
	580	92	5	5	390	510	1 920	2 270	232	840	1 000	**NU264**	**NJ264**	**NUP264**	**N264**	**NF264**	340	340	386	393	419	560	560	522	4	4	114
	670	112	7.5	7.5	425	—	2 460	2 880	287	650	870	**NU364**	—	—	—	—	352	352	419	428	—	638	638	575	6	6	199
340	520	82	5	5	385	—	1 370	1 750	183	980	1 200	**NU1068**	—	**NUP1068**	—	—	360	360	381	390	—	500	—	—	4	4	64.1
360	540	82	5	5	405	—	1 410	1 830	189	920	1 100	**NU1072**	—	**NUP1072**	—	—	380	380	401	410	—	520	—	—	4	4	67.1
380	560	82	5	5	425	—	1 440	1 920	195	860	1 000	**NU1076**	—	**NUP1076**	—	—	400	400	421	430	—	540	—	—	4	4	70.1
400	600	90	5	5	450	—	1 760	2 310	229	780	920	**NU1080**	—	**NUP1080**	—	—	420	420	446	455	—	580	—	—	4	4	91.0
420	620	90	5	5	470	—	1 750	2 320	228	730	860	**NU1084**	—	**NUP1084**	—	—	440	440	466	475	—	600	—	—	4	4	94.6
440	650	94	6	6	493	—	1 880	2 520	242	680	800	**NU1088**	—	**NUP1088**	—	—	464	464	489	498	—	626	—	—	5	5	109
460	680	100	6	6	516	—	2 000	2 730	259	630	750	**NU1092**	—	**NUP1092**	—	—	484	484	512	520	—	656	—	—	5	5	127

[Remarks] 1) Standard cage types used for the above bearings are shown in Table 1 earlier in this section. Please note that basic load ratings and limiting speeds shown above indicate the value applicable to machined cage. Consult JTEKT about bearings with pressed cage, since they may be different from bearings with machined cage in values above.

2) Bearing numbers of NU and NJ type bearings with mounted thrust collar (refer to specification table shown after this specification table) are NUJ and NH.

d 20 ~ (35) mm



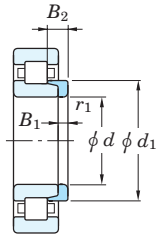
Thrust collar

Boundary dimensions (mm)					Thrust collar No.	(Refer.) Mass (kg)	Applicable bearing No.	
d	d_1	B_1	B_2	r_1 min.			NJ	NU
20	29.7	3	6.75	0.6	**HJ204**	0.012	NJ204	NU204
	29.8	3	5.5	0.6	**HJ204R**	0.011	NJ204R	NU204R
	30	3	7.5	0.6	**HJ2204**	0.012	NJ2204	NU2204
	29.8	3	6.5	0.6	**HJ2204R**	0.012	NJ2204R	NU2204R
	31.8	4	7.5	0.6	**HJ304**	0.017	NJ304	NU304
	31.4	4	6.5	0.6	**HJ304R**	0.017	NJ304R	NU304R
	31.8	4	8.5	0.6	**HJ2304**	0.020	NJ2304	NU2304
	31.4	4	7.5	0.6	**HJ2304R**	0.018	NJ2304R	NU2304R
25	34.7	3	7.25	0.6	**HJ205**	0.015	NJ205	NU205
	34.8	3	6	0.6	**HJ205R**	0.014	NJ205R	NU205R
	34.7	3	7.5	0.6	**HJ2205**	0.015	NJ2205	NU2205
	34.8	3	6.5	0.6	**HJ2205R**	0.014	NJ2205R	NU2205R
	39	4	8	1.1	**HJ305**	0.025	NJ305	NU305
	38.2	4	7	1.1	**HJ305R**	0.025	NJ305R	NU305R
	39	4	9	1.1	**HJ2305**	0.025	NJ2305	NU2305
	38.2	4	8	1.1	**HJ2305R**	0.026	NJ2305R	NU2305R
30	41.8	4	8.25	0.6	**HJ206**	0.025	NJ206	NU206
	41.4	4	7	0.6	**HJ206R**	0.025	NJ206R	NU206R
	41.8	4	8.5	0.6	**HJ2206**	0.025	NJ2206	NU2206
	41.4	4	7.5	0.6	**HJ2206R**	0.025	NJ2206R	NU2206R
	45.9	5	9.5	1.1	**HJ306**	0.039	NJ306	NU306
	45.1	5	8.5	1.1	**HJ306R**	0.042	NJ306R	NU306R
	45.9	5	11.5	1.1	**HJ2306**	0.039	NJ2306	NU2306
	45.1	5	9.5	1.1	**HJ2306R**	0.043	NJ2306R	NU2306R
	50.5	7	11.5	1.5	**HJ406**	0.080	NJ406	NU406
35	47.6	4	8	0.6	**HJ207**	0.030	NJ207	NU207
	48.2	4	7	0.6	**HJ207R**	0.033	NJ207R	NU207R
	47.6	4	8.5	0.6	**HJ2207**	0.030	NJ2207	NU2207

d (35) ~ (50) mm

Boundary dimensions (mm)					Thrust collar No.	(Refer.) Mass (kg)	Applicable bearing No.	
d	d_1	B_1	B_2	r_1 min.			NJ	NU
35	48.2	4	8.5	0.6	**HJ2207R**	0.035	NJ2207R	NU2207R
	50.8	6	11	1.1	**HJ307**	0.056	NJ307	NU307
	51.1	6	9.5	1.1	**HJ307R**	0.060	NJ307R	NU307R
	50.8	6	14	1.1	**HJ2307**	0.056	NJ2307	NU2307
	51.1	6	11	1.1	**HJ2307R**	0.062	NJ2307R	NU2307R
	59	8	13	1.5	**HJ407**	0.120	NJ407	NU407
40	54.2	5	9	1.1	**HJ208**	0.046	NJ208	NU208
	54.1	5	8.5	1.1	**HJ208R**	0.049	NJ208R	NU208R
	54.2	5	9.5	1.1	**HJ2208**	0.046	NJ2208	NU2208
	54.1	5	9	1.1	**HJ2208R**	0.050	NJ2208R	NU2208R
	58.4	7	12.5	1.5	**HJ308**	0.083	NJ308	NU308
	57.7	7	11	1.5	**HJ308R**	0.088	NJ308R	NU308R
	58.4	7	14.5	1.5	**HJ2308**	0.083	NJ2308	NU2308
	57.7	7	12.5	1.5	**HJ2308R**	0.091	NJ2308R	NU2308R
	64.8	8	13	2	**HJ408**	0.140	NJ408	NU408
45	59	5	9.5	1.1	**HJ209**	0.053	NJ209	NU209
	59.1	5	8.5	1.1	**HJ209R**	0.055	NJ209R	NU209R
	59	5	9.5	1.1	**HJ2209**	0.053	NJ2209	NU2209
	59.1	5	9	1.1	**HJ2209R**	0.055	NJ2209R	NU2209R
	64	7	12.5	1.5	**HJ309**	0.099	NJ309	NU309
	64.5	7	11.5	1.5	**HJ309R**	0.110	NJ309R	NU309R
	64	7	15	1.5	**HJ2309**	0.099	NJ2309	NU2309
	64.5	7	13	1.5	**HJ2309R**	0.113	NJ2309R	NU2309R
	71.8	8	13.5	2	**HJ409**	0.175	NJ409	NU409
50	64.6	5	10	1.1	**HJ210**	0.063	NJ210	NU210
	64.1	5	9	1.1	**HJ210R**	0.061	NJ210R	NU210R
	64.6	5	9.5	1.1	**HJ2210**	0.063	NJ2210	NU2210
	64.1	5	9	1.1	**HJ2210R**	0.061	NJ2210R	NU2210R
	71	8	14	2	**HJ310**	0.142	NJ310	NU310

d (50) ~ (65) mm



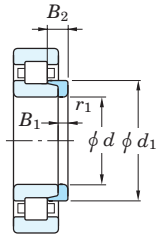
Thrust collar

Boundary dimensions (mm)					Thrust collar No.	(Refer.) Mass (kg)	Applicable bearing No.	
d	d_1	B_1	B_2	r_1 min.			NJ	NU
50	71.4	8	13	2	**HJ310R**	0.151	NJ310R	NU310R
	71	8	17	2	**HJ2310**	0.142	NJ2310	NU2310
	71.4	8	14.5	2	**HJ2310R**	0.155	NJ2310R	NU2310R
	78.8	9	14.5	2.1	**HJ410**	0.230	NJ410	NU410
55	70.8	6	11	1.1	**HJ211**	0.084	NJ211	NU211
	70.9	6	9.5	1.1	**HJ211R**	0.087	NJ211R	NU211R
	70.8	6	11	1.1	**HJ2211**	0.084	NJ2211	NU2211
	70.9	6	10	1.1	**HJ2211R**	0.088	NJ2211R	NU2211R
	77.2	9	15	2	**HJ311**	0.182	NJ311	NU311
	77.6	9	14	2	**HJ311R**	0.195	NJ311R	NU311R
	77.2	9	18.5	2	**HJ2311**	0.182	NJ2311	NU2311
	77.6	9	15.5	2	**HJ2311R**	0.200	NJ2311R	NU2311R
	85.2	10	16.5	2.1	**HJ411**	0.290	NJ411	NU411
60	78.4	6	11	1.5	**HJ212**	0.108	NJ212	NU212
	77.7	6	10	1.5	**HJ212R**	0.108	NJ212R	NU212R
	78.4	6	11	1.5	**HJ2212**	0.108	NJ2212	NU2212
	77.7	6	10	1.5	**HJ2212R**	0.108	NJ2212R	NU2212R
	84.2	9	15.5	2.1	**HJ312**	0.220	NJ312	NU312
	84.5	9	14.5	2.1	**HJ312R**	0.231	NJ312R	NU312R
	84.2	9	19	2.1	**HJ2312**	0.220	NJ2312	NU2312
	84.5	9	16	2.1	**HJ2312R**	0.237	NJ2312R	NU2312R
	91.8	10	16.5	2.1	**HJ412**	0.340	NJ412	NU412
65	84.8	6	11	1.5	**HJ213**	0.123	NJ213	NU213
	84.5	6	10	1.5	**HJ213R**	0.129	NJ213R	NU213R
	84.8	6	11.5	1.5	**HJ2213**	0.123	NJ2213	NU2213
	84.5	6	10.5	1.5	**HJ2213R**	0.131	NJ2213R	NU2213R
	91	10	17	2.1	**HJ313**	0.280	NJ313	NU313
	90.6	10	15.5	2.1	**HJ313R**	0.288	NJ313R	NU313R

d (65) ~ (80) mm

Boundary dimensions (mm)					Thrust collar No.	(Refer.) Mass (kg)	Applicable bearing No.	
d	d_1	B_1	B_2	r_1 min.			NJ	NU
65	91	10	20	2.1	**HJ2313**	0.280	NJ2313	NU2313
	90.6	10	18	2.1	**HJ2313R**	0.298	NJ2313R	NU2313R
	98.5	11	18	2.1	**HJ413**	0.420	NJ413	NU413
70	89.6	7	12.5	1.5	**HJ214**	0.150	NJ214	NU214
	89.5	7	11	1.5	**HJ214R**	0.157	NJ214R	NU214R
	89.6	7	12.5	1.5	**HJ2214**	0.150	NJ2214	NU2214
	89.5	7	11.5	1.5	**HJ2214R**	0.158	NJ2214R	NU2214R
	98	10	17.5	2.1	**HJ314**	0.330	NJ314	NU314
	97.5	10	15.5	2.1	**HJ314R**	0.330	NJ314R	NU314R
	98	10	20.5	2.1	**HJ2314**	0.330	NJ2314	NU2314
	97.5	10	18.5	2.1	**HJ2314R**	0.345	NJ2314R	NU2314R
	110.5	12	20	3	**HJ414**	0.605	NJ414	NU414
75	94	7	12.5	1.5	**HJ215**	0.156	NJ215	NU215
	94.5	7	11	1.5	**HJ215R**	0.166	NJ215R	NU215R
	94	7	12.5	1.5	**HJ2215**	0.156	NJ2215	NU2215
	94.5	7	11.5	1.5	**HJ2215R**	0.167	NJ2215R	NU2215R
	104.2	11	18.5	2.1	**HJ315**	0.400	NJ315	NU315
	104.2	11	16.5	2.1	**HJ315R**	0.410	NJ315R	NU315R
	104.2	11	21.5	2.1	**HJ2315**	0.400	NJ2315	NU2315
	104.2	11	19.5	2.1	**HJ2315R**	0.430	NJ2315R	NU2315R
	116	13	21.5	3	**HJ415**	0.710	NJ415	NU415
80	101.2	8	13.5	2	**HJ216**	0.207	NJ216	NU216
	101.6	8	12.5	2	**HJ216R**	0.222	NJ216R	NU216R
	101.2	8	13.5	2	**HJ2216**	0.207	NJ2216	NU2216
	101.6	8	12.5	2	**HJ2216R**	0.222	NJ2216R	NU2216R
	111.8	11	19.5	2.1	**HJ316**	0.470	NJ316	NU316
	110.6	11	17	2.1	**HJ316R**	0.460	NJ316R	NU316R
	111.8	11	23	2.1	**HJ2316**	0.470	NJ2316	NU2316
	110.6	11	20	2.1	**HJ2316R**	0.480	NJ2316R	NU2316R

d (80) ~ (100) mm



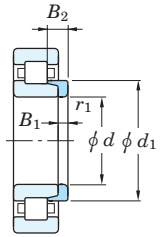
Thrust collar

Boundary dimensions (mm)					Thrust collar No.	(Refer.) Mass (kg)	Applicable bearing No.	
d	d_1	B_1	B_2	r_1 min.			NJ	NU
80	122	13	22	3	**HJ416**	0.780	NJ416	NU416
85	108.2	8	14	2	**HJ217**	0.250	NJ217	NU217
	107.6	8	12.5	2	**HJ217R**	0.250	NJ217R	NU217R
	108.2	8	14	2	**HJ2217**	0.250	NJ2217	NU2217
	107.6	8	13	2	**HJ2217R**	0.252	NJ2217R	NU2217R
	117.5	12	20.5	3	**HJ317**	0.560	NJ317	NU317
	117.9	12	18.5	3	**HJ317R**	0.575	NJ317R	NU317R
	117.5	12	24	3	**HJ2317**	0.560	NJ2317	NU2317
	117.9	12	22	3	**HJ2317R**	0.595	NJ2317R	NU2317R
	126	14	24	4	**HJ417**	0.880	NJ417	NU417
90	114.2	9	15	2	**HJ218**	0.305	NJ218	NU218
	114.4	9	14	2	**HJ218R**	0.320	NJ218R	NU218R
	114.2	9	16	2	**HJ2218**	0.305	NJ2218	NU2218
	114.4	9	15	2	**HJ2218R**	0.325	NJ2218R	NU2218R
	125	12	21	3	**HJ318**	0.630	NJ318	NU318
	124.2	12	18.5	3	**HJ318R**	0.630	NJ318R	NU318R
	125	12	26	3	**HJ2318**	0.630	NJ2318	NU2318
	124.2	12	22	3	**HJ2318R**	0.660	NJ2318R	NU2318R
	137	14	24	4	**HJ418**	1.05	NJ418	NU418
95	121	9	15.5	2.1	**HJ219**	0.352	NJ219	NU219
	120.6	9	14	2.1	**HJ219R**	0.355	NJ219R	NU219R
	121	9	16.5	2.1	**HJ2219**	0.352	NJ2219	NU2219
	120.6	9	15.5	2.1	**HJ2219R**	0.365	NJ2219R	NU2219R
	132	13	22.5	3	**HJ319**	0.760	NJ319	NU319
	132.2	13	20.5	3	**HJ319R**	0.785	NJ319R	NU319R
	132	13	26.5	3	**HJ2319**	0.760	NJ2319	NU2319
	147	15	25.5	4	**HJ419**	1.30	NJ419	NU419
100	128	10	17	2.1	**HJ220**	0.444	NJ220	NU220

d (100) ~ 120 mm

Boundary dimensions (mm)					Thrust collar No.	(Refer.) Mass (kg)	Applicable bearing No.	
d	d_1	B_1	B_2	r_1 min.			NJ	NU
100	127.5	10	15	2.1	**HJ220R**	0.435	NJ220R	NU220R
	128	10	18	2.1	**HJ2220**	0.444	NJ2220	NU2220
	127.5	10	16	2.1	**HJ2220R**	0.450	NJ2220R	NU2220R
	140.5	13	22.5	3	**HJ320**	0.895	NJ320	NU320
	139.6	13	20.5	3	**HJ320R**	0.890	NJ320R	NU320R
	140.5	13	27.5	3	**HJ2320**	0.895	NJ2320	NU2320
	139.6	13	23.5	3	**HJ2320R**	0.920	NJ2320R	NU2320R
	153.5	16	27	4	**HJ420**	1.50	NJ420	NU420
105	135	10	17.5	2.1	**HJ221**	0.505	NJ221	NU221
	147	13	22.5	3	**HJ321**	0.970	NJ321	NU321
	159.5	16	27	4	**HJ421**	1.65	NJ421	NU421
110	141.5	11	18.5	2.1	**HJ222**	0.615	NJ222	NU222
	141.7	11	17	2.1	**HJ222R**	0.620	NJ222R	NU222R
	141.5	11	20.5	2.1	**HJ2222**	0.615	NJ2222	NU2222
	141.7	11	19.5	2.1	**HJ2222R**	0.645	NJ2222R	NU2222R
	155.5	14	23	3	**HJ322**	1.17	NJ322	NU322
	155.8	14	22	3	**HJ322R**	1.21	NJ322R	NU322R
	155.5	14	28	3	**HJ2322**	1.17	NJ2322	NU2322
	155.8	14	26.5	3	**HJ2322R**	1.27	NJ2322R	NU2322R
	171	17	29.5	4	**HJ422**	2.10	NJ422	NU422
120	153	11	19	2.1	**HJ224**	0.715	NJ224	NU224
	153.4	11	17	2.1	**HJ224R**	0.710	NJ224R	NU224R
	153	11	22	2.1	**HJ2224**	0.715	NJ2224	NU2224
	153.4	11	20	2.1	**HJ2224R**	0.745	NJ2224R	NU2224R
	168.5	14	23.5	3	**HJ324**	1.40	NJ324	NU324
	168.6	14	22.5	3	**HJ324R**	1.41	NJ324R	NU324R
	168.5	14	28	3	**HJ2324**	1.40	NJ2324	NU2324
	168.6	14	26	3	**HJ2324R**	1.46	NJ2324R	NU2324R
	188	17	30.5	5	**HJ424**	2.60	NJ424	NU424

d 130 ~ (160) mm



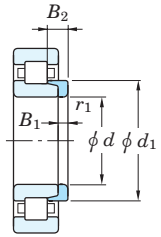
Thrust collar

Boundary dimensions (mm)					Thrust collar No.	(Refer.) Mass (kg)	Applicable bearing No.	
d	d_1	B_1	B_2	r_1 min.			NJ	NU
130	165.5	11	19	3	**HJ226**	0.840	NJ226	NU226
	164.2	11	17	3	**HJ226R**	0.790	NJ226R	NU226R
	165.5	11	25	3	**HJ2226**	0.840	NJ2226	NU2226
	164.2	11	21	3	**HJ2226R**	0.840	NJ2226R	NU2226R
	182	14	24	4	**HJ326**	1.62	NJ326	NU326
	182.3	14	23	4	**HJ326R**	1.65	NJ326R	NU326R
	182	14	29.5	4	**HJ2326**	1.62	NJ2326	NU2326
	182.3	14	28	4	**HJ2326R**	1.73	NJ2326R	NU2326R
	205	18	32	5	**HJ426**	3.30	NJ426	NU426
140	179.5	11	19	3	**HJ228**	1.00	NJ228	NU228
	180	11	18	3	**HJ228R**	0.990	NJ228R	NU228R
	179.5	11	25	3	**HJ2228**	1.00	NJ2228	NU2228
	180	11	23	3	**HJ2228R**	1.07	NJ2228R	NU2228R
	196	15	26	4	**HJ328**	1.93	NJ328	NU328
	196	15	25	4	**HJ328R**	2.04	NJ328R	NU328R
	196	15	33.5	4	**HJ2328**	1.98	NJ2328	NU2328
	196	15	31	4	**HJ2328R**	2.14	NJ2328R	NU2328R
	219	18	33	5	**HJ428**	3.75	NJ428	NU428
150	193	12	20.5	3	**HJ230**	1.24	NJ230	NU230
	193.7	12	19.5	3	**HJ230R**	1.26	NJ230R	NU230R
	193	12	26.5	3	**HJ2230**	1.24	NJ2230	NU2230
	193.7	12	24.5	3	**HJ2230R**	1.35	NJ2230R	NU2230R
	210	15	26.5	4	**HJ330**	2.37	NJ330	NU330
	210	15	25	4	**HJ330R**	2.35	NJ330R	NU330R
	210	15	34	4	**HJ2330**	2.37	NJ2330	NU2330
	210	15	31.5	4	**HJ2330R**	2.48	NJ2330R	NU2330R
	234	20	36.5	5	**HJ430**	4.70	NJ430	NU430
160	207	12	21	3	**HJ232**	1.48	NJ232	NU232
	207.3	12	20	3	**HJ232R**	1.48	NJ232R	NU232R

d (160) ~ (200) mm

Boundary dimensions (mm)					Thrust collar No.	(Refer.) Mass (kg)	Applicable bearing No.	
d	d_1	B_1	B_2	r_1 min.			NJ	NU
160	205	12	28	3	**HJ2232**	1.48	NJ2232	NU2232
	206.1	12	24.5	3	**HJ2232R**	1.55	NJ2232R	NU2232R
	225	15	28	4	**HJ332**	2.75	NJ332	NU332
	222.1	15	25	4	**HJ332R**	2.59	NJ332R	NU332R
	225	15	37	4	**HJ2332**	2.75	NJ2332	NU2332
	222.1	15	32	4	**HJ2332R**	2.76	NJ2332R	NU2332R
170	220.5	12	22	4	**HJ234**	1.70	NJ234	NU234
	220.8	12	20	4	**HJ234R**	1.70	NJ234R	NU234R
	219	12	29	4	**HJ2234**	1.70	NJ2234	NU2234
	219.5	12	24	4	**HJ2234R**	1.79	NJ2234R	NU2234R
	238	16	29.5	4	**HJ334**	3.25	NJ334	NU334
	238	16	38.5	4	**HJ2334**	3.25	NJ2334	NU2334
180	230.5	12	22	4	**HJ236**	1.80	NJ236	NU236
	230.8	12	20	4	**HJ236R**	1.79	NJ236R	NU236R
	229	12	29	4	**HJ2236**	1.80	NJ2236	NU2236
	229.5	12	24	4	**HJ2236R**	1.88	NJ2236R	NU2236R
	252	17	30.5	4	**HJ336**	3.85	NJ336	NU336
	252	17	40	4	**HJ2336**	3.85	NJ2336	NU2336
190	244.5	13	23.5	4	**HJ238**	2.20	NJ238	NU238
	244.5	13	21.5	4	**HJ238R**	2.19	NJ238R	NU238R
	243	13	31.5	4	**HJ2238**	2.20	NJ2238	NU2238
	243.2	13	26.5	4	**HJ2238R**	2.31	NJ2238R	NU2238R
	265	18	32	5	**HJ338**	4.45	NJ338	NU338
	265	18	41.5	5	**HJ2338**	4.45	NJ2338	NU2338
200	258	14	25	4	**HJ240**	2.60	NJ240	NU240
	258.2	14	23	4	**HJ240R**	2.65	NJ240R	NU240R
	258	14	34	4	**HJ2240**	2.60	NJ2240	NU2240
	256.9	14	28	4	**HJ2240R**	2.78	NJ2240R	NU2240R
	280	18	33	5	**HJ340**	5.00	NJ340	NU340

d (200) ~ 320 mm

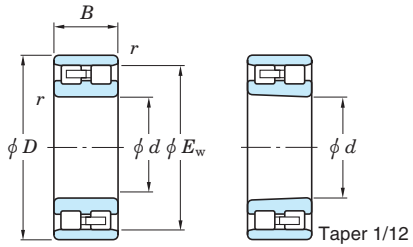


Thrust collar

Boundary dimensions (mm)					Thrust collar No.	(Refer.) Mass (kg)	Applicable bearing No.	
d	d_1	B_1	B_2	r_1 min.			NJ	NU
200	280	18	44.5	5	**HJ2340**	5.00	NJ2340	NU2340
220	286	15	27.5	4	**HJ244**	3.55	NJ244	NU244
	286	15	36.5	4	**HJ2244**	3.55	NJ2244	NU2244
	307	20	36	5	**HJ344**	7.05	NJ344	NU344
240	313	16	29.5	4	**HJ248**	4.65	NJ248	NU248
	313	16	38.5	4	**HJ2248**	4.65	NJ2248	NU2248
	335	22	39.5	5	**HJ348**	8.20	NJ348	NU348
260	340	18	33	5	**HJ252**	6.20	NJ252	NU252
	340	18	40.5	5	**HJ2252**	6.20	NJ2252	NU2252
280	360	18	33	5	**HJ256**	7.15	NJ256	NU256
300	387	20	34.5	5	**HJ260**	7.40	NJ260	NU260
320	415	21	37	5	**HJ264**	11.3	NJ264	NU264

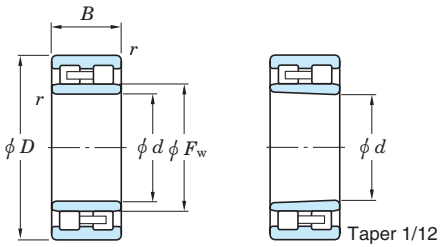
d 25 ~ (110) mm

NN

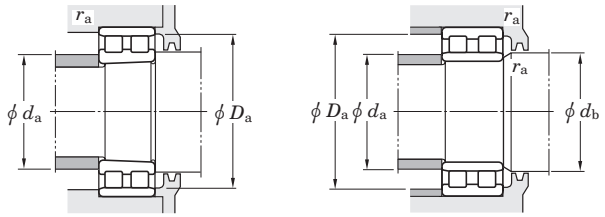


Cylindrical bore Tapered bore

NNU



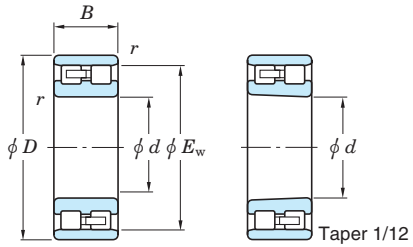
Cylindrical bore Tapered bore



Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. NN		Bearing No. NNU		Mounting dimensions (mm)						(Refer.) Mass (kg)	
d	D	B	r min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	Cylindrical bore	Tapered bore	d_a min.	d_a max.	d_b min.	D_a max.	D_a min.	r_a max.	Cylindrical bore	Tapered bore
25	47	16	0.6	—	41.3	32.2	30.0	5.20	14 000	17 000	**NN3005**	**NN3005K**	—	—	29	—	—	43	42	0.6	0.127	0.123
30	55	19	1	—	48.5	46.0	44.1	4.95	12 000	14 000	**NN3006**	**NN3006K**	—	—	35	—	—	50	49	1	0.198	0.192
35	62	20	1	—	55	49.1	50.0	5.65	10 000	12 000	**NN3007**	**NN3007K**	—	—	40	—	—	57	56	1	0.253	0.246
40	68	21	1	—	61	52.0	55.9	6.35	9 100	11 000	**NN3008**	**NN3008K**	—	—	45	—	—	63	62	1	0.307	0.298
45	75	23	1	—	67.5	67.1	71.9	8.75	8 300	9 900	**NN3009**	**NN3009K**	—	—	50	—	—	70	69	1	0.404	0.382
50	80	23	1	—	72.5	66.4	72.6	8.85	7 600	9 100	**NN3010**	**NN3010K**	—	—	55	—	—	75	74	1	0.429	0.415
55	90	26	1.1	—	81	89.6	101	13.2	6 800	8 200	**NN3011**	**NN3011K**	—	—	61.5	—	—	83.5	82	1	0.637	0.618
60	95	26	1.1	—	86.1	91.6	106	13.9	6 400	7 700	**NN3012**	**NN3012K**	—	—	66.5	—	—	88.5	87	1	0.685	0.664
65	100	26	1.1	—	91	93.6	111	14.6	6 000	7 200	**NN3013**	**NN3013K**	—	—	71.5	—	—	93.5	92	1	0.728	0.705
70	110	30	1.1	—	100	122	148	20.6	5 500	6 500	**NN3014**	**NN3014K**	—	—	76.5	—	—	103.5	101	1	1.04	1.02
75	115	30	1.1	—	105	124	155	21.5	5 200	6 200	**NN3015**	**NN3015K**	—	—	81.5	—	—	108.5	106	1	1.11	1.08
80	125	34	1.1	—	113	149	186	26.6	4 800	5 800	**NN3016**	**NN3016K**	—	—	86.5	—	—	118.5	114	1	1.55	1.50
85	130	34	1.1	—	118	152	194	27.3	4 600	5 500	**NN3017**	**NN3017K**	—	—	91.5	—	—	123.5	119	1	1.63	1.58
90	140	37	1.5	—	127	179	228	29.3	4 200	5 100	**NN3018**	**NN3018K**	—	—	98	—	—	132	129	1.5	2.07	2.01
95	145	37	1.5	—	132	188	246	31.3	4 100	4 900	**NN3019**	**NN3019K**	—	—	103	—	—	137	134	1.5	2.17	2.10
100	140	40	1.1	113	—	173	258	32.9	4 000	4 800	—	—	**NNU4920**	**NNU4920K**	106.5	111	115	133.5	—	1	1.95	1.87
	150	37	1.5	—	137	196	265	33.3	3 900	4 700	**NN3020**	**NN3020K**	—	—	108	—	—	142	139	1.5	2.28	2.21
105	145	40	1.1	118	—	196	306	40.2	3 900	4 600	—	—	**NNU4921**	**NNU4921K**	111.5	116	120	138.5	—	1	2.00	1.91
	160	41	2	—	146	247	322	42.5	3 700	4 400	**NN3021**	**NN3021K**	—	—	114	—	—	151	148	2	2.88	2.81
110	150	40	1.1	123	—	204	326	42.4	3 700	4 500	—	—	**NNU4922**	**NNU4922K**	116.5	121	125	143.5	—	1	2.10	2.01

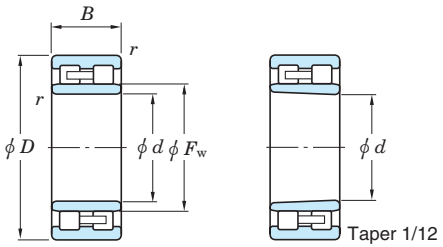
d (110) ~ (260) mm

NN

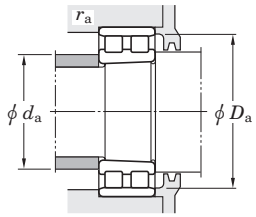


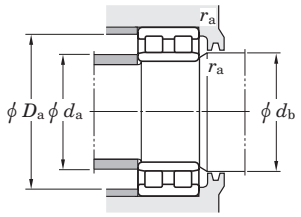
Cylindrical bore Tapered bore

NNU



Cylindrical bore Tapered bore

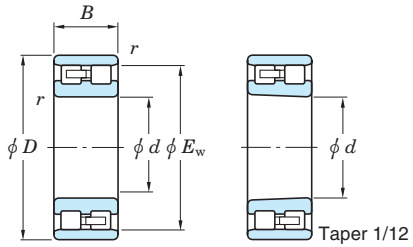




Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. NN		Bearing No. NNU		Mounting dimensions (mm)						(Refer.) Mass (kg)	
d	D	B	r min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	Cylindrical bore	Tapered bore	d_a min.	d_a max.	d_b min.	D_a max.	D_a min.	r_a max.	Cylindrical bore	Tapered bore
110	170	45	2	—	155	278	361	47.9	3 500	4 200	**NN3022**	**NN3022K**	—	—	119	—	—	161	157	2	3.65	3.56
120	165	45	1.1	134.5	—	234	373	47.6	3 400	4 000	—	—	**NNU4924**	**NNU4924K**	126.5	132	137	158.5	—	1	2.90	2.77
	180	46	2	—	165	291	392	51.1	3 200	3 900	**NN3024**	**NN3024K**	—	—	129	—	—	171	167	2	4.00	3.87
130	180	50	1.5	146	—	269	428	50.2	3 100	3 700	—	—	**NNU4926**	**NNU4926K**	138	143.5	148	172	—	1.5	3.90	3.73
	200	52	2	—	182	356	476	57.7	2 900	3 500	**NN3026**	**NN3026K**	—	—	139	—	—	191	183	2	5.94	5.76
140	190	50	1.5	156	—	277	456	52.5	2 900	3 500	—	—	**NNU4928**	**NNU4928K**	148	153.5	158	182	—	1.5	4.15	3.97
	210	53	2	—	192	372	516	61.5	2 700	3 300	**NN3028**	**NN3028K**	—	—	149	—	—	201	194	2	6.41	6.21
150	210	60	2	168.5	—	430	692	80.7	2 600	3 100	—	—	**NNU4930**	**NNU4930K**	159	166	171	201	—	2	6.50	6.22
	225	56	2.1	—	206	418	587	70.1	2 500	3 000	**NN3030**	**NN3030K**	—	—	161	—	—	214	208	2	7.74	7.50
160	220	60	2	178.5	—	425	695	79.8	2 500	3 000	—	—	**NNU4932**	**NNU4932K**	169	176	182	211	—	2	6.95	6.65
	240	60	2.1	—	219	499	695	79.6	2 400	2 800	**NN3032**	**NN3032K**	—	—	171	—	—	229	221	2	9.38	9.08
170	230	60	2	188.5	—	451	763	86.4	2 300	2 800	—	—	**NNU4934**	**NNU4934K**	179	186	192	221	—	2	7.20	6.88
	260	67	2.1	—	236	592	824	105	2 200	2 600	**NN3034**	**NN3034K**	—	—	181	—	—	249	238	2	12.8	12.4
180	250	69	2	202	—	572	964	117	2 100	2 600	—	—	**NNU4936**	**NNU4936K**	189	199.5	205	241	—	2	10.5	10.1
	280	74	2.1	—	255	705	958	118	2 000	2 400	**NN3036**	**NN3036K**	—	—	191	—	—	269	257	2	16.8	16.3
190	260	69	2	210	—	581	996	119	2 000	2 400	—	—	**NNU4938**	**NNU4938K**	199	207	215	251	—	2	11.0	10.5
	290	75	2.1	—	265	752	1 020	128	1 900	2 300	**NN3038**	**NN3038K**	—	—	201	—	—	279	267	2	17.6	17.1
200	280	80	2.1	223	—	636	1 050	125	1 900	2 300	—	—	**NNU4940**	**NNU4940K**	211	219.5	228	269	—	2	15.4	14.7
	310	82	2.1	—	282	793	1 120	137	1 700	2 100	**NN3040**	**NN3040K**	—	—	211	—	—	299	285	2	22.5	21.8
220	300	80	2.1	244	—	701	1 220	145	1 700	2 000	—	—	**NNU4944**	**NNU4944K**	231	241	248	289	—	2	16.7	16.0
	340	90	3	—	310	944	1 370	163	1 600	1 900	**NN3044**	**NN3044K**	—	—	233	—	—	327	313	2.5	29.3	28.4
240	320	80	2.1	263	—	736	1 340	155	1 600	1 900	—	—	**NNU4948**	**NNU4948K**	251	260	269	309	—	2	18.0	17.2
	360	92	3	—	330	1 090	1 590	184	1 400	1 700	**NN3048**	**NN3048K**	—	—	253	—	—	347	333	2.5	32.8	31.8
260	360	100	2.1	287	—	1 180	2 050	228	1 400	1 700	—	—	**NNU4952**	**NNU4952K**	271	284	296	349	—	2	31.4	30.0

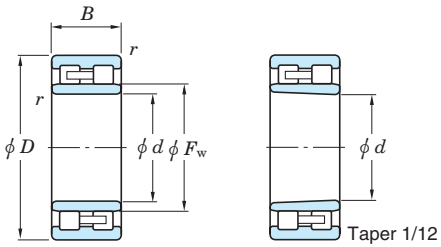
d (260) ~ 460 mm

NN

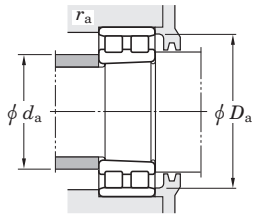


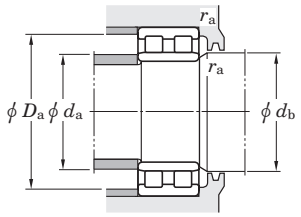
Cylindrical bore Tapered bore

NNU



Cylindrical bore Tapered bore





Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. NN		Bearing No. NNU		Mounting dimensions (mm)						(Refer.) Mass (kg)	
d	D	B	r min.	F_w	E_w	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	Cylindrical bore	Tapered bore	d_a min.	d_a max.	d_b min.	D_a max.	D_a min.	r_a max.	Cylindrical bore	Tapered bore
260	400	104	4	—	364	1 290	1 830	204	1 300	1 500	**NN3052**	**NN3052K**	—	—	276	—	—	384	367	3	47.4	46.0
280	380	100	2.1	308	—	1 220	2 200	239	1 300	1 500	—	—	**NNU4956**	**NNU4956K**	291	305	316	369	—	2	33.1	31.6
	420	106	4	—	384	1 370	2 010	220	1 200	1 400	**NN3056**	**NN3056K**	—	—	296	—	—	404	387	3	51.2	49.6
300	420	118	3	339	—	1 470	2 720	285	1 100	1 300	—	—	**NNU4960**	**NNU4960K**	313	335	343	407	—	2.5	51.9	49.7
	460	118	4	—	418	1 610	2 460	266	1 100	1 300	**NN3060**	**NN3060K**	—	—	316	—	—	444	421	3	70.8	68.7
320	440	118	3	352	—	1 530	2 750	286	1 100	1 300	—	—	**NNU4964**	**NNU4964K**	333	348	363	427	—	2.5	53.7	51.4
	480	121	4	—	438	1 690	2 670	283	980	1 200	**NN3064**	**NN3064K**	—	—	336	—	—	464	442	3	76.4	74.0
340	460	118	3	372	—	1 580	2 930	301	990	1 200	—	—	**NNU4968**	**NNU4968K**	353	368	383	447	—	2.5	56.8	54.3
	520	133	5	—	473	2 090	3 090	345	880	1 100	**NN3068**	**NN3068K**	—	—	360	—	—	500	477	4	101	97.8
360	540	134	5	—	493	1 950	3 090	315	830	990	**NN3072**	**NN3072K**	—	—	380	—	—	520	497	4	107	104
380	560	135	5	—	510	2 050	3 350	337	780	940	**NN3076**	**NN3076K**	—	—	400	—	—	540	514	4	113	109
400	600	148	5	—	548	2 550	4 140	414	700	850	**NN3080**	**NN3080K**	—	—	420	—	—	580	552	4	146	141
420	620	150	5	—	570	2 900	4 570	449	670	800	**NN3084**	**NN3084K**	—	—	440	—	—	600	574	4	154	149
440	650	157	6	—	597	3 160	5 060	489	620	740	**NN3088**	**NN3088K**	—	—	464	—	—	626	602	5	177	171
460	680	163	6	—	627	3 390	5 480	521	570	690	**NN3092**	**NN3092K**	—	—	484	—	—	656	632	5	201	195

Tapered roller bearings

Tapered roller bearings are designed such that outer ring, inner ring and rollers have tapered surfaces whose apexes converge at a common point on the bearing axis. Along with metric series bearings, inch series bearings are also available.

This type of bearing is suitable for applications that involve heavy or impact loading.

Single-row tapered roller bearings



Metric series

Bore diameter **15** – **360** mm

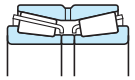


Inch series

(including J series metric bearing)

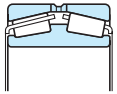
Bore diameter **9.525** – **292.100** mm

Double-row tapered roller bearings



TDO type

Bore diameter **25** – **500** mm

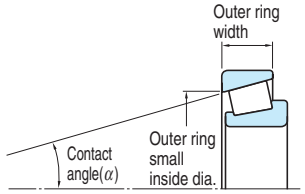


TDI type

Bore diameter **100** – **500** mm

■ Single-row tapered roller bearings

- Able to carry radial and axial load in one direction simultaneously.
  Because an axial component of force is produced when this type of bearing is loaded radially, two bearings are used together facing one another, or two or more bearings are matched and used.
- There are the standard, medium and steep type which are different in contact angle size.
  Medium-tapered metric series bearings are identified by the supplementary code "C" which is added as a suffix to bearing numbers.
- Bearings whose outer ring width, outer ring small inside diameter and contact angle are determined in accordance with ISO 355 specifications are identified by the supplementary code "J" as a suffix.
  Inner ring assemblies and the outer rings of such bearings are interchangeable with those of bearings produced abroad if the bearing numbers are the same.



ISO sub-unit specifications

[Note] When supplementary code "J" is added as a prefix (not a suffix) to bearing numbers (e.g. JHM720249/JHM720210), the bearings are not designed according to ISO 355.
Such bearings are called "J series metric tapered roller bearings," and are produced according to special tolerances.

■ Double-row tapered roller bearings

- These bearings are divided into the TDO type which has one double outer ring and two single-row inner rings, and the TDI type which has two single-row outer rings and one double inner ring. Both accommodate radial and axial loading in both directions.
  These two also carry moment loads, however, the TDO type is superior to the TDI type, because the distance between load centers (α) is longer in the TDO type.
- The spacer of the TDO type, or the TDI type, pre-adjusts the internal clearance to provide proper operating clearance after mounting.

Boundary dimensions

Metric single-row tapered roller bearings : as specified in JIS B 1512.

Reference JIS B 1512 specifies new dimension series which are based on ISO 355, as well as the conventional "3XX" dimension series. These new dimension series are as follows :

New dimension series

(1) Angle series

Angle series	Contact angle α	
	over	up to
2	10°	13° 52'
3	13° 52'	15° 59'
4	15° 59'	18° 55'
5	18° 55'	23°
6	23°	27°
7	27°	30°

(2) Diameter series

Diameter series	$D/(d^{0.77})$	
	over	up to
B	3.40	3.80
C	3.80	4.40
D	4.40	4.70
E	4.70	5.00
F	5.00	5.60
G	5.60	7.00

(3) Width series

Width series	$T/\{(D\text{-}d)^{0.95}\}$	
	over	up to
B	0.50	0.68
C	0.68	0.80
D	0.80	0.88
E	0.88	1.00

[Remarks]
1. Combine these series symbols in the listed order to make the dimension series numbers. (ex. 2BC)
2. Bearing numbers consist of a dimension series number and a bore diameter which is added as a suffix. (ex. 2BC080 : bore diameter 80 mm)

Tolerances

· Metric series single-row tapered roller bearings as specified in JIS B 1514-1.
..... (refer to Table 7-5 on pp. A 66 – A 68.)
· Metric series double-row tapered roller bearings as specified in BAS 1002.
..... (refer to Table 7-6 on p. A 69.)
· Inch series tapered roller bearings as specified in ABMA Section 19.
..... (refer to Table 7-7 on pp. A 70, 71.)
· J series metric tapered roller bearings the tolerance is specified separately.
..... (refer to Table 7-8 on pp. A 72, 73.)

Internal clearance

Radial internal clearance of double-row, four-row and matched pair tapered roller bearings ..... (refer to Table 10-10 on p. A 110.)

Recommended fits

· Metric series tapered roller bearings (classes 0, 6X and 6)
..... (refer to Table 9-4 on pp. A 91, 92.)
· Inch series tapered roller bearings ..... (refer to Table 9-7 on pp. A 96, 97.)
· J series metric tapered roller bearings
..... (refer to Table 9-6 on pp. A 94, 95.)

Standard cage

Pressed cage (supplementary code : //)
(Some large size bearings have a pin type cage (FP) instead. They are listed separately in the bearing specification table.)

Allowable misalignment

Single-row tapered roller bearings : 0.000 9 rad (3')
(If the misalignment exceeds this angle size, JTEKT is ready to design special bearings to order.)

Equivalent radial load

■ Single-row tapered roller bearings

Dynamic equivalent radial load
$\left(\text{when } \frac{F_a}{F_r} \leqq e\right) \quad P_r = F_r$
$\left(\text{when } \frac{F_a}{F_r} > e\right) \quad P_r = 0.4F_r + Y_1 F_a$

Static equivalent radial load $P_{0r} = 0.5F_r + Y_0 F_a$
when $P_{0r} < F_r$, $P_{0r} = F_r$

[Note]
Refer to the bearing specification table for the values of axial load factors Y_1, Y_2, Y_3 and Y_0 and constant e.

■ Double-row or four-row tapered roller bearings

Dynamic equivalent radial load
$\left(\text{when } \frac{F_a}{F_r} \leqq e\right) \quad P_r = F_r + Y_2 F_a$
$\left(\text{when } \frac{F_a}{F_r} > e\right) \quad P_r = 0.67F_r + Y_3 F_a$

Static equivalent radial load $P_{0r} = F_r + Y_0 F_a$

[Remarks]
1. When two single-row tapered roller bearings are used together facing one another, an axial component of force is produced under radial load. In this case, refer to pp. A 38, 39 for calculation of the dynamic equivalent radial load.
2. When the load is too small, slippage occurs between the rollers and raceways, causing smearing to develop. This also occurs to matched pair bearings when the ratio of axial load to radial load exceeds the value e shown in the specification table ($F_a/F_r > e$). Consult with JTEKT on use of bearings under such conditions.

[Series No. index]

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	342	B229		
	344	B227		
	344A	B229		
355	350A	B229	354A	B229,B231,
	355	B231		B233
	355A	B231		
	358	B231		
	359A	B233		
	359S	B233		
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	366	B235		
	368	B235		
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	368S	B237		
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	387	B239		
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	395	B241		
	395A	B243		
	395S	B243		
	396	B235		
	397	B241		
	399A	B243		
	399AS	B243		
415	418	B227	414	B227,B229
	419	B229		
	420	B229		
435	438	B231	432	B225
	449	B225	432A	B231
455	456	B237	453X	B233,B237,
(Continued)	462	B239		B239
	463	B233		
	466	B237		

series No.	inner ring	pages	outer ring	pages
455	467	B233		
	468	B237		
	469	B239		
475	477	B241	472	B241,B245
	482	B245	472A	B245
			472X	B245
495	495	B249	492A	B247,B249,
	495A	B247		B251
	495AX	B247	493	B247,B251
	496	B249		
	497	B251		
	497A	B251		
	498	B251		
525	525	B227	522	B227,B229,
	526	B229		B231,B233,
	527	B231		B235
	528	B233		
	529	B235		
	529X	B235		
535	535	B231	532A	B231
	537	B237	532X	B229,B237
	539	B237		
	539A	B237		
	543	B229		
555	557S	B237	552A	B237
565	565	B241	563	B241,B243,
	566	B245		B245,B247
	567	B245		
	567A	B245		
	568	B247		
	570	B243		
575R	575R	B247	572	B245,B247,
	575SR	B247		B249
	576R	B245	572X	B249
	577R	B247		
	580R	B249		
	581R	B249		
	582R	B249		
595	594A	B253	592A	B251
	596	B251	592XE	B253
615	615	B231	612	B231,B237,
	619	B237		B239
	621	B237		
	623	B239		
635	641	B243	633	B243
655	655	B245	652	B247
(Continued)	657	B247	653	B245,B247,
	659	B247		B249,B251
	661	B249		

series No.	inner ring	pages	outer ring	pages
655	663	B249		
	665	B251		
	665A	B251		
675	677	B251	672	B251,B253,
	679	B251		B255
	681	B253		
	681A	B253		
	683	B253		
	685	B253		
	687	B255		
745R	740R	B249	742	B245,B247,
	744R	B247		B249,B251
	745AR	B245		
	748SR	B247		
	749AR	B249		
	749R	B251		
	749SR	B251		
	750AR	B249		
755	756A	B249	752	B247,B249,
	757	B249		B251,B253
	758	B251		
	759	B251		
	760	B253		
	762	B247		
	766	B251		
775	778	B253	772	B253,B255
	780	B255		
	782	B255		
	786	B255		
	787	B255		
835R	835R	B245	832	B245,B251
	841R	B251		
855R	855R	B251	854	B251,B253,
	857R	B253		B255
	861R	B255		
	864R	B253		
935	936	B257	932	B255,B257
	938	B257		
	941	B255		
1200	1280	B219	1220	B219
1300	1380	B217	1328	B217
			1329	B217
1700	1755	B219	1729	B219
	1779	B219		
1900R	1986R	B219	1922	B221
	1988R	B221	1932	B219
A2000	A2037	B217	A2126	B217
	A2047	B217		
2500	2580	B223	2520	B223

series No.	inner ring	pages	outer ring	pages
2600	2682	B219	2631	B219,B221
	2684	B219		
	2687	B219		
	2688	B221		
	2689	B221		
	2690	B221		
2700R	2788R	B227	2720	B225
	2789R	B227	2729	B227
	2794R	B225	2735X	B225,B227
	2796R	B225		
2900	2984	B233	2924	B233
3100	3192	B221	3120	B221
	3198	B221		
3300	3382	B227	3320	B227
	3386	B227	3328	B227
3400	3478	B225	3420	B225,B227
	3479	B225		
	3490	B227		
3500R	3576R	B229	3520	B231
	3578R	B231	3525	B225,B229
	3581R	B225		
3700	3776	B231	3720	B231,B235
	3780	B235	3730	B235
			3732	B235
3800	3877	B229	3820	B225
	3878	B225	3821	B229
3900	3979	B239	3920	B239
	3984	B243	3925	B243
A4000	A4050	B217	A4138	B217
	A4059	B217		
4300	4375	B227	4335	B227,B229
	4388	B229		
	4395	B229		
4500	4580	B237	4535	B237
	4595	B237		
5500R	5566R	B239	5535	B237,B239,
	5578R	B237		B241,B243
	5583R	B241		
	5584R	B241		
	5595R	B243		
5700	5760	B247	5735	B247
6300	6379	B243	6320	B237,B241,
	6381	B237		B243
	6382	B241		
	6386	B243		
	6389	B243		
6400	6460	B247	6420	B247
	6461	B247		
	6461A	B247		

series No.	inner ring	pages	outer ring	pages
6500R	6580R	B251	6535	B251,B253
	6581XR	B253		
9100	9185	B243	9121	B243
02400	02473	B219	02420	B219,B221,
	02474	B221		B223
	02475	B223		
	02476	B223		
02800	02872	B221	02820	B221,B223,
	02875	B223		B225
	02876	B223		
	02877	B225		
	02878	B225		
03000	03062	B217	03162	B217
07000	07079	B217	07196	B217,B219
	07097	B219	07204	B219
	07098	B219		
	07100	B219		
	07100S	B219		
08000	08125	B223	08231	B223
09000	09062	B217	09195	B217
	09067	B217	09196	B217
	09078	B217		
11000R	11162R	B229	11300	B229
LM11700R	LM11749R	B217	LM11710	B217
LM11900	LM11949	B217	LM11910	B217
12000	12168	B231	12303	B231
	12175	B231		
12500	12580	B217	12520	B217
M12600	M12648	B217	M12610	B217
	M12649	B217		
LM12700	LM12749	B217	LM12711	B217
13600	13687	B227	13621	B227
13800	13889	B225,B227	13830	B225
			13836	B227
14000	14116	B223	14274	B223
	14117A	B221	14276	B221,B223
	14136A	B223		
15000	15100	B219	15243	B219
(Continued)	15101	B219	15245	B219,B221,
	15106	B221		B223
	15112	B221		
	15113	B221		
	15116	B221		
	15117	B221		
	15118	B223		
	15119	B223		
	15120	B223		
	15123	B223		
	15125	B223		

series No.	inner ring	pages	outer ring	pages
15000	15126	B223		
15500	15580	B221	15520	B221
	15590	B221	15523	B221
16000	16137	B225	16282	B227
	16150	B227	16283	B227
			16284	B225
17000	17098	B219	17244	B219,B221
	17118	B221		
	17119	B221		
17500R	17580R	B217	17520	B217
18000	18200	B235	18337	B235
18500	18587	B227	18520	B227,B229
	18590	B229		
18600	18685	B231	18620	B231,B233
	18690	B233		
18700	18790	B235	18724	B235
19000R	19150R	B227	19281	B227
			19283	B227
21000	21063	B217	21212	B217
L21500	L21549	B217	L21511	B217
23600	23690	B225	23620	B225
24700R	24780R	B229	24720	B229
25500	25572	B227	25520	B227,B231
	25577	B231	25526	B233
	25582	B231		
	25590	B233		
25800R	25877R	B225	25821	B225
	25880R	B225		
26000	26112	B221	26283	B221,B223
	26131	B223		
26800R	26877R	B225	26822	B225,B231
	26883R	B225		
	26884R	B231		
27600	27687	B249	27620	B249,B251
	27689	B249		
	27690	B251		
	27691	B251		
27800	27880	B227	27820	B227
	27881	B227		
28000	28137	B225	28300	B225,B227
	28150	B227	28317	B227
	28158	B227		
28500R	28579R	B235	28521	B235,B237
	28580R	B235		
	28584R	B237		
28600	28678	B235	28622	B235,B239
	28680	B239		
28900	28985	B241	28920	B241
			28921	B241

series No.	inner ring	pages	outer ring	pages
29500	29580	B241	29520	B241
	29585	B241	29521	B241
	29586	B241		
29600	29675	B245	29620	B245,B247
	29685	B245		
	29688	B247		
LM29700	LM29748	B227	LM29710	B227
	LM29749	B227	LM29711	B227
31500	31594	B225	31520	B225
33000	33225	B239	33462	B239,B243,
	33262	B243		B245
	33269	B243		
	33275	B245		
	33281	B245		
	33287	B245		
33800	33885	B231	33821	B231
	33889	B235	33822	B235,B237
	33895	B237		
34000	34274	B245	34478	B245,B247,
	34301	B247		B249
	34306	B249		
	34307	B249		
37000	37425	B255	37625	B255,B257
	37431	B257		
39500	39575	B237	39520	B237,B239,
	39580	B239		B241,B243
	39581	B239	39521	B243
	39585	B241		
	39586	B241		
	39590	B243		
41000	41125	B221	41286	B221
	41126	B221		
42600	42687	B247	42620	B247,B249
	42688	B247		
	42690	B249		
L44600R	L44640R	B219	L44610	B219
	L44643R	B219		
	L44649R	B219		
45200	45282	B233	45220	B233,B235
	45284	B235	45221	B239
	45291	B239		
46000	46162	B229	46368	B229,B231
	46175	B231		
	46176	B231		
47400R	47487R	B245	47420	B245
	47490R	B245	47423	B245
47600R	47678R	B247	47620	B247,B249
(Continued)	47680R	B247	47620A	B249
	47681R	B249		

series No.	inner ring	pages	outer ring	pages
47600R	47686R	B249		
47800R	47890R	B253	47820	B253
	47896R	B253		
48100	48190	B255	48120	B255
LM48500	LM48548	B223	LM48510	B223
48600	48684	B259	48620	B259
	48685	B259		
49000	49175	B231	49368	B231
49500	49576	B231	49520	B231,B235
	49585	B235		
52000	52375	B253	52618	B253,B255
	52393	B255		
	52400	B255		
	52401	B255		
56000R	56418R	B255	56650	B255
	56425R	B255	56662	B255
59000	59200	B235	59412	B235
64000R	64433R	B257	64700	B257
	64450R	B257		
65000	65200	B237	65500	B237,B239,
	65212	B237		B241
	65225	B239		
	65237	B241		
	65237A	B241		
65300	65390	B233	65320	B233
66000R	66212R	B237	66462	B237
66500	66584	B237	66520	B237,B239
	66589	B239		
LM67000	LM67048	B223	LM67010	B223
68000	68450	B257	68712	B257
	68462	B257		
	68463	B257		
L68100	L68149	B225	L68110	B225
			L68111	B225
71000	71412	B255	71750	B255,B257
	71425	B255		
	71450	B257		
	71453	B257		
	71455	B257		
LM72800	LM72849	B219	LM72810	B219
HM81600	HM81649	B217	HM81610	B217
M84200	M84249	B219	M84210	B219
M86600R	M86643R	B219	M86610	B219,B221
	M86647R	B221		
	M86649R	B221		
M88000	M88043	B223	M88010	B223
	M88046	B223		
	M88048	B223		

series No.	inner ring	pages	outer ring	pages
HM88500	HM88542	B223	HM88510	B223
	HM88547	B223	HM88512	B223
HM88600	HM88630	B219	HM88610	B219,B223,
	HM88648	B225		B225
	HM88649	B223		
HM89400	HM89443	B223	HM89410	B223
	HM89449	B225	HM89411	B225
98000	98316	B249	98788	B249,B251,
	98335	B251		B253,B255
	98350	B253		
	98400	B255		
L102800	L102849	B231	L102810	B231
LM102900	LM102949	B233	LM102910	B233
LM104900	LM104949	B235	LM104911	B235
HM212000	HM212046	B241	HM212010	B243
	HM212049	B243	HM212011	B241
L217800	L217849	B251	L217810	B251
HM218200	HM218248	B253	HM218210	B253
HH221400	HH221430	B247	HH221410	B247,B249,
	HH221431	B249		B253,B255
	HH221434	B253		
	HH221440	B253		
	HH221442	B253		
	HH221447	B255		
	HH221449	B255		
HH224300	HH224334	B253	HH224310	B253,B255,
	HH224335	B255		B257
	HH224340	B257		
	HH224346	B257		
	HH224349	B257		
HH228300	HH228340	B257	HH228310	B257
	HH228349	B257		
LM245800	LM245833	B259	LM245810	B259
	LM245846	B259		
	LM245848	B259		
M246900	M246942	B259	M246910	B259
M249700	M249732	B259	M249710	B259
	M249734	B259		
	M249749	B259		
L305600R	L305649R	B235	L305610	B235
L319200	L319249	B253	L319210	B253
LL319300	LL319349	B253	LL319310	B253
L327200	L327249	B259	L327210	B259
M349500	M349549	B259	M349510	B259
H414200	H414235	B241	H414210	B241,B243,
	H414242	B243		B245
	H414245	B243		
	H414249	B245		
L435000	L435049	B259	L435010	B259

series No.	inner ring	pages	outer ring	pages
LM501300	LM501349	B229	LM501310	B229
			LM501311	B229
			LM501314	B229
LM503300R	LM503349R	B233	LM503310	B233
HH506300	HH506348	B233	HH506310	B233
HM516400	HM516448	B249	HM516410	B249
HM518400	HM518445	B251	HM518410	B251
L521900R	L521949R	B255	L521910	B255
LM522500	LM522546	B255	LM522510	B255,B257
	LM522548	B257		
	LM522549	B257		
L540000	L540049	B259	L540010	B259
L555200	L555249	B259	L555210	B259
LM603000	LM603049	B233	LM603011	B233
			LM603012	B233
			LM603014	B233
LM613400	LM613449	B245	LM613410	B245
HM617000	HM617049	B251	HM617010	B251
HM624700	HM624749	B257	HM624710	B257
LL713000	LL713049	B245	LL713010	B245
H715300	H715332	B241	H715311	B241,B243,
	H715340	B243		B245
	H715341	B243		
	H715343	B243		
	H715345	B245		
HM801300	HM801346	B227	HM801310	B227,B229
	HM801346X	B227		
	HM801349	B229		
M802000	M802048	B229	M802011	B229
HM803100	HM803145	B229	HM803110	B229,B231
	HM803146	B229		
	HM803149	B231		
M804000	M804049	B233	M804010	B233
HM804800	HM804840	B229	HM804810	B229,B231,
	HM804842	B231		B233
	HM804843	B231		
	HM804846	B233		
	HM804848	B233		
	HM804849	B233		
LM806600	LM806649	B237	LM806610	B237
HM807000	HM807035	B229	HM807010	B229,B231,
	HM807040	B231		B233,B237
	HM807044	B233		
	HM807046	B237		
	HM807049	B237		
HM813800	HM813840	B239	HM813810	B239,B241
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	HM813841A	B241		B245
	HM813844	B243		

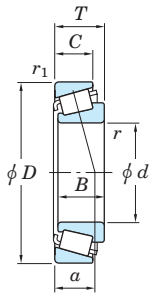
series No.	inner ring	pages	outer ring	pages
HM813800	HM813849	B245		
LM814800	LM814849	B249	LM814810	B249
HH926700	HH926744	B257	HH926710	B257

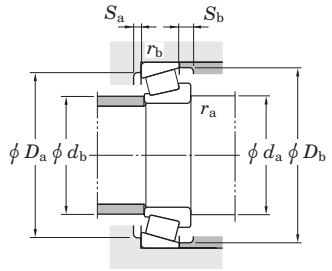
Metric J series

series No.	inner ring	pages	outer ring	pages
JL69300	JL69349	B225	JL69310	B225
JLM104900	JLM104948	B235	JLM104910	B235
JM205100	JM205149	B235	JM205110	B235
JM207000	JM207049	B239	JM207010	B239
JH211700	JH211749	B243	JH211710	B243
	JH211749A	B243		
JH217200	JH217249	B251	JH217210	B251
JH307700	JH307749	B239	JH307710	B239
JHM318400	JHM318448	B253	JHM318410	B253
JH415600	JH415647	B247	JH415610	B247
JLM506800	JLM506849	B237	JLM506810	B237
JLM508700	JLM508748	B239	JLM508710	B239
JM511900	JM511946	B241	JM511910	B241
JM515600	JM515649	B249	JM515610	B249
JHM516800	JHM516849	B251	JHM516810	B251
JHM522600	JHM522649	B257	JHM522610	B257
JHM534100	JHM534149	B259	JHM534110	B259
JM612900	JM612949	B245	JM612910	B245
JLM710900	JLM710949	B241	JLM710910	B241
JLM714100	JLM714149	B247	JLM714110	B247
JM714200	JM714249	B247	JM714210	B247
JM716600	JM716649	B251	JM716610	B251
JM718100	JM718149	B253	JM718110	B253
JM719100	JM719149	B253	JM719113	B253
JHM720200	JHM720249	B255	JHM720210	B255
JM720200	JM720249	B255	JM720210	B255
JM734400	JM734449	B259	JM734410	B259
JM736100	JM736149	B259	JM736110	B259
JM738200	JM738249	B259	JM738210	B259
JHM807000	JHM807045	B235	JHM807012	B235
JLM813000	JLM813049	B245	JLM813010	B245
JM822000	JM822049	B257	JM822010	B257
JHM840400	JHM840449	B259	JHM840410	B259

metric series

d 15 ~ 22 mm



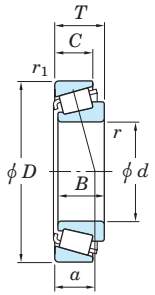


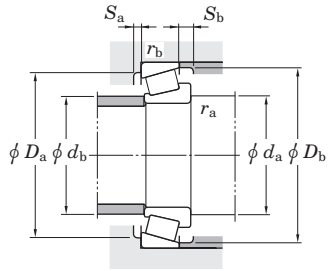
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Dimension series to ISO355	Load center (mm)	Mounting dimensions (mm)										Constant	Axial load factors		(Refer.) Mass
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.			(Refer.)	a	d_a min.	d_b max.	D_a max.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	(kg)
15	35	11.75	11	10	0.6	0.6	19.8	14.5	2.00	12 000	16 000	**30202R**		—	8.3	19.5	20	30.5	33	2	1.7	0.6	0.6	0.32	1.88	1.04	0.054
	42	14.25	13	11	1	1	27.4	19.2	2.65	10 000	14 000	**30302JR**		2FB	10.0	20.5	22	36.5	35	2	3	1	1	0.29	2.11	1.16	0.098
17	40	13.25	12	11	1	1	26.0	20.7	2.85	10 000	14 000	**30203JR**		2DB	10.1	22.5	23	34.5	33	2	2	1	1	0.35	1.74	0.96	0.081
	40	17.25	16	14	1	1	34.3	27.5	3.85	10 000	14 000	**32203JR**		2DD	11.4	22.5	23	34.5	33	2	3	1	1	0.31	1.92	1.06	0.104
	47	15.25	14	12	1	1	34.2	24.5	3.45	9 200	12 000	**30303JR**		2FB	11.0	22.5	25	41.5	40	2	3	1	1	0.29	2.11	1.16	0.133
	47	15.25	14	12	1	1	34.2	24.5	3.45	9 200	12 000	**30303R**		—	10.5	22.5	25	41.5	40	2	3	1	1	0.28	2.11	1.16	0.127
	47	20.25	19	16	1	1	39.9	29.9	4.25	9 400	13 000	**32303**		—	12.4	22.5	25	41.5	39	2	4	1	1	0.28	2.11	1.16	0.170
	47	20.25	19	16	1	1	45.7	35.9	5.10	9 400	13 000	**32303JR**		2FD	12.2	22.5	25	41.5	39	2	4	1	1	0.29	2.11	1.16	0.176
20	42	15	15	12	0.6	0.6	34.1	31.5	4.35	9 700	13 000	**32004JR**		3CC	10.5	24.5	25	37.5	35	3	3	0.6	0.6	0.37	1.60	0.88	0.102
	47	15.25	14	12	1	1	34.2	25.5	3.75	9 000	12 000	**57008R**		—	12.9	25.5	26	41.5	37	2	3	1	1	0.52	1.16	0.64	0.125
	47	15.25	14	12	1	1	33.8	27.2	3.80	8 700	12 000	**30204JR**		2DB	11.8	25.5	27	41.5	39	2	3	1	1	0.35	1.74	0.96	0.127
	47	19.25	18	15	1	1	41.4	34.7	4.90	8 900	12 000	**32204JR**		2DD	12.5	25.5	27	41.5	39	2	4	1	1	0.33	1.81	1.00	0.159
	47	19.25	18	16	1	1	41.6	37.0	5.00	9 100	12 000	**32204XR**		—	15.3	25.5	25	41.5	35	2	3	1	1	0.55	1.10	0.60	0.170
	52	16.25	16	12	1.5	1.5	43.3	28.4	4.65	8 300	11 000	**30304AC**		—	13.5	28.5	28	43.5	42	4	4	1.5	1.5	0.55	1.10	0.60	0.170
	52	16.25	16	13	1.5	1.5	45.3	35.1	5.05	8 300	11 000	**30304AJR**		—	11.1	28.5	28	44	44	2	3	1.5	1.5	0.30	2.00	1.10	0.179
	52	22.25	21	18	1.5	1.5	52.3	44.9	6.05	8 600	12 000	**32304CR**		—	16.5	28.5	25	43.5	37	3	4	1.5	1.5	0.55	1.10	0.60	0.250
	52	22.25	21	18	1.5	1.5	56.5	46.7	6.70	8 400	11 000	**32304JR**		2FD	14.4	28.5	27	43.5	43	3	4	1.5	1.5	0.30	2.00	1.10	0.244
22	44	15	15	11.5	0.6	0.6	35.4	33.6	4.65	9 100	12 000	**320/22JR**		3CC	11.0	26.5	27	39.5	38	3	3.5	0.6	0.6	0.40	1.51	0.83	0.108
	47	17	17.5	13.5	1	1	40.9	35.9	5.05	8 700	12 000	**T2CC022**		2CC	11.3	27.5	28	41.5	40	4	3.5	1	1	0.33	1.79	0.99	0.138
	50	15.25	14	12	1	1	32.1	25.7	3.50	8 400	11 000	**302/22CR**		—	13.9	27.5	28	44.5	40	2	3	1	1	0.55	1.10	0.60	0.140
	50	15.25	14	12	1	1	36.5	30.9	4.30	8 100	11 000	**302/22R**		—	12.2	27.5	30	44.5	41	2	3	1	1	0.37	1.60	0.88	0.144
	50	19.25	18	15	1	1	43.8	39.1	5.35	8 400	11 000	**322/22CR**		—	15.5	27.5	28	44.5	38	2	4	1	1	0.55	1.10	0.60	0.170
	50	19.25	18	15	1	1	46.0	41.6	5.85	8 100	11 000	**322/22R**		—	14.0	27.5	29	44.5	41	2	4	1	1	0.37	1.60	0.88	0.178
	56	17.25	16	13	1.5	1.5	43.0	33.9	4.70	7 700	10 000	**303/22XR**		—	15.7	30.5	31	47.5	44	3	4	1.5	1.5	0.59	1.02	0.56	0.210
	56	17.25	16	14	1.5	1.5	52.2	41.1	5.95	7 500	10 000	**303/22R**		—	12.2	30.5	32	47.5	47	2	3	1.5	1.5	0.31	1.97	1.08	0.216
	56	22.25	21	17	1.5	1.5	60.4	50.6	7.00	8 000	11 000	**323/22CR**		—	16.9	30.5	28	47.5	41	3	5	1.5	1.5	0.55	1.10	0.60	0.290
	56	22.25	21	18	1.5	1.5	63.3	52.7	7.70	7 600	10 000	**323/22R**		—	14.6	30.5	31	47.5	46	3	4	1.5	1.5	0.31	1.97	1.08	0.273

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metric series

d 25 ~ (30) mm



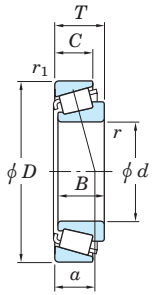


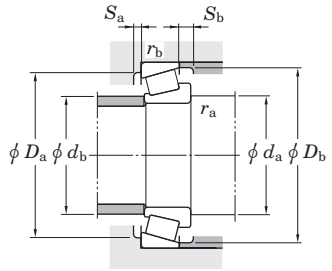
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)									Constant	Axial load factors		(Refer.) Mass (kg)
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	
25	47	15	15	11.5	0.6	0.6	37.8	37.7	5.20	8 300	11 000	**32005JR**		4CC	11.8	29.5	30	42.5	44	3	3.5	0.6	0.6	0.43	1.39	0.77	0.118
	47	17	17	14	0.6	0.6	42.0	42.3	5.95	8 300	11 000	**33005JR**		2CE	10.9	29.5	30	42.5	44	3	3	0.6	0.6	0.29	2.07	1.14	0.131
	52	16.25	15	12	1	1	38.0	32.4	4.45	7 900	11 000	**30205XR**		—	14.9	30.5	30	46.5	49	2	4	1	1	0.58	1.04	0.57	0.155
	52	16.25	15	13	1	1	39.3	33.7	4.75	7 800	10 000	**30205JR**		3CC	12.9	30.5	31	46.5	48	2	3	1	1	0.37	1.60	0.88	0.156
	52	19.25	18	16	1	1	45.5	43.2	5.90	7 900	11 000	**32205XR**		—	16.2	30.5	30	46.5	40	2	3	1	1	0.55	1.10	0.60	0.200
	52	19.25	18	16	1	1	49.7	44.8	6.35	7 900	11 000	**32205JR**		2CD	13.5	30.5	31	46.5	43	2	4	1	1	0.36	1.67	0.92	0.188
	52	22	22	18	1	1	61.1	58.5	8.25	7 900	10 000	**33205JR**		2DE	14.1	30.5	30	46.5	43	4	4	1	1	0.35	1.71	0.94	0.225
	62	18.25	17	13	1.5	1.5	49.7	42.5	5.80	5 700	8 000	**30305DJR**		7FB	20.4	33.5	34	53.5	47	3	5	1.5	1.5	0.83	0.73	0.40	0.269
	62	18.25	17	14	1.5	1.5	56.3	45.8	6.50	6 700	9 000	**TR0506R**		—	16.3	33.5	35	53.5	50	3	4	1.5	1.5	0.55	1.10	0.60	0.275
	62	18.25	17	15	1.5	1.5	60.3	46.9	6.90	6 800	9 000	**30305JR**		2FB	12.9	33.5	34	54	54	2	3	1.5	1.5	0.30	2.00	1.10	0.273
	62	25.25	24	19	1.5	1.5	71.6	65.8	9.20	7 000	9 300	**32305XR**		—	18.9	33.5	33	53.5	46	3	6	1.5	1.5	0.55	1.10	0.60	0.390
	62	25.25	24	20	1.5	1.5	76.6	64.1	9.50	6 900	9 100	**32305JR**		2FD	16.6	33.5	33	53.5	52	3	5	1.5	1.5	0.30	2.00	1.10	0.386
28	52	16	16	12	1	1	44.1	44.0	6.10	7 500	10 000	**320/28JR**		4CC	12.7	33.5	33	46.5	45	3	4	1	1	0.43	1.39	0.77	0.150
	58	17.25	16	13	1	1	48.5	41.7	5.85	7 000	9 300	**302/28CR**		—	16.0	33.5	34	52.5	47	2	4	1	1	0.55	1.10	0.60	0.205
	58	17.25	16	14	1	1	48.5	42.0	6.00	7 000	9 300	**302/28R**		—	13.4	33.5	35	52.5	49	2	3	1	1	0.37	1.60	0.88	0.209
	58	20.25	19	16	1	1	56.1	54.1	7.50	7 100	9 400	**322/28CR**		—	17.0	33.5	33	52.5	45	3	4	1	1	0.55	1.10	0.60	0.255
	58	20.25	19	16	1	1	61.5	55.2	7.95	6 900	9 100	**322/28R**		—	15.0	33.5	35	52.5	49	54.5	2	4	1	1	0.37	1.60	0.88	0.244
	58	24	24	19	1	1	71.9	69.5	10.0	7 000	9 300	**332/28JR**		2DE	15.4	33.5	34	52.5	49	55	4	5	1	1	0.34	1.77	0.97	0.302
	68	19.75	18	14	1.5	1.5	64.6	50.2	7.25	6 200	8 200	**303/28CR**		—	17.8	36.5	37	59.5	55	64	3	4.5	1.5	1.5	0.55	1.10	0.60	0.332
	68	19.75	18	16	1.5	1.5	66.9	54.0	8.00	6 100	8 200	**303/28R**		—	14.9	36.5	38	59.5	58	63	2	3.5	1.5	1.5	0.32	1.88	1.04	0.345
	68	25.75	24	20	1.5	1.5	83.2	72.9	10.5	6 300	8 500	**323/28CR**		—	20.5	36.5	35	59.5	51	64	3	5.5	1.5	1.5	0.55	1.10	0.60	0.480
	68	25.75	24	21	1.5	1.5	87.0	75.6	11.3	6 100	8 100	**323/28R**		—	17.6	36.5	38	59.5	57	63	3	4.5	1.5	1.5	0.32	1.88	1.04	0.469
30	55	17	17	13	1	1	47.9	48.0	6.75	7 000	9 400	**32006JR**		4CC	13.6	35.5	35	49.5	47	52	3	4	1	1	0.43	1.39	0.77	0.177
	55	20	20	16	1	1	54.1	55.2	7.90	7 000	9 400	**33006JR**		2CE	13.0	35.5	36	49.5	48	52	3	4	1	1	0.29	2.06	1.13	0.203
	62	17.25	16	13	1	1	52.9	45.1	6.35	6 500	8 700	**30206CR**		—	16.5	35.5	36	56.5	51	59	2	4	1	1	0.55	1.10	0.60	0.230
	62	17.25	16	14	1	1	51.8	44.8	6.45	6 500	8 700	**30206JR**		3DB	14.1	35.5	37	56.5	53	57	2	3	1	1	0.37	1.60	0.88	0.236
	62	21.25	20	16	1	1	64.6	59.0	8.30	6 600	8 900	**32206XR**		—	18.0	35.5	36	56.5	49	59	3	5	1	1	0.55	1.10	0.60	0.300
	62	21.25	20	17	1	1	63.3	57.9	8.40	6 500	8 700	**32206JR**		3DC	15.9	35.5	37	56.5	52	58	2	4	1	1	0.37	1.60	0.88	0.292

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metric series

d (30) ~ (35) mm



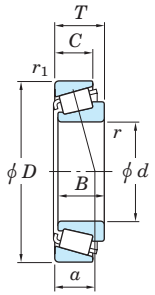


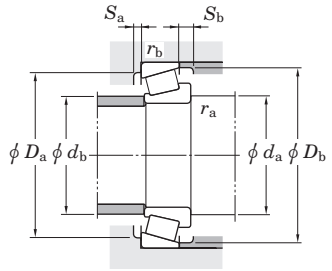
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)									Constant	Axial load factors		(Refer.) Mass (kg)
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	
30	62	25	25	19.5	1	1	83.1	79.4	11.6	6 500	8 700	**33206JR**		2DE	16.3	35.5	36	56.5	59	5	5.5	1	1	0.34	1.76	0.97	0.359
	72	20.75	19	14	1.5	1.5	63.5	54.9	7.70	4 900	6 800	**30306DJR**		7FB	23.7	38.5	40	63.5	68	3	6.5	1.5	1.5	0.83	0.73	0.40	0.400
	72	20.75	19	16	1.5	1.5	71.2	55.6	8.10	5 900	7 900	**TRA0607R**		—	18.6	38.5	39	63.5	68	3	4.5	1.5	1.5	0.55	1.10	0.60	0.405
	72	20.75	19	16	1.5	1.5	74.4	60.1	9.00	5 800	7 700	**30306JR**		2FB	15.7	38.5	40	63.5	66	3	4.5	1.5	1.5	0.31	1.90	1.05	0.411
	72	28.75	27	23	1.5	1.5	100	93.8	13.4	6 000	8 000	**32306CR**		5FD	22.0	38.5	37	63.5	68	3	5.5	1.5	1.5	0.55	1.10	0.60	0.610
	72	28.75	27	23	1.5	1.5	103	91.6	13.8	5 900	7 900	**32306JR**		2FD	18.9	38.5	39	63.5	66	3	5.5	1.5	1.5	0.31	1.90	1.05	0.588
32	58	17	17	13	1	1	49.2	50.6	7.10	6 700	8 900	**320/32JR**		4CC	14.3	37.5	38	52.5	55	3	4	1	1	0.45	1.32	0.73	0.196
	65	18.25	17	14	1	1	59.3	51.5	7.35	6 200	8 300	**302/32CR**		—	17.2	37.5	38	59.5	62	3	4	1	1	0.55	1.10	0.60	0.275
	65	18.25	17	15	1	1	60.1	51.4	7.45	6 200	8 200	**302/32R**		—	14.9	37.5	39	59.5	61	3	3	1	1	0.37	1.60	0.88	0.266
	65	22.25	21	17	1	1	69.6	65.1	9.20	6 300	8 400	**322/32CR**		—	18.7	37.5	37	59.5	62	3	5	1	1	0.55	1.10	0.60	0.340
	65	22.25	21	18	1	1	64.5	57.7	8.45	6 200	8 200	**322/32**		—	16.3	37.5	40	59.5	61	2	4	1	1	0.37	1.60	0.88	0.330
	65	26	26	20.5	1	1	89.7	86.9	12.8	6 200	8 300	**332/32JR**		2DE	16.9	37.5	38	59.5	55	5	5.5	1	1	0.35	1.73	0.95	0.404
	75	21.75	20	16	1.5	1.5	79.4	66.3	9.70	5 600	7 400	**303/32CR**		—	19.7	40.5	42	66.5	60	3	5.5	1.5	1.5	0.55	1.10	0.60	0.465
	75	21.75	20	18	1.5	1.5	80.5	65.6	9.90	5 500	7 300	**303/32R**		—	16.0	40.5	43	66.5	64	3	3.5	1.5	1.5	0.32	1.88	1.04	0.461
	75	29.75	28	23	1.5	1.5	93.8	87.1	12.6	5 600	7 400	**TR0608A**		5FD	23.7	40.5	41	66.5	57	3	6.5	1.5	1.5	0.55	1.10	0.60	0.649
	75	29.75	28	25	1.5	1.5	112	101	15.3	5 600	7 400	**323/32R**		—	19.6	40.5	42	66.5	63	3	4.5	1.5	1.5	0.32	1.88	1.04	0.650
35	55	14	14	11.5	0.6	0.6	32.8	36.5	5.10	6 600	8 800	**32907JR-2**		2BD	10.9	39.5	40	50.5	49	2.5	2.5	0.6	0.6	0.29	2.06	1.13	0.120
	62	18	18	14	1	1	57.0	59.4	8.40	6 200	8 200	**32007JR**		4CC	15.1	40.5	40	56.5	54	4	4	1	1	0.45	1.32	0.73	0.231
	62	21	20	16	1	1	51.3	53.8	7.70	6 200	8 200	**33007**		—	14.8	40.5	41	56.5	55	3	4	1	1	0.33	1.80	0.99	0.250
	62	21	21	17	1	1	64.3	68.0	9.85	6 200	8 200	**33007JR**		2CE	14.2	40.5	41	56.5	55	3	4	1	1	0.31	1.97	1.08	0.263
	72	18.25	17	15	1.5	1.5	66.1	56.2	8.10	5 700	7 600	**30207CR**		—	17.9	43.5	43	63.5	59	3	3	1.5	1.5	0.55	1.10	0.60	0.350
	72	18.25	17	15	1.5	1.5	68.8	60.9	8.95	5 600	7 400	**30207JR**		3DB	15.3	43.5	44	63.5	62	3	3	1.5	1.5	0.37	1.60	0.88	0.344
	72	24.25	23	19	1.5	1.5	86.3	86.6	12.3	5 700	7 600	**32207-1R**		—	21.1	43.5	42	63.5	56	3	5	1.5	1.5	0.58	1.04	0.57	0.465
	72	24.25	23	19	1.5	1.5	86.9	82.4	12.2	5 600	7 500	**32207JR**		3DC	18.2	43.5	43	63.5	61	3	5	1.5	1.5	0.37	1.60	0.88	0.453
	72	28	28	22	1.5	1.5	110	107	15.8	5 700	7 500	**33207JR**		2DE	18.4	43.5	42	63.5	61	5	6	1.5	1.5	0.35	1.70	0.93	0.551
	80	22.75	21	15	2	1.5	78.7	69.1	9.85	4 300	6 000	**30307DJR**		7FB	26.8	45	44	70	66	76.5	3	7.5	2	1.5	0.83	0.73	0.40	0.536
	80	22.75	21	18	2	1.5	87.2	77.8	11.4	5 200	7 000	**30307XR**		—	20.5	45	45	70	63	74	3	4.5	2	1.5	0.55	1.10	0.60	0.560
	80	22.75	21	18	2	1.5	95.2	78.9	12.0	5 200	6 900	**30307JR-1**		2FB	16.9	45	45	70	70	74	3	4.5	2	1.5	0.31	1.90	1.05	0.527

[Note] 1) Please consult with JTEKT when using the bearings identified by suffix C. They are medium-tapered types especially designed for special purposes.

metric series

d (35) ~ (45) mm



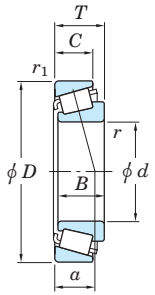


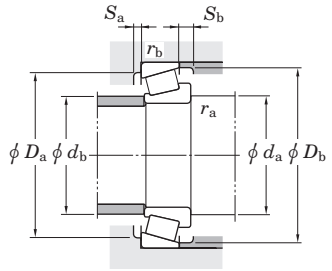
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)									Constant	Axial load factors		(Refer.) Mass (kg)
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_a min.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	
35	80	32.75	31	25	2	1.5	121	123	18.0	5 200	7 000	**TR0708-1R**		—	23.8	45	44	70	60	75	3	7.5	2	1.5	0.47	1.27	0.70	0.830
	80	32.75	31	25	2	1.5	126	114	17.3	5 300	7 000	**32307JR**		2FE	20.6	45	44	70	66	74	3	7.5	2	1.5	0.31	1.90	1.05	0.776
40	62	15	15	12	0.6	0.6	42.1	48.5	6.90	5 900	7 800	**32908JR**		2BC	11.9	44.5	45	57.5	55	59	3	3	0.6	0.6	0.29	2.07	1.14	0.164
	68	19	19	14.5	1	1	67.2	71.4	10.3	5 600	7 400	**32008JR**		3CD	15.1	45.5	46	62.5	60	65	4	4.5	1	1	0.38	1.58	0.87	0.282
	68	22	22	18	1	1	75.9	84.6	12.4	5 500	7 400	**33008JR**		2BE	14.7	45.5	46	62.5	60	65	3	4	1	1	0.28	2.12	1.17	0.326
	75	26	26	20.5	1.5	1.5	103	108	16.1	5 200	6 900	**33108JR**		2CE	18.3	48.5	47	66.5	65	71	4	5.5	1.5	1.5	0.36	1.69	0.93	0.508
	80	19.75	18	15	1.5	1.5	76.6	67.4	9.90	5 000	6 700	**30208CR**		—	20.2	48.5	49	71.5	66	76	3	4.5	1.5	1.5	0.55	1.10	0.60	0.445
	80	19.75	18	16	1.5	1.5	78.4	69.2	10.3	5 000	6 700	**30208JR**		3DB	17.0	48.5	49	71.5	69	75	3	3.5	1.5	1.5	0.37	1.60	0.88	0.434
	80	24.75	23	19	1.5	1.5	98.0	93.1	13.7	5 000	6 700	**32208CR**		5DC	22.0	48.5	48	71.5	64	76	3	5.5	1.5	1.5	0.55	1.10	0.60	0.570
	80	24.75	23	19	1.5	1.5	97.0	90.8	13.6	5 000	6 600	**32208JR**		3DC	19.4	48.5	48	71.5	68	75	3	5.5	1.5	1.5	0.37	1.60	0.88	0.554
	80	32	32	25	1.5	1.5	135	139	20.8	5 000	6 700	**33208JR**		2DE	20.7	48.5	47	71.5	67	76	5	7	1.5	1.5	0.36	1.68	0.92	0.758
	85	33	32.5	28	2.5	2	143	143	21.6	4 800	6 400	**T2EE040**		2EE	21.9	52	48	75	70	80	5	5	2	2	0.34	1.74	0.96	0.900
	90	25.25	23	17	2	1.5	100	90.2	13.1	3 800	5 300	**30308DJR**		7FB	29.9	50	51	80	71	86.5	3	8	2	1.5	0.83	0.73	0.40	0.757
	90	25.25	23	20	2	1.5	109	98.5	14.8	4 600	6 100	**30308XR**		—	23.8	50	53	80	72	84	3	5	2	1.5	0.55	1.10	0.60	0.780
	90	25.25	23	20	2	1.5	113	101	15.5	4 500	6 100	**30308JR**		2FB	19.9	50	52	80	77	82	3	5	2	1.5	0.35	1.74	0.96	0.757
	90	35.25	33	26	2	1.5	140	138	20.2	4 700	6 200	**TR0809AR**		—	27.5	50	49	80	67	85	3	9	2	1.5	0.55	1.10	0.60	1.10
	90	35.25	33	27	2	1.5	145	139	21.3	4 600	6 200	**32308JR**		2FD	24.3	50	50	80	73	82	3	8	2	1.5	0.35	1.74	0.96	1.06
45	68	15	15	12	0.6	0.6	43.5	52.4	7.45	5 300	7 100	**32909JR**		2BC	12.5	49.5	50	63.5	61	64	3	3	0.6	0.6	0.32	1.88	1.04	0.190
	75	20	20	15.5	1	1	78.8	86.5	12.6	5 000	6 600	**32009JR**		3CC	16.5	50.5	51	69.5	67	72	4	4.5	1	1	0.39	1.53	0.84	0.354
	75	24	24	19	1	1	87.4	101	14.9	5 000	6 700	**33009JR**		2CE	16.4	50.5	51	69.5	67	71	4	5	1	1	0.29	2.04	1.12	0.416
	80	26	26	20.5	1.5	1.5	110	120	17.9	4 800	6 400	**33109JR**		3CE	19.4	53.5	52	71.5	69	76.5	4	5.5	1.5	1.5	0.38	1.57	0.86	0.563
	85	20.75	19	15	1.5	1.5	83.1	77.0	11.4	4 600	6 100	**30209XR**		—	21.1	53.5	54	76.5	71	80	4	5.5	1.5	1.5	0.55	1.10	0.60	0.500
	85	20.75	19	16	1.5	1.5	83.9	77.4	11.6	4 600	6 100	**30209JR**		3DB	18.9	53.5	54	76.5	74	80	3	4.5	1.5	1.5	0.40	1.48	0.81	0.502
	85	24.75	23	19	1.5	1.5	101	102	15.1	4 600	6 200	**32209CR**		—	23.0	53.5	53	76.5	69	81	3	5.5	1.5	1.5	0.55	1.10	0.60	0.625
	85	24.75	23	19	1.5	1.5	105	104	15.6	4 600	6 100	**32209JR-1**		3DC	20.3	53.5	53	76.5	73	81	3	5.5	1.5	1.5	0.40	1.48	0.81	0.597
	85	32	32	25	1.5	1.5	139	149	22.3	4 600	6 200	**33209JR**		3DE	21.8	53.5	52	76.5	72	81	5	7	1.5	1.5	0.39	1.56	0.86	0.818
	95	29	26.5	20	2.5	2.5	118	118	17.0	3 600	5 100	**T7FC045**		7FC	32.6	57	54	83	71	91	3	9	2	2	0.87	0.69	0.38	0.943
	95	36	35	30	2.5	2.5	175	177	27.2	4 300	5 700	**T2ED045**		2ED	23.8	57	55	83	80	89	6	6	2	2	0.32	1.86	1.02	1.20

[Note] 1) Please consult with JTEKT when using the bearings identified by suffix C. They are medium-tapered types especially designed for special purposes.

metric series

d (45) ~ (55) mm



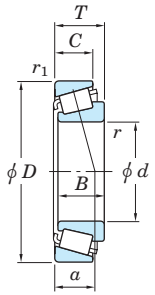


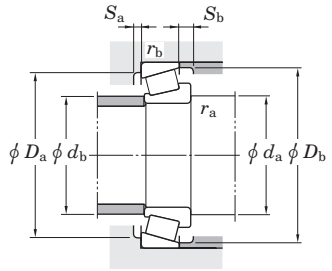
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)								Constant	Axial load factors		(Refer.) Mass (kg)
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	
45	100	27.25	25	18	2	1.5	119	107	15.8	3 400	4 700	**30309DJR**		7FB	32.9	55	56	90	79	96	3	9	2	1.5	0.83	0.73	0.40	0.973
	100	27.25	25	20	2	1.5	136	119	18.1	4 100	5 500	**30309CR**		—	25.7	55	57	90	81	94	4	7	2	1.5	0.55	1.10	0.60	1.00
	100	27.25	25	22	2	1.5	141	128	19.9	4 100	5 400	**30309JR**		2FB	21.3	55	59	90	86	93	3	5	2	1.5	0.35	1.74	0.96	1.01
	100	38.25	36	29	2	1.5	181	182	27.0	4 200	5 600	**32309CR**		—	30.3	55	56	90	76	95	4	9	2	1.5	0.55	1.10	0.60	1.45
	100	38.25	36	30	2	1.5	183	180	27.7	4 100	5 500	**32309JR**		2FD	26.8	55	56	90	82	93	3	8	2	1.5	0.35	1.74	0.96	1.43
50	72	15	15	12	0.6	0.6	45.0	56.3	8.00	4 900	6 600	**32910JR**		2BC	13.7	54.5	55	67.5	65	69	3	3	0.6	0.6	0.34	1.76	0.97	0.195
	80	20	20	15.5	1	1	82.7	94.5	13.8	4 600	6 100	**32010JR**		3CC	17.7	55.5	56	74.5	72	77	4	4.5	1	1	0.42	1.42	0.78	0.389
	80	24	24	19	1	1	91.8	110	16.3	4 600	6 100	**33010JR**		2CE	17.4	55.5	56	74.5	72	76	4	5	1	1	0.32	1.90	1.04	0.451
	85	26	26	20	1.5	1.5	112	127	18.9	4 400	5 900	**33110JR**		3CE	20.6	58.5	56	76.5	74	81.5	4	6	1.5	1.5	0.41	1.46	0.80	0.594
	90	21.75	20	16	1.5	1.5	96.7	95.8	14.3	4 300	5 700	**30210CR**		—	22.7	58.5	58	81.5	76	86	4	5.5	1.5	1.5	0.55	1.10	0.60	0.590
	90	21.75	20	17	1.5	1.5	95.6	91.7	13.8	4 300	5 700	**30210JR**		3DB	20.1	58.5	58	81.5	79	85	3	4.5	1.5	1.5	0.42	1.43	0.79	0.566
	90	24.75	23	19	1.5	1.5	106	113	16.7	4 300	5 700	**32210CR**		—	24.0	58.5	58	81.5	74	86	3	5.5	1.5	1.5	0.55	1.10	0.60	0.675
	90	24.75	23	19	1.5	1.5	106	105	15.9	4 300	5 700	**32210JR**		3DC	20.6	58.5	58	81.5	78	85	3	5.5	1.5	1.5	0.42	1.43	0.79	0.643
	90	32	32	24.5	1.5	1.5	150	167	25.0	4 300	5 700	**33210JR**		3DE	23.1	58.5	57	81.5	77	86.5	5	7.5	1.5	1.5	0.41	1.45	0.80	0.887
	100	36	35	30	2.5	2.5	196	196	30.2	4 100	5 400	**T2ED050**		2ED	24.5	62	58	88	84	94	6	6	2	2	0.34	1.75	0.96	1.28
	105	32	29	22	3	3	141	140	20.3	3 300	4 600	**T7FC050**		7FC	35.9	64	59	91	78	100	4	10	2.5	2.5	0.87	0.69	0.38	1.25
	110	29.25	27	19	2.5	2	144	133	19.8	3 100	4 300	**30310DJR**		7FB	35.0	62	62	98	87	105	3	10	2	2	0.83	0.73	0.40	1.25
	110	29.25	27	20	2.5	2	155	143	21.9	3 700	4 900	**30310CR**		—	27.5	62	64	98	90	103	4	9	2	2	0.55	1.10	0.60	1.25
	110	29.25	27	23	2.5	2	172	152	24.0	3 700	4 900	**30310JR**		2FB	22.9	62	65	98	95	102	3	6	2	2	0.35	1.74	0.96	1.32
	110	42.25	40	33	2.5	2	214	234	34.6	3 800	5 100	**32310CR**		5FD	33.4	62	61	98	81	103	4	9	2	2	0.55	1.10	0.60	2.00
	110	42.25	40	33	2.5	2	221	220	34.2	3 700	5 000	**32310JR**		2FD	29.4	62	62	98	90	102	3	9	2	2	0.35	1.74	0.96	1.89
55	80	17	17	14	1	1	55.8	73.3	10.6	4 400	5 900	**32911JR**		2BC	14.5	61	61	74	72	76	3	3	1	1	0.31	1.94	1.07	0.285
	90	23	23	17.5	1.5	1.5	106	121	18.2	4 100	5 500	**32011JR**		3CC	19.8	63.5	63	81.5	81	86	4	5.5	1.5	1.5	0.41	1.48	0.81	0.569
	90	27	27	21	1.5	1.5	121	149	22.6	4 100	5 400	**33011JR**		2CE	19.3	63.5	63	81.5	81	86	5	6	1.5	1.5	0.31	1.92	1.06	0.672
	95	30	30	23	1.5	1.5	145	161	24.6	4 000	5 300	**33111JR**		3CE	22.5	63.5	62	86.5	83	91	5	7	1.5	1.5	0.37	1.60	0.88	0.868
	100	22.75	21	17	2	1.5	112	108	16.2	3 900	5 200	**30211CR**		—	24.3	65	63	90	84	95	4	5.5	2	1.5	0.55	1.10	0.60	0.750
	100	22.75	21	18	2	1.5	118	113	17.3	3 900	5 200	**30211JR**		3DB	20.7	65	64	90	88	94	4	4.5	2	1.5	0.40	1.48	0.81	0.732
	100	26.75	25	21	2	1.5	134	135	20.4	3 900	5 200	**32211CR**		—	25.9	65	64	90	83	96	4	5.5	2	1.5	0.55	1.10	0.60	0.875

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metric series

d (55) ~ (65) mm



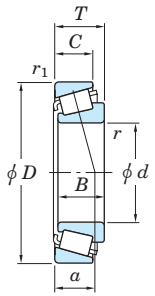


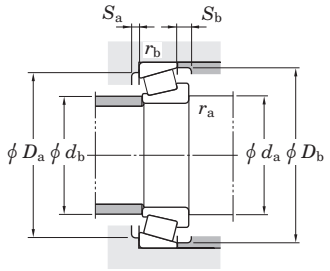
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Dimension series to ISO355	Load center (mm)	Mounting dimensions (mm)									Constant	Axial load factors		(Refer.) Mass
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.			(Refer.)	a	d_a min.	d_b max.	D_a max.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	(kg)
55	100	26.75	25	21	2	1.5	134	133	20.5	3 900	5 200	**32211JR-1**		3DC	23.0	65	63	90	87	95	4	5.5	2	1.5	0.40	1.48	0.81	0.863
	100	35	35	27	2	1.5	178	189	28.9	3 900	5 200	**33211JR**		3DE	25.3	65	62	90	85	96	6	8	2	1.5	0.40	1.50	0.83	1.18
	115	34	31	23.5	3	3	161	164	23.9	3 000	4 200	**T7FC055**		7FC	38.6	69	65	101	86	109	4	10.5	2.5	2.5	0.87	0.69	0.38	1.59
	120	31.5	29	21	2.5	2	161	148	22.3	2 900	4 000	**30311DJR**		7FB	38.4	67	68	108	94	113	4	10.5	2	2	0.83	0.73	0.40	1.59
	120	31.5	29	22	2.5	2	180	161	24.8	3 400	4 500	**30311CR**		—	29.8	67	70	108	97	112	4.5	9.5	2	2	0.55	1.10	0.60	1.58
	120	31.5	29	25	2.5	2	187	170	27.0	3 300	4 500	**30311JR**		2FB	25.5	67	71	108	104	111	4	6.5	2	2	0.35	1.74	0.96	1.65
	120	45.5	43	35	2.5	2	230	247	36.9	3 400	4 600	**32311C**		5FD	35.9	67	67	108	90	113	4	10	2	2	0.55	1.10	0.60	2.45
	120	45.5	43	35	2.5	2	214	203	31.8	3 400	4 500	**32311J**		2FD	32.4	67	68	108	99	111	4	10.5	2	2	0.35	1.74	0.96	2.24
	120	45.5	43	35	2.5	2	250	250	39.1	3 400	4 500	**32311JR**		2FD	32.4	67	68	108	99	111	4	10.5	2	2	0.35	1.74	0.96	2.38
60	85	17	17	14	1	1	57.6	78.2	11.3	4 100	5 500	**32912JR**		2BC	15.6	65.5	66	79.5	77	81	3	3	1	1	0.33	1.81	1.00	0.306
	95	23	23	17.5	1.5	1.5	108	127	19.0	3 900	5 200	**32012JR**		4CC	21.0	68.5	67	86.5	85	91	4	5.5	1.5	1.5	0.43	1.39	0.77	0.621
	95	27	27	21	1.5	1.5	127	162	24.5	3 900	5 200	**33012JR**		2CE	20.1	68.5	67	86.5	85	90	5	6	1.5	1.5	0.33	1.83	1.01	0.719
	100	30	30	23	1.5	1.5	149	170	25.9	3 700	5 000	**33112JR**		3CE	23.7	68.5	67	91.5	88	96	5	7	1.5	1.5	0.40	1.51	0.83	0.923
	110	23.75	22	17	2	1.5	127	123	18.8	3 500	4 700	**30212CR**		—	26.2	70	70	100	93	104	4	6.5	2	1.5	0.55	1.10	0.60	0.930
	110	23.75	22	19	2	1.5	133	127	19.7	3 500	4 700	**30212JR**		3EB	21.9	70	70	100	96	103	4	4.5	2	1.5	0.40	1.48	0.81	0.945
	110	29.75	28	22	2	1.5	160	164	25.1	3 600	4 700	**32212CR**		—	28.6	70	68	100	91	105	4	7.5	2	1.5	0.55	1.10	0.60	1.20
	110	29.75	28	24	2	1.5	164	167	25.9	3 500	4 700	**32212JR**		3EC	25.1	70	69	100	95	104	4	5.5	2	1.5	0.40	1.48	0.81	1.19
	110	38	38	29	2	1.5	217	239	36.6	3 600	4 700	**33212JR**		3EE	27.2	70	69	100	93	105	6	9	2	1.5	0.40	1.48	0.82	1.57
	115	39	38	31	4	2.5	198	227	34.0	3 400	4 600	**T5ED060**		5ED	32.4	78	70	103	92	110	5	8	3	2	0.53	1.13	0.62	1.81
	115	40	39	33	2.5	2.5	229	242	37.7	3 400	4 600	**T2EE060**		2EE	27.6	72	70	103	98	109	6	7	2	2	0.33	1.80	0.99	1.80
	125	37	33.5	26	3	3	191	194	28.8	2 800	3 900	**T7FC060**		7FC	40.8	74	71	111	94	119	4	11	2.5	2.5	0.82	0.73	0.40	2.03
	130	33.5	31	22	3	2.5	191	179	27.1	2 600	3 700	**30312DJR**		7FB	40.8	74	73	118	103	124	4	11.5	2.5	2	0.83	0.73	0.40	2.01
	130	33.5	31	23	3	2.5	211	196	30.5	3 100	4 200	**30312CR**		—	31.9	74	75	118	105	121	5	10.5	2.5	2	0.55	1.10	0.60	1.99
	130	33.5	31	26	3	2.5	217	201	31.9	3 100	4 100	**30312JR**		2FB	26.9	74	77	118	112	120	4	7.5	2.5	2	0.35	1.74	0.96	2.08
	130	48.5	46	37	3	2.5	286	310	41.4	3 200	4 300	**32312CR**		5FD	38.3	74	73	118	98	122	5	11	2.5	2	0.55	1.10	0.60	3.15
	130	48.5	46	37	3	2.5	277	275	38.6	3 100	4 200	**32312J**		2FD	32.3	74	74	118	107	120	4	11.5	2.5	2	0.35	1.74	0.96	2.87
	130	48.5	46	37	3	2.5	306	315	44.1	3 100	4 200	**32312JR**		2FD	32.3	74	74	118	107	120	4	11.5	2.5	2	0.35	1.74	0.96	2.99
65	90	17	17	14	1	1	59.2	83.1	12.0	3 900	5 200	**32913JR**		2BC	16.8	70.5	70	84.5	81	86	3	3	1	1	0.35	1.70	0.93	0.327

[Note] 1) Please consult with JTEKT when using the bearings identified by suffix C. They are medium-tapered types especially designed for special purposes.

metric series

d (65) ~ (70) mm



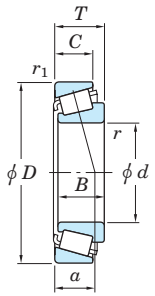


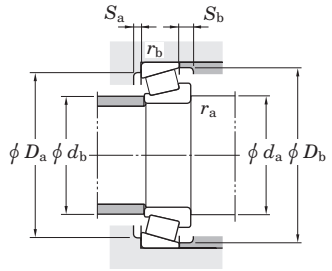
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)									Constant	Axial load factors		(Refer.) Mass (kg)
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_a min.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	
65	100	23	23	17.5	1.5	1.5	113	137	20.6	3 600	4 800	**32013JR**		4CC	22.5	73.5	72	91.5	90	97	4	5.5	1.5	1.5	0.46	1.31	0.72	0.664
	100	27	27	21	1.5	1.5	129	169	25.5	3 600	4 800	**33013JR**		2CE	21.1	73.5	72	91.5	89	96	5	6	1.5	1.5	0.35	1.72	0.95	0.762
	110	34	34	26.5	1.5	1.5	191	223	34.3	3 400	4 600	**33113JR**		3DE	25.9	73.5	73	101.5	96	106	6	7.5	1.5	1.5	0.39	1.55	0.85	1.33
	120	24.75	23	18	2	1.5	145	139	21.5	3 200	4 300	**30213CR**		—	28.1	75	77	110	102	114	4	6.5	2	1.5	0.55	1.10	0.60	1.15
	120	24.75	23	20	2	1.5	160	156	24.3	3 200	4 300	**30213JR**		3EB	24.2	75	77	110	106	113	4	4.5	2	1.5	0.40	1.48	0.81	1.18
	120	32.75	31	24	2	1.5	190	198	30.4	3 200	4 300	**32213CR**		—	31.3	75	75	110	99	114	4	8.5	2	1.5	0.55	1.10	0.60	1.55
	120	32.75	31	27	2	1.5	196	203	31.7	3 200	4 300	**32213JR**		3EC	26.6	75	76	110	104	115	4	5.5	2	1.5	0.40	1.48	0.81	1.58
	120	39	38	31	4	2.5	190	232	34.7	3 200	4 300	**T5ED065**		5ED	34.1	83	75	108	96	115	5	8	3	2	0.56	1.07	0.59	1.93
	120	41	41	32	2	1.5	250	277	43.0	3 200	4 300	**33213JR**		3EE	30.0	75	74	110	102	115	7	9	2	1.5	0.39	1.54	0.85	2.02
	130	37	33.5	26	3	3	186	211	31.2	2 600	3 600	**T7FC065**		7FC	44.4	79	78	116	98	124	4	11	2.5	2.5	0.87	0.69	0.38	2.17
	140	36	33	23	3	2.5	220	209	31.4	2 400	3 400	**30313DJR**		7GB	44.3	79	79	128	111	133	4	13	2.5	2	0.83	0.73	0.40	2.44
	140	36	33	25	3	2.5	241	227	35.1	2 900	3 900	**30313CR**		—	34.3	79	81	128	113	130	5	11	2.5	2	0.55	1.10	0.60	2.44
	140	36	33	28	3	2.5	255	239	37.6	2 800	3 800	**30313JR**		2GB	29.3	79	83	128	122	130	4	8	2.5	2	0.35	1.74	0.96	2.56
	140	51	48	39	3	2.5	322	361	49.0	2 900	3 900	**32313CR**		5GD	40.9	79	79	128	106	131	5	12	2.5	2	0.55	1.10	0.60	3.85
	140	51	48	39	3	2.5	313	312	43.4	2 900	3 900	**32313J**		2GD	34.7	79	80	128	117	130	4	12	2.5	2	0.35	1.74	0.96	3.49
	140	51	48	39	3	2.5	346	357	49.6	2 900	3 900	**32313JR**		2GD	34.7	79	80	128	117	130	4	12	2.5	2	0.35	1.74	0.96	3.64
70	100	20	20	16	1	1	89.0	115	17.2	3 500	4 700	**32914JR**		2BC	17.8	75.5	77	94.5	91	96	4	4	1	1	0.32	1.90	1.05	0.496
	110	25	25	19	1.5	1.5	136	163	24.8	3 300	4 400	**32014JR**		4CC	23.6	78.5	78	101.5	98	105	5	6	1.5	1.5	0.43	1.38	0.76	0.884
	110	31	31	25.5	1.5	1.5	168	208	32.3	3 300	4 400	**33014JR**		2CE	22.1	78.5	78	101.5	99	105	5	5.5	1.5	1.5	0.28	2.11	1.16	1.09
	120	37	37	29	2	1.5	227	266	41.2	3 100	4 200	**33114JR**		3DE	28.0	80	79	110	104	115	6	8	2	1.5	0.38	1.58	0.87	1.71
	125	26.25	24	19	2	1.5	158	158	24.5	3 000	4 000	**30214CR**		—	29.9	80	82	116.5	107	119	4	7	2	1.5	0.55	1.10	0.60	1.30
	125	26.25	24	21	2	1.5	173	173	27.1	3 100	4 100	**30214JR**		3EB	25.9	80	81	116.5	110	118	4	5	2	1.5	0.42	1.43	0.79	1.32
	125	33.25	31	24	2	1.5	197	212	32.6	3 100	4 100	**32214CR**		—	32.6	80	80	116.5	104	120	4	9.5	2	1.5	0.55	1.10	0.60	1.65
	125	33.25	31	27	2	1.5	212	225	35.2	3 100	4 100	**32214JR**		3EC	29.2	80	80	116.5	108	119	4	6	2	1.5	0.42	1.43	0.79	1.71
	125	41	41	32	2	1.5	258	294	45.5	3 100	4 100	**33214JR**		3EE	31.2	80	79	116.5	107	120	7	9	2	1.5	0.41	1.47	0.81	2.16
	130	43	42	35	3	2.5	291	319	50.0	3 000	4 000	**T2ED070**		2ED	30.2	84	81	118	111	123	1	1	2.5	2	0.33	1.80	0.99	2.48
	140	39	35.5	27	3	3	222	242	35.8	2 400	3 400	**T7FC070**		7FC	46.5	84	82	126	106	133	5	12	2.5	2.5	0.87	0.69	0.38	2.64
	140	52	51	43	5	3	330	382	51.6	2 900	3 800	**T4FE070**		4FE	37.7	92	82	126	111	133	7	9	4	2.5	0.45	1.34	0.74	3.69

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metric series

d (70) ~ (80) mm



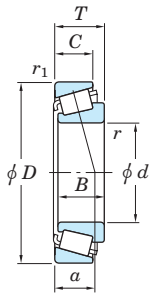


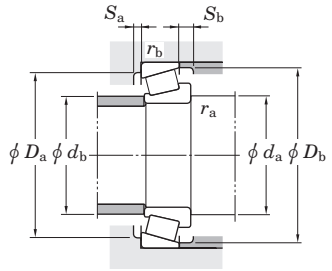
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Dimension series to ISO355	Load center (mm)	Mounting dimensions (mm)								Constant	Axial load factors		(Refer.) Mass
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.			(Refer.)	a	d_a min.	d_b max.	D_a max.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	(kg)
70	150	38	35	25	3	2.5	246	235	34.9	2 300	3 200	**30314DJR**		7GB	47.1	84	84	138	118	142	4	13	2.5	2	0.83	0.73	0.40	2.97
	150	38	35	30	3	2.5	280	256	36.0	2 700	3 600	**30314CR**		—	37.0	84	87	138	123	141	6	8	2.5	2	0.55	1.10	0.60	3.10
	150	38	35	30	3	2.5	288	273	42.2	2 600	3 500	**30314JR**		2GB	30.5	84	89	138	130	140	4	8	2.5	2	0.35	1.74	0.96	3.08
	150	54	51	42	3	2.5	321	315	44.1	2 700	3 600	**32314**		—	37.0	84	86	138	125	140	4	12	2.5	2	0.35	1.73	0.95	4.11
	150	54	51	42	3	2.5	371	391	51.4	2 700	3 600	**32314C**		5GD	44.4	84	84	138	115	142	5	12	2.5	2	0.55	1.10	0.60	4.50
	150	54	51	42	3	2.5	396	414	57.2	2 700	3 600	**32314JR**		2GD	37.4	84	86	138	125	140	4	12	2.5	2	0.35	1.74	0.96	4.50
75	105	20	20	16	1	1	92.2	123	18.4	3 300	4 400	**32915JR**		2BC	18.9	80.5	81	99.5	96	101	4	4	1	1	0.33	1.80	0.99	0.526
	115	25	25	19	1.5	1.5	139	169	25.8	3 100	4 200	**32015JR**		4CC	25.1	83.5	83	106.5	103	110	5	6	1.5	1.5	0.46	1.31	0.72	0.930
	115	31	31	25.5	1.5	1.5	177	225	35.0	3 200	4 200	**33015JR**		2CE	22.9	83.5	83	106.5	104	110	6	5.5	1.5	1.5	0.30	2.01	1.11	1.16
	125	37	37	29	2	1.5	234	280	43.4	3 000	4 000	**33115JR**		3DE	29.3	85	84	116.5	109	120	6	8	2	1.5	0.40	1.51	0.83	1.84
	130	27.25	25	20	2	1.5	171	178	27.4	2 900	3 800	**30215CR**		—	31.0	85	87	121.5	111	124	5	7	2	1.5	0.55	1.10	0.60	1.40
	130	27.25	25	22	2	1.5	178	181	28.2	2 900	3 900	**30215JR**		4DB	27.6	85	86	121.5	115	124	4	5	2	1.5	0.44	1.38	0.76	1.42
	130	33.25	31	24	2	1.5	204	225	34.5	2 900	3 900	**32215CR**		—	33.7	85	85	121.5	109	125	4	9	2	1.5	0.55	1.10	0.60	1.75
	130	33.25	31	27	2	1.5	218	234	36.4	2 900	3 900	**32215JR**		4DC	30.2	85	85	121.5	114	125	4	6	2	1.5	0.44	1.38	0.76	1.77
	130	41	41	31	2	1.5	266	310	47.7	2 900	3 900	**33215JR**		3EE	32.5	85	83	121.5	111	125	7	10	2	1.5	0.43	1.40	0.77	2.26
	150	42	38	29	3	3	240	270	39.0	2 200	3 100	**T7FC075**		7FC	50.6	89	89	136	114	143	5	13	2.5	2.5	0.87	0.69	0.38	3.24
	160	40	37	26	3	2.5	266	254	34.2	2 100	2 900	**30315DJR**		7GB	49.9	89	91	148	127	151	6	14	2.5	2	0.83	0.73	0.40	3.45
	160	40	37	26	3	2.5	277	266	36.9	2 100	2 900	**30315DR**		—	48.8	89	91	148	127	151	6	14	2.5	2	0.81	0.74	0.41	3.48
	160	40	37	31	3	2.5	310	296	42.1	2 500	3 400	**30315CR**		—	39.2	89	94	148	130	150	6	9	2.5	2	0.55	1.10	0.60	3.80
	160	40	37	31	3	2.5	325	311	44.9	2 500	3 300	**30315JR**		2GB	32.5	89	95	148	139	149	4	9	2.5	2	0.35	1.74	0.96	3.65
	160	40	37	31	3	2.5	313	298	43.3	2 500	3 300	**30315R**		—	31.9	89	95	148	139	149	4	9	2.5	2	0.35	1.73	0.95	3.52
	160	58	55	43	3	2.5	447	474	61.4	2 500	3 400	**32315CR**		—	46.6	89	90	148	125	154	6	15	2.5	2	0.55	1.10	0.60	5.50
	160	58	55	45	3	2.5	454	481	64.6	2 500	3 300	**32315JR**		2GD	40.0	89	91	148	133	149	4	13	2.5	2	0.35	1.74	0.96	5.41
	160	58	55	45	3	2.5	425	444	60.3	2 500	3 300	**32315R**		—	39.5	89	91	148	133	149	4	13	2.5	2	0.35	1.73	0.95	5.30
80	110	20	20	16	1	1	95.1	131	19.5	3 100	4 200	**32916JR**		2BC	20.1	85.5	86	104.5	101	106	4	4	1	1	0.35	1.71	0.94	0.556
	125	29	29	22	1.5	1.5	185	225	34.6	2 900	3 900	**32016JR**		3CC	26.7	88.5	89	116.5	112	120	6	7	1.5	1.5	0.42	1.42	0.78	1.32
	125	36	36	29.5	1.5	1.5	218	288	44.8	2 900	3 900	**33016JR**		2CE	25.1	88.5	90	116.5	112	119	6	6.5	1.5	1.5	0.28	2.16	1.19	1.63
	130	37	37	29	2	1.5	240	294	44.9	2 800	3 800	**33116JR**		3DE	30.5	90	89	121.5	114	126	6	8	2	1.5	0.42	1.44	0.79	1.93

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metric series

d (80) ~ (90) mm

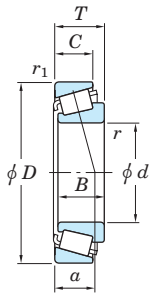


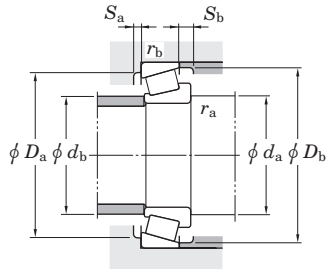


Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)								Constant	Axial load factors		(Refer.) Mass (kg)
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_a min.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	
80	140	28.25	26	22	2.5	2	202	202	31.2	2 700	3 600	**30216JR**		3EB	28.6	92	91	130	124	132	4	6	2	2	0.42	1.43	0.79	1.72
	140	35.25	33	28	2.5	2	253	271	41.5	2 700	3 600	**32216JR**		3EC	31.7	92	90	130	122	134	4	7	2	2	0.42	1.43	0.79	2.17
	140	46	46	35	2.5	2	313	371	56.1	2 700	3 600	**33216JR**		3EE	35.7	92	89	130	119	135	7	11	2	2	0.43	1.41	0.78	2.99
	145	46	45	38	3	2.5	333	381	52.0	2 600	3 500	**T2ED080**		2ED	32.7	94	92	133	125	137	7	8	2.5	2	0.32	1.88	1.03	3.20
	170	42.5	39	27	3	2.5	294	282	38.7	2 000	2 800	**30316DJR**		7GB	53.5	94	97	158	134	159	6	15.5	2.5	2	0.83	0.73	0.40	4.12
	170	42.5	39	33	3	2.5	368	355	49.9	2 300	3 100	**30316JR**		2GB	34.8	94	102	158	148	159	4	9.5	2.5	2	0.35	1.74	0.96	4.46
	170	42.5	39	33	3	2.5	345	330	47.1	2 300	3 100	**30316R**		—	33.9	94	102	158	148	159	4	9.5	2.5	2	0.35	1.73	0.95	4.26
	170	61.5	58	48	3	2.5	434	440	58.6	2 300	3 100	**32316J**		2GD	43.5	94	98	158	142	159	4	13.5	2.5	2	0.35	1.74	0.96	6.04
	170	61.5	58	48	3	2.5	480	503	67.0	2 300	3 100	**32316JR**		2GD	43.5	94	98	158	142	159	4	13.5	2.5	2	0.35	1.74	0.96	6.31
85	120	23	23	18	1.5	1.5	122	165	25.0	2 900	3 900	**32917JR**		2BC	21.2	93.5	93	111.5	109	115	5	5	1.5	1.5	0.33	1.83	1.01	0.794
	130	29	29	22	1.5	1.5	189	234	35.5	2 800	3 700	**32017JR**		4CC	28.0	93.5	94	121.5	117	125	6	7	1.5	1.5	0.44	1.36	0.75	1.38
	130	36	36	29.5	1.5	1.5	222	300	46.0	2 800	3 700	**33017JR**		2CE	26.3	93.5	94	121.5	118	125	6	6.5	1.5	1.5	0.29	2.06	1.13	1.72
	140	41	41	32	2.5	2	282	346	52.2	2 600	3 500	**33117JR**		3DE	33.2	97	95	130	122	135	7	9	2	2	0.41	1.48	0.81	2.43
	150	30.5	28	24	2.5	2	228	231	35.1	2 500	3 400	**30217JR**		3EB	30.4	97	97	140	132	141	5	6.5	2	2	0.42	1.43	0.79	2.17
	150	38.5	36	30	2.5	2	290	315	47.5	2 500	3 400	**32217JR**		3EC	34.2	97	96	140	130	142	5	8.5	2	2	0.42	1.43	0.79	2.80
	150	49	49	37	2.5	2	368	439	59.1	2 500	3 400	**33217JR**		3EE	37.1	97	95	140	128	144	7	12	2	2	0.42	1.43	0.79	3.63
	180	44.5	41	28	4	3	288	265	36.0	1 900	2 600	**30317D**		—	56.0	103	103	166	143	169	6	16.5	3	2.5	0.81	0.74	0.41	4.54
	180	44.5	41	28	4	3	328	317	42.6	1 900	2 600	**30317DJR**		7GB	56.3	103	103	166	143	169	6	16.5	3	2.5	0.83	0.73	0.40	4.81
	180	44.5	41	34	4	3	396	384	53.0	2 200	2 900	**30317JR**		2GB	36.0	103	107	166	156	167	5	10.5	3	2.5	0.35	1.74	0.96	5.15
	180	44.5	41	34	4	3	381	367	51.1	2 200	2 900	**30317R**		—	35.8	103	107	166	156	167	5	10.5	3	2.5	0.35	1.73	0.95	4.97
	180	63.5	60	49	4	3	549	587	77.6	2 200	3 000	**32317JR**		2GD	43.8	103	103	166	150	167	5	14.5	3	2.5	0.35	1.74	0.96	7.42
90	125	23	23	18	1.5	1.5	126	175	26.2	2 800	3 700	**32918JR**		2BC	22.3	98.5	97	116.5	114	120	5	5	1.5	1.5	0.34	1.75	0.96	0.834
	140	32	32	24	2	1.5	224	276	41.5	2 600	3 500	**32018JR**		3CC	29.8	100	100	131.5	125	134	6	8	2	1.5	0.42	1.42	0.78	1.80
	140	39	39	32.5	2	1.5	278	367	55.6	2 600	3 400	**33018JR**		2CE	27.1	100	100	131.5	127	135	7	6.5	2	1.5	0.27	2.23	1.23	2.22
	150	45	45	35	2.5	2	324	413	61.1	2 500	3 300	**33118JR**		3DE	35.4	102	100	140	130	144	7	10	2	2	0.40	1.51	0.83	3.13
	155	46	46	38	3	3	342	405	54.1	2 400	3 200	**T2ED090**		2ED	33.5	104	102	141	135	147	7	8	2.5	2.5	0.33	1.84	1.01	3.47
	160	32.5	30	26	2.5	2	255	261	39.0	2 400	3 200	**30218JR**		3FB	32.6	102	103	150	140	150	5	6.5	2	2	0.42	1.43	0.79	2.65
	160	42.5	40	34	2.5	2	329	362	53.7	2 400	3 200	**32218JR**		3FC	37.0	102	102	150	138	152	5	8.5	2	2	0.42	1.43	0.79	3.47

metric series

d (90) ~ (100) mm

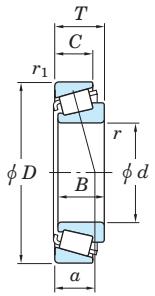


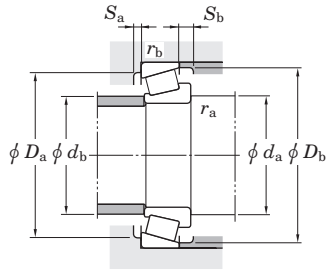


Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)									Constant	Axial load factors		(Refer.) Mass (kg)
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	
90	160	55	55	42	2.5	2	430	527	68.3	2 400	3 200	**33218JR**		3FE	40.8	102	101	150	135	154	9	13	2	2	0.42	1.43	0.78	4.76
	190	46.5	43	30	4	3	359	350	46.2	1 700	2 400	**30318DJR**		7GB	59.6	108	109	176	151	179	6	16.5	3	2.5	0.83	0.73	0.40	5.57
	190	46.5	43	30	4	3	352	336	44.9	1 700	2 400	**30318DR**		—	59.1	108	109	176	151	179	6	16.5	3	2.5	0.81	0.74	0.41	5.60
	190	46.5	43	36	4	3	432	420	57.1	2 100	2 700	**30318JR**		2GB	38.1	108	113	176	165	177	5	10.5	3	2.5	0.35	1.74	0.96	6.04
	190	46.5	43	36	4	3	421	407	55.5	2 100	2 700	**30318R**		—	37.2	108	113	176	165	177	5	10.5	3	2.5	0.35	1.73	0.95	5.78
	190	67.5	64	53	4	3	577	614	78.7	2 100	2 800	**32318JR**		2GD	46.6	108	108	176	157	177	5	14.5	3	2.5	0.35	1.74	0.96	8.61
95	130	23	23	18	1.5	1.5	130	186	27.4	2 600	3 500	**32919JR**		2BC	23.5	103.5	102	121.5	119	125	5	5	1.5	1.5	0.36	1.68	0.92	0.876
	145	32	32	24	2	1.5	229	287	42.6	2 500	3 300	**32019JR**		4CC	31.2	105	105	136.5	130	140	6	8	2	1.5	0.44	1.36	0.75	1.88
	145	39	39	32.5	2	1.5	284	382	57.3	2 500	3 300	**33019JR**		2CE	27.8	105	104	136.5	131	139	7	6.5	2	1.5	0.28	2.16	1.19	2.31
	160	46	46	38	3	3	353	427	56.4	2 300	3 100	**T2ED095**		2ED	34.6	109	107	146	140	152	7	8	2.5	2.5	0.34	1.77	0.97	3.62
	160	49	49	38	2.5	2	381	473	62.5	2 300	3 100	**33119JR**		3EE	37.3	107	106	150	138	154	8	11	2	2	0.39	1.54	0.85	3.89
	170	34.5	32	27	3	2.5	289	299	44.0	2 200	3 000	**30219JR**		3FB	34.9	109	110	158	149	159	5	7.5	2.5	2	0.42	1.43	0.79	3.20
	170	45.5	43	37	3	2.5	389	439	64.1	2 200	3 000	**32219JR**		3FC	38.9	109	108	158	145	161	5	8.5	2.5	2	0.42	1.43	0.79	4.34
	170	58	58	44	3	2.5	468	582	74.0	2 200	2 900	**33219JR**		3FE	42.8	109	107	158	144	163	9	14	2.5	2	0.41	1.47	0.81	5.66
	200	49.5	45	32	4	3	398	391	50.4	1 700	2 300	**30319DJR**		7GB	62.7	113	113	186	157	187	6	17.5	3	2.5	0.83	0.73	0.40	6.68
	200	49.5	45	38	4	3	396	368	49.2	2 000	2 600	**30319**		—	39.8	113	118	186	172	186	5	11.5	3	2.5	0.35	1.73	0.95	6.32
	200	49.5	45	38	4	3	465	455	60.9	2 000	2 600	**30319JR**		2GB	40.8	113	118	186	172	186	5	11.5	3	2.5	0.35	1.74	0.96	6.96
	200	71.5	67	55	4	3	534	544	70.2	2 000	2 600	**32319**		—	49.1	113	115	186	166	186	5	16.5	3	2.5	0.35	1.73	0.95	9.35
	200	71.5	67	55	4	3	646	695	89.2	2 000	2 600	**32319JR**		2GD	49.8	113	115	186	166	186	5	16.5	3	2.5	0.35	1.74	0.96	10.1
100	140	25	25	20	1.5	1.5	158	217	32.0	2 400	3 300	**32920JR**		2CC	24.0	109	108	131	128	135	5	5	1.5	1.5	0.33	1.82	1.00	1.19
	145	24	22.5	17.5	3	3	146	167	24.6	2 400	3 200	**T4CB100**		4CB	29.9	112	109	133	132	140	4	6.5	2.5	2.5	0.47	1.27	0.70	1.12
	150	32	32	24	2	1.5	233	298	43.8	2 400	3 200	**32020JR**		4CC	32.6	110	109	141	134	144	6	8	2	1.5	0.46	1.31	0.72	1.95
	150	39	39	32.5	2	1.5	290	397	59.0	2 400	3 200	**33020JR**		2CE	28.6	110	108	141	135	143	7	6.5	2	1.5	0.29	2.09	1.15	2.40
	165	47	46	39	3	3	368	458	59.5	2 200	3 000	**T2EE100**		2EE	35.1	114	112	151	145	157	7	8	2.5	2.5	0.32	1.88	1.04	3.86
	165	52	52	40	2.5	2	408	523	67.4	2 200	3 000	**33120JR**		3EE	40.1	112	111	155	142	159	8	12	2	2	0.41	1.48	0.81	4.29
	180	37	34	29	3	2.5	323	338	49.1	2 100	2 800	**30220JR**		3FB	36.8	114	116	168	157	168	5	8	2.5	2	0.42	1.43	0.79	3.83
	180	49	46	39	3	2.5	435	495	63.9	2 100	2 800	**32220JR**		3FC	42.1	114	114	168	154	171	5	10	2.5	2	0.42	1.43	0.79	5.21
	180	63	63	48	3	2.5	540	680	85.8	2 100	2 800	**33220JR**		3FE	45.7	114	112	168	151	172	10	15	2.5	2	0.40	1.48	0.82	6.92

metric series

d (100) ~ (110) mm

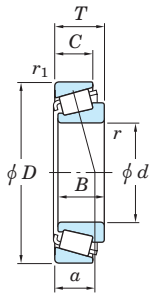


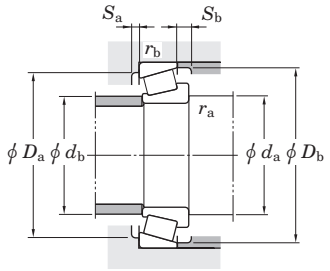


Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)									Constant	Axial load factors		(Refer.) Mass
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	(kg)
100	215	51.5	47	34	4	3	397	374	48.5	1 500	2 100	**30320D**		—	65.9	118	121	201	183	204	5	17	3	2.5	0.81	0.74	0.41	8.02
	215	51.5	47	39	4	3	430	400	52.5	1 800	2 400	**30320**		—	41.4	118	127	201	184	200	6	12.5	3	2.5	0.35	1.73	0.95	7.76
	215	51.5	47	39	4	3	528	521	68.0	1 800	2 400	**30320JR**		2GB	42.7	118	127	201	184	200	6	12.5	3	2.5	0.35	1.74	0.96	8.49
	215	56.5	51	35	4	3	465	459	56.4	1 500	2 200	**31320JR**		7GB	67.7	118	120	201	183	202	6	17.5	3	2.5	0.83	0.73	0.40	8.72
	215	77.5	73	60	4	3	614	637	79.6	1 800	2 400	**32320**		—	52.6	118	123	201	177	200	8	17.5	3	2.5	0.35	1.73	0.95	12.2
	215	77.5	73	60	4	3	725	783	96.9	1 800	2 400	**32320JR**		2GD	53.9	118	123	201	177	200	8	17.5	3	2.5	0.35	1.74	0.96	13.0
105	145	25	25	20	1.5	1.5	160	224	32.6	2 400	3 100	**32921JR**		2CC	25.1	113.5	113	136.5	133	140	5	5	1.5	1.5	0.34	1.75	0.96	1.23
	160	35	35	26	2.5	2	270	344	49.9	2 200	3 000	**32021JR**		4DC	34.5	117	116	150	143	154	6	9	2	2	0.44	1.35	0.74	2.45
	160	43	43	34	2.5	2	335	461	67.4	2 200	3 000	**33021JR**		2DE	30.9	117	116	150	145	153	7	9	2	2	0.28	2.12	1.17	3.08
	175	56	56	44	2.5	2	453	607	76.0	2 100	2 800	**33121JR**		3EE	43.2	117	116	165	150	169	9	12	2	2	0.40	1.48	0.82	5.33
	190	39	36	30	3	2.5	360	380	52.3	2 000	2 600	**30221JR**		3FB	39.0	119	122	178	165	178	6	9	2.5	2	0.42	1.43	0.79	4.49
	190	53	50	43	3	2.5	490	567	73.0	2 000	2 700	**32221JR**		3FC	44.8	119	120	178	161	180	6	10	2.5	2	0.42	1.43	0.79	6.37
	190	68	68	52	3	2.5	622	790	97.4	2 000	2 600	**33221JR**		3FE	48.8	119	117	178	159	182	10	16	2.5	2	0.40	1.49	0.82	8.43
	225	53.5	49	36	4	3	423	396	50.1	1 400	2 000	**30321D**		—	69.1	123	127	211	193	209	6	17	3	2.5	0.81	0.74	0.41	8.76
	225	53.5	49	41	4	3	464	432	56.0	1 700	2 300	**30321**		—	43.1	123	132	211	193	209	7	12.5	3	2.5	0.35	1.73	0.95	8.74
	225	53.5	49	41	4	3	581	578	73.6	1 700	2 300	**30321JR**		2GB	44.1	123	132	211	193	209	7	12.5	3	2.5	0.35	1.74	0.96	9.73
	225	58	53	36	4	3	495	489	59.4	1 500	2 100	**31321JR**		7GB	70.3	123	126	211	193	211	6	18	3	2.5	0.83	0.73	0.40	9.72
	225	81.5	77	63	4	3	679	707	86.7	1 800	2 300	**32321**		—	55.7	123	128	211	185	209	8	18.5	3	2.5	0.35	1.73	0.95	13.9
	225	81.5	77	63	4	3	794	866	107	1 800	2 300	**32321JR**		2GD	56.1	123	128	211	185	209	8	18.5	3	2.5	0.35	1.74	0.96	14.9
110	150	25	25	20	1.5	1.5	162	231	33.3	2 300	3 000	**32922JR**		2CC	26.3	119	118	141	138	145	5	5	1.5	1.5	0.36	1.69	0.93	1.28
	160	27	25.5	19.5	3	3	183	225	32.3	2 200	2 900	**T4CB110**		4CB	31.8	124	120	146	145	154	5	7.5	2.5	2.5	0.44	1.36	0.75	1.63
	170	38	38	29	2.5	2	312	395	56.7	2 100	2 800	**32022JR**		4DC	36.1	122	122	160	152	163	7	9	2	2	0.43	1.39	0.77	3.12
	170	47	47	37	2.5	2	360	502	64.9	2 100	2 800	**33022JR**		2DE	33.4	122	123	160	152	161	7	10	2	2	0.29	2.09	1.15	3.81
	180	56	56	43	2.5	2	464	634	78.6	2 000	2 700	**33122JR**		3EE	44.5	122	121	170	155	174	9	13	2	2	0.42	1.43	0.79	5.52
	200	41	38	32	3	2.5	405	434	58.1	1 900	2 500	**30222JR**		3FB	40.8	124	129	188	174	188	6	9	2.5	2	0.42	1.43	0.79	5.33
	200	56	53	46	3	2.5	547	640	80.4	1 900	2 500	**32222JR**		3FC	46.7	124	126	188	170	190	6	10	2.5	2	0.42	1.43	0.79	7.45
	240	54.5	50	36	4	3	456	429	53.5	1 400	1 900	**30322D**		—	71.5	128	135	226	205	222	6	18	3	2.5	0.81	0.74	0.41	10.2
	240	54.5	50	42	4	3	509	475	60.5	1 600	2 100	**30322**		—	44.8	128	141	226	206	222	8	12.5	3	2.5	0.35	1.73	0.95	10.4

metric series

d (110) ~ 130 mm

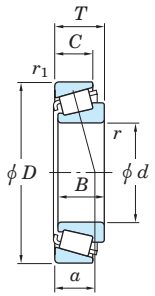


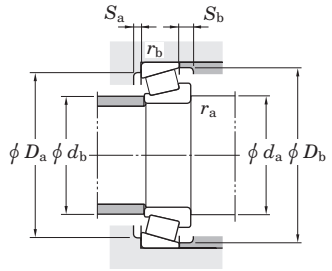


Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)									Constant	Axial load factors		(Refer.) Mass
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	(kg)
110	240	54.5	50	42	4	3	601	590	75.2	1 600	2 100	**30322JR**		2GB	46.3	128	141	226	206	222	8	12.5	3	2.5	0.35	1.74	0.96	11.4
	240	63	57	38	4	3	564	563	68.4	1 400	1 900	**31322JR**		7GB	76.2	128	135	226	205	224	6	21	3	2.5	0.83	0.73	0.40	12.2
	240	84.5	80	65	4	3	759	797	97.4	1 600	2 200	**32322**		—	57.3	128	137	226	198	222	9	19.5	3	2.5	0.35	1.73	0.95	16.6
	240	84.5	80	65	4	3	865	943	115	1 600	2 200	**32322JR**		2GD	59.3	128	137	226	198	222	9	19.5	3	2.5	0.35	1.74	0.96	17.8
120	165	29	29	23	1.5	1.5	215	298	42.5	2 100	2 700	**32924JR**		2CC	29.4	129	128	156	152	160	6	6	1.5	1.5	0.35	1.72	0.95	1.77
	170	27	25	19.5	3	3	206	262	37.0	2 000	2 700	**T4CB120**		4CB	34.6	134	130	156	155	164	4	7.5	2.5	2.5	0.47	1.27	0.70	1.76
	180	38	38	29	2.5	2	325	427	60.0	2 000	2 600	**32024JR**		4DC	38.8	132	131	170	161	173	7	9	2	2	0.46	1.31	0.72	3.34
	180	48	48	38	2.5	2	375	540	68.5	2 000	2 600	**33024JR**		2DE	36.2	132	132	170	160	171	6	10	2	2	0.31	1.97	1.08	4.16
	200	62	62	48	2.5	2	581	785	96.1	1 800	2 400	**33124JR**		3FE	47.8	132	133	190	172	192	9	14	2	2	0.40	1.51	0.83	7.73
	215	43.5	40	34	3	2.5	435	473	61.7	1 700	2 300	**30224JR**		4FB	44.2	134	140	203	187	203	6	9.5	2.5	2	0.44	1.38	0.76	6.36
	215	61.5	58	50	3	2.5	589	691	84.0	1 700	2 300	**32224JR**		4FD	51.6	134	136	203	181	204	7	11.5	2.5	2	0.44	1.38	0.76	9.04
	260	59.5	55	38	4	3	536	512	61.5	1 200	1 700	**30324D**		—	77.8	138	145	246	219	239	6	21	3	2.5	0.81	0.74	0.41	13.0
	260	59.5	55	46	4	3	631	611	76.9	1 500	2 000	**30324**		—	48.9	138	152	246	221	239	10	13.5	3	2.5	0.35	1.73	0.95	13.7
	260	59.5	55	46	4	3	712	714	89.9	1 500	2 000	**30324JR**		2GB	50.2	138	152	246	221	239	10	13.5	3	2.5	0.35	1.74	0.96	14.5
	260	68	62	42	4	3	657	665	77.8	1 300	1 800	**31324JR**		7GB	81.9	138	145	246	221	244	6	21	3	2.5	0.83	0.73	0.40	15.4
	260	90.5	86	69	4	3	1 000	1 110	131	1 500	2 000	**32324JR**		2GD	62.7	138	148	246	213	239	9	21.5	3	2.5	0.35	1.74	0.96	22.2
	260	90.5	86	69	4	3	997	1 110	132	1 500	2 000	**32324R**		—	61.1	138	148	246	213	239	9	21.5	3	2.5	0.35	1.73	0.95	21.8
130	180	32	32	25	2	1.5	251	368	51.2	1 900	2 500	**32926JR**		2CC	31.4	140	141	171	165	174	6	7	2	1.5	0.34	1.77	0.97	2.42
	185	29	27	21	3	3	230	282	39.2	1 800	2 500	**T4CB130**		4CB	37.8	144	141	171	170	179	5	8	2.5	2.5	0.47	1.27	0.70	2.22
	200	45	45	34	2.5	2	428	563	77.4	1 800	2 300	**32026JR**		4EC	42.9	142	144	190	178	192	8	11	2	2	0.43	1.38	0.76	5.04
	200	55	55	43	2.5	2	489	705	85.8	1 700	2 300	**33026JR**		2EE	42.5	142	143	190	178	192	8	12	2	2	0.34	1.76	0.97	6.19
	230	43.75	40	34	4	3	472	511	65.7	1 600	2 100	**30226JR**		4FB	46.2	148	152	216	203	218	7	9.5	3	2.5	0.44	1.38	0.76	7.24
	230	67.75	64	54	4	3	693	830	99.9	1 600	2 200	**32226JR**		4FD	56.0	148	146	216	193	219	7	13.5	3	2.5	0.44	1.38	0.76	11.5
	280	63.75	58	41	5	4	604	582	69.9	1 200	1 600	**30326D**		—	84.0	152	155	262	240	261	7	22	4	3	0.81	0.74	0.41	16.3
	280	63.75	58	49	5	4	823	834	102	1 400	1 800	**30326JR**		2GB	54.0	152	164	262	239	255	8	14.5	4	3	0.35	1.74	0.96	18.1
	280	72	66	44	5	4	734	748	85.7	1 200	1 600	**31326JR**		7GB	87.3	152	155	262	236	261	7	23	4	3	0.83	0.73	0.40	18.9
	280	98.75	93	78	5	4	1 070	1 160	134	1 400	1 800	**32326**		—	69.1	152	163	262	226	259	10	15	4	3	0.35	1.73	0.95	26.5

metric series

d **140 ~ (170) mm**

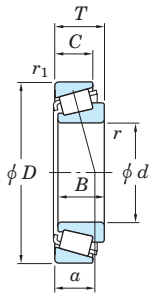


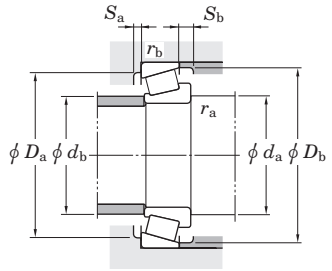


Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)									Constant	Axial load factors		(Refer.) Mass
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	(kg)
140	190	32	32	25	2	1.5	258	390	53.2	1 800	2 300	**32928JR**		2CC	33.6	150	150	181	184	6	7	2	1.5	0.36	1.67	0.92	2.57
	195	29	27	21	3	3	232	293	39.9	1 700	2 300	**T4CB140**		4CB	40.9	154	151	181	189	5	8	2.5	2.5	0.50	1.19	0.66	2.36
	210	45	45	34	2.5	2	435	585	79.2	1 700	2 200	**32028JR**		4DC	45.6	152	153	200	187	202	8	11	2	2	0.46	1.31	0.72	5.28
	210	56	56	44	2.5	2	510	758	90.9	1 600	2 200	**33028JR**		2DE	45.6	152	152	200	186	202	7	12	2	2	0.36	1.67	0.92	6.61
	250	45.75	42	36	4	3	526	570	71.8	1 500	1 900	**30228JR**		4FB	49.4	158	163	236	219	237	9	9.5	3	2.5	0.44	1.38	0.76	8.97
	250	71.75	68	58	4	3	796	961	112	1 500	2 000	**32228JR**		4FD	60.0	158	158	236	210	238	9	13.5	3	2.5	0.44	1.38	0.76	14.7
	300	67.75	62	44	5	4	655	627	74.5	1 100	1 500	**30328D**		—	90.2	162	169	282	254	280	7	23	4	3	0.81	0.74	0.41	20.0
	300	67.75	62	53	5	4	938	962	114	1 300	1 700	**30328JR**		2GB	56.9	162	179	282	254	273	10	14.5	4	3	0.35	1.74	0.96	22.6
	300	77	70	47	5	4	841	865	99.1	1 100	1 500	**31328JR**		7GB	93.8	162	167	282	254	280	8	26	4	3	0.83	0.73	0.40	23.3
	300	107.75	102	85	5	4	1 370	1 570	175	1 300	1 700	**32328R**		—	74.2	162	175	282	246	280	10	17	4	3	0.35	1.74	0.96	35.1
150	210	38	38	30	2.5	2	358	536	72.1	1 600	2 100	**32930JR**		2DC	36.1	162	163	200	194	202	7	8	2	2	0.33	1.83	1.01	3.96
	225	48	48	36	3	2.5	492	668	79.6	1 500	2 000	**32030JR**		4EC	48.8	164	164	213	200	216	8	12	2.5	2	0.46	1.31	0.72	6.41
	225	59	59	46	3	2.5	575	869	101	1 500	2 000	**33030JR**		2EE	47.8	164	164	213	200	217	8	13	2.5	2	0.36	1.65	0.90	8.09
	270	49	45	38	4	3	604	664	80.9	1 300	1 800	**30230JR**		4GB	52.4	168	175	256	234	255	9	11	3	2.5	0.44	1.38	0.76	11.6
	270	77	73	60	4	3	881	1 070	122	1 300	1 800	**32230JR**		4GD	65.2	168	170	256	226	254	8	17	3	2.5	0.44	1.38	0.76	18.2
	320	72	65	46	5	4	768	750	85.7	970	1 400	**30330D**		—	96.0	172	183	302	270	301	9	26	4	3	0.81	0.74	0.41	23.9
	320	72	65	55	5	4	1 050	1 080	129	1 200	1 500	**30330JR**		2GB	60.8	172	193	302	272	292	12	17	4	3	0.35	1.74	0.96	26.6
	320	82	75	50	5	4	952	989	110	980	1 400	**31330JR**		7GB	100.1	172	179	302	272	301	9	27	4	3	0.83	0.73	0.40	28.0
	320	114	108	90	5	4	1 550	1 790	195	1 200	1 600	**32330R**		—	78.4	172	187	302	263	298	10	17	4	3	0.35	1.74	0.96	42.0
160	220	32	30	23	3	3	282	379	50.2	1 500	2 000	**T4DB160**		4DB	44.7	174	172	206	204	213	5	9	2.5	2.5	0.49	1.23	0.68	3.23
	220	38	38	30	2.5	2	368	568	75.2	1 500	2 000	**32932JR**		2DC	38.4	172	173	210	204	212	7	8	2	2	0.35	1.73	0.95	4.19
	240	51	51	38	3	2.5	553	758	90.3	1 400	1 900	**32032JR**		4EC	52.1	174	175	228	213	231	8	13	2.5	2	0.46	1.31	0.72	7.75
	290	52	48	40	4	3	679	750	89.3	1 200	1 600	**30232JR**		4GB	56.3	178	189	276	252	269	8	12	3	2.5	0.44	1.38	0.76	14.1
	290	84	80	67	4	3	994	1 210	137	1 200	1 700	**32232JR**		4GD	70.3	178	182	276	242	274	10	17	3	2.5	0.44	1.38	0.76	23.2
	340	75	68	48	5	4	926	933	104	900	1 300	**30332D**		—	101.8	182	195	322	290	320	9	27	4	3	0.81	0.74	0.41	29.1
	340	75	68	58	5	4	1 170	1 220	142	1 100	1 400	**30332JR**		2GB	63.3	182	205	322	289	310	12	17	4	3	0.35	1.74	0.96	31.8
	340	121	114	95	5	4	1 530	1 720	187	1 100	1 400	**32332**		—	83.0	182	200	322	277	316	10	18	4	3	0.35	1.73	0.95	47.9
170	230	38	38	30	2.5	2	370	606	78.8	1 400	1 900	**32934JR**		3DC	42.0	182	183	220	213	222	7	8	2	2	0.38	1.57	0.86	4.49

metric series

d (170) ~ 200 mm

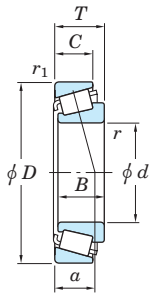


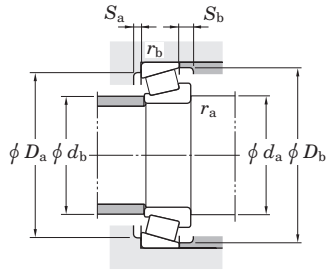


Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)									Constant	Axial load factors		(Refer.) Mass (kg)
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_a min.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	
170	260	57	57	43	3	2.5	661	905	105	1 300	1 700	**32034JR**		4EC	55.8	184	187	248	230	249	10	14	2.5	2	0.44	1.35	0.74	10.5
	310	57	52	43	5	4	776	867	103	1 100	1 500	**30234JR**		4GB	61.2	192	202	292	269	288	8	14	4	3	0.44	1.38	0.76	17.8
	310	91	86	71	5	4	1 120	1 380	152	1 100	1 500	**32234JR**		4GD	76.2	192	195	292	259	294	10	20	4	3	0.44	1.38	0.76	28.9
	360	80	72	50	5	4	953	1 040	115	830	1 200	**30334D**		—	108.3	192	211	342	310	333	9	30	4	3	0.81	0.74	0.41	34.3
	360	80	72	62	5	4	1 300	1 370	155	1 000	1 300	**30334JR**		2GB	67.9	192	218	342	306	329	13	18	4	3	0.35	1.74	0.96	37.5
	360	127	120	100	5	4	1 640	1 830	193	1 000	1 300	**32334**		—	86.1	192	200	342	295	337	14	26	4	3	0.35	1.73	0.95	56.9
180	250	45	45	34	2.5	2	447	735	93.4	1 300	1 700	**32936JR**		4DC	53.5	192	193	240	225	241	8	11	2	2	0.48	1.25	0.69	6.64
	280	64	64	48	3	2.5	810	1 100	127	1 200	1 600	**32036JR**		3FD	59.5	194	199	268	247	268	10	16	2.5	2	0.42	1.42	0.78	14.1
	320	57	52	43	5	4	771	870	102	1 100	1 400	**30236JR**		4GB	63.6	202	211	302	278	297	9	14	4	3	0.45	1.33	0.73	18.3
	320	91	86	71	5	4	1 200	1 520	164	1 100	1 500	**32236JR**		4GD	77.8	202	204	302	267	303	10	20	4	3	0.45	1.33	0.73	29.9
	380	83	75	52	5	4	1 040	1 150	125	780	1 100	**30336D**		—	112.8	202	225	362	330	351	10	31	4	3	0.81	0.74	0.41	40.1
	380	83	75	64	5	4	1 130	1 110	126	940	1 300	**30336**		—	71.0	202	227	362	318	346	13	19	4	3	0.35	1.73	0.95	39.7
	380	134	126	106	5	4	1 760	1 980	206	960	1 300	**32336**		—	91.8	202	215	362	310	355	14	27	4	3	0.35	1.73	0.95	67.0
190	260	45	45	34	2.5	2	459	789	88.6	1 200	1 600	**32938JR**		4DC	55.0	202	204	250	235	252	8	11	2	2	0.48	1.26	0.69	6.89
	290	64	64	48	3	2.5	823	1 170	131	1 100	1 500	**32038JR**		4FD	62.9	204	209	278	257	279	10	16	2.5	2	0.44	1.36	0.75	14.7
	340	60	55	46	5	4	912	1 030	118	1 000	1 300	**30238JR**		4GB	66.4	212	225	322	298	318	12	13	4	3	0.44	1.38	0.76	21.9
	340	97	92	75	5	4	1 370	1 740	187	1 000	1 300	**32238JR**		4GD	81.9	212	216	322	286	323	12	22	4	3	0.44	1.38	0.76	36.6
	400	86	78	52	6	5	1 190	1 210	131	740	1 000	**30338D**		—	119.2	218	232	378	350	372	11	34	5	4	0.81	0.74	0.41	44.8
	400	86	78	65	6	5	1 260	1 250	139	880	1 200	**30338**		—	73.2	218	241	378	342	370	10	20	5	4	0.35	1.73	0.95	46.2
	400	140	132	109	6	5	1 940	2 190	224	890	1 200	**32338**		—	96.5	218	225	378	330	375	14	30	5	4	0.35	1.73	0.95	76.6
200	280	51	51	39	3	2.5	608	958	109	1 100	1 500	**32940JR**		3EC	53.6	214	216	268	257	271	9	12	2.5	2	0.39	1.52	0.84	9.44
	310	70	70	53	3	2.5	949	1 340	146	1 100	1 400	**32040JR**		4FD	66.9	214	221	298	273	297	11	17	2.5	2	0.43	1.39	0.77	19.1
	360	64	58	48	5	4	991	1 120	126	940	1 200	**30240JR**		4GB	70.3	222	238	342	315	336	12	15	4	3	0.44	1.38	0.76	26.4
	360	104	98	82	5	4	1 550	1 880	200	960	1 300	**32240JR**		3GD	84.6	222	225	342	302	340	11	22	4	3	0.41	1.48	0.81	44.2
	420	89	80	56	6	5	1 130	1 230	132	690	970	**30340D**		—	122.6	228	248	398	365	385	11	33	5	4	0.81	0.74	0.41	50.6
	420	89	80	67	6	5	1 400	1 450	159	820	1 100	**30340**		—	79.8	228	255	398	354	385	11	21	5	4	0.35	1.73	0.95	53.5
	420	146	138	115	6	5	2 240	2 580	260	830	1 100	**32340**		—	102.9	228	240	398	345	395	16	30	5	4	0.35	1.73	0.95	91.0

metric series

d 220 ~ 360 mm

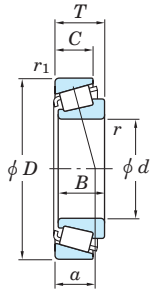


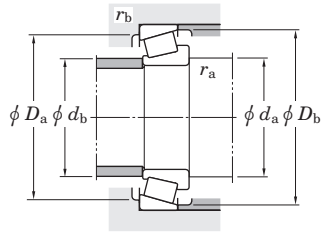


Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Dimension series to ISO355 (Refer.)	Load center (mm)	Mounting dimensions (mm)									Constant	Axial load factors		(Refer.) Mass (kg)
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.				a	d_a min.	d_b max.	D_a max.	D_b min.	S_a min.	S_b min.	r_a max.	r_b max.	e	Y_1	Y_0	
220	300	51	51	39	3	2.5	621	1 010	112	1 000	1 400	**32944JR**		3EC	58.6	234	234	288	275	290	9	12	2.5	2	0.43	1.41	0.78	10.1
	340	76	76	57	4	3	1 120	1 620	175	940	1 300	**32044JR**		4FD	72.8	238	243	326	300	326	12	19	3	2.5	0.43	1.39	0.77	25.2
	400	72	65	54	5	4	1 260	1 440	160	830	1 100	**30244JR**		—	76.5	242	263	382	344	371	14	17	4	3	0.44	1.43	0.79	35.9
	400	114	108	90	5	4	1 500	1 930	198	830	1 100	**32244**		—	95.9	242	260	382	333	377	16	14	4	3	0.43	1.39	0.77	56.8
	460	97	88	73	6	5	1 570	1 680	181	730	980	**30344**		—	84.6	248	282	438	386	420	12	23	5	4	0.35	1.73	0.95	69.0
240	320	51	51	39	3	2.5	645	1 090	119	940	1 300	**32948JR**		4EC	64.5	254	254	308	294	311	9	12	2.5	2	0.46	1.31	0.72	10.9
	360	76	76	57	4	3	1 160	1 720	180	870	1 200	**32048JR**		4FD	78.5	258	261	346	318	346	12	19	3	2.5	0.46	1.31	0.72	26.8
	440	79	72	60	5	4	1 540	1 790	191	730	980	**30248R**		—	82.7	262	287	422	377	409	14	18	4	3	0.42	1.43	0.79	49.5
	440	127	120	100	5	4	1 920	2 480	245	740	980	**32248**		—	106.1	262	282	422	365	415	16	14	4	3	0.43	1.39	0.77	76.4
260	360	63.5	63.5	48	3	2.5	926	1 550	163	830	1 100	**32952JR**		3EC	69.6	274	279	348	328	347	11	15.5	2.5	2	0.41	1.48	0.81	18.9
	400	87	87	65	5	4	1 470	2 170	221	770	1 000	**32052JR**		4FC	85.0	282	287	382	352	383	14	22	4	3	0.43	1.38	0.76	39.5
	480	89	80	67	6	5	1 510	1 860	190	650	870	**30252**		—	93.6	288	310	458	415	450	14	21	5	4	0.42	1.44	0.79	64.9
	480	137	130	106	6	5	2 200	2 870	276	660	880	**32252**		—	115.2	288	300	458	400	455	16	30	5	4	0.43	1.39	0.77	102
280	380	63.5	63.5	48	3	2.5	949	1 630	168	770	1 000	**32956JR**		4EC	75.1	294	298	368	347	368	11	15.5	2.5	2	0.43	1.39	0.76	20.1
	420	87	87	65	5	4	1 510	2 280	230	720	960	**32056JR**		4FC	91.1	302	305	402	370	402	14	22	4	3	0.46	1.31	0.72	41.7
	500	89	80	67	6	5	1 580	1 920	196	610	810	**30256**		—	96.2	308	325	478	440	475	14	21	5	4	0.42	1.44	0.79	67.6
	500	137	130	106	6	5	2 340	3 150	297	610	810	**32256**		—	117.2	308	325	478	420	474	16	30	5	4	0.43	1.39	0.77	108
300	420	76	76	57	4	3	1 320	2 210	223	680	910	**32960JR**		3FD	79.9	318	324	406	383	405	12	19	3	2.5	0.39	1.52	0.84	32.4
	460	100	100	74	5	4	1 800	2 660	263	640	850	**32060JR**		4GD	97.9	322	329	442	404	439	15	26	4	3	0.43	1.38	0.76	57.5
	540	96	85	71	6	5	1 890	2 360	240	550	730	**30260**		—	103.9	328	350	518	475	505	14	24	5	4	0.42	1.44	0.79	84.7
320	440	76	76	57	4	3	1 330	2 270	226	640	850	**32964JR**		3FD	85.0	338	342	426	401	426	12	19	3	2.5	0.42	1.44	0.79	34.0
	480	100	100	74	5	4	1 900	2 810	273	600	800	**32064JR**		4GD	103.0	342	344	462	418	461	16	26	4	3	0.46	1.31	0.72	58.7
	580	104	92	75	6	5	2 190	2 770	273	490	660	**30264**		—	111.9	348	370	558	505	540	14	28	5	4	0.42	1.44	0.79	108
340	460	76	76	57	4	3	1 340	2 340	229	590	790	**32968JR**		4FD	90.5	358	361	446	420	446	12	19	3	2.5	0.44	1.37	0.75	35.6
360	480	76	76	57	4	3	1 350	2 400	231	560	740	**32972JR**		4FD	96.2	378	379	466	438	466	12	19	3	2.5	0.46	1.31	0.72	37.1

inch series

d 9.525 ~ (22.225) mm



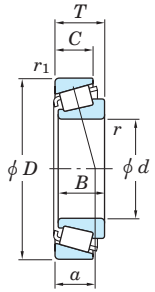


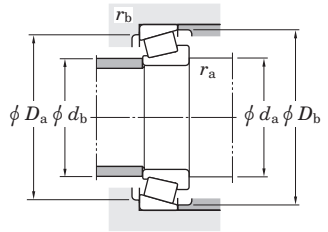
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
9.525	31.991	10.008	10.785	7.938	1.2	1.2	13.4	9.30	1.25	14 000	19 000	**A2037**	**A2126**	7.1	15.0	13.5	26.0	29.0	1.2	1.2	0.40	1.48	0.82	0.029	0.017
11.986	31.991	10.008	10.785	7.938	0.8	1.2	13.4	9.30	1.25	14 000	19 000	**A2047**	**A2126**	7.1	16.5	15.5	26.0	29.0	0.8	1.2	0.40	1.48	0.82	0.023	0.017
12.700	34.988	10.998	10.988	8.730	1.2	1.2	15.7	11.9	1.55	12 000	17 000	**A4050**	**A4138**	8.3	18.5	17.0	29.0	32.0	1.2	1.2	0.45	1.33	0.73	0.033	0.022
14.989	34.988	10.998	10.988	8.730	0.8	1.2	15.7	11.9	1.55	12 000	17 000	**A4059**	**A4138**	8.3	19.5	19.0	29.0	32.0	0.8	1.2	0.45	1.33	0.73	0.029	0.022
15.875	34.988	10.998	10.998	8.712	1.2	1.2	18.1	14.3	1.90	12 000	16 000	**L21549**	**L21511**	7.6	21.5	19.5	29.0	32.5	1.2	1.2	0.32	1.88	1.04	0.031	0.018
	41.275	14.288	14.681	11.112	1.2	2.0	27.3	20.5	2.85	11 000	14 000	**03062**	**03162**	9.3	21.5	20.0	34.0	37.5	1.2	2.0	0.31	1.93	1.06	0.060	0.035
	42.862	16.670	16.670	13.495	1.6	1.6	38.2	29.5	4.15	10 000	14 000	**17580R**	**17520**	10.9	23.0	21.0	36.5	39.0	1.6	1.6	0.33	1.81	1.00	0.078	0.048
	49.225	19.845	21.539	14.288	0.8	1.2	47.2	37.7	5.40	8 900	12 000	**09062**	**09195**	10.6	22.0	21.5	42.0	44.5	0.8	1.2	0.27	2.26	1.24	0.139	0.065
	53.975	22.225	21.839	15.875	0.8	2.4	52.6	41.2	5.65	8 400	11 000	**21063**	**21212**	16.6	29.0	26.5	43.0	50.0	0.8	2.4	0.59	1.02	0.56	0.163	0.097
16.000	47.000	21.000	21.000	16.000	1.0	2.0	45.4	37.7	5.05	9 800	13 000	**HM81649**	**HM81610**	15.0	27.5	23.0	37.5	43.0	1.0	2.0	0.55	1.10	0.60	0.111	0.080
17.462	39.878	13.843	14.605	10.668	1.2	1.2	31.8	26.0	3.60	11 000	14 000	**LM11749R**	**LM11710**	8.6	23.0	21.5	34.0	37.0	1.2	1.2	0.29	2.10	1.15	0.058	0.028
19.050	45.237	15.494	16.637	12.065	1.2	1.2	36.8	30.1	4.25	9 400	13 000	**LM11949**	**LM11910**	10.0	25.0	23.5	39.5	41.5	1.2	1.2	0.30	2.00	1.10	0.081	0.044
	49.225	19.845	21.539	14.288	1.2	1.2	47.2	37.7	5.40	8 900	12 000	**09078**	**09195**	10.6	25.5	24.0	42.0	44.5	1.2	1.2	0.27	2.26	1.24	0.124	0.065
	49.225	21.209	19.050	17.462	1.2	1.6	47.2	37.7	5.40	8 900	12 000	**09067**	**09196**	13.8	25.5	24.0	41.5	44.5	1.2	1.6	0.27	2.26	1.24	0.114	0.084
20.000	50.005	13.495	14.260	9.525	1.6	1.0	33.3	28.8	4.05	7 900	11 000	**07079**	**07196**	10.8	27.5	26.0	44.5	47.0	1.6	1.0	0.40	1.49	0.82	0.104	0.034
20.638	49.225	19.845	19.845	15.875	1.6	1.6	45.5	37.7	5.35	8 600	12 000	**12580**	**12520**	12.7	28.5	26.0	42.5	45.5	1.6	1.6	0.32	1.86	1.02	0.116	0.067
21.430	50.005	17.526	18.288	13.970	1.2	1.2	48.8	40.7	5.80	8 500	11 000	**M12649**	**M12610**	11.1	27.5	25.5	44.0	46.0	1.2	1.2	0.28	2.16	1.19	0.119	0.058
21.987	45.974	15.494	16.637	12.065	1.2	1.2	37.5	34.6	4.85	8 900	12 000	**LM12749**	**LM12711**	10.0	27.5	26.0	40.0	42.5	1.2	1.2	0.31	1.96	1.08	0.078	0.043
22.225	50.005	17.526	18.288	13.970	1.2	1.2	48.8	40.7	5.80	8 500	11 000	**M12648**	**M12610**	11.1	28.5	26.5	44.0	46.0	1.2	1.2	0.28	2.16	1.19	0.115	0.058
	52.388	19.368	20.168	14.288	1.6	1.6	45.9	37.9	5.45	8 000	11 000	**1380**	**1328**	11.6	29.5	29.5	45.0	48.5	1.6	1.6	0.29	2.05	1.13	0.132	0.066
	53.975	19.368	20.168	14.288	1.6	1.6	45.9	37.9	5.45	8 000	11 000	**1380**	**1329**	11.6	29.5	29.5	46.0	49.0	1.6	1.6	0.29	2.05	1.13	0.137	0.082

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (22.225) ~ (26.988) mm



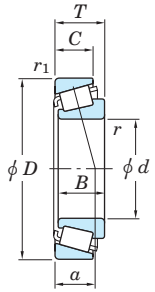


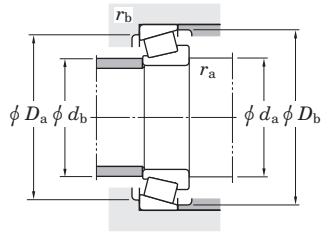
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
22.225	56.896	19.368	19.837	15.875	1.2	1.2	50.0	43.1	6.20	7 600	10 000	**1755**	**1729**	12.5	29.0	27.5	49.0	51.0	1.2	1.2	0.31	1.95	1.07	0.150	0.100
	57.150	22.225	22.225	17.462	0.8	1.6	65.8	55.7	8.05	7 600	10 000	**1280**	**1220**	15.3	29.5	29.0	49.0	52.0	0.8	1.6	0.35	1.73	0.95	0.189	0.105
	66.421	23.812	25.433	19.050	1.6	1.2	83.8	75.2	11.2	6 500	8 700	**2684**	**2631**	13.9	31.5	29.0	58.0	60.0	1.6	1.2	0.25	2.36	1.30	0.295	0.163
22.606	47.000	15.500	15.500	12.000	1.6	1.0	35.0	32.8	4.45	8 700	12 000	**LM72849**	**LM72810**	12.3	30.0	28.0	40.5	44.0	1.6	1.0	0.47	1.27	0.70	0.076	0.047
23.812	50.292	14.224	14.732	10.668	1.6	1.2	39.1	37.0	5.15	7 800	10 000	**L44640R**	**L44610**	10.8	30.5	28.5	44.5	47.0	1.6	1.2	0.37	1.60	0.88	0.099	0.034
	56.896	19.368	19.837	15.875	0.8	1.2	50.0	43.1	6.20	7 600	10 000	**1779**	**1729**	12.5	29.5	28.5	49.0	51.0	0.8	1.2	0.31	1.95	1.07	0.141	0.100
24.981	50.005	13.495	14.260	9.525	1.6	1.0	33.3	28.8	4.05	7 900	11 000	**07098**	**07196**	10.8	31.0	29.0	44.5	47.0	1.6	1.0	0.40	1.49	0.82	0.084	0.034
	62.000	16.002	16.566	14.288	1.6	1.6	47.4	40.6	5.80	6 700	8 900	**17098**	**17244**	12.7	33.0	30.5	54.0	57.0	1.6	1.6	0.38	1.57	0.86	0.162	0.090
25.000	50.005	13.495	14.260	9.525	1.6	1.0	33.3	28.8	4.05	7 900	11 000	**07097**	**07196**	10.8	31.0	29.0	44.5	47.0	1.6	1.0	0.40	1.49	0.82	0.085	0.035
25.400	50.005	13.495	14.260	9.525	1.0	1.0	33.3	28.8	4.05	7 900	11 000	**07100**	**07196**	10.8	30.5	29.5	44.5	47.0	1.0	1.0	0.40	1.49	0.82	0.084	0.035
	50.005	13.495	14.260	9.525	1.6	1.0	33.3	28.8	4.05	7 900	11 000	**07100S**	**07196**	10.8	31.5	29.5	44.5	47.0	1.6	1.0	0.40	1.49	0.82	0.082	0.035
	50.292	14.224	14.732	10.668	1.2	1.2	39.1	37.0	5.15	7 800	10 000	**L44643R**	**L44610**	10.8	31.5	29.5	44.5	47.0	1.2	1.2	0.37	1.60	0.88	0.092	0.039
	51.994	15.011	14.260	12.700	1.0	1.2	33.3	28.8	4.05	7 900	11 000	**07100**	**07204**	12.3	30.5	29.5	45.0	48.0	1.0	1.2	0.40	1.49	0.82	0.075	0.065
	58.738	19.050	19.355	15.080	1.2	1.2	60.8	57.1	8.25	7 000	9 300	**1986R**	**1932**	13.1	32.5	30.5	52.0	54.0	1.2	1.2	0.33	1.82	1.00	0.179	0.088
	59.530	23.368	23.114	18.288	0.8	1.6	63.0	57.1	7.95	7 200	9 600	**M84249**	**M84210**	18.2	36.0	32.5	49.5	56.0	0.8	1.6	0.55	1.10	0.60	0.194	0.128
	61.912	19.050	20.638	14.288	0.8	2.0	55.7	50.7	7.30	6 400	8 600	**15101**	**15243**	13.2	32.5	31.5	55.0	58.0	0.8	2.0	0.35	1.71	0.94	0.215	0.080
	62.000	19.050	20.638	14.288	3.6	1.2	55.7	50.7	7.30	6 400	8 600	**15100**	**15245**	13.2	38.0	31.5	55.0	58.0	3.6	1.2	0.35	1.71	0.94	0.215	0.081
	63.500	19.050	20.638	14.288	0.8	1.2	55.7	50.7	7.30	6 400	8 600	**15101**	**15250R**	13.2	32.5	31.5	55.0	59.0	0.8	1.2	0.35	1.71	0.94	0.215	0.097
	64.292	21.432	21.432	16.670	1.6	1.6	69.1	70.7	9.90	6 400	8 500	**M86643R**	**M86610**	18.0	38.0	36.5	54.0	61.0	1.6	1.6	0.55	1.10	0.60	0.248	0.127
	66.421	23.812	25.433	19.050	1.2	1.2	83.8	75.2	11.2	6 500	8 700	**2687**	**2631**	13.9	33.5	31.5	58.0	60.0	1.2	1.2	0.25	2.36	1.30	0.272	0.163
	68.262	22.225	22.225	17.462	0.8	1.6	63.7	61.1	8.80	6 000	8 000	**02473**	**02420**	17.1	34.5	33.5	59.0	63.0	0.8	1.6	0.42	1.44	0.79	0.275	0.150
	72.233	25.400	25.400	19.842	0.8	2.4	83.8	87.4	12.4	5 700	7 600	**HM88630**	**HM88610**	20.7	39.5	39.5	60.0	69.0	0.8	2.4	0.55	1.10	0.60	0.391	0.185
26.162	66.421	23.812	25.433	19.050	1.6	1.2	83.8	75.2	11.2	6 500	8 700	**2682**	**2631**	13.9	34.5	32.0	58.0	60.0	1.6	1.2	0.25	2.36	1.30	0.268	0.163
26.988	50.292	14.224	14.732	10.668	3.6	1.2	39.1	37.0	5.15	7 800	10 000	**L44649R**	**L44610**	10.8	37.5	31.0	44.5	47.0	3.6	1.2	0.37	1.60	0.88	0.083	0.039

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (26.988) ~ (30.162) mm



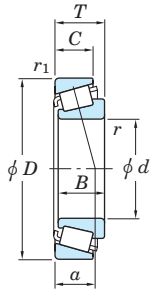


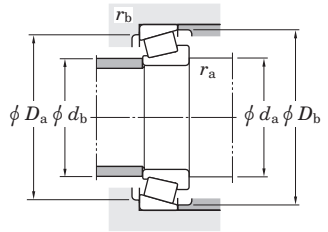
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
26.988	60.325	19.842	17.462	15.875	3.6	1.6	47.2	42.7	6.10	7 000	9 400	**15580**	**15523**	15.1	38.5	32.0	51.0	54.0	3.6	1.6	0.35	1.73	0.95	0.140	0.122
	62.000	19.050	20.638	14.288	0.8	1.2	55.7	50.7	7.30	6 400	8 600	**15106**	**15245**	13.2	33.5	33.0	55.0	58.0	0.8	1.2	0.35	1.71	0.94	0.206	0.081
	66.421	23.812	25.433	19.050	1.6	1.2	83.8	75.2	11.2	6 500	8 700	**2688**	**2631**	13.9	35.0	33.0	58.0	60.0	1.6	1.2	0.25	2.36	1.30	0.262	0.163
28.575	57.150	17.462	17.462	13.495	3.6	1.6	47.2	42.7	6.10	7 000	9 400	**15590**	**15520**	12.7	39.0	33.5	51.0	53.0	3.6	1.6	0.35	1.73	0.95	0.131	0.069
	57.150	19.845	19.355	15.875	3.6	1.6	60.8	57.1	8.25	7 000	9 300	**1988R**	**1922**	13.9	39.5	33.5	51.0	53.5	3.6	1.6	0.33	1.82	1.00	0.151	0.076
	62.000	19.050	20.638	14.288	3.6	1.2	55.7	50.7	7.30	6 400	8 600	**15112**	**15245**	13.2	40.0	34.0	55.0	58.0	3.6	1.2	0.35	1.71	0.94	0.193	0.081
	62.000	19.050	20.638	14.288	0.8	1.2	55.7	50.7	7.30	6 400	8 600	**15113**	**15245**	13.2	34.5	34.0	55.0	58.0	0.8	1.2	0.35	1.71	0.94	0.195	0.081
	64.292	21.432	21.432	16.670	1.6	1.6	69.1	70.7	9.90	6 400	8 500	**M86647R**	**M86610**	18.0	40.0	38.0	54.0	61.0	1.6	1.6	0.55	1.10	0.60	0.225	0.127
	66.421	23.812	25.433	19.050	1.2	1.2	83.8	75.2	11.2	6 500	8 700	**2689**	**2631**	13.9	36.0	34.0	58.0	60.0	1.2	1.2	0.25	2.36	1.30	0.249	0.165
	68.262	22.225	22.225	17.462	0.8	1.6	63.7	61.1	8.80	6 000	8 000	**02474**	**02420**	17.1	36.5	36.0	59.0	63.0	0.8	1.6	0.42	1.44	0.79	0.252	0.150
	72.000	19.000	18.923	15.875	1.6	1.6	59.4	49.6	7.25	5 900	7 800	**26112**	**26283**	15.3	37.0	35.0	62.0	65.0	1.6	1.6	0.36	1.67	0.92	0.217	0.163
	72.626	24.608	24.257	17.462	4.8	1.6	77.3	60.5	8.75	6 100	8 100	**41125**	**41286**	20.7	48.0	36.5	61.0	68.0	4.8	1.6	0.60	1.00	0.55	0.292	0.177
	72.626	24.608	24.257	17.462	1.6	1.6	77.3	60.5	8.75	6 100	8 100	**41126**	**41286**	20.7	41.5	36.5	61.0	68.0	1.6	1.6	0.60	1.00	0.55	0.295	0.177
	72.626	30.162	29.997	23.812	3.6	3.2	98.6	89.3	13.3	5 800	7 700	**3192**	**3120**	20.3	42.5	37.0	61.0	67.0	3.6	3.2	0.33	1.80	0.99	0.401	0.222
	72.626	30.162	29.997	23.812	1.2	3.2	98.6	89.3	13.3	5 800	7 700	**3198**	**3120**	20.3	39.0	37.0	61.0	67.0	1.2	3.2	0.33	1.80	0.99	0.410	0.222
	73.025	22.225	22.225	17.462	0.8	3.2	68.8	65.7	9.55	5 500	7 400	**02872**	**02820**	18.4	37.5	37.0	62.0	68.0	0.8	3.2	0.45	1.32	0.73	0.319	0.158
29.000	50.292	14.224	14.732	10.668	3.6	1.2	36.3	37.2	5.15	7 600	10 000	**L45449**	**L45410**	10.9	39.5	33.0	44.5	48.0	3.6	1.2	0.37	1.62	0.89	0.079	0.036
29.367	66.421	23.812	25.433	19.050	3.6	1.2	83.8	75.2	11.2	6 500	8 700	**2690**	**2631**	13.9	41.0	35.0	58.0	60.0	3.6	1.2	0.25	2.36	1.30	0.242	0.165
29.987	62.000	16.002	16.566	14.288	1.6	1.6	47.4	40.6	5.80	6 700	8 900	**17118**	**17244**	12.7	37.0	34.5	54.0	57.0	1.6	1.6	0.38	1.57	0.86	0.135	0.090
	62.000	19.050	20.638	14.288	1.2	1.2	55.7	50.7	7.30	6 400	8 600	**15117**	**15245**	13.2	36.5	35.0	55.0	58.0	1.2	1.2	0.35	1.71	0.94	0.184	0.081
30.000	69.012	19.845	19.583	15.875	3.6	1.2	57.7	55.0	7.95	5 900	7 800	**14117A**	**14276**	15.5	42.5	39.5	60.0	63.0	3.6	1.2	0.38	1.57	0.86	0.225	0.135
30.112	62.000	19.050	20.638	14.288	0.8	1.2	55.7	50.7	7.30	6 400	8 600	**15116**	**15245**	13.2	36.0	35.5	55.0	58.0	0.8	1.2	0.35	1.71	0.94	0.184	0.081
30.162	62.000	16.002	16.566	14.288	1.6	1.6	47.4	40.6	5.80	6 700	8 900	**17119**	**17244**	12.7	37.0	34.5	54.0	57.0	1.6	1.6	0.38	1.57	0.86	0.139	0.091
	64.292	21.432	21.432	16.670	1.6	1.6	69.1	70.7	9.90	6 400	8 500	**M86649R**	**M86610**	18.0	41.0	38.0	54.0	61.0	1.6	1.6	0.55	1.10	0.60	0.213	0.127

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (30.162) ~ (34.925) mm





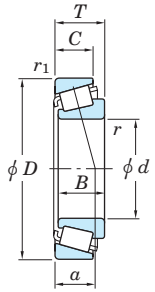
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r 1) min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
30.162	68.262	22.225	22.225	17.462	2.4	1.6	70.2	71.1	10.0	6 000	7 900	**M88043**	**M88010**	19.2	43.5	39.5	58.0	65.0	2.4	1.6	0.55	1.10	0.60	0.258	0.144
30.213	62.000	19.050	20.638	14.288	3.6	1.2	55.7	50.7	7.30	6 400	8 600	**15118**	**15245**	13.2	41.5	35.5	55.0	58.0	3.6	1.2	0.35	1.71	0.94	0.181	0.081
	62.000	19.050	20.638	14.288	1.6	1.2	55.7	50.7	7.30	6 400	8 600	**15119**	**15245**	13.2	37.5	35.5	55.0	58.0	1.6	1.2	0.35	1.71	0.94	0.183	0.081
	62.000	19.050	20.638	14.288	0.8	1.2	55.7	50.7	7.30	6 400	8 600	**15120**	**15245**	13.2	36.0	35.5	55.0	58.0	0.8	1.2	0.35	1.71	0.94	0.183	0.081
30.226	69.012	19.845	19.583	15.875	0.8	3.2	57.7	55.0	7.95	5 900	7 800	**14116**	**14274**	15.5	37.0	36.5	59.0	63.0	0.8	3.2	0.38	1.57	0.86	0.226	0.131
31.750	58.738	14.684	15.080	10.716	1.0	1.0	37.0	33.3	4.60	6 600	8 900	**08125**	**08231**	13.5	37.5	36.0	52.0	55.0	1.0	1.0	0.48	1.26	0.69	0.109	0.056
	59.131	15.875	16.764	11.811	SP	1.2	44.8	43.1	6.05	6 600	8 800	**LM67048**	**LM67010**	13.0	42.5	36.0	52.0	56.0	3.5	1.2	0.41	1.46	0.80	0.120	0.062
	62.000	18.161	19.050	14.288	SP	1.2	55.7	50.7	7.30	6 400	8 600	**15123**	**15245**	13.2	42.5	36.5	55.0	58.0	3.5	1.2	0.35	1.71	0.94	0.157	0.081
	62.000	19.050	20.638	14.288	3.6	1.2	55.7	50.7	7.30	6 400	8 600	**15125**	**15245**	13.2	42.5	36.5	55.0	58.0	3.6	1.2	0.35	1.71	0.94	0.169	0.081
	62.000	19.050	20.638	14.288	0.8	1.2	55.7	50.7	7.30	6 400	8 600	**15126**	**15245**	13.2	37.0	36.5	55.0	58.0	0.8	1.2	0.35	1.71	0.94	0.171	0.081
	66.421	25.400	25.357	20.638	0.8	3.2	89.2	85.1	12.7	6 000	8 000	**2580**	**2520**	16.0	38.5	37.5	57.0	62.5	0.8	3.2	0.27	2.19	1.21	0.281	0.123
	68.262	22.225	22.225	17.462	3.6	1.6	63.7	61.1	8.80	6 000	8 000	**02475**	**02420**	17.1	44.5	38.5	59.0	63.0	3.6	1.6	0.42	1.44	0.79	0.224	0.150
	68.262	22.225	22.225	17.462	0.8	1.6	63.7	61.1	8.80	6 000	8 000	**02476**	**02420**	17.1	39.0	38.5	59.0	63.0	0.8	1.6	0.42	1.44	0.79	0.226	0.150
	68.262	22.225	22.225	17.462	1.6	1.6	70.2	71.1	10.0	6 000	7 900	**M88046**	**M88010**	19.2	43.0	40.5	58.0	65.0	1.6	1.6	0.55	1.10	0.60	0.245	0.144
	73.025	22.225	22.225	17.462	3.6	3.2	68.8	65.7	9.55	5 600	7 400	**02875**	**02820**	17.1	45.5	39.5	62.0	68.0	3.6	3.2	0.45	1.32	0.73	0.293	0.158
	73.025	22.225	22.225	17.462	0.8	3.2	68.8	65.7	9.55	5 500	7 400	**02876**	**02820**	17.1	40.0	39.5	62.0	68.0	0.8	3.2	0.45	1.32	0.73	0.293	0.158
	73.025	29.370	27.783	23.020	1.2	3.2	93.0	101	14.2	5 600	7 500	**HM88542**	**HM88510**	23.4	45.5	42.5	59.0	70.0	1.2	3.2	0.55	1.10	0.60	0.377	0.238
	73.812	29.370	27.783	23.020	1.2	3.2	93.0	101	14.2	5 600	7 500	**HM88542**	**HM88512**	23.4	45.5	42.5	59.0	70.0	1.2	3.2	0.55	1.10	0.60	0.377	0.254
33.338	68.262	22.225	22.225	17.462	0.8	1.6	70.2	71.1	10.0	6 000	7 900	**M88048**	**M88010**	19.2	42.5	41.0	58.0	65.0	0.8	1.6	0.55	1.10	0.60	0.231	0.144
	72.000	19.000	18.923	15.875	3.6	1.6	69.8	60.0	8.85	5 900	7 800	**26131**	**26283**	14.3	44.5	38.5	62.0	65.0	3.6	1.6	0.36	1.67	0.92	0.200	0.163
	73.025	29.370	27.783	23.020	0.8	3.2	93.0	101	14.2	5 600	7 500	**HM88547**	**HM88510**	23.4	45.5	42.6	59.0	70.0	0.8	3.2	0.55	1.10	0.60	0.360	0.238
	76.200	29.370	28.575	23.020	0.8	3.2	99.5	107	15.2	5 400	7 200	**HM89443**	**HM89410**	23.9	46.5	44.6	62.0	73.0	0.8	3.2	0.55	1.10	0.60	0.415	0.254
34.925	65.088	18.034	18.288	13.970	SP	1.2	60.0	58.5	8.40	6 000	8 000	**LM48548**	**LM48510**	14.3	46.0	40.0	58.0	61.0	3.5	1.2	0.38	1.59	0.88	0.164	0.086
	69.012	26.982	26.721	15.875	0.8	1.2	57.7	55.0	7.95	5 900	7 800	**14136A**	**14276**	22.6	40.0	38.0	60.0	63.0	0.8	1.2	0.38	1.57	0.86	0.254	0.133
	72.233	25.400	25.400	19.842	2.4	2.4	83.8	87.4	12.4	5 700	7 600	**HM88649**	**HM88610**	20.7	48.5	42.5	60.0	69.0	2.4	2.4	0.55	1.10	0.60	0.301	0.185

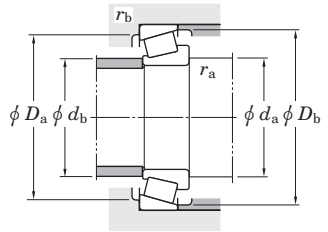
[Note] 1) SP indicates the specially chamfered from.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (34.925) ~ (38.100) mm





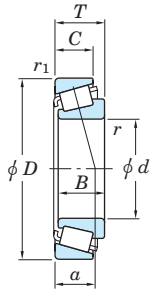
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [2]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r [1] min.	r_1 [1] min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
34.925	72.238	20.638	20.638	15.875	3.6	1.2	62.3	61.3	8.90	5 600	7 400	**16137**	**16284**	16.6	46.5	40.5	63.0	67.0	3.6	1.2	0.40	1.49	0.82	0.236	0.144
	73.025	22.225	22.225	17.462	3.6	3.2	68.8	65.7	9.55	5 500	7 400	**02877**	**02820**	18.4	48.5	42.0	62.0	68.0	3.6	3.2	0.45	1.32	0.73	0.262	0.158
	73.025	22.225	22.225	17.462	0.8	3.2	68.8	65.7	9.55	5 500	7 400	**02878**	**02820**	18.4	42.5	42.0	62.0	68.0	0.8	3.2	0.45	1.32	0.73	0.265	0.158
	73.025	23.812	24.608	19.050	1.6	0.8	90.1	87.3	13.1	5 600	7 400	**25877R**	**25821**	15.8	43.0	40.5	65.0	68.0	1.6	0.8	0.29	2.07	1.14	0.310	0.165
	73.025	26.988	26.975	22.225	3.6	1.6	97.2	94.1	13.9	5 700	7 600	**23690**	**23620**	18.8	49.0	42.0	64.0	68.0	3.6	1.6	0.37	1.62	0.89	0.326	0.212
	76.200	20.638	20.940	15.507	1.6	1.2	71.6	65.9	9.70	5 300	7 000	**28137**	**28300**	16.5	43.5	41.0	68.0	71.0	1.6	1.2	0.40	1.49	0.82	0.315	0.137
	76.200	23.812	25.654	19.050	3.6	3.2	92.6	92.2	13.8	5 400	7 200	**2796R**	**2720**	15.9	47.5	41.0	66.0	70.0	3.6	3.2	0.30	1.98	1.09	0.344	0.185
	76.200	29.370	28.575	23.812	1.6	3.2	101	97.4	14.4	5 400	7 200	**31594**	**31520**	21.6	46.0	43.5	64.0	72.0	1.6	3.2	0.40	1.49	0.82	0.388	0.232
	79.375	29.370	29.771	23.812	3.6	3.2	109	105	15.7	5 200	6 900	**3478**	**3420**	20.8	50.0	43.5	67.0	74.0	3.6	3.2	0.37	1.64	0.90	0.462	0.256
	87.312	30.162	30.886	23.812	3.6	3.2	120	120	18.2	4 600	6 200	**3581R**	**3525**	20.5	48.0	45.5	75.0	81.0	3.6	3.2	0.31	1.96	1.08	0.622	0.300
	95.250	27.783	29.901	22.225	0.8	2.4	129	122	18.8	4 500	5 900	**449**	**432**	18.4	44.0	43.5	83.0	87.0	0.8	2.4	0.28	2.11	1.16	0.686	0.384
34.980	59.131	15.875	16.764	11.938	SP	1.2	44.9	48.5	6.85	6 400	8 500	**L68149**	**L68110**	13.2	45.5	39.0	53.0	56.0	3.5	1.2	0.42	1.44	0.79	0.112	0.056
	59.975	15.875	16.764	11.938	SP	1.2	44.9	48.5	6.85	6 400	8 500	**L68149**	**L68111**	13.2	45.5	39.0	53.0	56.0	3.5	1.2	0.42	1.44	0.79	0.112	0.063
35.000	79.375	23.812	25.400	19.050	0.8	0.8	101	105	15.8	5 000	6 700	**26883R**	**26822**	16.4	42.5	42.0	71.0	74.0	0.8	0.8	0.32	1.88	1.04	0.414	0.186
	80.000	21.000	22.403	17.826	0.8	1.2	85.0	74.8	11.4	4 900	6 600	**339**	**332**	15.1	42.5	41.5	73.0	75.0	0.8	1.2	0.27	2.20	1.21	0.385	0.144
35.717	72.233	25.400	25.400	19.842	3.6	2.4	83.8	87.4	12.4	5 700	7 600	**HM88648**	**HM88610**	20.7	52.0	42.5	60.0	69.0	3.6	2.4	0.55	1.10	0.60	0.291	0.185
36.487	73.025	23.812	24.608	19.050	1.6	0.8	90.1	87.3	13.1	5 600	7 400	**25880R**	**25821**	15.8	44.0	42.0	65.0	68.0	1.6	0.8	0.29	2.07	1.14	0.294	0.165
	73.025	23.812	25.654	19.050	3.6	0.8	92.6	92.2	13.8	5 400	7 200	**2794R**	**2735X**	15.9	49.0	42.5	66.0	69.0	3.6	0.8	0.30	1.98	1.09	0.344	0.134
36.512	76.200	29.370	28.575	23.020	3.6	0.8	99.5	107	15.2	5 400	7 200	**HM89449**	**HM89411**	23.9	54.0	44.5	65.0	73.0	3.6	0.8	0.55	1.10	0.60	0.386	0.258
	79.375	23.812	25.400	19.050	0.8	0.8	101	105	15.8	5 000	6 700	**26877R**	**26822**	16.4	44.0	43.0	71.0	74.0	0.8	0.8	0.32	1.88	1.04	0.404	0.186
	79.375	29.370	29.771	23.812	0.8	3.2	109	105	15.7	5 200	6 900	**3479**	**3420**	20.8	45.5	44.5	67.0	74.0	0.8	3.2	0.37	1.64	0.90	0.429	0.259
	85.725	30.162	30.162	23.812	0.8	3.2	135	136	20.3	4 800	6 400	**3878**	**3820**	22.9	48.0	47.0	73.0	81.0	0.8	3.2	0.40	1.49	0.82	0.605	0.285
38.000	63.000	17.000	17.000	13.500	SP	SP	54.7	58.2	8.25	6 000	8 000	**JL69349**	**JL69310**	14.6	49.0	41.0	60.0	56.5	3.5	1.2	0.42	1.44	0.79	0.128	0.070
38.100	63.500	12.700	11.908	9.525	1.6	0.8	32.1	33.1	4.60	5 800	7 700	**13889**	**13830**	11.9	45.0	42.5	59.0	60.0	1.6	0.8	0.35	1.73	0.95	0.104	0.045

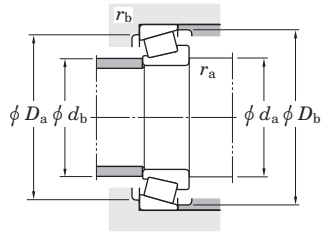
[Notes] 1) SP indicates the specially chamfered from.
2) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (38.100) ~ (40.000) mm





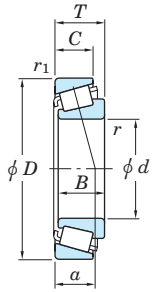
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r 1) min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
38.100	65.088	12.700	11.908	9.525	1.6	0.8	32.1	33.1	4.60	5 800	7 700	**13889**	**13836**	11.9	45.0	42.5	59.0	61.0	1.6	0.8	0.35	1.73	0.95	0.104	0.046
	65.088	18.034	18.288	13.970	SP	1.2	53.9	56.5	8.15	5 800	7 800	**LM29748**	**LM29710**	13.8	49.0	42.5	59.0	62.0	3.5	1.2	0.33	1.80	0.99	0.154	0.079
	65.088	19.812	18.288	15.748	2.4	1.2	53.9	56.5	8.15	5 800	7 800	**LM29749**	**LM29711**	15.6	46.0	42.5	58.0	62.0	2.4	1.2	0.33	1.80	0.99	0.159	0.092
	69.012	19.050	19.050	15.083	2.0	2.4	61.7	62.0	8.95	5 600	7 500	**13687**	**13621**	16.1	46.5	43.0	61.0	65.0	2.0	2.4	0.40	1.49	0.82	0.191	0.102
	71.438	15.875	16.520	11.908	1.6	1.0	57.6	53.8	7.70	5 700	7 600	**19150R**	**19281**	14.5	45.0	43.0	63.0	66.0	1.6	1.0	0.44	1.35	0.74	0.167	0.105
	71.996	17.018	16.520	14.288	1.6	1.6	57.6	53.8	7.70	5 700	7 600	**19150R**	**19283**	15.7	45.0	43.0	63.0	66.0	1.6	1.6	0.44	1.35	0.74	0.167	0.132
	71.996	19.000	20.638	14.237	3.6	1.6	62.3	61.3	8.90	5 600	7 400	**16150**	**16282**	15.0	49.5	43.0	63.0	67.0	3.6	1.6	0.40	1.49	0.82	0.207	0.121
	72.238	20.638	20.638	15.875	3.6	1.2	62.3	61.3	8.90	5 600	7 400	**16150**	**16284**	16.6	49.5	43.0	63.0	67.0	3.6	1.2	0.40	1.49	0.82	0.207	0.144
	72.238	23.812	20.638	19.050	3.6	2.4	62.3	61.3	8.90	5 600	7 400	**16150**	**16283**	19.8	49.5	43.0	61.0	67.0	3.6	2.4	0.40	1.49	0.82	0.207	0.183
	73.025	23.812	25.654	19.050	3.6	0.8	92.6	92.2	13.8	5 400	7 200	**2788R**	**2735X**	15.9	50.0	43.5	66.0	69.0	3.6	0.8	0.30	1.98	1.09	0.308	0.134
	76.200	23.812	25.654	19.050	3.6	0.8	92.6	92.2	13.8	5 400	7 200	**2788R**	**2729**	15.9	50.0	43.5	68.0	70.0	3.6	0.8	0.30	1.98	1.09	0.308	0.189
	79.375	29.370	29.771	23.812	3.6	3.2	109	105	15.7	5 200	6 900	**3490**	**3420**	20.8	52.0	45.9	67.0	74.0	3.6	3.2	0.37	1.64	0.90	0.419	0.256
	80.035	21.432	20.940	15.875	1.6	1.6	71.6	65.9	9.70	5 300	7 000	**28150**	**28317**	16.9	45.5	43.5	69.0	73.0	1.6	1.6	0.40	1.49	0.82	0.285	0.201
	80.035	24.608	23.698	18.512	0.8	1.6	91.6	91.6	13.3	5 200	6 900	**27880**	**27820**	22.2	48.0	47.0	68.0	75.0	0.8	1.6	0.56	1.07	0.59	0.378	0.208
	80.035	24.608	23.698	18.512	3.6	1.6	91.6	91.6	13.3	5 200	6 900	**27881**	**27820**	22.2	53.0	47.0	68.0	75.0	3.6	1.6	0.56	1.07	0.59	0.378	0.208
	82.550	29.370	28.575	23.020	0.8	3.2	109	117	16.9	4 900	6 600	**HM801346**	**HM801310**	24.4	51.0	49.0	68.0	78.0	0.8	3.2	0.55	1.10	0.60	0.483	0.282
	82.550	29.370	28.575	23.020	2.4	3.2	109	117	16.9	4 900	6 600	**HM801346X**	**HM801310**	24.4	54.0	49.0	68.0	78.0	2.4	3.2	0.55	1.10	0.60	0.483	0.282
	82.931	23.812	25.400	19.050	0.8	0.8	96.8	100	15.1	4 800	6 300	**25572**	**25520**	17.5	46.0	46.0	74.0	77.0	0.8	0.8	0.33	1.79	0.99	0.437	0.203
	88.501	26.988	29.083	22.225	3.6	1.6	123	112	17.2	4 900	6 500	**418**	**414**	16.9	51.0	44.5	77.0	80.0	3.6	1.6	0.26	2.28	1.25	0.523	0.325
	90.488	39.688	40.386	33.338	1.6	3.2	166	169	25.9	4 500	6 000	**4375**	**4335**	25.6	51.0	48.5	77.0	85.0	1.6	3.2	0.28	2.11	1.16	0.841	0.459
	101.600	34.925	36.068	26.988	3.6	3.2	164	159	24.8	4 000	5 300	**525**	**522**	22.2	54.0	48.0	89.0	95.0	3.6	3.2	0.29	2.10	1.16	1.05	0.411
39.688	73.025	16.667	17.462	12.700	0.8	1.6	57.6	55.8	8.15	5 200	6 900	**18587**	**18520**	14.5	46.0	46.0	66.0	69.0	0.8	1.6	0.35	1.71	0.94	0.215	0.085
	73.025	23.812	25.654	19.050	3.6	0.8	92.6	92.2	13.8	5 400	7 200	**2789R**	**2735X**	15.9	52.0	45.0	66.0	69.0	3.6	0.8	0.30	1.98	1.09	0.288	0.134
	80.167	29.370	30.391	23.812	0.8	3.2	114	106	16.2	5 000	6 700	**3386**	**3320**	18.7	46.5	45.5	70.0	75.0	0.8	3.2	0.27	2.20	1.21	0.442	0.217
	84.138	29.370	30.391	23.812	3.6	3.2	114	106	16.2	5 000	6 700	**3382**	**3328**	18.7	52.0	45.5	72.0	76.0	3.6	3.2	0.27	2.20	1.21	0.438	0.312
40.000	76.200	20.638	20.940	15.507	1.6	1.2	71.6	65.9	9.70	5 300	7 000	**28158**	**28300**	16.5	47.5	45.0	68.0	71.0	1.6	1.2	0.40	1.49	0.82	0.266	0.137
	80.000	21.000	22.403	17.826	3.6	1.2	85.0	74.8	11.4	4 900	6 600	**344**	**332**	15.1	52.0	45.5	73.0	75.0	3.6	1.2	0.27	2.20	1.21	0.334	0.144

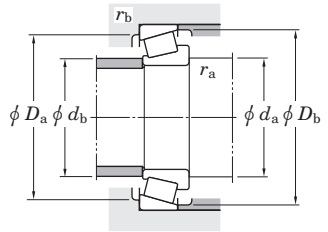
[Note] 1) SP indicates the specially chamfered from.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (40.000) ~ 42.070 mm



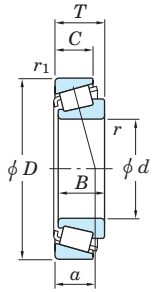


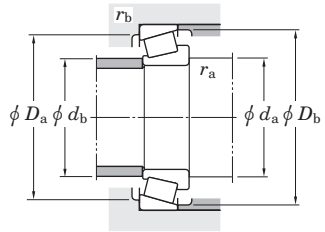
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
40.000	80.000	21.000	22.403	17.826	0.8	1.2	85.0	74.8	11.4	4 900	6 600	**344A**	**332**	15.1	46.0	45.5	73.0	75.0	0.8	1.2	0.27	2.20	1.21	0.334	0.144
	85.000	20.638	21.692	17.462	0.8	1.2	89.6	81.7	12.4	4 600	6 200	**350A**	**354A**	15.5	47.5	46.5	77.0	80.0	0.8	1.2	0.31	1.96	1.08	0.416	0.162
	88.501	26.988	29.083	22.225	3.6	1.6	123	112	17.2	4 900	6 500	**420**	**414**	16.9	52.0	46.0	77.0	80.0	3.6	1.6	0.26	2.28	1.25	0.465	0.325
	107.950	36.512	36.957	28.575	3.6	3.2	172	172	26.8	3 800	5 100	**543**	**532X**	23.9	57.0	50.0	94.0	100.0	3.6	3.2	0.30	2.03	1.11	1.17	0.570
40.483	82.550	29.370	28.575	23.020	3.6	3.2	109	117	16.9	4 900	6 600	**HM801349**	**HM801310**	24.4	58.0	49.0	68.0	78.0	3.6	3.2	0.55	1.10	0.60	0.450	0.282
41.275	73.025	16.667	17.462	12.700	3.6	1.6	57.6	55.8	8.15	5 200	6 900	**18590**	**18520**	14.5	53.0	46.0	66.0	69.0	3.6	1.6	0.35	1.71	0.94	0.199	0.085
	73.431	19.558	19.812	14.732	3.6	0.8	72.5	73.0	10.6	5 200	7 000	**LM501349**	**LM501310**	16.1	53.0	46.5	67.0	70.0	3.6	0.8	0.40	1.50	0.83	0.227	0.107
	73.431	21.430	19.812	16.604	3.6	0.8	72.5	73.0	10.6	5 200	7 000	**LM501349**	**LM501314**	18.0	53.0	46.5	66.0	70.0	3.6	0.8	0.40	1.50	0.83	0.227	0.126
	73.431	23.012	19.812	18.186	3.6	2.4	72.5	73.0	10.6	5 200	7 000	**LM501349**	**LM501311**	16.1	53.0	46.5	64.0	70.0	3.6	2.4	0.40	1.50	0.83	0.227	0.140
	76.200	18.009	17.384	14.288	1.6	1.6	64.7	63.3	9.15	5 200	6 900	**11162R**	**11300**	17.5	49.0	46.5	67.0	72.0	1.6	1.6	0.49	1.23	0.68	0.221	0.127
	76.200	22.225	23.020	17.462	3.6	0.8	82.9	83.3	12.3	5 200	6 900	**24780R**	**24720**	17.4	54.0	47.0	68.0	72.0	3.6	0.8	0.39	1.53	0.84	0.275	0.148
	80.000	21.000	22.403	17.826	0.8	1.2	85.0	74.8	11.4	4 900	6 600	**336**	**332**	15.1	47.0	46.0	73.0	75.0	0.8	1.2	0.27	2.20	1.21	0.325	0.144
	80.000	21.000	22.403	17.826	3.6	1.2	85.0	74.8	11.4	4 900	6 600	**342**	**332**	15.1	53.0	46.0	73.0	75.0	3.6	1.2	0.27	2.20	1.21	0.317	0.144
	82.550	26.543	25.654	20.193	3.6	3.2	105	105	15.4	4 900	6 500	**M802048**	**M802011**	23.3	57.0	50.6	70.0	79.0	3.6	3.2	0.55	1.10	0.60	0.403	0.227
	85.725	30.162	30.162	23.812	3.6	1.2	135	136	20.3	4 800	6 400	**3877**	**3821**	22.9	57.0	50.3	75.0	81.0	3.6	1.2	0.40	1.49	0.82	0.506	0.324
	87.312	30.162	30.886	23.812	0.8	3.2	120	120	18.2	4 600	6 200	**3576R**	**3525**	20.5	49.0	48.0	75.0	81.0	0.8	3.2	0.31	1.96	1.08	0.533	0.300
	88.501	26.988	29.083	22.225	3.6	1.6	123	112	17.2	4 900	6 500	**419**	**414**	16.9	54.0	47.0	77.0	80.0	3.6	1.6	0.26	2.28	1.25	0.441	0.325
	88.900	20.638	22.225	16.513	3.6	1.2	92.9	87.3	13.3	4 400	5 800	**365A**	**362A**	16.1	55.0	48.5	81.0	84.0	3.6	1.2	0.32	1.88	1.03	0.458	0.164
	88.900	30.162	29.370	23.020	0.8	3.2	124	125	18.5	4 600	6 100	**HM803145**	**HM803110**	26.1	54.0	53.0	74.0	85.0	0.8	3.2	0.55	1.10	0.60	0.577	0.318
	88.900	30.162	29.370	23.020	3.6	3.2	124	125	18.5	4 600	6 100	**HM803146**	**HM803110**	26.1	60.0	53.0	74.0	85.0	3.6	3.2	0.55	1.10	0.60	0.574	0.318
	90.488	39.688	40.386	33.338	3.6	3.2	166	169	25.9	4 500	6 000	**4388**	**4335**	25.6	57.0	51.0	77.0	85.0	3.6	3.2	0.28	2.11	1.16	0.775	0.454
	93.662	31.750	31.750	26.195	0.8	3.2	132	134	20.2	4 400	5 800	**46162**	**46368**	24.0	52.0	51.0	79.0	87.0	0.8	3.2	0.40	1.49	0.82	0.695	0.403
	95.250	30.162	29.370	23.020	3.6	3.2	130	140	20.7	3 300	4 400	**HM804840**	**HM804810**	26.5	61.0	54.0	81.0	91.0	3.6	3.2	0.55	1.10	0.60	0.719	0.351
	101.600	34.925	36.068	26.988	3.6	3.2	164	159	24.8	4 000	5 300	**526**	**522**	22.2	57.0	50.0	89.0	95.0	3.6	3.2	0.29	2.10	1.16	1.02	0.411
	104.775	36.512	36.512	28.575	1.6	3.2	176	195	29.3	3 800	5 100	**HM807035**	**HM807010**	29.3	60.0	57.0	89.0	100.0	1.6	3.2	0.49	1.23	0.68	1.19	0.497
42.070	90.488	39.688	40.386	33.338	3.6	3.2	166	169	25.9	4 500	6 000	**4395**	**4335**	25.6	58.0	51.0	77.0	85.0	3.6	3.2	0.28	2.11	1.16	0.751	0.459

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d 42.862 ~ 45.000 mm



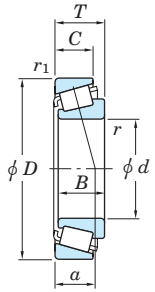


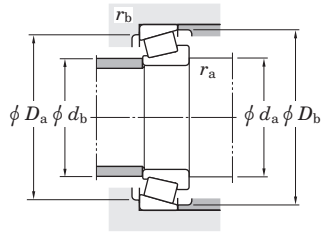
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
42.862	76.992	17.463	17.145	11.908	1.6	1.6	60.8	62.2	8.95	5 000	6 600	**12168**	**12303**	17.5	51.0	48.5	68.0	73.0	1.6	1.6	0.51	1.19	0.65	0.220	0.097
42.875	79.375	23.812	25.400	19.050	3.6	0.8	101	105	15.8	5 000	6 700	**26884R**	**26822**	16.1	55.0	48.5	71.0	74.0	3.6	0.8	0.32	1.88	1.04	0.314	0.186
	82.931	23.812	25.400	19.050	3.6	0.8	96.8	100	15.1	4 800	6 300	**25577**	**25520**	17.5	55.0	49.0	74.0	77.0	3.6	0.8	0.33	1.79	0.99	0.382	0.200
44.450	73.025	18.258	18.258	15.083	1.6	1.6	59.4	65.5	9.50	5 100	6 800	**L102849**	**L102810**	14.6	51.0	49.0	66.0	69.0	1.6	1.6	0.32	1.88	1.04	0.183	0.102
	76.992	17.463	17.145	11.908	1.6	1.6	60.8	62.2	8.95	5 000	6 600	**12175**	**12303**	17.5	52.0	49.5	68.0	73.0	1.6	1.6	0.51	1.19	0.65	0.206	0.097
	79.375	17.462	17.462	13.495	2.8	1.6	59.2	59.1	8.65	4 800	6 400	**18685**	**18620**	16.0	54.0	49.5	71.0	74.0	2.8	1.6	0.37	1.60	0.88	0.214	0.126
	82.931	23.812	25.400	19.050	5.2	0.8	96.8	100	15.1	4 800	6 300	**25582**	**25520**	17.5	59.0	51.0	74.0	77.0	5.2	0.8	0.33	1.79	0.99	0.361	0.200
	84.138	30.162	30.886	23.812	3.6	3.2	120	120	18.2	4 600	6 200	**3578R**	**3520**	20.5	57.0	51.0	74.0	79.5	3.6	3.2	0.31	1.96	1.08	0.479	0.221
	85.000	20.638	21.692	17.462	2.4	1.2	89.6	81.7	12.4	4 600	6 200	**355**	**354A**	15.5	54.0	50.0	77.0	80.0	2.4	1.2	0.31	1.96	1.08	0.344	0.160
	85.000	20.638	21.692	17.462	0.8	1.2	89.6	81.7	12.4	4 600	6 200	**355A**	**354A**	15.5	51.0	50.0	77.0	80.0	0.8	1.2	0.31	1.96	1.08	0.344	0.160
	88.900	30.162	29.370	23.020	3.6	3.2	124	125	18.5	4 600	6 100	**HM803149**	**HM803110**	26.1	62.0	53.4	74.0	85.0	3.6	3.2	0.55	1.10	0.60	0.525	0.318
	93.662	31.750	31.750	25.400	3.6	3.2	131	123	18.8	4 400	5 900	**49175**	**49368**	22.9	59.0	53.0	82.0	87.0	3.6	3.2	0.36	1.67	0.92	0.645	0.371
	93.662	31.750	31.750	26.195	0.8	3.2	132	134	20.2	4 400	5 800	**46175**	**46368**	24.0	55.0	54.0	79.0	87.0	0.8	3.2	0.40	1.49	0.82	0.609	0.403
	93.662	31.750	31.750	26.195	3.6	3.2	132	134	20.2	4 400	5 800	**46176**	**46368**	24.0	60.0	54.0	79.0	87.0	3.6	3.2	0.40	1.49	0.82	0.609	0.403
	95.250	27.783	28.575	22.225	0.8	2.4	135	141	21.6	4 100	5 400	**33885**	**33821**	20.4	53.0	53.0	85.0	90.0	0.8	2.4	0.33	1.82	1.00	0.714	0.264
	95.250	27.783	29.901	22.225	3.6	0.8	129	122	18.8	4 500	5 900	**438**	**432A**	18.4	57.0	51.0	84.0	87.0	3.6	0.8	0.28	2.11	1.16	0.555	0.375
	95.250	30.162	29.370	23.020	0.8	2.4	130	140	20.7	3 300	4 400	**HM804842**	**HM804810**	26.5	57.0	57.0	81.0	91.0	0.8	2.4	0.55	1.10	0.60	0.673	0.351
	95.250	30.162	29.370	23.020	3.6	2.4	130	140	20.7	3 300	4 400	**HM804843**	**HM804810**	26.5	63.0	57.0	81.0	91.0	3.6	2.4	0.55	1.10	0.60	0.670	0.351
	98.425	30.162	31.750	25.400	0.8	3.2	143	143	21.9	3 900	5 200	**49576**	**49520**	24.1	55.0	54.0	88.0	96.0	0.8	3.2	0.40	1.50	0.82	0.856	0.384
	101.600	34.925	36.068	26.988	3.6	3.2	164	159	24.8	4 000	5 300	**527**	**522**	22.2	59.0	53.0	89.0	95.0	3.6	3.2	0.29	2.10	1.16	0.939	0.411
	104.775	36.512	36.512	28.575	3.6	3.2	176	195	29.3	3 800	5 100	**HM807040**	**HM807010**	29.3	66.0	59.0	89.0	100.0	3.6	3.2	0.49	1.23	0.68	1.13	0.497
	111.125	38.100	36.957	30.162	3.6	3.2	172	172	26.8	3 800	5 100	**535**	**532A**	23.9	60.0	54.0	95.0	100.0	3.6	3.2	0.30	2.03	1.11	1.09	0.746
	120.650	41.275	41.275	31.750	3.6	3.2	218	217	34.0	3 500	4 600	**615**	**612**	27.3	62.0	56.0	105.0	110.0	3.6	3.2	0.31	1.91	1.05	1.48	0.853
44.983	93.264	30.162	30.302	23.812	3.6	3.2	129	137	20.9	4 200	5 500	**3776**	**3720**	22.2	59.0	53.0	82.0	88.0	3.6	3.2	0.34	1.77	0.97	0.650	0.288
45.000	85.000	20.638	21.692	17.462	1.6	1.2	89.6	81.7	12.4	4 600	6 200	**358**	**354A**	15.5	52.5	50.0	77.0	80.0	1.6	1.2	0.31	1.96	1.08	0.338	0.162

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d 45.242 ~ 49.212 mm



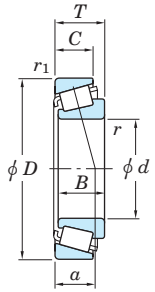


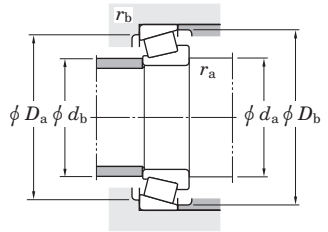
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
45.242	73.431	19.558	19.812	15.748	3.6	0.8	70.0	78.1	11.4	5 100	6 700	**LM102949**	**LM102910**	14.7	56.0	50.0	68.0	70.0	3.6	0.8	0.31	1.97	1.08	0.209	0.100
	77.788	19.842	19.842	15.080	3.6	0.8	71.7	73.5	10.7	4 900	6 500	**LM603049**	**LM603011**	17.5	57.0	50.0	71.0	74.0	3.6	0.8	0.43	1.41	0.77	0.243	0.120
	77.788	21.430	19.842	16.667	3.6	0.8	71.7	73.5	10.7	4 900	6 500	**LM603049**	**LM603012**	19.1	57.0	50.0	71.0	74.0	3.6	0.8	0.43	1.41	0.77	0.243	0.138
	79.974	19.842	19.842	15.080	3.6	0.8	71.7	73.5	10.7	4 900	6 500	**LM603049**	**LM603014**	17.5	57.0	50.0	71.0	74.0	3.6	0.8	0.43	1.41	0.77	0.243	0.152
45.618	85.000	23.812	25.400	19.050	3.6	2.4	96.8	100	15.1	4 800	6 300	**25590**	**25526**	17.5	58.0	51.0	74.0	78.0	3.6	2.4	0.33	1.79	0.99	0.344	0.241
45.987	74.976	18.000	18.000	14.000	2.4	1.6	66.2	74.6	10.8	5 000	6 600	**LM503349R**	**LM503310**	16.0	53.0	51.0	67.0	72.0	2.4	1.6	0.40	1.49	0.82	0.207	0.095
46.038	79.375	17.462	17.462	13.495	2.8	1.6	59.2	59.1	8.65	4 800	6 400	**18690**	**18620**	16.0	56.0	51.0	71.0	74.0	2.8	1.6	0.37	1.60	0.88	0.208	0.123
	85.000	20.638	21.692	17.462	3.6	1.2	89.6	81.7	12.4	4 600	6 200	**359A**	**354A**	15.5	57.0	51.0	77.0	80.0	3.6	1.2	0.31	1.96	1.08	0.323	0.160
	85.000	20.638	21.692	17.462	2.4	1.2	89.6	81.7	12.4	4 600	6 200	**359S**	**354A**	15.5	55.0	51.0	77.0	80.0	2.4	1.2	0.31	1.96	1.08	0.323	0.160
	85.000	25.400	25.608	20.638	3.6	1.2	100	106	16.0	4 600	6 100	**2984**	**2924**	18.9	58.0	52.0	76.0	80.0	3.6	1.2	0.35	1.73	0.95	0.389	0.220
47.625	88.900	20.638	22.225	16.513	3.6	1.2	92.9	87.3	13.3	4 400	5 800	**369A**	**362A**	16.1	60.0	53.0	81.0	84.0	3.6	1.2	0.32	1.88	1.03	0.373	0.164
	88.900	25.400	25.400	19.050	3.6	3.2	109	112	16.6	4 400	5 900	**M804049**	**M804010**	23.6	62.0	55.0	76.0	85.0	3.6	3.2	0.55	1.10	0.60	0.450	0.216
	95.250	30.162	29.370	23.020	3.6	3.2	130	140	20.7	3 300	4 400	**HM804846**	**HM804810**	26.5	64.0	57.0	81.0	91.0	3.6	3.2	0.55	1.10	0.60	0.617	0.351
	96.838	21.000	21.946	15.875	0.8	0.8	101	101	15.3	3 900	5 200	**386A**	**382A**	17.4	56.0	55.0	89.0	92.0	0.8	0.8	0.35	1.69	0.93	0.563	0.177
	101.600	34.925	36.068	26.988	3.6	3.2	164	159	24.8	4 000	5 300	**528**	**522**	22.2	62.0	55.0	89.0	95.0	3.6	3.2	0.29	2.10	1.16	0.871	0.411
	104.775	30.162	29.317	24.605	4.8	3.2	136	144	22.2	3 700	4 900	**463**	**453X**	23.6	65.0	56.0	92.0	98.0	4.8	3.2	0.34	1.79	0.98	0.838	0.372
	104.775	30.162	29.317	24.605	0.8	3.2	136	144	22.2	3 700	4 900	**467**	**453X**	23.6	57.0	56.0	92.0	98.0	0.8	3.2	0.34	1.79	0.98	0.844	0.372
	104.775	30.162	30.958	23.812	3.6	3.2	157	165	25.6	3 700	4 900	**45282**	**45220**	22.2	64.0	59.0	93.0	99.0	3.6	3.2	0.33	1.80	0.99	0.940	0.345
48.412	95.250	30.162	29.370	23.020	2.4	3.2	130	140	20.7	3 300	4 400	**HM804848**	**HM804810**	26.5	63.0	57.5	81.0	91.0	2.4	3.2	0.55	1.10	0.60	0.606	0.351
	95.250	30.162	29.370	23.020	3.6	3.2	130	140	20.7	3 300	4 400	**HM804849**	**HM804810**	26.5	66.0	57.5	81.0	91.0	3.6	3.2	0.55	1.10	0.60	0.604	0.351
49.212	88.900	20.638	22.225	16.513	0.8	1.2	92.9	87.3	13.3	4 400	5 800	**365S**	**362A**	16.1	55.0	54.0	81.0	84.0	0.8	1.2	0.32	1.88	1.03	0.366	0.164
	104.775	36.512	36.512	28.575	3.6	3.2	176	195	29.3	3 800	5 100	**HM807044**	**HM807010**	29.3	69.0	63.0	89.0	100.0	3.6	3.2	0.49	1.23	0.68	1.03	0.497
	114.300	44.450	44.450	34.925	3.6	3.2	237	230	35.1	3 800	5 000	**65390**	**65320**	31.7	70.0	60.0	97.0	107.0	3.6	3.2	0.43	1.40	0.77	1.28	0.894
	114.300	44.450	44.450	36.068	3.6	3.2	265	263	35.4	3 700	5 000	**HH506348**	**HH506310**	30.6	71.0	61.0	97.0	107.0	3.6	3.2	0.40	1.49	0.82	1.49	0.834

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d 49.987 ~ (50.800) mm





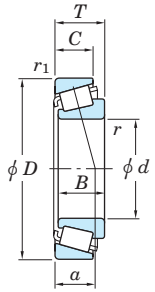
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
49.987	92.075	24.608	25.400	19.845	2.4	0.8	107	119	17.9	4 200	5 600	**28579R**	**28521**	19.9	60.0	56.0	83.0	87.0	2.4	0.8	0.38	1.59	0.87	0.463	0.247
50.000	82.000	21.501	21.501	17.000	3.0	0.5	90.0	97.9	14.7	4 500	6 000	**JLM104948**	**JLM104910**	16.2	60.0	55.0	76.0	78.0	3.0	0.5	0.31	1.97	1.08	0.304	0.128
	88.900	20.638	22.225	16.513	2.0	1.2	92.9	87.3	13.3	4 400	5 800	**365**	**362A**	16.1	58.0	55.0	81.0	84.0	2.0	1.2	0.32	1.88	1.03	0.346	0.164
	88.900	20.638	22.225	16.513	2.4	1.2	92.9	87.3	13.3	4 400	5 800	**366**	**362A**	16.1	59.0	55.0	81.0	84.0	2.4	1.2	0.32	1.88	1.03	0.351	0.166
	90.000	28.000	28.000	23.000	3.0	2.5	132	138	21.1	4 300	5 800	**JM205149**	**JM205110**	20.2	62.0	57.0	80.0	85.0	3.0	2.5	0.33	1.82	1.00	0.508	0.243
	105.000	37.000	36.000	29.000	3.0	2.8	186	205	30.6	3 800	5 100	**JHM807045**	**JHM807012**	29.4	69.0	63.0	90.0	100.0	3.0	2.8	0.49	1.23	0.68	1.01	0.523
	110.000	22.000	21.996	18.824	0.8	1.2	109	116	17.7	3 400	4 500	**396**	**394A**	21.3	61.0	60.0	101.0	105.0	0.8	1.2	0.40	1.49	0.82	0.777	0.264
50.800	80.962	18.258	18.258	14.288	1.6	1.6	67.8	81.1	11.8	4 600	6 100	**L305649R**	**L305610**	16.0	58.0	56.0	73.0	77.0	1.6	1.6	0.35	1.69	0.93	0.228	0.119
	82.550	21.590	22.225	16.510	3.6	1.2	77.0	84.3	12.5	4 500	6 000	**LM104949**	**LM104911**	16.4	62.0	55.0	75.0	78.0	3.6	1.2	0.31	1.97	1.08	0.287	0.131
	85.725	19.050	18.263	12.700	1.6	1.6	63.8	66.4	9.55	4 400	5 900	**18200**	**18337**	22.7	59.0	56.0	76.0	81.0	1.6	1.6	0.57	1.06	0.58	0.268	0.134
	88.900	17.462	17.462	13.495	3.6	1.2	62.5	65.5	9.55	4 400	5 900	**18790**	**18724**	17.4	62.0	56.0	78.0	82.0	3.6	1.2	0.41	1.48	0.81	0.226	0.190
	88.900	20.638	22.225	16.513	1.6	1.2	92.9	87.3	13.3	4 400	5 800	**368**	**362A**	16.1	58.0	56.0	81.0	84.0	1.6	1.2	0.32	1.88	1.03	0.333	0.164
	88.900	20.638	22.225	16.513	3.6	1.2	92.9	87.3	13.3	4 400	5 800	**368A**	**362A**	16.1	62.0	56.0	81.0	84.0	3.6	1.2	0.32	1.88	1.03	0.331	0.164
	88.900	20.638	22.225	16.513	5.2	1.2	92.9	87.3	13.3	4 400	5 800	**370A**	**362A**	16.1	65.0	56.0	81.0	84.0	5.2	1.2	0.32	1.88	1.03	0.326	0.164
	92.075	24.608	25.400	19.845	3.6	0.8	107	119	17.9	4 200	5 600	**28580R**	**28521**	19.9	63.0	57.0	83.0	87.0	3.6	0.8	0.38	1.59	0.87	0.453	0.247
	93.264	20.638	22.225	15.083	2.4	1.2	105	98.5	15.1	4 200	5 600	**375**	**374**	17.1	60.0	57.0	85.0	88.0	2.4	1.2	0.34	1.77	0.97	0.416	0.174
	93.264	30.162	30.302	23.812	3.6	3.2	129	137	20.9	4 200	5 500	**3780**	**3720**	22.2	64.0	58.0	82.0	88.0	3.6	3.2	0.34	1.77	0.97	0.547	0.288
	93.264	30.162	30.302	23.812	3.6	0.8	129	137	20.9	4 200	5 500	**3780**	**3730**	22.2	64.0	58.0	84.0	88.0	3.6	0.8	0.34	1.77	0.97	0.547	0.293
	95.250	27.783	28.575	22.225	3.6	0.8	135	141	21.6	4 100	5 400	**33889**	**33822**	20.4	64.0	58.0	86.0	90.0	3.6	0.8	0.33	1.82	1.00	0.604	0.267
	96.838	21.000	21.946	15.875	0.8	0.8	101	101	15.3	3 900	5 200	**385AX**	**382A**	17.4	59.0	58.0	89.0	92.0	0.8	0.8	0.35	1.69	0.93	0.521	0.177
	97.630	24.608	24.608	19.446	3.6	0.8	113	131	19.7	3 900	5 200	**28678**	**28622**	21.2	65.0	58.0	88.0	92.0	3.6	0.8	0.40	1.49	0.82	0.569	0.267
	98.425	30.162	30.302	23.812	3.6	3.2	129	137	20.9	4 200	5 500	**3780**	**3732**	22.2	64.0	58.0	84.0	90.0	3.6	3.2	0.34	1.77	0.97	0.547	0.433
	101.600	31.750	31.750	25.400	3.6	3.2	143	143	21.9	3 900	5 200	**49585**	**49520**	24.1	66.0	59.0	88.0	96.0	3.6	3.2	0.40	1.50	0.82	0.736	0.384
	101.600	34.925	36.068	26.988	0.8	3.2	164	159	24.8	4 000	5 300	**529**	**522**	22.2	59.0	58.0	89.0	95.0	0.8	3.2	0.29	2.10	1.16	0.806	0.411
	101.600	34.925	36.068	26.988	3.6	3.2	164	159	24.8	4 000	5 300	**529X**	**522**	22.2	65.0	58.0	89.0	95.0	3.6	3.2	0.29	2.10	1.16	0.802	0.411
	104.775	30.162	30.958	23.812	6.4	3.2	157	165	25.6	3 700	4 900	**45284**	**45220**	22.2	71.0	59.0	93.0	99.0	6.4	3.2	0.33	1.80	0.99	0.873	0.345
	104.775	36.512	36.512	28.575	3.6	3.2	185	187	28.6	3 900	5 100	**59200**	**59412**	26.9	68.0	61.0	92.0	99.0	3.6	3.2	0.40	1.49	0.82	0.767	0.623

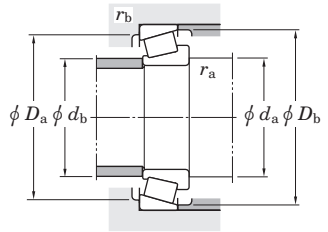
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (50.800) ~ (55.000) mm





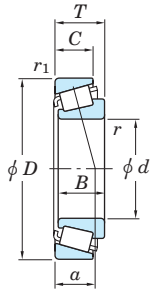
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
50.800	104.775	36.512	36.512	28.575	3.6	3.2	176	195	29.3	3 800	5 100	**HM807046**	**HM807010**	29.3	70.0	63.0	89.0	100.0	3.6	3.2	0.49	1.23	0.68	0.995	0.497
	104.775	39.688	40.157	33.338	3.6	3.2	189	211	32.3	3 800	5 100	**4580**	**4535**	27.3	67.0	61.0	90.0	99.0	3.6	3.2	0.34	1.79	0.98	1.06	0.576
	107.950	36.512	36.957	28.575	3.6	3.2	172	172	26.8	3 800	5 100	**537**	**532X**	23.9	65.0	59.0	94.0	100.0	3.6	3.2	0.30	2.03	1.11	0.969	0.569
	112.712	30.162	30.162	23.812	3.6	3.2	184	207	32.1	3 300	4 500	**39575**	**39520**	23.3	68.0	61.0	101.0	107.0	3.6	3.2	0.34	1.77	0.97	1.13	0.355
	120.650	41.275	41.275	31.750	3.6	3.2	218	217	34.0	3 500	4 600	**619**	**612**	27.3	67.0	61.0	105.0	110.0	3.6	3.2	0.31	1.91	1.05	1.44	0.853
	127.000	44.450	44.450	34.925	3.6	3.2	259	269	41.0	3 300	4 400	**65200**	**65500**	35.2	75.0	69.0	107.0	119.0	3.6	3.2	0.49	1.23	0.68	1.86	1.03
51.592	88.900	20.638	22.225	16.513	2.0	1.2	92.9	87.3	13.3	4 400	5 800	**368S**	**362A**	16.1	59.0	56.0	81.0	84.0	2.0	1.2	0.32	1.88	1.03	0.321	0.164
52.388	92.075	24.608	25.400	19.845	3.6	0.8	107	119	17.9	4 200	5 600	**28584R**	**28521**	19.9	65.0	58.0	83.0	87.0	3.6	0.8	0.38	1.59	0.87	0.435	0.247
	104.775	30.162	29.317	24.605	1.6	3.2	136	144	22.2	3 700	4 900	**468**	**453X**	23.6	62.0	60.0	92.0	98.0	1.6	3.2	0.34	1.79	0.98	0.748	0.372
53.975	88.900	19.050	19.050	13.492	2.4	2.0	79.1	86.8	12.6	4 200	5 600	**LM806649**	**LM806610**	21.5	63.0	60.0	80.0	85.0	2.4	2.0	0.55	1.10	0.60	0.312	0.135
	95.250	27.783	28.575	22.225	1.6	0.8	135	141	21.6	4 100	5 400	**33895**	**33822**	20.4	63.0	60.0	86.0	90.0	1.6	0.8	0.33	1.82	1.00	0.550	0.267
	104.775	30.162	29.317	24.605	3.6	3.2	136	144	22.2	3 700	4 900	**456**	**453X**	23.6	68.0	61.0	92.0	98.0	3.6	3.2	0.34	1.79	0.98	0.728	0.372
	104.775	36.512	36.512	28.575	3.6	3.2	176	195	29.3	3 800	5 100	**HM807049**	**HM807010**	29.3	73.0	63.0	89.0	100.0	3.6	3.2	0.49	1.23	0.68	0.921	0.497
	104.775	39.688	40.157	33.338	3.6	3.2	189	211	32.3	3 800	5 100	**4595**	**4535**	27.3	70.0	63.0	90.0	99.0	3.6	3.2	0.34	1.79	0.98	0.981	0.576
	107.950	36.512	36.957	28.575	3.6	3.2	172	172	26.8	3 800	5 100	**539**	**532X**	23.9	68.0	61.0	94.0	100.0	3.6	3.2	0.30	2.03	1.11	0.894	0.569
	107.950	36.512	36.957	28.575	5.6	3.2	172	172	26.8	3 800	5 100	**539A**	**532X**	23.9	72.0	61.0	94.0	100.0	5.6	3.2	0.30	2.03	1.11	0.861	0.569
	117.475	33.338	31.750	23.812	3.6	3.2	162	152	23.2	3 500	4 600	**66212R**	**66462**	33.2	73.0	67.0	100.0	111.0	3.6	3.2	0.63	0.96	0.53	1.03	0.552
	120.650	41.275	41.275	31.750	3.6	3.2	218	217	34.0	3 500	4 600	**621**	**612**	27.3	70.0	63.0	105.0	110.0	3.6	3.2	0.31	1.91	1.05	1.36	0.853
	122.238	33.338	31.750	23.812	3.6	3.2	160	153	23.3	3 300	4 300	**66584**	**66520**	35.4	75.0	68.0	105.0	116.0	3.6	3.2	0.67	0.90	0.50	1.25	0.551
	122.238	43.658	43.764	36.512	3.6	3.2	276	318	43.6	3 200	4 300	**5578R**	**5535**	31.1	73.0	67.0	106.0	116.0	3.6	3.2	0.36	1.67	0.92	1.84	0.807
	123.825	38.100	36.678	30.162	3.6	3.2	202	223	34.8	3 200	4 200	**557S**	**552A**	28.7	71.0	65.0	109.0	116.0	3.6	3.2	0.35	1.73	0.95	1.47	0.756
	127.000	44.450	44.450	34.925	3.6	3.2	259	269	41.0	3 300	4 400	**65212**	**65500**	35.2	77.0	71.0	107.0	119.0	3.6	3.2	0.49	1.23	0.68	1.78	1.02
54.988	104.775	30.162	29.317	24.605	2.4	3.2	136	144	22.2	3 700	4 900	**466**	**453X**	23.6	67.0	61.0	92.0	98.0	2.4	3.2	0.34	1.79	0.98	0.708	0.372
54.991	135.755	53.975	56.007	44.450	3.6	3.2	333	357	49.3	3 000	4 000	**6381**	**6320**	34.8	76.0	70.0	117.0	126.0	3.6	3.2	0.32	1.85	1.02	2.75	1.37
55.000	90.000	23.000	23.000	18.500	1.6	0.5	102	115	17.2	4 200	5 500	**JLM506849**	**JLM506810**	20.1	63.0	61.0	82.0	86.0	1.6	0.5	0.40	1.49	0.82	0.370	0.183

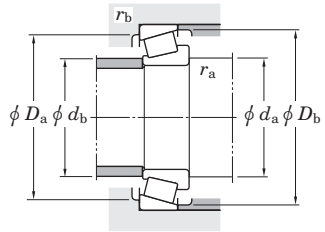
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (55.000) ~ (60.000) mm





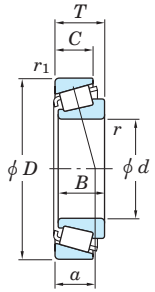
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
55.000	95.000	29.000	29.000	23.500	1.6	2.8	138	150	23.0	4 000	5 300	**JM207049**	**JM207010**	21.3	64.0	62.0	85.0	91.0	1.6	2.8	0.33	1.79	0.99	0.567	0.256
	96.838	21.000	21.946	15.875	2.4	0.8	101	101	15.3	3 900	5 200	**385**	**382A**	17.4	65.0	61.0	89.0	92.0	2.4	0.8	0.35	1.69	0.93	0.461	0.177
	96.838	21.000	21.946	15.875	3.6	0.8	101	101	15.3	3 900	5 200	**385X**	**382A**	17.4	67.0	61.0	89.0	92.0	3.6	0.8	0.35	1.69	0.93	0.459	0.177
	110.000	39.000	39.000	32.000	3.0	2.5	220	224	34.7	3 600	4 900	**JH307749**	**JH307710**	26.8	71.0	64.0	97.0	104.0	3.0	2.5	0.35	1.73	0.95	1.16	0.560
55.562	97.630	24.608	24.608	19.446	3.6	0.8	113	131	19.7	3 900	5 200	**28680**	**28622**	21.2	68.0	62.0	88.0	92.0	3.6	0.8	0.40	1.49	0.82	0.492	0.267
	122.238	43.658	43.764	36.512	1.2	3.2	276	318	43.6	3 200	4 300	**5566R**	**5535**	31.1	70.0	68.0	106.0	116.0	1.2	3.2	0.36	1.67	0.92	1.82	0.807
	127.000	36.512	36.512	26.988	3.6	3.2	209	235	36.2	3 000	4 000	**HM813840**	**HM813810**	32.9	76.0	70.0	111.0	121.0	3.6	3.2	0.50	1.20	0.66	1.72	0.606
55.575	96.838	21.000	21.946	15.875	2.4	0.8	101	101	15.3	3 900	5 200	**389**	**382A**	17.4	65.0	61.0	89.0	92.0	2.4	0.8	0.35	1.69	0.93	0.452	0.177
57.150	96.838	21.000	21.946	15.875	2.4	0.8	101	101	15.3	3 900	5 200	**387**	**382A**	17.4	66.0	62.0	89.0	92.0	2.4	0.8	0.35	1.69	0.93	0.428	0.177
	96.838	21.000	21.946	15.875	3.6	0.8	101	101	15.3	3 900	5 200	**387A**	**382A**	17.4	69.0	62.0	89.0	92.0	3.6	0.8	0.35	1.69	0.93	0.426	0.177
	96.838	21.000	21.946	15.875	5.2	0.8	101	101	15.3	3 900	5 200	**387AS**	**382A**	17.4	72.0	62.0	89.0	92.0	5.2	0.8	0.35	1.69	0.93	0.422	0.177
	96.838	21.000	21.946	15.875	0.8	0.8	101	101	15.3	3 900	5 200	**387S**	**382A**	17.4	63.0	62.0	89.0	92.0	0.8	0.8	0.35	1.69	0.93	0.431	0.177
	98.425	21.000	21.946	17.826	2.4	0.8	101	101	15.3	3 900	5 200	**387**	**382**	17.4	66.0	62.0	89.0	92.0	2.4	0.8	0.35	1.69	0.93	0.428	0.223
	104.775	30.162	29.317	24.605	2.4	3.2	136	144	22.2	3 700	4 900	**462**	**453X**	23.6	67.0	63.0	92.0	98.0	2.4	3.2	0.34	1.79	0.98	0.685	0.372
	104.775	30.162	29.317	24.605	3.6	3.2	136	144	22.2	3 700	4 900	**469**	**453X**	23.6	70.0	63.0	92.0	98.0	3.6	3.2	0.34	1.79	0.98	0.682	0.372
	104.775	30.162	30.958	23.812	6.4	0.8	157	165	25.6	3 700	4 900	**45291**	**45221**	22.2	76.0	65.0	95.0	99.0	6.4	0.8	0.33	1.80	0.99	0.742	0.350
	112.712	30.162	30.048	23.812	3.6	3.2	139	164	25.1	3 400	4 500	**3979**	**3920**	25.9	72.0	66.0	99.0	106.0	3.6	3.2	0.40	1.49	0.82	0.916	0.448
	112.712	30.162	30.162	23.812	3.6	3.2	184	207	32.1	3 300	4 500	**39580**	**39520**	23.3	72.0	66.0	101.0	107.0	3.6	3.2	0.34	1.77	0.97	1.05	0.355
	112.712	30.162	30.162	23.812	7.9	3.2	184	207	32.1	3 300	4 500	**39581**	**39520**	23.3	81.0	66.0	101.0	107.0	7.9	3.2	0.34	1.77	0.97	1.03	0.355
	117.475	30.162	30.162	23.812	3.6	3.2	148	179	27.4	3 200	4 200	**33225**	**33462**	27.8	74.0	68.0	104.0	112.0	3.6	3.2	0.44	1.38	0.76	1.13	0.442
	120.650	41.275	41.275	31.750	3.6	3.2	218	217	34.0	3 500	4 600	**623**	**612**	27.3	72.0	66.0	105.0	110.0	3.6	3.2	0.31	1.91	1.05	1.27	0.853
	127.000	44.450	44.450	34.925	3.6	3.2	259	269	41.0	3 300	4 400	**65225**	**65500**	35.2	80.0	71.0	107.0	119.0	3.6	3.2	0.49	1.23	0.68	1.69	1.02
57.531	96.838	21.000	21.946	15.875	3.6	0.8	101	101	15.3	3 900	5 200	**388A**	**382A**	17.4	69.0	63.0	89.0	92.0	3.6	0.8	0.35	1.69	0.93	0.420	0.177
59.972	122.238	33.338	31.750	23.812	0.8	3.2	160	153	23.3	3 300	4 300	**66589**	**66520**	35.4	74.0	73.0	105.0	116.0	0.8	3.2	0.67	0.90	0.50	1.11	0.551
60.000	95.000	24.000	24.000	19.000	5.0	2.5	108	125	18.9	3 900	5 200	**JLM508748**	**JLM508710**	21.2	75.0	66.0	85.0	91.0	5.0	2.5	0.40	1.49	0.82	0.402	0.196

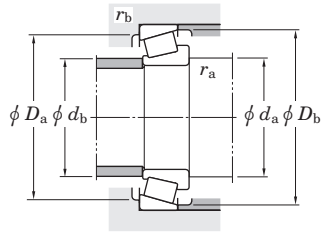
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (60.000) ~ (65.000) mm





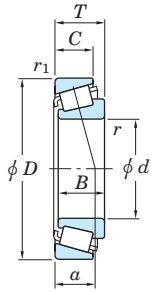
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
60.000	107.950	25.400	25.400	19.050	3.6	3.2	116	143	21.6	3 400	4 500	**29580**	**29520**	24.7	74.0	68.0	96.0	103.0	3.6	3.2	0.46	1.31	0.72	0.713	0.277
	110.000	22.000	21.996	18.824	0.8	1.2	109	116	17.7	3 400	4 500	**397**	**394A**	21.3	69.0	68.0	101.0	104.5	0.8	1.2	0.40	1.49	0.82	0.637	0.259
60.325	100.000	25.400	25.400	19.845	3.6	3.2	115	137	20.6	3 700	4 900	**28985**	**28921**	22.8	73.0	67.0	89.0	96.0	3.6	3.2	0.43	1.41	0.78	0.533	0.230
	101.600	25.400	25.400	19.845	3.6	3.2	115	137	20.6	3 700	4 900	**28985**	**28920**	22.8	73.0	67.0	89.0	96.0	3.6	3.2	0.43	1.41	0.78	0.533	0.269
	122.238	43.658	43.764	36.512	3.6	3.2	276	318	43.6	3 200	4 300	**5583R**	**5535**	31.1	78.0	72.0	106.0	116.0	3.6	3.2	0.36	1.67	0.92	1.66	0.807
	127.000	36.512	36.512	26.988	3.6	1.6	209	235	36.2	3 000	4 000	**HM813841**	**HM813811**	32.9	80.0	73.0	113.0	121.0	3.6	1.6	0.50	1.20	0.66	1.60	0.622
	127.000	36.512	36.512	26.988	1.6	3.2	209	235	36.2	3 000	4 000	**HM813841A**	**HM813810**	32.9	74.0	71.0	110.0	121.0	1.6	3.2	0.50	1.20	0.66	1.62	0.606
	127.000	44.450	44.450	34.925	3.6	3.2	259	269	41.0	3 300	4 400	**65237**	**65500**	35.2	82.0	71.0	107.0	119.0	3.6	3.2	0.49	1.23	0.68	1.59	1.02
	127.000	44.450	44.450	34.925	1.6	3.2	259	269	41.0	3 300	4 400	**65237A**	**65500**	35.2	78.0	71.0	107.0	119.0	1.6	3.2	0.49	1.23	0.68	1.59	1.02
	136.525	46.038	46.038	36.512	3.6	3.2	290	369	49.6	2 800	3 700	**H715332**	**H715311**	37.0	84.0	78.0	118.0	132.0	3.6	3.2	0.47	1.27	0.70	2.56	0.950
61.912	110.000	22.000	21.996	18.824	0.8	1.2	109	116	17.7	3 400	4 500	**392**	**394A**	21.3	70.0	69.0	101.0	104.5	0.8	1.2	0.40	1.49	0.82	0.606	0.259
63.500	107.950	25.400	25.400	19.050	1.6	3.2	116	143	21.6	3 400	4 500	**29586**	**29520**	24.7	73.0	71.0	96.0	103.0	1.6	3.2	0.46	1.31	0.72	0.649	0.277
	110.000	22.000	21.996	18.824	1.6	1.2	109	116	17.7	3 400	4 500	**390A**	**394A**	21.3	73.0	70.0	101.0	104.5	1.6	1.2	0.40	1.49	0.82	0.579	0.259
	110.000	22.000	21.996	18.824	3.6	1.2	109	116	17.7	3 400	4 500	**395**	**394A**	21.3	77.0	70.0	101.0	104.5	3.6	1.2	0.40	1.49	0.82	0.575	0.259
	110.000	25.400	25.400	19.050	3.6	1.2	116	143	21.6	3 400	4 500	**29585**	**29521**	24.7	77.0	71.0	99.0	104.0	3.6	1.2	0.46	1.31	0.72	0.644	0.333
	112.712	30.162	30.162	23.812	3.6	3.2	184	207	32.1	3 300	4 500	**39585**	**39520**	23.3	77.0	71.0	101.0	107.0	3.6	3.2	0.34	1.77	0.97	0.908	0.355
	120.000	29.794	29.007	24.237	0.8	2.0	148	161	25.0	3 200	4 200	**477**	**472**	25.7	73.0	72.0	108.0	113.0	0.8	2.0	0.38	1.56	0.86	0.967	0.493
	122.238	38.354	38.100	29.718	3.6	3.2	238	249	39.1	3 200	4 300	**HM212046**	**HM212011**	27.6	80.0	73.0	108.0	116.0	3.6	3.2	0.34	1.78	0.98	1.36	0.591
	122.238	43.658	43.764	36.512	3.6	3.2	276	318	43.6	3 200	4 300	**5584R**	**5535**	31.1	81.0	75.0	106.0	116.0	3.6	3.2	0.36	1.67	0.92	1.56	0.807
	127.000	36.512	36.170	28.575	3.6	3.2	196	226	35.3	3 000	4 000	**565**	**563**	28.6	80.0	73.0	112.0	120.0	3.6	3.2	0.36	1.65	0.91	1.43	0.648
	135.755	53.975	56.007	44.450	4.3	3.2	333	357	49.3	3 000	4 000	**6382**	**6320**	34.8	84.0	77.0	117.0	126.0	4.3	3.2	0.32	1.85	1.02	2.29	1.39
	136.525	41.275	41.275	31.750	3.6	3.2	302	308	48.1	2 900	3 800	**H414235**	**H414210**	30.3	82.0	78.0	121.0	129.0	3.6	3.2	0.36	1.67	0.92	2.11	0.796
64.986	112.712	30.162	30.924	23.812	2.4	3.2	184	207	32.1	3 300	4 500	**39586**	**39520**	23.3	76.0	72.0	101.0	107.0	2.4	3.2	0.34	1.77	0.97	0.845	0.355
65.000	105.000	24.000	23.000	18.500	3.0	1.0	120	129	19.6	3 500	4 700	**JLM710949**	**JLM710910**	23.8	77.0	71.0	96.0	100.5	3.0	1.0	0.45	1.32	0.73	0.513	0.234
	110.000	28.000	28.000	22.500	3.0	2.8	170	191	29.4	3 400	4 600	**JM511946**	**JM511910**	24.5	78.0	72.0	99.0	105.0	3.0	2.8	0.40	1.49	0.82	0.733	0.338

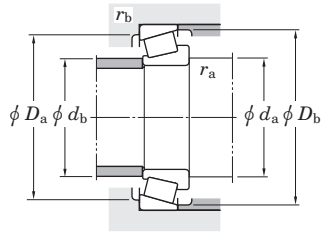
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (65.000) ~ 68.262 mm





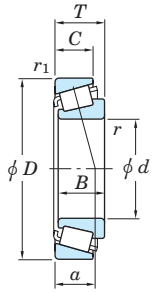
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
65.000	120.000	39.000	38.500	32.000	3.0	2.8	236	255	39.7	3 200	4 300	**JH211749**	**JH211710**	27.9	80.0	74.0	107.0	114.0	3.0	2.8	0.34	1.78	0.98	1.27	0.618
	120.000	39.000	38.500	32.000	7.1	2.8	236	255	39.7	3 200	4 300	**JH211749A**	**JH211710**	27.9	88.0	74.0	107.0	114.0	7.1	2.8	0.34	1.78	0.98	1.27	0.618
65.088	135.755	53.975	56.007	44.450	3.6	3.2	333	357	49.3	3 000	4 000	**6379**	**6320**	34.8	84.0	77.5	117.0	126.0	3.6	3.2	0.32	1.85	1.02	2.34	1.37
	136.525	46.038	46.038	36.512	3.6	3.2	290	369	49.6	2 800	3 700	**H715340**	**H715311**	37.0	88.0	82.0	118.0	132.0	3.6	3.2	0.47	1.27	0.70	2.39	0.950
65.883	122.238	43.658	43.764	36.512	3.6	3.2	276	318	43.6	3 200	4 300	**5595R**	**5535**	31.1	83.0	77.0	106.0	116.0	3.6	3.2	0.36	1.67	0.92	1.48	0.807
66.675	110.000	22.000	21.996	18.824	0.8	1.2	109	116	17.7	3 400	4 500	**395A**	**394A**	21.3	73.0	73.0	101.0	104.5	0.8	1.2	0.40	1.49	0.82	0.524	0.259
	110.000	22.000	21.996	18.824	3.6	1.2	109	116	17.7	3 400	4 500	**395S**	**394A**	21.3	79.0	73.0	101.0	104.5	3.6	1.2	0.40	1.49	0.82	0.519	0.259
	112.712	30.162	30.048	23.812	3.6	0.8	139	164	25.1	3 400	4 500	**3984**	**3925**	25.9	80.0	74.0	101.0	106.0	3.6	0.8	0.40	1.49	0.82	0.700	0.454
	112.712	30.162	30.162	23.812	3.6	3.2	184	207	32.1	3 300	4 500	**39590**	**39520**	23.3	80.0	74.0	101.0	107.0	3.6	3.2	0.34	1.77	0.97	0.832	0.355
	112.712	30.162	30.162	23.812	3.6	0.8	184	207	32.1	3 300	4 500	**39590**	**39521**	23.3	80.0	74.0	103.0	107.0	3.6	0.8	0.34	1.77	0.97	0.832	0.360
	117.475	30.162	30.162	23.812	3.6	3.2	148	179	27.4	3 200	4 200	**33262**	**33462**	27.8	81.0	75.0	104.0	112.0	3.6	3.2	0.44	1.38	0.76	0.910	0.436
	122.238	38.100	38.354	29.718	3.6	1.6	238	249	39.1	3 200	4 300	**HM212049**	**HM212010**	27.3	82.0	75.5	110.0	116.0	3.6	1.6	0.34	1.78	0.98	1.26	0.596
	127.000	36.512	36.512	26.988	3.6	1.6	209	235	36.2	3 000	4 000	**HM813844**	**HM813811**	32.9	85.0	78.0	113.0	121.0	3.6	1.6	0.50	1.20	0.66	1.42	0.622
	130.175	41.275	41.275	31.750	3.6	3.2	246	267	41.8	3 000	3 900	**641**	**633**	30.3	83.0	77.0	116.0	124.0	3.6	3.2	0.36	1.66	0.91	1.68	0.703
	135.755	53.975	56.007	44.450	4.3	3.2	333	357	49.3	3 000	4 000	**6386**	**6320**	34.8	87.0	77.5	117.0	126.0	4.3	3.2	0.32	1.85	1.02	2.27	1.37
	135.755	53.975	56.007	44.450	6.4	3.2	333	357	49.3	3 000	4 000	**6389**	**6320**	34.8	91.0	77.5	117.0	126.0	6.4	3.2	0.32	1.85	1.02	2.15	1.37
	136.525	41.275	41.275	31.750	3.6	3.2	302	308	48.1	2 900	3 800	**H414242**	**H414210**	30.3	85.0	81.0	121.0	129.0	3.6	3.2	0.36	1.67	0.92	2.01	0.796
	136.525	46.038	46.038	36.512	3.6	3.2	290	369	49.6	2 800	3 700	**H715341**	**H715311**	37.0	89.0	83.0	118.0	132.0	3.6	3.2	0.47	1.27	0.70	2.33	0.950
68.262	110.000	22.000	21.996	18.824	2.4	1.2	109	116	17.7	3 400	4 500	**399A**	**394A**	21.3	78.0	74.0	101.0	104.5	2.4	1.2	0.40	1.49	0.82	0.493	0.259
	110.000	22.000	21.996	18.824	5.2	1.2	109	116	17.7	3 400	4 500	**399AS**	**394A**	21.3	83.0	74.0	101.0	104.5	5.2	1.2	0.40	1.49	0.82	0.485	0.259
	117.475	30.162	30.162	23.812	3.6	3.2	148	179	27.4	3 200	4 200	**33269**	**33462**	27.8	82.0	76.0	104.0	112.0	3.6	3.2	0.44	1.38	0.76	0.870	0.436
	127.000	36.512	36.170	28.575	3.6	3.2	196	226	35.3	3 000	4 000	**570**	**563**	28.6	83.0	77.0	112.0	120.0	3.6	3.2	0.36	1.65	0.91	1.29	0.648
	136.525	41.275	41.275	31.750	3.6	3.2	284	308	46.1	2 900	3 800	**H414245**	**H414210**	30.3	86.0	82.0	121.0	129.0	3.6	3.2	0.36	1.67	0.92	1.92	0.788
	136.525	46.038	46.038	36.512	3.6	3.2	290	369	49.6	2 800	3 700	**H715343**	**H715311**	37.0	90.0	84.0	118.0	132.0	3.6	3.2	0.47	1.27	0.70	2.27	0.950
	152.400	47.625	46.038	31.750	3.6	3.2	306	278	38.3	2 700	3 600	**9185**	**9121**	44.5	94.0	81.5	130.0	145.0	3.6	3.2	0.66	0.91	0.50	2.67	1.20

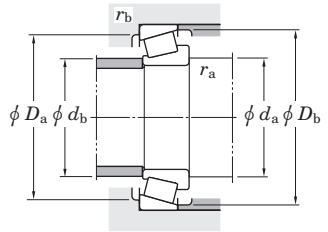
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d 69.850 ~ (73.025) mm





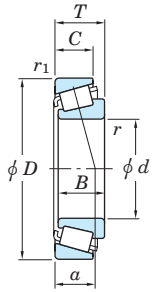
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
69.850	98.425	13.495	13.495	9.525	1.6	1.6	49.1	59.8	8.45	3 500	4 700	**LL713049**	**LL713010**	18.4	77.0	74.0	92.0	94.5	1.6	1.6	0.44	1.37	0.75	0.205	0.086
	112.712	22.225	21.996	15.875	1.6	0.8	115	127	19.4	3 300	4 400	**LM613449**	**LM613410**	21.9	78.0	76.0	104.0	107.0	1.6	0.8	0.42	1.44	0.79	0.562	0.238
	112.712	25.400	25.400	19.050	1.6	3.2	122	155	23.3	3 200	4 300	**29675**	**29620**	26.2	80.0	77.0	101.0	109.0	1.6	3.2	0.49	1.23	0.68	0.676	0.270
	117.475	30.162	30.162	23.812	3.6	3.2	148	179	27.4	3 200	4 200	**33275**	**33462**	27.8	84.0	77.0	104.0	112.0	3.6	3.2	0.44	1.38	0.76	0.830	0.436
	120.000	29.002	29.007	23.444	3.6	3.2	148	161	25.0	3 200	4 200	**482**	**472A**	24.9	83.0	77.0	106.0	114.0	3.6	3.2	0.38	1.56	0.86	0.791	0.462
	120.000	29.794	29.007	24.237	3.6	2.0	148	161	25.0	3 200	4 200	**482**	**472**	25.7	83.0	77.0	108.0	113.0	3.6	2.0	0.38	1.56	0.86	0.791	0.487
	120.000	32.545	32.545	26.195	3.6	3.2	189	218	33.9	3 100	4 200	**47487R**	**47420**	26.6	84.0	78.0	107.0	114.0	3.6	3.2	0.36	1.67	0.92	1.01	0.476
	120.650	32.545	32.545	26.195	3.6	0.8	189	218	33.9	3 100	4 200	**47487R**	**47423**	26.6	84.0	78.0	109.0	114.0	3.6	0.8	0.36	1.67	0.92	1.01	0.513
	123.825	30.162	29.007	24.605	3.6	3.2	148	161	25.0	3 200	4 200	**482**	**472X**	26.0	83.0	77.0	109.0	114.0	3.6	3.2	0.38	1.56	0.86	0.791	0.625
	127.000	36.512	36.170	28.575	3.6	3.2	196	226	35.3	3 000	4 000	**566**	**563**	28.6	85.0	78.0	112.0	120.0	3.6	3.2	0.36	1.65	0.91	1.24	0.648
	146.050	41.275	41.275	31.750	3.6	3.2	261	301	45.3	2 600	3 400	**655**	**653**	33.4	88.0	82.0	131.0	139.0	3.6	3.2	0.41	1.47	0.81	2.35	0.891
	150.089	44.450	46.672	36.512	3.6	3.2	330	368	50.1	2 500	3 400	**745AR**	**742**	32.4	88.0	82.0	134.0	142.0	3.6	3.2	0.33	1.84	1.01	2.79	1.07
	168.275	53.975	56.363	41.275	3.6	3.2	429	467	62.1	2 300	3 100	**835R**	**832**	35.0	91.0	84.0	149.0	155.0	3.6	3.2	0.30	2.00	1.10	4.32	1.72
69.952	121.442	24.608	23.012	17.462	2.0	2.0	113	127	19.4	3 000	4 000	**34274**	**34478**	26.8	81.0	78.0	110.0	116.0	2.0	2.0	0.45	1.33	0.73	0.764	0.316
70.000	110.000	26.000	25.000	20.500	1.0	2.5	129	158	23.9	3 300	4 400	**JLM813049**	**JLM813010**	26.1	78.0	77.0	98.0	105.0	1.0	2.5	0.49	1.23	0.68	0.590	0.300
	115.000	29.000	29.000	23.000	3.0	2.5	155	173	26.6	3 200	4 300	**JM612949**	**JM612910**	26.2	83.0	77.0	103.0	110.0	3.0	2.5	0.43	1.39	0.77	0.776	0.358
71.438	117.475	30.162	30.162	23.812	3.6	3.2	148	179	27.4	3 200	4 200	**33281**	**33462**	27.8	85.0	79.0	104.0	112.0	3.6	3.2	0.44	1.38	0.76	0.789	0.436
	120.000	32.545	32.545	26.195	3.6	3.2	189	218	33.9	3 100	4 200	**47490R**	**47420**	26.6	86.0	79.0	107.0	114.0	3.6	3.2	0.36	1.67	0.92	0.964	0.476
	127.000	36.512	36.170	28.575	3.6	3.2	196	226	35.3	3 000	4 000	**567A**	**563**	28.6	86.0	80.0	112.0	120.0	3.6	3.2	0.36	1.65	0.91	1.19	0.648
	127.000	36.512	36.512	26.988	3.6	1.6	209	235	36.2	3 000	4 000	**HM813849**	**HM813811**	32.9	89.0	81.9	113.0	121.0	3.6	1.6	0.50	1.20	0.66	1.28	0.622
	136.525	41.275	41.275	31.750	3.6	3.2	284	308	46.1	2 900	3 800	**H414249**	**H414210**	30.3	89.0	83.3	121.0	129.0	3.6	3.2	0.36	1.67	0.92	1.80	0.788
	136.525	46.038	46.038	36.512	3.6	3.2	290	369	49.6	2 800	3 700	**H715345**	**H715311**	37.0	93.0	87.0	118.0	132.0	3.6	3.2	0.47	1.27	0.70	2.15	0.950
73.025	112.712	25.400	25.400	19.050	3.6	3.2	122	155	23.3	3 200	4 300	**29685**	**29620**	26.2	86.0	80.0	101.0	109.0	3.6	3.2	0.49	1.23	0.68	0.602	0.270
	117.475	30.162	30.162	23.812	3.6	3.2	148	179	27.4	3 200	4 200	**33287**	**33462**	27.8	87.0	80.0	104.0	112.0	3.6	3.2	0.44	1.38	0.76	0.747	0.436
	127.000	36.512	36.170	28.575	3.6	3.2	196	226	35.3	3 000	4 000	**567**	**563**	28.6	88.0	81.0	112.0	120.0	3.6	3.2	0.36	1.65	0.91	1.14	0.648
	139.992	36.512	36.098	28.575	3.6	3.2	220	262	39.8	2 700	3 600	**576R**	**572**	31.0	90.0	83.0	125.0	133.0	3.6	3.2	0.40	1.49	0.82	1.74	0.779

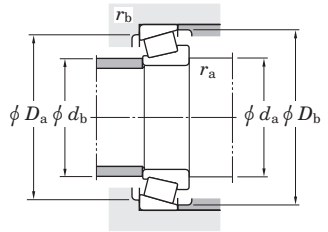
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (73.025) ~ 76.200 mm





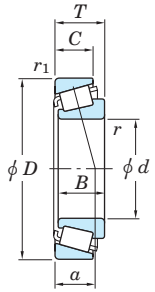
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
73.025	146.050	41.275	41.275	31.750	3.6	3.2	261	301	45.3	2 600	3 400	**657**	**653**	33.4	90.0	85.0	131.0	139.0	3.6	3.2	0.41	1.47	0.81	2.28	0.880
	149.225	53.975	54.229	44.450	3.6	3.2	357	404	54.4	2 700	3 500	**6460**	**6420**	39.3	93.0	87.0	129.0	141.0	3.6	3.2	0.36	1.66	0.91	2.79	1.61
	150.089	44.450	46.672	36.512	3.6	3.2	330	368	50.1	2 500	3 400	**744R**	**742**	32.4	91.0	85.0	134.0	142.0	3.6	3.2	0.33	1.84	1.01	2.66	1.07
	161.925	47.625	48.260	38.100	3.6	3.2	342	391	52.4	2 400	3 200	**762**	**752**	35.5	92.0	97.0	144.0	150.0	3.6	3.2	0.34	1.76	0.97	3.18	1.61
73.817	112.712	25.400	25.400	19.050	1.6	3.2	122	155	23.3	3 200	4 300	**29688**	**29620**	26.2	83.0	81.0	101.0	109.0	1.6	3.2	0.49	1.23	0.68	0.588	0.270
	127.000	36.512	36.170	28.575	0.8	3.2	196	226	35.3	3 000	4 000	**568**	**563**	28.6	83.0	82.0	112.0	120.0	0.8	3.2	0.36	1.65	0.91	1.12	0.648
74.612	139.992	36.512	36.098	28.575	3.6	3.2	220	262	39.8	2 700	3 600	**577R**	**572**	31.0	91.0	85.0	125.0	133.0	3.6	3.2	0.40	1.49	0.82	1.69	0.779
75.000	115.000	25.000	25.000	19.000	3.0	2.8	127	151	23.0	3 100	4 200	**JLM714149**	**JLM714110**	25.5	87.0	81.0	104.0	110.0	3.0	2.8	0.46	1.31	0.72	0.612	0.269
	120.000	31.000	29.500	25.000	3.0	2.8	182	216	33.2	3 100	4 100	**JM714249**	**JM714210**	30.0	88.0	82.9	108.0	115.0	3.0	2.8	0.44	1.35	0.74	0.846	0.430
	145.000	51.000	51.000	42.000	3.0	2.5	362	412	55.2	2 700	3 600	**JH415647**	**JH415610**	36.6	94.0	89.0	129.0	139.0	3.0	2.5	0.36	1.66	0.91	2.66	1.18
76.200	121.442	24.608	23.012	17.462	3.6	2.0	113	127	19.4	3 000	4 000	**34301**	**34478**	26.8	89.0	83.0	110.0	116.0	3.6	2.0	0.45	1.33	0.73	0.617	0.313
	127.000	30.162	31.000	22.225	3.6	3.2	179	225	32.3	2 400	3 200	**42687**	**42620**	27.1	90.0	84.0	114.0	121.0	3.6	3.2	0.42	1.43	0.79	1.05	0.434
	127.000	30.162	31.000	22.225	6.4	3.2	179	225	32.3	2 400	3 200	**42688**	**42620**	27.1	96.0	84.0	114.0	121.0	6.4	3.2	0.42	1.43	0.79	1.04	0.434
	133.350	30.162	29.769	22.225	6.4	3.2	167	198	30.0	2 700	3 600	**495AX**	**492A**	29.8	98.0	86.0	120.0	128.0	6.4	3.2	0.44	1.35	0.74	1.20	0.430
	133.350	33.338	33.338	26.195	6.4	3.2	193	245	37.2	2 700	3 700	**47678R**	**47620**	29.2	97.0	90.0	119.0	128.0	6.4	3.2	0.40	1.48	0.82	1.29	0.577
	133.350	33.338	33.338	26.195	0.8	3.2	193	245	37.2	2 700	3 700	**47680R**	**47620**	29.2	86.0	85.0	119.0	128.0	0.8	3.2	0.40	1.48	0.82	1.39	0.577
	135.733	44.450	46.101	34.925	3.6	3.2	267	337	51.0	2 800	3 700	**5760**	**5735**	33.0	94.0	88.0	119.0	130.0	3.6	3.2	0.41	1.48	0.81	1.85	0.877
	136.525	30.162	29.769	22.225	3.6	3.2	167	198	30.0	2 700	3 600	**495A**	**493**	29.8	92.0	86.0	122.0	130.0	3.6	3.2	0.44	1.35	0.74	1.26	0.544
	139.992	36.512	36.098	28.575	3.6	3.2	220	262	39.8	2 700	3 600	**575R**	**572**	31.0	92.0	86.0	125.0	133.0	3.6	3.2	0.40	1.49	0.82	1.64	0.779
	139.992	36.512	36.098	28.575	6.7	3.2	220	262	39.8	2 700	3 600	**575SR**	**572**	31.0	99.0	86.0	125.0	133.0	6.7	3.2	0.40	1.49	0.82	1.61	0.779
	149.225	53.975	54.229	44.450	3.6	3.2	357	404	54.4	2 700	3 500	**6461**	**6420**	39.3	96.0	89.5	129.0	141.0	3.6	3.2	0.36	1.66	0.91	2.64	1.61
	149.225	53.975	54.229	44.450	9.5	3.2	357	404	54.4	2 700	3 500	**6461A**	**6420**	39.3	105.0	90.0	129.0	141.0	9.5	3.2	0.36	1.66	0.91	2.60	1.61
	150.089	44.450	46.672	36.512	3.6	3.2	330	368	50.1	2 500	3 400	**748SR**	**742**	32.4	93.0	87.0	134.0	142.0	3.6	3.2	0.33	1.84	1.01	2.51	1.06
	152.400	41.275	41.275	31.750	3.6	3.2	261	301	45.3	2 600	3 400	**659**	**652**	33.4	93.0	87.0	134.0	141.0	3.6	3.2	0.41	1.47	0.81	2.16	1.25
	190.500	57.150	57.531	46.038	3.6	3.2	549	602	76.9	2 000	2 700	**HH221430**	**HH221410**	42.5	101.0	95.0	171.0	179.0	3.6	3.2	0.33	1.79	0.99	6.33	2.21

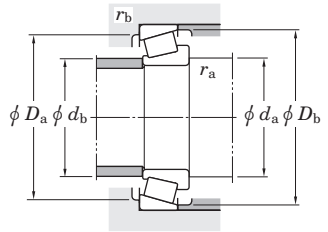
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d 77.788 ~ (83.345) mm





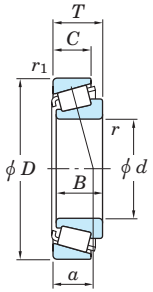
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors			(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
77.788	117.475	25.400	25.400	19.050	3.6	3.2	127	166	25.1	3 100	4 100	**LM814849**	**LM814810**	27.6	91.0	85.0	105.0	113.0	3.6	3.2	0.51	1.18	0.65	0.619	0.295
	121.442	24.608	23.012	17.462	3.6	2.0	113	127	19.4	3 000	4 000	**34306**	**34478**	26.8	90.0	84.0	110.0	116.0	3.6	2.0	0.45	1.33	0.73	0.583	0.313
	121.442	24.608	23.012	17.462	6.4	2.0	113	127	19.4	3 000	4 000	**34307**	**34478**	26.8	96.0	84.0	110.0	116.0	6.4	2.0	0.45	1.33	0.73	0.571	0.313
	127.000	30.162	31.000	22.225	3.6	3.2	179	225	32.3	2 400	3 200	**42690**	**42620**	27.1	91.0	85.0	114.0	121.0	3.6	3.2	0.42	1.43	0.79	1.00	0.434
79.375	146.050	41.275	41.275	31.750	3.6	3.2	261	301	45.3	2 600	3 400	**661**	**653**	33.4	96.0	90.0	131.0	139.0	3.6	3.2	0.41	1.47	0.81	2.04	0.880
	161.925	47.625	48.260	38.100	7.9	3.2	342	391	52.4	2 400	3 200	**756A**	**752**	35.5	106.0	91.0	144.0	150.0	7.9	3.2	0.34	1.76	0.97	2.95	1.59
	190.500	57.150	57.531	46.038	3.6	3.2	549	602	76.9	2 000	2 700	**HH221431**	**HH221410**	42.5	103.0	97.0	171.0	179.0	3.6	3.2	0.33	1.79	0.99	6.16	2.21
80.000	130.000	35.000	34.000	28.500	3.2	2.5	211	256	39.3	2 800	3 800	**JM515649**	**JM515610**	29.6	94.0	88.0	117.0	125.0	3.2	2.5	0.39	1.54	0.85	1.19	0.575
	200.000	52.761	49.212	34.925	3.6	3.2	433	471	58.8	1 400	1 900	**98316**	**98788**	54.5	111.0	105.0	174.0	188.0	3.6	3.2	0.63	0.95	0.52	5.73	2.28
80.962	133.350	30.162	29.769	22.225	3.6	3.2	167	198	30.0	2 700	3 600	**496**	**492A**	29.8	95.0	89.0	120.0	128.0	3.6	3.2	0.44	1.35	0.74	1.12	0.429
	133.350	33.338	33.338	26.195	3.6	3.2	193	245	37.2	2 700	3 700	**47681R**	**47620**	29.2	95.0	89.0	119.0	128.0	3.6	3.2	0.40	1.48	0.82	1.17	0.577
	139.992	36.512	36.098	28.575	3.6	3.2	220	262	39.8	2 700	3 600	**581R**	**572**	31.0	96.0	90.0	125.0	133.0	3.6	3.2	0.40	1.49	0.82	1.47	0.779
	150.089	44.450	46.672	36.512	5.2	3.2	330	368	50.1	2 500	3 400	**740R**	**742**	32.4	101.0	91.0	134.0	142.0	5.2	3.2	0.33	1.84	1.01	2.30	1.06
82.550	125.412	25.400	25.400	19.845	3.6	1.6	126	162	24.4	2 900	3 800	**27687**	**27620**	24.7	96.0	89.0	115.0	120.0	3.6	1.6	0.42	1.44	0.79	0.710	0.344
	133.350	30.162	29.769	22.225	3.6	3.2	167	198	30.0	2 700	3 600	**495**	**492A**	29.8	97.0	90.0	120.0	128.0	3.6	3.2	0.44	1.35	0.74	1.08	0.429
	133.350	33.338	33.338	26.195	3.6	0.8	193	245	37.2	2 700	3 700	**47686R**	**47620A**	29.2	97.0	90.0	121.0	128.0	3.6	0.8	0.40	1.48	0.82	1.13	0.577
	133.350	39.688	39.688	32.545	6.7	3.2	222	306	45.9	2 800	3 700	**HM516448**	**HM516410**	32.2	105.0	92.0	118.0	128.0	6.7	3.2	0.40	1.49	0.82	1.33	0.763
	139.700	36.512	36.098	28.575	3.6	3.2	220	262	39.8	2 700	3 600	**580R**	**572X**	31.0	98.0	91.0	125.0	133.0	3.6	3.2	0.40	1.49	0.82	1.41	0.765
	139.992	36.512	36.098	28.575	3.6	3.2	220	262	39.8	2 700	3 600	**580R**	**572**	31.0	98.0	91.0	125.0	133.0	3.6	3.2	0.40	1.49	0.82	1.41	0.779
	139.992	36.512	36.098	28.575	6.7	3.2	220	262	39.8	2 700	3 600	**582R**	**572**	31.0	104.0	91.0	125.0	133.0	6.7	3.2	0.40	1.49	0.82	1.40	0.779
	146.050	41.275	41.275	31.750	3.6	3.2	261	301	45.3	2 600	3 400	**663**	**653**	33.4	99.0	92.0	131.0	139.0	3.6	3.2	0.41	1.47	0.81	1.91	0.880
	150.089	44.450	46.672	36.512	3.6	3.2	330	368	50.1	2 500	3 400	**749AR**	**742**	32.4	99.0	93.0	134.0	142.0	3.6	3.2	0.33	1.84	1.01	2.23	1.06
	150.089	44.450	46.672	36.512	6.7	3.2	330	368	50.1	2 500	3 400	**750AR**	**742**	32.4	106.0	93.0	134.0	142.0	6.7	3.2	0.33	1.84	1.01	2.19	1.06
	161.925	47.625	48.260	38.100	3.6	3.2	342	391	52.4	2 400	3 200	**757**	**752**	35.5	100.0	94.0	144.0	150.0	3.6	3.2	0.34	1.76	0.97	2.83	1.59
83.345	125.412	25.400	25.400	19.845	0.8	1.6	126	162	24.4	2 900	3 800	**27689**	**27620**	24.7	90.0	90.0	115.0	120.0	0.8	1.6	0.42	1.44	0.79	0.746	0.344

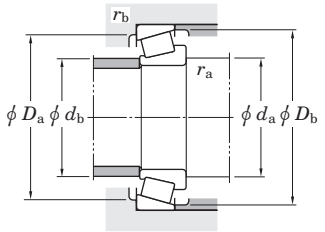
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (83.345) ~ (88.900) mm





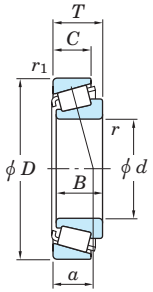
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors			(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
83.345	125.412	25.400	25.400	19.845	3.6	1.6	126	162	24.4	2 900	3 800	**27690**	**27620**	24.7	96.0	90.0	115.0	120.0	3.6	1.6	0.42	1.44	0.79	0.689	0.344
	125.412	25.400	25.400	19.845	6.4	1.6	126	162	24.4	2 900	3 800	**27691**	**27620**	24.7	102.0	90.0	115.0	120.0	6.4	1.6	0.42	1.44	0.79	0.646	0.344
84.138	133.350	30.162	29.769	22.225	3.6	3.2	167	198	30.0	2 700	3 600	**498**	**492A**	29.8	98.0	91.0	120.0	128.0	3.6	3.2	0.44	1.35	0.74	1.04	0.429
85.000	130.000	30.000	29.000	24.000	3.0	2.5	179	228	34.5	2 800	3 700	**JM716649**	**JM716610**	29.1	98.0	92.0	117.0	125.0	3.0	2.5	0.44	1.35	0.74	0.937	0.456
	140.000	39.000	38.000	31.500	3.0	2.5	254	308	46.4	2 700	3 500	**JHM516849**	**JHM516810**	32.8	100.0	93.9	125.0	134.0	3.0	2.5	0.41	1.47	0.81	1.54	0.759
	150.000	46.000	46.000	38.000	3.0	2.5	342	390	53.1	2 500	3 400	**JH217249**	**JH217210**	33.6	101.0	95.2	134.0	142.0	3.0	2.5	0.33	1.80	0.99	2.28	1.08
	200.000	52.761	49.212	34.925	3.6	3.2	433	471	58.8	1 400	1 900	**98335**	**98788**	54.5	115.0	109.0	174.0	188.0	3.6	3.2	0.63	0.95	0.52	5.47	2.28
85.026	150.089	44.450	46.672	36.512	3.6	3.2	330	368	50.1	2 500	3 400	**749R**	**742**	32.4	101.0	95.0	134.0	142.0	3.6	3.2	0.33	1.84	1.01	2.12	1.06
	150.089	44.450	46.672	36.512	5.2	3.2	330	368	50.1	2 500	3 400	**749SR**	**742**	32.4	104.0	95.0	134.0	142.0	5.2	3.2	0.33	1.84	1.01	2.08	1.06
85.725	133.350	30.162	29.769	22.225	3.6	3.2	167	198	30.0	2 700	3 600	**497**	**492A**	29.8	99.0	93.0	120.0	128.0	3.6	3.2	0.44	1.35	0.74	0.978	0.429
	136.525	30.162	29.769	22.225	6.4	3.2	167	198	30.0	2 700	3 600	**497A**	**493**	29.8	105.0	93.0	122.0	130.0	6.4	3.2	0.44	1.35	0.74	0.965	0.544
	142.138	42.862	42.862	34.133	4.8	3.2	276	351	52.4	2 600	3 500	**HM617049**	**HM617010**	35.2	106.0	95.7	125.0	137.0	4.8	3.2	0.43	1.39	0.76	1.72	0.902
	146.050	41.275	41.275	31.750	3.6	3.2	261	301	45.3	2 600	3 400	**665**	**653**	33.4	102.0	95.0	131.0	139.0	3.6	3.2	0.41	1.47	0.81	1.77	0.880
	146.050	41.275	41.275	31.750	6.4	3.2	261	301	45.3	2 600	3 400	**665A**	**653**	33.4	107.0	95.0	131.0	139.0	6.4	3.2	0.41	1.47	0.81	1.76	0.880
	152.400	39.688	36.322	30.162	3.6	3.2	230	287	42.5	2 400	3 300	**596**	**592A**	37.1	102.0	96.0	135.0	144.0	3.6	3.2	0.44	1.36	0.75	1.83	1.04
	161.925	47.625	48.260	38.100	3.6	3.2	342	391	52.4	2 400	3 200	**758**	**752**	35.5	103.0	97.0	144.0	150.0	3.6	3.2	0.34	1.76	0.97	2.67	1.59
	168.275	41.275	41.275	30.162	3.6	3.2	282	349	50.4	2 200	3 000	**677**	**672**	38.6	105.0	99.0	149.0	160.0	3.6	3.2	0.47	1.28	0.70	2.89	1.22
	168.275	53.975	56.363	41.275	3.6	3.2	429	467	62.1	2 300	3 100	**841R**	**832**	35.0	104.0	97.0	149.0	155.0	3.6	3.2	0.30	2.00	1.10	3.47	1.72
88.900	123.825	20.638	20.638	16.670	1.6	1.6	102	145	21.5	2 800	3 700	**L217849**	**L217810**	20.7	97.0	94.0	116.0	119.0	1.6	1.6	0.33	1.82	1.00	0.507	0.235
	152.400	39.688	39.688	30.162	6.4	3.2	311	359	53.5	2 400	3 200	**HM518445**	**HM518410**	33.1	110.0	98.0	135.0	146.0	6.4	3.2	0.40	1.49	0.82	2.10	0.768
	161.925	47.625	48.260	38.100	3.6	3.2	342	391	52.4	2 400	3 200	**759**	**752**	35.5	106.0	99.0	144.0	150.0	3.6	3.2	0.34	1.76	0.97	2.50	1.59
	161.925	47.625	48.260	38.100	7.1	3.2	342	391	52.4	2 400	3 200	**766**	**752**	35.5	113.0	99.0	144.0	150.0	7.1	3.2	0.34	1.76	0.97	2.48	1.59
	161.925	53.975	55.100	42.862	3.6	3.2	395	471	61.4	2 400	3 200	**6580R**	**6535**	49.8	109.0	98.0	141.0	154.0	3.6	3.2	0.40	1.50	0.82	3.09	1.65
	168.275	41.275	41.275	30.162	3.6	3.2	282	349	50.4	2 200	3 000	**679**	**672**	38.6	107.0	101.0	149.0	160.0	3.6	3.2	0.47	1.28	0.70	2.75	1.22
	190.500	57.150	57.531	44.450	7.9	3.2	482	565	72.4	2 100	2 700	**855R**	**854**	40.0	118.0	103.0	170.0	174.0	7.9	3.2	0.33	1.79	0.99	5.05	2.66

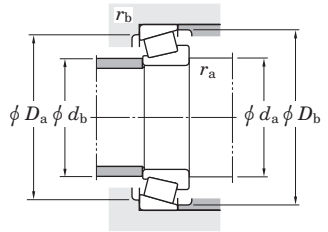
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (88.900) ~ 99.975 mm





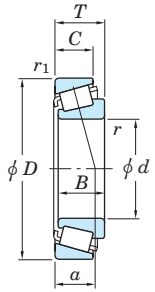
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors			(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
88.900	190.500	57.150	57.531	46.038	7.9	3.2	549	602	76.9	2 000	2 700	**HH221434**	**HH221410**	42.5	120.0	105.0	171.0	179.0	7.9	3.2	0.33	1.79	0.99	5.57	2.21
	200.000	52.761	49.212	34.925	3.6	3.2	433	471	58.8	1 400	1 900	**98350**	**98788**	54.5	118.0	112.0	174.0	188.0	3.6	3.2	0.63	0.95	0.52	5.27	2.28
89.974	146.975	40.000	40.000	32.500	7.1	3.6	259	310	46.6	2 500	3 300	**HM218248**	**HM218210**	30.8	112.0	99.0	133.0	141.0	7.1	3.6	0.33	1.80	0.99	1.66	0.784
90.000	145.000	35.000	34.000	27.000	3.0	2.5	244	291	43.5	2 500	3 400	**JM718149**	**JM718110**	32.7	105.0	99.0	131.0	139.0	3.0	2.5	0.44	1.35	0.74	1.47	0.652
	155.000	44.000	44.000	35.500	3.0	2.5	363	407	54.8	2 400	3 200	**JHM318448**	**JHM318410**	34.5	106.0	100.0	140.0	148.0	3.0	2.5	0.34	1.76	0.97	2.37	1.00
	161.925	53.975	55.100	42.862	3.0	3.2	395	471	61.4	2 400	3 200	**6581XR**	**6535**	41.0	102.0	98.0	141.0	154.0	3.0	3.2	0.40	1.50	0.82	3.02	1.65
90.488	161.925	47.625	48.260	38.100	3.6	3.2	342	391	52.4	2 400	3 200	**760**	**752**	35.5	107.0	101.0	144.0	150.0	3.6	3.2	0.34	1.76	0.97	2.42	1.59
92.075	146.050	33.338	34.925	26.195	3.6	3.2	223	293	43.2	2 500	3 300	**47890R**	**47820**	32.6	107.0	101.0	131.0	140.0	3.6	3.2	0.45	1.34	0.74	1.46	0.657
	168.275	41.275	41.275	30.162	3.6	3.2	282	349	50.4	2 200	3 000	**681**	**672**	38.6	110.0	104.0	149.0	160.0	3.6	3.2	0.47	1.28	0.70	2.61	1.22
	168.275	41.275	41.275	30.162	6.4	3.2	282	349	50.4	2 200	3 000	**681A**	**672**	38.6	116.0	104.0	149.0	160.0	6.4	3.2	0.47	1.28	0.70	2.60	1.22
	180.975	47.625	48.006	38.100	3.6	3.2	362	438	56.6	2 100	2 800	**778**	**772**	39.5	111.0	105.0	161.0	168.0	3.6	3.2	0.39	1.56	0.86	3.65	1.92
	190.500	57.150	57.531	44.450	7.9	3.2	482	565	72.4	2 100	2 700	**857R**	**854**	39.9	121.0	106.0	170.0	174.0	7.9	3.2	0.33	1.79	0.99	4.86	2.66
95.000	150.000	35.000	34.000	27.000	3.0	2.5	235	294	43.4	2 400	3 300	**JM719149**	**JM719113**	33.5	109.0	104.0	135.0	143.0	3.0	2.5	0.44	1.36	0.75	1.43	0.766
95.250	128.588	15.875	15.083	11.908	1.6	1.6	72.6	93.0	13.1	2 600	3 500	**LL319349**	**LL319310**	20.3	103.0	100.0	122.0	125.0	1.6	1.6	0.35	1.71	0.94	0.393	0.147
	130.175	20.638	21.432	16.670	1.6	1.6	121	167	24.7	2 600	3 500	**L319249**	**L319210**	22.2	107.0	101.0	122.0	125.0	1.6	1.6	0.35	1.72	0.95	0.548	0.246
	146.050	33.338	34.925	26.195	3.6	3.2	223	293	43.2	2 500	3 300	**47896R**	**47820**	32.6	110.0	103.0	131.0	140.0	3.6	3.2	0.45	1.34	0.74	1.34	0.657
	147.638	35.717	36.322	26.192	5.2	0.8	230	287	42.5	2 400	3 300	**594A**	**592XE**	33.4	113.0	104.0	135.0	142.0	5.2	0.8	0.44	1.36	0.75	1.45	0.620
	157.162	36.512	36.116	26.195	3.6	3.2	227	288	41.7	2 300	3 000	**52375**	**52618**	36.0	112.0	105.0	142.0	153.0	3.6	3.2	0.47	1.26	0.69	1.94	0.694
	168.275	41.275	41.275	30.162	3.6	3.2	282	349	50.4	2 200	3 000	**683**	**672**	38.6	113.0	106.0	149.0	160.0	3.6	3.2	0.47	1.28	0.70	2.46	1.22
	190.500	57.150	57.531	44.450	7.9	3.2	482	565	72.4	2 100	2 700	**864R**	**854**	39.9	123.0	108.0	170.0	174.0	7.9	3.2	0.33	1.79	0.99	4.64	2.66
	190.500	57.150	57.531	46.038	7.9	3.2	549	602	76.9	2 000	2 700	**HH221440**	**HH221410**	42.5	125.0	110.0	171.0	179.0	7.9	3.2	0.33	1.79	0.99	5.16	2.21
98.425	168.275	41.275	41.275	30.162	3.6	3.2	282	349	50.4	2 200	3 000	**685**	**672**	38.6	116.0	109.0	149.0	160.0	3.6	3.2	0.47	1.28	0.70	2.29	1.22
	190.500	57.150	57.531	46.038	3.6	3.2	549	602	76.9	2 000	2 700	**HH221442**	**HH221410**	42.5	119.0	113.0	171.0	179.0	3.6	3.2	0.33	1.79	0.99	4.97	2.21
99.975	212.725	66.675	66.675	53.975	3.6	3.2	641	699	87.1	1 800	2 400	**HH224334**	**HH224310**	47.6	122.0	117.0	192.0	202.0	3.6	3.2	0.33	1.84	1.01	7.91	3.03

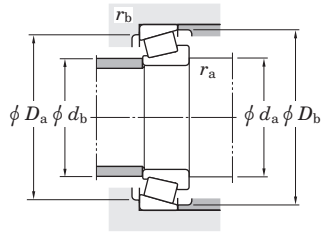
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d 99.982 ~ (107.950) mm





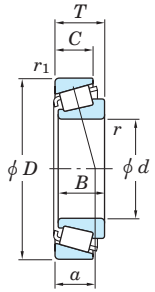
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
99.982	190.500	57.150	57.531	46.038	6.4	3.2	549	602	76.9	2 000	2 700	**HH221447**	**HH221410**	42.5	126.0	114.0	171.0	179.0	6.4	3.2	0.33	1.79	0.99	4.84	2.21
100.000	155.000	36.000	35.000	28.000	3.0	2.5	256	328	47.7	2 300	3 100	**JM720249**	**JM720210**	35.6	110.0	110.0	139.0	148.0	3.0	2.5	0.47	1.27	0.70	1.64	0.763
	160.000	41.000	40.000	32.000	3.0	2.5	298	378	54.6	2 300	3 000	**JHM720249**	**JHM720210**	38.3	110.0	111.0	143.0	153.0	3.0	2.5	0.47	1.28	0.70	2.11	0.964
100.012	157.162	36.512	36.116	26.195	3.6	3.2	227	288	41.7	2 300	3 000	**52393**	**52618**	36.0	113.0	115.0	142.0	153.0	3.6	3.2	0.47	1.26	0.69	1.74	0.694
101.600	157.162	36.512	36.116	26.195	3.6	3.2	227	288	41.7	2 300	3 000	**52400**	**52618**	36.0	114.0	115.0	142.0	153.0	3.6	3.2	0.47	1.26	0.69	1.67	0.694
	157.162	36.512	36.116	26.195	7.9	3.2	227	288	41.7	2 300	3 000	**52401**	**52618**	36.0	126.0	111.0	142.0	153.0	7.9	3.2	0.47	1.26	0.69	1.64	0.694
	168.275	41.275	41.275	30.162	3.6	3.2	282	349	50.4	2 200	3 000	**687**	**672**	38.6	114.0	115.0	146.0	157.0	3.6	3.2	0.47	1.28	0.70	2.15	1.22
	180.975	47.625	48.006	38.100	3.6	3.2	362	438	56.6	2 100	2 800	**780**	**772**	39.5	114.0	120.0	156.0	165.0	3.6	3.2	0.39	1.56	0.86	3.09	1.92
	190.500	57.150	57.531	44.450	7.9	3.2	482	565	72.4	2 100	2 700	**861R**	**854**	39.9	129.0	114.0	170.0	174.0	7.9	3.2	0.33	1.79	0.99	4.20	2.66
	190.500	57.150	57.531	46.038	7.9	3.2	549	602	76.9	2 000	2 700	**HH221449**	**HH221410**	42.5	123.0	119.0	168.0	178.0	7.9	3.2	0.33	1.79	0.99	4.72	2.21
	200.000	52.761	49.212	34.925	3.6	3.2	433	471	58.8	1 400	1 900	**98400**	**98788**	54.5	114.0	123.0	170.0	185.0	3.6	3.2	0.63	0.95	0.52	4.55	2.28
	212.725	66.675	66.675	53.975	7.1	3.2	563	674	84.1	1 800	2 400	**941**	**932**	47.6	121.0	135.0	181.0	192.0	7.1	3.2	0.33	1.84	1.01	7.07	4.07
	212.725	66.675	66.675	53.975	7.1	3.2	641	699	87.1	1 800	2 400	**HH224335**	**HH224310**	47.6	121.0	134.0	189.0	201.0	7.1	3.2	0.33	1.84	1.01	7.76	3.03
104.775	180.975	47.625	48.006	38.100	3.6	3.2	362	438	56.6	2 100	2 800	**782**	**772**	39.5	117.0	120.0	156.0	165.0	3.6	3.2	0.39	1.56	0.86	2.90	1.92
	180.975	47.625	48.006	38.100	6.4	3.2	362	438	56.6	2 100	2 800	**786**	**772**	39.5	123.0	120.0	156.0	165.0	6.4	3.2	0.39	1.56	0.86	2.88	1.92
	180.975	47.625	48.006	38.100	7.1	3.2	362	438	56.6	2 100	2 800	**787**	**772**	39.5	129.0	116.0	161.0	168.0	7.1	3.2	0.39	1.56	0.86	2.87	1.92
	190.500	47.625	49.212	34.925	3.6	3.2	381	483	60.9	1 900	2 600	**71412**	**71750**	40.9	117.0	131.0	167.0	177.0	3.6	3.2	0.42	1.44	0.79	3.96	1.72
106.362	165.100	36.512	36.512	26.988	3.6	3.2	245	325	46.3	2 200	2 900	**56418R**	**56650**	38.6	122.0	116.0	149.0	159.0	3.6	3.2	0.50	1.21	0.66	1.84	0.852
107.950	146.050	21.432	21.432	16.670	1.6	1.6	108	167	23.5	2 300	3 100	**L521949R**	**L521910**	26.2	116.0	114.0	136.0	141.0	1.6	1.6	0.39	1.53	0.84	0.665	0.325
	158.750	23.020	21.438	15.875	3.6	3.2	130	169	23.9	2 200	3 000	**37425**	**37625**	36.5	121.0	121.0	141.0	148.0	3.6	3.2	0.61	0.99	0.54	0.893	0.484
	159.987	34.925	34.925	26.988	3.6	3.2	231	319	45.8	2 200	2 900	**LM522546**	**LM522510**	32.9	122.0	116.0	146.0	154.0	3.6	3.2	0.40	1.50	0.82	1.64	0.784
	161.925	34.925	34.925	26.988	3.6	3.2	216	293	41.8	2 200	2 900	**48190**	**48120**	39.1	121.0	120.0	145.0	154.0	3.6	3.2	0.51	1.19	0.65	1.57	0.820
	165.100	36.512	36.512	26.988	3.6	3.2	245	325	46.3	2 200	2 900	**56425R**	**56650**	38.6	123.0	117.0	149.0	159.0	3.6	3.2	0.50	1.21	0.66	1.76	0.852
	168.275	36.512	36.512	26.988	3.6	3.2	245	325	46.3	2 200	2 900	**56425R**	**56662**	38.6	123.0	117.0	150.0	160.0	3.6	3.2	0.50	1.21	0.66	1.76	1.03
	190.500	47.625	49.212	34.925	3.6	3.2	381	483	60.9	1 900	2 600	**71425**	**71750**	40.9	121.0	131.0	167.0	177.0	3.6	3.2	0.42	1.44	0.79	3.76	1.72

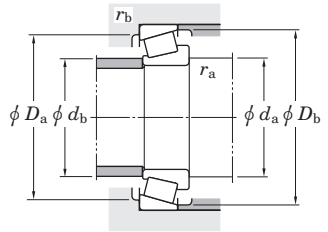
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d (107.950) ~ 127.000 mm





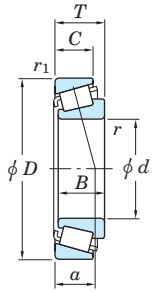
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors			(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
107.950	212.725	66.675	66.675	53.975	7.9	3.2	563	674	84.1	1 800	2 400	**936**	**932**	47.6	137.0	122.0	187.0	193.0	7.9	3.2	0.33	1.84	1.01	6.52	4.07
	212.725	66.675	66.675	53.975	7.9	3.2	641	699	87.1	1 800	2 400	**HH224340**	**HH224310**	47.6	129.0	134.0	189.0	201.0	7.9	3.2	0.33	1.84	1.01	7.21	3.03
109.538	158.750	23.020	21.438	15.875	3.6	3.2	130	169	23.9	2 200	3 000	**37431**	**37625**	36.5	123.0	116.0	143.0	152.0	6.4	6.4	0.61	0.99	0.54	0.848	0.484
109.987	159.987	34.925	34.925	26.988	7.9	3.2	231	319	45.8	2 200	2 900	**LM522548**	**LM522510**	32.9	131.0	121.0	146.0	154.0	7.9	3.2	0.40	1.50	0.82	1.52	0.784
	159.987	34.925	34.925	26.988	3.6	3.2	231	319	45.8	2 200	2 900	**LM522549**	**LM522510**	32.9	123.0	121.0	146.0	154.0	3.6	3.2	0.40	1.50	0.82	1.55	0.784
109.992	177.800	41.275	41.275	30.162	3.6	3.2	294	380	53.4	2 000	2 700	**64433R**	**64700**	42.8	128.0	121.0	160.0	172.6	3.6	3.2	0.52	1.16	0.64	2.69	1.10
110.000	165.000	35.000	35.000	26.500	3.0	2.5	245	325	46.3	2 200	2 900	**JM822049**	**JM822010**	38.1	121.0	121.0	148.0	157.0	3.0	2.5	0.50	1.21	0.66	1.64	0.826
	180.000	47.000	46.000	38.000	3.0	2.5	385	487	62.3	2 000	2 700	**JHM522649**	**JHM522610**	40.6	121.0	125.0	160.0	171.0	3.0	2.5	0.41	1.48	0.81	3.08	1.49
114.300	177.800	41.275	41.275	30.162	3.6	3.2	294	380	53.4	2 000	2 700	**64450R**	**64700**	42.8	131.0	125.0	160.0	172.0	3.6	3.2	0.52	1.16	0.64	2.45	1.10
	180.975	34.925	31.750	25.400	3.6	3.2	216	247	35.1	2 000	2 700	**68450**	**68712**	40.6	127.0	131.0	161.0	170.0	3.6	3.2	0.50	1.21	0.66	1.89	1.04
	190.500	47.625	49.212	34.925	3.6	3.2	381	483	60.9	1 900	2 600	**71450**	**71750**	40.9	127.0	131.0	167.0	177.0	3.6	3.2	0.42	1.44	0.79	3.33	1.72
	212.725	66.675	66.675	53.975	7.1	3.2	563	674	84.1	1 800	2 400	**938**	**932**	47.6	141.0	128.0	187.0	193.0	7.1	3.2	0.33	1.84	1.01	5.96	4.07
	212.725	66.675	66.675	53.975	7.1	3.2	641	699	87.1	1 800	2 400	**HH224346**	**HH224310**	47.6	134.0	134.0	189.0	201.0	7.1	3.2	0.33	1.84	1.01	6.64	3.03
	273.050	82.550	82.550	53.975	6.4	6.4	885	898	104	1 500	1 900	**HH926744**	**HH926710**	76.1	133.0	151.0	230.0	252.0	6.4	6.4	0.63	0.95	0.52	15.0	6.97
114.976	212.725	66.675	66.675	53.975	7.1	3.2	641	699	87.1	1 800	2 400	**HH224349**	**HH224310**	47.6	135.0	134.0	189.0	201.0	7.1	3.2	0.33	1.84	1.01	6.58	3.03
115.087	190.500	47.625	49.212	34.925	3.6	3.2	381	483	60.9	1 900	2 600	**71453**	**71750**	40.9	133.0	126.0	171.0	181.0	3.6	3.2	0.42	1.44	0.79	3.28	1.72
	190.500	47.625	49.212	34.925	7.9	3.2	381	483	60.9	1 900	2 600	**71455**	**71750**	40.9	136.0	131.0	167.0	177.0	7.9	3.2	0.42	1.44	0.79	3.25	1.72
117.475	180.975	34.925	31.750	25.400	3.6	3.2	216	247	35.1	2 000	2 700	**68462**	**68712**	40.6	130.0	131.0	161.0	170.0	3.6	3.2	0.50	1.21	0.66	1.75	1.04
	180.975	34.925	31.750	25.400	7.9	3.2	216	247	35.1	2 000	2 700	**68463**	**68712**	40.6	141.0	125.0	163.0	172.0	7.9	3.2	0.50	1.21	0.66	1.61	1.05
120.650	190.500	46.038	46.038	34.925	3.6	1.6	393	512	63.9	1 900	2 500	**HM624749**	**HM624710**	41.6	146.0	132.0	174.0	184.0	3.6	1.6	0.43	1.41	0.77	3.20	1.44
	254.000	77.788	82.550	61.912	9.5	6.4	895	1 050	125	1 500	2 000	**HH228340**	**HH228310**	54.3	158.0	142.0	223.0	234.0	9.5	6.4	0.32	1.87	1.03	12.6	6.00
127.000	254.000	77.788	82.550	61.912	9.5	6.4	895	1 050	125	1 500	2 000	**HH228349**	**HH228310**	54.3	164.0	148.0	223.0	234.0	9.5	6.4	0.32	1.87	1.03	11.8	6.00

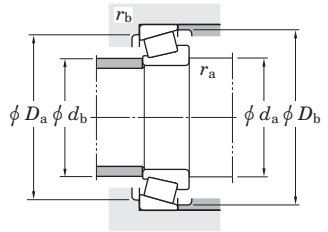
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

inch series

d 133.350 ~ 292.100 mm





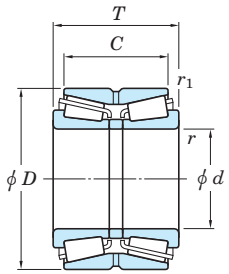
Boundary dimensions (mm)							Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No. [1]		Load center (mm)	Mounting dimensions (mm)						Constant	Axial load factors		(Refer.) Mass (kg)	
d	D	T	B	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Inner ring	Outer ring	a	d_a	d_b	D_a	D_b	r_a max.	r_b max.	e	Y_1	Y_0	Inner ring	Outer ring
133.350	177.008	25.400	26.195	20.638	1.6	1.6	176	278	38.2	1 900	2 500	**L327249**	**L327210**	29.1	142.0	145.0	164.0	171.0	1.6	1.6	0.35	1.72	0.95	1.14	0.543
142.875	200.025	41.275	39.688	34.130	7.9	3.3	307	491	66.5	1 700	2 200	**48684**	**48620**	38.4	166.0	151.0	185.0	193.0	7.9	3.3	0.34	1.78	0.98	2.43	1.38
	200.025	41.275	39.688	34.130	3.6	3.3	307	491	66.5	1 700	2 200	**48685**	**48620**	38.4	156.0	157.0	182.0	192.0	3.6	3.3	0.34	1.78	0.98	2.46	1.38
170.000	230.000	39.000	38.000	31.000	3.0	2.5	363	558	72.8	1 400	1 900	**JHM534149**	**JHM534110**	43.6	181.0	184.0	214.0	222.0	3.0	2.5	0.38	1.57	0.86	3.17	1.29
	240.000	46.000	44.500	37.000	3.0	2.5	443	666	77.1	1 400	1 800	**JM734449**	**JM734410**	50.6	181.0	184.0	220.0	231.0	3.0	2.5	0.44	1.37	0.75	4.31	2.00
171.450	222.250	25.400	24.608	19.050	1.6	1.6	197	299	38.7	1 400	1 900	**L435049**	**L435010**	36.0	181.0	179.0	211.0	215.0	1.6	1.6	0.38	1.60	0.88	1.63	0.697
180.000	250.000	47.000	45.000	37.000	3.0	2.5	456	705	81.7	1 300	1 700	**JM736149**	**JM736110**	55.2	191.0	193.0	230.0	242.0	3.0	2.5	0.48	1.25	0.69	4.47	2.10
190.000	260.000	46.000	44.000	36.500	3.0	2.5	461	723	81.4	1 200	1 700	**JM738249**	**JM738210**	56.0	201.0	203.0	240.0	251.0	3.0	2.5	0.48	1.26	0.69	4.71	2.18
196.850	254.000	28.575	27.783	21.433	1.6	1.6	236	387	48.2	1 200	1 600	**L540049**	**L540010**	43.1	206.0	214.0	238.0	245.0	1.6	1.6	0.40	1.51	0.83	2.34	1.02
200.000	300.000	65.000	62.000	51.000	3.6	2.5	773	1 140	124	1 100	1 500	**JHM840449**	**JHM840410**	72.1	213.0	218.0	270.0	288.0	3.6	2.5	0.52	1.15	0.63	9.97	5.13
220.878	317.500	47.625	52.388	36.513	3.2	3.2	611	928	103	970	1 300	**LM245833**	**LM245810**	50.5	234.0	253.0	296.0	304.0	3.2	3.2	0.33	1.80	0.99	9.56	2.78
228.600	358.775	71.438	71.438	53.975	3.6	3.2	968	1 590	166	840	1 100	**M249732**	**M249710**	64.4	242.0	279.0	330.0	343.0	3.6	3.2	0.33	1.80	0.99	20.1	6.44
230.188	317.500	47.625	52.388	36.513	3.2	3.2	611	928	103	970	1 300	**LM245846**	**LM245810**	50.5	242.0	238.0	309.0	312.0	3.2	3.2	0.33	1.80	0.99	8.25	2.78
231.775	317.500	47.625	52.388	36.513	3.2	3.2	611	928	103	970	1 300	**LM245848**	**LM245810**	50.5	244.0	240.0	309.0	312.0	3.2	3.2	0.33	1.80	0.99	8.02	2.78
	336.550	65.088	65.088	50.800	6.4	3.2	887	1 380	150	920	1 200	**M246942**	**M246910**	59.9	258.0	249.0	313.0	322.0	6.4	3.2	0.33	1.80	0.99	13.1	5.44
	358.775	71.438	71.438	53.975	6.4	3.2	968	1 590	166	920	1 200	**M249734**	**M249710**	64.4	258.0	253.0	335.0	343.0	6.4	3.2	0.33	1.80	0.99	19.9	6.44
254.000	358.775	71.438	71.438	53.975	3.6	3.2	968	1 590	166	840	1 100	**M249749**	**M249710**	64.4	268.0	279.0	330.0	343.0	3.6	3.2	0.33	1.80	0.99	14.8	6.44
257.175	342.900	57.150	57.150	44.450	6.4	3.2	764	1 280	135	870	1 200	**M349549**	**M349510**	60.1	276.0	276.0	320.0	330.0	6.4	3.2	0.35	1.73	0.95	9.27	3.99
292.100	374.650	47.625	47.625	34.925	3.6	3.2	587	971	111	760	1 000	**L555249**	**L555210**	64.7	306.0	309.0	351.0	360.0	3.6	3.2	0.40	1.49	0.82	7.97	3.53

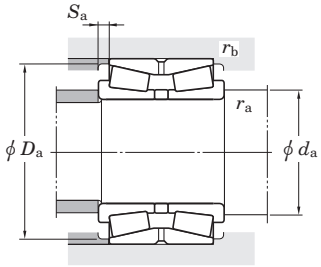
[Note] 1) To the bearings with supplementary code "J" attached at the front of bearing number, tolerances shown in table 7-8 on page A72 are applied.

[Remark] Inch series tapered roller bearings with bore diameter larger than 100 mm are shown in catalog "large size ball & roller bearings".

TDO type

d 25 ~ (60) mm



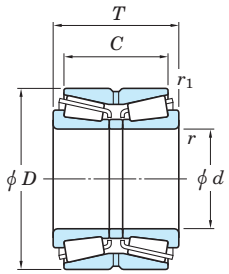


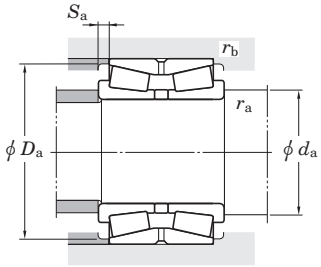
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass
d	D	T	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.		d_a min.	D_a min.	S_a min.	r_a max.	r_b max.	e	Y_2	Y_3	Y_0	(kg)
25	62	40	29.5	1.5	0.6	85.2	84.9	11.6	4 500	6 400	**46T30305DJR/29.5**	33.5	58.5	5	1.5	0.6	0.83	0.82	1.22	0.8	0.592
30	72	45	31.5	1.5	0.6	109	110	15.4	3 900	5 400	**46T30306DJR/31.5**	38.5	68	6.5	1.5	0.6	0.83	0.82	1.22	0.8	0.872
35	80	51	35.5	2	0.6	135	138	19.7	3 400	4 800	**46T30307DJR/35.5**	45	76.5	7.5	2	0.6	0.83	0.82	1.22	0.8	1.2
40	80	45	37.5	1.5	0.6	134	138	20.6	4 000	5 300	**46T30208JR/37.5**	48.5	75	3.5	1.5	0.6	0.37	1.8	2.68	1.76	0.954
	80	55	43.5	1.5	0.6	166	182	27.3	4 000	5 300	**46T32208JR/43.5**	48.5	75	5.5	1.5	0.6	0.37	1.8	2.68	1.76	1.19
	90	56	39.5	2	0.6	172	180	26.2	3 000	4 200	**46T30308DJR/39.5**	50	86.5	8	2	0.6	0.83	0.82	1.22	0.8	1.67
	90	56	45.5	2	0.6	194	202	31.0	3 600	4 900	**46T30308JR/45.5**	50	82	5	2	0.6	0.35	1.96	2.91	1.91	1.67
45	85	47	37.5	1.5	0.6	144	155	23.1	3 700	4 900	**46T30209JR/37.5**	53.5	80	4.5	1.5	0.6	0.4	1.67	2.48	1.63	1.1
	85	55	43.5	1.5	0.6	180	207	31.2	3 700	4 900	**46T32209JR-1/43.5**	53.5	81	5.5	1.5	0.6	0.4	1.67	2.48	1.63	1.31
	100	60	41.5	2	0.6	204	214	31.7	2 700	3 800	**46T30309DJR/41.5**	55	96	9	2	0.6	0.83	0.82	1.22	0.8	2.15
	100	60	49.5	2	0.6	242	256	39.8	3 300	4 300	**46T30309JR/49.5**	55	93	5	2	0.6	0.35	1.96	2.91	1.91	2.2
50	90	49	39.5	1.5	0.6	164	183	27.6	3 400	4 600	**46T30210JR/39.5**	58.5	85	4.5	1.5	0.6	0.42	1.61	2.39	1.57	1.22
	90	55	43.5	1.5	0.6	182	211	31.8	3 500	4 600	**46T32210JR/43.5**	58.5	85	5.5	1.5	0.6	0.42	1.61	2.39	1.57	1.39
	110	64	51.5	2	0.6	295	305	47.9	3 000	4 000	**46T30310JR/51.5**	62	102	6	2	0.6	0.35	1.96	2.91	1.91	2.68
	110	73	52.5	2	0.6	247	266	39.5	2 500	3 500	**46T30310DJR/52.5**	62	105	10	2	0.6	0.83	0.82	1.22	0.8	3.11
	110	90	71.5	2	0.6	378	440	68.4	3 000	4 000	**46T32310JR/71.5**	62	102	9	2	0.6	0.35	1.96	2.91	1.91	3.95
55	100	51	41.5	2	0.6	203	226	34.6	3 100	4 100	**46T30211JR/41.5**	65	94	4.5	2	0.6	0.4	1.67	2.48	1.63	1.6
	100	60	48.5	2	0.6	230	266	41.0	3 100	4 100	**46T32211JR-1/48.5**	65	95	5.5	2	0.6	0.4	1.67	2.48	1.63	1.87
	120	70	49	2	0.6	276	297	44.6	2 300	3 200	**46T30311DJR/49**	67	113	10.5	2	0.6	0.83	0.82	1.22	0.8	3.54
	120	70	57	2	0.6	320	341	53.9	2 700	3 600	**46T30311JR/57**	67	111	6.5	2	0.6	0.35	1.96	2.91	1.91	3.57
	120	97	76	2	0.6	429	500	78.2	2 700	3 600	**46T32311JR/76**	67	111	10.5	2	0.6	0.35	1.96	2.91	1.91	4.98
60	110	53	43.5	2	0.6	228	254	39.4	2 800	3 800	**46T30212JR/43.5**	70	103	4.5	2	0.6	0.4	1.67	2.48	1.63	2.04
	110	66	54.5	2	0.6	282	334	51.8	2 800	3 800	**46T32212JR/54.5**	70	104	5.5	2	0.6	0.4	1.67	2.48	1.63	—
	130	74	51	2.5	1	327	359	54.2	2 100	2 900	**46T30312DJR/51**	74	124	11.5	2.5	1	0.83	0.82	1.22	0.8	4.45
	130	74	59	2.5	1	372	401	63.8	2 500	3 300	**46T30312JR/59**	74	120	7.5	2.5	1	0.35	1.96	2.91	1.91	4.46

[Remark] Bearings not shown above (e.g. inch series) are shown in catalog "large size ball & roller bearings".

TDO type

d (60) ~ (90) mm



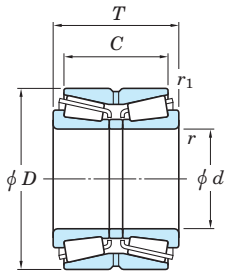


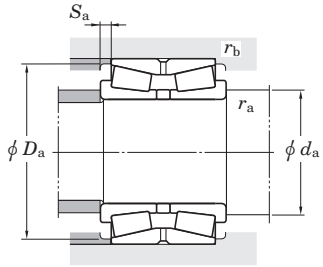
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass
d	D	T	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.		d_a min.	D_a min.	S_a min.	r_a max.	r_b max.	e	Y_2	Y_3	Y_0	(kg)
60	130	104	81	2.5	1	524	629	88.3	2 500	3 300	**46T32312JR/81**	74	120	11.5	2.5	1	0.35	1.96	2.91	1.91	6.45
65	120	56	46.5	2	0.6	275	311	48.7	2 600	3 400	**46T30213JR/46.5**	75	113	4.5	2	0.6	0.4	1.67	2.48	1.63	—
	120	73	61.5	2	0.6	337	406	63.3	2 600	3 400	**46T32213JR/61.5**	75	115	5.5	2	0.6	0.4	1.67	2.48	1.63	3.4
	140	79	53	2.5	1	377	417	62.8	1 900	2 700	**46T30313DJR/53**	79	133	13	2.5	1	0.83	0.82	1.22	0.8	5.3
	140	79	63	2.5	1	437	478	75.3	2 300	3 000	**46T30313JR/63**	79	130	8	2.5	1	0.35	1.96	2.91	1.91	5.51
	140	108	84	2.5	1	593	714	99.2	2 300	3 100	**46T32313JR/84**	79	130	12	2.5	1	0.35	1.96	2.91	1.91	7.71
70	125	59	48.5	2	0.6	296	346	54.2	2 400	3 300	**46T30214JR/48.5**	80	118	5	2	0.6	0.42	1.61	2.39	1.57	—
	125	74	61.5	2	0.6	363	450	70.4	2 400	3 300	**46T32214JR/61.5**	80	119	6	2	0.6	0.42	1.61	2.39	1.57	3.7
	150	83	57	2.5	1	421	470	69.8	1 800	2 500	**46T30314DJR/57**	84	142	13	2.5	1	0.83	0.82	1.22	0.8	6.48
	150	83	67	2.5	1	493	546	84.4	2 100	2 800	**46T30314JR/67**	84	140	8	2.5	1	0.35	1.96	2.91	1.91	6.65
	150	116	92	2.5	1	679	829	114	2 200	2 900	**46T32314JR/92**	84	140	12	2.5	1	0.35	1.96	2.91	1.91	9.46
75	115	30	26	1.5	0.6	89.9	105	14.6	2 500	3 300	**46215**	83.5	106.5	2	1.5	0.6	0.32	2.12	3.15	2.07	0.994
	115	38	30	1.5	0.6	153	207	31.2	2 500	3 300	**46215A**	83.5	107.4	4	1.5	0.6	0.32	2.12	3.15	2.07	1.32
	130	62	51.5	2	0.6	305	362	56.4	2 300	3 100	**46T30215JR/51.5**	85	124	5	2	0.6	0.44	1.55	2.31	1.52	3.12
	130	74	61.5	2	0.6	373	469	72.7	2 300	3 100	**46T32215JR/61.5**	85	125	6	2	0.6	0.44	1.55	2.31	1.52	3.85
	160	87	69	2.5	1	557	621	89.8	2 000	2 600	**46T30315JR/69**	89	149	9	2.5	1	0.35	1.96	2.91	1.91	7.8
	160	125	99	2.5	1	779	963	129	2 000	2 700	**46T32315JR/99**	89	149	13	2.5	1	0.35	1.96	2.91	1.91	11.5
80	125	34	30	1.5	0.6	136	155	22.6	2 300	3 100	**46216**	88.5	116.9	2	1.5	0.6	0.35	1.95	2.90	1.91	1.38
	140	64	51.5	2	0.6	346	405	62.3	2 200	2 900	**46T30216JR/51.5**	92	132	6	2	0.6	0.42	1.61	2.39	1.57	3.76
	140	78	63.5	2	0.6	434	542	83.1	2 200	2 900	**46T32216JR/63.5**	92	134	7	2	0.6	0.42	1.61	2.39	1.57	4.71
	170	92	73	2.5	1	630	711	99.8	1 800	2 500	**46T30316JR/73**	94	159	9.5	2.5	1	0.35	1.96	2.91	1.91	9.44
85	150	70	57	2	0.6	391	463	70.3	2 000	2 700	**46T30217JR/57**	97	141	6.5	2	0.6	0.42	1.61	2.39	1.57	4.79
	150	86	69	2	0.6	498	630	95.1	2 000	2 700	**46T32217JR/69**	97	142	8.5	2	0.6	0.42	1.61	2.39	1.57	6.05
	180	98	77	3	1	679	768	106	1 700	2 300	**46T30317JR/77**	103	167	10.5	3	1	0.35	1.96	2.91	1.91	11
	180	137	108	3	1	941	1 170	155	1 800	2 400	**46T32317JR/108**	103	167	14.5	3	1	0.35	1.96	2.91	1.91	16
90	140	37	33	2	0.6	171	199	28.8	2 100	2 800	**46218**	100	130.6	2	2	0.6	0.35	1.95	2.90	1.91	1.89

[Remark] Bearings not shown above (e.g. inch series) are shown in catalog "large size ball & roller bearings".

TDO type

d (90) ~ 110 mm



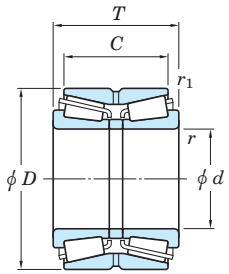


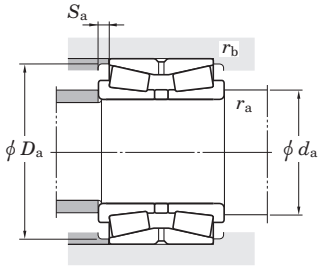
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass
d	D	T	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.		d_a min.	D_a min.	S_a min.	r_a max.	r_b max.	e	Y_2	Y_3	Y_0	(kg)
90	140	46	37	2	0.6	196	266	39.3	2 000	2 700	**46218A**	100	129.9	4.5	2	0.6	0.32	2.12	3.15	2.07	2.37
	160	74	61	2	0.6	438	522	78.1	1 900	2 500	**46T30218JR/61**	102	150	6.5	2	0.6	0.42	1.61	2.39	1.57	5.85
	160	94	77	2	0.6	565	724	107	1 900	2 500	**46T32218JR/77**	102	152	8.5	2	0.6	0.42	1.61	2.39	1.57	7.53
	190	102	81	3	1	741	841	114	1 600	2 200	**46T30318JR/81**	108	177	10.5	3	1	0.35	1.96	2.91	1.91	13
	190	144	115	3	1	989	1 230	157	1 700	2 200	**46T32318JR/115**	108	177	14.5	3	1	0.35	1.96	2.91	1.91	18.6
95	170	78	63	2.5	1	496	598	88.1	1 800	2 400	**46T30219JR/63**	109	159	7.5	2.5	1	0.42	1.61	2.39	1.57	7.01
	170	100	83	2.5	1	667	877	128	1 800	2 400	**46T32219JR/83**	109	161	8.5	2.5	1	0.42	1.61	2.39	1.57	9.25
	200	108	85	3	1	798	909	122	1 600	2 100	**46T30319JR/85**	113	186	11.5	3	1	0.35	1.96	2.91	1.91	14.8
	200	151	118	3	1	1 110	1 390	178	1 600	2 100	**46T32319JR/118**	113	186	16.5	3	1	0.35	1.96	2.91	1.91	21.4
100	150	46	37	2	0.6	226	293	42.6	1 900	2 500	**46220A**	110	142	4.5	2	0.6	0.35	1.95	2.90	1.91	2.53
	165	52	46	2.5	0.6	249	305	44.1	1 700	2 300	**46320**	112	154	3	2	0.6	0.35	1.95	2.90	1.91	4.03
	165	65	52	2.5	0.6	333	443	64.7	1 800	2 300	**46320A**	112	153	6.5	2	0.6	0.35	1.95	2.90	1.91	4.97
	180	83	67	2.5	1	554	676	98.2	1 700	2 200	**46T30220JR/67**	114	168	8	2.5	1	0.42	1.61	2.39	1.57	8.33
	180	107	87	2.5	1	745	990	128	1 700	2 200	**46T32220JR/87**	114	171	10	2.5	1	0.42	1.61	2.39	1.57	11.1
	215	112	87	3	1	906	1 040	136	1 500	1 900	**46T30320JR/87**	118	200	12.5	3	1	0.35	1.96	2.91	1.91	18.1
	215	162	127	3	1	1 240	1 570	194	1 500	2 000	**46T32320JR/127**	118	200	17.5	3	1	0.35	1.96	2.91	1.91	27.2
105	190	88	70	2.5	1	618	761	105	1 600	2 100	**46T30221JR/70**	119	178	9	2.5	1	0.42	1.61	2.39	1.57	9.87
	190	115	95	2.5	1	840	1 130	146	1 600	2 100	**46T32221JR/95**	119	180	10	2.5	1	0.42	1.61	2.39	1.57	13.5
	225	116	91	3	1	995	1 160	147	1 400	1 800	**46T30321JR/91**	123	209	12.5	3	1	0.35	1.96	2.91	1.91	20.7
	225	170	133	3	1	1 360	1 730	214	1 400	1 900	**46T32321JR/133**	123	209	18.5	3	1	0.35	1.96	2.91	1.91	30.9
110	170	45	40	2.5	0.6	219	304	42.5	1 700	2 200	**46222**	122	158	2.5	2	0.6	0.35	1.95	2.90	1.91	3.58
	180	56	50	2.5	0.6	308	388	55.3	1 600	2 100	**46322**	122	168	3	2	0.6	0.35	1.95	2.90	1.91	5.13
	180	70	56	2.5	0.6	391	533	76.1	1 600	2 100	**46322A**	122	168	7	2	0.6	0.35	1.92	2.86	1.88	6.43
	200	92	74	2.5	1	695	868	116	1 500	2 000	**46T30222JR/74**	124	188	9	2.5	1	0.42	1.61	2.39	1.57	11.6
	200	121	101	2.5	1	938	1 280	161	1 500	2 000	**46T32222JR/101**	124	190	10	2.5	1	0.42	1.61	2.39	1.57	15.9
	240	118	93	3	1	1 030	1 180	150	1 300	1 700	**46T30322JR/93**	128	222	12.5	3	1	0.35	1.96	2.91	1.91	23.8
	240	181	142	3	1	1 480	1 890	230	1 300	1 700	**46T32322JR/142**	128	222	19.5	3	1	0.35	1.96	2.91	1.91	37.3

[Remark] Bearings not shown above (e.g. inch series) are shown in catalog "large size ball & roller bearings".

TDO type

d 120 ~ (150) mm



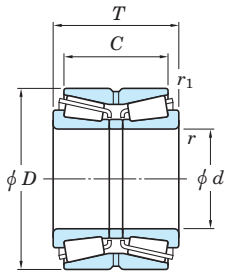


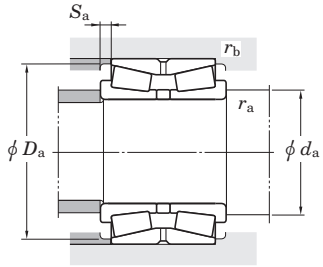
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass
d	D	T	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.		d_a min.	D_a min.	S_a min.	r_a max.	r_b max.	e	Y_2	Y_3	Y_0	(kg)
120	180	46	41	2.5	0.6	232	317	43.6	1 500	2 000	**46224**	132	170	2.5	2	0.6	0.35	1.95	2.90	1.91	3.81
	180	58	46	2.5	0.6	309	460	64.4	1 500	2 100	**46224A**	132	169	6	2	0.6	0.35	1.95	2.90	1.91	4.66
	200	62	55	2.5	0.6	367	470	65.7	1 400	1 900	**46324**	132	184	3.5	2	0.6	0.35	1.95	2.90	1.91	7.28
	200	78	62	2.5	0.6	486	672	93.9	1 400	1 900	**46324A**	132	185	8	2	0.6	0.35	1.95	2.90	1.91	9.14
	200	100	84	2.5	0.6	670	1 010	125	1 400	1 900	**46324AS**	132	190	8	2	0.6	0.35	1.95	2.90	1.91	12.0
	215	97	78	2.5	1	745	945	123	1 400	1 800	**46T30224JR/78**	134	203	9.5	2.5	1	0.44	1.55	2.31	1.52	13.9
	215	132	109	2.5	1	1 010	1 380	168	1 400	1 900	**46T32224JR/109**	134	204	11.5	2.5	1	0.44	1.55	2.31	1.52	19.8
	260	128	101	3	1	1 220	1 430	180	1 200	1 600	**46T30324JR/101**	138	239	13.5	3	1	0.35	1.96	2.91	1.91	30.6
	260	188	145	4	1.5	1 720	2 210	261	1 200	1 600	**46T32324JR/145**	142	239	21.5	4	1.5	0.35	1.96	2.91	1.91	45.9
130	200	52	46	2.5	0.6	299	425	57.8	1 400	1 800	**46226**	142	187	3	2	0.6	0.35	1.95	2.90	1.91	5.57
	200	65	52	2.5	0.6	400	618	85.0	1 400	1 900	**46226A**	142	185	6.5	2	0.6	0.35	1.95	2.90	1.91	7.06
	210	64	57	2.5	0.6	404	535	73.6	1 400	1 800	**46326**	142	196	3.5	2	0.6	0.36	1.87	2.79	1.83	7.81
	210	80	64	2.5	0.6	513	723	99.3	1 300	1 800	**46326A**	142	198	8	2	0.6	0.36	1.87	2.79	1.83	9.57
	230	98	78.5	3	1	809	1 020	131	1 300	1 700	**46T30226JR/78.5**	148	218	9.5	3	1	0.44	1.55	2.31	1.52	15.7
	230	145	117.5	3	1	1 190	1 660	200	1 300	1 700	**46T32226JR/117.5**	148	219	14	3	1	0.44	1.55	2.31	1.52	24.1
	280	137	107.5	4	1.5	1 410	1 670	203	1 100	1 400	**46T30326JR/107.5**	152	255	15	4	1.5	0.35	1.96	2.91	1.91	38.1
140	210	53	47	2.5	0.6	299	404	54.5	1 300	1 800	**46228**	152	196	3	2	0.6	0.33	2.03	3.02	1.98	5.85
	210	66	53	2.5	0.6	452	639	86.9	1 300	1 800	**46228A**	152	199	6.5	2	0.6	0.47	1.43	2.12	1.40	7.18
	225	68	61	3	1	423	564	76.1	1 200	1 700	**46328**	154	210	3.5	2.5	1	0.35	1.95	2.90	1.91	9.56
	225	85	68	3	1	597	836	113	1 200	1 700	**46328A**	154	212	8	2.5	1	0.35	1.95	2.90	1.91	11.8
	250	102	82.5	3	1	902	1 140	144	1 200	1 500	**46T30228JR/82.5**	158	237	9.5	3	1	0.44	1.55	2.31	1.52	19.7
	250	153	125.5	3	1	1 360	1 920	224	1 200	1 600	**46T32228JR/125.5**	158	238	14	3	1	0.44	1.55	2.31	1.52	30.2
	300	145	115.5	4	1.5	1 610	1 920	228	1 000	1 300	**46T30328JR/115.5**	162	273	15	4	1.5	0.35	1.96	2.91	1.91	46.6
150	225	56	50	3	1	348	476	63.2	1 200	1 600	**46230**	164	213	3	2.5	1	0.33	2.03	3.02	1.98	7.09
	225	70	56	3	1	472	703	94.1	1 200	1 600	**46230A**	164	213	7	2.5	1	0.33	2.03	3.02	1.98	8.82
	250	80	71	3	1	587	786	98.4	1 100	1 500	**46330**	164	233	4.5	2.5	1	0.35	1.95	2.90	1.91	14.6
	250	100	80	3	1	748	1 070	132	1 100	1 500	**46330A**	164	234	10	2.5	1	0.35	1.95	2.90	1.91	17.6
	270	109	87	3	1	1 040	1 330	162	1 100	1 400	**46T30230JR/87**	168	255	11	3	1	0.44	1.55	2.31	1.52	24.6

[Remark] Bearings not shown above (e.g. inch series) are shown in catalog "large size ball & roller bearings".

TDO type

d (150) ~ (200) mm



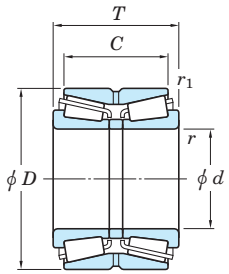


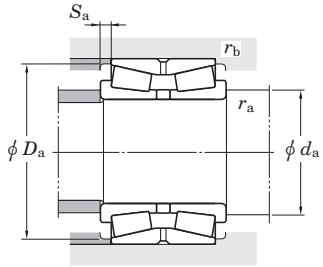
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass
d	D	T	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.		d_a min.	D_a min.	S_a min.	r_a max.	r_b max.	e	Y_2	Y_3	Y_0	(kg)
150	270	164	130	3	1	1 510	2 130	245	1 100	1 400	**46T32230JR/130**	168	254	17	3	1	0.44	1.55	2.31	1.52	38
	320	154	120	4	1.5	1 800	2 160	257	930	1 200	**46T30330JR/120**	172	292	17	4	1.5	0.35	1.96	2.91	1.91	56
160	240	60	53	3	1	405	565	74.0	1 100	1 500	**46232**	174	228	3.5	2.5	1	0.33	2.03	3.02	1.98	8.71
	240	75	60	3	1	508	756	99.6	1 100	1 500	**46232A**	174	226	7.5	2.5	1	0.33	2.03	3.02	1.98	10.6
	270	86	76	3	1	695	950	115	1 000	1 400	**46332**	174	252	5	2.5	1	0.35	1.95	2.90	1.91	18.8
	270	108	86	3	1	871	1 270	150	1 000	1 400	**46332A**	174	252	11	2.5	1	0.35	1.95	2.90	1.91	23.1
	290	115	91	3	1	1 160	1 500	179	980	1 300	**46T30232JR/91**	178	269	12	3	1	0.44	1.55	2.31	1.52	29.9
	290	178	144	3	1	1 700	2 420	273	1 000	1 300	**46T32232JR/144**	178	274	17	3	1	0.44	1.55	2.31	1.52	47.6
170	260	67	60	3	1	480	642	83.4	1 000	1 400	**46234**	184	243	3.5	2.5	1	0.33	2.03	3.02	1.98	11.4
	260	84	67	3	1	629	969	125	1 000	1 400	**46234A**	184	244	8.5	2.5	1	0.33	2.03	3.02	1.98	14.7
	280	88	78	3	1	754	1 050	125	970	1 300	**46334**	184	263	5	2.5	1	0.33	2.06	3.06	2.01	19.8
	280	110	88	3	1	938	1 390	163	980	1 300	**46334A**	184	260	11	2.5	1	0.33	2.06	3.06	2.01	24.7
	310	125	97	4	1.5	1 330	1 730	205	900	1 200	**46T30234JR/97**	192	288	14	4	1.5	0.44	1.55	2.31	1.52	37.5
	310	192	152	4	1.5	1 930	2 760	303	910	1 200	**46T32234JR/152**	192	294	20	4	1.5	0.44	1.55	2.31	1.52	58.8
180	280	74	66	3	1	582	801	98.9	950	1 300	**46236**	194	263	4	2.5	1	0.33	2.03	3.02	1.98	15.5
	280	93	74	3	1	732	1 080	131	960	1 300	**46236A**	194	261	9.5	2.5	1	0.33	2.03	3.02	1.98	19.0
	300	96	85	4	1.5	872	1 240	149	910	1 200	**46336**	198	277	5.5	3	1.5	0.33	2.06	3.06	2.01	25.8
	300	120	96	4	1.5	1 080	1 630	190	900	1 200	**46336A**	198	279	12	3	1.5	0.33	2.06	3.06	2.01	31.3
	320	127	99	4	1.5	1 320	1 740	204	860	1 200	**46T30236JR/99**	202	297	14	4	1.5	0.45	1.5	2.23	1.47	40.1
	320	192	152	4	1.5	2 060	3 030	328	880	1 200	**46T32236JR/152**	202	303	20	4	1.5	0.45	1.5	2.23	1.47	62.5
190	290	75	67	3	1	610	866	106	910	1 200	**46238**	204	272	4	2.5	1	0.32	2.12	3.15	2.07	16.5
	290	94	75	3	1	793	1 170	140	900	1 200	**46238A**	204	274	9.5	2.5	1	0.33	2.03	3.02	1.98	20.0
	320	104	92	4	1.5	1 020	1 450	168	830	1 100	**46338**	208	298	6	3	1.5	0.35	1.95	2.90	1.91	31.9
	320	130	104	4	1.5	1 230	1 860	212	840	1 100	**46338A**	208	298	13	3	1.5	0.35	1.95	2.90	1.91	39.0
	340	133	105	4	1.5	1 560	2 060	235	800	1 100	**46T30238JR/105**	212	318	14	4	1.5	0.44	1.55	2.31	1.52	47.8
	340	204	160	4	1.5	2 340	3 480	373	810	1 100	**46T32238JR/160**	212	323	22	4	1.5	0.44	1.55	2.31	1.52	75.1
200	310	82	73	3	1	716	1 040	123	850	1 100	**46240**	214	288	4.5	2.5	1	0.32	2.12	3.15	2.07	21.4

[Remark] Bearings not shown above (e.g. inch series) are shown in catalog "large size ball & roller bearings".

TDO type

d (200) ~ (300) mm



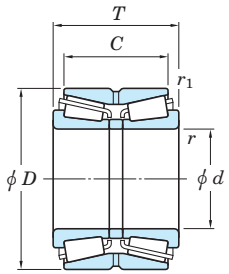


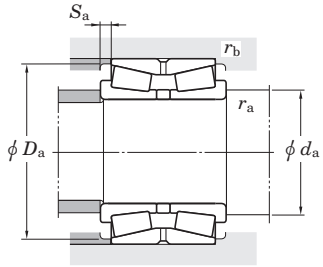
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass
d	D	T	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.		d_a min.	D_a min.	S_a min.	r_a max.	r_b max.	e	Y_2	Y_3	Y_0	(kg)
200	310	103	82	3	1	893	1 380	160	840	1 100	**46240A**	214	289	10.5	2.5	1	0.32	2.12	3.15	2.07	26.3
	340	112	100	4	1.5	1 100	1 580	180	780	1 000	**46340**	218	316	6	3	1.5	0.35	1.95	2.90	1.91	39.6
	340	140	112	4	1.5	1 350	2 040	226	770	1 000	**46340A**	218	319	14	3	1.5	0.35	1.95	2.90	1.91	48.2
	360	142	110	4	1.5	1 700	2 240	252	750	1 000	**46T30240JR/110**	222	336	16	4	1.5	0.44	1.55	2.31	1.52	56.5
	360	218	174	4	1.5	2 660	3 760	399	770	1 000	**46T32240JR/174**	222	340	22	4	1.5	0.41	1.66	2.47	1.62	88.2
220	340	90	80	4	1.5	849	1 240	142	750	990	**46244**	238	319	5	3	1.5	0.32	2.12	3.15	2.07	27.8
	340	113	90	4	1.5	1 040	1 620	183	750	1 000	**46244A**	238	318	11.5	3	1.5	0.32	2.12	3.15	2.07	34.2
	370	120	107	5	1.5	1 260	1 810	202	700	930	**46344**	242	346	6.5	4	1.5	0.35	1.95	2.90	1.91	49.1
	370	150	120	5	1.5	1 600	2 470	272	710	940	**46344A**	242	343	15	4	1.5	0.35	1.95	2.90	1.91	60.1
	400	150	114	4	1.5	2 170	2 880	320	660	890	**46T30244JR/114**	242	371	18	4	1.5	0.42	1.61	2.39	1.57	75.8
240	360	92	82	4	1.5	962	1 430	159	690	920	**46248**	258	338	5	3	1.5	0.32	2.12	3.15	2.07	29.6
	360	115	92	4	1.5	1 240	1 980	216	690	920	**46248A**	258	341	11.5	3	1.5	0.32	2.12	3.15	2.07	36.9
	400	128	114	5	1.5	1 490	2 180	241	630	840	**46348**	262	377	7	4	1.5	0.35	1.95	2.90	1.91	59.0
	400	160	128	5	1.5	1 940	3 060	325	630	850	**46348A**	262	373	16	4	1.5	0.35	1.95	2.90	1.91	76.2
260	400	104	92	5	1.5	1 170	1 830	200	610	820	**46252**	282	373	6	4	1.5	0.33	2.03	3.02	1.98	44.6
	400	130	104	5	1.5	1 520	2 480	265	610	810	**46252A**	282	376	13	4	1.5	0.32	2.12	3.15	2.07	54.8
	440	144	128	5	1.5	1 900	2 880	302	560	750	**46352**	282	410	8	4	1.5	0.35	1.95	2.90	1.91	83.8
	440	180	144	5	1.5	2 430	3 960	408	570	760	**46352A**	282	409	18	4	1.5	0.35	1.95	2.90	1.91	105
280	420	106	94	5	1.5	1 260	1 970	213	570	760	**46256**	302	395	6	4	1.5	0.33	2.03	3.02	1.98	46.9
	420	133	106	5	1.5	1 570	2 610	277	570	760	**46256A**	302	394	13.5	4	1.5	0.33	2.03	3.02	1.98	58.9
	460	146	130	6	2	1 950	2 930	308	530	700	**46356**	308	430	8	5	2	0.35	1.95	2.90	1.91	90.0
	460	183	146	6	2	2 470	3 940	407	520	690	**46356A**	308	434	18.5	5	2	0.35	1.95	2.90	1.91	111
300	460	118	105	5	1.5	1 630	2 400	254	500	670	**46260**	322	436	6.5	4	1.5	0.32	2.12	3.15	2.07	64.6
	460	148	118	5	1.5	2 050	3 230	331	510	680	**46260A**	322	433	15	4	1.5	0.32	2.12	3.15	2.07	80.2
	500	160	142	6	2	2 320	3 540	366	470	620	**46360**	328	469	9	5	2	0.35	1.95	2.90	1.91	116
	500	200	160	6	2	2 860	4 630	463	470	630	**46360A**	328	466	20	5	2	0.35	1.95	2.90	1.91	144

[Remark] Bearings not shown above (e.g. inch series) are shown in catalog "large size ball & roller bearings".

TDO type

d (300) ~420 mm



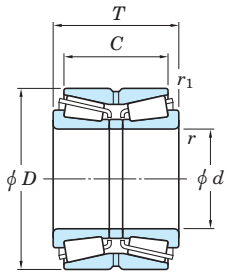


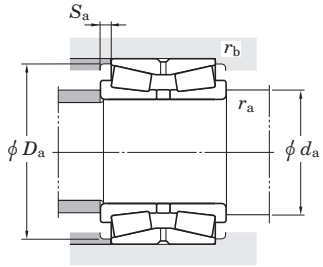
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass
d	D	T	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.		d_a min.	D_a min.	S_a min.	r_a max.	r_b max.	e	Y_2	Y_3	Y_0	(kg)
300	500	200	160	6	1.5	3 140	4 650	474	—	—	**46360D**	328	475	20	5	1.5	0.40	1.68	2.50	1.64	139
320	480	121	108	5	1.5	1 800	2 700	283	480	640	**46264**	342	452	6.5	4	1.5	0.32	2.12	3.15	2.07	71.6
	480	151	121	5	1.5	2 060	3 410	342	470	630	**46264A**	342	454	15	4	1.5	0.32	2.12	3.15	2.07	87.7
	540	176	157	6	2	2 880	4 570	457	420	560	**46364**	348	502	9.5	5	2	0.35	1.95	2.90	1.91	154
	540	220	176	6	2	3 280	5 390	528	430	570	**46364A**	348	497	22	5	2	0.35	1.95	2.90	1.91	190
340	520	133	118	6	2	1 940	3 070	314	420	570	**46268**	368	489	7.5	5	2	0.32	2.12	3.15	2.07	95.3
	520	165	133	6	2	2 420	4 060	406	420	560	**46268A**	368	491	16	5	2	0.32	2.12	3.15	2.07	117
	580	190	169	6	2	2 980	4 620	454	380	510	**46368**	368	539	10.5	5	2	0.35	1.95	2.90	1.91	198
	580	238	190	6	2	3 820	6 340	606	370	500	**46368A**	368	543	24	5	2	0.35	1.95	2.90	1.91	244
360	540	134	120	6	2	2 070	3 290	332	400	530	**46272**	388	510	7	5	2	0.32	2.12	3.15	2.07	93.0
	540	169	134	6	2	2 530	4 230	419	390	530	**46272A**	388	512	17.5	5	2	0.32	2.12	3.15	2.07	124
	600	192	171	6	2	3 600	4 880	473	360	480	**46372**	388	557	10.5	5	2	0.35	1.95	2.90	1.91	206
	600	240	192	6	2	4 590	7 230	689	360	480	**46372A**	388	568	24	5	2	0.39	1.74	2.59	1.70	254
380	560	135	122	6	2	2 190	3 560	355	370	500	**46276**	408	530	6.5	5	2	0.32	2.12	3.15	2.07	100
	560	171	135	6	2	2 810	4 670	456	380	500	**46276A**	408	531	18	5	2	0.39	1.74	2.59	1.70	129
	620	194	173	6	2	3 380	5 220	500	340	450	**46376**	408	582	10.5	5	2	0.39	1.74	2.59	1.70	215
	620	243	194	6	2	4 390	7 360	683	330	440	**46376A**	408	587	24.5	5	2	0.35	1.95	2.90	1.91	265
400	600	148	132	6	2	2 350	3 720	366	340	460	**46280**	428	560	8	5	2	0.32	2.12	3.15	2.07	135
	600	185	148	6	2	3 030	5 150	491	340	460	**46280A**	428	563	18.5	5	2	0.32	2.12	3.15	2.07	167
	650	200	178	6	3	3 740	5 920	565	320	420	**46380**	428	605	11	5	2.5	0.35	1.95	2.90	1.91	243
	650	250	200	6	3	5 110	8 850	811	310	420	**46380A**	428	610	25	5	2.5	0.35	1.95	2.90	1.91	306
420	620	150	134	6	2	2 520	4 130	399	320	420	**46284**	448	590	8	5	2	0.33	2.03	3.02	1.98	142
	620	188	150	6	2	3 390	5 660	534	320	430	**46284A**	448	589	19	5	2	0.39	1.74	2.59	1.70	176
	700	224	200	6	3	4 650	6 880	647	290	380	**46384**	448	656	12	5	2.5	0.39	1.74	2.59	1.70	325
	700	280	224	6	3	6 040	9 620	861	290	380	**46384A**	448	659	28	5	2.5	0.39	1.74	2.59	1.70	400

[Remark] Bearings not shown above (e.g. inch series) are shown in catalog "large size ball & roller bearings".

TDO type

d 440 ~ 500 mm



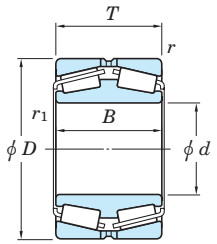


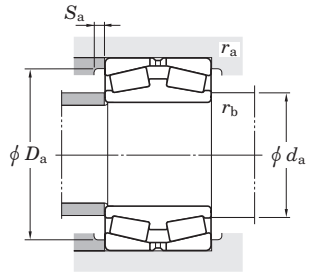
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)					Con-stant	Axial load factors			(Refer.) Mass
d	D	T	C	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.		d_a min.	D_a min.	S_a min.	r_a max.	r_b max.	e	Y_2	Y_3	Y_0	(kg)
440	650	157	140	6	3	2 840	4 430	423	300	390	**46288**	468	622	8.5	5	2.5	0.33	2.03	3.02	1.98	156
	650	196	157	6	3	3 770	6 370	600	300	400	**46288A**	468	620	19.5	5	2.5	0.39	1.74	2.59	1.70	198
	720	226	201	6	3	4 950	8 110	744	270	360	**46388**	468	676	12.5	5	2.5	0.39	1.74	2.59	1.70	354
	720	283	226	6	3	6 210	10 100	893	270	360	**46388A**	468	679	28.5	5	2.5	0.40	1.68	2.51	1.65	418
460	680	163	145	6	3	3 130	5 340	507	280	370	**46292**	488	637	9	5	2.5	0.37	1.83	2.72	1.78	196
	680	204	163	6	3	4 040	6 850	635	280	370	**46292A**	488	646	20.5	5	2.5	0.39	1.74	2.59	1.70	232
	760	240	214	7.5	4	5 460	9 000	817	250	330	**46392**	496	710	13	6	3	0.39	1.74	2.59	1.70	424
	760	300	240	7.5	4	7 130	11 600	1 010	250	330	**46392A**	496	718	30	6	3	0.39	1.74	2.59	1.70	506
480	700	165	147	6	3	3 180	5 300	494	260	340	**46296**	508	672	9	5	2.5	0.33	2.03	3.02	1.98	186
	700	206	165	6	3	4 040	7 230	666	260	340	**46296A**	508	666	20.5	5	2.5	0.33	2.03	3.02	1.98	240
	790	248	221	7.5	4	5 820	8 920	810	230	310	**46396**	516	742	13.5	6	3	0.39	1.74	2.59	1.70	457
	790	310	248	7.5	4	7 530	12 400	1 060	230	310	**46396A**	516	749	31	6	3	0.39	1.74	2.59	1.70	560
500	720	167	149	6	3	3 230	5 690	529	250	330	**462/500**	528	679	9	5	2.5	0.40	1.71	2.54	1.67	210
	720	209	167	6	3	4 390	7 850	712	250	330	**462/500A**	528	690	21	5	2.5	0.42	1.62	2.41	1.58	258
	830	264	235	7.5	4	6 570	10 900	955	210	280	**463/500**	536	776	14.5	6	3	0.39	1.74	2.59	1.70	559
	830	330	264	7.5	4	8 510	14 000	1 170	210	280	**463/500A**	536	784	33	6	3	0.39	1.74	2.59	1.70	669

[Remark] Bearings not shown above (e.g. inch series) are shown in catalog "large size ball & roller bearings".

TDI type

d 100 ~ (220) mm



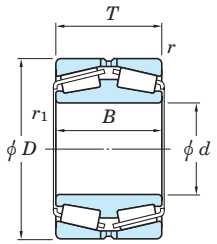


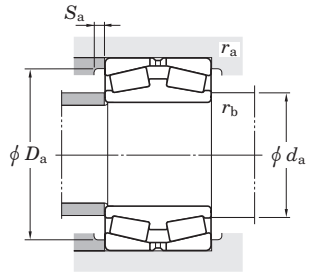
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)						Constant	Axial load factors			(Refer.) Mass
d	D	B	T	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.		d_a max.	D_a max.	D_a min.	S_a min.	r_a max.	r_b max.	e	Y_2	Y_3	Y_0	(kg)
100	165	52	52	2	2.5	298	384	55.9	1 800	2 300	**45320**	119	155	148	3.9	2	2	0.35	1.95	2.90	1.91	4.26
110	180	56	56	2	2.5	378	505	72.2	1 600	2 100	**45322**	128	170	160	4	2	2	0.35	1.95	2.90	1.91	5.40
120	180	46	46	2	2.5	286	424	59.4	1 500	2 100	**45224**	138	170	163	4	2	2	0.26	2.55	3.80	2.50	4.08
	200	62	62	2	2.5	444	598	83.4	1 400	1 900	**45324**	142	190	178	4	2	2	0.35	1.95	2.90	1.91	7.92
130	200	52	52	2	2.5	376	548	75.6	1 400	1 800	**45226**	152	190	179	4	2	2	0.27	2.47	3.67	2.41	5.96
	210	64	64	2	2.5	476	657	90.3	1 300	1 800	**45326**	153	200	185	4	2	2	0.36	1.87	2.79	1.83	8.41
140	210	53	53	2	2.5	390	564	76.9	1 300	1 800	**45228**	159	200	188	4	2	2	0.27	2.47	3.67	2.41	6.45
	225	68	68	2.5	3	611	807	103	1 200	1 700	**45328**	160	213	210	4	2	2.5	0.40	1.68	2.50	1.64	10.0
150	225	56	56	2.5	3	445	686	91.6	1 200	1 600	**45230**	174	213	203	4	2	2.5	0.26	2.55	3.80	2.50	7.87
	250	80	80	2.5	3	684	955	120	1 100	1 500	**45330**	179	238	220	4	2	2.5	0.35	1.95	2.90	1.91	15.5
160	240	60	60	2.5	3	488	705	93.1	1 100	1 500	**45232**	184	228	217	5	2	2.5	0.24	2.79	4.15	2.73	9.22
	270	86	86	2.5	3	832	1 100	146	1 000	1 400	**45332**	193	258	237	4	2	2.5	0.35	1.95	2.90	1.91	19.8
170	260	67	67	2.5	3	654	956	124	1 000	1 400	**45234**	195	248	233	5	2	2.5	0.31	2.21	3.29	2.16	12.4
	280	88	88	2.5	3	834	1 210	145	970	1 300	**45334**	201	268	247	5	2	2.5	0.33	2.03	3.02	1.98	21.6
180	280	74	74	2.5	3	722	1 050	125	950	1 300	**45236**	208	268	250	5	2	2.5	0.28	2.43	3.61	2.37	16.8
	300	96	96	3	4	992	1 370	162	910	1 200	**45336**	210	286	263	5	2.5	3	0.35	1.95	2.90	1.91	26.5
190	290	75	75	2.5	3	751	1 130	133	900	1 200	**45238**	219	278	260	5	2	2.5	0.26	2.55	3.80	2.50	17.7
	320	104	104	3	4	1 130	1 590	183	840	1 100	**45338**	224	306	280	5	2.5	3	0.35	1.95	2.90	1.91	34.0
200	310	82	82	2.5	3	913	1 410	166	830	1 100	**45240**	234	298	280	5	2	2.5	0.26	2.55	3.80	2.50	22.9
	340	112	112	3	4	1 250	1 840	208	770	1 000	**45340**	244	326	300	5	2.5	3	0.35	1.95	2.90	1.91	41.9
220	340	90	90	3	4	933	1 460	167	740	990	**45244**	259	326	306	5	2.5	3	0.28	2.43	3.61	2.37	28.5

[Remark] Bearings not shown above (e.g. inch series) are shown in catalog "large size ball & roller bearings".

TDI type

d (220) ~ (420) mm



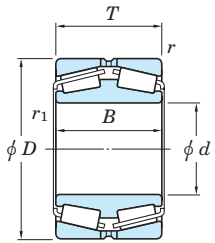


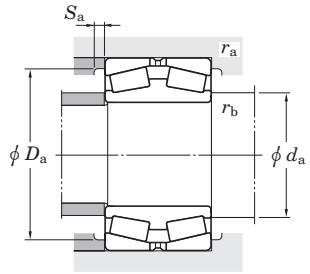
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)						Constant	Axial load factors			(Refer.) Mass
d	D	B	T	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.		d_a max.	D_a max.	D_a min.	S_a min.	r_a max.	r_b max.	e	Y_2	Y_3	Y_0	(kg)
220	370	120	120	4	5	1 400	2 060	226	700	930	**45344**	263	352	324	5	3	4	0.35	1.95	2.90	1.91	50.8
230	350	90	90	3	4	991	1 560	177	710	950	**45246**	267	336	318	6	2.5	3	0.28	2.43	3.61	2.37	30.6
240	360	92	92	3	4	1 150	1 790	200	690	920	**45248**	271	346	325	5	2.5	3	0.32	2.12	3.15	2.07	32.2
	400	128	128	4	5	1 650	2 470	265	630	840	**45348**	286	382	354	5	3	4	0.35	1.95	2.90	1.91	65.4
260	400	104	104	4	5	1 320	2 120	227	610	810	**45252**	302	382	360	6	3	4	0.25	2.74	4.08	2.68	48.1
	440	144	144	4	5	2 180	3 440	357	560	750	**45352**	313	422	386	6	3	4	0.35	1.95	2.90	1.91	92.2
280	420	106	106	4	5	1 490	2 470	265	560	750	**45256**	321	402	370	6	3	4	0.25	2.69	4.00	2.63	51.9
	460	146	146	5	6	2 310	3 320	351	520	700	**45356**	323	438	409	6	4	5	0.39	1.74	2.59	1.70	93.1
300	460	118	118	4	5	1 870	3 150	325	500	670	**45260**	350	442	418	6	3	4	0.25	2.74	4.08	2.68	78.5
	500	160	160	5	6	2 670	4 240	431	470	630	**45360**	356	478	440	6	4	5	0.35	1.95	2.90	1.91	129
320	480	121	121	4	5	1 830	3 180	322	470	630	**45264**	368	462	434	6	3	4	0.26	2.55	3.80	2.50	77.8
	540	176	176	5	6	3 380	5 280	528	430	570	**45364R**	378	518	474	6	4	5	0.32	2.12	3.15	2.07	167
340	520	133	133	5	6	2 380	3 850	372	420	570	**45268**	398	498	464	6	4	5	0.26	2.55	3.80	2.50	104
	580	190	190	5	6	3 790	5 470	537	390	510	**45368**	401	558	515	6	4	5	0.32	2.12	3.15	2.07	202
360	540	134	134	5	6	2 370	3 910	393	400	540	**45272**	408	518	488	11	4	5	0.32	2.12	3.15	2.07	101
	600	192	192	5	6	4 230	6 750	648	360	490	**45372**	419	578	528	10	4	5	0.32	2.12	3.15	2.07	228
380	560	135	135	5	6	2 300	3 790	371	380	500	**45276**	428	538	510	6	4	5	0.27	2.47	3.67	2.41	112
	620	194	194	5	6	3 860	6 360	606	340	450	**45376**	445	598	545	6	4	5	0.32	2.12	3.15	2.07	234
400	600	148	148	5	6	3 020	4 960	478	340	450	**45280**	452	578	545	6	4	5	0.33	2.03	3.02	1.98	143
	650	200	200	6	6	4 840	7 810	735	320	420	**45380**	458	622	580	11	5	5	0.39	1.74	2.59	1.70	265
420	620	150	150	5	6	3 010	5 200	496	320	430	**45284**	475	598	564	6	4	5	0.33	2.03	3.02	1.98	152

[Remark] Bearings not shown above (e.g. inch series) are shown in catalog "large size ball & roller bearings".

TDI type

d (420) ~ 500 mm





Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.	Mounting dimensions (mm)						Con-stant	Axial load factors			(Refer.) Mass (kg)
d	D	B	T	r min.	r_1 min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.		d_a max.	D_a max.	D_a min.	S_a min.	r_a max.	r_b max.	e	Y_2	Y_3	Y_0	
420	700	224	224	6	6	5 430	8 380	777	280	380	**45384**	488	672	623	7	5	5	0.39	1.74	2.59	1.70	352
440	650	157	157	6	6	3 190	5 500	512	300	390	**45288**	500	622	592	10	5	5	0.28	2.43	3.61	2.37	182
	720	226	226	6	6	5 750	9 130	834	270	360	**45388**	506	692	642	7	5	5	0.39	1.74	2.59	1.70	367
460	680	163	163	6	6	3 480	5 660	531	280	370	**45292**	510	652	616	6	5	5	0.39	1.74	2.59	1.70	197
	760	240	240	7.5	7.5	6 570	10 400	927	250	330	**45392**	532	724	677	7	6	6	0.39	1.74	2.59	1.70	444
480	700	165	165	6	6	3 830	6 710	614	260	350	**45296**	531	672	625	6	5	5	0.40	1.68	2.50	1.64	215
500	720	167	167	6	6	4 300	7 350	681	250	330	**452/500**	545	692	645	8	5	5	0.39	1.74	2.59	1.70	222
	830	264	264	7.5	7.5	7 970	12 300	1 110	210	280	**453/500**	587	794	729	7	6	6	0.33	2.03	3.02	1.98	586

[Remark] Bearings not shown above (e.g. inch series) are shown in catalog "large size ball & roller bearings".

Spherical roller bearings

Spherical roller bearings feature a large load rating capacity and self-aligning capability.

This type of bearing is suitable for low- or medium-speed applications which involve heavy or impact loading.

- These bearings are divided into R(RR), RZ and RHA types, which differ in internal structure. (refer to Table 1.)
- Each type can be produced with a cylindrical bore or tapered bore.

Bearings with a tapered bore can be fit and removed easily using an adapter assembly or withdrawal sleeve.

The rate of taper is equivalent among all bearing series.

240 and 241 series ... 1 : 30 (supplementary code "K30")

Others ... 1 : 12 (supplementary code "K")

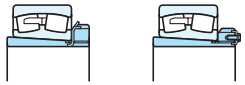
Spherical roller bearings



Cylindrical bore Tapered bore

Bore diameter **25 – 500** mm

Adapter assemblies



Bore diameter **20 – 470** mm

Withdrawal sleeves



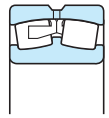
Bore diameter **35 – 480** mm

Table 1 Spherical roller bearings : types and structures

Structure	 R, RR type	 RZ type	 RHA type
Roller	Convex asymmetrical roller	Convex symmetrical roller	Convex symmetrical roller
Cage	Copper alloy prong type machined cage	Pressed cage	Copper alloy integral type machined cage
Inner ring (with or without rib)	With center rib	Without center rib (guide ring)	Without center rib (guide ring)
	With ribs on both sides (to prevent rollers from falling)	Without ribs on both sides	With ribs on both sides (to prevent rollers from falling)
Characteristics	Excellent high-speed properties	Excellent high-speed properties Large load rating capacity Usable at high temperatures (up to 200°C)	Large load rating capacity

■ Spherical roller bearings for shaker screens





- These bearings consist of convex asymmetric rollers and a prong type, copper alloy, outer ring guided, machined cage. This cage possesses optimum characteristics for use with shaker screens.
- The bearings most commonly used with shaker screens are 223 series spherical roller bearings. They are identified by the supplementary code "ROVS W502." The outer ring outside diameter tolerance of these bearings is held to a small allowable variation.

■ Bearings with lubrication holes and a lubrication groove

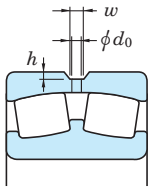
- Outer rings can be provided with lubrication holes, a lubrication groove and an anti-rotation pin hole. (Specifications are given in Table 4.)
- Inner rings can also be provided with lubrication holes and a lubrication groove.

Table 2 Supplementary codes for identification of bearings with lubrication holes, lubrication groove and anti-rotation pin hole (outer ring)

Supplementary code		Number of lubrication holes	Hole layout
With lubrication holes and lubrication groove	With lubrication holes, lubrication groove and anti-rotation pin hole		
W33	**W3N**	3[1)]	3 equally spaced positions[1)]
W33A	W3NA	4	4 equally spaced positions
–	W3NB	5	6 equally spaced positions[2)]
W33C	W3NC	6	6 equally spaced positions
–	W3ND	7	8 equally spaced positions[2)]
W33T	–	8	8 equally spaced positions

Table 3 Supplementary codes for identification of bearings with lubrication holes and/or lubrication groove

Supplementary code	Inner ring		Outer ring	
	Number of lubrication holes	Lubrication groove	Number of lubrication holes	Lubrication groove
W513	3	–	3	○
W518	3	–	3	–
W26	3	–	–	–



[Notes] 1) Also 4 or 6 holes are provided in smaller size bearings, consult with JTEKT.
2) One hole is used for the antirotation pin.
[Remark] Boldfaced codes indicate JTEKT standards.

Table 4 (1) Lubrication hole and lubrication groove dimensions Unit : mm

Bore diameter number	Nominal bore diameter d	239			230			240			231			241			222			232			213			223		
		d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h
5	25	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	0.7	–	–	–	–	–	–	–	–	–
6	30	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	0.7	–	–	–	3	4	0.7	–	–	–
7	35	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	0.8	–	–	–	3	4	0.7	3	4	1
8	40	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	0.8	–	–	–	3	4	0.7	4	5	1
9	45	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	0.8	–	–	–	3	4	0.7	4	6	1
10	50	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	0.8	–	–	–	3	4	0.7	4	6	1
11	55	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	0.8	–	–	–	3	4	0.7	4	6	1.1
12	60	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	1	–	–	–	3	4	1	4	6	1.1
13	65	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	1	–	–	–	3	4	1	4	6	1.2
14	70	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	1	–	–	–	3	4	1	5	7	1.3
15	75	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	1	–	–	–	3	4	1	5	7	1.3
16	80	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	3	4	1	4	6	1.2	4	6	1	5	7	1.3
17	85	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	4	5	1	4	6	1.2	4	6	1.2	6	8	1.3
18	90	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	4	5	1	5	8	1.2	4	6	1.2	6	8	1.3
19	95	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	5	6	1.2	5	8	1.2	4	6	1.2	6	8	2
20	100	–	–	–	4	5	1	–	–	–	5	6	1.4	–	–	–	5	6	1.2	5	8	1.2	4	6	1.2	6	8	2
22	110	–	–	–	5	7	1	–	–	–	5	6	1.4	6	8	1.5	5	7	1.5	6	8	1.7	4	6	1.2	6	8	2
24	120	–	–	–	5	7	1	5	6	1.4	5	6	1.4	6	8	1.5	5	7	1.5	6	10	1.7	–	–	–	8	10	2.5
26	130	–	–	–	5	7	1.2	6	8	1.5	5	6	1.5	6	8	1.5	5	7	1.5	6	10	1.7	–	–	–	8	12	2.5
28	140	4	5	1	5	7	1.2	6	8	1.5	6	8	1.5	8	10	2	6	8	1.8	8	10	2.5	–	–	–	12	14	3
30	150	5	7	1	5	8	1.2	6	8	1.5	6	10	1.5	8	10	2	6	10	1.8	8	10	2.5	–	–	–	12	14	3
32	160	5	7	1.2	5	8	1.2	6	8	1.5	8	12	2	10	12	2	10	12	2.5	10	12	2.5	–	–	–	12	14	3
34	170	5	7	1.2	6	10	1.5	8	10	2	8	12	2	10	12	2	12	14	3	10	12	2.5	–	–	–	12	14	3
36	180	6	7	1.3	8	12	1.5	10	12	2.5	10	12	2.5	10	12	2	12	14	3	10	12	2.5	–	–	–	14	16.5	4
38	190	5	7	1.2	10	12	2.5	10	12	2.5	10	12	2.5	10	12	2	12	14	3	12	14	3	–	–	–	14	16.5	4
40	200	6	8	1.5	10	12	2.5	10	12	2.5	12	14	3	12	14	3	12	14	3	12	14	3	–	–	–	14	16.5	4
44	220	6	8	1.5	10	12	2.5	10	12	2.5	12	14	3	12	14	3	12	14	3	12	14	3	–	–	–	14	16.5	4
48	240	6	8	1.5	10	12	2.5	10	12	2.5	12	14	3	12	14	3	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4
52	260	10	12	2.5	12	14	3	12	14	3	12	14	3	12	14	3	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4
56	280	10	12	2.5	12	14	3	12	14	3	12	14	3	12	14	3	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4
60	300	10	12	2.5	12	14	3	12	14	3	12	14	3	12	14	3	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4
64	320	10	12	2.5	12	14	3	12	14	3	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4
68	340	12	14	3	14	16.5	4	14	16.5	3	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4
72	360	12	14	3	14	16.5	4	14	16.5	3	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4

Table 4 (2) Lubrication hole and lubrication groove dimensions Unit : mm

Bore diameter number	Nominal bore diameter d	239			230			240			231			241			222			232			213			223		
		d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h	d_0	w	h
76	380	12	14	3	14	16.5	4	14	16.5	3	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4	–	–	–	–	–	–
80	400	12	14	3	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4	–	–	–	–	–	–
84	420	12	14	3	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4	–	–	–	–	–	–
88	440	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4	–	–	–	–	–	–
92	460	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4	–	–	–	–	–	–
96	480	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	–	–	–	14	16.5	4	–	–	–	–	–	–
/500	500	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	4	14	16.5	5	–	–	–	14	16.5	4	–	–	–	–	–	–

Boundary dimensions	As specified in JIS B 1512.
Tolerances	As specified in JIS B 1514-1, class 0. (refer to Table 7-3 on pp. A 60 – A 63.) Refer to Table 7-11 on p. A 76 for the tolerance of tapered bores.
Radial internal clearance	As specified in JIS B 1520. (refer to Table 10-9 on p. A 108.)
Recommended fits	Refer to Table 9-4 on pp. A 91, 92.
Standard cages	Refer to Table 5.
Allowable aligning angle	Refer to Table 5. (varies depending on bearing series.)
Equivalent radial load	Dynamic equivalent radial load $\left(\text{When } \frac{F_a}{F_r} \leqq e\right) \quad P_r = F_r + Y_1 F_a \qquad \left(\text{When } \frac{F_a}{F_r} > e\right) \quad P_r = 0.67 F_r + Y_2 F_a$ Static equivalent radial load $P_{0r} = F_r + Y_0 F_a$ [Note] Refer to the specification table for the values of axial load factors Y_1, Y_2 and Y_0 and of constant e.

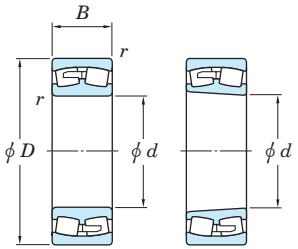
[Remark] If the ratio of axial load to radial load exceeds the value e given in the specification table ($F_a / F_r > e$), slippage occurs between rollers in rows that are not axial-loaded and the raceway.
This may cause smearing, especially when the bearing is large. Consult with JTEKT on the use of bearings under such conditions.

Table 5 Application of standard cages and allowable aligning angle

Bearing series	Standard cages		Allowable aligning angle
	Pressed cage	Machined cage	
239 R	——	23930R – 239/500R	0.026 rad (1.5°)
230 R	——	23038R – 230/500R	0.026 rad (1.5°)
RZ	23020RZ – 23036RZ	——	
RHA	——	23038RHA – 23096RHA	
240 R(RR)	——	24036RR – 240/500R	0.035 rad (2°)
RZ	24022RZ – 24034RZ	——	
RHA	——	24038RHA – 24096RHA	
231 R	——	23136R – 231/500R	0.026 rad (1.5°)
RZ	23120RZ – 23134RZ	——	
RHA	——	23136RHA – 23196RHA	
241 R(RR)	——	24132RR – 241/500R	0.044 rad (2.5°)
RZ	24122RZ – 24130RZ	——	
RHA	——	24136RHA – 24196RHA	
222 R(RR)	——	22232RR – 22272R	0.026 rad (1.5°)
RZ	22205RZ – 22230RZ	——	
RHA	——	22232RHA – 22260RHA	
232 R	——	23232R – 232/500R	0.044 rad (2.5°)
RZ	23216RZ – 23230RZ	——	
RHA	——	23232RHA – 23296RHA	
213 R	——	——	0.017 rad (1°)
RZ	21306RZ – 21322RZ	——	
223 R(RR)	——	22330R – 22360R	0.035 rad (2°)
RZ	22308RZ – 22326RZ	——	
RHA	——	22330RHA – 22356RHA	

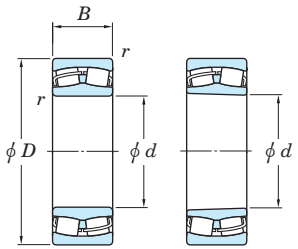
d 25 ~ 70 mm

R, RR



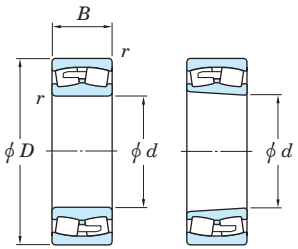
Cylindrical bore | Tapered bore

RZ

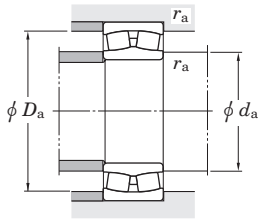


Cylindrical bore | Tapered bore

RHA



Cylindrical bore | Tapered bore

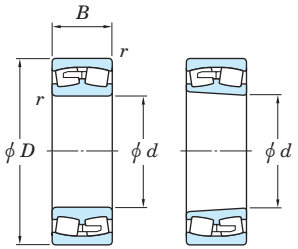


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Con-stant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	d_a max.	D_a max.	D_a min.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
25	52	18	1	56.8	48.1	3.90	9 600	12 800	**22205RZ**	**22205RZK**	31	31	46	46	1	0.35	1.91	2.85	1.87	0.188	0.184
30	62	20	1	76.6	65.9	5.30	8 100	10 900	**22206RZ**	**22206RZK**	36	36.5	56	55.5	1	0.33	2.04	3.04	2.00	0.296	0.290
	72	19	1.1	74.2	62.7	4.80	7 200	9 600	**21306RZ**	**21306RZK**	37	41.5	65	61.5	1	0.27	2.49	3.71	2.43	0.430	0.424
35	72	23	1.1	100	88.7	7.75	6 900	9 200	**22207RZ**	**22207RZK**	42	42.5	65	64	1	0.32	2.09	3.11	2.04	0.459	0.449
	80	21	1.5	86.8	75.8	5.90	6 200	8 300	**21307RZ**	**21307RZK**	43.5	46.5	71.5	68.5	1.5	0.27	2.49	3.71	2.43	0.572	0.564
40	80	23	1.1	114	102	9.55	6 200	8 300	**22208RZ**	**22208RZK**	47	49	73	72.5	1	0.28	2.37	3.53	2.32	0.602	0.591
	90	23	1.5	105	95.5	7.55	5 600	7 600	**21308RZ**	**21308RZK**	48.5	53.5	81.5	77	1.5	0.26	2.55	3.80	2.50	0.781	0.770
	90	33	1.5	170	152	11.8	5 600	7 600	**22308RZ**	**22308RZK**	48.5	51	81.5	78.5	1.5	0.37	1.83	2.72	1.79	1.08	1.06
45	85	23	1.1	119	110	10.2	5 800	7 700	**22209RZ**	**22209RZK**	52	53.5	78	77.5	1	0.26	2.55	3.80	2.50	0.602	0.590
	100	25	1.5	132	124	9.95	5 000	6 700	**21309RZ**	**21309RZK**	53.5	60	91.5	86	1.5	0.26	2.62	3.90	2.56	1.05	1.04
	100	36	1.5	208	183	13.8	5 100	6 700	**22309RZ**	**22309RZK**	53.5	55.5	91.5	87	1.5	0.37	1.83	2.72	1.79	1.42	1.39
50	90	23	1.1	128	122	12.7	5 400	7 200	**22210RZ**	**22210RZK**	57	58.5	83	82.5	1	0.24	2.79	4.15	2.73	0.648	0.634
	110	27	2	157	151	12.0	4 500	6 100	**21310RZ**	**21310RZK**	60	67	100	94.5	2	0.25	2.71	4.04	2.65	1.37	1.35
	110	40	2	255	237	17.5	4 500	6 200	**22310RZ**	**22310RZK**	60	62.5	100	95.5	2	0.36	1.85	2.76	1.81	1.92	1.88
55	100	25	1.5	154	144	15.0	4 700	6 300	**22211RZ**	**22211RZK**	63.5	64	91.5	91.5	1.5	0.24	2.84	4.23	2.78	0.867	0.849
	120	29	2	180	165	13.0	4 100	5 600	**21311RZ**	**21311RZK**	65	71.5	110	101.5	2	0.25	2.71	4.03	2.65	1.69	1.67
	120	43	2	296	264	21.1	4 100	5 500	**22311RZ**	**22311RZK**	65	66	110	104	2	0.36	1.85	2.76	1.81	2.40	2.35
60	110	28	1.5	190	181	18.7	4 300	5 800	**22212RZ**	**22212RZK**	68.5	70	101.5	100	1.5	0.25	2.74	4.08	2.68	1.19	1.17
	130	31	2.1	210	193	15.1	3 900	5 100	**21312RZ**	**21312RZK**	72	77.5	118	110	2	0.24	2.78	4.14	2.72	2.11	2.08
	130	46	2.1	354	334	24.9	3 900	5 100	**22312RZ**	**22312RZK**	72	73.5	118	113	2	0.36	1.86	2.77	1.82	3.06	2.99
65	120	31	1.5	222	211	20.7	4 000	5 200	**22213RZ**	**22213RZK**	73.5	76	111.5	109	1.5	0.25	2.69	4.00	2.63	1.55	1.52
	140	33	2.1	242	232	19.8	3 600	4 700	**21313RZ**	**21313RZK**	77	85.5	128	119	2	0.24	2.83	4.21	2.76	2.62	2.58
	140	48	2.1	382	360	30.8	3 600	4 700	**22313RZ**	**22313RZK**	77	79.5	128	122	2	0.34	1.98	2.94	1.93	3.66	3.58
70	125	31	1.5	233	222	24.4	3 700	5 000	**22214RZ**	**22214RZK**	78.5	80	116.5	114	1.5	0.24	2.87	4.27	2.80	1.64	1.61
	150	35	2.1	268	260	21.6	3 300	4 400	**21314RZ**	**21314RZK**	82	91	138	126.5	2	0.24	2.84	4.23	2.78	3.19	3.15
	150	51	2.1	435	413	35.0	3 300	4 400	**22314RZ**	**22314RZK**	82	85.5	138	131	2	0.34	1.98	2.94	1.93	4.45	4.36

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

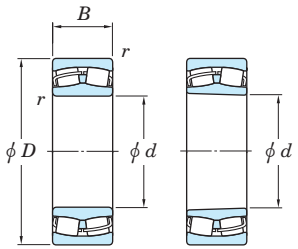
d 75 ~ (110) mm

R, RR



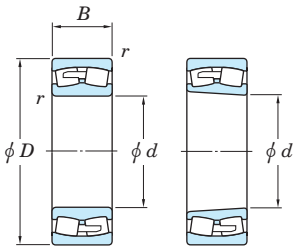
Cylindrical bore Tapered bore

RZ

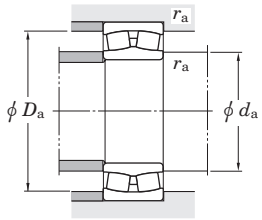


Cylindrical bore Tapered bore

RHA



Cylindrical bore Tapered bore

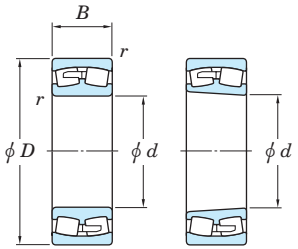


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Con-stant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	d_a max.	D_a max.	D_a min.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
75	130	31	1.5	241	236	28.2	3 600	4 700	**22215RZ**	**22215RZK**	83.5	85.5	121.5	119	1.5	0.22	3.07	4.57	3.00	1.73	1.69
	160	37	2.1	306	298	24.3	3 000	4 100	**21315RZ**	**21315RZK**	87	98	148	138	2	0.24	2.87	4.27	2.80	3.81	3.76
	160	55	2.1	492	473	38.4	3 000	4 100	**22315RZ**	**22315RZK**	87	91	148	139.5	2	0.35	1.95	2.90	1.91	5.45	5.33
80	140	33	2	271	271	30.5	3 300	4 400	**22216RZ**	**22216RZK**	90	92	130	128	2	0.22	3.07	4.57	3.00	2.17	2.13
	140	44.4	2	305	342	31.2	3 300	4 400	**23216RZ**	**23216RZK**	90	93	130	124	2	0.29	2.35	3.50	2.30	2.95	2.86
	170	39	2.1	344	339	27.5	2 900	3 900	**21316RZ**	**21316RZK**	92	104	158	146	2	0.23	2.88	4.29	2.82	4.53	4.47
	170	58	2.1	539	521	41.7	2 900	3 900	**22316RZ**	**22316RZK**	92	97	158	148	2	0.35	1.95	2.90	1.91	6.44	6.30
85	150	36	2	322	324	35.7	3 000	4 100	**22217RZ**	**22217RZK**	95	97	140	137	2	0.22	3.01	4.48	2.94	2.75	2.69
	150	49.2	2	358	410	36.2	3 000	4 100	**23217RZ**	**23217RZK**	95	99	140	134	2	0.30	2.25	3.34	2.20	3.78	3.67
	180	41	3	374	372	29.6	2 800	3 600	**21317RZ**	**21317RZK**	99	109	166	154	2.5	0.23	2.89	4.33	2.83	5.32	5.25
	180	60	3	601	586	47.8	2 800	3 600	**22317RZ**	**22317RZK**	99	103	166	157	2.5	0.33	2.02	3.00	1.97	7.47	7.31
90	160	40	2	372	381	39.2	2 900	3 900	**22218RZ**	**22218RZK**	100	104	150	145	2	0.24	2.79	4.15	2.73	3.50	3.43
	160	52.4	2	421	482	42.9	2 900	3 900	**23218RZ**	**23218RZK**	100	103	150	141	2	0.32	2.14	3.19	2.09	4.63	4.50
	190	43	3	413	416	32.9	2 600	3 400	**21318RZ**	**21318RZK**	104	116	176	162	2.5	0.23	2.91	4.30	2.84	6.20	6.11
	190	64	3	672	662	50.5	2 600	3 400	**22318RZ**	**22318RZK**	104	108	176	166	2.5	0.34	2.00	2.98	1.96	8.82	8.63
95	170	43	2.1	417	422	42.7	2 800	3 600	**22219RZ**	**22219RZK**	107	109	158	154	2	0.24	2.76	4.11	2.70	4.24	4.15
	170	55.6	2.1	457	516	43.9	2 800	3 600	**23219RZ**	**23219RZK**	107	110	158	150	2	0.30	2.25	3.34	2.20	5.50	5.35
	200	45	3	452	461	36.3	2 500	3 200	**21319RZ**	**21319RZK**	109	123	186	171	2.5	0.23	2.92	4.35	2.86	7.16	7.06
	200	67	3	733	726	55.6	2 500	3 200	**22319RZ**	**22319RZK**	109	114	186	174	2.5	0.33	2.02	3.00	1.97	10.2	9.98
100	150	37	1.5	262	332	33.7	2 900	3 900	**23020RZ**	**23020RZK**	109	110	141	138	1.5	0.22	3.01	4.48	2.94	2.34	2.27
	165	52	2	412	510	48.5	2 800	3 600	**23120RZ**	**23120RZK**	110	114	155	147	2	0.29	2.33	3.47	2.28	4.52	4.38
	180	46	2.1	470	481	47.6	2 600	3 400	**22220RZ**	**22220RZK**	112	115	168	163	2	0.25	2.74	4.08	2.68	5.11	5.00
	180	60.3	2.1	533	629	53.5	2 600	3 400	**23220RZ**	**23220RZK**	112	116	168	157	2	0.32	2.09	3.11	2.04	6.85	6.66
	215	47	3	519	524	40.2	2 200	3 000	**21320RZ**	**21320RZK**	114	131	201	184	2.5	0.22	3.02	4.49	2.95	8.79	8.68
	215	73	3	875	877	63.9	2 200	3 000	**22320RZ**	**22320RZK**	114	121	201	187	2.5	0.35	1.95	2.90	1.91	13.2	12.9
110	170	45	2	377	486	48.4	2 600	3 400	**23022RZ**	**23022RZK**	120	123	160	156	2	0.24	2.84	4.23	2.78	3.85	3.74

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

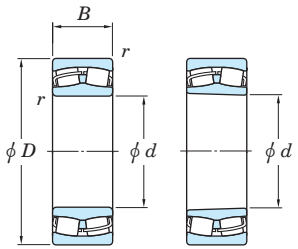
d (110) ~ 140 mm

R, RR



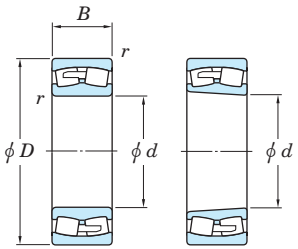
Cylindrical bore | Tapered bore

RZ

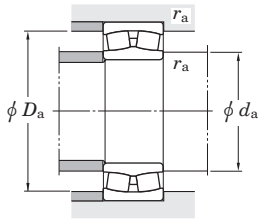


Cylindrical bore | Tapered bore

RHA



Cylindrical bore | Tapered bore

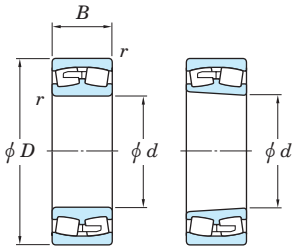


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	d_a max.	D_a max.	D_a min.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
110	170	60	2	472	647	58.6	2 600	3 600	**24022RZ**	**24022RZK30**	120	120	160	151	2	0.32	2.08	3.10	2.04	5.07	4.99
	180	56	2	484	605	53.7	2 500	3 300	**23122RZ**	**23122RZK**	120	125	170	161	2	0.29	2.36	3.51	2.31	5.72	5.54
	180	69	2	569	778	63.4	2 500	3 300	**24122RZ**	**24122RZK30**	120	120	170	154	2	0.37	1.84	2.74	1.80	6.98	6.87
	200	53	2.1	612	642	58.7	2 300	3 000	**22222RZ**	**22222RZK**	122	127	188	180	2	0.26	2.64	3.93	2.58	7.37	7.21
	200	69.8	2.1	672	792	65.4	2 300	3 000	**23222RZ**	**23222RZK**	122	127	188	173	2	0.34	1.99	2.96	1.94	9.76	9.48
	240	50	3	604	616	46.0	1 900	2 600	**21322RZ**	**21322RZK**	124	147	226	205	2.5	0.21	3.19	4.75	3.12	11.8	11.7
	240	80	3	1 040	1 040	77.7	1 900	2 600	**22322RZ**	**22322RZK**	124	136	226	208	2.5	0.33	2.03	3.02	1.98	18.1	17.7
120	180	46	2	394	524	51.6	2 300	3 200	**23024RZ**	**23024RZK**	130	132	170	165	2	0.23	2.95	4.40	2.89	4.20	4.07
	180	60	2	484	709	61.8	2 300	3 200	**24024RZ**	**24024RZK30**	130	130	170	160	2	0.30	2.23	3.32	2.18	5.43	5.34
	200	62	2	571	714	61.2	2 200	3 000	**23124RZ**	**23124RZK**	130	137	190	176	2	0.29	2.34	3.49	2.29	7.98	7.74
	200	80	2	733	1 020	78.6	2 200	3 000	**24124RZ**	**24124RZK30**	130	133	190	172	2	0.38	1.75	2.61	1.72	10.2	10.0
	215	58	2.1	706	764	67.2	2 100	2 800	**22224RZ**	**22224RZK**	132	138	203	193	2	0.26	2.60	3.87	2.54	9.31	9.10
	215	76	2.1	772	956	78.9	2 100	2 900	**23224RZ**	**23224RZK**	132	139	203	185	2	0.34	1.97	2.94	1.93	12.2	11.8
	260	86	3	1 120	1 130	87.2	1 800	2 500	**22324RZ**	**22324RZK**	134	149	246	228	2.5	0.33	2.03	3.02	1.98	22.8	22.3
130	200	52	2	509	674	63.6	2 200	2 900	**23026RZ**	**23026RZK**	140	145	190	182	2	0.24	2.87	4.27	2.80	6.15	5.97
	200	69	2	625	914	77.3	2 200	2 900	**24026RZ**	**24026RZK30**	140	143	190	177	2	0.32	2.14	3.18	2.09	8.03	7.90
	210	64	2	621	799	68.4	2 100	2 800	**23126RZ**	**23126RZK**	140	147	200	187	2	0.28	2.42	3.61	2.37	8.71	8.44
	210	80	2	754	1 080	91.8	2 100	2 800	**24126RZ**	**24126RZK30**	140	145	200	184	2	0.36	1.90	2.83	1.86	10.8	10.6
	230	64	3	821	914	74.4	1 900	2 600	**22226RZ**	**22226RZK**	144	148	216	206	2.5	0.26	2.55	3.80	2.50	11.6	11.3
	230	80	3	880	1 090	89.4	1 900	2 600	**23226RZ**	**23226RZK**	144	151	216	201	2.5	0.33	2.05	3.05	2.00	14.4	14.0
	280	93	4	1 310	1 340	98.6	1 700	2 200	**22326RZ**	**22326RZK**	148	160	262	245	3	0.33	2.03	3.02	1.98	28.5	27.9
140	210	53	2	530	723	67.9	2 100	2 800	**23028RZ**	**23028RZK**	150	155	200	192	2	0.23	2.98	4.44	2.92	6.62	6.42
	210	69	2	640	957	81.7	2 100	2 800	**24028RZ**	**24028RZK30**	150	153	200	188	2	0.30	2.28	3.39	2.23	8.49	8.35
	225	68	2.1	710	940	79.6	1 900	2 600	**23128RZ**	**23128RZK**	152	158	213	201	2	0.28	2.45	3.65	2.40	10.6	10.3
	225	85	2.1	853	1 170	90.7	1 900	2 600	**24128RZ**	**24128RZK30**	152	153	213	194	2	0.36	1.89	2.82	1.85	13.1	12.9
	250	68	3	947	1 030	85.2	1 800	2 300	**22228RZ**	**22228RZK**	154	158	236	224	2.5	0.26	2.60	3.87	2.54	14.5	14.2
	250	88	3	1 020	1 290	103	1 800	2 300	**23228RZ**	**23228RZK**	154	161	236	214	2.5	0.34	1.99	2.96	1.95	19.0	18.4
	300	102	4	1 470	1 570	105	1 500	2 100	**22328RZ**	**22328RZK**	158	172	282	255	3	0.35	1.95	2.90	1.90	35.7	34.9

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

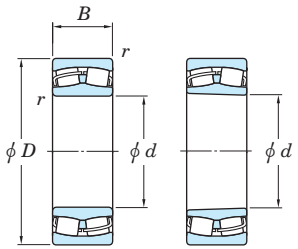
d 150 ~ (170) mm

R, RR



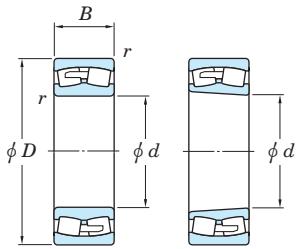
Cylindrical bore | Tapered bore

RZ

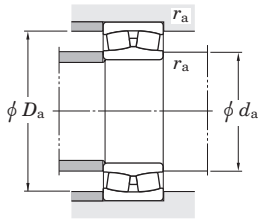


Cylindrical bore | Tapered bore

RHA



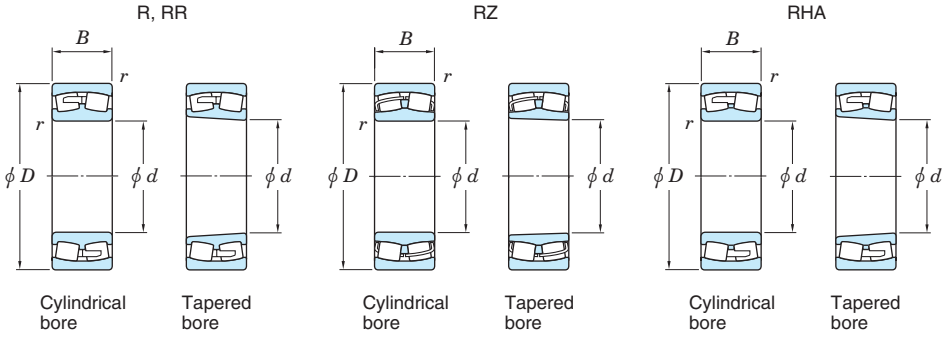
Cylindrical bore | Tapered bore

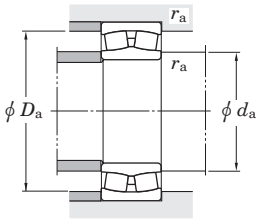


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	d_a max.	D_a max.	D_a min.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
150	210	45	2	418	622	62.5	1 600	2 100	**23930R**	**23930RK**	160	170	200	195	2	0.20	3.44	5.12	3.36	5.09	4.93
	225	56	2.1	579	797	76.3	1 900	2 500	**23030RZ**	**23030RZK**	162	166	213	205	2	0.22	3.04	4.53	2.97	8.01	7.77
	225	75	2.1	724	1 100	90.3	1 900	2 500	**24030RZ**	**24030RZK30**	162	163	213	199	2	0.30	2.23	3.32	2.18	10.6	10.4
	250	80	2.1	902	1 230	102	1 800	2 300	**23130RZ**	**23130RZK**	162	171	238	216	2	0.30	2.24	3.34	2.19	16.4	15.9
	250	100	2.1	1 110	1 590	116	1 800	2 300	**24130RZ**	**24130RZK30**	162	166	238	213	2	0.38	1.77	2.64	1.73	19.9	19.6
	270	73	3	1 080	1 200	102	1 700	2 200	**22230RZ**	**22230RZK**	164	172	256	243	2.5	0.25	2.69	4.00	2.63	18.9	18.5
	270	96	3	1 200	1 540	121	1 700	2 200	**23230RZ**	**23230RZK**	164	173	256	230	2.5	0.34	1.96	2.93	1.92	24.5	23.8
	320	108	4	1 540	1 600	175	1 200	1 500	**22330R**	**22330RK**	168	195	302	273	3	0.38	1.78	2.64	1.74	43.6	42.7
	320	108	4	1 620	1 740	121	1 200	1 500	**22330RHA**	**22330RHAK**	168	196	302	273	3	0.35	1.93	2.87	1.88	40.3	39.4
160	220	45	2	426	649	65.4	1 500	2 000	**23932R**	**23932RK**	170	179	210	204	2	0.19	3.60	5.37	3.52	5.37	5.20
	240	60	2.1	667	924	86.0	1 800	2 300	**23032RZ**	**23032RZK**	172	177	228	219	2	0.22	3.01	4.48	2.94	9.74	9.44
	240	80	2.1	829	1 270	103	1 800	2 300	**24032RZ**	**24032RZK30**	172	175	228	215	2	0.30	2.24	3.34	2.19	12.9	12.7
	270	86	2.1	1 070	1 430	117	1 700	2 200	**23132RZ**	**23132RZK**	172	182	258	234	2	0.30	2.22	3.30	2.17	20.8	20.2
	270	109	2.1	1 270	1 720	145	1 300	1 700	**24132RR**	**24132RRK30**	172	188	258	230	2	0.39	1.72	2.56	1.68	25.9	25.5
	290	80	3	1 110	1 270	127	1 200	1 600	**22232R**	**22232RK**	174	199	276	257	2.5	0.28	2.40	3.57	2.35	23.4	22.9
	290	80	3	1 120	1 320	97.1	1 200	1 600	**22232RHA**	**22232RHAK**	174	200	276	257	2.5	0.27	2.49	3.71	2.44	21.9	21.4
	290	104	3	1 290	1 650	163	1 200	1 600	**23232R**	**23232RK**	174	194	276	245	2.5	0.38	1.79	2.66	1.75	31.0	30.1
	290	104	3	1 370	1 780	139	1 200	1 600	**23232RHA**	**23232RHAK**	174	193	276	245	2.5	0.36	1.87	2.78	1.83	29.4	28.5
	340	114	4	1 720	1 790	188	1 100	1 400	**22332R**	**22332RK**	178	207	322	290	3	0.38	1.76	2.62	1.72	51.9	51.0
	340	114	4	1 780	1 940	135	1 100	1 400	**22332RHA**	**22332RHAK**	178	210	322	290	3	0.35	1.94	2.89	1.90	48.0	47.1
170	230	45	2	441	691	69.6	1 400	1 900	**23934R**	**23934RK**	180	189	220	214	2	0.18	3.78	5.63	3.70	5.67	5.49
	260	67	2.1	795	1 090	97.9	1 700	2 200	**23034RZ**	**23034RZK**	182	189	248	236	2	0.23	2.90	4.31	2.83	13.2	12.8
	260	90	2.1	1 010	1 540	120	1 700	2 200	**24034RZ**	**24034RZK30**	182	184	248	227	2	0.32	2.11	3.15	2.07	17.5	17.2
	280	88	2.1	1 150	1 550	124	1 500	2 100	**23134RZ**	**23134RZK**	182	194	268	249	2	0.29	2.30	3.43	2.25	21.9	21.2
	280	109	2.1	1 320	1 820	154	1 200	1 600	**24134RR**	**24134RRK30**	182	198	268	241	2	0.37	1.80	2.68	1.76	27.2	26.8
	310	86	4	1 190	1 390	141	1 100	1 500	**22234R**	**22234RK**	188	212	292	271	3	0.29	2.29	3.41	2.24	29.0	28.4
	310	86	4	1 260	1 490	109	1 100	1 500	**22234RHA**	**22234RHAK**	188	210	292	271	3	0.28	2.45	3.64	2.39	27.1	26.5
	310	110	4	1 560	1 920	127	1 100	1 500	**23234RR**	**23234RRK**	188	209	292	268	3	0.37	1.85	2.75	1.80	37.2	36.1
	310	110	4	1 520	1 940	147	1 100	1 500	**23234RHA**	**23234RHAK**	188	207	292	261	3	0.36	1.89	2.82	1.85	35.6	34.6

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

d (170) ~ (190) mm



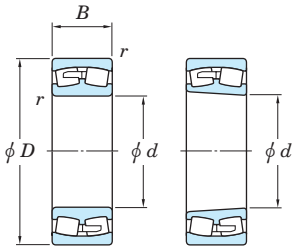


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Con-stant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	d_a max.	D_a max.	D_a min.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
170	360	120	4	1 830	1 920	206	1 000	1 300	**22334R**	**22334RK**	188	221	342	307	3	0.38	1.77	2.64	1.73	62.0	60.8
	360	120	4	1 990	2 200	150	1 000	1 300	**22334RHA**	**22334RHAK**	188	222	342	307	3	0.35	1.95	2.91	1.91	57.3	56.1
180	250	52	2	599	939	88.9	1 300	1 700	**23936R**	**23936RK**	190	203	240	232	2	0.19	3.55	5.29	3.48	8.22	7.97
	280	74	2.1	966	1 330	118	1 500	1 900	**23036RZ**	**23036RZK**	192	202	268	253	2	0.24	2.84	4.23	2.78	17.4	16.9
	280	100	2.1	1 170	1 710	138	1 200	1 600	**24036RR**	**24036RRK30**	192	206	268	246	2	0.34	2.00	2.98	1.96	23.4	23.0
	300	96	3	1 260	1 800	165	1 100	1 500	**23136R**	**23136RK**	194	214	286	259	2.5	0.33	2.04	3.04	2.00	28.4	27.5
	300	96	3	1 330	1 790	139	1 100	1 500	**23136RHA**	**23136RHAK**	194	215	286	265	2.5	0.31	2.19	3.25	2.14	26.5	25.6
	300	118	3	1 530	2 120	176	1 100	1 500	**24136RR**	**24136RRK30**	194	211	286	258	2.5	0.38	1.78	2.65	1.74	34.4	33.9
	300	118	3	1 510	2 240	155	1 100	1 500	**24136RHA**	**24136RHAK30**	194	207	286	255	2.5	0.38	1.79	2.66	1.75	31.8	31.2
	320	86	4	1 220	1 450	165	1 100	1 400	**22236R**	**22236RK**	198	222	302	281	3	0.28	2.37	3.53	2.32	30.5	29.8
	320	86	4	1 320	1 610	118	1 100	1 400	**22236RHA**	**22236RHAK**	198	221	302	281	3	0.26	2.55	3.80	2.50	28.5	27.8
	320	112	4	1 640	2 100	134	1 100	1 400	**23236RR**	**23236RRK**	198	219	302	279	3	0.36	1.87	2.78	1.83	39.8	38.6
	320	112	4	1 660	2 170	166	1 100	1 400	**23236RHA**	**23236RHAK**	198	220	302	277	3	0.34	1.97	2.93	1.92	37.7	36.5
	380	126	4	2 180	2 360	263	920	1 200	**22336R**	**22336RK**	198	237	362	327	3	0.36	1.89	2.81	1.84	71.4	69.9
	380	126	4	2 180	2 410	163	930	1 200	**22336RHA**	**22336RHAK**	198	235	362	323	3	0.34	1.97	2.94	1.93	66.0	64.5
190	260	52	2	608	969	90.7	1 200	1 600	**23938R**	**23938RK**	200	212	250	241	2	0.18	3.69	5.50	3.61	8.40	8.10
	290	75	2.1	923	1 370	132	1 100	1 500	**23038R**	**23038RK**	202	221	278	260	2	0.25	2.67	3.97	2.61	18.8	18.2
	290	75	2.1	992	1 430	115	1 100	1 500	**23038RHA**	**23038RHAK**	202	219	278	260	2	0.25	2.75	4.10	2.69	17.2	16.6
	290	100	2.1	1 240	1 840	161	1 100	1 500	**24038RR**	**24038RRK30**	202	215	278	257	2	0.33	2.06	3.07	2.02	24.5	24.1
	290	100	2.1	1 230	1 920	152	1 100	1 500	**24038RHA**	**24038RHAK30**	202	215	278	256	2	0.32	2.14	3.19	2.09	22.4	22.0
	320	104	3	1 370	2 000	162	1 000	1 400	**23138R**	**23138RK**	204	229	306	275	2.5	0.34	1.96	2.92	1.92	35.5	34.4
	320	104	3	1 520	2 080	161	1 000	1 400	**23138RHA**	**23138RHAK**	204	227	306	281	2.5	0.31	2.14	3.19	2.10	33.2	32.1
	320	128	3	1 750	2 470	198	1 000	1 400	**24138RR**	**24138RRK30**	204	223	306	272	2.5	0.39	1.74	2.59	1.70	43.0	42.4
	320	128	3	1 770	2 630	179	1 000	1 400	**24138RHA**	**24138RHAK30**	204	222	306	272	2.5	0.38	1.76	2.63	1.72	40.1	39.5
	340	92	4	1 390	1 730	172	1 000	1 300	**22238R**	**22238RK**	208	236	322	296	3	0.29	2.29	3.41	2.24	37.4	36.6
	340	92	4	1 420	1 770	128	1 000	1 300	**22238RHA**	**22238RHAK**	208	234	322	296	3	0.27	2.52	3.76	2.46	34.9	34.1
	340	120	4	1 830	2 370	160	1 000	1 300	**23238RR**	**23238RRK**	208	233	322	294	3	0.36	1.86	2.76	1.81	48.5	47.1
	340	120	4	1 870	2 470	185	990	1 300	**23238RHA**	**23238RHAK**	208	233	322	293	3	0.35	1.94	2.89	1.90	44.9	43.5
	400	132	5	2 380	2 610	258	880	1 200	**22338R**	**22338RK**	212	248	378	342	4	0.38	1.79	2.66	1.75	84.1	82.4

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

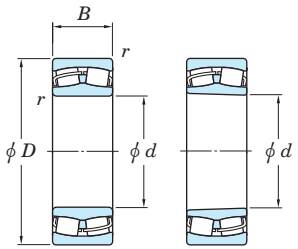
d (190) ~ (220) mm

R, RR



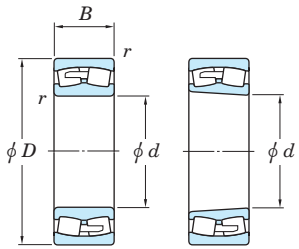
Cylindrical bore | Tapered bore

RZ

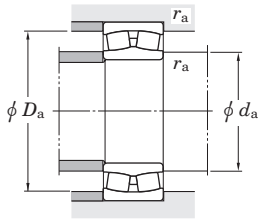


Cylindrical bore | Tapered bore

RHA



Cylindrical bore | Tapered bore

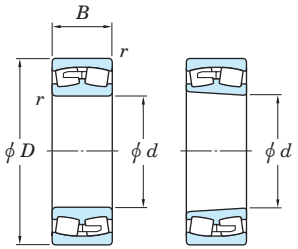


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Con-stant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	d_a max.	D_a max.	D_a min.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
190	400	132	5	2 430	2 810	192	870	1 200	**22338RHA**	**22338RHAK**	212	251	378	342	4	0.34	1.99	2.97	1.95	77.7	76.0
200	280	60	2.1	753	1 190	109	1 100	1 500	**23940R**	**23940RK**	212	226	268	259	2	0.20	3.44	5.13	3.37	12.0	11.6
	310	82	2.1	1 120	1 670	155	1 000	1 400	**23040R**	**23040RK**	212	235	298	278	2	0.26	2.62	3.90	2.56	24.1	23.4
	310	82	2.1	1 180	1 680	133	1 100	1 400	**23040RHA**	**23040RHAK**	212	231	298	278	2	0.25	2.68	3.99	2.62	22.0	21.3
	310	109	2.1	1 430	2 110	180	1 100	1 400	**24040RR**	**24040RRK30**	212	228	298	273	2	0.33	2.02	3.00	1.97	31.2	30.7
	310	109	2.1	1 440	2 230	173	1 100	1 400	**24040RHA**	**24040RHAK30**	212	227	298	272	2	0.33	2.06	3.07	2.02	28.5	28.0
	340	112	3	1 740	2 350	186	980	1 300	**23140RR**	**23140RRK**	214	241	326	298	2.5	0.33	2.04	3.03	1.99	43.3	42.0
	340	112	3	1 730	2 340	178	970	1 300	**23140RHA**	**23140RHAK**	214	239	326	297	2.5	0.32	2.10	3.13	2.06	40.8	39.5
	340	140	3	2 030	2 820	222	990	1 300	**24140RR**	**24140RRK30**	214	234	326	289	2.5	0.40	1.68	2.49	1.64	53.3	52.5
	340	140	3	2 000	2 970	196	990	1 300	**24140RHA**	**24140RHAK30**	214	232	326	286	2.5	0.41	1.65	2.46	1.62	49.5	48.7
	360	98	4	1 620	2 050	138	930	1 200	**22240RR**	**22240RRK**	218	252	342	316	3	0.30	2.26	3.36	2.21	45.0	44.0
	360	98	4	1 630	2 030	146	940	1 300	**22240RHA**	**22240RHAK**	218	247	342	316	3	0.27	2.50	3.72	2.45	42.0	41.0
	360	128	4	1 950	2 610	228	940	1 300	**23240R**	**23240RK**	218	244	342	306	3	0.38	1.79	2.67	1.75	58.1	56.4
	360	128	4	2 080	2 780	209	930	1 200	**23240RHA**	**23240RHAK**	218	245	342	309	3	0.35	1.92	2.86	1.88	55.1	53.4
	420	138	5	2 510	2 750	288	830	1 100	**22340R**	**22340RK**	222	260	398	359	4	0.38	1.80	2.68	1.76	95.4	93.5
	420	138	5	2 570	2 920	193	820	1 100	**22340RHA**	**22340RHAK**	222	262	398	356	4	0.34	1.99	2.97	1.95	88.1	86.2
220	300	60	2.1	792	1 300	119	1 000	1 400	**23944R**	**23944RK**	232	246	288	279	2	0.18	3.70	5.50	3.61	13.0	12.6
	340	90	3	1 230	1 890	173	940	1 300	**23044R**	**23044RK**	234	256	326	301	2.5	0.26	2.55	3.80	2.50	31.5	30.6
	340	90	3	1 370	1 950	148	940	1 200	**23044RHA**	**23044RHAK**	234	255	326	307	2.5	0.25	2.69	4.01	2.63	28.8	27.9
	340	118	3	1 660	2 480	208	950	1 300	**24044RR**	**24044RRK30**	234	251	326	300	2.5	0.33	2.04	3.04	2.00	40.5	39.8
	340	118	3	1 680	2 630	199	950	1 300	**24044RHA**	**24044RHAK30**	234	248	326	297	2.5	0.33	2.08	3.09	2.03	37.0	36.4
	370	120	4	1 810	2 700	205	880	1 200	**23144R**	**23144RK**	238	266	352	319	3	0.34	2.00	2.98	1.96	54.8	53.2
	370	120	4	2 000	2 790	208	870	1 200	**23144RHA**	**23144RHAK**	238	263	352	324	3	0.31	2.15	3.20	2.10	51.2	49.6
	370	150	4	2 360	3 390	258	880	1 200	**24144RR**	**24144RRK30**	238	258	352	315	3	0.39	1.71	2.55	1.67	67.3	66.2
	370	150	4	2 330	3 550	229	880	1 200	**24144RHA**	**24144RHAK30**	238	255	352	313	3	0.40	1.69	2.52	1.65	62.0	61.0
	400	108	4	2 000	2 410	257	820	1 100	**22244RR**	**22244RRK**	238	276	382	355	3	0.28	2.40	3.57	2.34	60.3	59.0
	400	108	4	1 980	2 440	168	820	1 100	**22244RHA**	**22244RHAK**	238	274	382	349	3	0.27	2.52	3.76	2.47	58.8	57.5
	400	144	4	2 350	3 200	259	830	1 100	**23244R**	**23244RK**	238	268	382	336	3	0.39	1.71	2.55	1.68	81.6	79.2
	400	144	4	2 520	3 350	239	810	1 100	**23244RHA**	**23244RHAK**	238	272	382	346	3	0.36	1.89	2.81	1.85	77.4	75.0

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

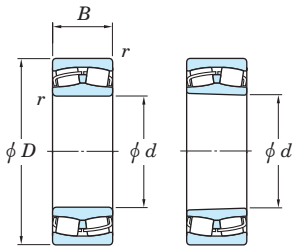
d (220) ~ (260) mm

R, RR



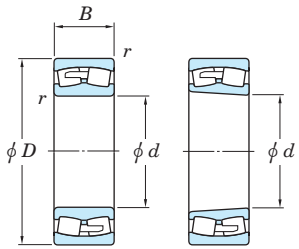
Cylindrical bore | Tapered bore

RZ

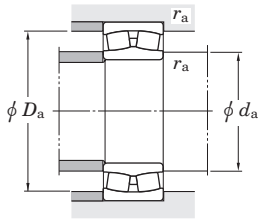


Cylindrical bore | Tapered bore

RHA



Cylindrical bore | Tapered bore

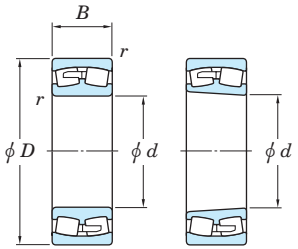


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	d_a max.	D_a max.	D_a min.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
220	460	145	5	2 980	3 380	359	720	960	**22344R**	**22344RK**	242	290	438	393	4	0.34	2.00	2.99	1.96	124	122
	460	145	5	2 960	3 470	226	730	970	**22344RHA**	**22344RHAK**	242	290	438	390	4	0.32	2.08	3.09	2.03	115	113
240	320	60	2.1	814	1 380	128	940	1 300	**23948R**	**23948RK**	252	265	308	298	2	0.17	3.95	5.88	3.86	14.0	13.5
	360	92	3	1 480	2 190	161	860	1 100	**23048RR**	**23048RRK**	254	276	346	327	2.5	0.25	2.73	4.07	2.67	33.9	32.9
	360	92	3	1 470	2 180	166	860	1 100	**23048RHA**	**23048RHAK**	254	275	346	327	2.5	0.24	2.83	4.21	2.77	31.9	30.9
	360	118	3	1 750	2 710	228	870	1 200	**24048RR**	**24048RRK30**	254	272	346	321	2.5	0.31	2.20	3.27	2.15	43.5	42.9
	360	118	3	1 750	2 840	215	870	1 200	**24048RHA**	**24048RHAK30**	254	269	346	321	2.5	0.30	2.24	3.33	2.19	39.6	39.0
	400	128	4	2 280	3 220	213	790	1 100	**23148RR**	**23148RRK**	258	287	382	353	3	0.32	2.11	3.14	2.06	67.2	65.1
	400	128	4	2 270	3 200	233	790	1 000	**23148RHA**	**23148RHAK**	258	286	382	353	3	0.31	2.19	3.25	2.14	63.1	61.1
	400	160	4	2 640	3 850	287	800	1 100	**24148RR**	**24148RRK30**	258	280	382	340	3	0.39	1.75	2.60	1.71	82.7	81.4
	400	160	4	2 670	4 130	262	800	1 100	**24148RHA**	**24148RHAK30**	258	278	382	340	3	0.39	1.72	2.56	1.68	76.6	75.3
	440	120	4	2 390	2 940	295	730	970	**22248R**	**22248RK**	258	299	422	384	3	0.29	2.35	3.50	2.30	85.0	83.2
	440	120	4	2 400	2 990	202	730	970	**22248RHA**	**22248RHAK**	258	299	422	384	3	0.27	2.49	3.71	2.43	79.4	77.6
	440	160	4	3 050	3 970	310	730	970	**23248RR**	**23248RRK**	258	295	422	376	3	0.38	1.78	2.64	1.74	110	107
	440	160	4	3 080	4 130	289	730	970	**23248RHA**	**23248RHAK**	258	295	422	376	3	0.36	1.87	2.78	1.83	104	101
	500	155	5	3 360	4 020	347	650	870	**22348R**	**22348RK**	262	320	478	420	4	0.35	1.94	2.89	1.90	157	154
	500	155	5	3 400	3 990	255	650	870	**22348RHA**	**22348RHAK**	262	315	478	426	4	0.32	2.12	3.16	2.07	145	142
260	360	75	2.1	1 140	1 880	160	820	1 100	**23952R**	**23952RK**	272	292	348	333	2	0.19	3.54	5.27	3.46	24.0	23.3
	400	104	4	1 670	2 570	212	760	1 000	**23052R**	**23052RK**	278	304	382	359	3	0.25	2.65	3.95	2.59	50.7	49.3
	400	104	4	1 850	2 720	201	760	1 000	**23052RHA**	**23052RHAK**	278	302	382	359	3	0.25	2.75	4.10	2.69	46.3	44.9
	400	140	4	2 280	3 570	282	770	1 000	**24052RR**	**24052RRK30**	278	296	382	352	3	0.33	2.02	3.01	1.98	66.3	65.2
	400	140	4	2 270	3 670	265	770	1 000	**24052RHA**	**24052RHAK30**	278	292	382	347	3	0.33	2.06	3.07	2.02	60.3	59.4
	440	144	4	2 760	3 850	231	710	940	**23152RR**	**23152RRK**	278	313	422	387	3	0.33	2.05	3.06	2.01	92.2	89.4
	440	144	4	2 790	4 000	285	700	930	**23152RHA**	**23152RHAK**	278	311	422	384	3	0.32	2.12	3.16	2.08	87.4	84.6
	440	180	4	3 250	4 700	345	720	950	**24152RR**	**24152RRK30**	278	304	422	374	3	0.40	1.69	2.51	1.65	114	112
	440	180	4	3 210	4 950	309	720	950	**24152RHA**	**24152RHAK30**	278	299	422	368	3	0.41	1.66	2.47	1.62	106	105
	480	130	5	2 800	3 460	347	650	870	**22252R**	**22252RK**	282	326	458	419	4	0.28	2.40	3.57	2.35	110	108
	480	130	5	2 790	3 430	226	650	870	**22252RHA**	**22252RHAK**	282	324	458	418	4	0.27	2.50	3.72	2.44	103	101
	480	174	5	3 440	4 640	326	640	860	**23252R**	**23252RK**	282	325	458	408	4	0.40	1.69	2.51	1.65	144	140

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

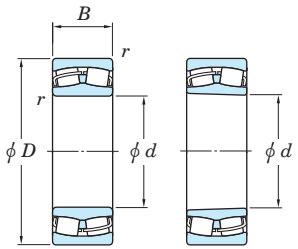
d (260) ~ (300) mm

R, RR



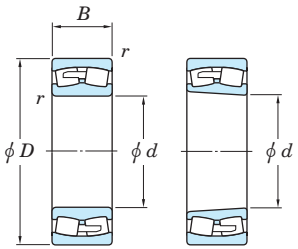
Cylindrical bore Tapered bore

RZ

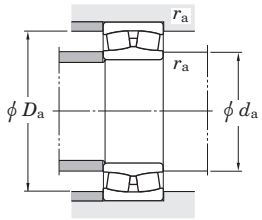


Cylindrical bore Tapered bore

RHA



Cylindrical bore Tapered bore

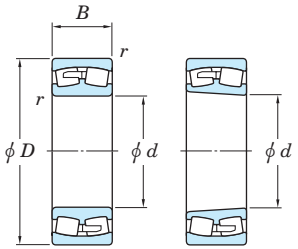


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	d_a max.	D_a max.	D_a min.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
260	480	174	5	3 590	4 900	280	650	860	**23252RHA**	**23252RHAK**	282	322	458	408	4	0.36	1.87	2.78	1.83	137	133
	540	165	6	3 540	4 380	363	590	780	**22352R**	**22352RK**	288	346	512	453	5	0.35	1.94	2.89	1.90	196	192
	540	165	6	3 900	4 620	290	580	780	**22352RHA**	**22352RHAK**	288	342	512	461	5	0.31	2.15	3.21	2.11	181	177
280	380	75	2.1	1 160	1 960	165	760	1 000	**23956R**	**23956RK**	292	312	368	353	2	0.18	3.74	5.57	3.66	26.0	25.2
	420	106	4	1 790	2 860	235	710	950	**23056R**	**23056RK**	298	322	402	377	3	0.25	2.74	4.08	2.68	54.5	52.9
	420	106	4	1 940	2 950	218	700	940	**23056RHA**	**23056RHAK**	298	322	402	380	3	0.24	2.87	4.27	2.80	49.8	48.2
	420	140	4	2 370	3 780	291	710	950	**24056RR**	**24056RRK30**	298	316	402	373	3	0.31	2.15	3.21	2.11	70.2	69.1
	420	140	4	2 390	4 000	287	710	950	**24056RHA**	**24056RHAK30**	298	314	402	372	3	0.31	2.20	3.28	2.15	64.0	62.9
	460	146	5	2 910	4 160	250	660	880	**23156RR**	**23156RRK**	302	332	438	407	4	0.32	2.14	3.18	2.09	98.8	95.7
	460	146	5	2 940	4 290	304	650	870	**23156RHA**	**23156RHAK**	302	331	438	406	4	0.30	2.22	3.30	2.17	93.4	90.3
	460	180	5	3 390	5 140	370	660	880	**24156RR**	**24156RRK30**	302	326	438	396	4	0.38	1.79	2.67	1.75	122	120
	460	180	5	3 320	5 240	322	660	880	**24156RHA**	**24156RHAK30**	302	321	438	390	4	0.38	1.76	2.62	1.72	113	112
	500	130	5	2 640	3 380	308	610	810	**22256R**	**22256RK**	302	347	478	438	4	0.28	2.42	3.60	2.37	114	112
	500	130	5	2 900	3 670	240	610	810	**22256RHA**	**22256RHAK**	302	346	478	440	4	0.26	2.64	3.93	2.58	106	104
	500	176	5	3 370	4 910	323	610	820	**23256R**	**23256RK**	302	345	478	421	4	0.37	1.83	2.72	1.79	153	149
	500	176	5	3 770	5 300	365	600	800	**23256RHA**	**23256RHAK**	302	343	478	430	4	0.35	1.95	2.91	1.91	145	141
	580	175	6	3 930	4 910	407	530	710	**22356R**	**22356RK**	308	372	552	486	5	0.34	1.98	2.95	1.93	229	225
	580	175	6	4 390	5 260	325	530	700	**22356RHA**	**22356RHAK**	308	367	552	495	5	0.31	2.19	3.25	2.14	212	208
300	420	90	3	1 610	2 610	220	680	910	**23960R**	**23960RK**	314	336	406	387	2.5	0.20	3.42	5.09	3.34	40.0	38.8
	460	118	4	2 190	3 480	286	630	840	**23060R**	**23060RK**	318	351	442	412	3	0.25	2.69	4.00	2.63	75.8	73.7
	460	118	4	2 370	3 700	255	630	840	**23060RHA**	**23060RHAK**	318	347	442	416	3	0.24	2.79	4.16	2.73	68.9	66.8
	460	160	4	2 950	4 690	354	640	850	**24060RR**	**24060RRK30**	318	342	442	406	3	0.33	2.04	3.04	2.00	99.5	97.9
	460	160	4	2 950	4 910	350	640	850	**24060RHA**	**24060RHAK30**	318	338	442	404	3	0.32	2.09	3.11	2.04	90.7	89.1
	500	160	5	3 450	5 030	351	590	790	**23160RR**	**23160RRK**	322	358	478	439	4	0.32	2.09	3.11	2.04	131	127
	500	160	5	3 430	4 970	345	580	780	**23160RHA**	**23160RHAK**	322	357	478	439	4	0.31	2.18	3.25	2.13	123	119
	500	200	5	4 160	6 280	433	590	790	**24160RR**	**24160RRK30**	322	349	478	425	4	0.40	1.67	2.49	1.63	162	160
	500	200	5	4 030	6 420	385	590	790	**24160RHA**	**24160RHAK30**	322	347	478	424	4	0.39	1.72	2.56	1.68	150	148
	540	140	5	3 360	4 330	412	550	740	**22260R**	**22260RK**	322	368	518	467	4	0.27	2.48	3.69	2.43	145	142
	540	140	5	3 320	4 360	284	550	740	**22260RHA**	**22260RHAK**	322	370	518	467	4	0.26	2.62	3.90	2.56	135	132

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

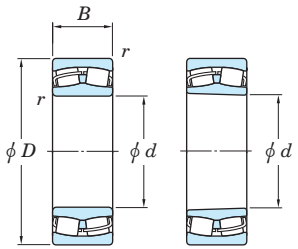
d (300) ~ (360) mm

R, RR



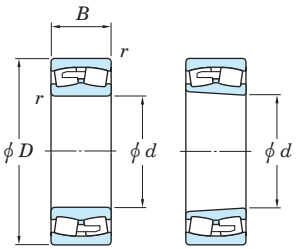
Cylindrical bore | Tapered bore

RZ

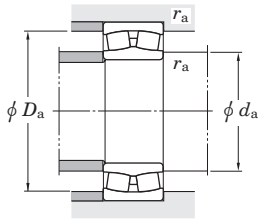


Cylindrical bore | Tapered bore

RHA



Cylindrical bore | Tapered bore

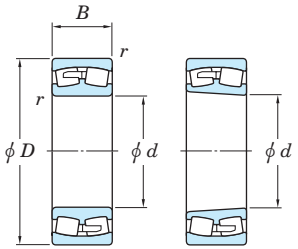


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass (kg)	
											d_a		D_a		r_a						
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	min.	max.	max.	min.	max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
300	540	192	5	4 300	5 910	401	540	720	**23260R**	**23260RK**	322	370	518	464	4	0.37	1.83	2.72	1.79	197	192
	540	192	5	4 440	6 310	429	540	720	**23260RHA**	**23260RHAK**	322	371	518	464	4	0.35	1.93	2.88	1.89	187	182
	620	185	7.5	4 890	5 430	555	470	630	**22360R**	**22360RK**	336	390	584	547	6	0.32	2.09	3.10	2.04	289	284
320	440	90	3	1 670	2 870	233	630	840	**23964R**	**23964RK**	334	358	426	408	2.5	0.19	3.61	5.38	3.53	43.0	41.7
	480	121	4	2 290	3 740	295	590	790	**23064R**	**23064RK**	338	369	462	431	3	0.24	2.76	4.11	2.70	81.2	78.8
	480	121	4	2 490	3 850	278	590	780	**23064RHA**	**23064RHAK**	338	367	462	436	3	0.24	2.87	4.27	2.80	74.5	72.1
	480	160	4	3 020	4 920	382	590	790	**24064RR**	**24064RRK30**	338	363	462	427	3	0.31	2.16	3.22	2.11	105	103
	480	160	4	3 060	5 230	363	590	790	**24064RHA**	**24064RHAK30**	338	360	462	425	3	0.31	2.21	3.29	2.16	93.4	91.4
	540	176	5	3 650	5 700	366	530	700	**23164R**	**23164RK**	342	389	518	467	4	0.33	2.04	3.04	2.00	171	166
	540	176	5	4 040	5 960	404	530	700	**23164RHA**	**23164RHAK**	342	383	518	472	4	0.32	2.13	3.17	2.08	160	155
	540	218	5	4 680	6 950	486	530	710	**24164RR**	**24164RRK30**	342	373	518	460	4	0.39	1.72	2.56	1.68	208	205
	540	218	5	4 550	7 190	429	530	710	**24164RHA**	**24164RHAK30**	342	371	518	458	4	0.40	1.70	2.52	1.66	199	196
	580	150	5	3 420	4 540	385	490	660	**22264R**	**22264RK**	342	402	558	504	4	0.28	2.41	3.59	2.35	175	171
	580	208	5	4 550	6 550	496	500	670	**23264R**	**23264RK**	342	394	558	488	4	0.38	1.76	2.62	1.72	249	242
	580	208	5	5 020	7 030	464	490	650	**23264RHA**	**23264RHAK**	342	392	558	495	4	0.36	1.90	2.83	1.86	236	229
340	460	90	3	1 680	2 980	242	590	790	**23968R**	**23968RK**	354	377	446	426	2.5	0.18	3.82	5.69	3.74	45.0	43.6
	520	133	5	2 670	4 330	353	530	710	**23068R**	**23068RK**	362	397	498	465	4	0.25	2.69	4.00	2.63	108	105
	520	133	5	2 930	4 470	312	530	710	**23068RHA**	**23068RHAK**	362	393	498	468	4	0.24	2.80	4.18	2.74	98.7	95.7
	520	180	5	3 680	5 970	432	530	710	**24068RR**	**24068RRK30**	362	387	498	460	4	0.33	2.06	3.06	2.01	142	140
	520	180	5	3 720	6 330	430	530	710	**24068RHA**	**24068RHAK30**	362	385	498	459	4	0.32	2.11	3.14	2.06	130	128
	580	190	5	4 130	6 430	472	480	640	**23168R**	**23168RK**	362	413	558	497	4	0.34	1.97	2.93	1.93	216	210
	580	190	5	4 620	6 720	449	480	640	**23168RHA**	**23168RHAK**	362	407	558	503	4	0.32	2.11	3.14	2.06	202	196
	580	243	5	5 570	8 400	564	490	650	**24168RR**	**24168RRK30**	362	396	558	490	4	0.41	1.64	2.45	1.61	270	266
	580	243	5	5 490	8 810	449	490	650	**24168RHA**	**24168RHAK30**	362	390	558	482	4	0.42	1.61	2.39	1.57	259	255
	620	165	6	4 430	5 430	551	440	590	**22268R**	**22268RK**	368	424	592	551	5	0.28	2.43	3.61	2.37	221	216
	620	224	6	5 130	7 560	526	450	600	**23268R**	**23268RK**	368	423	592	521	5	0.38	1.77	2.63	1.73	306	297
	620	224	6	5 690	8 030	517	440	590	**23268RHA**	**23268RHAK**	368	418	592	532	5	0.36	1.88	2.81	1.84	290	281
360	480	90	3	1 710	3 060	248	550	730	**23972R**	**23972RK**	374	399	466	447	2.5	0.17	3.95	5.88	3.86	46.5	45.0

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

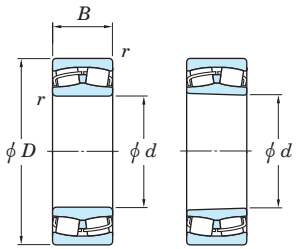
d (360) ~ (400) mm

R, RR



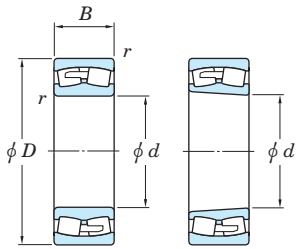
Cylindrical bore | Tapered bore

RZ

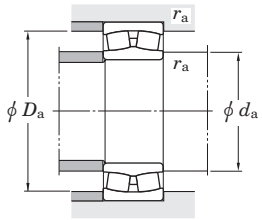


Cylindrical bore | Tapered bore

RHA



Cylindrical bore | Tapered bore

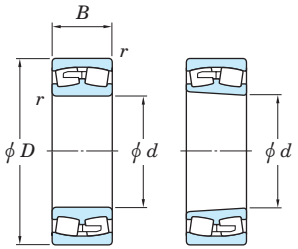


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Con-stant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	d_a max.	D_a max.	D_a min.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
360	540	134	5	2 860	4 800	375	500	660	**23072R**	**23072RK**	382	416	518	484	4	0.24	2.76	4.11	2.70	115	111
	540	134	5	3 040	4 770	334	500	660	**23072RHA**	**23072RHAK**	382	414	518	489	4	0.23	2.92	4.34	2.85	105	101
	540	180	5	3 810	6 300	465	500	660	**24072RR**	**24072RRK30**	382	407	518	481	4	0.31	2.15	3.21	2.11	149	147
	540	180	5	3 810	6 620	446	500	660	**24072RHA**	**24072RHAK30**	382	406	518	480	4	0.30	2.22	3.30	2.17	135	133
	600	192	5	4 740	7 040	459	440	590	**23172R**	**23172RK**	382	431	578	527	4	0.33	2.07	3.09	2.03	228	221
	600	192	5	4 830	7 210	474	450	590	**23172RHA**	**23172RHAK**	382	429	578	527	4	0.31	2.19	3.25	2.14	213	206
	600	243	5	5 080	7 690	437	450	600	**24172R**	**24172RK30**	382	420	578	512	4	0.39	1.74	2.59	1.70	287	283
	600	243	5	5 580	9 180	517	460	610	**24172RHA**	**24172RHAK30**	382	413	578	505	4	0.40	1.69	2.51	1.65	274	270
	650	170	6	4 710	5 830	583	410	550	**22272R**	**22272RK**	388	447	622	579	5	0.27	2.47	3.68	2.42	248	243
	650	232	6	6 080	8 810	548	410	540	**23272R**	**23272RK**	388	446	622	555	5	0.37	1.83	2.72	1.79	346	336
	650	232	6	6 220	9 050	591	410	550	**23272RHA**	**23272RHAK**	388	442	622	558	5	0.35	1.92	2.85	1.87	328	318
380	520	106	4	2 220	3 940	295	500	660	**23976R**	**23976RK**	398	425	502	481	3	0.19	3.62	5.39	3.54	70.0	67.9
	560	135	5	2 910	4 970	355	470	630	**23076R**	**23076RK**	402	433	538	503	4	0.24	2.79	4.16	2.73	122	118
	560	135	5	3 160	5 080	354	460	620	**23076RHA**	**23076RHAK**	402	434	538	512	4	0.22	3.03	4.51	2.96	112	108
	560	180	5	3 900	6 590	486	470	620	**24076RR**	**24076RRK30**	402	428	538	502	4	0.30	2.26	3.36	2.21	156	154
	560	180	5	3 900	6 910	454	470	620	**24076RHA**	**24076RHAK30**	402	426	538	502	4	0.29	2.32	3.45	2.27	142	139
	620	194	5	4 520	7 320	442	420	560	**23176R**	**23176RK**	402	454	598	540	4	0.31	2.18	3.24	2.13	240	233
	620	194	5	5 030	7 700	503	420	560	**23176RHA**	**23176RHAK**	402	450	598	547	4	0.30	2.26	3.36	2.21	224	217
	620	243	5	5 300	8 220	467	430	570	**24176R**	**24176RK30**	402	439	598	529	4	0.38	1.78	2.65	1.74	302	297
	620	243	5	5 870	9 840	561	420	560	**24176RHA**	**24176RHAK30**	402	438	598	534	4	0.38	1.78	2.65	1.74	288	283
	680	240	6	6 510	9 500	590	380	500	**23276R**	**23276RK**	408	469	652	583	5	0.36	1.85	2.76	1.81	386	375
	680	240	6	6 660	9 760	622	380	510	**23276RHA**	**23276RHAK**	408	466	652	586	5	0.35	1.94	2.89	1.90	365	354
400	540	106	4	2 350	4 300	320	470	620	**23980R**	**23980RK**	418	443	522	500	3	0.18	3.76	5.59	3.67	73.0	70.7
	600	148	5	3 390	5 790	408	420	560	**23080R**	**23080RK**	422	462	578	540	4	0.24	2.84	4.23	2.78	155	151
	600	148	5	3 690	5 860	398	420	560	**23080RHA**	**23080RHAK**	422	460	578	543	4	0.23	2.94	4.37	2.87	142	138
	600	200	5	4 820	8 110	444	430	570	**24080R**	**24080RK30**	422	450	578	531	4	0.32	2.09	3.12	2.05	206	203
	600	200	5	4 620	8 140	535	420	570	**24080RHA**	**24080RHAK30**	422	450	578	534	4	0.31	2.21	3.29	2.16	192	189
	650	200	6	4 730	7 780	521	390	520	**23180R**	**23180RK**	428	476	622	564	5	0.31	2.19	3.25	2.14	273	265

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

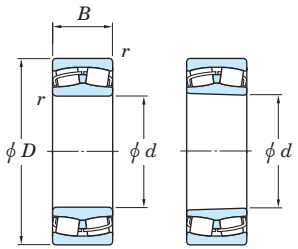
d (400) ~ (460) mm

R, RR



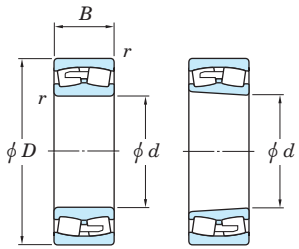
Cylindrical bore | Tapered bore

RZ

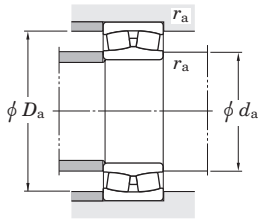


Cylindrical bore | Tapered bore

RHA



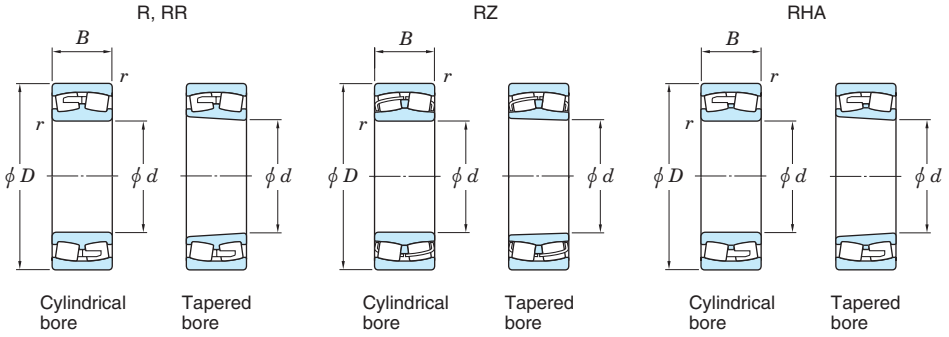
Cylindrical bore | Tapered bore

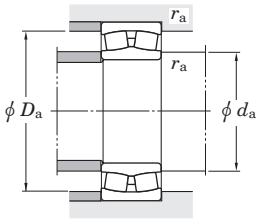


Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Constant	Axial load factors			(Refer.) Mass (kg)	
											d_a		D_a		r_a						
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	min.	max.	max.	min.	max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
400	650	200	6	5 410	8 300	542	390	520	**23180RHA**	**23180RHAK**	428	473	622	574	5	0.29	2.30	3.43	2.25	255	247
	650	250	6	5 840	9 140	499	390	530	**24180R**	**24180RK30**	428	461	622	558	5	0.37	1.82	2.70	1.78	338	333
	650	250	6	6 290	10 600	600	390	520	**24180RHA**	**24180RHAK30**	428	462	622	558	5	0.37	1.82	2.71	1.78	322	317
	720	256	6	6 540	9 850	590	350	470	**23280R**	**23280RK**	428	496	692	605	5	0.37	1.80	2.69	1.76	468	454
	720	256	6	7 320	10 600	665	350	460	**23280RHA**	**23280RHAK**	428	489	692	619	5	0.35	1.92	2.86	1.88	441	427
420	560	106	4	2 330	4 320	331	430	580	**23984R**	**23984RK**	438	465	542	522	3	0.17	3.91	5.82	3.82	76.0	73.6
	620	150	5	3 500	6 120	412	400	530	**23084R**	**23084RK**	442	483	598	560	4	0.23	2.90	4.31	2.83	164	159
	620	150	5	3 820	6 230	425	400	530	**23084RHA**	**23084RHAK**	442	480	598	563	4	0.22	3.02	4.49	2.95	150	145
	620	200	5	4 510	7 600	438	400	530	**24084R**	**24084RK30**	442	471	598	554	4	0.30	2.23	3.32	2.18	212	209
	620	200	5	4 730	8 490	555	400	530	**24084RHA**	**24084RHAK30**	442	471	598	554	4	0.29	2.31	3.44	2.26	198	195
	700	224	6	5 620	9 110	583	350	470	**23184R**	**23184RK**	448	506	672	604	5	0.33	2.03	3.02	1.98	363	352
	700	224	6	6 330	9 630	616	350	470	**23184RHA**	**23184RHAK**	448	500	672	615	5	0.31	2.19	3.25	2.14	339	328
	700	280	6	6 840	10 600	574	360	480	**24184R**	**24184RK30**	448	486	672	593	5	0.40	1.71	2.54	1.67	445	438
	700	280	6	7 420	12 400	685	350	470	**24184RHA**	**24184RHAK30**	448	486	672	596	5	0.39	1.72	2.56	1.68	425	418
	760	272	7.5	8 130	11 500	754	320	430	**23284R**	**23284RK**	456	514	724	652	6	0.37	1.84	2.74	1.80	556	540
	760	272	7.5	8 230	11 900	735	320	430	**23284RHA**	**23284RHAK**	456	512	724	652	6	0.36	1.90	2.83	1.86	525	508
440	600	118	4	2 910	5 330	387	400	530	**23988R**	**23988RK**	458	490	582	554	3	0.18	3.75	5.58	3.66	101	97.8
	650	157	6	3 790	6 540	455	370	500	**23088R**	**23088RK**	468	501	622	584	5	0.24	2.76	4.11	2.70	188	183
	650	157	6	4 230	6 910	465	370	490	**23088RHA**	**23088RHAK**	468	504	622	591	5	0.22	3.04	4.53	2.97	172	167
	650	212	6	4 910	8 320	475	370	490	**24088R**	**24088RK30**	468	494	622	579	5	0.29	2.35	3.50	2.30	247	243
	650	212	6	5 290	9 560	618	370	490	**24088RHA**	**24088RHAK30**	468	492	622	575	5	0.30	2.28	3.39	2.23	231	227
	720	226	6	5 800	9 600	591	330	440	**23188R**	**23188RK**	468	526	692	625	5	0.33	2.08	3.09	2.03	378	366
	720	226	6	6 590	10 300	655	330	440	**23188RHA**	**23188RHAK**	468	521	692	636	5	0.30	2.25	3.34	2.20	353	341
	720	280	6	7 080	11 200	589	340	450	**24188R**	**24188RK30**	468	507	692	615	5	0.38	1.76	2.62	1.72	460	453
	720	280	6	7 540	12 900	707	330	440	**24188RHA**	**24188RHAK30**	468	509	692	616	5	0.38	1.79	2.67	1.75	439	432
	790	280	7.5	8 580	12 300	793	300	400	**23288R**	**23288RK**	476	540	754	684	6	0.36	1.86	2.77	1.82	613	595
	790	280	7.5	8 670	12 700	776	300	390	**23288RHA**	**23288RHAK**	476	539	754	682	6	0.35	1.93	2.88	1.89	580	562
460	600	90	3	1 800	3 660	306	350	460	**23896R**	**23896RK**	476	519	586	568	2.5	0.13	5.06	7.53	4.95	60.4	58.4

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

d (460) ~ 500 mm

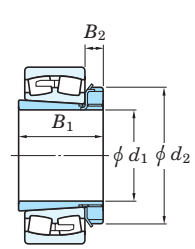




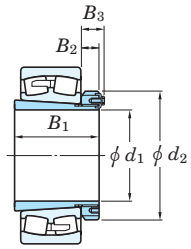
Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.		Mounting dimensions (mm)					Con-stant	Axial load factors			(Refer.) Mass (kg)	
d	D	B	r min.	C_r	C_{0r}	C_u	Grease lub.	Oil lub.	Cylindrical bore	Tapered bore	d_a min.	d_a max.	D_a max.	D_a min.	r_a max.	e	Y_1	Y_2	Y_0	Cylindrical bore	Tapered bore
460	620	118	4	2 890	5 350	404	370	500	**23992R**	**23992RK**	478	512	602	577	3	0.17	3.89	5.79	3.80	107	104
	680	163	6	4 060	7 170	480	340	460	**23092R**	**23092RK**	488	529	652	613	5	0.23	2.92	4.34	2.85	215	209
	680	163	6	4 520	7 430	497	340	460	**23092RHA**	**23092RHAK**	488	527	652	618	5	0.22	3.04	4.53	2.97	197	191
	680	218	6	5 740	10 100	536	340	460	**24092R**	**24092RK30**	488	519	652	607	5	0.30	2.23	3.32	2.18	277	272
	680	218	6	5 660	10 300	656	340	460	**24092RHA**	**24092RHAK30**	488	518	652	604	5	0.29	2.33	3.46	2.27	259	254
	760	240	7.5	6 510	10 800	648	310	410	**23192R**	**23192RK**	496	552	724	656	6	0.33	2.07	3.09	2.03	450	436
	760	240	7.5	7 240	11 200	697	300	400	**23192RHA**	**23192RHAK**	496	546	724	669	6	0.30	2.22	3.31	2.17	420	406
	760	300	7.5	7 320	12 200	597	310	410	**24192R**	**24192RK30**	496	537	724	647	6	0.35	1.95	2.90	1.91	550	541
	760	300	7.5	8 390	14 200	746	310	410	**24192RHA**	**24192RHAK30**	496	535	724	651	6	0.38	1.75	2.61	1.72	525	516
	830	296	7.5	9 520	13 700	867	270	370	**23292R**	**23292RK**	496	567	794	718	6	0.36	1.85	2.76	1.81	720	699
	830	296	7.5	9 600	14 200	856	270	360	**23292RHA**	**23292RHAK**	496	564	794	714	6	0.35	1.92	2.85	1.87	679	658
480	650	128	5	3 290	6 130	446	350	460	**23996R**	**23996RK**	502	534	628	603	4	0.18	3.75	5.59	3.67	123	119
	700	165	6	4 190	7 540	505	320	430	**23096R**	**23096RK**	508	549	672	633	5	0.22	3.01	4.47	2.94	225	218
	700	165	6	4 670	7 860	532	320	430	**23096RHA**	**23096RHAK**	508	548	672	639	5	0.22	3.12	4.64	3.05	206	199
	700	218	6	5 540	9 650	514	320	430	**24096R**	**24096RK30**	508	539	672	626	5	0.29	2.32	3.45	2.26	287	282
	700	218	6	5 800	10 700	492	320	430	**24096RHA**	**24096RHAK30**	508	537	672	626	5	0.28	2.41	3.59	2.35	268	263
	790	248	7.5	6 840	11 500	698	280	380	**23196R**	**23196RK**	516	579	754	685	6	0.32	2.09	3.12	2.05	503	488
	790	248	7.5	7 740	12 000	638	280	380	**23196RHA**	**23196RHAK**	516	570	754	697	6	0.30	2.24	3.34	2.19	470	455
	790	308	7.5	8 730	14 800	707	280	380	**24196R**	**24196RK30**	516	560	754	678	6	0.39	1.74	2.59	1.70	606	597
	790	308	7.5	9 880	15 900	792	290	380	**24196RHA**	**24196RHAK30**	516	553	754	684	6	0.38	1.78	2.65	1.74	580	568
	870	310	7.5	10 500	15 100	953	250	340	**23296R**	**23296RK**	516	588	834	745	6	0.36	1.85	2.75	1.81	831	807
	870	310	7.5	10 600	15 700	791	250	340	**23296RHA**	**23296RHAK**	516	589	834	748	6	0.35	1.91	2.85	1.87	785	761
500	670	128	5	3 330	6 310	447	330	440	**239/500R**	**239/500RK**	522	553	648	622	4	0.17	3.87	5.76	3.79	131	127
	720	167	6	4 490	8 090	561	310	410	**230/500R**	**230/500RK**	528	568	692	656	5	0.23	2.94	4.37	2.87	235	228
	720	218	6	5 620	10 300	545	310	410	**240/500R**	**240/500RK30**	528	561	692	647	5	0.28	2.39	3.56	2.34	297	292
	830	264	7.5	7 750	13 000	708	260	350	**231/500R**	**231/500RK**	536	601	794	714	6	0.33	2.05	3.05	2.00	595	577
	830	325	7.5	9 350	15 900	763	260	350	**241/500R**	**241/500RK30**	536	591	794	712	6	0.36	1.85	2.76	1.81	712	701
	920	336	7.5	11 000	16 700	908	230	310	**232/500R**	**232/500RK**	536	622	884	774	6	0.39	1.74	2.59	1.70	1 020	992

[Remark] Standard cage types used for the above bearings are shown in Table 5 earlier in this section.

d_1 20 ~ 65 mm



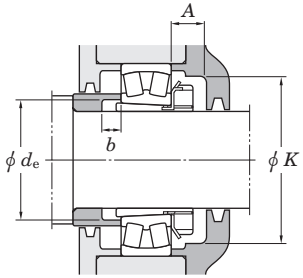
$d_1 \leq 180$ mm



$d_1 \geq 200$ mm

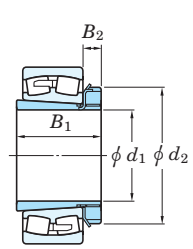
Boundary dimensions (mm)					Brg. bore	Designations	Mounting dimensions (mm)				Mass	(Refer.)	
d_1	B_1	d_2	B_2	B_3	d (mm)	Bearing + **adapter ass'y**	A min.	K min.	d_e min.	b min.	Brg.+adapter ass'y (kg)	Adapter sleeve No.	Locknut No.
20	29	38	8	—	25	22205RZK+**H305X**	15	45	29	5	0.269	A305X	AN05
25	31	45	8	—	30	22206RZK+**H306X**	15	50	34	5	0.404	A306X	AN06
	31	45	8	—	30	21306RZK+**H306X**	15	50	34	6	0.538	A306X	AN06
30	35	52	9	—	35	22207RZK+**H307X**	17	58	39	5	0.610	A307X	AN07
	35	52	9	—	35	21307RZK+**H307X**	17	58	39	7	0.725	A307X	AN07
35	36	58	10	—	40	22208RZK+**H308X**	17	65	44	5	0.793	A308X	AN08
	36	58	10	—	40	21308RZK+**H308X**	17	65	44	5	0.972	A308X	AN08
	46	58	10	—	40	22308RZK+**H2308X**	17	65	45	5	1.30	A2308X	AN08
40	39	65	11	—	45	22209RZK+**H309X**	17	72	49	8	0.855	A309X	AN09
	39	65	11	—	45	21309RZK+**H309X**	17	72	49	5	1.31	A309X	AN09
	50	65	11	—	45	22309RZK+**H2309X**	17	72	50	5	1.70	A2309X	AN09
45	42	70	12	—	50	22210RZK+**H310X**	19	76	54	10	0.953	A310X	AN10
	42	70	12	—	50	21310RZK+**H310X**	19	76	54	5	1.67	A310X	AN10
	55	70	12	—	50	22310RZK+**H2310X**	19	76	56	5	2.26	A2310X	AN10
50	45	75	12	—	55	22211RZK+**H311X**	19	85	60	11	1.22	A311X	AN11
	45	75	12	—	55	21311RZK+**H311X**	19	85	60	6	2.04	A311X	AN11
	59	75	12	—	55	22311RZK+**H2311X**	19	85	61	6	2.80	A2311X	AN11
55	47	80	13	—	60	22212RZK+**H312X**	20	90	65	9	1.59	A312X	AN12
	47	80	13	—	60	21312RZK+**H312X**	20	90	65	5	2.50	A312X	AN12
	62	80	13	—	60	22312RZK+**H2312X**	20	90	66	5	3.50	A2312X	AN12
60	50	85	14	—	65	22213RZK+**H313X**	21	96	70	8	2.01	A313X	AN13
	50	85	14	—	65	21313RZK+**H313X**	21	96	70	5	3.07	A313X	AN13
	65	85	14	—	65	22313RZK+**H2313X**	21	96	72	5	4.17	A2313X	AN13
65	55	98	15	—	75	22215RZK+**H315X**	23	110	80	12	2.58	A315X	AN15
	55	98	15	—	75	21315RZK+**H315X**	23	110	80	5	4.65	A315X	AN15
	73	98	15	—	75	22315RZK+**H2315X**	23	110	82	5	6.44	A2315X	AN15

d_1 70 ~ 110 mm

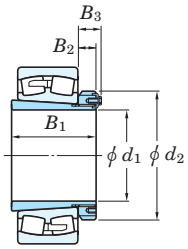


Boundary dimensions (mm)					Brg. bore	Designations	Mounting dimensions (mm)				Mass	(Refer.)	
d_1	B_1	d_2	B_2	B_3	d (mm)	Bearing + **adapter ass'y**	A min.	K min.	d_e min.	b min.	Brg.+adapter ass'y (kg)	Adapter sleeve No.	Locknut No.
70	59	105	17	—	80	22216RZK+**H316X**	25	120	86	12	3.22	A316X	AN16
	59	105	17	—	80	21316RZK+**H316X**	25	120	86	5	5.56	A316X	AN16
	78	105	17	—	80	22316RZK+**H2316X**	25	120	87	5	7.64	A2316X	AN16
75	63	110	18	—	85	22217RZK+**H317X**	27	128	91	12	3.93	A317X	AN17
	63	110	18	—	85	21317RZK+**H317X**	27	128	91	6	6.49	A317X	AN17
	82	110	18	—	85	22317RZK+**H2317X**	27	128	94	6	8.83	A2317X	AN17
80	65	120	18	—	90	22218RZK+**H318X**	28	139	96	10	4.88	A318X	AN18
	86	120	18	—	90	23218RZK+**H2318X**	28	139	99	18	6.20	A2318X	AN18
	65	120	18	—	90	21318RZK+**H318X**	28	139	96	6	7.56	A318X	AN18
	86	120	18	—	90	22318RZK+**H2318X**	28	139	99	6	10.3	A2318X	AN18
85	68	125	19	—	95	22219RZK+**H319X**	29	145	102	9	5.77	A319X	AN19
	68	125	19	—	95	21319RZK+**H319X**	29	145	102	7	8.68	A319X	AN19
	90	125	19	—	95	22319RZK+**H2319X**	29	145	105	7	12.0	A2319X	AN19
90	71	130	20	—	100	22220RZK+**H320X**	30	150	107	8	6.80	A320X	AN20
	97	130	20	—	100	23220RZK+**H2320X**	30	150	110	19	8.94	A2320X	AN20
	71	130	20	—	100	21320RZK+**H320X**	30	150	107	7	10.5	A320X	AN20
	97	130	20	—	100	22320RZK+**H2320X**	30	150	110	7	15.2	A2320X	AN20
100	81	145	21	—	110	23122RZK+**H3122X**	32	170	117	7	7.91	A3122X	AN22
	77	145	21	—	110	22222RZK+**H322X**	32	170	117	6	9.50	A322X	AN22
	105	145	21	—	110	23222RZK+**H2322X**	32	170	121	17	12.4	A2322X	AN22
	77	145	21	—	110	21322RZK+**H322X**	32	170	117	9	14.0	A322X	AN22
	105	145	21	—	110	22322RZK+**H2322X**	32	170	121	7	20.6	A2322X	AN22
110	72	145	22	—	120	23024RZK+**H3024X**	33	180	127	7	6.12	A3024	ANL24
	88	155	22	—	120	23124RZK+**H3124X**	33	180	128	7	10.5	A3124	AN24
	88	155	22	—	120	22224RZK+**H3124X**	33	180	128	11	11.9	A3124	AN24
	112	155	22	—	120	23224RZK+**H2324X**	33	180	131	17	15.1	A2324	AN24
	112	155	22	—	120	22324RZK+**H2324X**	33	180	131	7	25.6	A2324	AN24

d_1 115 ~ (150) mm



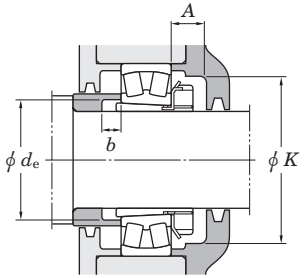
$d_1 \leq 180$ mm



$d_1 \geq 200$ mm

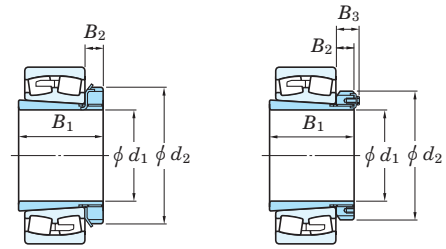
Boundary dimensions (mm)					Brg. bore	Designations	Mounting dimensions (mm)				Mass	(Refer.)	
d_1	B_1	d_2	B_2	B_3	d (mm)	Bearing + adapter ass'y	A min.	K min.	d_e min.	b min.	Brg.+adapter ass'y (kg)	Adapter sleeve No.	Locknut No.
115	80	155	23	—	130	23026RZK+**H3026**	34	190	137	8	9.01	A3026	ANL26
	92	165	23	—	130	23126RZK+**H3126**	34	190	138	8	12.3	A3126	AN26
	92	165	23	—	130	22226RZK+**H3126**	34	190	138	8	15.1	A3126	AN26
	121	165	23	—	130	23226RZK+**H2326**	34	190	142	21	18.8	A2326	AN26
	121	165	23	—	130	22326RZK+**H2326**	34	190	142	8	32.7	A2326	AN26
125	82	165	24	—	140	23028RZK+**H3028**	36	205	147	8	9.79	A3028	ANL28
	97	180	24	—	140	23128RZK+**H3128**	36	205	149	8	14.9	A3128	AN28
	97	180	24	—	140	22228RZK+**H3128**	36	205	149	8	18.8	A3128	AN28
	131	180	24	—	140	23228RZK+**H2328**	36	205	152	22	24.3	A2328	AN28
	131	180	24	—	140	22328RZK+**H2328**	36	205	152	8	40.8	A2328	AN28
135	87	180	26	—	150	23030RZK+**H3030**	37	220	158	8	11.9	A3030	ANL30
	111	195	26	—	150	23130RZK+**H3130**	37	220	160	8	21.7	A3130	AN30
	111	195	26	—	150	22230RZK+**H3130**	37	220	160	15	24.3	A3130	AN30
	139	195	26	—	150	23230RZK+**H2330**	37	220	163	20	30.8	A2330	AN30
	139	195	26	—	150	22330RK+**H2330**	37	220	163	8	49.7	A2330	AN30
	139	195	26	—	150	22330RHAK+**H2330**	37	220	163	8	46.4	A2330	AN30
140	93	190	28	—	160	23032RZK+**H3032**	39	230	168	8	15.0	A3032	ANL32
	119	210	28	—	160	23132RZK+**H3132**	39	230	170	8	27.9	A3132	AN32
	119	210	28	—	160	22232RK+**H3132**	39	230	170	14	30.6	A3132	AN32
	119	210	28	—	160	22232RHAK+**H3132**	39	230	170	14	29.1	A3132	AN32
	147	210	28	—	160	23232RK+**H2332**	39	230	174	18	39.6	A2332	AN32
	147	210	28	—	160	23232RHAK+**H2332**	39	230	174	18	38.0	A2332	AN32
	147	210	28	—	160	22332RK+**H2332**	39	230	174	8	60.5	A2332	AN32
	147	210	28	—	160	22332RHAK+**H2332**	39	230	174	8	56.6	A2332	AN32
150	101	200	29	—	170	23034RZK+**H3034**	40	250	179	8	19.2	A3034	ANL34
	122	220	29	—	170	23134RZK+**H3134**	40	250	180	8	30.0	A3134	AN34
	122	220	29	—	170	22234RK+**H3134**	40	250	180	10	37.2	A3134	AN34
	122	220	29	—	170	22234RHAK+**H3134**	40	250	180	10	35.3	A3134	AN34

d_1 (150) ~ (180) mm



Boundary dimensions (mm)					Brg. bore	Designations	Mounting dimensions (mm)				Mass	(Refer.)	
d_1	B_1	d_2	B_2	B_3	d (mm)	Bearing + adapter ass'y	A min.	K min.	d_e min.	b min.	Brg.+adapter ass'y (kg)	Adapter sleeve No.	Locknut No.
150	154	220	29	—	170	23234RRK+**H2334**	40	250	185	18	47.2	A2334	AN34
	154	220	29	—	170	23234RHAK+**H2334**	40	250	185	18	45.3	A2334	AN34
	154	220	29	—	170	22334RK+**H2334**	40	250	185	8	71.5	A2334	AN34
	154	220	29	—	170	22334RHAK+**H2334**	40	250	185	8	66.8	A2334	AN34
160	109	210	30	—	180	23036RZK+**H3036**	41	260	189	8	24.2	A3036	ANL36
	131	230	30	—	180	23136RK+**H3136**	41	260	191	8	37.1	A3136	AN36
	131	230	30	—	180	23136RHAK+**H3136**	41	260	191	8	35.2	A3136	AN36
	131	230	30	—	180	22236RK+**H3136**	41	260	191	18	39.4	A3136	AN36
	131	230	30	—	180	22236RHAK+**H3136**	41	260	191	18	37.4	A3136	AN36
	161	230	30	—	180	23236RRK+**H2336**	41	260	195	22	50.5	A2336	AN36
	161	230	30	—	180	23236RHAK+**H2336**	41	260	195	22	48.4	A2336	AN36
	161	230	30	—	180	22336RK+**H2336**	41	260	195	8	81.8	A2336	AN36
	161	230	30	—	180	22336RHAK+**H2336**	41	260	195	8	76.4	A2336	AN36
170	112	220	31	—	190	23038RK+**H3038**	43	270	199	9	26.1	A3038	ANL38
	112	220	31	—	190	23038RHAK+**H3038**	43	270	199	9	24.5	A3038	ANL38
	141	240	31	—	190	23138RK+**H3138**	43	270	202	9	45.3	A3138	AN38
	141	240	31	—	190	23138RHAK+**H3138**	43	270	202	9	43.0	A3138	AN38
	141	240	31	—	190	22238RK+**H3138**	43	270	202	21	47.5	A3138	AN38
	141	240	31	—	190	22238RHAK+**H3138**	43	270	202	21	45.0	A3138	AN38
	169	240	31	—	190	23238RRK+**H2338**	43	270	206	21	59.2	A2338	AN38
	169	240	31	—	190	23238RHAK+**H2338**	43	270	206	21	56.7	A2338	AN38
	169	240	31	—	190	22338RK+**H2338**	43	270	206	9	95.6	A2338	AN38
	169	240	31	—	190	22338RHAK+**H2338**	43	270	206	9	89.2	A2338	AN38
180	120	240	32	—	200	23040RK+**H3040**	46	280	210	10	32.8	A3040	ANL40
	120	240	32	—	200	23040RHAK+**H3040**	46	280	210	10	30.7	A3040	ANL40
	150	250	32	—	200	23140RRK+**H3140**	46	280	212	10	54.7	A3140	AN40
	150	250	32	—	200	23140RHAK+**H3140**	46	280	212	10	51.8	A3140	AN40
	150	250	32	—	200	22240RRK+**H3140**	46	280	212	24	56.3	A3140	AN40

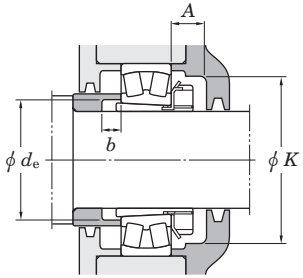
d_1 (180) ~ (240) mm



$d_1 \leq 180$ mm $\quad$ $d_1 \geq 200$ mm

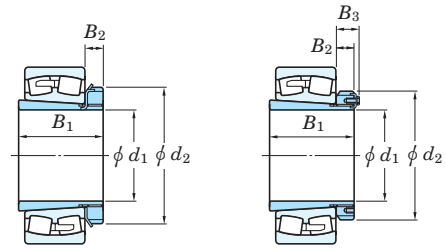
Boundary dimensions (mm)					Brg. bore	Designations	Mounting dimensions (mm)				Mass	(Refer.)	
d_1	B_1	d_2	B_2	B_3	d (mm)	Bearing + **adapter ass'y**	A min.	K min.	d_e min.	b min.	Brg.+adapter ass'y (kg)	Adapter sleeve No.	Locknut No.
180	150	250	32	—	200	22240RHAK+**H3140**	46	280	212	24	53.3	A3140	AN40
	176	250	32	—	200	23240RK+**H2340**	46	280	216	20	71.0	A2340	AN40
	176	250	32	—	200	23240RHAK+**H2340**	46	280	216	20	68.0	A2340	AN40
	176	250	32	—	200	22340RK+**H2340**	46	280	216	10	108	A2340	AN40
	176	250	32	—	200	22340RHAK+**H2340**	46	280	216	10	101	A2340	AN40
200	128	260	30	41	220	23044RK+**H3044**	—	—	231	12	41.4	A3044	ANL44
	128	260	30	41	220	23044RHAK+**H3044**	—	—	231	12	38.7	A3044	ANL44
	158	280	32	44	220	23144RK+**H3144**	—	—	233	10	68.4	A3144	AN44
	158	280	32	44	220	23144RHAK+**H3144**	—	—	233	10	64.8	A3144	AN44
	158	280	32	44	220	22244RRK+**H3144**	—	—	233	22	76.9	A3144	AN44
	158	280	32	44	220	22244RHAK+**H3144**	—	—	233	22	72.7	A3144	AN44
	183	280	32	44	220	23244RK+**H2344**	—	—	236	11	96.5	A2344	AN44
	183	280	32	44	220	23244RHAK+**H2344**	—	—	236	11	92.3	A2344	AN44
	183	280	32	44	220	22344RK+**H2344**	—	—	236	10	139	A2344	AN44
	183	280	32	44	220	22344RHAK+**H2344**	—	—	236	10	130	A2344	AN44
220	133	290	34	46	240	23048RRK+**H3048**	—	—	251	11	47.7	A3048	ANL48
	133	290	34	46	240	23048RHAK+**H3048**	—	—	251	11	44.8	A3048	ANL48
	169	300	34	46	240	23148RRK+**H3148**	—	—	254	11	83.6	A3148	AN48
	169	300	34	46	240	23148RHAK+**H3148**	—	—	254	11	79.1	A3148	AN48
	169	300	34	46	240	22248RK+**H3148**	—	—	254	19	101	A3148	AN48
	169	300	34	46	240	22248RHAK+**H3148**	—	—	254	19	95.6	A3148	AN48
	196	300	34	46	240	23248RRK+**H2348**	—	—	257	6	128	A2348	AN48
	196	300	34	46	240	23248RHAK+**H2348**	—	—	257	6	122	A2348	AN48
	196	300	34	46	240	22348RK+**H2348**	—	—	257	11	175	A2348	AN48
	196	300	34	46	240	22348RHAK+**H2348**	—	—	257	11	163	A2348	AN48
240	147	310	34	46	260	23052RK+**H3052**	—	—	272	13	65.4	A3052	ANL52
	147	310	34	46	260	23052RHAK+**H3052**	—	—	272	13	61.0	A3052	ANL52
	187	330	36	49	260	23152RRK+**H3152**	—	—	276	11	114	A3152	AN52

d_1 (240) ~ (300) mm



Boundary dimensions (mm)					Brg. bore	Designations	Mounting dimensions (mm)				Mass	(Refer.)	
d_1	B_1	d_2	B_2	B_3	d (mm)	Bearing + **adapter ass'y**	A min.	K min.	d_e min.	b min.	Brg.+adapter ass'y (kg)	Adapter sleeve No.	Locknut No.
240	187	330	36	49	260	23152RHAK+**H3152**	—	—	276	11	108	A3152	AN52
	187	330	36	49	260	22252RK+**H3152**	—	—	276	25	131	A3152	AN52
	187	330	36	49	260	22252RHAK+**H3152**	—	—	276	25	124	A3152	AN52
	208	330	36	49	260	23252RK+**H2352**	—	—	278	2	165	A2352	AN52
	208	330	36	49	260	23252RHAK+**H2352**	—	—	278	2	158	A2352	AN52
	208	330	36	49	260	22352RK+**H2352**	—	—	278	11	217	A2352	AN52
	208	330	36	49	260	22352RHAK+**H2352**	—	—	278	11	202	A2352	AN52
260	152	330	38	50	280	23056RK+**H3056**	—	—	292	12	71.5	A3056	ANL56
	152	330	38	50	280	23056RHAK+**H3056**	—	—	292	12	66.8	A3056	ANL56
	192	350	38	51	280	23156RRK+**H3156**	—	—	296	12	123	A3156	AN56
	192	350	38	51	280	23156RHAK+**H3156**	—	—	296	12	116	A3156	AN56
	192	350	38	51	280	22256RK+**H3156**	—	—	296	28	138	A3156	AN56
	192	350	38	51	280	22256RHAK+**H3156**	—	—	296	28	130	A3156	AN56
	221	350	38	51	280	23256RK+**H2356**	—	—	299	11	178	A2356	AN56
	221	350	38	51	280	23256RHAK+**H2356**	—	—	299	11	170	A2356	AN56
	221	350	38	51	280	22356RK+**H2356**	—	—	299	12	254	A2356	AN56
	221	350	38	51	280	22356RHAK+**H2356**	—	—	299	12	237	A2356	AN56
280	168	360	42	54	300	23060RK+**H3060**	—	—	313	12	97.7	A3060	ANL60
	168	360	42	54	300	23060RHAK+**H3060**	—	—	313	12	90.8	A3060	ANL60
	208	380	40	53	300	23160RRK+**H3160**	—	—	317	12	159	A3160	AN60
	208	380	40	53	300	23160RHAK+**H3160**	—	—	317	12	150	A3160	AN60
	208	380	40	53	300	22260RK+**H3160**	—	—	317	32	173	A3160	AN60
	208	380	40	53	300	22260RHAK+**H3160**	—	—	317	32	163	A3160	AN60
	240	380	40	53	300	23260RK+**H3260**	—	—	321	12	227	A3260	AN60
	240	380	40	53	300	23260RHAK+**H3260**	—	—	321	12	217	A3260	AN60
300	171	380	42	55	320	23064RK+**H3064**	—	—	334	13	105	A3064	ANL64
	171	380	42	55	320	23064RHAK+**H3064**	—	—	334	13	98.1	A3064	ANL64
	226	400	42	56	320	23164RK+**H3164**	—	—	339	13	202	A3164	AN64

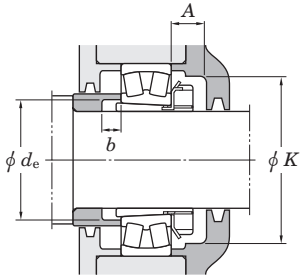
d_1 (300) ~ 380 mm



$d_1 \leq 180$ mm $\quad$ $d_1 \geq 200$ mm

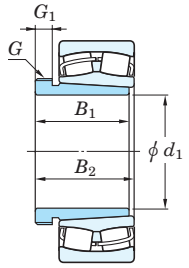
Boundary dimensions (mm)					Brg. bore	Designations	Mounting dimensions (mm)				Mass	(Refer.)	
d_1	B_1	d_2	B_2	B_3	d (mm)	Bearing + **adapter ass'y**	A min.	K min.	d_e min.	b min.	Brg.+adapter ass'y (kg)	Adapter sleeve No.	Locknut No.
300	226	400	42	56	320	23164RHAK+**H3164**	—	—	339	13	191	A3164	AN64
	226	400	42	56	320	22264RK+**H3164**	—	—	339	39	207	A3164	AN64
	258	400	42	56	320	23264RK+**H3264**	—	—	343	13	283	A3264	AN64
	258	400	42	56	320	23264RHAK+**H3264**	—	—	343	13	270	A3264	AN64
320	187	400	45	58	340	23068RK+**H3068**	—	—	355	14	135	A3068	ANL68
	187	400	45	58	340	23068RHAK+**H3068**	—	—	355	14	126	A3068	ANL68
	254	440	55	72	340	23168RK+**H3168**	—	—	360	14	262	A3168	AN68
	254	440	55	72	340	23168RHAK+**H3168**	—	—	360	14	248	A3168	AN68
	288	440	55	72	340	23268RK+**H3268**	—	—	364	14	355	A3268	AN68
	288	440	55	72	340	23268RHAK+**H3268**	—	—	364	14	339	A3268	AN68
340	188	420	45	58	360	23072RK+**H3072**	—	—	375	14	143	A3072	ANL72
	188	420	45	58	360	23072RHAK+**H3072**	—	—	375	14	133	A3072	ANL72
	259	460	58	75	360	23172RK+**H3172**	—	—	380	14	278	A3172	AN72
	259	460	58	75	360	23172RHAK+**H3172**	—	—	380	14	263	A3172	AN72
	299	460	58	75	360	23272RK+**H3272**	—	—	385	14	400	A3272	AN72
	299	460	58	75	360	23272RHAK+**H3272**	—	—	385	14	382	A3272	AN72
360	193	450	48	62	380	23076RK+**H3076**	—	—	396	15	156	A3076	ANL76
	193	450	48	62	380	23076RHAK+**H3076**	—	—	396	15	146	A3076	ANL76
	264	490	60	77	380	23176RK+**H3176**	—	—	401	15	298	A3176	AN76
	264	490	60	77	380	23176RHAK+**H3176**	—	—	401	15	282	A3176	AN76
	310	490	60	77	380	23276RK+**H3276**	—	—	405	15	448	A3276	AN76
	310	490	60	77	380	23276RHAK+**H3276**	—	—	405	15	427	A3276	AN76
380	210	470	52	66	400	23080RK+**H3080**	—	—	417	15	195	A3080	ANL80
	210	470	52	66	400	23080RHAK+**H3080**	—	—	417	15	182	A3080	ANL80
	272	520	62	82	400	23180RK+**H3180**	—	—	421	15	339	A3180	AN80
	272	520	62	82	400	23180RHAK+**H3180**	—	—	421	15	321	A3180	AN80
	328	520	62	82	400	23280RK+**H3280**	—	—	427	15	539	A3280	AN80
	328	520	62	82	400	23280RHAK+**H3280**	—	—	427	15	512	A3280	AN80

d_1 400 ~ 470 mm



Boundary dimensions (mm)					Brg. bore	Designations	Mounting dimensions (mm)				Mass	(Refer.)	
d_1	B_1	d_2	B_2	B_3	d (mm)	Bearing + **adapter ass'y**	A min.	K min.	d_e min.	b min.	Brg.+adapter ass'y (kg)	Adapter sleeve No.	Locknut No.
400	212	490	52	66	420	23084RK+**H3084**	—	—	437	16	205	A3084	ANL84
	212	490	52	66	420	23084RHAK+**H3084**	—	—	437	16	191	A3084	ANL84
	304	540	70	90	420	23184RK+**H3184**	—	—	443	16	441	A3184	AN84
	304	540	70	90	420	23184RHAK+**H3184**	—	—	443	16	417	A3184	AN84
	352	540	70	90	420	23284RK+**H3284**	—	—	448	16	639	A3284	AN84
	352	540	70	90	420	23284RHAK+**H3284**	—	—	448	16	607	A3284	AN84
410	228	520	60	77	440	23088RK+**H3088**	—	—	458	17	252	A3088	ANL88
	228	520	60	77	440	23088RHAK+**H3088**	—	—	458	17	236	A3088	ANL88
	307	560	70	90	440	23188RK+**H3188**	—	—	464	17	474	A3188	AN88
	307	560	70	90	440	23188RHAK+**H3188**	—	—	464	17	449	A3188	AN88
	361	560	70	90	440	23288RK+**H3288**	—	—	469	17	718	A3288	AN88
	361	560	70	90	440	23288RHAK+**H3288**	—	—	469	17	685	A3288	AN88
430	234	540	60	77	460	23092RK+**H3092**	—	—	478	17	283	A3092	ANL92
	234	540	60	77	460	23092RHAK+**H3092**	—	—	478	17	265	A3092	ANL92
	326	580	75	95	460	23192RK+**H3192**	—	—	485	17	559	A3192	AN92
	326	580	75	95	460	23192RHAK+**H3192**	—	—	485	17	529	A3192	AN92
	382	580	75	95	460	23292RK+**H3292**	—	—	491	17	838	A3292	AN92
	382	580	75	95	460	23292RHAK+**H3292**	—	—	491	17	797	A3292	AN92
450	237	560	60	77	480	23096RK+**H3096**	—	—	499	18	295	A3096	ANL96
	237	560	60	77	480	23096RHAK+**H3096**	—	—	499	18	276	A3096	ANL96
	335	620	75	95	480	23196RK+**H3196**	—	—	505	18	628	A3196	AN96
	335	620	75	95	480	23196RHAK+**H3196**	—	—	505	18	595	A3196	AN96
	397	620	75	95	480	23296RK+**H3296**	—	—	512	18	966	A3296	AN96
	397	620	75	95	480	23296RHAK+**H3296**	—	—	512	18	920	A3296	AN96
470	247	580	68	85	500	230/500RK+**H30/500**	—	—	519	18	315	A30/500	ANL100
	356	630	80	100	500	231/500RK+**H31/500**	—	—	527	18	727	A31/500	AN100
	428	630	80	100	500	232/500RK+**H32/500**	—	—	534	18	1 167	A32/500	AN100

d_1 35 ~ (75) mm



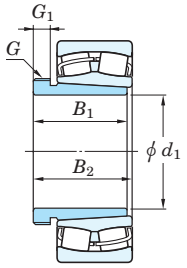
Boundary dimensions (mm) d_1	B_1	B_2	G [1] Screw size	G_1	Brg. bore d (mm)	Designations Bearing + withdrawal sleeve	Mass Brg.+withdrawal sleeve (kg)	(Refer.) Applicable locknut No.
35	29	32	M45×1.5	6	40	22208RZK+**AH308**	0.681	AN09
	29	32	M45×1.5	6	40	21308RZK+**AH308**	0.860	AN09
	40	43	M45×1.5	7	40	22308RZK+**AH2308**	1.19	AN09
40	31	34	M50×1.5	6	45	22209RZK+**AH309**	0.699	AN10
	31	34	M50×1.5	6	45	21309RZK+**AH309**	1.14	AN10
	44	47	M50×1.5	7	45	22309RZK+**AH2309**	1.55	AN10
45	35	38	M55×2	7	50	22210RZK+**AHX310**	0.771	AN11
	35	38	M55×2	7	50	21310RZK+**AHX310**	1.49	AN11
	50	53	M55×2	9	50	22310RZK+**AHX2310**	2.09	AN11
50	37	40	M60×2	7	55	22211RZK+**AHX311**	1.01	AN12
	37	40	M60×2	7	55	21311RZK+**AHX311**	1.83	AN12
	54	57	M60×2	10	55	22311RZK+**AHX2311**	2.60	AN12
55	40	43	M65×2	8	60	22212RZK+**AHX312**	1.35	AN13
	40	43	M65×2	8	60	21312RZK+**AHX312**	2.27	AN13
	58	61	M65×2	11	60	22312RZK+**AHX2312**	3.29	AN13
60	42	45	M75×2	8	65	22213RZK+**AH313**	1.77	AN15
	42	45	M75×2	8	65	21313RZK+**AH313**	2.84	AN15
	61	64	M75×2	12	65	22313RZK+**AH2313**	3.98	AN15
65	43	47	M80×2	8	70	22214RZK+**AH314**	1.89	AN16
	43	47	M80×2	8	70	21314RZK+**AH314**	3.43	AN16
	64	68	M80×2	12	70	22314RZK+**AHX2314**	4.82	AN16
70	45	49	M85×2	8	75	22215RZK+**AH315**	2.01	AN17
	45	49	M85×2	8	75	21315RZK+**AH315**	4.07	AN17
	68	72	M85×2	12	75	22315RZK+**AHX2315**	5.87	AN17
75	48	52	M90×2	8	80	22216RZK+**AH316**	2.49	AN18
	48	52	M90×2	8	80	21316RZK+**AH316**	4.83	AN18

[Note] 1) Basic profile and dimensions of screw thread identified by prefix M are in accordance with JIS B 0205.
Basic profile and dimensions of screw thread identified by prefix Tr are in accordance with JIS B 0216.

d_1 (75) ~ (115) mm

Boundary dimensions (mm) d_1	B_1	B_2	G [1] Screw size	G_1	Brg. bore d (mm)	Designations Bearing + withdrawal sleeve	Mass Brg.+withdrawal sleeve (kg)	(Refer.) Applicable locknut No.
75	71	75	M90×2	12	80	22316RZK+**AHX2316**	6.90	AN18
80	52	56	M95×2	9	85	22217RZK+**AHX317**	3.12	AN19
	52	56	M95×2	9	85	21317RZK+**AHX317**	5.68	AN19
	74	78	M95×2	13	85	22317RZK+**AHX2317**	7.98	AN19
85	53	57	M100×2	9	90	22218RZK+**AHX318**	3.89	AN20
	63	67	M100×2	10	90	23218RZK+**AHX3218**	5.08	AN20
	53	57	M100×2	9	90	21318RZK+**AHX318**	6.58	AN20
	79	83	M100×2	14	90	22318RZK+**AHX2318**	9.41	AN20
90	57	61	M105×2	10	95	22219RZK+**AHX319**	4.68	AN21
	57	61	M105×2	10	95	21319RZK+**AHX319**	7.59	AN21
	85	89	M105×2	16	95	22319RZK+**AHX2319**	10.9	AN21
95	59	63	M110×2	10	100	22220RZK+**AHX320**	5.58	AN22
	73	77	M110×2	11	100	23220RZK+**AHX3220**	7.43	AN22
	59	63	M110×2	10	100	21320RZK+**AHX320**	9.26	AN22
	90	94	M110×2	16	100	22320RZK+**AHX2320**	13.9	AN22
105	68	72	M120×2	11	110	23122RZK+**AHX3122**	6.30	AN24
	82	91	M115×2	13	110	24122RZK30+**AH24122**	7.60	AN23
	68	72	M120×2	11	110	22222RZK+**AHX3122**	7.97	AN24
	82	86	M125×2	11	110	23222RZK+**AHX3222**	10.5	AN25
	63	67	M120×2	12	110	21322RZK+**AHX322**	12.3	AN24
	98	102	M125×2	16	110	22322RZK+**AHX2322**	19.1	AN25
115	60	64	M130×2	13	120	23024RZK+**AHX3024**	4.82	AN26
	73	82	M125×2	13	120	24024RZK30+**AH24024**	5.99	AN25
	75	79	M130×2	12	120	23124RZK+**AHX3124**	8.69	AN26
	93	102	M130×2	13	120	24124RZK30+**AH24124**	11.0	AN26
	75	79	M130×2	12	120	22224RZK+**AHX3124**	10.1	AN26

d_1 (115) ~ (150) mm



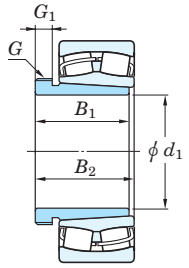
Boundary dimensions (mm)					Brg. bore	Designations	Mass	(Refer.)
d_1	B_1	B_2	G [1] Screw size	G_1	d (mm)	Bearing + **withdrawal sleeve**	Brg.+withdrawal sleeve (kg)	**Applicable locknut No.**
115	90	94	M135×2	13	120	23224RZK+**AHX3224**	13.1	AN27
	105	109	M135×2	17	120	22324RZK+**AHX2324**	23.9	AN27
125	67	71	M140×2	14	130	23026RZK+**AHX3026**	6.90	AN28
	83	93	M135×2	14	130	24026RZK30+**AH24026**	8.74	AN27
	78	82	M140×2	12	130	23126RZK+**AHX3126**	9.52	AN28
	94	104	M140×2	14	130	24126RZK30+**AH24126**	11.7	AN28
	78	82	M140×2	12	130	22226RZK+**AHX3126**	12.4	AN28
	98	102	M145×2	15	130	23226RZK+**AHX3226**	15.6	AN29
	115	119	M145×2	19	130	22326RZK+**AHX2326**	29.9	AN29
135	68	73	M150×2	14	140	23028RZK+**AHX3028**	7.43	AN30
	83	93	M145×2	14	140	24028RZK30+**AH24028**	9.26	AN29
	83	88	M150×2	14	140	23128RZK+**AHX3128**	11.5	AN30
	99	109	M150×2	14	140	24128RZK30+**AH24128**	14.1	AN30
	83	88	M150×2	14	140	22228RZK+**AHX3128**	15.4	AN30
	104	109	M155×3	15	140	23228RZK+**AHX3228**	20.3	AN31
	125	130	M155×3	20	140	22328RZK+**AHX2328**	35.0	AN31
145	72	77	M160×3	15	150	23030RZK+**AHX3030**	8.92	AN32
	90	101	M155×3	15	150	24030RZK30+**AH24030**	11.4	AN31
	96	101	M165×3	15	150	23130RZK+**AHX3130**	17.7	AN33
	115	126	M160×3	15	150	24130RZK30+**AH24130**	21.2	AN32
	96	101	M165×3	15	150	22230RZK+**AHX3130**	20.3	AN33
	114	119	M165×3	17	150	23230RZK+**AHX3230**	26.0	AN33
	135	140	M165×3	24	150	22330RK+**AHX2330**	45.5	AN33
	135	140	M165×3	24	150	22330RHAK+**AHX2330**	42.2	AN33
150	77	82	M170×3	16	160	23032RZK+**AH3032**	11.5	AN34
	95	106	M170×3	15	160	24032RZK30+**AH24032**	15.0	AN34
	103	108	M180×3	16	160	23132RZK+**AH3132**	23.4	AN36

[Note] 1) Basic profile and dimensions of screw thread identified by prefix M are in accordance with JIS B 0205.
Basic profile and dimensions of screw thread identified by prefix Tr are in accordance with JIS B 0216.

d_1 (150) ~ 170 mm

Boundary dimensions (mm)					Brg. bore	Designations	Mass	(Refer.)
d_1	B_1	B_2	G [1] Screw size	G_1	d (mm)	Bearing + **withdrawal sleeve**	Brg.+withdrawal sleeve (kg)	**Applicable locknut No.**
150	103	108	M180×3	16	160	22232RK+**AH3132**	26.1	AN36
	103	108	M180×3	16	160	22232RHAK+**AH3132**	24.6	AN36
	124	130	M180×3	20	160	23232RK+**AH3232**	35.1	AN36
	124	130	M180×3	20	160	23232RHAK+**AH3232**	32.6	AN36
	140	146	M180×3	24	160	22332RK+**AH2332**	55.7	AN36
	140	146	M180×3	24	160	22332RHAK+**AH2332**	51.8	AN36
160	85	90	M180×3	17	170	23034RZK+**AH3034**	15.2	AN36
	106	117	M180×3	16	170	24034RZK30+**AH24034**	20.0	AN36
	104	109	M190×3	16	170	23134RZK+**AH3134**	24.6	AN38
	125	136	M180×3	16	170	24134RRK30+**AH24134**	30.0	AN36
	104	109	M190×3	16	170	22234RK+**AH3134**	31.8	AN38
	104	109	M190×3	16	170	22234RHAK+**AH3134**	29.9	AN38
	134	140	M190×3	24	170	23234RRK+**AH3234**	42.3	AN38
	134	140	M190×3	24	170	23234RHAK+**AH3234**	39.4	AN38
	146	152	M190×3	24	170	22334RK+**AH2334**	66.1	AN38
	146	152	M190×3	24	170	22334RHAK+**AH2334**	61.4	AN38
170	92	98	M190×3	17	180	23036RZK+**AH3036**	19.7	AN38
	116	127	M190×3	16	180	24036RRK30+**AH24036**	26.1	AN38
	116	122	M200×3	19	180	23136RK+**AH3136**	31.7	AN40
	116	122	M200×3	19	180	23136RHAK+**AH3136**	29.8	AN40
	134	145	M190×3	16	180	24136RRK30+**AH24136**	37.6	AN38
	134	145	M190×3	16	180	24136RHAK30+**AH24136**	34.9	AN38
	105	110	M200×3	17	180	22236RK+**AH2236**	33.5	AN40
	105	110	M200×3	17	180	22236RHAK+**AH2236**	31.5	AN40
	140	146	M200×3	24	180	23236RRK+**AH3236**	45.1	AN40
	140	146	M200×3	24	180	23236RHAK+**AH3236**	41.8	AN40
	154	160	M200×3	24	180	22336RK+**AH2336**	75.7	AN40
	154	160	M200×3	24	180	22336RHAK+**AH2336**	70.3	AN40

d_1 180 ~ 190 mm



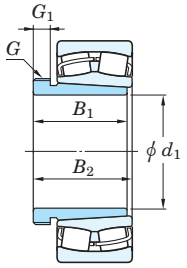
Boundary dimensions (mm) d_1	B_1	B_2	G [1] Screw size	G_1	Brg. bore d (mm)	Designations Bearing + withdrawal sleeve	Mass Brg.+withdrawal sleeve (kg)	(Refer.) Applicable locknut No.
180	96	102	Tr205×4	18	190	23038RK+**AH3038**	21.5	HNL41
	96	102	Tr205×4	18	190	23038RHAK+**AH3038**	19.9	HNL41
	118	131	M200×3	18	190	24038RRK30+**AH24038**	27.6	AN40
	118	131	M200×3	18	190	24038RHAK30+**AH24038**	25.5	AN40
	125	131	Tr210×4	20	190	23138RK+**AH3138**	39.3	HN42
	125	131	Tr210×4	20	190	23138RHAK+**AH3138**	37.0	HN42
	146	159	M200×3	18	190	24138RRK30+**AH24138**	46.7	AN40
	146	159	M200×3	18	190	24138RHAK30+**AH24138**	43.8	AN40
	112	117	Tr210×4	18	190	22238RK+**AH2238**	40.9	HN42
	112	117	Tr210×4	18	190	22238RHAK+**AH2238**	38.4	HN42
	145	152	Tr210×4	25	190	23238RRK+**AH3238**	53.3	HN42
	145	152	Tr210×4	25	190	23238RHAK+**AH3238**	49.4	HN42
	160	167	Tr210×4	26	190	22338RK+**AH2338**	89.0	HN42
	160	167	Tr210×4	26	190	22338RHAK+**AH2338**	82.6	HN42
190	102	108	Tr215×4	19	200	23040RK+**AH3040**	27.2	HNL43
	102	108	Tr215×4	19	200	23040RHAK+**AH3040**	25.1	HNL43
	127	140	Tr210×4	18	200	24040RRK30+**AH24040**	34.6	HN42
	127	140	Tr210×4	18	200	24040RHAK30+**AH24040**	31.9	HN42
	134	140	Tr220×4	21	200	23140RRK+**AH3140**	47.9	HN44
	134	140	Tr220×4	21	200	23140RHAK+**AH3140**	45.0	HN44
	158	171	Tr210×4	18	200	24140RRK30+**AH24140**	57.6	HN42
	158	171	Tr210×4	18	200	24140RHAK30+**AH24140**	53.8	HN42
	118	123	Tr220×4	19	200	22240RRK+**AH2240**	48.7	HN44
	118	123	Tr220×4	19	200	22240RHAK+**AH2240**	45.7	HN44
	153	160	Tr220×4	25	200	23240RK+**AH3240**	64.7	HN44
	153	160	Tr220×4	25	200	23240RHAK+**AH3240**	60.1	HN44
	170	177	Tr220×4	26	200	22340RK+**AH2340**	101	HN44
	170	177	Tr220×4	26	200	22340RHAK+**AH2340**	93.4	HN44

[Note] 1) Basic profile and dimensions of screw thread identified by prefix M are in accordance with JIS B 0205.
Basic profile and dimensions of screw thread identified by prefix Tr are in accordance with JIS B 0216.

d_1 200 ~ 220 mm

Boundary dimensions (mm) d_1	B_1	B_2	G [1] Screw size	G_1	Brg. bore d (mm)	Designations Bearing + withdrawal sleeve	Mass Brg.+withdrawal sleeve (kg)	(Refer.) Applicable locknut No.
200	111	117	Tr235×4	20	220	23044RK+**AH3044**	38.0	HNL47
	111	117	Tr235×4	20	220	23044RHAK+**AH3044**	35.3	HNL47
	138	152	Tr230×4	20	220	24044RRK30+**AH24044**	48.1	—
	138	152	Tr230×4	20	220	24044RHAK30+**AH24044**	44.7	—
	145	151	Tr240×4	23	220	23144RK+**AH3144**	63.6	HN48
	145	151	Tr240×4	23	220	23144RHAK+**AH3144**	60.0	HN48
	170	184	Tr230×4	20	220	24144RRK30+**AH24144**	76.4	—
	170	184	Tr230×4	20	220	24144RHAK30+**AH24144**	71.2	—
	130	136	Tr240×4	20	220	22244RRK+**AH2244**	70.8	HN48
	130	136	Tr240×4	20	220	22244RHAK+**AH2244**	66.6	HN48
	181	189	Tr240×4	30	220	23244RK+**AH2344**	95.1	HN48
	181	189	Tr240×4	30	220	23244RHAK+**AH2344**	88.5	HN48
	181	189	Tr240×4	30	220	22344RK+**AH2344**	136	HN48
	181	189	Tr240×4	30	220	22344RHAK+**AH2344**	127	HN48
220	116	123	Tr260×4	21	240	23048RRK+**AH3048**	42.6	HNL52
	116	123	Tr260×4	21	240	23048RHAK+**AH3048**	39.7	HNL52
	138	153	Tr250×4	20	240	24048RRK30+**AH24048**	51.9	—
	138	153	Tr250×4	20	240	24048RHAK30+**AH24048**	48.0	—
	154	161	Tr260×4	25	240	23148RRK+**AH3148**	77.6	HN52
	154	161	Tr260×4	25	240	23148RHAK+**AH3148**	73.1	HN52
	180	195	Tr260×4	20	240	24148RRK30+**AH24148**	94.0	HN52
	180	195	Tr260×4	20	240	24148RHAK30+**AH24148**	87.9	HN52
	144	150	Tr260×4	21	240	22248RK+**AH2248**	94.3	HN52
	144	150	Tr260×4	21	240	22248RHAK+**AH2248**	88.7	HN52
	189	197	Tr260×4	30	240	23248RRK+**AH2348**	126	HN52
	189	197	Tr260×4	30	240	23248RHAK+**AH2348**	117	HN52
	189	197	Tr260×4	30	240	22348RK+**AH2348**	170	HN52
	189	197	Tr260×4	30	240	22348RHAK+**AH2348**	158	HN52

d_1 240 ~ 260 mm



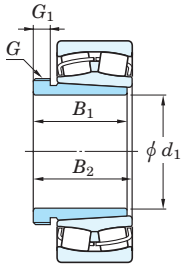
Boundary dimensions (mm) d_1	B_1	B_2	G [1] Screw size	G_1	Brg. bore d (mm)	Designations Bearing + withdrawal sleeve	Mass Brg.+withdrawal sleeve (kg)	(Refer.) Applicable locknut No.
240	128	135	Tr280×4	23	260	23052RK+**AH3052**	60.0	HNL56
	128	135	Tr280×4	23	260	23052RHAK+**AH3052**	55.6	HNL56
	162	178	Tr270×4	22	260	24052RRK30+**AH24052**	77.0	—
	162	178	Tr270×4	22	260	24052RHAK30+**AH24052**	71.2	—
	172	179	Tr290×4	26	260	23152RRK+**AH3152**	107	HN58
	172	179	Tr290×4	26	260	23152RHAK+**AH3152**	101	HN58
	202	218	Tr280×4	22	260	24152RRK30+**AH24152**	128	—
	202	218	Tr280×4	22	260	24152RHAK30+**AH24152**	120	—
	155	161	Tr290×4	23	260	22252RK+**AH2252**	122	HN58
	155	161	Tr290×4	23	260	22252RHAK+**AH2252**	115	HN58
	205	213	Tr290×4	30	260	23252RK+**AH2352**	164	HN58
	205	213	Tr290×4	30	260	23252RHAK+**AH2352**	153	HN58
	205	213	Tr290×4	30	260	22352RK+**AH2352**	212	HN58
	205	213	Tr290×4	30	260	22352RHAK+**AH2352**	197	HN58
260	131	139	Tr300×4	24	280	23056RK+**AH3056**	64.9	HNL60
	131	139	Tr300×4	24	280	23056RHAK+**AH3056**	60.2	HNL60
	162	179	Tr290×4	22	280	24056RRK30+**AH24056**	81.9	HN58
	162	179	Tr290×4	22	280	24056RHAK30+**AH24056**	75.7	HN58
	175	183	Tr310×5	28	280	23156RRK+**AH3156**	114	HN62
	175	183	Tr310×5	28	280	23156RHAK+**AH3156**	108	HN62
	202	219	Tr300×4	22	280	24156RRK30+**AH24156**	136	—
	202	219	Tr300×4	22	280	24156RHAK30+**AH24156**	128	—
	155	163	Tr310×5	24	280	22256RK+**AH2256**	127	HN62
	155	163	Tr310×5	24	280	22256RHAK+**AH2256**	119	HN62
	212	220	Tr310×5	30	280	23256RK+**AH2356**	175	HN62
	212	220	Tr310×5	30	280	23256RHAK+**AH2356**	163	HN62
	212	220	Tr310×5	30	280	22356RK+**AH2356**	247	HN62
	212	220	Tr310×5	30	280	22356RHAK+**AH2356**	230	HN62

[Note] 1) Basic profile and dimensions of screw thread identified by prefix M are in accordance with JIS B 0205.
Basic profile and dimensions of screw thread identified by prefix Tr are in accordance with JIS B 0216.

d_1 280 ~ (320) mm

Boundary dimensions (mm) d_1	B_1	B_2	G [1] Screw size	G_1	Brg. bore d (mm)	Designations Bearing + withdrawal sleeve	Mass Brg.+withdrawal sleeve (kg)	(Refer.) Applicable locknut No.
280	145	153	Tr320×5	26	300	23060RK+**AH3060**	88.1	HNL64
	145	153	Tr320×5	26	300	23060RHAK+**AH3060**	81.2	HNL64
	184	202	Tr310×5	24	300	24060RRK30+**AH24060**	112	HN62
	184	202	Tr310×5	24	300	24060RHAK30+**AH24060**	105	HN62
	192	200	Tr330×5	30	300	23160RRK+**AH3160**	149	HN66
	192	200	Tr330×5	30	300	23160RHAK+**AH3160**	140	HN66
	224	242	Tr320×5	24	300	24160RRK30+**AH24160**	180	—
	224	242	Tr320×5	24	300	24160RHAK30+**AH24160**	168	—
	170	178	Tr330×5	26	300	22260RK+**AH2260**	160	HN66
	170	178	Tr330×5	26	300	22260RHAK+**AH2260**	150	HN66
	228	236	Tr330×5	34	300	23260RK+**AH3260**	223	HN66
	228	236	Tr330×5	34	300	23260RHAK+**AH3260**	208	HN66
300	149	157	Tr345×5	27	320	23064RK+**AH3064**	94.8	HNL69
	149	157	Tr345×5	27	320	23064RHAK+**AH3064**	88.1	HNL69
	184	202	Tr330×5	24	320	24064RRK30+**AH24064**	120	HN66
	184	202	Tr330×5	24	320	24064RHAK30+**AH24064**	108	HN66
	209	217	Tr350×5	31	320	23164RK+**AH3164**	191	HN70
	209	217	Tr350×5	31	320	23164RHAK+**AH3164**	180	HN70
	242	260	Tr340×5	24	320	24164RRK30+**AH24164**	226	—
	242	260	Tr340×5	24	320	24164RHAK30+**AH24164**	217	—
	180	190	Tr350×5	27	320	22264RK+**AH2264**	191	HN70
	246	254	Tr350×5	36	320	23264RK+**AH3264**	280	HN70
	246	254	Tr350×5	36	320	23264RHAK+**AH3264**	260	HN70
320	162	171	Tr365×5	28	340	23068RK+**AH3068**	125	HNL73
	162	171	Tr365×5	28	340	23068RHAK+**AH3068**	115	HNL73
	225	234	Tr370×5	33	340	23168RK+**AH3168**	239	HN74
	225	234	Tr370×5	33	340	23168RHAK+**AH3168**	225	HN74
	269	288	Tr360×5	26	340	24168RRK30+**AH24168**	293	—

d_1 (320) ~ 380 mm



Boundary dimensions (mm) d_1	B_1	B_2	G [1] Screw size	G_1	Brg. bore d (mm)	Designations Bearing + withdrawal sleeve	Mass Brg.+withdrawal sleeve (kg)	(Refer.) Applicable locknut No.
320	269	288	Tr360×5	26	340	24168RHAK30+**AH24168**	282	—
	264	273	Tr370×5	38	340	23268RK+**AH3268**	342	HN74
	264	273	Tr370×5	38	340	23268RHAK+**AH3268**	317	HN74
340	167	176	Tr385×5	30	360	23072RK+**AH3072**	132	HNL77
	167	176	Tr385×5	30	360	23072RHAK+**AH3072**	122	HNL77
	229	238	Tr400×5	35	360	23172RK+**AH3172**	254	HN80
	232	238	Tr400×5	35	360	23172RHAK+**AH3172**	239	HN80
	269	289	Tr380×5	26	360	24172RK30+**AH24172**	313	—
	269	289	Tr380×5	26	360	24172RHAK30+**AH24172**	300	—
	274	283	Tr400×5	40	360	23272RK+**AH3272**	388	HN80
	274	283	Tr400×5	40	360	23272RHAK+**AH3272**	360	HN80
360	170	180	Tr410×5	31	380	23076RK+**AH3076**	141	HNL82
	170	180	Tr410×5	31	380	23076RHAK+**AH3076**	131	HNL82
	232	242	Tr420×5	36	380	23176RK+**AH3176**	269	HN84
	240	242	Tr420×5	36	380	23176RHAK+**AH3176**	253	HN84
	271	291	Tr400×5	28	380	24176RK30+**AH24176**	328	HN80
	271	291	Tr400×5	28	380	24176RHAK30+**AH24176**	314	HN80
	284	294	Tr420×5	42	380	23276RK+**AH3276**	432	HN84
	284	294	Tr420×5	42	380	23276RHAK+**AH3276**	400	HN84
380	183	193	Tr430×5	33	400	23080RK+**AH3080**	178	HNL86
	183	193	Tr430×5	33	400	23080RHAK+**AH3080**	165	HNL86
	240	250	Tr440×5	38	400	23180RK+**AH3180**	305	HN88
	266	250	Tr440×5	38	400	23180RHAK+**AH3180**	287	HN88
	278	298	Tr420×5	28	400	24180RK30+**AH24180**	368	HN84
	278	298	Tr420×5	28	400	24180RHAK30+**AH24180**	352	HN84
	302	312	Tr440×5	44	400	23280RK+**AH3280**	521	HN88
	302	312	Tr440×5	44	400	23280RHAK+**AH3280**	480	HN88

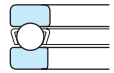
[Note] 1) Basic profile and dimensions of screw thread identified by prefix M are in accordance with JIS B 0205.
Basic profile and dimensions of screw thread identified by prefix Tr are in accordance with JIS B 0216.

d_1 400 ~ 480 mm

Boundary dimensions (mm) d_1	B_1	B_2	G [1] Screw size	G_1	Brg. bore d (mm)	Designations Bearing + withdrawal sleeve	Mass Brg.+withdrawal sleeve (kg)	(Refer.) Applicable locknut No.
400	186	196	Tr450×5	34	420	23084RK+**AH3084**	188	HNL90
	186	196	Tr450×5	34	420	23084RHAK+**AH3084**	174	HNL90
	266	276	Tr460×5	40	420	23184RK+**AH3184**	399	HN92
	270	276	Tr460×5	40	420	23184RHAK+**AH3184**	375	HN92
	321	331	Tr460×5	46	420	23284RK+**AH3284**	673	HN92
	321	331	Tr460×5	46	420	23284RHAK+**AH3284**	568	HN92
420	194	205	Tr470×5	35	440	23088RK+**AHX3088**	215	HNL94
	194	205	Tr470×5	35	440	23088RHAK+**AHX3088**	199	HNL94
	270	281	Tr480×5	42	440	23188RK+**AHX3188**	416	HN96
	285	281	Tr480×5	42	440	23188RHAK+**AHX3188**	391	HN96
	330	341	Tr480×5	48	440	23288RK+**AHX3288**	678	HN96
	330	341	Tr480×5	48	440	23288RHAK+**AHX3288**	627	HN96
440	202	213	Tr490×5	37	460	23092RK+**AHX3092**	244	HNL98
	202	213	Tr490×5	37	460	23092RHAK+**AHX3092**	226	HNL98
	285	296	Tr510×6	43	460	23192RK+**AHX3192**	494	HN102
	295	296	Tr510×6	43	460	23192RHAK+**AHX3192**	464	HN102
	349	360	Tr510×6	50	460	23292RK+**AHX3292**	795	HN102
	349	360	Tr510×6	50	460	23292RHAK+**AHX3292**	733	HN102
460	205	217	Tr520×6	38	480	23096RK+**AHX3096**	257	HNL104
	205	217	Tr520×6	38	480	23096RHAK+**AHX3096**	238	HNL104
	295	307	Tr530×6	45	480	23196RK+**AHX3196**	551	HN106
	313	307	Tr530×6	45	480	23196RHAK+**AHX3196**	518	HN106
	364	376	Tr530×6	52	480	23296RK+**AHX3296**	914	HN106
	364	376	Tr530×6	52	480	23296RHAK+**AHX3296**	844	HN106
480	209	221	Tr540×6	40	500	230/500RK+**AHX30/500**	271	HNL108
	313	325	Tr550×6	47	500	231/500RK+**AHX31/500**	648	HN110
	393	405	Tr550×6	54	500	232/500RK+**AHX32/500**	1 015	HN110

Thrust ball bearings

Single direction thrust ball bearings



Bore diameter **10** – **360** mm

Single direction thrust ball bearings can accommodate axial load in one direction, but they have almost no radial load capacity and are not suitable for high-speed rotation.

Bearings whose housing race back face is spherical (with a spherical back face or aligning seat race) are designed with a self-aligning capability and can accommodate the effects of inaccurate mounting.

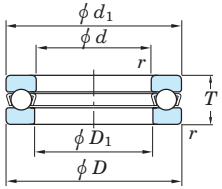


Boundary dimensions	As specified in JIS B 1512.
Tolerances	As specified in JIS B 1514-2. (refer to Table 7-9 on p. A 74.)
Recommended fits	Refer to Table 9-8 on p. A 98.
Standard cages	• Pressed cage (supplementary code : //) • Copper alloy or carbon steel machined cage (supplementary code : FY or FC) • Polyamide resin molded cage (supplementary code : MG)
Required minimum axial load	A certain degree of load is necessary in order for bearings to operate satisfactorily. (refer to p. A 116.)
Allowable misalignment	Misalignment not allowed. (for flat back face type.)
Equivalent axial load	Dynamic equivalent axial load $P_a = F_a$ Static equivalent axial load $P_{0a} = F_a$

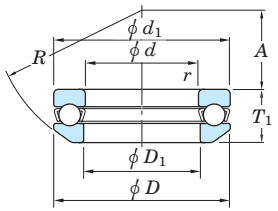
Application of standard cages

Bearing series	Molded cage	Pressed cage	Machined cage
511	51100 – 51107	51108 – 51132	51134 – 51172
512	51200 – 51207	51208 – 51224	51226 – 51272
532	53200 – 53207	53208 – 53224	53226 – 53272
532 U	53200U – 53207U	53208U – 53224U	53226U – 53272U
513	–	51305 – 51313	51314 – 51340
533	–	53305 – 53313	53314 – 53340
533 U	–	53305U – 53313U	53314U – 53340U
514	–	51405 – 51416	51417 – 51436
534	–	53405 – 53416	53417 – 53420
534 U	–	53405U – 53416U	53417U – 53420U

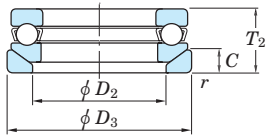
d 10 ~ (40) mm



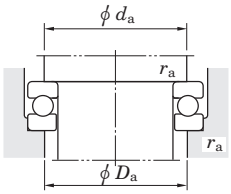
With flat back faces

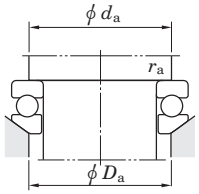


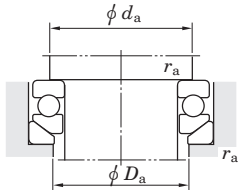
With spherical back face



With aligning seat race



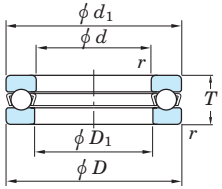




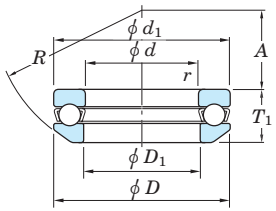
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.			Dimensions (mm)							Mounting dimensions (mm)			(Refer.) Mass (kg)		
d	D	T	T_1	T_2	r min.	C_a	C_{0a}	C_u	Grease lub.	Oil lub.	With flat back faces	With spherical back face	With aligning seat race	d_1 max.	D_1 min.	D_2	D_3	A	R	C	d_a min.	D_a max.	r_a max.	With flat back faces	With spherical back face	With aligning seat race
10	24	9	—	—	0.3	12.5	14.0	0.630	6 500	10 000	**51100**	—	—	24	11	—	—	—	—	—	18	16	0.3	0.020	—	—
	26	11	11.6	13	0.6	15.8	17.1	0.770	5 700	8 800	**51200**	**53200**	**53200U**	26	12	18	28	8.5	22	3.5	20	16	0.6	0.030	0.029	0.037
12	26	9	—	—	0.3	12.9	14.0	0.690	6 500	10 000	**51101**	—	—	26	13	—	—	—	—	—	20	18	0.3	0.022	—	—
	28	11	11.4	13	0.6	16.5	19.0	0.860	5 400	8 300	**51201**	**53201**	**53201U**	28	14	20	30	11.5	25	3.5	22	18	0.6	0.034	0.031	0.043
15	28	9	—	—	0.3	13.2	15.4	0.760	6 100	9 400	**51102**	—	—	28	16	—	—	—	—	—	23	20	0.3	0.024	—	—
	32	12	13.3	15	0.6	20.8	24.8	1.10	4 900	7 500	**51202**	**53202**	**53202U**	32	17	24	35	12	28	4	25	22	0.6	0.046	0.048	0.062
17	30	9	—	—	0.3	13.5	18.2	0.820	6 100	9 400	**51103**	—	—	30	18	—	—	—	—	—	25	22	0.3	0.028	—	—
	35	12	13.2	15	0.6	21.5	27.3	1.25	4 900	7 500	**51203**	**53203**	**53203U**	35	19	26	38	16	32	4	28	24	0.6	0.053	0.055	0.070
20	35	10	—	—	0.3	17.8	24.7	1.10	5 100	7 900	**51104**	—	—	35	21	—	—	—	—	—	29	26	0.3	0.040	—	—
	40	14	14.7	17	0.6	27.9	37.7	1.70	3 900	6 000	**51204**	**53204**	**53204U**	40	22	30	42	18	36	5	32	28	0.6	0.082	0.080	0.100
25	42	11	—	—	0.6	24.4	37.2	1.70	4 400	6 800	**51105**	—	—	42	26	—	—	—	—	—	35	32	0.6	0.059	—	—
	47	15	16.7	19	0.6	34.6	50.4	2.30	3 600	5 500	**51205**	**53205**	**53205U**	47	27	36	50	19	40	5.5	38	34	0.6	0.120	0.120	0.152
	52	18	19.8	22	1	44.7	61.4	2.75	3 100	4 800	**51305**	**53305**	**53305U**	52	27	38	55	21	45	6	41	36	1	0.180	0.180	0.224
	60	24	26.4	29	1	69.5	89.4	4.05	2 600	4 000	**51405**	**53405**	**53405U**	60	27	42	62	19	50	8	46	39	1	0.340	0.350	0.442
30	47	11	—	—	0.6	25.5	42.2	1.90	4 300	6 600	**51106**	—	—	47	32	—	—	—	—	—	40	37	0.6	0.068	—	—
	52	16	17.8	20	0.6	36.7	58.2	2.65	3 400	5 200	**51206**	**53206**	**53206U**	52	32	42	55	22	45	5.5	43	39	0.6	0.150	0.160	0.193
	60	21	22.6	25	1	53.5	78.7	3.55	2 700	4 200	**51306**	**53306**	**53306U**	60	32	45	62	22	50	7	48	42	1	0.270	0.270	0.326
	70	28	30.1	33	1	91.0	126	5.70	2 200	3 400	**51406**	**53406**	**53406U**	70	32	50	75	20	56	9	54	46	1	0.530	0.530	0.660
35	52	12	—	—	0.6	25.5	47.2	2.00	3 900	6 000	**51107**	—	—	52	37	—	—	—	—	—	45	42	0.6	0.090	—	—
	62	18	19.9	22	1	48.9	78.2	3.55	2 900	4 500	**51207**	**53207**	**53207U**	62	37	48	65	24	50	7	51	46	1	0.220	0.220	0.277
	68	24	25.6	28	1	69.3	105	4.75	2 400	3 700	**51307**	**53307**	**53307U**	68	37	52	72	24	56	7.5	55	48	1	0.390	0.400	0.484
	80	32	34	37	1.1	109	155	7.00	1 900	2 900	**51407**	**53407**	**53407U**	80	37	58	85	23	64	10	62	53	1	0.790	0.790	0.960
40	60	13	—	—	0.6	33.6	62.8	2.85	3 400	5 300	**51108**	—	—	60	42	—	—	—	—	—	52	48	0.6	0.120	—	—
	68	19	20.3	23	1	58.7	98.3	4.45	2 700	4 200	**51208**	**53208**	**53208U**	68	42	55	72	28.5	56	7	57	51	1	0.270	0.270	0.340
	78	26	28.5	31	1	86.6	135	6.05	2 100	3 300	**51308**	**53308**	**53308U**	78	42	60	82	28	64	8.5	63	55	1	0.550	0.570	0.690

[Remark] Standard cage types used for the above bearings are described earlier in this section.

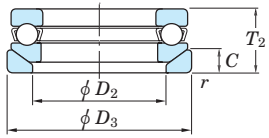
d (40) ~ 70 mm



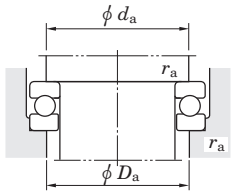
With flat back faces

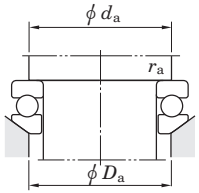


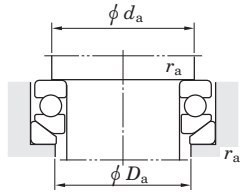
With spherical back face



With aligning seat race



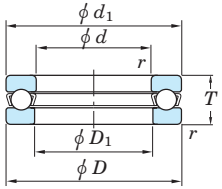




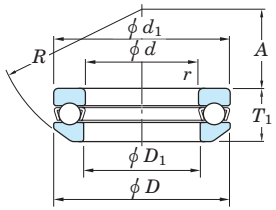
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.			Dimensions (mm)							Mounting dimensions (mm)			(Refer.) Mass (kg)		
d	D	T	T_1	T_2	r min.	C_a	C_{0a}	C_u	Grease lub.	Oil lub.	With flat back faces	With spherical back face	With aligning seat race	d_1 max.	D_1 min.	D_2	D_3	A	R	C	d_a min.	D_a max.	r_a max.	With flat back faces	With spherical back face	With aligning seat race
40	90	36	38.2	42	1.1	141	205	9.25	1 700	2 600	**51408**	**53408**	**53408U**	90	42	65	95	26	72	12	70	60	1	1.14	1.12	1.37
45	65	14	—	—	0.6	34.8	69.1	3.10	3 200	5 000	**51109**	—	—	65	47	—	—	—	—	—	57	53	0.6	0.150	—	—
	73	20	21.3	24	1	59.7	105	4.75	2 600	4 000	**51209**	**53209**	**53209U**	73	47	60	78	26	56	7.5	62	56	1	0.320	0.310	0.397
	85	28	30.1	33	1	100	163	7.40	1 900	3 000	**51309**	**53309**	**53309U**	85	47	65	90	25	64	10	69	61	1	0.690	0.680	0.850
	100	39	42.4	46	1.1	162	242	10.9	1 500	2 300	**51409**	**53409**	**53409U**	100	47	72	105	29	80	12.5	78	67	1	1.47	1.50	1.82
50	70	14	—	—	0.6	35.9	75.4	3.40	3 100	4 800	**51110**	—	—	70	52	—	—	—	—	—	62	58	0.6	0.160	—	—
	78	22	23.5	26	1	60.6	111	5.05	2 300	3 600	**51210**	**53210**	**53210U**	78	52	62	82	32.5	64	7.5	67	61	1	0.390	0.380	0.480
	95	31	34.3	37	1.1	121	202	9.10	1 800	2 700	**51310**	**53310**	**53310U**	95	52	72	100	28	72	11	77	68	1	1.00	1.01	1.24
	110	43	45.6	50	1.5	185	283	12.8	1 400	2 100	**51410**	**53410**	**53410U**	110	52	80	115	35	90	14	86	74	1.5	1.99	1.97	2.38
55	78	16	—	—	0.6	43.5	93.1	4.20	2 800	4 300	**51111**	—	—	78	57	—	—	—	—	—	69	64	0.6	0.240	—	—
	90	25	27.3	30	1	86.7	159	7.20	2 100	3 200	**51211**	**53211**	**53211U**	90	57	72	95	35	72	9	76	69	1	0.610	0.620	0.770
	105	35	39.3	42	1.1	149	246	11.1	1 600	2 400	**51311**	**53311**	**53311U**	105	57	80	110	30	80	11.5	85	75	1	1.34	1.41	1.69
	120	48	50.5	55	1.5	223	359	16.2	1 200	1 900	**51411**	**53411**	**53411U**	120	57	88	125	28	90	15.5	94	81	1.5	2.64	2.57	3.10
60	85	17	—	—	1	51.8	113	5.10	2 600	4 000	**51112**	—	—	85	62	—	—	—	—	—	75	70	1	0.290	—	—
	95	26	28	31	1	92.0	179	8.05	1 900	3 000	**51212**	**53212**	**53212U**	95	62	78	100	32.5	72	9	81	74	1	0.690	0.690	0.850
	110	35	38.3	42	1.1	154	267	12.1	1 500	2 300	**51312**	**53312**	**53312U**	110	62	85	115	41	90	11.5	90	80	1	1.43	1.47	1.78
	130	51	54	58	1.5	267	437	19.7	1 100	1 700	**51412**	**53412**	**53412U**	130	62	95	135	34	100	16	102	88	1.5	3.51	3.44	4.13
65	90	18	—	—	1	52.1	117	5.30	2 400	3 700	**51113**	—	—	90	67	—	—	—	—	—	80	75	1	0.340	—	—
	100	27	28.7	32	1	93.6	189	8.50	1 900	2 900	**51213**	**53213**	**53213U**	100	67	82	105	40	80	9	86	79	1	0.770	0.750	0.930
	115	36	39.4	43	1.1	159	287	13.0	1 400	2 200	**51313**	**53313**	**53313U**	115	67	90	120	38.5	90	12.5	95	85	1	1.57	1.61	1.95
	140	56	60.2	65	2	290	493	22.0	1 000	1 600	**51413**	**53413**	**53413U**	140	68	100	145	40	112	17.5	110	95	2	4.47	4.47	5.28
70	95	18	—	—	1	53.8	127	5.70	2 300	3 600	**51114**	—	—	95	72	—	—	—	—	—	85	80	1	0.360	—	—
	105	27	28.8	32	1	95.2	199	8.95	1 800	2 800	**51214**	**53214**	**53214U**	105	72	88	110	38	80	9	91	84	1	0.810	0.800	0.990
	125	40	44.2	48	1.1	167	291	13.1	1 300	2 000	**51314**	**53314**	**53314U**	125	72	98	130	43	100	13	103	92	1	2.06	2.15	2.56
	150	60	63.6	69	2	312	553	23.8	940	1 450	**51414**	**53414**	**53414U**	150	73	110	155	34	112	19.5	118	102	2	5.48	5.38	6.37

[Remark] Standard cage types used for the above bearings are described earlier in this section.

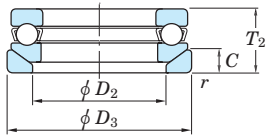
d 75 ~ (120) mm



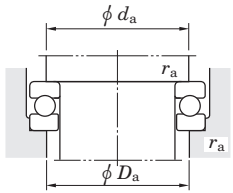
With flat back faces

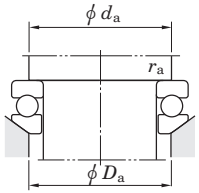


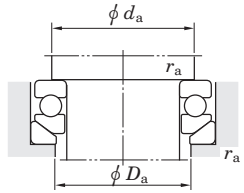
With spherical back face



With aligning seat race



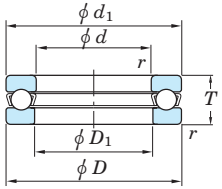




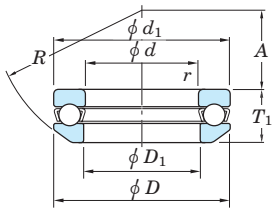
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.			Dimensions (mm)							Mounting dimensions (mm)			(Refer.) Mass (kg)		
d	D	T	T_1	T_2	r min.	C_a	C_{0a}	C_u	Grease lub.	Oil lub.	With flat back faces	With spherical back face	With aligning seat race	d_1 max.	D_1 min.	D_2	D_3	A	R	C	d_a min.	D_a max.	r_a max.	With flat back faces	With spherical back face	With aligning seat race
75	100	19	—	—	1	55.5	136	6.15	2 200	3 400	**51115**	—	—	100	77	—	—	—	—	—	90	85	1	0.420	—	—
	110	27	28.3	32	1	96.7	209	9.40	1 800	2 700	**51215**	**53215**	**53215U**	110	77	92	115	49	90	9.5	96	89	1	0.860	0.850	1.06
	135	44	48.1	52	1.5	192	339	15.0	1 200	1 900	**51315**	**53315**	**53315U**	135	77	105	140	37	100	15	111	99	1.5	2.68	2.72	3.27
	160	65	69	75	2	315	560	23.3	880	1 350	**51415**	**53415**	**53415U**	160	78	115	165	42	125	21	125	110	2	6.75	6.64	7.87
80	105	19	—	—	1	55.8	141	6.35	2 100	3 300	**51116**	—	—	105	82	—	—	—	—	—	95	90	1	0.430	—	—
	115	28	29.5	33	1	98.1	218	9.85	1 700	2 600	**51216**	**53216**	**53216U**	115	82	98	120	46	90	10	101	94	1	0.950	0.930	1.15
	140	44	47.6	52	1.5	200	368	15.8	1 200	1 800	**51316**	**53316**	**53316U**	140	82	110	145	50	112	15	116	104	1.5	2.82	2.86	3.43
	170	68	72.2	78	2.1	337	621	25.1	810	1 250	**51416**	**53416**	**53416U**	170	83	125	175	36	125	22	133	117	2	7.97	7.84	9.22
85	110	19	—	—	1	57.4	150	6.80	2 100	3 200	**51117**	—	—	110	87	—	—	—	—	—	100	95	1	0.460	—	—
	125	31	33.1	37	1	119	264	11.6	1 500	2 300	**51217**	**53217**	**53217U**	125	88	105	130	52	100	11	109	101	1	1.29	1.28	1.57
	150	49	53.1	58	1.5	232	419	17.5	1 100	1 700	**51317**	**53317**	**53317U**	150	88	115	155	43	112	17.5	124	111	1.5	3.66	3.63	4.44
	180	72	77	83	2.1	384	753	29.5	780	1 200	**51417**	**53417**	**53417U**	177	88	130	185	47	140	23	141	124	2	9.29	9.20	10.8
90	120	22	—	—	1	74.6	190	8.40	1 900	2 900	**51118**	—	—	120	92	—	—	—	—	—	108	102	1	0.680	—	—
	135	35	38.5	42	1.1	146	326	13.9	1 400	2 100	**51218**	**53218**	**53218U**	135	93	110	140	45	100	13.5	117	108	1	1.77	1.77	2.19
	155	50	54.6	59	1.5	242	454	18.5	1 000	1 600	**51318**	**53318**	**53318U**	155	93	120	160	40	112	18	129	116	1.5	3.88	3.87	4.71
	190	77	81.2	88	2.1	409	826	31.5	710	1 100	**51418**	**53418**	**53418U**	187	93	140	195	40	140	25.5	149	131	2	11.0	10.7	12.6
100	135	25	—	—	1	106	268	11.2	1 600	2 500	**51120**	—	—	135	102	—	—	—	—	—	121	114	1	0.990	—	—
	150	38	40.9	45	1.1	183	410	16.6	1 200	1 900	**51220**	**53220**	**53220U**	150	103	125	155	52	112	14	130	120	1	2.36	2.34	2.84
	170	55	59.2	64	1.5	296	595	23.2	940	1 450	**51320**	**53320**	**53320U**	170	103	135	175	46	125	18	142	128	1.5	5.11	5.10	6.05
	210	85	90	98	3	460	983	35.7	620	950	**51420**	**53420**	**53420U**	205	103	155	220	50	160	27	165	145	2.5	14.6	14.5	17.4
110	145	25	—	—	1	109	288	11.5	1 600	2 400	**51122**	—	—	145	112	—	—	—	—	—	131	124	1	1.08	—	—
	160	38	40.2	45	1.1	191	450	17.6	1 200	1 800	**51222**	**53222**	**53222U**	160	113	135	165	65	125	14	140	130	1	2.57	2.50	3.06
	190	63	67.2	72	2	334	704	25.9	810	1 250	**51322**	**53322**	**53322U**	187	113	150	195	51	140	20.5	158	142	2	7.72	7.63	8.90
	230	95	—	—	3	474	1 070	37.1	550	850	**51422**	—	—	225	113	—	—	—	—	—	181	159	2.5	19.8	—	—
120	155	25	—	—	1	111	305	11.9	1 500	2 300	**51124**	—	—	155	122	—	—	—	—	—	141	134	1	1.16	—	—

[Remark] Standard cage types used for the above bearings are described earlier in this section.

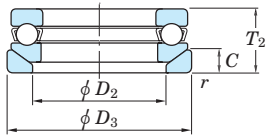
d (120) ~ (180) mm



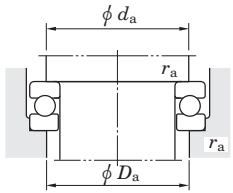
With flat back faces

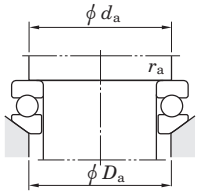


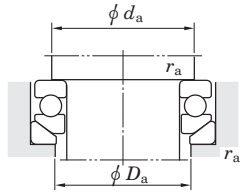
With spherical back face



With aligning seat race



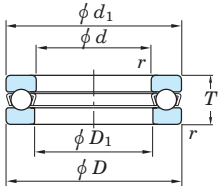




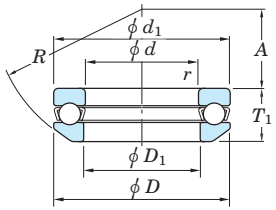
Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.			Dimensions (mm)							Mounting dimensions (mm)			(Refer.) Mass (kg)		
d	D	T	T_1	T_2	r min.	C_a	C_{0a}	C_u	Grease lub.	Oil lub.	With flat back faces	With spherical back face	With aligning seat race	d_1 max.	D_1 min.	D_2	D_3	A	R	C	d_a min.	D_a max.	r_a max.	With flat back faces	With spherical back face	With aligning seat race
120	170	39	40.8	46	1.1	192	470	17.7	1 100	1 700	**51224**	**53224**	**53224U**	170	123	145	175	61	125	15	150	140	1	2.86	2.81	3.46
	210	70	74.1	80	2.1	389	869	30.5	710	1 100	**51324**	**53324**	**53324U**	205	123	165	220	63	160	22	173	157	2	10.6	10.4	12.4
	250	102	—	—	4	601	1 460	48.5	520	800	**51424**	—	—	245	123	—	—	—	—	—	196	174	3	25.0	—	—
130	170	30	—	—	1	130	350	13.0	1 300	2 000	**51126**	—	—	170	132	—	—	—	—	—	154	146	1	1.87	—	—
	190	45	47.9	53	1.5	254	620	22.2	970	1 500	**51226**	**53226**	**53226U**	187	133	160	195	67	140	17	166	154	1.5	4.09	3.98	4.88
	225	75	80.3	86	2.1	413	958	32.5	650	1 000	**51326**	**53326**	**53326U**	220	134	177	235	53	160	26	186	169	2	13.0	12.7	15.2
	270	110	—	—	4	623	1 540	49.0	490	750	**51426**	—	—	265	134	—	—	—	—	—	212	188	3	31.4	—	—
140	180	31	—	—	1	133	375	13.5	1 200	1 900	**51128**	—	—	178	142	—	—	—	—	—	164	156	1	2.02	—	—
	200	46	48.6	55	1.5	234	650	19.6	940	1 450	**51228**	**53228**	**53228U**	197	143	170	210	87	160	17	176	164	1.5	4.46	4.35	5.89
	240	80	84.9	92	2.1	458	1 130	36.9	620	950	**51328**	**53328**	**53328U**	235	144	190	250	68	180	26	199	181	2	15.5	15.1	18.0
	280	112	—	—	4	650	1 680	52.2	450	700	**51428**	—	—	275	144	—	—	—	—	—	222	198	3	33.9	—	—
150	190	31	—	—	1	137	400	13.9	1 200	1 900	**51130**	—	—	188	152	—	—	—	—	—	174	166	1	2.15	—	—
	215	50	53.3	60	1.5	266	652	21.8	840	1 300	**51230**	**53230**	**53230U**	212	153	180	225	79	160	20.5	189	176	1.5	5.64	5.45	7.14
	250	80	83.7	92	2.1	451	1 130	36.0	580	900	**51330**	**53330**	**53330U**	245	154	200	260	89.5	200	26	209	191	2	16.3	15.7	18.8
	300	120	—	—	4	711	1 910	57.4	420	650	**51430**	—	—	295	154	—	—	—	—	—	238	212	3	41.6	—	—
160	200	31	—	—	1	140	425	14.4	1 200	1 800	**51132**	—	—	198	162	—	—	—	—	—	184	176	1	2.28	—	—
	225	51	54.7	61	1.5	279	718	23.4	810	1 250	**51232**	**53232**	**53232U**	222	163	190	235	74	160	21	199	186	1.5	6.53	6.09	7.90
	270	87	91.7	100	3	512	1 340	41.3	550	850	**51332**	**53332**	**53332U**	265	164	215	280	77	200	29	225	205	2.5	21.0	21.0	23.4
	320	130	—	—	5	852	2 410	70.3	390	600	**51432**	—	—	315	164	—	—	—	—	—	254	226	4	51.2	—	—
170	215	34	—	—	1.1	168	510	16.7	1 100	1 700	**51134**	—	—	213	172	—	—	—	—	—	197	188	1	3.25	—	—
	240	55	58.7	65	1.5	326	834	26.3	750	1 150	**51234**	**53234**	**53234U**	237	173	200	250	91	180	21.5	212	198	1.5	8.12	7.69	9.83
	280	87	91.3	100	3	579	1 570	47.4	520	800	**51334**	**53334**	**53334U**	275	174	220	290	105	225	29	235	215	2.5	22.0	22.0	24.5
	340	135	—	—	5	943	2 730	77.2	360	550	**51434**	—	—	335	174	—	—	—	—	—	270	240	4	60.0	—	—
180	225	34	—	—	1.1	168	525	16.7	1 000	1 600	**51136**	—	—	222	183	—	—	—	—	—	207	198	1	3.39	—	—
	250	56	58.2	66	1.5	332	874	26.9	710	1 100	**51236**	**53236**	**53236U**	247	183	210	260	112	200	21.5	222	208	1.5	8.68	8.08	10.4
	300	95	99.3	109	3	578	1 580	46.2	490	750	**51336**	**53336**	**53336U**	295	184	240	310	91	225	32	251	229	2.5	28.1	26.9	29.9

[Remark] Standard cage types used for the above bearings are described earlier in this section.

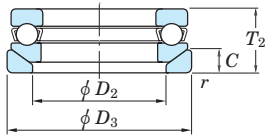
d (180) ~ 360 mm



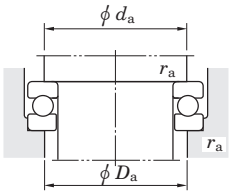
With flat back faces

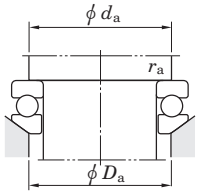


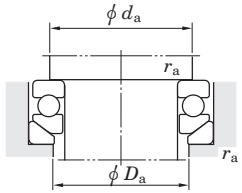
With spherical back face



With aligning seat race





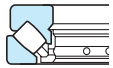


Boundary dimensions (mm)						Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})		Bearing No.			Dimensions (mm)							Mounting dimensions (mm)			(Refer.) Mass (kg)		
d	D	T	T_1	T_2	r min.	C_a	C_{0a}	C_u	Grease lub.	Oil lub.	With flat back faces	With spherical back face	With aligning seat race	d_1 max.	D_1 min.	D_2	D_3	A	R	C	d_a min.	D_a max.	r_a max.	With flat back faces	With spherical back face	With aligning seat race
180	360	140	—	—	5	928	2 730	75.1	320	500	**51436**	—	—	355	184	—	—	—	—	—	286	254	4	69.5	—	—
190	240	37	—	—	1.1	213	655	20.2	970	1 500	**51138**	—	—	237	193	—	—	—	—	—	220	210	1	3.95	—	—
	270	62	65.7	73	2	385	1 060	31.4	650	1 000	**51238**	**53238**	**53238U**	267	194	230	280	98	200	23	238	222	2	11.7	11.2	13.9
	320	105	111	121	4	679	1 950	55.3	440	680	**51338**	**53338**	**53338U**	315	195	255	330	104	250	33	266	244	3	36.0	36.3	39.7
200	250	37	—	—	1.1	215	675	20.4	940	1 450	**51140**	—	—	247	203	—	—	—	—	—	230	220	1	4.13	—	—
	280	62	65.3	74	2	392	1 110	32.2	620	950	**51240**	**53240**	**53240U**	277	204	240	290	125	225	23	248	232	2	12.2	11.6	14.8
	340	110	118.4	130	4	745	2 220	61.1	420	650	**51340**	**53340**	**53340U**	335	205	270	350	92	250	38	282	258	3	42.9	42.7	46.7
220	270	37	—	—	1.1	221	740	21.3	880	1 350	**51144**	—	—	267	223	—	—	—	—	—	250	240	1	4.50	—	—
	300	63	65.6	75	2	428	1 310	36.6	580	900	**51244**	**53244**	**53244U**	297	224	260	310	118	225	25	268	252	2	13.5	12.6	15.9
240	300	45	—	—	1.5	301	1 020	28.0	750	1 150	**51148**	—	—	297	243	—	—	—	—	—	276	264	1.5	7.38	—	—
	340	78	81.6	92	2.1	553	1 800	47.8	520	800	**51248**	**53248**	**53248U**	335	244	290	350	122	250	30	299	281	2	23.1	20.9	25.6
260	320	45	—	—	1.5	289	990	26.2	710	1 100	**51152**	—	—	317	263	—	—	—	—	—	296	284	1.5	7.93	—	—
	360	79	82.8	93	2.1	556	1 880	48.1	490	750	**51252**	**53252**	**53252U**	355	264	305	370	152	280	30	319	301	2	25.0	22.6	28.5
280	350	53	—	—	1.5	411	1 430	36.4	640	900	**51156**	—	—	347	283	—	—	—	—	—	322	308	1.5	12.0	—	—
300	380	62	—	—	2	454	1 610	39.4	540	810	**51160**	—	—	376	304	—	—	—	—	—	348	332	2	17.5	—	—
	420	95	100.5	112	3	713	2 600	61.9	400	600	**51260**	**53260**	**53260U**	415	304	360	430	164	320	34	371	349	2.5	42.5	39.5	48.0
320	400	63	—	—	2	474	1 760	41.9	540	810	**51164**	—	—	396	324	—	—	—	—	—	368	352	2	19.0	—	—
	440	95	100.5	112	3	721	2 710	62.9	400	600	**51264**	**53264**	**53264U**	435	325	380	450	157	320	36	391	369	2.5	45.0	42.0	52.0
340	420	64	—	—	2	483	1 860	43.1	500	770	**51168**	—	—	416	344	—	—	—	—	—	388	372	2	20.5	—	—
	460	96	100.3	113	3	730	2 830	63.8	380	570	**51268**	**53268**	**53268U**	455	345	400	470	199	360	36	411	389	2.5	48.0	45.0	55.0
360	440	65	—	—	2	493	1 960	44.3	500	720	**51172**	—	—	436	364	—	—	—	—	—	408	392	2	21.5	—	—
	500	110	116.7	130	4	876	3 500	76.1	340	500	**51272**	**53272**	**53272U**	495	365	430	510	172	360	43	443	417	3	70.0	65.0	82.0

[Remark] Standard cage types used for the above bearings are described earlier in this section.

Spherical thrust roller bearings

Spherical thrust roller bearings



Bore diameter **60** – **500** mm

Spherical thrust roller bearings are designed to carry high axial loads. They can also support radial load if magnitude is no more than 55 % of the axial load being carried.

These bearings are not suitable for high-speed rotation.

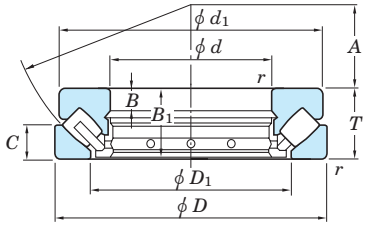
Having a spherical housing race raceway surface, these bearings are self-alignings, adjusting to axial inclination.

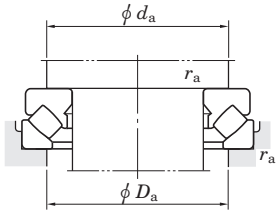
They are usually used with oil lubrication.



Boundary dimensions	As specified in JIS B 1512.
Tolerances	As specified in JIS B 1514-2, class 0. (refer to table 7-10 on p. A 75.)
Recommended fits	Refer to Table 9-8 on p. A 98.
Required minimum axial load	A certain degree of load is necessary in order for bearings to operate satisfactorily. (refer to p. A 116.)
Standard cage	Copper alloy machined cage (supplementary code : FY)
Allowable aligning angle	0.035 – 0.052 rad (2° – 3°) in general, depending on bearing series.
Equivalent axial load	Dynamic equivalent axial load $P_a = 1.2F_r + F_a$ Static equivalent axial load $P_{0a} \fallingdotseq 2.7F_r + F_a$ (Note : $F_r/F_a \leqq 0.55$)

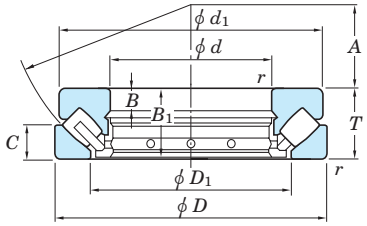
d 60 ~ 160 mm

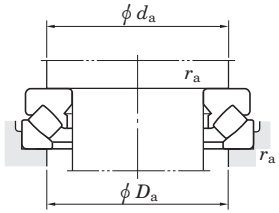




Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speed (min^{-1})	Bearing No.		Dimensions (mm)						Mounting dimensions (mm)			(Refer.) Mass (kg)
d	D	T	r min.	C_a	C_{0a}	C_u	Oil lub.			d_1	D_1	B	B_1	C	A	d_a min.	D_a max.	r_a max.	
60	130	42	1.5	399	884	73.7	2 700	**29412R**		123	89	15	39.5	20	38	90	108	1.5	2.75
65	140	45	2	450	1 020	73.4	2 500	**29413R**		133	96	16	42.5	21	42	100	115	2	3.41
70	150	48	2	485	1 100	105	2 300	**29414R**		142	103	17	45.5	23	44	105	125	2	4.16
75	160	51	2	584	1 360	102	2 100	**29415R**		152	109	18	48	24	47	115	132	2	4.98
80	170	54	2.1	631	1 480	128	2 000	**29416R**		162	117	19	51	26	50	120	140	2	5.95
85	150	39	1.5	400	1 000	100	2 600	**29317R**		143.5	114	13	37	19	50	115	135	1.5	2.87
	180	58	2.1	714	1 700	124	1 900	**29417R**		170	125	21	55	28	54	130	150	2	7.19
90	155	39	1.5	412	1 050	103	2 500	**29318R**		148.5	117	13	37	19	52	120	140	1.5	3.06
	190	60	2.1	821	2 010	158	1 800	**29418R**		180	132	22	57	29	56	135	157	2	8.28
100	170	42	1.5	481	1 270	118	2 300	**29320R**		163	129	14	40	20.8	58	130	150	1.5	3.91
	210	67	3	911	2 220	166	1 650	**29420R**		200	146	24	64	32	62	150	175	2.5	11.2
110	190	48	2	628	1 690	147	2 000	**29322R**		182	143	16	45.5	23	64	145	165	2	5.67
	230	73	3	1 120	2 810	203	1 500	**29422R**		220	162	26	69	35	69	165	190	2.5	14.7
120	210	54	2.1	759	2 030	182	1 800	**29324R**		200	159	18	51	26	70	160	180	2	7.90
	250	78	4	1 300	3 270	241	1 350	**29424R**		236	174	29	74	37	74	180	205	3	18.5
130	225	58	2.1	894	2 440	209	1 700	**29326R**		215	171	19	55	28	76	170	195	2	9.45
	270	85	4	1 490	3 870	270	1 250	**29426R**		255	189	31	81	41	81	195	225	3	23.5
140	240	60	2.1	898	2 490	206	1 600	**29328R**		230	183	20	57	29	82	185	205	2	11.1
	280	85	4	1 560	4 080	289	1 250	**29428R**		268	199	31	81	41	86	205	235	3	24.6
150	250	60	2.1	965	2 740	233	1 550	**29330R**		240	194	20	57	29	87	195	215	2	11.7
	300	90	4	1 730	4 620	334	1 100	**29430R**		285	214	32	86	44	92	220	250	3	29.6
160	270	67	3	1 150	3 070	272	1 400	**29332R**		260	208	23	64	32	92	210	235	2.5	15.4
	320	95	5	1 990	5 370	375	1 050	**29432R**		306	229	34	91	45	99	230	265	4	35.9

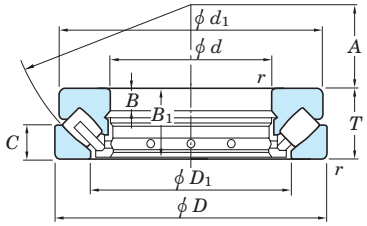
d 170 ~ 320 mm

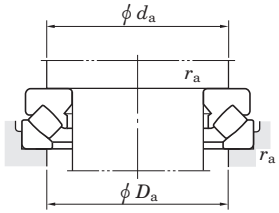




Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speed (min^{-1})	Bearing No.		Dimensions (mm)						Mounting dimensions (mm)			(Refer.) Mass
d	D	T	r min.	C_a	C_{0a}	C_u	Oil lub.			d_1	D_1	B	B_1	C	A	d_a min.	D_a max.	r_a max.	(kg)
170	280	67	3	1 190	3 180	286	1 350	**29334R**		270	216	23	64	32	96	220	245	2.5	15.4
	340	103	5	2 120	5 880	389	950	**29434R**		324	243	37	99	50	104	245	285	4	44.0
180	300	73	3	1 380	3 170	330	1 250	**29336R**		290	232	25	69	35	103	235	260	2.5	20.7
	360	109	5	2 450	6 590	447	900	**29436R**		342	255	39	105	52	110	260	300	4	52.2
190	320	78	4	1 570	4 230	369	1 150	**29338R**		308	246	27	74	38	110	250	275	3	25.1
	380	115	5	2 790	7 690	504	850	**29438R**		360	271	41	111	55	117	275	320	4	61.4
200	280	48	2	641	2 170	151	1 600	**29240**		271	236	15	45	24	108	235	255	2	8.90
	340	85	4	1 810	5 040	415	1 050	**29340R**		325	261	29	81	41	116	265	295	3	31.2
	400	122	5	3 060	8 470	575	800	**29440R**		380	286	43	117	59	122	290	335	4	73.0
220	300	48	2	670	2 340	148	1 550	**29244**		292	254	15	45	24	117	260	275	2	10.0
	360	85	4	1 840	5 240	439	1 000	**29344R**		345	280	29	81	41	125	285	315	3	33.3
	420	122	6	3 160	8 990	619	750	**29444R**		400	308	43	117	58	132	310	355	5	74.2
240	340	60	2.1	1 030	3 670	233	1 250	**29248**		330	283	19	57	30	130	285	305	2	16.7
	380	85	4	1 790	5 330	99.3	950	**29348A**		365	300	29	81	41	135	300	330	3	35.5
	440	122	6	3 260	9 510	659	700	**29448R**		420	326	43	117	59	142	330	375	5	83.0
260	360	60	2.1	1 050	3 720	240	1 200	**29252**		350	302	19	57	30	139	305	325	2	18.5
	420	95	5	1 960	6 040	389	850	**29352**		405	329	32	91	45	148	330	365	4	51.5
	480	132	6	3 760	11 100	764	650	**29452R**		460	357	48	127	64	154	360	405	5	110
280	380	60	2.1	1 030	3 730	225	1 150	**29256**		370	323	19	57	30	150	325	345	2	19.5
	440	95	5	2 200	6 870	439	800	**29356**		423	348	32	91	46	158	350	390	4	53.2
	520	145	6	4 560	13 600	907	550	**29456R**		495	387	52	140	68	166	390	440	5	137
300	420	73	3	1 330	4 880	302	950	**29260**		405	353	21	69	38	162	355	380	2.5	30.5
	480	109	5	2 470	7 780	496	700	**29360**		460	379	37	105	50	168	380	420	4	74.9
	540	145	6	4 670	14 900	925	550	**29460R**		515	402	52	140	70	175	410	460	5	146
320	440	73	3	1 780	6 480	321	900	**29264R**		430	372	21	69	38	172	375	400	2.5	32.7
	500	109	5	2 890	9 380	573	650	**29364**		482	399	37	105	53	180	400	440	4	78.0
	580	155	7.5	5 190	16 100	1 040	500	**29464R**		555	435	55	149	75	191	435	495	6	179

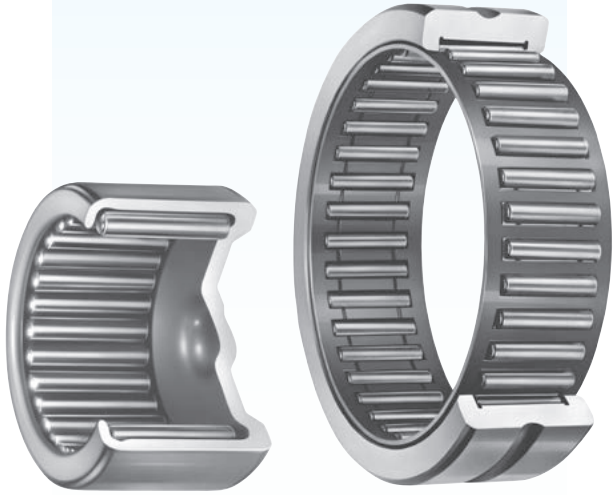
d 340 ~ 500 mm





Boundary dimensions (mm)				Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speed (min^{-1})	Bearing No.		Dimensions (mm)						Mounting dimensions (mm)			(Refer.) Mass
d	D	T	r min.	C_a	C_{0a}	C_u	Oil lub.			d_1	D_1	B	B_1	C	A	d_a min.	D_a max.	r_a max.	(kg)
340	460	73	3	1 800	6 420	307	900	**29268R**		445	395	21	69	37	183	395	420	2.5	34.7
	540	122	5	3 810	12 700	890	600	**29368R**		520	428	41	117	59	192	430	470	4	106
	620	170	7.5	6 190	19 400	1 210	450	**29468R**		590	462	61	164	82	201	465	530	6	224
360	500	85	4	1 650	6 080	332	750	**29272**		485	423	25	81	44	194	420	455	3	51.8
	560	122	5	3 890	13 200	923	550	**29372R**		540	448	41	117	59	202	450	495	4	110
	640	170	7.5	6 440	20 600	1 300	450	**29472R**		610	480	61	164	82	210	485	550	6	231
380	520	85	4	1 750	6 610	343	700	**29276**		505	441	27	81	42	202	440	475	3	52.8
	600	132	6	4 430	15 000	1 030	500	**29376R**		580	477	44	127	63	216	480	525	5	141
	670	175	7.5	6 780	22 000	1 300	410	**29476R**		640	504	63	168	85	230	510	575	6	263
400	540	85	4	1 980	7 610	377	700	**29280**		526	460	27	81	42	212	460	490	3	55.3
	620	132	6	4 630	16 100	1 080	500	**29380R**		596	494	44	127	64	225	500	550	5	144
	710	185	7.5	7 750	25 300	1 530	380	**29480R**		680	534	67	178	89	236	540	610	6	315
420	580	95	5	2 310	8 750	463	600	**29284**		564	489	30	91	46	225	490	525	4	75.4
	650	140	6	5 070	17 700	1 160	450	**29384R**		626	520	48	135	68	235	525	575	5	169
	730	185	7.5	7 960	26 500	1 630	370	**29484R**		700	556	67	178	89	244	560	630	6	330
440	600	95	5	2 340	8 970	441	600	**29288**		585	508	30	91	49	235	510	545	4	77.9
	680	145	6	5 360	18 800	1 250	420	**29388R**		655	548	49	140	70	245	550	600	5	190
	780	206	9.5	9 100	30 000	1 800	320	**29488R**		745	588	74	199	100	260	595	670	8	423
460	620	95	5	2 460	9 620	440	550	**29292**		605	530	30	91	46	245	530	570	4	81.0
	710	150	6	4 580	15 800	875	400	**29392**		685	567	51	144	72	257	575	630	5	216
	800	206	9.5	9 360	31 600	1 870	300	**29492R**		765	608	74	199	100	272	615	690	8	438
480	650	103	5	2 880	11 600	531	500	**29296**		635	556	33	99	55	259	555	595	4	89.0
	850	224	9.5	10 900	36 300	2 100	270	**29496R**		810	638	81	216	108	280	645	730	8	548
500	870	224	9.5	10 800	36 400	2 120	270	**294/500R**		830	661	81	216	107	290	670	750	8	562

Needle roller bearings



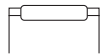
Needle roller bearings are small in sectional height, therefore useful in making machinery smaller and lighter. This type of bearing is used in a wide range of machinery, such as automobiles, motor cycles, electric machines, machine tools, aerospace and office equipment.

- Compact, highly rigid and superior in load carrying performance, compared with other types of bearings.
- Excellent for carrying oscillating loads; contains many small diameter rollers.

- Widely employed in stud type and yoke type track rollers used as guide rollers in cam mechanisms or linear motion units. Allowable loads of these truck rollers are examined with load ratings different form those of general bearings. For detailed information, contact us.
  Also used in miniature one-way clutches in the clutch mechanisms of office equipment, such as copying machines.

The catalog also covers bearings employing rollers other than those prescribed in JIS B 1506 "rollers for roller bearings".

Needle roller and cage assemblies

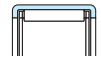


Bore diameter of a needle roller and cage assembly

Metric series **3 – 110** mm

Inch series **9.525 – 127.000** mm

Drawn cup needle roller bearings



Metric series Roller set bore dia. **3 – 60** mm

Inch series Roller set bore dia. **3.175 – 69.850** mm

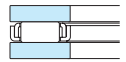
Heavy-duty needle roller bearings



Metric series Roller set bore dia. **5 – 175** mm

Inch series Roller set bore dia. **15.875 – 88.900** mm

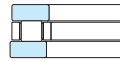
Needle roller thrust bearings



Metric series Bore dia. **6 – 160** mm

Inch series Bore dia. **6.350 – 104.780** mm

Cylindrical roller thrust bearings



Bore dia. **15 – 90** mm

Miniature one-way clutches (Refer.)



Roller set bore dia. **4 – 12** mm

For details, refer to JTEKT separate catalog for Europe “Needle Roller Bearings” (CAT. NO. BS010EN) and catalog for regions excluding Europe “Needle Roller Bearings” (CAT. NO. BS007EN)

(CAT. NO. BS010EN)

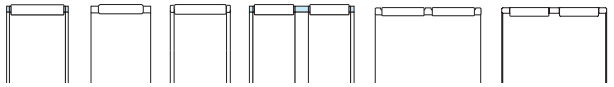
(CAT. NO. BS007EN)

Table 1 (1) Types of needle roller bearing

(1) Radial Needle Roller and Cage Assemblies

Metric Series | **Inch Series**

Single-Row, Double-Row



K, RF, RFN | RS, R, RP | RV, V, VS | K ... ZW, WRF | WR, WRS | WRP

Single-Row



WJ, WJC

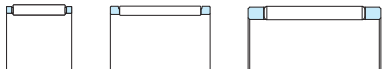
Metric Series

Assemblies for Crank Pin End Applications



K.BE | BE, GS | VE, VS P

Assemblies for Wrist Pin End Applications

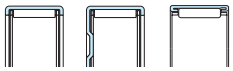


K.SE | R P | RE, UR P

(2) Drawn Cup Needle Roller Bearings

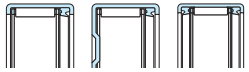
Metric Series (Caged)

Open Ends, Closed One End



BKM, BSM, HK | BK | BTM, BHTM

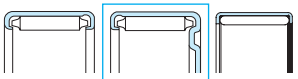
Sealed



HK RS | BK RS | BKM UU, BHKM UU, HK .2RS

(Full Complement)

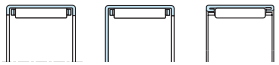
Open Ends, Closed One End



BM, BHM, DL | DLF | YM

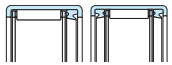
Inch Series (Caged)

Open Ends, Closed One End



J,JH | MJ-1,MJH-1 | BT

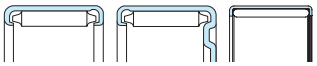
Sealed



JT | JTT

(Full Complement)

Open Ends, Closed One End



B,BH | M-1, MH-1 | Y

Extra-Precision



GB,GBH

Inner Rings

Inch Series



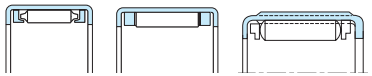
IRA, IR (4 digit or less)

The needle roller bearings explained in this catalog are indicated in blue.
For additional details on JTEKT's Needle Roller Bearing product line please refer to Catalogs BS007EN.
Bearings surrounded by blue lines are listed in the European catalog (CAT. NO. BS010EN) and not in the non-European catalog (BS007EN).

(3) Drawn Cup Roller Clutches

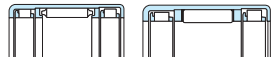
Metric Series

Clutches



FC | FCS FCL-K FC-K | 1WC EWC

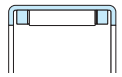
Clutch and Bearing Assemblies



FCB | FCBL-K FCBN-K

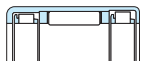
Inch Series

Clutches



RC RC-FS

Clutch and Bearing Assemblies

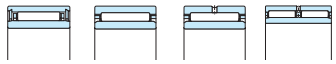


RCB RCB-FS

(4) Heavy-Duty Needle Roller Bearings

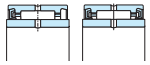
Metric Series (Caged, With Inner Ring)

Unsealed



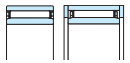
NKJ | NQI | NKJ,NKJS NA48,49,69 | NA69

Sealed



NA49RS | NA49.2RS

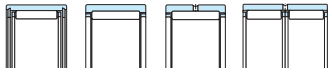
Without Flanges



NAO | +SNSH

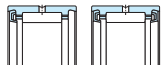
(Without Inner Ring)

Unsealed



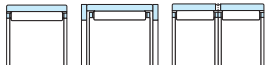
NK | NQ | NK,NKS RNA48,49,69 | RNA69

Sealed



RNA49RS | RNA49.2RS

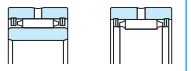
Without Flanges



RNAO | +SNSH | RNAO

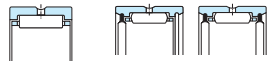
(Full Complement) | **Inch Series (Without Inner Ring)** | **Inner Rings**

With Inner Ring | Without Inner Ring



NA | RNA

Unsealed | Sealed



HJ | HJ-RS | HJ-.2RS

Inch Series



IR

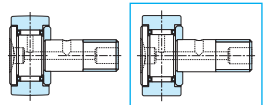
Table 1 (2) Types of needle roller bearing

(5) Track Rollers

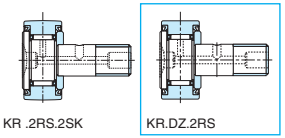
[Stud-Type]

Metric Series (Caged)

Unsealed

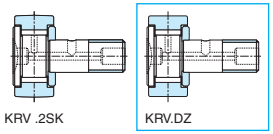


Sealed



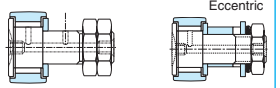
(Full Complement)

Unsealed



Standard Series

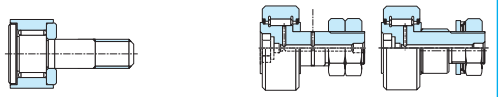
Eccentric



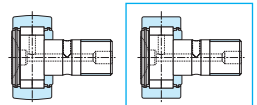
Small Series

Light Series

Eccentric



Cylindrical Rollers



[Yoke-Type]

Metric Series (Caged, Without End Washers)

Unsealed, Without Inner Ring



Unsealed, With Inner Ring



Sealed, Without Inner Ring



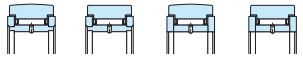
(Caged, Without End Washers)

Sealed, With Inner Ring



(Caged, With End Washers)

With Inner Ring

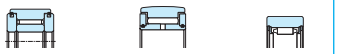


(Full Complement, With End Washers)

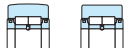
Small Series

Non-Separable

Without Inner Ring Unsealed



With Inner Ring, Cylindrical Rollers



(Full Complement, With Metal Seals)

Non-Separable, Light Series, Heavy Series

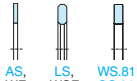


(6) Thrust Bearings, Assemblies, Washers

Metric Series

Thrust Needle Roller and Cage Assemblies and Thrust Washers





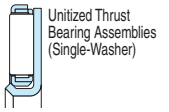
Unitized Thrust Bearing Assemblies (Double-Washer)



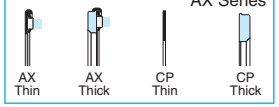
Unitized Thrust Bearing Assemblies (Single-Washer)



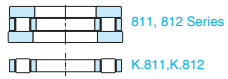
Unitized Thrust Bearing Assemblies (Single-Washer)



Unitized Thrust Bearing Assembly AX Series

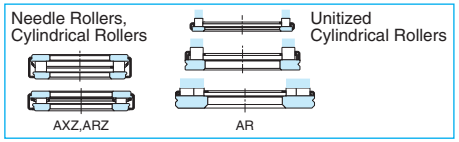


Thrust Cylindrical Roller and Cage Assemblies and Thrust Washers



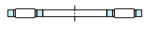
Needle Rollers, Cylindrical Rollers

Unitized Cylindrical Rollers



Inch Series

Thrust Needle Roller and Cage Assemblies and Thrust Washers



Thrust Cylindrical Roller and Cage Assembly



Cylindrical Roller Thrust Bearing



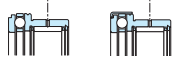
(7) Combined Needle Roller Bearings

Metric Series (Heavy-Duty, Without Inner Ring)

Ball Thrust Series

Separable

Non-Separable



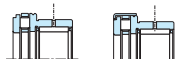
Cylindrical Roller Thrust Series

Non-Separable



Separable

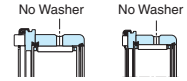
Non-Separable



Needle Roller and Cylindrical Roller Thrust Series

No Washer

No Washer

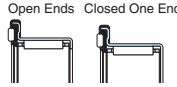


(Drawn Cup, Without Inner Ring)

Needle Roller Thrust Series

Open Ends

Closed One End

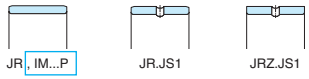


(8) Needle Rollers, Accessories

Inner Rings (Caged)

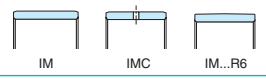
<Metric Series>

For Drawn Cup Needle Roller Bearings, Heavy-Duty



(Full Complement)

For Drawn Cup Needle Roller Bearings



End Washer

For Machine-Tool Quality Precision-Combined Bearings

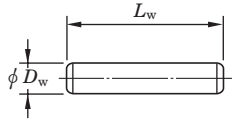
For RNA Bearings (With Oil Holes, Extra Wide)



For Metric Series NAO and RNAO Bearings

[Tolerances of needle roller bearings]

Table 2 Tolerance grades of needle rollers (JIS B 1506)



Unit : μm

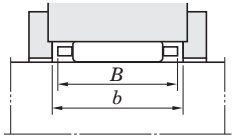
Class	Single [1] plane diameter variation V_{Dwp} max.	Deviation [1] from circular form Δ_{Rw} max.	Gauge lot [1] diameter variation V_{DwL} max.
2	1	1	2
3	1.5	1.5	3
5	2	2.5	5

Class	Actual [2] length deviation Δ_{Lws}	Recommended gauge S
2	h 13	0/– 2, – 1/– 3, – 2/– 4, – 3/– 5, – 4/– 6, – 5/– 7, – 6/– 8, – 7/– 9, – 8/– 10
3		0/– 3, – 1.5/– 4.5, – 3/– 6, – 4.5/– 7.5, – 6/– 9,– 7/– 10
5		0/– 5, – 3/– 8, – 5/– 10

[Notes] 1) Values apply only at middle of roller length.
2) Applied tolerance differs according to L_w division.

[Remark] Along the entire length of the roller, all the actually measured diameters should not exceed the actual maximum diameter at the middle of the entire length of the roller by the lengths shown below.
a) Class 2 : 0.5 μm b) Class 3 : 0.8 μm
c) Class 5 : 1 μm

Table 3 Tolerance of needle roller cage width B (JIS B 1536-3)



Bearing type	B deviation (mm)	
	upper	lower
K, K...ZW	– 0.2	– 0.55
WJ, WJC	*0*	*– 0.38*

[Remark] Values in Italics are prescribed in JTEKT standards.

[Reference] The guide width (b) should satisfy the equation : $b = B +$ (H11)

Table 4 Metric series drawn cup needle roller bearing gauge specifications (caged)

Unit : mm

Nominal bore diameter	Ring gage [1]	Needle roller complement bore diameter	
		max.	min.
3	6.484	3.024	3.006
4	7.984	4.028	4.010
5	8.984	5.028	5.010
6	9.984	6.028	6.010
7	10.980	7.031	7.013
8	11.980	8.031	8.013
9	12.980	9.031	9.013
10	13.980	10.031	10.013
12	15.980	12.034	12.016
12	17.980	12.034	12.016
13	18.976	13.034	13.016
14	19.976	14.034	14.016
15	20.976	15.034	15.016
16	21.976	16.034	16.016
17	22.976	17.034	17.016
18	23.976	18.034	18.016
20	25.976	20.041	20.020
22	27.976	22.041	22.020
25	31.972	25.041	25.020
28	34.972	28.041	28.020
30	36.972	30.041	30.020
35	41.972	35.050	35.025
40	46.972	40.050	40.025
45	51.967	45.050	45.025
50	57.967	50.050	50.025
60	67.967	60.060	60.030

[Note] 1) The ring gage sizes are in accordance with ISO N6 lower limit.

Table 5 Inch series drawn cup needle roller bearing gauge specifications (caged)

Unit : mm

Bearing bore designation	Nominal shaft diameter	Nominal bore diameter	Ring gage	Needle roller complement bore diameter	
				max.	min.
2	3.175	3.175	6.363	3.218	3.195
2 1/2	3.970	3.967	7.155	4.013	3.99
3	4.763	4.763	8.730	4.806	4.783
4	6.350	6.350	11.125	6.411	6.388
5	7.938	7.938	12.713	7.998	7.976
H 5	H 7.938	7.938	14.300	7.998	7.976
6	9.525	9.525	14.300	9.586	9.563
H 6	H 9.525	9.525	15.888	9.586	9.563
7	11.113	11.113	15.888	11.174	11.151
H 7	H 11.113	11.113	17.475	11.174	11.151
8	12.700	12.700	17.475	12.761	12.738
H 8	H12.700	12.700	19.063	12.761	12.738
9	14.288	14.288	19.063	14.349	14.326
H 9	H 14.288	14.288	20.650	14.349	14.326
10	15.875	15.875	20.650	14.349	15.913
H 10	H 15.875	15.875	22.238	14.349	15.913
11	17.463	17.463	22.238	17.524	17.501
H 11	H 17.463	17.463	23.825	17.524	17.501
12	19.050	19.050	25.387	19.086	19.063
H 12	H 19.050	19.050	26.975	19.086	19.063
13	20.638	20.638	26.975	20.673	20.650
H 13	H 20.638	20.638	28.562	20.673	20.650
14	22.225	22.225	28.562	22.261	22.238
H 14	H 22.225	22.225	30.150	22.261	22.238
15	23.813	23.813	30.150	23.848	23.825
16	25.400	25.400	31.737	25.436	25.413
H 16	H 25.400	25.400	33.325	25.436	25.413
17	26.988	26.988	33.325	27.023	27.000
18	28.575	28.575	34.912	28.611	28.588
H 18	H 28.575	28.575	38.087	28.611	28.588
19	30.163	30.163	38.087	30.198	30.175
20	31.750	31.750	38.087	31.786	31.763
H 20	H 31.750	31.750	41.262	31.786	31.763
21	33.338	33.338	41.262	33.376	33.350
22	34.925	34.925	41.262	34.963	34.938
H 22	H 34.925	34.925	44.437	34.963	34.938
24	38.100	38.100	47.612	38.141	38.113
26	41.275	41.275	50.787	41.316	41.288
28	44.450	44.450	53.962	44.493	44.463
30	47.625	47.625	57.137	47.668	47.638
32	50.800	50.800	60.312	50.846	50.815
H 33	H 52.388	52.388	64.280	52.436	52.400
34	53.975	53.975	63.487	54.026	53.990
36	57.150	57.150	66.662	57.201	57.165
42	66.675	66.675	76.187	66.736	66.700
44	69.850	69.850	79.362	69.911	69.875
56	88.900	88.900	101.587	88.961	88.925
88	139.700	139.700	152.375	139.774	139.725

[Remark] Bearing bore should be checked with "go" and "no go" plug gages. The "go" gage size is the minimum needle roller complement bore diameter. The "no go" gage size is larger than the maximum needle roller complement bore diameter by 0.0001 in.

Table 6 Metric series caged needle roller complement bore diameter for bearings without inner rings

Unit : mm

F_w		$\Delta F_{w\,min.}$	
over	up to	max.	min.
3	**6**	+ 0.018	+ 0.010
6	**10**	+ 0.022	+ 0.013
10	**18**	+ 0.027	+ 0.016
18	**30**	+ 0.033	+ 0.020
30	**50**	+ 0.041	+ 0.025
50	**80**	+ 0.049	+ 0.030
80	**120**	+ 0.058	+ 0.036
120	**180**	+ 0.068	+ 0.043
180	**250**	+ 0.079	+ 0.050
250	**315**	+ 0.088	+ 0.056
315	**400**	+ 0.098	+ 0.062

Table 7 Metric series heavy-duty needle roller bearing tolerances = JIS B 1415 (ISO 492) =

(1) Inner ring

Unit : μm

Nominal bore diameter d (mm)		Single plane mean bore diameter deviation Δ_{dmp}						Single plane bore diameter variation V_{dsp} Diameter series 9			Mean bore diameter variation V_{dmp}			Radial runout of assembled bearing inner ring K_{ia}			S_d	Single inner ring width deviation Δ_{Bs}						Inner ring width variation V_{Bs}		
		class 0		class 6		class 5		class 0	class 6	class 5	class 0	class 6	class 5	class 0	class 6	class 5	class 5	class 0		class 6		class 5		class 0	class 6	class 5
over	up to	upper	lower	upper	lower	upper	lower	max.			max.			max.			max.	upper	lower	upper	lower	upper	lower	max.		
2.5	10	0	− 8	0	− 7	0	− 5	10	9	5	6	5	3	10	6	4	7	0	−120	0	−120	0	− 40	15	15	5
10	18	0	− 8	0	− 7	0	− 5	10	9	5	6	5	3	10	7	4	7	0	−120	0	−120	0	− 80	20	20	5
18	30	0	−10	0	− 8	0	− 6	13	10	6	8	6	3	13	8	4	8	0	−120	0	−120	0	−120	20	20	5
30	50	0	−12	0	−10	0	− 8	15	13	8	9	8	4	15	10	5	8	0	−120	0	−120	0	−120	20	20	5
50	80	0	−15	0	−12	0	− 9	19	15	9	11	9	5	20	10	5	8	0	−150	0	−150	0	−150	25	25	6
80	120	0	−20	0	−15	0	−10	25	19	10	15	11	5	25	13	6	9	0	−200	0	−200	0	−200	25	25	7
120	150	0	−25	0	−18	0	−13	31	23	13	19	14	7	30	18	8	10	0	−250	0	−250	0	−250	30	30	8
150	180	0	−25	0	−18	0	−13	31	23	13	19	14	7	30	18	8	10	0	−250	0	−250	0	−250	30	30	8
180	250	0	−30	0	−22	0	−15	38	28	15	23	17	8	40	20	10	11	0	−300	0	−300	0	−300	30	30	10

S_d : Perpendicularity of inner ring face with respect to the bore

(2) Outer ring

Unit : μm

Nominal outside diameter D (mm)		Single plane mean outside diameter deviation Δ_{Dmp}						Single plane outside diameter variation V_{Dsp} Diameter series 9			Mean outside diameter variation V_{Dmp}			Radial runout of assembled bearing outer ring K_{ea}			S_D	Δ_{Cs}		Ring width variation V_{Cs}		
		class 0		class 6		class 5		class 0[1]	class 6[1]	class 5	class 0[1]	class 6[1]	class 5	class 0	class 6	class 5	class 5	class 0, 6, 5		class 0	class 6	class 5
over	up to	upper	lower	upper	lower	upper	lower	max.			max.			max.			max.	upper	lower	max.		
6	18	0	− 8	0	− 7	0	− 5	10	9	5	6	5	3	15	8	5	8					5
18	30	0	− 9	0	− 8	0	− 6	12	10	6	7	6	3	15	9	6	8					5
30	50	0	−11	0	− 9	0	− 7	14	11	7	8	7	4	20	10	7	8					5
50	80	0	−13	0	−11	0	− 9	16	14	9	10	8	5	25	13	8	8	Shall conform to the tolerance Δ_{Bs} on d of the same bearing		Shall conform to the tolerance V_{Bs} on d of the same bearing		6
80	120	0	−15	0	−13	0	−10	19	16	10	11	10	5	35	18	10	9					8
120	150	0	−18	0	−15	0	−11	23	19	11	14	11	6	40	20	11	10					8
150	180	0	−25	0	−18	0	−13	31	23	13	19	14	7	45	23	13	10					8
180	250	0	−30	0	−20	0	−15	38	25	15	23	15	8	50	25	15	11					10
250	315	0	−35	0	−25	0	−18	44	31	18	26	19	9	60	30	18	13					11

[Note] 1) Shall be applied when locating snap ring is not fitted.

S_D : Perpendicularity of outer ring outside surface with respect to the face

Δ_{Cs} : Deviation of a single outer ring width

Table 8 Inch series heavy-duty needle roller bearing (HJ outer ring)

(1) Outside diameter and width tolerances Unit : mm

Nominal outside diameter D		Single plane mean outside diameter deviation (D_{mp}) [1]		Deviaiton from nominal of width (C)	
over	up to	max.	min.	max.	min.
19.050	50.800	+0	−0.013		
50.800	82.550	+0	−0.015	+0	−0.013
82.550	120.650	+0	−0.020		

[Note] 1) "Single mean diameter" is defined as the mean diameter in a single radial plane.

(2) Roller complement bore tolerance Unit : mm

Nominal roller complement bore diameter F_w		Deviaiton from nominal of the smallest single diameter of the roller complement bore (F_m) [1]	
over	up to	max.	min.
12.700	15.875	+0.043	+0.020
15.875	28.575	+0.046	+0.023
28.575	41.275	+0.048	+0.025
41.275	47.625	+0.050	+0.025
47.625	69.850	+0.053	+0.028
69.850	76.200	+0.058	+0.028
76.200	101.600	+0.060	+0.030

[Note] 1) "The smallest single diameter of the roller complement bore" is defined as the diameter of the cylinder which, when used as a bearing inner ring, results in zero radial internal clearance in the bearing on at least one diameter.

Table 9 Inch series heavy-duty needle roller bearing (IR inner ring)

(1) Bore and width tolerances Unit : mm

Nominal bore diameter d		Single plane mean bore diameter deviation (d_{mp}) [1]		Deviaiton from nominal of width (B)	
over	up to	max.	min.	max.	min.
7.938	19.050	+0	−0.010		
19.050	50.800	+0	−0.013	+0.25	+0.12
50.800	82.550	+0	−0.015		

[Note] 1) "Single mean diameter" is defined as the mean diameter in a single radial plane.

(2) Outside diameter tolerance Unit : mm

Nominal outside diameter F		Single plane mean outside diameter deviation (F_{mp}) [1]	
over	up to	max.	min.
12.700	15.875	−0.013	−0.023
15.875	25.400	−0.018	−0.031
25.400	28.575	−0.023	−0.036
28.575	34.925	−0.023	−0.036
34.925	47.625	−0.025	−0.038
47.625	76.200	−0.028	−0.040
76.200	95.250	−0.033	−0.046

[Note] 1) "Single mean diameter" is defined as the mean diameter in a single radial plane.

Table 10 Tolerance for metric series thrust needle roller and cage assemblies (type code : FNT)

(1) Bore diameter Unit : mm

Nominal bore diameter D_{c1}		Smallest single bore diameter deviation (E11)	
over	up to	upper	lower
3	6	+0.095	+0.020
6	10	+0.115	+0.025
10	18	+0.142	+0.032
18	30	+0.170	+0.040
30	50	+0.210	+0.050
50	80	+0.250	+0.060
80	120	+0.292	+0.072
120	180	+0.335	+0.085

(2) Outside diameter Unit : mm

Nominal outside diameter D_c		Largest single outside diameter deviation (c12)	
over	up to	upper	lower
18	30	−0.110	−0.320
30	40	−0.120	−0.370
40	50	−0.130	−0.380
50	65	−0.140	−0.440
65	80	−0.150	−0.450
80	100	−0.170	−0.520
100	120	−0.180	−0.530
120	140	−0.200	−0.600
140	160	−0.210	−0.610
160	180	−0.230	−0.630
180	200	−0.240	−0.700

Table 11 Tolerance for metric series thrust needle roller and cage assemblies (type code : AXK)

(1) Bore diameter Unit : mm

Nominal bore diameter D_{c1}		Smallest single bore diameter deviation (E12)	
over	up to	upper	lower
3	6	+0.140	+0.020
6	10	+0.175	+0.025
10	18	+0.212	+0.032
18	30	+0.250	+0.040
30	50	+0.300	+0.050
50	80	+0.360	+0.060
80	120	+0.422	+0.072
120	180	+0.485	+0.085

(2) Outside diameter Unit : mm

Nominal outside diameter D_c		Largest single outside diameter deviation (c13)	
over	up to	upper	lower
18	30	−0.110	−0.440
30	40	−0.120	−0.510
40	50	−0.130	−0.520
50	65	−0.140	−0.600
65	80	−0.150	−0.610
80	100	−0.170	−0.710
100	120	−0.180	−0.720
120	140	−0.200	−0.830
140	160	−0.210	−0.840
160	180	−0.230	−0.860
180	200	−0.240	−0.960

Table 12 Tolerance for metric series thrust washers (type code : AS series)

(1) Bore diameter Unit : mm

Nominal bore diameter d		Smallest single bore diameter deviation (E13)	
over	up to	upper	lower
3	6	+0.200	+0.020
6	10	+0.245	+0.025
10	18	+0.302	+0.032
18	30	+0.370	+0.040
30	50	+0.440	+0.050
50	80	+0.520	+0.060
80	120	+0.612	+0.072
120	180	+0.715	+0.085

(2) Outside diameter Unit : mm

Nominal outside diameter d_1		Largest single outside diameter deviation (e13)	
over	up to	upper	lower
18	30	−0.040	−0.370
30	50	−0.050	−0.440
50	80	−0.060	−0.520
80	120	−0.072	−0.612
120	180	−0.085	−0.715
180	250	−0.100	−0.820

Table 13 Tolerance for metric series thrust washers (type code : LS series)

(1) Bore diameter Unit : mm

Nominal bore diameter d		Smallest single bore diameter deviation (E12)	
over	up to	upper	lower
3	6	+0.140	+0.020
6	10	+0.175	+0.025
10	18	+0.212	+0.032
18	30	+0.250	+0.040
30	50	+0.300	+0.050
50	80	+0.360	+0.060
80	120	+0.422	+0.072
120	180	+0.485	+0.085

(2) Outside diameter Unit : mm

Nominal outside diameter d_1		Largest single outside diameter deviation (a12)	
over	up to	upper	lower
18	30	−0.300	−0.510
30	40	−0.310	−0.560
40	50	−0.320	−0.570
50	65	−0.340	−0.640
65	80	−0.360	−0.660
80	100	−0.380	−0.730
100	120	−0.410	−0.760
120	140	−0.460	−0.860
140	160	−0.520	−0.920
160	180	−0.580	−0.980
180	200	−0.660	−1.120

[Remark] Thickness tolerances for series LS heavy thrust washers are given in bearing tables.

Table 14 Tolerance for metric series shaft piloted washers of thrust bearings (type code : WS.811 and WS.812)

Unit : mm

Nominal bore diameter d		Tolerance class P0				Tolerance class P6				Tolerance class P5			
		Deviations Δ_{dmp}		Variation V_{dsp}	S_i 1)	Deviations Δ_{dmp}		Variation V_{dsp}	S_i 1)	Deviations Δ_{dmp}		Variation V_{dsp}	S_i 1)
over	up to	upper	lower	max.	min.	upper	lower	max.	min.	upper	lower	max.	min.
	18	+0	−0.008	0.006	0.010	+0	−0.008	0.006	0.005	+0	−0.008	0.006	0.003
18	30	+0	−0.010	0.008	0.010	+0	−0.010	0.008	0.005	+0	−0.010	0.008	0.003
30	50	+0	−0.012	0.009	0.010	+0	−0.012	0.009	0.006	+0	−0.012	0.009	0.003
50	80	+0	−0.015	0.011	0.010	+0	−0.015	0.011	0.007	+0	−0.015	0.011	0.004
80	120	+0	−0.020	0.015	0.015	+0	−0.020	0.015	0.008	+0	−0.020	0.015	0.004
120	180	+0	−0.025	0.019	0.015	+0	−0.025	0.019	0.009	+0	−0.025	0.019	0.005
180	250	+0	−0.030	0.023	0.020	+0	−0.030	0.023	0.010	+0	−0.030	0.023	0.005
250	315	+0	−0.035	0.026	0.025	+0	−0.035	0.026	0.013	+0	−0.035	0.026	0.007
315	400	+0	−0.040	0.030	0.030	+0	−0.040	0.030	0.015	+0	−0.040	0.030	0.007
400	500	+0	−0.045	0.034	0.030	+0	−0.045	0.034	0.018	+0	−0.045	0.034	0.009

[Note] 1) The values of the wall thickness variation S_e, for the housing piloted washer are identical to S_i for the shaft - piloted washers.
Δ_{dmp} : Single plane mean bore diameter deviation
V_{dsp} : Single plane bore diameter variation
S_i : Wall thickness variation

Table 15 Tolerance for metric series housing piloted washers of thrust bearings (type code : GS.811 and GS.812)

Unit : mm

Nominal outside diameter D		Tolerance class P0			Tolerance class P6			Tolerance class P5		
		Deviations Δ_{Dmp}		Variation V_{Dsp}	Deviations Δ_{Dmp}		Variation V_{Dsp}	Deviations Δ_{Dmp}		Variation V_{Dsp}
over	up to	upper	lower	max.	upper	lower	max.	upper	lower	max.
	30	+0	−0.013	0.010	+0	−0.013	0.010	+0	−0.013	0.010
30	50	+0	−0.016	0.012	+0	−0.016	0.012	+0	−0.016	0.012
50	80	+0	−0.019	0.014	+0	−0.019	0.014	+0	−0.019	0.014
80	120	+0	−0.022	0.017	+0	−0.022	0.017	+0	−0.022	0.017
120	180	+0	−0.025	0.019	+0	−0.025	0.019	+0	−0.025	0.019
180	250	+0	−0.030	0.023	+0	−0.030	0.023	+0	−0.030	0.023
250	315	+0	−0.035	0.026	+0	−0.035	0.026	+0	−0.035	0.026
315	400	+0	−0.040	0.030	+0	−0.040	0.030	+0	−0.040	0.030
400	500	+0	−0.045	0.034	+0	−0.045	0.034	+0	−0.045	0.034

[Note] Δ_{Dmp} : Single plane mean outside diameter deviation
V_{Dsp} : Single plane outside diameter variation

Table 16 Tolerance for inch series thrust needle roller and cage assemblies (type code : NTA)

Unit : mm

Needle roller diameter (nominal) D_w	Deviations			
	Bore diameter D_{c1}		Outside diameter D_c	
	upper	lower	upper	lower
1.981	+0.178	+0.051	−0.254	−0.508
3.175	+0.254	+0.051	−0.254	−0.635

Table 17 Tolerance for inch series thrust washers (type code : TRA, TRB, etc.)

(1) Bore diameter

Unit : mm

Nominal bore diameter d		Deviations	
over	up to	upper	lower
6.000	57.200	+0.300	+0.050
57.200	133.400	+0.430	+0.050

(2) Outside diameter

Unit : mm

Nominal Outside diameter d_1		Deviations	
over	up to	upper	lower
6.000	133.400	−0.025	−0.760

[Recommended fit and internal clearance]

Table 18 Recommended fit for metric series radial needle roller and cage assemblies

Condition	Tolerance zone class		
	Shaft		Housing bore
Radial clearance	$F_w \leqq 50$ mm	$F_w > 50$ mm	
Smaller than normal	j 5	h 5	G 6
Normal	h 5	g 5	
Larger than normal	g 6	f 6	

Table 19 Recommended fit for metric series drawn cup needle roller bearings

Bearing type	Operating condition	Shaft fit (recommended internal radial clearances)	Housing fit (recommended internal radial clearances)
HK, BK (caged)	One piece heavy section steel or cast iron housing	h5 (h6)	N6 (N7)
	Housing material of low rigidity	h5 (h6)	R6 (R7)
	Outer ring rotation (one piece heavy section steel or cast iron housing)	f5 (f6)	R6 (R7)
	Oscillating Motion	j5 (j6)	1)

[Note] 1) Tolerance dependent on housing design.
[Remark] When the bearing is provided with an inner ring, the shaft tolerance class h5 should be selected.

Table 20 Recommended fit for inch series drawn cup needle roller bearings

Bearing design	Shaft			Housing		
	Classification	max.	min.	Classification	max.	min.
J, JTT 1)	$F_w \leqq 5.556$ mm (7/32")	0	−0.008 mm (−0.0003")	$D \leqq 7.144$ mm (9/32")	+0.013 mm (−0.0005")	0
	5.556 mm (7/32") $< F_w \leqq$ 50.006 mm (1 31/32")	0	−0.013 mm (−0.0005")			
	50.006 mm (1 31/32") $< F_w \leqq$ 119.856 mm (4 23/32")	0	−0.015 mm (−0.0006")	7.144 mm (9/32") $< D$	+0.013 mm (−0.0005")	+0.013 mm (−0.0005")
	119.856 mm (4 23/32") $< F_w \leqq$ 180.181 mm (7 3/32")	0	−0.018 mm (−0.0007")			

[Note] 1) Special fits
When D = 8.733 mm (0.3438") : housing fit : −0.003 mm (−0.0001") maximum, −0.015 mm (−0.0006") minimum
When D = 22.212 mm (0.8745") : housing fit : +0.025 mm (+0.0010") maximum, 0 mm (0") minimum
When D = 152.400 mm (6.0000") : housing fit : +0.025 mm (+0.0010") maximum, −0.025 mm (−0.0010") minimum

Table 21 Recommended fit for metric series heavy-duty needle roller bearings

Without inner ring

Rotation conditions	Nominal housing bore diameter D (mm)	ISO tolerance zone for housing		Nominal shaft diameter F shaft (mm)	ISO tolerance zone for shaft	
		caged	full		caged	full
Load stationary relative to housing	all diameters	H7 (J7)	J6	all diameters	h6 (h5)	h5
General work with larger clearance		K7	—		g6	—
Load rotates relative to housing		N7	M6		f6	g5

[Remark] Care should be taken that the selected bearing internal clearance is appropriate for the operating conditions.

Table 22 Recommended fit for metric series needle roller thrust bearings

Bearing components	Shaft tolerance (shaft piloting)	Housing tolerance (housing piloting)
Needle roller and cage assembly. Types : AXK, FNT	h8	H8
Thin thrust washer. Type : AS	h8	H8
Heavy thrust washer. Type : LS	h8	H8
Shaft-piloted thrust washer. Type : WS.811	h6 (j6)	Clearance
Housing-piloted thrust washer. Type : GS.811	Clearance	H7 (K7)

Table 23 Recommended fit for metric series cylindrical roller thrust bearings

Bearing components	Shaft tolerance (shaft piloting)	Housing tolerance (housing piloting)	Piloting components
Thrust cylindrical roller and cage assembly. Types : K.811 and K.812	h8	H10	Shaft
Heavy thrust washer. Type : LS	h10	H11	Shaft
Shaft-piloted thrust washer. Types : WS.811, WS.812	h6 (j6)	Clearance	Shaft
Housing-piloted thrust washer. Types : GS.811, GS.812	Clearance	H7 (K7)	Housing

[Shaft and housing specifications]

Table 24 Specifications of shafts and housings (needle roller and cage assemblies, drawn cup needle roller bearings, and heavy-duty needle roller bearings)

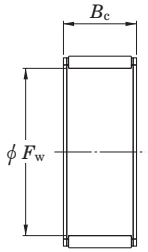
	Shaft		Housing bore	
	Raceway surface	Fitting surface	Raceway surface	Fitting surface
Roundness	· Shaft diameter ≦ 25 mm : 2.5 μm or less · Shaft diameter > 25 mm : 2.5 μm × (shaft diameter /25 mm) or less	One-half the shaft diameter tolerance or less	8 μm or one-half the bore tolerance or less	One-half the bore tolerance or less
Variation of mean diameter (Cylindricity)	Within the range of the bearing width, 5 μm or less per 25 mm or one-half the bearing tolerance or less (whichever is smaller)	One-half the shaft diameter tolerance or less	Within the range of the bearing width, 5 μm or less per 25 mm or one-half the bearing tolerance or less (whichever is smaller)	Within the length of the outer ring, 13 μm or one-half the diameter tolerance (whichever is smaller) or less
Surface roughness	0.2 a or less	0.8 a or less	0.2 a or less	1.6 a or less
Hardness	58 HRC or harder [1)]	—	58 HRC or harder [1)]	—
Shaft slope	13 μm or less per 25 mm		—	

[Note] 1) During the carburizing or induction-hardening of case hardened steel, not only must the surface hardness requirement specified above be met, but the case depth of HV 550 (52.3 HRC) must be met in the range of 0.08 D_w to 0.10 D_w mm. (D_w : roller diameter)

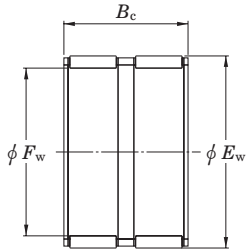
Table 25 Needle roller thrust bearing mounting surface specifications

Flatness (dishing or coning)		Maximum angle : Arctan 0.001
Squareness		Maximum angle : Arctan 0.0005
Roughness (Ra)		0.2 a or less
Hardness		58 HRC or harder (refer to the note for Table 25 above regarding depth.)

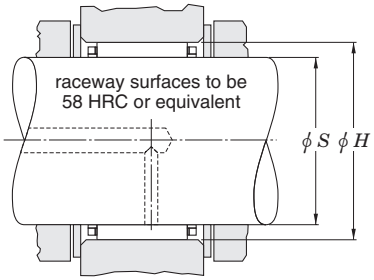
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. 3 ~ (10) mm



K



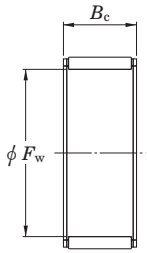
K ZW



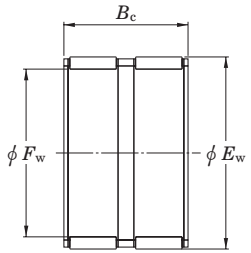
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
3	3	5	7	**K3X5X7TN**	1.56	1.29	0.200	P	48 000	74 000		0.0002	3.000	2.996	5.004	5.012
4	4	7	7	**K4X7X7TN**	1.83	1.32	0.200	P	34 000	52 000		0.0005	4.000	3.995	7.014	7.005
5	5	8	8	**K5X8X8TN**	2.18	1.71	0.260	P	31 000	47 000		0.0007	5.000	4.995	8.014	8.005
	5	8	10	**K5X8X10TN**	3.04	2.63	0.400	P	31 000	47 000		0.0008	5.000	4.995	8.014	8.005
	5	9	13	**K5X9X13TN**	4.29	3.55	0.540	P	26 000	40 000		0.002	5.000	4.995	9.014	9.005
6	6	9	8	**K6X9X8H**	3.19	2.90	0.420	S	29 000	44 000		0.0008	6.000	5.995	9.014	9.005
	6	9	8	**K6X9X8TN**	2.47	2.07	0.310	P	29 000	44 000		0.001	6.000	5.995	9.014	9.005
	6	9	10	**K6X9X10TN**	3.07	2.74	0.420	P	29 000	44 000		0.001	6.000	5.995	9.014	9.005
7	7	10	8	**K7X10X8TN**	2.74	2.44	0.370	P	28 000	42 000		0.001	7.000	6.994	10.014	10.005
	7	10	10	**K7X10X10TN**	3.40	3.22	0.490	P	28 000	42 000		0.001	7.000	6.994	10.014	10.005
	7	11	15	**K7X11X15TN**	6.44	6.24	0.940	P	23 000	35 000		0.003	7.000	6.994	11.017	11.006
8	8	11	8	**K8X11X8FV**	3.23	3.11	0.470	S	26 000	41 000		0.002	8.000	7.994	11.017	11.006
	8	11	8	**K8X11X8TN**	2.34	2.05	0.300	P	26 000	41 000		0.001	8.000	7.994	11.017	11.006
	8	11	10	**K8X11X10H**	4.57	4.89	0.740	S	26 000	41 000		0.002	8.000	7.994	11.017	11.006
	8	11	10	**K8X11X10FV**	4.01	4.11	0.630	S	26 000	41 000		0.002	8.000	7.994	11.017	11.006
	8	11	10	**K8x11x10TN**	3.84	3.91	0.600	P	26 000	41 000		0.001	8.000	7.994	11.006	11.017
	8	11	13	**K8x11x13TN**	5.18	5.75	0.870	P	26 000	41 000		0.002	8.000	7.994	11.006	11.017
	8	11	13	**K8X11X13H**	5.22	5.78	0.880	S	26 000	41 000		0.003	8.000	7.994	11.017	11.006
9	9	12	10	**K9X12X10FH**	4.27	4.60	0.700	S	26 000	40 000		0.003	9.000	8.994	12.017	12.006
	9	12	10	**K9X12X10FV**	4.27	4.60	0.700	S	26 000	40 000		0.002	9.000	8.994	12.017	12.006
	9	12	13	**K9X12X13FH**	5.57	6.47	0.980	S	26 000	40 000		0.003	9.000	8.994	12.017	12.006
	9	12	13	**K9X12X13FV**	5.57	6.47	0.980	S	26 000	40 000		0.003	9.000	8.994	12.017	12.006
	9	13	8	**K9X13X8H**	3.96	3.50	0.530	S	21 000	32 000		0.003	9.000	8.994	13.017	13.006
10	10	13	10	**K10X13X10H**	5.40	6.43	0.980	S	25 000	39 000		0.002	10.000	9.994	13.017	13.006
	10	13	10	**K10X13X10TN**	4.29	4.77	0.730	P	25 000	39 000		0.002	10.000	9.994	13.017	13.006
	10	13	13	**K10X13X13**	5.90	7.16	1.10	S	25 000	39 000		0.003	10.000	9.994	13.017	13.006

[Note] 1) Cage material: P: polymer cage, S: steel cage

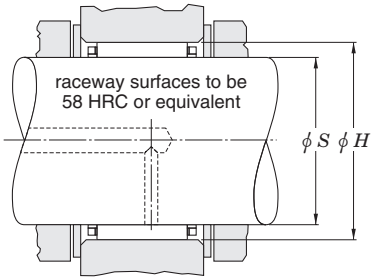
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. (10) ~ (15) mm



K



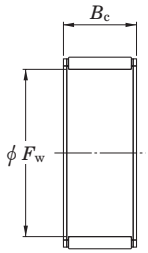
K ZW



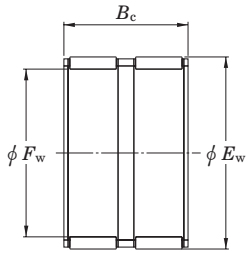
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
10	10	13	16	**K10X13X16**	7.43	9.64	1.50	S	25 000	39 000		0.004	10.000	9.994	13.017	13.006
	10	14	10	**K10X14X10H**	6.12	6.29	0.960	S	20 000	31 000		0.003	10.000	9.994	14.017	14.006
	10	14	13	**K10X14X13H**	7.88	8.71	1.35	S	20 000	31 000		0.004	10.000	9.994	14.017	14.006
	10	16	12	**K10X16X12F**	8.39	7.47	1.15	S	15 000	24 000		0.006	10.000	9.994	16.017	16.006
	10	16	12	**K10X16X12TN**	7.50	6.40	0.970	P	15 000	24 000		0.005	10.000	9.994	16.017	16.006
12	12	15	10	**K12X15X10H**	5.85	7.51	1.15	S	24 000	37 000		0.003	12.000	11.992	15.017	15.006
	12	15	13	**K12X15X13H**	6.78	9.03	1.40	S	24 000	37 000		0.004	12.000	11.992	15.017	15.006
	12	16	13	**K12X16X13H**	7.49	8.51	1.60	S	19 000	30 000		0.006	12.000	11.992	16.017	16.006
	12	17	13	**K12X17X13**	8.93	9.29	1.20	S	16 000	25 000		0.008	12.000	11.992	17.017	17.006
	12	18	12	**K12X18X12H**	9.76	9.40	1.40	S	14 000	22 000		0.009	12.000	11.992	18.017	18.006
13	13	17	10	**K13X17X10**	7.22	8.33	1.25	S	19 000	29 000		0.004	13.000	12.992	17.017	17.006
	13	18	15	**K13X18X15F**	10.8	12.1	1.85	S	16 000	25 000		0.008	13.000	12.992	18.017	18.006
14	14	18	8	**K14X18X8**	5.39	5.82	0.880	S	19 000	29 000		0.004	14.000	13.992	18.017	18.006
	14	18	10	**K14X18X10**	7.17	8.41	1.30	S	19 000	29 000		0.005	14.000	13.992	18.017	18.006
	14	18	13	**K14X18X13**	9.73	12.5	1.90	S	19 000	29 000		0.006	14.000	13.992	18.017	18.006
	14	18	15	**K14X18X15**	10.5	13.8	2.15	S	19 000	29 000		0.007	14.000	13.992	18.017	18.006
	14	18	17	**K14X18X17H**	12.4	17.1	2.65	S	19 000	29 000		0.008	14.000	13.992	18.017	18.006
	14	19	13	**K14X19X13H**	10.2	11.4	1.75	S	16 000	24 000		0.008	14.000	13.992	19.020	19.007
	14	19	18	**K14X19X18F**	13.2	16.0	2.50	S	16 000	24 000		0.011	14.000	13.992	19.020	19.007
	14	20	12	**K14X20X12**	10.5	10.6	1.60	S	14 000	21 000		0.009	14.000	13.992	20.020	20.007
15	15	18	14	**K15X18X14TN**	7.92	11.9	1.80	P	13 000	23 000		0.003	15.000	14.992	18.017	18.006
	15	18	16	**K15X18X16F**	8.36	12.6	1.95	S	13 000	23 000		0.005	15.000	14.992	18.017	18.006
	15	18	17	**K15X18X17**	8.08	12.1	1.85	S	23 000	36 000		0.005	15.000	14.992	18.017	18.006
	15	19	10	**K15X19X10**	7.87	9.69	1.45	S	18 000	28 000		0.005	15.000	14.992	19.020	19.007
	15	19	13	**K15X19X13**	9.66	12.6	1.90	S	18 000	28 000		0.007	15.000	14.992	19.020	19.007
	15	19	17	**K15X19X17H**	12.3	17.2	2.65	S	18 000	28 000		0.009	15.000	14.992	19.020	19.007
	15	19	22	**K15X19X22ZW**	12.2	17.0	2.60	S	18 000	28 000		0.010	15.000	14.992	19.020	19.007

[Note] 1) Cage material: P: polymer cage, S: steel cage

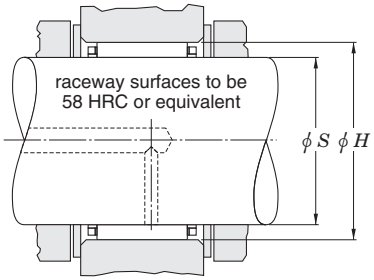
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. (15) ~ (18) mm



K



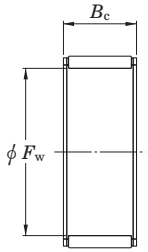
K ZW



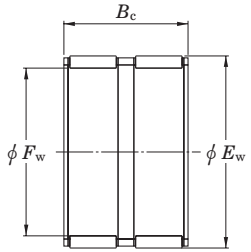
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
15	15	20	13	**K15X20X13H**	9.93	11.3	1.80	S	16 000	24 000		0.008	15.000	14.992	20.020	20.007
	15	21	15	**K15X21X15**	13.4	14.8	2.30	S	14 000	21 000		0.013	15.000	14.992	21.020	21.007
	15	21	21	**K15X21X21H**	18.0	21.7	3.40	S	14 000	21 000		0.018	15.000	14.992	21.020	21.007
16	16	20	8	**K16X20X8F**	6.37	7.51	1.15	S	18 000	28 000		0.005	16.000	15.992	20.020	20.007
	16	20	10	**K16X20X10H**	7.82	9.76	1.50	S	18 000	28 000		0.006	16.000	15.992	20.020	20.007
	16	20	13	**K16X20X13**	10.1	13.5	2.05	S	18 000	28 000		0.007	16.000	15.992	20.020	20.007
	16	20	14	**K16X20X14**	10.8	14.8	2.25	S	18 000	28 000		0.007	16.000	15.992	20.020	20.007
	16	20	17	**K16X20X17H**	12.9	18.5	2.85	S	18 000	28 000		0.008	16.000	15.992	20.020	20.007
	16	20	20	**K16X20X20**	13.4	19.5	3.05	S	18 000	28 000		0.011	16.000	15.992	20.020	20.007
	16	22	12	**K16X22X12**	11.2	11.9	1.80	S	19 000	29 000		0.010	16.000	15.992	22.020	22.007
	16	22	16	**K16X22X16H**	14.9	17.2	2.70	S	19 000	29 000		0.014	16.000	15.992	22.020	22.007
	16	22	20	**K16X22X20**	18.6	22.9	3.60	S	19 000	29 000		0.017	16.000	15.992	22.020	22.007
	16	24	20	**K16X24X20**	20.2	21.4	3.45	S	20 000	30 000		0.025	16.000	15.992	24.020	24.007
17	17	20	10	**K17X20X10**	5.96	8.53	1.30	S	16 000	25 000		0.004	17.000	16.992	20.020	20.007
	17	21	10	**K17X21X10**	8.12	10.4	1.60	S	17 000	26 000		0.006	17.000	16.992	21.020	21.007
	17	21	12.8	**K17X21X13H**	10.5	14.5	2.20	S	17 000	26 000		0.008	17.000	16.992	21.020	21.007
	17	21	15	**K17X21X15**	11.4	16.1	2.50	S	17 000	26 000		0.008	17.000	16.992	21.020	21.007
	17	21	17	**K17X21X17H**	13.4	19.8	3.05	S	17 000	26 000		0.011	17.000	16.992	21.020	21.007
	17	22	20	**K17X22X20FH**	17.0	23.3	3.65	S	17 000	27 000		0.015	17.000	16.992	22.020	22.007
	17	23	15	**K17X23X15F**	14.1	16.3	2.55	S	18 000	27 000		0.010	17.000	16.992	23.020	23.007
18	18	22	8	**K18X22X8F**	6.32	7.70	1.15	S	16 000	24 000		0.005	18.000	17.992	22.020	22.007
	18	22	10	**K18X22X10H**	8.41	11.1	1.70	S	16 000	24 000		0.006	18.000	17.992	22.020	22.007
	18	22	13	**K18X22X13H**	10.8	15.4	2.35	S	16 000	24 000		0.008	18.000	17.992	22.020	22.007
	18	22	14	**K18X22X14**	11.6	16.8	2.55	S	16 000	24 000		0.009	18.000	17.992	22.020	22.007
	18	22	14	**K18X22X14FV**	11.3	16.3	2.45	S	16 000	24 000		0.009	18.000	17.992	22.020	22.007
	18	22	17	**K18X22X17H**	13.3	19.9	3.10	S	16 000	24 000		0.009	18.000	17.992	22.020	22.007
	18	22	20	**K18X22X20F**	15.0	23.4	3.65	S	16 000	24 000		0.011	18.000	17.992	22.020	22.007
	18	24	12	**K18X24X12**	11.8	13.1	1.95	S	17 000	25 000		0.011	18.000	17.992	24.020	24.007

[Note] 1) Cage material: P: polymer cage, S: steel cage

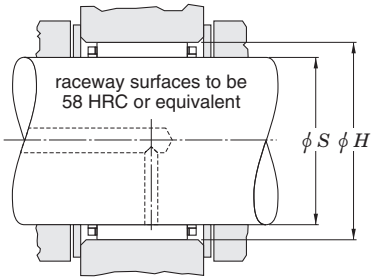
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. (18) ~ (22) mm



K



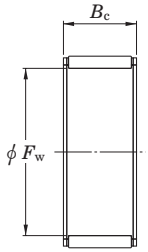
K ZW



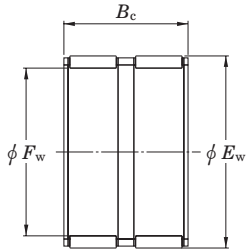
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
18	18	24	20	**K18X24X20H**	19.4	24.9	3.90	S	16 000	25 000		0.019	18.000	17.992	24.020	24.007
	18	25	22	**K18X25X22H**	23.3	28.6	4.50	S	17 000	26 000		0.025	18.000	17.992	25.020	25.007
	18	26	12	**K18X26X12FV**	13.8	13.5	2.10	S	11 000	17 000		0.020	18.000	17.992	26.020	26.007
	18	26	20	**K18X26X20F**	21.7	24.1	3.85	S	17 000	26 000		0.027	18.000	17.992	26.020	26.007
19	19	23	13	**K19X23X13**	10.8	15.5	2.35	S	15 000	23 000		0.008	19.000	18.991	23.020	23.007
	19	23	17	**K19X23X17**	13.4	20.6	3.20	S	15 000	23 000		0.011	19.000	18.991	23.020	23.007
20	20	24	8	**K20X24X8F**	7.31	9.60	1.50	S	14 000	22 000		0.005	20.000	19.991	24.020	24.007
	20	24	10	**K20X24X10H**	8.97	12.5	2.05	S	14 000	22 000		0.006	20.000	19.991	24.020	24.007
	20	24	12	**K20X24X12**	10.7	15.7	2.40	S	14 000	22 000		0.008	20.000	19.991	24.020	24.007
	20	24	13	**K20X24X13H**	11.5	17.3	2.65	S	14 000	22 000		0.009	20.000	19.991	24.020	24.007
	20	24	14	**K20X24X14**	12.4	18.9	2.85	S	14 000	22 000		0.009	20.000	19.991	24.020	24.007
	20	24	17	**K20X24X17H**	14.8	23.7	3.65	S	14 000	22 000		0.011	20.000	19.991	24.020	24.007
	20	26	12	**K20X26X12**	13.0	15.3	2.30	S	15 000	23 000		0.012	20.000	19.991	26.020	26.007
	20	26	13	**K20X26X13H**	13.4	15.9	2.35	S	15 000	23 000		0.014	20.000	19.991	26.020	26.007
	20	26	17	**K20X26X17H**	19.3	25.5	4.00	S	15 000	23 000		0.017	20.000	19.991	26.020	26.007
	20	26	20	**K20X26X20**	20.3	27.2	4.25	S	15 000	23 000		0.020	20.000	19.991	26.020	26.007
	20	28	20	**K20X28X20H**	24.6	29.0	2.70	S	15 000	23 000		0.028	20.000	19.991	28.020	28.007
	20	28	25	**K20X28X25H**	29.7	37.0	5.80	S	15 000	23 000		0.036	20.000	19.991	28.020	28.007
	20	30	30	**K20X30X30H**	38.9	45.8	7.20	S	16 000	24 000		0.055	20.000	19.991	30.020	30.007
	20	32	36	**K20X32X36H**	49.9	57.0	9.15	S	16 000	25 000		0.082	20.000	19.991	32.025	32.009
21	21	25	17	**K21X25X17H**	14.3	23.1	3.60	S	14 000	21 000		0.013	21.000	20.991	25.020	25.007
22	22	26	10	**K22X26X10H**	9.81	14.5	2.20	S	13 000	20 000		0.007	22.000	21.991	26.020	26.007
	22	26	13	**K22X26X13H**	11.8	18.3	2.95	S	13 000	20 000		0.012	22.000	21.991	26.020	26.007
	22	26	17	**K22X26X17H**	15.6	26.3	4.05	S	13 000	20 000		0.012	22.000	21.991	26.020	26.007
	22	26	18	**K22X26X18H**	15.3	25.5	4.00	S	13 000	20 000		0.017	22.000	21.991	26.020	26.007
	22	28	13	**K22X28X13**	13.9	17.1	2.60	S	13 000	20 000		0.015	22.000	21.991	28.020	28.007
	22	28	17	**K22X28X17H**	18.2	24.2	3.80	S	13 000	20 000		0.020	22.000	21.991	28.020	28.007

[Note] 1) Cage material: P: polymer cage, S: steel cage

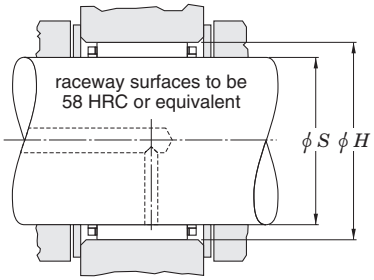
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. (22) ~ (25) mm



K



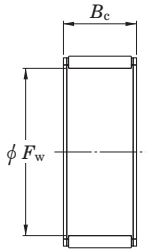
K ZW



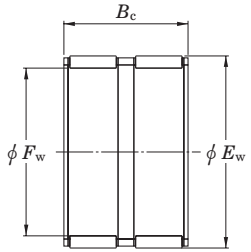
Shaft dia.	Boundary dimensions (mm) F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
22	22	30	15	**K22X30X15H**	19.7	22.3	3.45	S	14 000	21 000		0.023	22.000	21.991	30.020	30.007
	22	30	20	**K22X30X20FV**	24.4	29.4	4.70	S	14 000	21 000		0.031	22.000	21.991	30.020	30.007
	22	32	24	**K22X32X24F**	33.1	37.9	6.05	S	14 000	22 000		0.046	22.000	21.991	32.025	32.009
	22	32	30	**K22X32X30H**	41.8	51.3	8.05	S	14 000	22 000		0.057	22.000	21.991	32.025	32.009
23	23	28	24	**K23X28X24F**	22.4	36.2	5.70	S	12 000	19 000		0.023	23.000	22.991	28.020	28.007
	23	35	16	**K23X35X16H**	25.9	25.1	3.90	S	14 000	21 000		0.040	23.000	22.991	35.025	35.009
24	24	28	10	**K24X28X10H**	9.67	14.6	2.20	S	12 000	18 000		0.027	24.000	23.991	28.020	28.007
	24	28	13	**K24X28X13H**	12.5	20.2	3.05	S	12 000	18 000		0.010	24.000	23.991	28.020	28.007
	24	28	16	**K24X28X16F**	12.6	20.4	3.10	S	12 000	18 000		0.012	24.000	23.991	28.020	28.007
	24	28	17	**K24X28X17H**	15.4	26.4	4.10	S	12 000	18 000		0.013	24.000	23.991	28.020	28.007
	24	30	10	**K24X30X10TN**	11.3	13.5	2.05	P	12 000	19 000		0.008	24.000	23.991	30.020	30.007
	24	30	17	**K24X30X17H**	19.8	27.7	4.35	S	12 000	19 000		0.020	24.000	23.991	30.020	30.007
	24	30	22	**K24X30X22**	25.0	37.3	5.80	S	12 000	19 000		0.024	24.000	23.991	30.020	30.007
	24	36	23	**K24X36X23H**	37.1	40.1	6.40	S	13 000	20 000		0.070	24.000	23.991	36.025	36.009
25	25	29	10	**K25X29X10H**	9.61	14.6	2.25	S	11 000	17 000		0.008	25.000	24.991	29.020	29.007
	25	29	13	**K25X29X13H**	12.8	21.1	3.20	S	11 000	17 000		0.010	25.000	24.991	29.020	29.007
	25	29	17	**K25X29X17H**	15.1	26.2	4.10	S	11 000	17 000		0.016	25.000	24.991	29.020	29.007
	25	30	13	**K25X30X13**	14.6	21.4	3.25	S	11 000	17 000		0.012	25.000	24.991	30.020	30.007
	25	30	17	**K25X30X17H**	18.8	29.8	4.60	S	11 000	17 000		0.016	25.000	24.991	30.020	30.007
	25	30	18	**K25X30X18**	20.6	33.4	5.30	S	11 000	17 000		0.017	25.000	24.991	30.020	30.007
	25	30	20	**K25X30X20H**	21.9	36.1	5.65	S	11 000	17 000		0.019	25.000	24.991	30.020	30.007
	25	30	24	**K25X30X24H**	24.8	42.4	6.60	S	11 000	17 000		0.024	25.000	24.991	30.020	30.007
	25	30	26	**K25X30X26ZW**	23.0	38.6	5.90	S	11 000	17 000		0.027	25.000	24.991	30.020	30.007
	25	31	14	**K25X31X14H**	16.8	22.7	3.45	S	12 000	18 000		0.017	25.000	24.991	31.025	31.009
	25	31	17	**K25X31X17H**	19.7	27.8	4.35	S	12 000	18 000		0.020	25.000	24.991	31.025	31.009
	25	31	21	**K25X31X21H**	25.1	38.0	5.95	S	12 000	18 000		0.026	25.000	24.991	31.025	31.009
	25	31	24	**K25X31X24FH**	25.3	38.5	6.05	S	12 000	18 000		0.031	25.000	24.991	31.025	31.009
	25	32	16	**K25X32X16**	19.8	25.3	4.00	S	12 000	18 000		0.027	25.000	24.991	32.025	32.009

[Note] 1) Cage material: P: polymer cage, S: steel cage

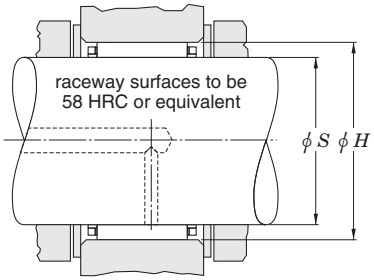
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. (25) ~ 29 mm



K



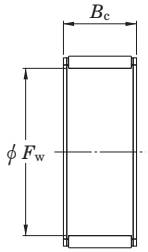
K ZW



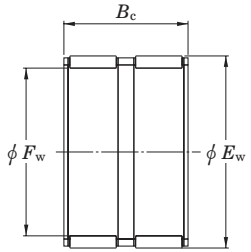
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
25	25	33	20	**K25X33X20H**	28.8	37.6	5.95	S	12 000	18 000		0.035	25.000	24.991	33.025	33.009
	25	33	24	**K25X33X24H**	32.3	43.5	6.85	S	12 000	18 000		0.038	25.000	24.991	33.025	33.009
	25	33	25	**K25X33X25H**	33.0	44.6	7.00	S	12 000	18 000		0.041	25.000	24.991	33.025	33.009
	25	35	23.7	**K25X35X23,7H**	35.9	42.3	6.90	S	12 000	19 000		0.050	25.000	24.991	35.025	35.009
	25	35	25	**K25X35X25H**	37.8	46.2	7.25	S	12 000	19 000		0.054	25.000	24.991	35.025	35.009
	25	35	30	**K25X35X30H**	44.6	57.2	9.00	S	12 000	19 000		0.060	25.000	24.991	35.025	35.009
	25	35	36	**K25X35X36H**	52.4	70.4	11.0	S	12 000	19 000		0.074	25.000	24.991	35.025	35.009
	25	37	20	**K25X37X20H**	32.5	34.1	5.45	S	12 000	19 000		0.055	25.000	24.991	37.025	37.009
26	26	30	10	**K26X30X10F**	9.46	14.5	2.20	S	11 000	16 000		0.007	26.000	25.991	30.020	30.007
	26	30	13	**K26X30X13**	12.3	20.4	3.10	S	10 000	16 000		0.011	26.000	25.991	30.020	30.007
	26	30	17	**K26X30X17**	15.0	26.3	3.10	S	10 000	16 000		0.014	26.000	25.991	30.020	30.007
	26	30	22	**K26X30X22ZW**	16.7	30.2	4.60	S	10 000	16 000		0.018	26.000	25.991	30.020	30.007
28	28	32	21	**K28X32X21F**	18.7	35.7	5.55	S	9 900	15 000		0.018	28.000	27.991	32.025	32.009
	28	33	13	**K28X33X13F**	14.1	21.4	3.25	S	10 000	15 000		0.015	28.000	27.991	33.025	33.009
	28	33	17	**K28X33X17H**	19.8	33.0	5.10	S	10 000	15 000		0.018	28.000	27.991	33.025	33.009
	28	33	27	**K28X33X27**	29.0	53.8	8.30	S	10 000	15 000		0.027	28.000	27.991	33.025	33.009
	28	34	17	**K28X34X17**	21.1	31.5	6.30	S	10 000	16 000		0.022	28.000	27.991	34.025	34.009
	28	34	20	**K28X34X20H**	24.4	37.8	7.65	S	10 000	16 000		0.025	28.000	27.991	34.025	34.009
	28	35	15	**K28X35X15H**	19.5	25.6	3.95	S	10 000	16 000		0.025	28.000	27.991	35.025	35.009
	28	35	16	**K28X35X16H**	21.5	29.1	4.60	S	10 000	16 000		0.026	28.000	27.991	35.025	35.009
	28	35	27	**K28X35X27H**	35.2	54.7	8.50	S	10 000	16 000		0.042	28.000	27.991	35.025	35.009
	28	36	20	**K28X36X20FV**	27.8	37.0	5.95	S	10 000	16 000		0.039	28.000	27.991	36.025	36.009
	28	38	25	**K28X38X25,5**	40.9	52.7	8.25	S	11 000	16 000		0.059	28.000	27.991	38.025	38.009
	28	40	18	**K28X40X18H**	33.6	36.5	5.90	S	11 000	17 000		0.060	28.000	27.991	40.025	40.009
	28	40	25	**K28X40X25H**	45.5	54.0	8.55	S	11 000	17 000		0.072	28.000	27.991	40.025	40.009
	28	40	30	**K28X40X30H**	54.3	67.8	10.7	S	11 000	17 000		0.100	28.000	27.991	40.025	40.009
	28	41	25	**K28X41X25H**	49.2	57.1	9.05	S	11 000	17 000		0.082	28.000	27.991	41.025	41.009
29	29	34	27	**K29X34X27F**	28.9	54.0	8.40	S	9 700	15 000		0.033	29.000	28.991	34.025	34.009

[Note] 1) Cage material: P: polymer cage, S: steel cage

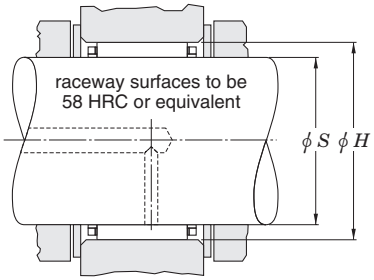
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. 30 ~ (34) mm



K



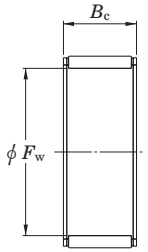
K ZW



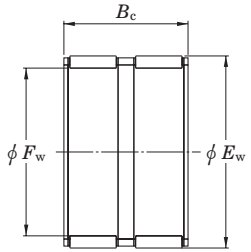
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
30	30	34	13	**K30X34X13**	13.5	24.1	3.65	S	9 200	14 000		0.011	30.000	29.991	34.025	34.009
	30	35	13	**K30X35X13H**	15.6	24.9	3.80	S	9 300	14 000		0.017	30.000	29.991	35.025	35.009
	30	35	17	**K30X35X17H**	20.2	34.6	5.35	S	9 300	14 000		0.022	30.000	29.991	35.025	35.009
	30	35	20	**K30X35X20H**	23.5	41.9	6.55	S	9 300	14 000		0.023	30.000	29.991	35.025	35.009
	30	35	22.8	**K30X35X23F**	25.6	46.8	7.40	S	9 300	14 000		0.028	30.000	29.991	35.025	35.009
	30	35	27	**K30X35X27H**	30.6	59.0	9.10	S	9 300	14 000		0.032	30.000	29.991	35.025	35.009
	30	35	27	**K30X35X27HZW**	19.9	33.6	5.10	S	9 300	14 000		0.033	30.000	29.991	35.025	35.009
	30	36	14	**K30X36X14**	18.0	26.2	4.00	S	9 500	15 000		0.020	30.000	29.991	36.025	36.009
	30	37	17.8	**K30X37X18**	24.3	34.8	6.00	S	9 600	15 000		0.033	30.000	29.991	37.025	37.009
	30	40	30	**K30X40X30H**	49.2	67.8	10.6	S	9 900	15 000		0.077	30.000	29.991	40.025	40.009
	30	42	30	**K30X42X30H**	54.2	68.6	10.8	S	10 000	16 000		0.096	30.000	29.991	42.025	42.009
	30	44	26	**K30X44X26H**	52.4	59.9	9.55	S	10 000	16 000		0.095	30.000	29.991	44.025	44.009
32	32	36	15	**K32X36X15F**	11.6	20.2	3.10	S	8 600	13 000		0.015	32.000	31.989	36.025	36.009
	32	37	13	**K32X37X13**	15.2	24.4	4.00	S	8 700	13 000		0.018	32.000	31.989	37.025	37.009
	32	37	17	**K32X37X17H**	20.0	34.8	5.40	S	8 700	13 000		0.020	32.000	31.989	37.025	37.009
	32	37	27	**K32X37X27**	29.3	56.8	8.85	S	8 700	13 000		0.035	32.000	31.989	37.025	37.009
	32	38	20	**K32X38X20H**	27.3	45.7	7.15	S	8 800	14 000		0.030	32.000	31.989	38.025	38.009
	32	38	26	**K32X38X26H**	33.2	58.8	9.15	S	8 800	14 000		0.037	32.000	31.989	38.025	38.009
	32	39	16	**K32X39X16H**	23.0	33.0	5.20	S	8 900	14 000		0.030	32.000	31.989	39.025	39.009
	32	39	18	**K32X39X18H**	25.8	38.2	6.05	S	8 900	14 000		0.033	32.000	31.989	39.025	39.009
	32	40	25	**K32X40X25H**	37.9	57.2	8.90	S	9 000	14 000		0.052	32.000	31.989	40.025	40.009
	32	40	36	**K32X40X36H**	52.3	86.4	13.6	S	9 000	14 000		0.080	32.000	31.989	40.025	40.009
	32	42	42	**K32X42X42H**	69.2	108	17.1	S	9 200	14 000		0.110	32.000	31.989	42.025	42.009
	32	46	18	**K32X46X18H**	39.2	41.9	6.80	S	9 600	15 000		0.075	32.000	31.989	46.025	46.009
	32	46	32	**K32X46X32H**	67.0	83.4	13.1	S	9 600	15 000		0.140	32.000	31.989	46.025	46.009
	32	46	40	**K32X46X40H**	81.7	108	12.2	S	9 600	15 000		0.158	32.000	31.989	46.025	46.009
33	33	51	23	**K33X51X23H**	55.9	57.6	9.35	S	9 600	15 000		0.140	33.000	32.989	51.029	51.010
34	34	38	11	**K34X38X11**	12.2	21.9	3.35	S	8 100	12 000		0.011	34.000	33.989	38.025	38.009

[Note] 1) Cage material: P: polymer cage, S: steel cage

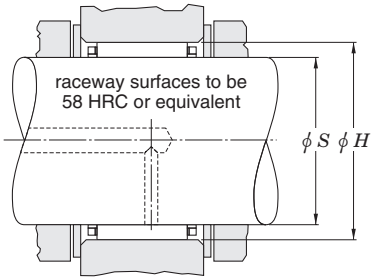
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. (34) ~ (38) mm



K



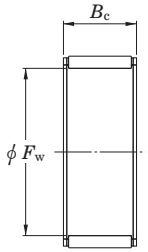
K ZW



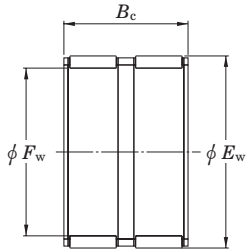
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
34	34	44	26	**K34X44X26FH**	42.9	58.9	9.40	S	8 600	13 000		0.080	34.000	33.989	44.025	44.009
35	35	40	13	**K35X40X13H**	16.2	27.2	4.15	S	7 900	12 000		0.018	35.000	34.989	40.025	40.009
	35	40	17	**K35X40X17H**	22.1	40.8	6.35	S	7 900	12 000		0.025	35.000	34.989	40.025	40.009
	35	40	19	**K35X40X19H**	23.2	43.2	6.80	S	7 900	12 000		0.025	35.000	34.989	40.025	40.009
	35	40	25	**K35X40X25H**	28.4	56.2	8.70	S	7 900	12 000		0.035	35.000	34.989	40.025	40.009
	35	40	27	**K35X40X27H**	29.8	59.6	9.20	S	7 900	12 000		0.037	35.000	34.989	40.025	40.009
	35	42	16	**K35X42X16AH**	24.5	36.8	5.80	S	8 100	12 000		0.031	35.000	34.989	42.025	42.009
	35	42	18	**K35X42X18**	27.5	42.6	6.75	S	8 100	12 000		0.035	35.000	34.989	42.025	42.009
	35	42	20	**K35X42X20H**	30.4	48.5	7.65	S	8 100	12 000		0.037	35.000	34.989	42.025	42.009
	35	42	30	**K35X42X30FH**	40.5	70.0	10.9	S	8 100	12 000		0.061	35.000	34.989	42.025	42.009
	35	45	20	**K35X45X20FH**	36.5	49.9	8.00	S	8 400	13 000		0.059	35.000	34.989	45.025	45.009
	35	45	30	**K35X45X30F**	51.2	74.5	11.7	S	8 400	13 000		0.100	35.000	34.989	45.025	45.009
	35	45	35	**K35X45X35H**	62.1	95.5	15.0	S	8 400	13 000		0.085	35.000	34.989	45.025	45.009
	35	45	41	**K35X45X41**	70.8	113	17.7	S	8 400	13 000		0.120	35.000	34.989	45.025	45.009
	35	45	49	**K35X45X49H**	82.5	138	21.4	S	8 400	13 000		0.143	35.000	34.989	45.025	45.009
	35	45	49	**K35X45X49HZW**	71.8	115	18.1	S	8 400	13 000		0.143	35.000	34.989	45.025	45.009
	35	50	23	**K35X50X23H**	53.0	60.3	9.75	S	8 700	13 000		0.110	35.000	34.989	50.025	50.009
	35	50	40	**K35X50X40F**	79.7	102	16.2	S	8 700	13 000		0.200	35.000	34.989	50.025	50.009
36	36	40	29	**K36X40X29TN**	21.2	45.2	7.15	P	7 600	12 000		0.029	36.000	35.989	40.025	40.009
	36	42	16	**K36X42X16**	22.8	37.7	5.95	S	7 800	12 000		0.027	36.000	35.989	42.025	42.009
37	37	42	13	**K37X42X13H**	16.9	29.4	4.50	S	7 500	11 000		0.017	37.000	36.989	42.025	42.009
	37	42	17	**K37X42X17H**	21.9	41.0	6.35	S	7 500	11 000		0.025	37.000	36.989	42.025	42.009
	37	42	27	**K37X42X27F**	32.1	66.9	10.4	S	7 500	11 000		0.039	37.000	36.989	42.025	42.009
	37	44	19	**K37X44X19H**	29.7	48.0	7.65	S	7 600	12 000		0.039	37.000	36.989	44.025	44.009
38	38	41	9	**K38X41X9TN**	5.93	11.0	1.65	P	7 100	11 000		0.004	38.000	37.989	41.025	41.009
	38	43	17	**K38X43X17H**	21.8	41.0	6.35	S	7 300	11 000		0.032	38.000	37.989	43.025	43.009
	38	43	27	**K38X43X27**	31.9	67.0	10.4	S	7 300	11 000		0.041	38.000	37.989	43.025	43.009

[Note] 1) Cage material: P: polymer cage, S: steel cage

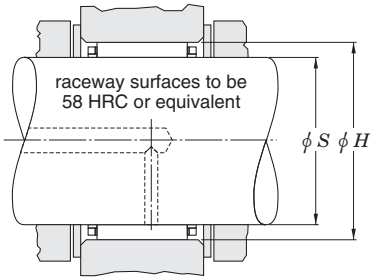
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. (38) ~ 42 mm



K



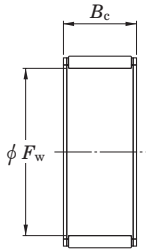
K ZW



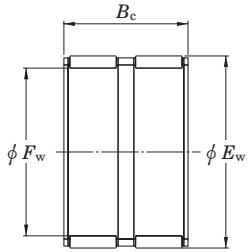
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
38	38	46	19.8	**K38X46X20H**	33.3	51.0	8.10	S	7 500	12 000		0.055	38.000	37.989	46.025	46.009
	38	46	32	**K38X46X32H**	55.2	98.1	15.3	S	7 500	12 000		0.090	38.000	37.989	46.025	46.009
	38	50	25	**K38X50X25**	53.0	70.8	11.2	S	7 800	12 000		0.100	38.000	37.989	50.025	50.009
	38	50	33	**K38X50X33H**	68.3	98.2	15.4	S	7 800	12 000		0.126	38.000	37.989	50.025	50.009
	38	50	40	**K38X50X40FH**	76.2	113	17.8	S	7 800	12 000		0.170	38.000	37.989	50.025	50.009
40	40	45	13	**K40X45X13H**	17.6	31.7	4.80	S	6 900	11 000		0.022	40.000	39.989	45.025	45.009
	40	45	18	**K40X45X18H**	25.1	50.4	8.00	S	6 900	11 000		0.031	40.000	39.989	45.025	45.009
	40	45	21	**K40X45X21H**	23.3	45.2	8.50	S	6 900	11 000		0.033	40.000	39.989	45.025	45.009
	40	45	27	**K40X45X27H**	32.7	70.2	10.8	S	6 900	11 000		0.040	40.000	39.989	45.025	45.009
	40	45	27	**K40X45X27TN**	33.3	72.1	11.2	P	6 900	11 000		0.030	40.000	39.989	45.025	45.009
	40	45	29	**K40X45X29H**	34.7	75.9	11.7	S	6 900	11 000		0.050	40.000	39.989	45.025	45.009
	40	46	17	**K40X46X17**	25.2	44.0	6.95	S	7 000	11 000		0.033	40.000	39.989	46.025	46.009
	40	47	18	**K40X47X18**	28.0	45.6	7.25	S	7 000	11 000		0.041	40.000	39.989	47.025	47.009
	40	47	20	**K40X47X20**	31.1	52.1	8.25	S	7 000	11 000		0.042	40.000	39.989	47.025	47.009
	40	48	20	**K40X48X20FV1**	35.5	56.3	8.45	S	7 100	11 000		0.052	40.000	39.989	48.025	48.009
	40	48	20	**K40X48X20H**	35.5	56.3	8.95	S	7 100	11 000		0.050	40.000	39.989	48.025	48.009
	40	48	35	**K40X48X35H**	57.3	104	16.3	S	7 100	11 000		0.098	40.000	39.989	48.025	48.009
	40	50	27	**K40X50X27H**	53.0	81.0	12.7	S	7 200	11 000		0.084	40.000	39.989	50.025	50.009
	40	55	45	**K40X55X45H**	103	146	23.0	S	7 500	12 000		0.221	40.000	39.989	55.029	55.010
	40	56	26	**K40X56X26H**	63.7	75.7	12.0	S	7 600	12 000		0.138	40.000	39.989	56.029	56.010
41	41	48	31	**K41X48X31HZW**	38.0	68.1	10.6	S	6 800	11 000		0.067	41.000	40.989	48.025	48.009
42	42	47	13	**K42X47X13H**	18.7	34.9	5.30	S	6 500	10 000		0.027	42.000	41.989	47.025	47.009
	42	47	17	**K42X47X17H**	22.8	45.2	7.30	S	6 500	10 000		0.028	42.000	41.989	47.025	47.009
	42	47	27	**K42X47X27H**	33.8	74.7	11.6	S	6 500	10 000		0.041	42.000	41.989	47.025	47.009
	42	48	24	**K42X48X24F**	33.1	63.9	10.1	S	6 600	10 000		0.046	42.000	41.989	48.025	48.009
	42	50	13	**K42X50X13H**	20.9	28.9	4.45	S	6 700	10 000		0.035	42.000	41.989	50.025	50.009
	42	50	20	**K42X50X20H**	35.2	56.6	9.00	S	6 700	10 000		0.054	42.000	41.989	50.025	50.009
	42	50	30	**K42X50X30H**	51.3	91.9	14.4	S	6 700	10 000		0.080	42.000	41.989	50.025	50.009

[Note] 1) Cage material: P: polymer cage, S: steel cage

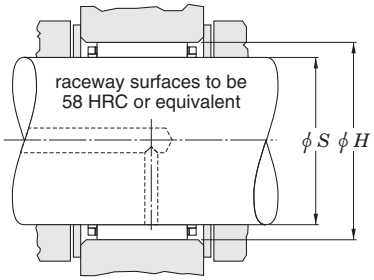
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. 43 ~ (47) mm



K



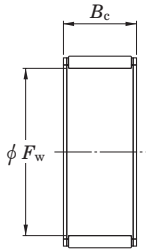
K ZW



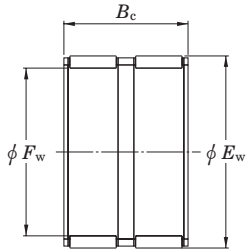
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
43	43	48	17	**K43X48X17FH**	23.0	45.8	6.85	S	6 400	9 800		0.036	43.000	42.989	48.025	48.009
	43	48	27	**K43X48X27H**	34.8	78.0	12.1	S	6 400	9 800		0.050	43.000	42.989	48.025	48.009
44	44	50	22	**K44X50X22H**	31.6	60.6	9.45	S	6 400	9 900		0.046	44.000	43.989	50.025	50.009
	44	50	30	**K44X50X30,5HZW**	35.5	70.5	10.7	S	6 400	9 900		0.068	44.000	43.989	50.025	50.009
45	45	50	13	**K45X50X13H**	18.4	35.1	5.35	S	6 100	9 400		0.022	45.000	44.989	50.025	50.009
	45	50	15	**K45X50X15H**	19.4	37.3	5.75	S	6 100	9 400		0.028	45.000	44.989	50.025	50.009
	45	50	17	**K45X50X17H**	24.9	51.8	8.05	S	6 100	9 400		0.030	45.000	44.989	50.025	50.009
	45	50	20	**K45X50X20F**	27.0	57.4	9.00	S	6 100	9 400		0.040	45.000	44.989	50.025	50.009
	45	50	21	**K45X50X21CH**	24.6	50.4	7.85	S	6 100	9 400		0.036	45.000	44.989	50.025	50.009
	45	50	27	**K45X50X27FH**	34.2	77.4	12.0	S	6 100	9 400		0.043	45.000	44.989	50.025	50.009
	45	50	27	**K45X50X27TN**	31.8	70.7	11.0	P	6 100	9 400		0.048	45.000	44.989	50.025	50.009
	45	52	18	**K45X52X18H**	30.1	52.0	8.25	S	6 200	9 500		0.045	45.000	44.989	52.029	52.010
	45	52	21	**K45X52X21F**	35.0	63.2	9.90	S	6 200	9 500		0.055	45.000	44.989	52.029	52.010
	45	53	20	**K45X53X20H**	36.0	59.5	9.45	S	6 200	9 600		0.054	45.000	44.989	53.029	53.010
	45	53	24.8	**K45X53X25H**	45.9	81.5	12.7	S	6 200	9 600		0.072	45.000	44.989	53.029	53.010
	45	53	25	**K45X53X25F**	42.5	73.7	11.7	S	6 200	9 600		0.075	45.000	44.989	53.029	53.010
	45	53	28	**K45X53X28H**	49.3	89.2	13.9	S	6 200	9 600		0.078	45.000	44.989	53.029	53.010
	45	55	20	**K45X55X20H**	42.0	62.2	10.0	S	6 400	9 800		0.074	45.000	44.989	55.029	55.010
	45	59	18	**K45X59X18H**	47.8	58.9	9.60	S	6 600	10 000		0.107	45.000	44.989	59.029	59.010
	45	59	18	**K45X59X18TN**	45.7	55.4	9.00	P	6 600	10 000		0.097	45.000	44.989	59.029	59.010
	45	59	36	**K45X59X36H**	82.4	118	18.6	S	6 600	10 000		0.181	45.000	44.989	59.029	59.010
	45	60	30	**K45X60X30H**	75.5	101	16.0	S	6 600	10 000		0.171	45.000	44.989	60.029	60.010
	45	60	45	**K45X60X45H**	108	160	25.2	S	6 600	10 000		0.280	45.000	44.989	60.029	60.010
46	46	53	36	**K46X53X36HZW**	48.6	96.7	15.3	S	6 100	9 300		0.100	46.000	45.989	53.029	53.010
47	47	52	15	**K47X52X15FH**	20.1	39.8	6.15	S	5 800	8 900		0.030	47.000	46.989	52.029	52.010
	47	52	17	**K47X52X17H**	24.2	50.4	7.85	S	5 800	8 900		0.032	47.000	46.989	52.029	52.010
	47	52	27	**K47X52X27H**	36.6	85.9	13.3	S	5 800	8 900		0.045	47.000	46.989	52.029	52.010

[Note] 1) Cage material: P: polymer cage, S: steel cage

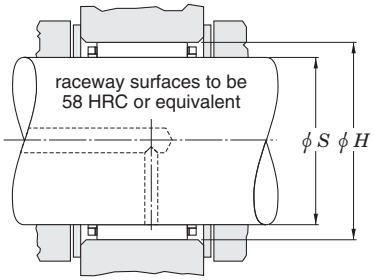
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. (47) ~ (55) mm



K



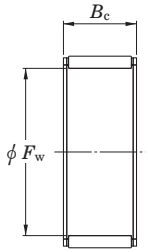
K ZW



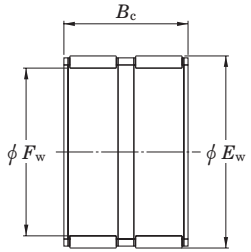
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
47	47	55	28	**K47X55X28FV1**	48.9	89.5	14.0	S	6 000	9 200		0.092	47.000	46.989	55.029	55.010
48	48	53	17	**K48X53X17H**	25.7	54.9	8.55	S	5 700	8 700		0.032	48.000	47.989	53.029	53.010
	48	54	19	**K48X54X19H**	30.9	61.2	9.85	S	5 700	8 800		0.042	48.000	47.989	54.029	54.010
49	49	55	32	**K49X55X32HZW**	40.2	86.4	13.4	S	5 600	8 600		0.080	49.000	48.989	55.029	55.010
	49	65	38	**K49X65X38H**	100	142	22.7	S	6 100	9 300		0.244	49.000	48.989	65.029	65.010
50	50	55	17	**K50X55X17H**	25.5	55.0	8.55	S	5 400	8 400		0.032	50.000	49.989	55.029	55.010
	50	55	20	**K50X55X20H**	30.2	68.5	10.7	S	5 400	8 400		0.038	50.000	49.989	55.029	55.010
	50	55	30	**K50X55X30**	38.2	92.4	14.4	S	5 400	8 400		0.057	50.000	49.989	55.029	55.010
	50	55	30	**K50X55X30FV1**	38.2	92.4	14.4	S	5 400	8 400		0.057	50.000	49.989	55.029	55.010
	50	56	23	**K50X56X23**	35.5	74.1	11.7	S	5 500	8 500		0.051	50.000	49.989	56.029	56.010
	50	57	18	**K50X57X18FH**	31.3	56.4	8.95	S	5 500	8 500		0.050	50.000	49.989	57.029	57.010
	50	58	20	**K50X58X20H**	38.8	67.8	10.8	S	5 600	8 600		0.065	50.000	49.989	58.029	58.010
	50	58	25	**K50X58X25H**	46.5	85.6	13.4	S	5 600	8 600		0.081	50.000	49.989	58.029	58.010
	50	58	35	**K50X58X35H**	64.9	131	20.6	S	5 600	8 600		0.105	50.000	49.989	58.029	58.010
	50	62	30	**K50X62X30H**	64.6	98.1	15.5	S	5 800	8 900		0.136	50.000	49.989	62.029	62.010
	50	66	30	**K50X66X30H**	80.9	109	17.4	S	5 900	9 100		0.192	50.000	49.989	66.029	66.010
	50	70	32	**K50X70X32H**	103	129	20.6	S	6 100	9 300		0.224	50.000	49.989	70.029	70.010
52	52	57	12	**K52X57X12**	18.4	36.7	5.60	S	5 200	8 000		0.022	52.000	51.987	57.029	57.010
	52	57	17	**K52X57X17H**	21.4	44.3	6.90	S	5 200	8 000		0.035	52.000	51.987	57.029	57.010
	52	60	24	**K52X60X24**	47.1	88.3	13.9	S	5 400	8 200		0.078	52.000	51.987	60.029	60.010
55	55	60	17	**K55X60X17**	26.0	58.3	9.10	S	4 900	7 600		0.037	55.000	54.987	60.029	60.010
	55	60	20	**K55X60X20H**	30.7	72.4	11.3	S	4 900	7 600		0.042	55.000	54.987	60.029	60.010
	55	60	27	**K55X60X27H**	40.1	102	15.7	S	4 900	7 600		0.055	55.000	54.987	60.029	60.010
	55	60	30	**K55X60X30FH**	40.6	103	16.1	S	4 900	7 600		0.068	55.000	54.987	60.029	60.010
	55	61	26	**K55X61X26H**	44.3	102	15.9	S	5 000	7 600		0.063	55.000	54.987	61.029	61.010
	55	62	18	**K55X62X18H**	33.2	62.8	10.0	S	5 000	7 700		0.055	55.000	54.987	62.029	62.010
	55	63	15	**K55X63X15F**	30.5	51.5	8.00	S	5 000	7 800		0.054	55.000	54.987	63.029	63.010

[Note] 1) Cage material: P: polymer cage, S: steel cage

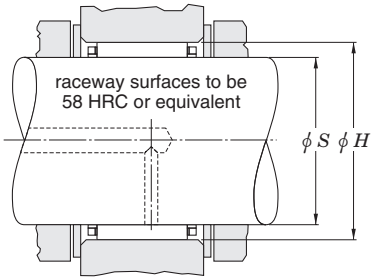
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. (55) ~ 68 mm



K



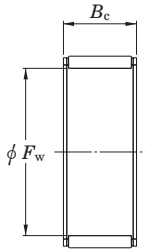
K ZW



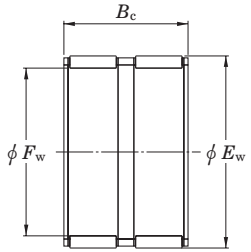
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
55	55	63	20	**K55X63X20**	40.3	73.5	11.7	S	5 000	7 800		0.072	55.000	54.987	63.029	63.010
	55	63	25	**K55X63X25**	49.8	96.5	15.1	S	5 000	7 800		0.080	55.000	54.987	63.029	63.010
	55	63	32	**K55X63X32**	62.3	129	20.0	S	5 000	7 800		0.108	55.000	54.987	63.029	63.010
58	58	63	17	**K58X63X17F**	27.0	62.6	9.80	S	4 700	7 200		0.037	58.000	57.987	63.029	63.010
	58	64	19	**K58X64X19H**	32.9	70.6	11.3	S	4 700	7 200		0.037	58.000	57.987	64.029	64.010
	58	65	18	**K58X65X18H**	34.3	67.1	10.7	S	4 700	7 300		0.058	58.000	57.987	65.029	65.010
60	60	65	20	**K60X65X20H**	31.9	78.1	12.2	S	4 500	6 900		0.046	60.000	59.987	65.029	65.010
	60	65	26.8	**K60X65X27FH**	39.5	103	16.0	S	4 500	6 900		0.059	60.000	59.987	65.029	65.010
	60	65	29.8	**K60X65X30FH**	42.9	114	17.8	S	4 500	6 900		0.085	60.000	59.987	65.029	65.010
	60	65	30	**K60X65X30**	42.9	114	17.8	S	4 500	6 900		0.070	60.000	59.987	65.029	65.010
	60	68	17	**K60X68X17F**	34.2	61.4	9.50	S	4 600	7 100		0.066	60.000	59.987	68.029	68.010
	60	68	20	**K60X68X20H**	41.8	79.2	12.6	S	4 600	7 100		0.066	60.000	59.987	68.029	68.010
	60	68	23	**K60X68X23H**	49.0	97.2	15.4	S	4 600	7 100		0.089	60.000	59.987	68.029	68.010
	60	68	25	**K60X68X25**	51.6	104	16.3	S	4 600	7 100		0.091	60.000	59.987	68.029	68.010
	60	68	30	**K60X68X30ZW**	46.4	90.1	13.9	S	4 600	7 100		0.119	60.000	59.987	68.029	68.010
63	63	71	20	**K63X71X20**	41.4	79.4	12.7	S	4 400	6 700		0.070	63.000	62.987	71.029	71.010
64	64	70	16	**K64X70X16**	26.4	55.1	8.55	S	4 200	6 500		0.049	64.000	63.987	70.029	70.010
65	65	70	20	**K65X70X20CH**	28.6	69.2	10.8	S	4 100	6 400		0.050	65.000	64.987	70.029	70.010
	65	70	30	**K65X70X30**	44.4	123	19.1	S	4 100	6 400		0.075	65.000	64.987	70.029	70.010
	65	73	23	**K65X73X23H**	48.2	97.7	15.5	S	4 200	6 500		0.091	65.000	64.987	73.029	73.010
	65	73	30	**K65X73X30H**	60.1	129	20.3	S	4 200	6 500		0.116	65.000	64.987	73.029	73.010
68	68	74	20	**K68X74X20FH**	37.5	88.1	13.2	S	4 000	6 100		0.062	68.000	67.987	74.029	74.010
	68	74	28	**K68X74X28CH**	44.8	110	17.1	S	4 000	6 100		0.082	68.000	67.987	74.029	74.010
	68	74	30	**K68X74X30H**	47.6	119	18.5	S	4 000	6 100		0.098	68.000	67.987	74.029	74.010
	68	74	35	**K68X74X35HZW**	45.1	111	17.1	S	4 000	6 100		0.120	68.000	67.987	74.029	74.010
	68	76	20	**K68X76X20**	43.8	87.8	14.0	S	4 000	6 200		0.086	68.000	67.987	76.029	76.010

[Note] 1) Cage material: P: polymer cage, S: steel cage

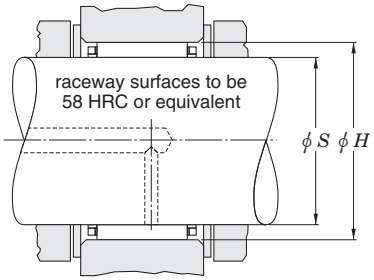
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. 70 ~ 95 mm



K



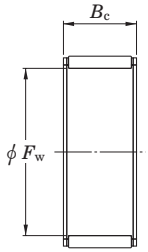
K ZW



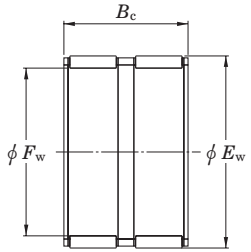
Boundary dimensions (mm) Shaft dia.	F_w	E_w	B_c −0.20 −0.55	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Cage material 1) P / S	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Recommended dimensions (mm) S (Shaft) max.	min.	H (Housing) max.	min.
70	70	76	20	**K70X76X20**	36.1	84.7	13.5	S	3 900	5 900		0.065	70.000	69.987	76.029	76.010
	70	76	30	**K70X76X30**	51.6	134.0	20.9	S	3 900	5 900		0.097	70.000	69.987	76.029	76.010
	70	78	20	**K70X78X20H**	43.6	87.9	14.0	S	3 900	6 000		0.090	70.000	69.987	78.029	78.010
	70	78	23	**K70X78X23F**	49.8	104.0	16.6	S	3 900	6 000		0.115	70.000	69.987	78.029	78.010
	70	78	24.8	**K70X78X25F**	49.8	104.0	16.6	S	3 900	6 000		0.115	70.000	69.987	78.029	78.010
	70	78	30	**K70X78X30H**	62.2	139.0	21.8	S	3 900	6 000		0.140	70.000	69.987	78.029	78.010
	70	78	46	**K70X78X46ZW**	78.4	187.0	29.5	S	3 900	6 000		0.188	70.000	69.987	78.029	78.010
	70	85	40	**K70X85X40F**	118	203	32.4	S	4 100	6 300		0.338	70.000	69.987	85.034	85.012
	70	88	30	**K70X88X30H**	115	175	28.1	S	4 100	6 400		0.205	70.000	69.987	88.034	88.012
72	72	80	20	**K72X80X20**	44.4	90.7	14.5	S	3 800	5 800		0.084	72.000	71.987	80.029	80.010
73	73	79	20	**K73X79X20**	37.0	88.7	14.1	S	3 700	5 700		0.068	73.000	72.987	79.029	79.010
75	75	81	20	**K75X81X20F**	37.4	90.7	14.5	S	3 600	5 500		0.075	75.000	74.987	81.034	81.012
	75	83	23	**K75X83X23**	52.5	114.0	18.2	S	3 600	5 600		0.104	75.000	74.987	83.034	83.012
	75	83	30	**K75X83X30**	60.9	138	21.7	S	3 600	5 600		0.141	75.000	74.987	83.034	83.012
	75	83	30	**K75X83X30FH**	60.9	138	21.7	S	3 600	5 600		0.141	75.000	74.987	83.034	83.012
80	80	86	20	**K80X86X20H**	38.6	96.7	15.4	S	3 400	5 200		0.072	80.000	79.987	86.034	86.012
	80	88	25	**K80X88X25FV1**	54.0	121	19.2	S	3 400	5 200		0.134	80.000	79.987	88.034	88.012
	80	88	30	**K80X88X30**	67.5	161	25.4	S	3 400	5 200		0.153	80.000	79.987	88.034	88.012
85	85	92	20	**K85X92X20H**	39.9	91.7	14.6	S	3 200	4 900		0.085	84.988	84.973	92.034	92.012
	85	93	25	**K85X93X25F**	58.8	138	21.7	S	3 200	4 900		0.128	84.988	84.973	93.034	93.012
	85	93	30	**K85X93X30H**	69.4	170.4	26.8	S	3 200	4 900		0.166	84.988	84.973	93.034	93.012
90	90	97	20	**K90X97X20**	46.3	114	18.1	S	3 000	4 600		0.095	89.988	89.973	97.034	97.012
	90	98	25	**K90X98X25F**	54.8	128	20.3	S	3 000	4 600		0.134	89.988	89.973	98.034	98.012
	90	98	30	**K90X98X30**	63.6	155	24.3	S	3 000	4 600		0.168	89.988	89.973	98.034	98.012
95	95	103	20	**K95X103X20**	49.3	114	18.3	S	2 800	4 400		0.130	94.988	94.973	103.034	103.012
	95	103	30	**K95X103X30F**	71.0	183	28.6	S	2 800	4 400		0.180	94.988	94.973	103.034	103.012

[Note] 1) Cage material: P: polymer cage, S: steel cage

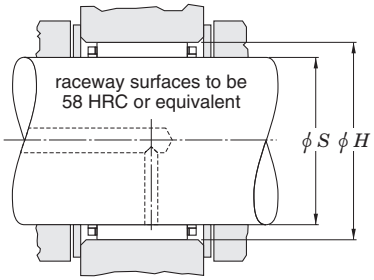
single-row, double-row assemblies
metric series
K, K ZW series
Shaft dia. 100 ~ 110 mm



K



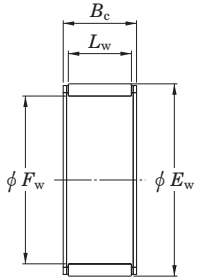
K ZW



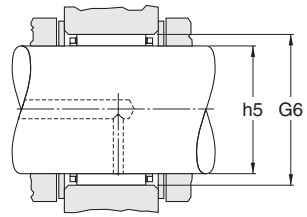
Boundary dimensions (mm)				Bearing No.	Basic load ratings (kN)		Fatigue load limit (kN)	Cage material [1]	Limiting speeds (min^{-1})			(Refer.) Mass (kg)	Recommended dimensions (mm)			
													S (Shaft)		H (Housing)	
Shaft dia.	F_w	E_w	B_c −0.20 −0.55		C_r	C_{0r}	C_u	P / S	Grease lub.	Oil lub.			max.	min.	max.	min.
100	100	108	30	**K100X108X30**	72.4	191	29.5	S	2 700	4 200		0.210	99.988	99.973	108.034	108.012
110	110	118	24	**K110X118X24**	64.0	168	25.6	S	2 400	3 800		0.165	109.988	109.973	118.034	118.012
	110	118	30	**K110X118X30H**	75.3	207	31.2	S	2 400	3 800		0.200	109.988	109.973	118.034	118.012

[Note] 1) Cage material: P: polymer cage, S: steel cage

single-row assemblies
inch series
Shaft dia. $^{3}/_{8} \sim (1\ ^{1}/_{2})$ in
(9.525 ~ (38.100) mm)



WJ, WJC

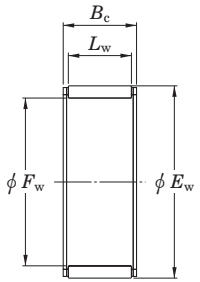


raceway surfaces to be
58 HRC or equivalent

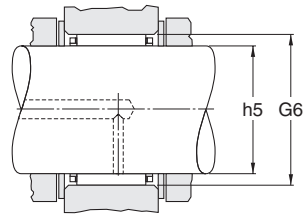
Boundary dimensions (mm) Shaft dia. (in)	F_w	E_w	B_c $^{+0}_{-0.38}$	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		Recommended dimensions (mm) Shaft dia. (h5) max.	min.	Housing bore dia. (G6) max.	min.	(Refer.) Mass (kg)
$^{3}/_{8}$	9.525	12.700	9.53	**WJC-060806**	3.87	4.00	0.600	24 000	37 000		9.525	9.520	12.715	12.705	0.003
$^{1}/_{2}$	12.700	15.875	12.70	**WJC-081008**	6.23	8.01	1.65	23 000	35 000		12.700	12.692	15.890	15.880	0.005
$^{9}/_{16}$	14.288	17.463	12.70	**WJC-091108**	6.81	9.25	1.40	22 000	34 000		14.288	14.280	17.478	17.468	0.006
$^{5}/_{8}$	15.875	19.050	12.70	**WJC-101208**	7.03	9.96	1.50	18 000	27 000		15.875	15.867	19.070	19.058	0.006
	15.875	22.225	15.88	**WJ-101410**	15.6	17.8	2.80	19 000	29 000		15.875	15.867	22.245	22.233	0.012
	15.875	22.225	22.23	**WJ-101414**	21.3	26.4	4.10	19 000	29 000		15.875	15.867	22.245	22.233	0.017
$^{3}/_{4}$	19.050	25.400	25.40	**WJ-121616**	26.8	37.2	5.80	16 000	24 000		19.050	19.040	25.420	25.408	0.023
$^{13}/_{16}$	20.638	26.988	22.23	**WJ-131714**	25.1	35.0	5.50	14 000	22 000		20.638	20.627	27.008	26.995	0.021
$^{7}/_{8}$	22.225	28.575	25.40	**WJ-141816**	29.2	43.5	6.75	13 000	20 000		22.225	22.215	28.595	28.583	0.026
1	25.400	33.338	19.05	**WJ-162112**	28.1	37.1	5.90	12 000	18 000		25.400	25.390	33.363	33.348	0.029
	25.400	33.338	25.40	**WJ-162116**	36.8	52.5	8.20	12 000	18 000		25.400	25.390	33.363	33.348	0.038
	25.400	33.338	31.75	**WJ-162120**	44.5	67.2	10.5	12 000	18 000		25.400	25.390	33.363	33.348	0.048
$1\ ^{1}/_{8}$	28.575	38.100	25.40	**WJ-182416**	42.4	57.8	9.05	10 000	16 000		28.575	28.565	38.125	38.110	0.041
	28.575	38.100	31.75	**WJ-182420**	52.0	74.7	11.7	10 000	16 000		28.575	28.565	38.125	38.110	0.065
$1\ ^{1}/_{4}$	31.750	41.275	19.05	**WJ-202612**	33.4	43.7	7.05	9 300	14 000		31.750	31.740	41.300	41.285	0.043
	31.750	41.275	25.40	**WJ-202616**	44.1	62.3	9.80	9 300	14 000		31.750	31.740	41.300	41.285	0.061
	31.750	41.275	31.75	**WJ-202620**	53.8	81.0	12.6	9 300	14 000		31.750	31.740	41.300	41.285	0.071
	31.750	41.275	38.10	**WJ-202624**	63.6	99.6	15.6	9 300	14 000		31.750	31.740	41.300	41.285	0.085
$1\ ^{3}/_{8}$	34.925	44.450	25.40	**WJ-222816**	45.8	67.2	10.5	8 300	13 000		34.925	34.915	44.475	44.460	0.067
	34.925	44.450	31.75	**WJ-222820**	56.0	87.2	13.6	8 300	13 000		34.925	34.915	44.475	44.460	0.077
$1\ ^{1}/_{2}$	38.100	47.625	25.40	**WJ-243016**	47.2	71.6	11.3	7 600	12 000		38.100	38.090	47.650	47.635	0.078
	38.100	47.625	31.75	**WJ-243020**	57.8	93.0	14.5	7 600	12 000		38.100	38.090	47.650	47.635	0.083

[Remarks] 1) Load ratings are based on a minimum raceway hardness of 58 HRC or equivalent.
2) Minimum axial clearance should be 0.02 mm (0.008 in).

single-row assemblies
inch series
Shaft dia. (1 1/2) ~ 3 in ((38.100) ~ 76.200 mm)



WJ, WJC

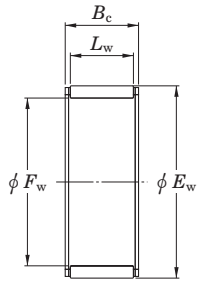


raceway surfaces to be 58 HRC or equivalent

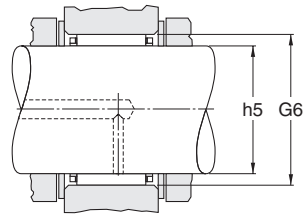
Boundary dimensions (mm) Shaft dia. (in)	F_w	E_w	B_c +0 −0.38	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.	Recommended dimensions (mm) Shaft dia. (h5) max.	min.	Housing bore dia. (G6) max.	min.	(Refer.) Mass (kg)
1 1/2	38.100	47.625	38.10	**WJ-243024**	68.1	114.8	18.0	7 600	12 000	38.100	38.090	47.650	47.635	0.100
	38.100	47.625	44.45	**WJ-243028**	77.4	135.7	21.2	7 600	12 000	38.100	38.090	47.650	47.635	0.134
1 3/4	44.450	53.975	19.05	**WJ-283412**	39.5	59.6	9.60	6 400	9 900	44.450	44.440	54.003	53.985	0.058
	44.450	53.975	25.40	**WJ-283416**	52.0	85.0	13.4	6 400	9 900	44.450	44.440	54.003	53.985	0.084
	44.450	53.975	38.10	**WJ-283424**	74.7	136	21.3	6 400	9 900	44.450	44.440	54.003	53.985	0.115
2	50.800	60.325	19.05	**WJ-323812**	42.8	69.0	11.1	5 600	8 600	50.800	50.787	60.353	60.335	0.065
	50.800	60.325	25.40	**WJ-323816**	56.5	98.0	15.5	5 600	8 600	50.800	50.787	60.353	60.335	0.105
	50.800	60.325	31.75	**WJ-323820**	69.0	127	20.0	5 600	8 600	50.800	50.787	60.353	60.335	0.108
	50.800	60.325	38.10	**WJ-323824**	81.0	157	24.6	5 600	8 600	50.800	50.787	60.353	60.335	0.130
2 1/16	52.388	61.913	25.40	**WJ-333916**	57.8	102	16.2	5 400	8 300	52.388	52.375	61.940	61.923	0.099
2 1/8	53.975	63.500	25.40	**WJ-344016**	52.5	92.08	14.6	5 200	8 000	53.975	53.962	63.528	63.510	0.089
	53.975	63.500	38.10	**WJ-344024**	78.3	153	24.0	5 200	8 000	53.975	53.962	63.528	63.510	0.137
2 3/16	55.563	65.088	19.05	**WJ-354112**	44.5	75.17	12.2	5 000	7 800	55.563	55.550	65.115	65.098	0.070
	55.563	65.088	25.40	**WJ-354116**	57.8	107	16.9	5 000	7 800	55.563	55.550	65.115	65.098	0.094
2 1/4	57.150	66.675	25.40	**WJ-364216**	53.8	96.08	15.2	4 900	7 500	57.150	57.137	66.703	66.685	0.096
	57.150	66.675	31.75	**WJ-364220**	67.6	128	20.1	4 900	7 500	57.150	57.137	66.703	66.685	0.120
2 3/8	60.325	69.850	38.10	**WJ-384424**	81.4	167	26.1	4 600	7 100	60.325	60.312	69.878	69.860	0.151
2 1/2	63.500	73.025	25.40	**WJ-404616**	55.6	104	16.5	4 400	6 700	63.500	63.487	73.053	73.035	0.106
	63.500	73.025	31.75	**WJ-404620**	69.8	139	21.8	4 400	6 700	63.500	63.487	73.053	73.035	0.132
	63.500	73.025	38.10	**WJ-404624**	83.2	173	27.2	4 400	6 700	63.500	63.487	73.053	73.035	0.179
2 3/4	69.850	79.375	25.40	**WJ-445016**	57.8	112.54	17.8	4 000	6 100	69.850	69.837	79.403	79.385	0.116
3	76.200	85.725	25.40	**WJ-485416**	59.6	120.55	19.1	3 600	5 600	76.200	76.187	85.761	85.738	0.126
	76.200	85.725	38.10	**WJ-485424**	85.4	191.72	29.9	3 600	5 600	76.200	76.187	85.761	85.738	0.189

[Remarks] 1) Load ratings are based on a minimum raceway hardness of 58 HRC or equivalent.
2) Minimum axial clearance should be 0.02 mm (0.008 in).

single-row assemblies
inch series
Shaft dia. 3 1/4 ~ 5 in
(82.550 ~ 127.000 mm)



WJ, WJC



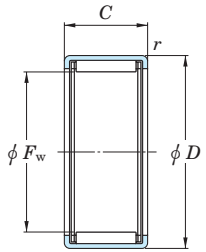
raceway surfaces to be
58 HRC or equivalent

Boundary dimensions (mm)				Bearing No.	Basic load ratings (kN)		Fatigue load limit (kN)	Limiting speeds (min^{-1})			Recommended dimensions (mm)				(Refer.) Mass (kg)
Shaft dia. (in)	F_w	E_w	B_c +0 −0.38		C_r	C_{0r}	C_u	Grease lub.	Oil lub.		Shaft dia. (h5) max.	Shaft dia. (h5) min.	Housing bore dia. (G6) max.	Housing bore dia. (G6) min.	
3 1/4	82.550	92.075	25.40	**WJ-525816**	61.4	128.55	20.4	3 300	5 100		82.550	82.535	92.111	92.088	0.136
	82.550	92.075	38.10	**WJ-525824**	88.1	204.62	31.9	3 300	5 100		82.550	82.535	92.111	92.088	0.220
3 1/2	88.900	98.425	25.40	**WJ-566216**	63.2	136.56	21.7	3 100	4 700		88.900	88.885	98.461	98.438	0.146
	88.900	101.600	25.40	**WJ-566416**	79.6	150.35	23.9	3 100	4 800		88.900	88.885	101.636	101.613	0.197
	88.900	101.600	38.10	**WJ-566424**	113	237.53	37.4	3 100	4 800		88.900	88.885	101.636	101.613	0.296
4	101.600	114.300	25.40	**WJ-647216**	83.6	166.59	30.9	2 700	4 200		101.600	101.585	114.336	114.313	0.224
	101.600	114.300	38.10	**WJ-647224**	119	263.33	40.6	2 700	4 200		101.600	101.585	114.336	114.313	0.335
5	127.000	152.400	38.10	**WJ-809624**	211	365.20	51.9	2 200	3 400		127.000	126.982	152.438	152.415	1.018

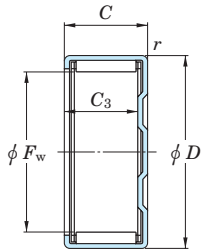
[Remarks] 1) Load ratings are based on a minimum raceway hardness of 58 HRC or equivalent.
2) Minimum axial clearance should be 0.02 mm (0.008 in).

caged, open ends, closed one end metric series HK, BK series

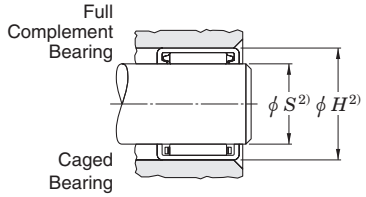
Shaft dia. 3 ~ (10) mm



HK



BK



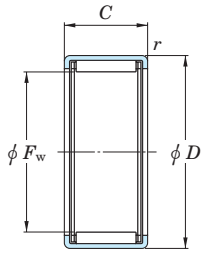
Shaft surface to be 58 HRC or equivalent

Shaft dia.	Boundary dimensions (mm) F_w	D	C $^{+0}_{-0.3}$	C_3 min.	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Inspection gage	Mounting inner ring
3	3	6.5	6	5.20	0.30	**BK0306**	1.20	0.78	0.130	30 000	46 000		0.001	Table 4	
	3	6.5	6	—	0.30	**HK0306**	1.20	0.78	0.130	30 000	46 000		0.001	Table 4	
4	4	8	8	6.40	0.40	**BK0408**	1.88	1.38	0.200	25 000	39 000		0.002	Table 4	
	4	8	8	—	0.40	**HK0408**	1.88	1.38	0.200	25 000	39 000		0.002	Table 4	
5	5	9	9	7.40	0.40	**BK0509**	2.52	2.07	0.320	23 000	36 000		0.002	Table 4	
	5	9	9	—	0.40	**HK0509**	2.52	2.07	0.320	23 000	36 000		0.002	Table 4	
6	6	10	8	6.40	0.40	**BK0608**	2.34	1.95	0.290	22 000	33 000		0.002	Table 4	
	6	10	8	—	0.40	**HK0608**	2.34	1.95	0.290	22 000	33 000		0.002	Table 4	
	6	10	9	7.40	0.40	**BK0609**	3.14	2.85	0.290	22 000	33 000		0.003	Table 4	
	6	10	9	—	0.40	**HK0609**	3.14	2.85	0.290	22 000	33 000		0.002	Table 4	
7	7	11	9	7.40	0.40	**BK0709**	3.23	3.05	0.470	21 000	32 000		0.003	Table 4	
	7	11	9	—	0.40	**HK0709**	3.23	3.05	0.470	21 000	32 000		0.003	Table 4	
8	8	12	8	6.40	0.40	**BK0808**	2.90	2.73	0.400	20 000	31 000		0.003	Table 4	
	8	12	8	—	0.40	**HK0808**	2.90	2.73	0.400	20 000	31 000		0.003	Table 4	
	8	12	10	8.40	0.40	**BK0810**	3.95	4.07	0.600	20 000	31 000		0.004	Table 4	JR5x8x12
	8	12	10	—	0.40	**HK0810**	3.95	4.07	0.600	20 000	31 000		0.004	Table 4	JR5x8x12
9	9	13	10	8.40	0.40	**BK0910**	4.57	5.07	0.770	19 000	30 000		0.004	Table 4	JR6x9x12
	9	13	10	—	0.40	**HK0910**	4.57	5.07	0.770	19 000	30 000		0.004	Table 4	JR6x9x12
	9	13	12	10.40	0.40	**BK0912**	5.65	6.65	1.00	19 000	30 000		0.005	Table 4	JR6x9x12
	9	13	12	—	0.40	**HK0912**	5.65	6.65	1.00	19 000	30 000		0.005	Table 4	JR6x9x12
10	10	14	10	8.40	0.40	**BK1010**	4.78	5.51	0.840	19 000	29 000		0.004	Table 4	JR7x10x10,5
	10	14	10	—	0.40	**HK1010**	4.78	5.51	0.840	19 000	29 000		0.004	Table 4	JR7x10x10,5
	10	14	12	10.40	0.40	**BK1012**	5.90	7.23	1.10	19 000	29 000		0.006	Table 4	JR7x10x12
	10	14	12	—	0.40	**HK1012**	5.90	7.23	1.10	19 000	29 000		0.005	Table 4	JR7x10x12

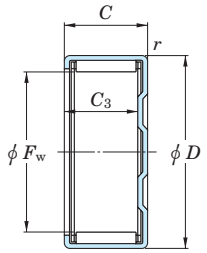
[Notes] 1) Drawn cup needle roller bearings with two needle roller and cage assemblies and one lubricating hole.
2) For the recommended mounting dimensions see Table 19.
3) For the mounting inner ring, please refer to JTEKT special catalog “Needle Roller Bearings” (CAT. NO. BS007EN).

caged, open ends, closed one end metric series HK, BK series

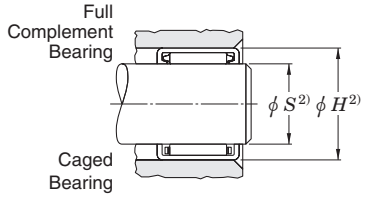
Shaft dia. (10) ~ 17 mm



HK



BK



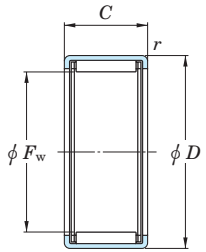
Shaft surface to be 58 HRC or equivalent

Shaft dia.	Boundary dimensions (mm) F_w	D	C $^{+0}_{-0.3}$	C_3 min.	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Inspection gage	Mounting inner ring
10	10	14	15	13.40	0.40	**BK1015**	7.49	9.81	1.50	19 000	29 000		0.006	Table 4	JR7x10x16
	10	14	15	—	0.40	**HK1015**	7.49	9.81	1.50	19 000	29 000		0.006	Table 4	JR7x10x16
12	12	16	10	8.40	0.4	**BK1210**	5.24	6.55	0.890	18 000	28 000		0.006	Table 4	JR8x12x10,5
	12	16	10	—	0.4	**HK1210**	5.24	6.55	0.890	18 000	28 000		0.006	Table 4	JR8x12x10,5
	12	18	12	9.30	1	**BK1212**	6.61	7.29	1.10	14 000	22 000		0.012	Table 4	JR8x12x12,5
	12	18	12	—	1	**HK1212**	6.61	7.29	1.10	14 000	22 000		0.01	Table 4	JR8x12x12,5
13	13	19	12	9.30	1	**BK1312**	6.92	7.89	1.20	14 000	22 000		0.012	Table 4	JR10x13x12,5
	13	19	12	—	1	**HK1312**	6.92	7.89	1.20	14 000	22 000		0.01	Table 4	JR10x13x12,5
14	14	20	12	9.30	1	**BK1412**	7.21	8.50	1.30	14 000	21 000		0.014	Table 4	JR10x14x12
	14	20	12	—	1	**HK1412**	7.21	8.50	1.30	14 000	21 000		0.011	Table 4	JR10x14x12
15	15	21	12	9.30	1	**BK1512**	7.49	9.11	1.40	14 000	21 000		0.015	Table 4	JR12x15x12,5
	15	21	12	—	1	**HK1512**	7.49	9.11	1.40	14 000	21 000		0.012	Table 4	JR12x15x12,5
	15	21	16	13.30	1	**BK1516**	10.70	14.4	2.20	14 000	21 000		0.019	Table 4	JR12x15x16,5
	15	21	16	—	1	**HK1516**	10.70	14.4	2.20	14 000	21 000		0.018	Table 4	JR12x15x16,5
	15	21	22	19.30	1	**BK1522** [1)]	13.50	19.4	2.95	14 000	21 000		0.022	Table 4	JR12x15x22,5
	15	21	22	—	1	**HK1522** [1)]	13.50	19.4	2.95	14 000	21 000		0.024	Table 4	JR12x15x22,5
16	16	22	12	9.30	1	**BK1612**	7.76	9.72	1.50	14 000	21 000		0.016	Table 4	JR12x16x12
	16	22	12	—	1	**HK1612**	7.76	9.72	1.50	14 000	21 000		0.012	Table 4	JR12x16x12
	16	22	16	13.30	1	**BK1616**	11.1	15.3	2.35	14 000	21 000		0.02	Table 4	JR12x16x16
	16	22	16	—	1	**HK1616**	11.1	15.3	2.35	14 000	21 000		0.016	Table 4	JR12x16x16
	16	22	22	19.30	1	**BK1622** [1)]	13.4	19.5	2.95	14 000	21 000		0.028	Table 4	JR12x16x22
	16	22	22	—	1	**HK1622** [1)]	13.40	19.5	2.95	14 000	21 000		0.022	Table 4	JR12x16x22
17	17	23	12	9.30	1	**BK1712**	8.12	10.4	1.60	13 000	20 000		0.018	Table 4	
	17	23	12	—	1	**HK1712**	8.12	10.4	1.60	13 000	20 000		0.013	Table 4	

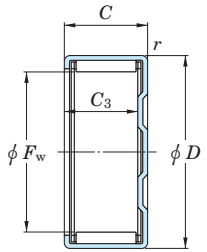
[Notes] 1) Drawn cup needle roller bearings with two needle roller and cage assemblies and one lubricating hole.
2) For the recommended mounting dimensions see Table 19.
3) For the mounting inner ring, please refer to JTEKT special catalog "Needle Roller Bearings" (CAT. NO. BS007EN).

caged, open ends, closed one end metric series HK, BK series

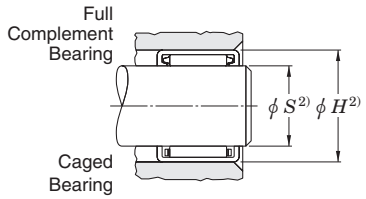
Shaft dia. 18 ~ (25) mm



HK



BK



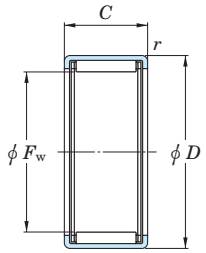
Shaft surface to be 58 HRC or equivalent

Shaft dia.	Boundary dimensions (mm) F_w	D	C $^{+0}_{-0.3}$	C_3 min.	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Inspection gage	Mounting inner ring
18	18	24	12	9.30	1	**BK1812**	8.41	11.1	1.70	12 000	18 000		0.017	Table 4	
	18	24	12	—	1	**HK1812**	8.41	11.1	1.70	12 000	18 000		0.015	Table 4	
	18	24	16	13.30	1	**BK1816**	11.6	16.8	2.55	12 000	18 000		0.022	Table 4	JR15x18x16,5
	18	24	16	—	1	**HK1816**	11.6	16.8	2.55	12 000	18 000		0.018	Table 4	JR15x18x16,5
20	20	26	12	9.3	1	**BK2012**	8.97	12.5	1.90	11 000	16 000		0.017	Table 4	JR15x20x12
	20	26	12	—	1	**HK2012**	8.97	12.5	1.90	11 000	16 000		0.015	Table 4	JR15x20x12
	20	26	16	13.3	1	**BK2016**	12.40	18.90	2.85	11 000	16 000		0.024	Table 4	JR17x20x16,5
	20	26	16	—	1	**HK2016**	12.40	18.90	2.85	11 000	16 000		0.022	Table 4	JR17x20x16,5
	20	26	20	17.3	1	**BK2020**	15.50	25.30	3.95	11 000	16 000		0.027	Table 4	JR17x20x20,5
	20	26	20	—	1	**HK2020**	15.50	25.30	3.95	11 000	16 000		0.025	Table 4	JR17x20x20,5
	20	26	30	27.3	1	**BK2030** [1]	21.20	37.80	5.75	11 000	16 000		0.043	Table 4	JR17x20x30,5
	20	26	30	—	1	**HK2030** [1]	21.20	37.80	5.75	11 000	16 000		0.041	Table 4	JR17x20x30,5
22	22	28	10	—	1	**HK2210**	7.06	9.49	1.45	9 600	15 000		0.013	Table 4	
	22	28	12	9.3	1	**BK2212**	9.81	14.50	2.20	9 600	15 000		0.02	Table 4	JR17x22x13
	22	28	12	—	1	**HK2212**	9.81	14.50	2.20	9 600	15 000		0.015	Table 4	JR17x22x13
	22	28	16	13.3	1	**BK2216**	13.10	20.90	3.20	9 600	15 000		0.027	Table 4	JR17x22x16
	22	28	16	—	1	**HK2216**	13.10	20.90	3.20	9 600	15 000		0.022	Table 4	JR17x22x16
	22	28	20	17.3	1	**BK2220**	15.30	25.50	4.00	9 600	15 000		0.028	Table 4	JR17x22x23
	22	28	20	—	1	**HK2220**	15.30	25.50	4.00	9 600	15 000		0.026	Table 4	JR17x22x23
25	25	32	12	9.30	1	**BK2512**	10.90	14.70	2.25	8 500	13 000		0.025	Table 4	
	25	32	12	—	1	**HK2512**	10.90	14.70	2.25	8 500	13 000		0.021	Table 4	
	25	32	16	13.3	1	**BK2516**	15.60	23.50	3.55	8 500	13 000		0.031	Table 4	JR20x25x17
	25	32	16	—	1	**HK2516**	15.60	23.50	3.55	8 500	13 000		0.028	Table 4	JR20x25x17
	25	32	20	17.3	1	**BK2520**	20.60	33.40	5.30	8 500	13 000		0.043	Table 4	JR20x25x20,5
	25	32	20	—	1	**HK2520**	20.60	33.40	5.30	8 500	13 000		0.040	Table 4	JR20x25x20,5
	25	32	26	23.3	1	**BK2526**	25.70	44.40	6.95	8 500	13 000		0.051	Table 4	JR20x25x26,5

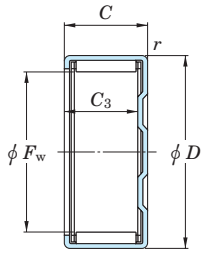
[Notes] 1) Drawn cup needle roller bearings with two needle roller and cage assemblies and one lubricating hole.
2) For the recommended mounting dimensions see Table 19.
3) For the mounting inner ring, please refer to JTEKT special catalog "Needle Roller Bearings" (CAT. NO. BS007EN).

caged, open ends, closed one end metric series HK, BK series

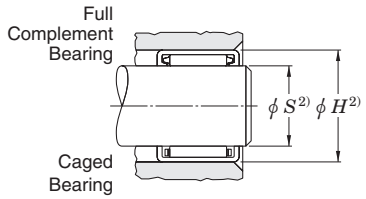
Shaft dia. (25) ~ (45) mm



HK



BK



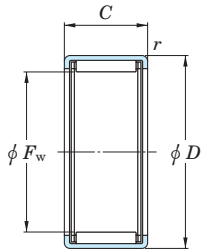
Shaft surface to be 58 HRC or equivalent

Shaft dia.	Boundary dimensions (mm) F_w	D	C $^{+0}_{-0.3}$	C_3 min.	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Inspection gage	Mounting inner ring
25	25	32	26	—	1	**HK2526**	25.70	44.40	6.95	8 500	13 000		0.046	Table 4	JR20x25x26,5
	25	32	38	35.3	1	**BK2538** [1]	35.30	66.90	10.6	8 500	13 000		0.077	Table 4	JR20x25x38,5
	25	32	38	—	1	**HK2538** [1]	35.30	66.90	10.6	8 500	13 000		0.068	Table 4	JR20x25x38,5
28	28	35	16	—	1	**HK2816**	17.4	28.1	3.85	7 500	12 000		0.032	Table 4	JR22x28x17
	28	35	20	17.3	1	**BK2820**	20.9	35.3	5.60	7 500	12 000		0.047	Table 4	JR22x28x20,5
	28	35	20	—	1	**HK2820**	20.9	35.3	5.60	7 500	12 000		0.040	Table 4	JR22x28x20,5
30	30	37	12	9.3	1	**BK3012**	12.0	17.7	2.90	7 000	11 000		0.031	Table 4	
	30	37	12	—	1	**HK3012**	12.0	17.7	2.70	7 000	11 000		0.024	Table 4	
	30	37	16	13.30	1	**BK3016**	16.8	27.3	4.20	7 000	11 000		0.041	Table 4	JR25x30x17
	30	37	16	—	1	**HK3016**	16.8	27.3	4.20	7 000	11 000		0.032	Table 4	JR25x30x17
	30	37	20	17.3	1	**BK3020**	22.4	39.6	6.25	7 000	11 000		0.053	Table 4	JR25x30x20,5
	30	37	20	—	1	**HK3020**	22.4	39.6	6.25	7 000	11 000		0.042	Table 4	JR25x30x20,5
	30	37	26	23.3	1	**BK3026**	27.4	51.2	7.95	7 000	11 000		0.067	Table 4	JR25x30x26,5
	30	37	26	—	1	**HK3026**	27.4	51.2	7.95	7 000	11 000		0.054	Table 4	JR25x30x26,5
	30	37	38	35.3	1	**BK3038** [1]	38.4	79.2	12.5	7 000	11 000		0.093	Table 4	JR25x30x38,5
	30	37	38	—	1	**HK3038** [1]	38.4	79.2	12.5	7 000	11 000		0.075	Table 4	JR25x30x38,5
35	35	42	12	—	1	**HK3512**	12.3	19.2	2.90	5 900	9 100		0.028	Table 4	
	35	42	16	—	1	**HK3516**	18.7	33.0	4.60	5 900	9 100		0.037	Table 4	JR30x35x17
	35	42	20	17.3	1	**BK3520**	24.5	46.8	7.40	5 900	9 100		0.065	Table 4	JR30x35x20,5
	35	42	20	—	1	**HK3520**	24.5	46.8	7.40	5 900	9 100		0.049	Table 4	JR30x35x20,5
40	40	47	12	—	1	**HK4012**	13.4	22.4	3.40	5 200	7 900		0.033	Table 4	
	40	47	16	—	1	**HK4016**	18.9	34.8	5.35	5 200	7 900		0.042	Table 4	JR35x40x17
	40	47	20	17.3	1	**BK4020**	25.1	50.4	8.00	5 200	7 900		0.070	Table 4	JR35x40x20,5
	40	47	20	—	1	**HK4020**	25.1	50.4	8.00	5 200	7 900		0.060	Table 4	JR35x40x20,5
45	45	52	12	—	1	**HK4512**	14.1	24.8	3.75	4 600	7 000		0.036	Table 4	

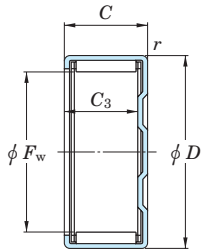
[Notes] 1) Drawn cup needle roller bearings with two needle roller and cage assemblies and one lubricating hole.
2) For the recommended mounting dimensions see Table 19.
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caged, open ends, closed one end metric series HK, BK series

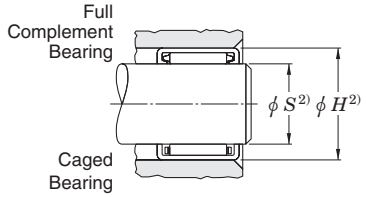
Shaft dia. (45) ~ 60 mm



HK



BK



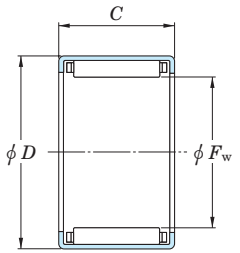
Shaft surface to be 58 HRC or equivalent

Shaft dia.	Boundary dimensions (mm) F_w	D	C $^{+0}_{-0.3}$	C_3 min.	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.		(Refer.) Mass (kg)	Inspection gage	Mounting inner ring
45	45	52	16	—	1	**HK4516**	19.8	38.5	5.95	4 600	7 000		0.048	Table 4	JR40x45x17
	45	52	20	17.3	1	**BK4520**	27.2	58.2	8.80	4 600	7 000		0.079	Table 4	JR40x45x20,5
	45	52	20	—	1	**HK4520**	27.2	58.2	8.80	4 600	7 000		0.059	Table 4	JR40x45x20,5
50	50	58	12	—	1	**HK5012**	17.0	28.7	4.40	4 100	6 300		0.045	Table 4	
	50	58	20	—	1	**HK5020**	30.9	62.2	8.80	4 100	6 300		0.072	Table 4	JR45x50x20
	50	58	25	—	1	**HK5025**	35.5	74.1	11.7	4 100	6 300		0.092	Table 4	JR45x50x25,5
55	55	63	20	—	1	**HK5520**	31.0	64.4	10.0	3 700	5 700		0.079	Table 4	
60	60	68	12	—	1	**HK6012**	18.6	34.4	5.25	3 400	5 200		0.060	Table 4	
	60	68	20	—	1	**HK6020**	35.6	79.5	10.9	3 400	5 200		0.090	Table 4	

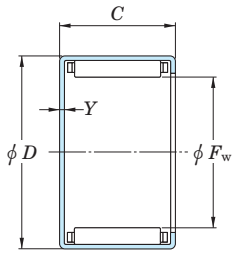
[Notes] 1) Drawn cup needle roller bearings with two needle roller and cage assemblies and one lubricating hole.
2) For the recommended mounting dimensions see Table 19.
3) For the mounting inner ring, please refer to JTEKT special catalog “Needle Roller Bearings” (CAT. NO. BS007EN).

caged, open ends, closed one end inch series J, JH, MJ-1, MJH-1 series

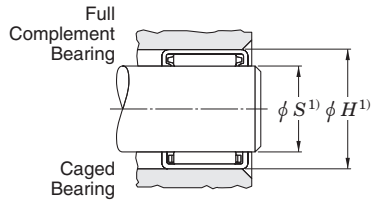
Shaft dia. $^1/_8 \sim {}^1/_2$ in (3.175 ~ 12.700 mm)



J, JH



MJ-1, MJH-1



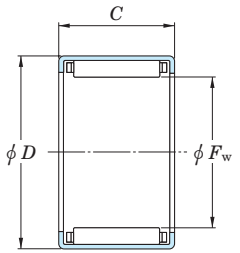
Shaft surface to be 58 HRC or equivalent

Shaft dia. (in)	Boundary dimensions (mm) F_w	D	C $^{+0}_{-0.3}$	Y max.	Bearing No. With open ends	With closed end	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u		Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg) With open ends	With closed end	Inspection gage
$^1/_8$	3.175	6.350	4.78	—	**JP-23-F**	—	0.90	0.61	0.100		33 000	51 000	0.001	—	Table 5
	3.175	6.350	6.35	—	**JP-24-F**	—	1.33	1.01	0.150		33 000	51 000	0.001	—	Table 5
$^5/_{32}$	3.970	7.142	4.78	—	**JP-2 1/2 3F**	—	0.91	0.62	0.110		31 000	47 000	0.001	—	Table 5
$^3/_{16}$	4.763	8.733	4.77	—	**JP-33-F**	—	1.07	0.73	0.120		25 000	38 000	0.001	—	Table 5
	4.763	8.733	6.35	—	**JP-34-F**	—	1.72	1.34	0.200		25 000	38 000	0.001	—	Table 5
	4.763	8.733	9.53	1.02	**J-36**	**MJ-361**	2.28	1.92	0.290		25 000	38 000	0.002	0.002	Table 5
$^1/_4$	6.350	11.113	7.92	1.02	**J-45**	**MJ-451**	2.21	1.74	0.300		20 000	30 000	0.003	0.003	Table 5
	6.350	11.113	11.13	1.02	**J-47**	**MJ-471**	3.40	3.01	0.450		20 000	30 000	0.004	0.004	Table 5
$^5/_{16}$	7.938	12.700	7.92	—	**J-55**	—	2.40	2.01	0.340		18 000	28 000	0.003	—	Table 5
	7.938	12.700	11.13	1.02	**J-57**	**MJ-571**	4.03	3.92	0.590		18 000	28 000	0.004	0.005	Table 5
	7.938	14.288	11.13	1.02	**JH-57**	**MJH-571**	4.65	3.76	0.570		14 000	22 000	0.006	0.007	Table 5
$^3/_8$	9.525	14.288	7.92	1.02	**J-65**	**MJ-651**	2.73	2.49	0.430		18 000	27 000	0.004	0.004	Table 5
	9.525	14.288	9.53	1.02	**J-66**	**MJ-661**	3.53	3.46	0.530		18 000	27 000	0.004	0.005	Table 5
	9.525	14.288	12.70	1.02	**J-68**	**MJ-681**	5.22	5.72	0.860		18 000	27 000	0.005	0.006	Table 5
	9.525	15.875	12.70	—	**JH-68**	—	6.59	6.08	0.920		13 000	20 000	0.008	—	Table 5
$^7/_{16}$	11.113	15.875	12.70	1.02	**J-78**	**MJ-781**	6.34	7.67	1.15		17 000	26 000	0.006	0.007	Table 5
	11.113	17.463	12.70	—	**JH-78**	—	7.10	6.89	1.05		13 000	19 000	0.009	—	Table 5
$^1/_2$	12.700	17.463	7.92	1.02	**J-85**	**MJ-851**	3.46	3.66	0.630		16 000	25 000	0.005	0.005	Table 5
	12.700	17.463	9.53	1.02	**J-86**	**MJ-861**	4.67	5.39	0.830		16 000	25 000	0.005	0.006	Table 5
	12.700	17.463	12.70	1.02	**J-88**	**MJ-881**	6.32	7.92	1.20		16 000	25 000	0.007	0.008	Table 5
	12.700	17.463	19.05	—	**J-812**	—	10.23	14.72	2.25		16 000	25 000	0.010	—	Table 5
	12.700	19.050	11.13	1.02	**JH-87**	**MJH-871**	6.39	6.20	0.950		12 000	19 000	0.009	0.010	Table 5
	12.700	19.050	12.70	1.02	**JH-88**	**MJH-881**	7.56	7.69	1.15		12 000	19 000	0.010	0.012	Table 5
	12.700	19.050	19.05	—	**JH-812**	—	12.32	14.41	2.25		12 000	19 000	0.015	—	Table 5

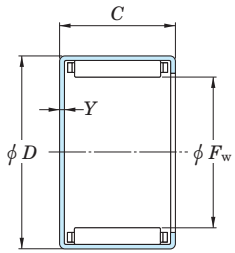
[Note] 1) For the recommended mounting dimensions see Table 20.

caged, open ends, closed one end inch series J, JH, MJ-1, MJH-1 series

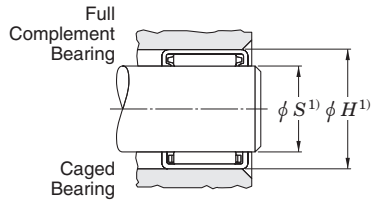
Shaft dia. $^{9}/_{16} \sim {}^{7}/_{8}$ in (14.288 ~ 22.225 mm)



J, JH



MJ-1, MJH-1



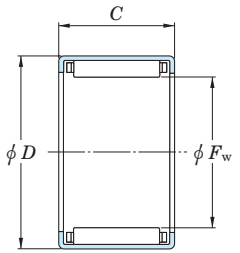
Shaft surface to be 58 HRC or equivalent

Shaft dia. (in)	Boundary dimensions (mm) F_w	D	C $^{+0}_{-0.3}$	Y max.	Bearing No. With open ends	With closed end	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u		Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg) With open ends	With closed end	Inspection gage
$^{9}/_{16}$	14.288	19.050	11.13	1.02	**J-97**	**MJ-971**	5.47	6.80	1.05		16 000	25 000	0.007	0.009	Table 5
	14.288	19.050	12.70	1.02	**J-98**	**MJ-981**	6.23	8.03	1.20		16 000	25 000	0.008	0.009	Table 5
	14.288	19.050	15.88	—	**J-910**	—	8.27	11.60	1.75		16 000	25 000	0.010	—	Table 5
	14.288	20.638	12.70	1.02	**JH-98**	**MJH-981**	7.98	8.49	1.30		12 000	18 000	0.011	0.014	Table 5
$^{5}/_{8}$	15.875	20.638	12.70	1.02	**J-108**	**MJ-1081**	6.71	9.13	1.40		13 000	21 000	0.009	0.010	Table 5
	15.875	20.638	15.88	1.02	**J-1010**	**MJ-10101**	8.80	12.94	1.95		13 000	21 000	0.010	0.013	Table 5
	15.875	20.638	19.05	1.02	**J-1012**	**MJ-10121**	11.80	18.86	2.90		13 000	21 000	0.013	0.015	Table 5
	15.875	22.212	15.88	1.02	**JH-1010**	**MJH-10101**	11.57	14.10	2.15		14 000	21 000	0.015	0.017	Table 5
	15.875	22.212	25.40	1.02	**JH-1016**	**MJH-10161**	19.79	28.11	4.35		14 000	21 000	0.024	0.028	Table 5
$^{11}/_{16}$	17.463	22.212	19.05	1.02	**J-1112**	**MJ-11121**	12.46	20.91	3.20		12 000	19 000	0.014	0.016	Table 5
	17.463	23.813	15.88	1.02	**JH-1110**	**MJH-11101**	12.05	15.21	2.30		13 000	19 000	0.016	0.019	Table 5
	17.463	23.813	19.05	—	**JH-1112**	—	16.10	22.20	3.10		13 000	19 000	0.019	—	Table 5
$^{3}/_{4}$	19.050	25.400	9.53	—	**J-126**	—	6.49	7.05	1.10		11 000	18 000	0.010	—	Table 5
	19.050	25.400	12.70	—	**J-128**	—	9.94	12.19	1.85		11 000	18 000	0.014	—	Table 5
	19.050	25.400	15.88	1.02	**J-1210**	**MJ-12101**	12.50	16.32	2.50		11 000	18 000	0.017	0.020	Table 5
	19.050	25.400	19.05	1.02	**J-1212**	**MJ-12121**	15.52	21.62	3.35		11 000	18 000	0.020	0.025	Table 5
	19.050	26.988	19.05	1.02	**JH-1212**	**MJH-12121**	19.08	23.58	3.70		12 000	18 000	0.026	0.031	Table 5
$^{13}/_{16}$	20.638	26.988	22.23	—	**J-1314**	—	19.31	29.31	4.55		10 000	16 000	0.025	—	Table 5
	20.638	28.575	19.05	1.27	**JH-1312**	**MJH-13121**	18.77	24.50	3.85		11 000	16 000	0.028	0.034	Table 5
$^{7}/_{8}$	22.225	28.575	9.53	—	**J-146**	—	7.20	8.43	1.30		9 700	15 000	0.012	—	Table 5
	22.225	28.575	12.70	—	**J-148**	—	10.94	14.50	2.20		9 700	15 000	0.015	—	Table 5
	22.225	28.575	19.05	1.02	**J-1412**	**MJ-14121**	17.88	27.18	4.20		9 700	15 000	0.024	0.028	Table 5
	22.225	28.575	25.40	1.02	**J-1416**	**MJ-14161**	23.66	38.97	6.05		9 700	15 000	0.031	0.059	Table 5
	22.225	30.163	19.05	1.27	**JH-1412**	**MJH-14121**	18.33	24.50	3.75		9 800	15 000	0.030	0.036	Table 5
	22.225	30.163	25.40	1.27	**JH-1416**	**MJH-14161**	25.40	37.37	5.80		9 800	15 000	0.040	0.048	Table 5

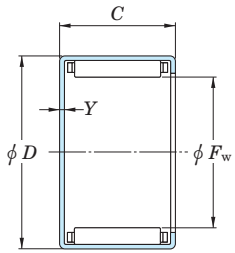
[Note] 1) For the recommended mounting dimensions see Table 20.

caged, open ends, closed one end inch series J, JH, MJ-1, MJH-1 series

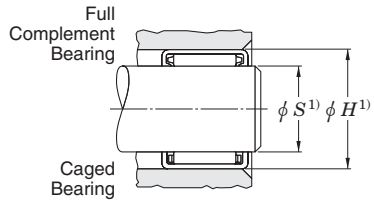
Shaft dia. 1 ~ (1 3/4) in (25.400 ~ (44.450) mm)



J, JH



MJ-1, MJH-1



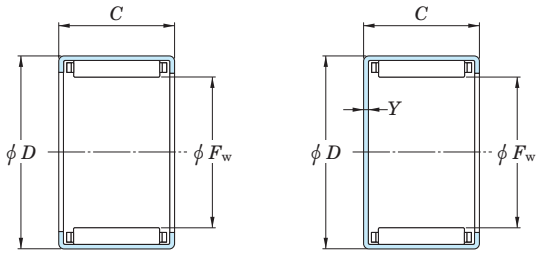
Shaft surface to be 58 HRC or equivalent

Shaft dia. (in)	Boundary dimensions (mm) F_w	D	C $^{+0}_{-0.3}$	Y max.	Bearing No. With open ends	With closed end	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u		Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg) With open ends	With closed end	Inspection gage
1	25.400	31.750	19.05	—	**J-1612**	—	18.15	28.82	4.45		8 400	13 000	0.026	—	Table 5
	25.400	31.750	25.40	1.02	**J-1616**	**MJ-16161**	24.95	43.41	6.75		8 400	13 000	0.035	0.042	Table 5
	25.400	33.338	19.05	1.27	**JH-1612**	**MJH-16121**	20.68	29.58	4.60		8 500	13 000	0.034	0.040	Table 5
	25.400	33.338	25.40	1.27	**JH-1616**	**MJH-16161**	27.58	42.88	6.65		8 500	13 000	0.045	0.054	Table 5
1 1/8	28.575	34.925	12.70	1.02	**J-188**	**MJ-1881**	11.65	16.95	2.55		7 400	11 000	0.020	0.023	Table 5
	28.575	34.925	19.05	1.02	**J-1812**	**MJ-18121**	19.04	31.76	4.90		7 400	11 000	0.029	0.035	Table 5
	28.575	34.925	25.40	1.02	**J-1816**	**MJ-18161**	26.16	48.04	7.40		7 400	11 000	0.039	0.047	Table 5
	28.575	38.100	19.05	1.27	**JH-1812**	**MJH-18121**	23.35	31.32	4.75		7 600	12 000	0.046	0.055	Table 5
	28.575	38.100	25.40	1.27	**JH-1816**	**MJH-18161**	33.14	49.38	7.70		7 600	12 000	0.061	0.074	Table 5
	28.575	38.100	28.58	1.27	**JH-1818**	**MJH-18181**	36.30	55.16	8.60		7 600	12 000	0.069	0.082	Table 5
1 1/4	31.750	38.100	19.05	1.02	**J-2012**	**MJ-20121**	19.84	34.70	5.35		6 600	10 000	0.036	0.043	Table 5
	31.750	38.100	25.40	1.02	**J-2016**	**MJ-20161**	28.82	56.49	8.70		6 600	10 000	0.043	0.051	Table 5
	31.750	41.275	19.05	—	**JH-2012**	—	24.11	33.94	5.80		6 800	10 000	0.050	—	Table 5
	31.750	41.275	25.40	—	**JH-2016**	—	33.94	52.93	8.20		6 800	10 000	0.067	—	Table 5
	31.750	41.275	31.75	—	**JH-2020**	—	43.37	72.51	10.8		6 800	10 000	0.084	—	Table 5
1 3/8	34.925	41.275	12.70	1.02	**J-228**	**MJ-2281**	13.97	22.91	3.50		6 000	9 200	0.024	0.028	Table 5
	34.925	41.275	19.05	—	**J-2212**	—	22.82	42.97	6.65		6 000	9 200	0.035	—	Table 5
	34.925	44.450	19.05	1.27	**JH-2212**	**MJH-22121**	26.24	38.43	5.90		6 100	9 400	0.055	0.065	Table 5
	34.925	44.450	25.40	1.27	**JH-2216**	**MJH-22161**	36.52	58.72	9.20		6 100	9 400	0.073	0.087	Table 5
1 1/2	38.100	47.625	19.05	1.27	**J-2412**	**MJ-24121**	29.89	47.15	7.40		5 600	8 600	0.059	0.094	Table 5
	38.100	47.625	25.40	1.27	**J-2416**	**MJ-24161**	39.32	66.72	10.4		5 600	8 600	0.079	0.094	Table 5
	38.100	47.625	31.75	—	**J-2420**	—	49.38	89.85	14.0		5 600	8 600	0.099	—	Table 5
1 5/8	41.275	50.800	15.88	—	**J-2610**	—	26.11	40.97	6.25		5 100	7 900	0.053	—	Table 5
	41.275	50.800	25.40	1.27	**J-2616**	**M-26161**	39.28	68.95	10.8		5 100	7 900	0.085	0.101	Table 5
1 3/4	44.450	53.975	19.05	1.27	**J-2812**	**MJ-28121**	29.58	49.38	7.45		4 700	7 300	0.068	0.081	Table 5

[Note] 1) For the recommended mounting dimensions see Table 20.

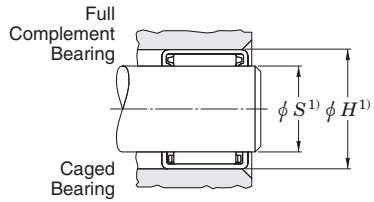
caged, open ends, closed one end inch series J, JH, MJ-1, MJH-1 series

Shaft dia. (1 3/4) ~ 2 3/4 in ((44.450) ~ 69.850 mm)



J, JH

MJ-1, MJH-1

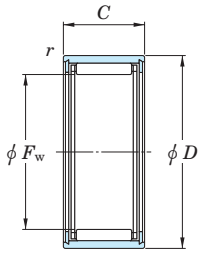


Shaft surface to be 58 HRC or equivalent

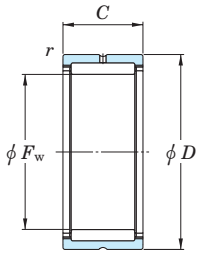
Shaft dia. (in)	Boundary dimensions (mm) F_w	D	C $^{+0}_{-0.3}$	Y max.	Bearing No. With open ends	With closed end	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u		Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg) With open ends	With closed end	Inspection gage
1 3/4	44.450	53.975	25.40	1.27	**J-2816**	**MJ-28161**	40.08	72.95	11.4		4 700	7 300	0.091	0.108	Table 5
	44.450	53.975	38.10	1.27	**J-2824**	**MJ-28241**	59.61	121.88	18.9		4 700	7 300	0.136	0.162	Table 5
1 7/8	47.625	57.150	25.40	1.27	**J-3016**	**MJ-30161**	41.10	76.06	11.9		4 400	6 800	0.097	0.115	Table 5
2	50.800	60.325	25.40	1.27	**J-3216**	**MJ-32161**	42.39	81.40	12.7		4 100	6 300	0.103	0.137	Table 5
2 1/4	57.150	66.675	19.05	—	**J-3612**	—	35.41	65.83	10.0		3 600	5 600	0.086	—	Table 5
	57.150	66.675	25.40	—	**J-3616**	—	46.26	92.52	14.4		3 600	5 600	0.114	—	Table 5
2 3/4	69.850	79.375	19.05	—	**J-4412**	—	36.25	72.95	11.3		2 900	4 500	0.103	—	Table 5

[Note] 1) For the recommended mounting dimensions see Table 20.

without inner rings
metric series
NK, NKS, RNA48, RNA49
RNA69 series
Shaft dia. 5 ~ (17) mm



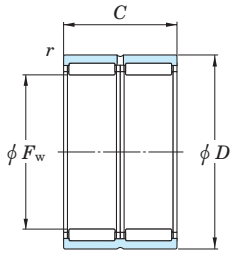
NK ($\phi F_w \leq 10$)



NK ($\phi F_w \geq 12$), NKS, RNA48, RNA49, RNA69 ($\phi F_w \leq 35$)

Shaft dia.	Boundary dimensions (mm) F_w	D	C	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg)
5	5	10	10	0.2	**NK5/10TN**	2.18	1.71	0.260	31 000	47 000	0.004
	5	10	12	0.2	**NK5/12TN**	3.04	2.63	0.400	31 000	47 000	0.004
6	6	12	10	0.2	**NK6/10**	3.19	2.90	0.420	29 000	44 000	0.005
	6	12	12	0.2	**NK6/12TN**	3.07	2.74	0.420	29 000	44 000	0.006
7	7	14	10	0.3	**NK7/10TN**	2.74	2.44	0.370	28 000	42 000	0.007
	7	14	12	0.3	**NK7/12TN**	3.40	3.22	0.490	28 000	42 000	0.009
8	8	15	12	0.3	**NK8/12**	4.57	4.89	0.740	26 000	41 000	0.011
	8	15	12	0.3	**NK8/12ASR1**	4.57	4.89	0.740	26 000	41 000	0.011
	8	15	16	0.3	**NK8/16**	5.22	5.78	0.880	26 000	41 000	0.013
9	9	16	12	0.3	**NK9/12**	4.27	4.60	0.700	26 000	40 000	0.012
	9	16	16	0.3	**NK9/16**	5.57	6.47	0.980	26 000	40 000	0.015
10	10	17	12	0.3	**NK10/12**	5.40	6.43	0.980	25 000	39 000	0.013
	10	17	16	0.3	**NK10/16TN**	5.30	6.27	0.940	25 000	39 000	0.015
12	12	19	12	0.3	**NK12/12**	6.86	7.60	1.15	19 000	30 000	0.013
	12	19	16	0.3	**NK12/16**	6.78	9.03	1.40	24 000	37 000	0.018
14	14	22	13	0.3	**RNA4900**	9.39	10.3	1.55	16 000	24 000	0.018
	14	22	16	0.3	**NK14/16**	12.4	14.8	2.25	16 000	24 000	0.023
	14	22	20	0.3	**NK14/20**	14.7	18.4	2.90	16 000	24 000	0.028
15	15	23	16	0.3	**NK15/16**	12.4	15.0	2.30	15 000	24 000	0.024
	15	23	20	0.3	**NK15/20**	14.7	18.6	2.95	15 000	24 000	0.031
16	16	24	13	0.3	**RNA4901**	10.5	12.3	1.85	18 000	28 000	0.020
	16	24	16	0.3	**NK16/16**	15.4	20.2	2.50	18 000	28 000	0.025
	16	24	20	0.3	**NK16/20**	16.1	21.3	3.20	18 000	28 000	0.036
	16	24	22	0.3	**RNA6901**	16.1	21.3	3.30	18 000	28 000	0.036
17	17	25	16	0.3	**NK17/16**	13.6	17.5	2.70	17 000	27 000	0.027

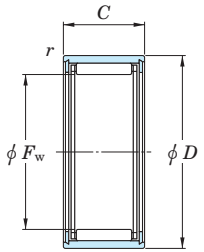
Shaft dia. (17) ~ 25 mm



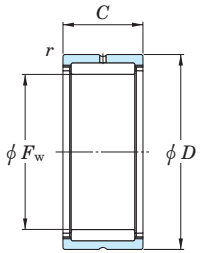
RNA69 ($\phi F_w \geq 40$)

Shaft dia.	Boundary dimensions (mm) F_w	D	C	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg)
17	17	25	20	0.3	**NK17/20**	15.4	20.4	3.25	17 000	27 000	0.034
18	18	26	16	0.3	**NK18/16**	13.6	17.7	2.70	16 000	25 000	0.028
	18	26	20	0.3	**NK18/20**	16.1	22.0	3.50	16 000	25 000	0.035
	18	30	16	0.3	**NKS18**	15.9	16.2	2.45	17 000	26 000	0.045
19	19	27	16	0.3	**NK19/16**	14.1	19.0	2.90	15 000	24 000	0.029
	19	27	20	0.3	**NK19/20**	18.8	23.6	3.75	15 000	24 000	0.037
20	20	28	13	0.3	**RNA4902**	11.8	15.3	2.35	14 000	22 000	0.023
	20	28	16	0.3	**NK20/16**	14.1	19.1	2.90	14 000	22 000	0.030
	20	28	20	0.3	**NK20/20**	17.5	25.3	4.00	14 000	22 000	0.038
	20	28	23	0.3	**RNA6902**	18.4	26.9	4.20	14 000	22 000	0.042
	20	32	20	0.6	**NKS20**	24.4	26.7	4.30	15 000	24 000	0.058
21	21	29	16	0.3	**NK21/16**	15.3	21.6	3.30	14 000	21 000	0.032
	21	29	20	0.3	**NK21/20**	18.1	26.9	4.25	14 000	21 000	0.040
22	22	30	13	0.3	**RNA4903**	12.2	16.4	2.50	13 000	20 000	0.025
	22	30	16	0.3	**NK22/16**	15.2	21.7	3.30	13 000	20 000	0.033
	22	30	20	0.3	**NK22/20**	18.0	27.0	4.30	13 000	20 000	0.041
	22	30	23	0.3	**RNA6903**	19.8	30.6	4.75	13 000	20 000	0.056
	22	35	20	0.6	**NKS22**	22.9	27.1	4.30	14 000	21 000	0.069
24	24	32	16	0.3	**NK24/16**	16.2	24.3	3.70	12 000	18 000	0.035
	24	32	20	0.3	**NK24/20**	19.3	30.3	4.80	12 000	18 000	0.045
	24	37	20	0.6	**NKS24**	29.1	32.8	5.30	13 000	20 000	0.073
25	25	33	16	0.3	**NK25/16**	16.1	24.4	3.75	11 000	17 000	0.037
	25	33	20	0.3	**NK25/20**	19.1	30.4	4.80	11 000	17 000	0.047
	25	37	17	0.3	**RNA4904**	21.3	25.5	3.95	12 000	18 000	0.061
	25	37	30	0.3	**RNA6904**	36.6	51.0	7.95	12 000	18 000	0.091
	25	38	20	0.6	**NKS25**	29.1	33.0	5.30	12 000	19 000	0.076

without inner rings
metric series
NK, NKS, RNA48, RNA49
RNA69 series
Shaft dia. 26 ~ 37 mm



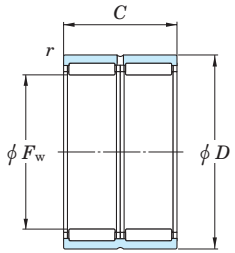
NK ($\phi F_w \leq 10$)



NK ($\phi F_w \geq 12$), NKS, RNA48, RNA49, RNA69 ($\phi F_w \leq 35$)

Shaft dia.	Boundary dimensions (mm) F_w	D	C	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg)
26	26	34	16	0.3	**NK26/16**	16.6	25.7	3.95	11 000	17 000	0.039
	26	34	20	0.3	**NK26/20**	19.7	32.0	5.05	11 000	17 000	0.048
28	28	37	20	0.3	**NK28/20**	22.6	34.4	5.50	10 000	16 000	0.057
	28	37	30	0.3	**NK28/30**	29.0	53.8	8.30	10 000	16 000	0.088
	28	39	17	0.3	**RNA49/22**	23.3	29.6	4.55	10 000	16 000	0.059
	28	39	30	0.3	**RNA69/22**	30.6	50.7	3.95	10 000	16 000	0.107
	28	42	20	0.6	**NKS28**	30.3	38.4	6.15	11 000	16 000	0.094
29	29	38	20	0.3	**NK29/20**	23.4	36.4	5.80	9 800	15 000	0.059
	29	38	30	0.3	**NK29/30**	29.8	56.4	8.70	9 700	15 000	0.090
30	30	40	20	0.3	**NK30/20**	24.2	38.3	6.10	9 500	15 000	0.071
	30	40	30	0.3	**NK30/30**	34.7	61.0	9.45	9 500	15 000	0.107
	30	42	17	0.3	**RNA4905**	24.3	31.7	4.90	9 700	15 000	0.071
	30	42	30	0.3	**RNA6905**	39.7	59.6	9.30	9 700	15 000	0.127
	30	45	22	0.6	**NKS30**	34.3	42.8	6.85	9 900	15 000	0.114
32	32	42	20	0.3	**NK32/20**	24.8	40.4	6.45	8 800	14 000	0.074
	32	42	30	0.3	**NK32/30**	35.6	64.3	9.95	8 800	14 000	0.112
	32	45	17	0.3	**RNA49/28**	25.1	33.8	5.20	9 000	14 000	0.080
	32	45	30	0.3	**RNA69/28**	43.2	62.5	9.75	9 100	14 000	0.140
	32	47	22	0.6	**NKS32**	36.0	46.2	7.40	9 200	14 000	0.120
35	35	45	20	0.3	**NK35/20**	26.1	44.4	7.05	8 000	12 000	0.081
	35	45	30	0.3	**NK35/30**	37.4	70.6	11.0	8 000	12 000	0.122
	35	47	18	0.3	**RNA4906**	25.9	36.0	5.55	8 200	13 000	0.081
	35	47	30	0.3	**RNA6906**	42.6	68.2	10.6	8 200	13 000	0.148
	35	50	22	0.6	**NKS35**	37.5	49.9	8.00	8 400	13 000	0.130
37	37	47	20	0.3	**NK37/20**	26.6	46.4	7.40	7 600	12 000	0.084
	37	47	30	0.3	**NK37/30**	38.2	73.9	11.5	7 600	12 000	0.128
	37	52	22	0.6	**NKS37**	39.0	53.4	8.55	7 900	12 000	0.134

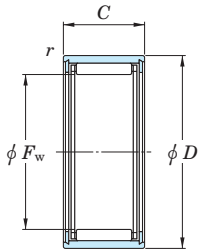
Shaft dia. 38 ~ 52 mm



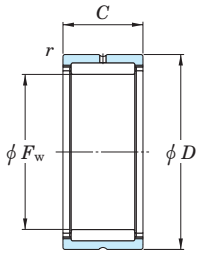
RNA69 ($\phi F_w \geq 40$)

Shaft dia.	Boundary dimensions (mm) F_w	D	C	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg)
38	38	48	20	0.3	**NK38/20**	21.7	40.9	6.40	7 300	11 000	0.087
	38	48	30	0.3	**NK38/30**	31.9	67.0	10.4	7 300	11 000	0.131
40	40	50	20	0.3	**NK40/20**	27.8	50.4	8.05	7 000	11 000	0.089
	40	50	30	0.3	**NK40/30**	40.0	80.2	12.4	7 000	11 000	0.137
	40	52	20	0.6	**RNA49/32**	32.0	49.3	7.85	7 100	11 000	0.100
	40	52	36	0.6	**RNA69/32**	48.6	84.5	26.1	7 100	11 000	0.185
	40	55	22	0.6	**NKS40**	40.3	57.0	9.15	7 200	11 000	0.140
42	42	52	20	0.3	**NK42/20**	28.3	52.4	8.35	6 600	10 000	0.085
	42	52	30	0.3	**NK42/30**	40.7	83.5	13.0	6 600	10 000	0.141
	42	55	20	0.6	**RNA4907**	32.8	51.7	8.25	6 700	10 000	0.114
	42	55	36	0.6	**RNA6907**	49.9	88.7	13.7	6 700	10 000	0.218
43	43	53	20	0.3	**NK43/20**	29.0	54.4	8.65	6 400	9 900	0.096
	43	53	30	0.3	**NK43/30**	41.6	86.6	13.4	6 400	9 900	0.134
	43	58	22	0.6	**NKS43**	41.6	60.7	9.75	6 700	10 000	0.150
45	45	55	20	0.3	**NK45/20**	29.5	56.4	9.00	6 100	9 400	0.100
	45	55	30	0.3	**NK45/30**	42.3	89.8	13.9	6 100	9 400	0.151
	45	60	22	0.6	**NKS45**	43.0	64.2	10.3	6 400	9 800	0.156
47	47	57	20	0.3	**NK47/20**	30.0	58.5	9.30	5 900	9 000	0.104
	47	57	30	0.3	**NK47/30**	43.0	93.1	14.4	5 900	9 000	0.158
48	48	62	22	0.6	**RNA4908**	44.2	67.8	10.9	5 900	9 100	0.154
	48	62	40	0.6	**RNA6908**	70.8	124	19.8	5 900	9 100	0.300
50	50	62	25	0.3	**NK50/25**	40.7	79.3	12.5	5 500	8 500	0.171
	50	62	35	0.6	**NK50/35**	55.0	117	18.2	5 500	8 500	0.242
	50	65	22	1	**NKS50**	45.5	71.3	11.4	5 700	8 700	0.170
52	52	68	22	0.6	**RNA4909**	46.8	74.8	12.0	5 400	8 400	0.201
	52	68	40	0.6	**RNA6909**	74.7	137	21.7	5 400	8 400	0.392

without inner rings
metric series
NK, NKS, RNA48, RNA49
RNA69 series
Shaft dia. 55 ~ (75) mm



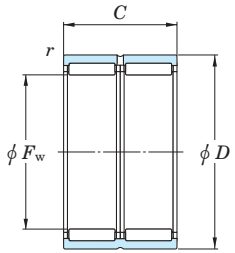
NK ($\phi F_w \leq 10$)



NK ($\phi F_w \geq 12$), NKS, RNA48, RNA49, RNA69 ($\phi F_w \leq 35$)

Shaft dia.	Boundary dimensions (mm) F_w	D	C	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg)
55	55	68	25	0.6	**NK55/25**	46.1	87.3	13.9	5 000	7 800	0.207
	55	68	35	0.6	**NK55/35**	62.3	129	20.0	5 000	7 800	0.293
	55	72	22	1	**NKS55**	47.9	78.4	12.6	5 100	7 900	0.225
58	58	72	22	0.6	**RNA4910**	48.9	82.0	13.2	4 800	7 400	0.179
	58	72	40	0.6	**RNA6910**	75.7	144	22.8	4 800	7 400	0.364
60	60	72	25	0.6	**NK60/25**	44.3	94.0	14.9	4 400	7 000	0.202
	60	72	35	0.6	**NK60/35**	59.9	139	21.5	4 400	7 000	0.286
	60	80	28	1.1	**NKS60**	66.9	103	16.5	4 800	7 300	0.337
63	63	80	25	1	**RNA4911**	62.0	107	17.1	4 500	6 900	0.285
	63	80	45	1	**RNA6911**	94.2	172	27.8	4 500	6 900	0.540
65	65	78	25	0.6	**NK65/25**	48.2	97.7	15.5	4 200	6 500	0.257
	65	78	35	0.6	**NK65/35**	65.2	144	22.4	4 200	6 500	0.298
	65	85	28	1.1	**NKS65**	71.0	114	18.3	4 200	6 700	0.362
68	68	82	25	0.6	**NK68/25**	49.0	101	16.1	4 000	6 200	0.287
	68	82	35	0.6	**NK68/35**	66.2	149	23.2	4 000	6 200	0.350
	68	85	25	1	**RNA4912**	64.8	116	18.6	4 100	6 300	0.304
	68	85	45	1	**RNA6912**	99.3	189	30.5	4 100	6 300	0.546
70	70	85	25	0.6	**NK70/25**	43.6	87.9	16.6	3 900	6 000	0.298
	70	85	35	0.6	**NK70/35**	62.2	139	24.0	3 900	6 000	0.411
	70	90	28	1.1	**NKS70**	72.6	120	19.3	4 000	6 200	0.383
72	72	90	25	1	**RNA4913**	66.0	121	19.4	3 900	5 900	0.346
	72	90	45	1	**RNA6913**	107	213	34.5	3 900	5 900	0.679
73	73	90	25	0.6	**NK73/25**	61.5	119	19.0	3 800	5 800	0.320
	73	90	35	0.6	**NK73/35**	82.5	173	27.1	3 800	5 800	0.450
75	75	92	25	0.6	**NK75/25**	43.7	90.2	19.0	3 600	5 600	0.364
	75	92	35	0.6	**NK75/35**	60.9	138	27.1	3 600	5 600	0.518

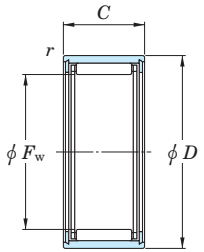
Shaft dia. (75) ~ 110 mm



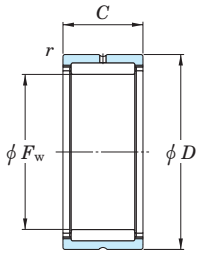
RNA69 ($\phi F_w \geq 40$)

Shaft dia.	Boundary dimensions (mm) F_w	D	C	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg)
75	75	95	28	1.1	**NKS75**	76.5	132	21.1	3 700	5 800	0.413
80	80	95	25	1	**NK80/25**	65.0	131	21.0	3 400	5 300	0.331
	80	95	35	1	**NK80/35**	79.7	184	28.7	3 400	5 300	0.380
	80	100	30	1	**RNA4914**	86.3	157	25.1	3 500	5 400	0.502
	80	100	54	1	**RNA6914**	137	286	45.7	3 500	5 400	0.946
85	85	105	25	1	**NK85/25**	76.4	137	22.2	3 300	5 000	0.506
	85	105	30	1	**RNA4915**	92.4	175	28.0	3 300	5 000	0.528
	85	105	35	1	**NK85/35**	108	214	34.7	3 300	5 000	0.610
	85	105	54	1	**RNA6915**	143	308	49.3	3 300	5 000	1.020
90	90	110	25	1	**NK90/25**	79.5	147	23.8	3 100	4 700	0.450
	90	110	30	1	**RNA4916**	91.5	176	28.1	3 100	4 700	0.556
	90	110	35	1	**NK90/35**	113	230	36.1	3 100	4 700	0.745
	90	110	54	1	**RNA6916**	126	320	50.8	3 100	4 700	1.050
95	95	115	26	1	**NK95/26**	49.3	114	24.6	2 800	4 400	0.572
	95	115	36	1	**NK95/36**	114	238	37.3	2 900	4 500	0.803
100	100	120	26	1	**NK100/26**	83.6	163	25.8	2 800	4 200	0.530
	100	120	35	1.1	**RNA4917**	110	230	36.0	2 800	4 200	0.715
	100	120	36	1	**NK100/36**	118	254	39.1	2 800	4 200	0.658
	100	120	63	1.1	**RNA6917**	150	416	63.0	2 800	4 200	1.350
105	105	125	26	1	**NK105/26**	52.2	127	19.9	2 600	3 900	0.595
	105	125	35	1.1	**RNA4918**	114	245	37.8	2 600	4 000	0.746
	105	125	63	1.1	**RNA6918**	154	437	66.0	2 600	4 000	1.500
110	110	130	30	1.1	**NK110/30**	103	220	33.6	2 500	3 800	0.660
	110	130	35	1.1	**RNA4919**	115	253	38.4	2 500	3 800	0.777
	110	130	40	1.1	**NK110/40**	132	301	45.7	2 500	3 800	0.900
	110	130	63	1.1	**RNA6919**	158	458	68.8	2 500	3 800	1.470

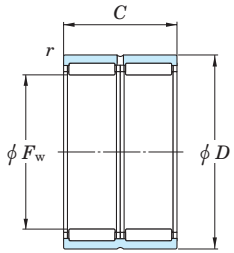
without inner rings
metric series
NK, NKS, RNA48, RNA49
RNA69 series
Shaft dia. 115 ~ 175 mm



NK ($\phi F_w \leq 10$)



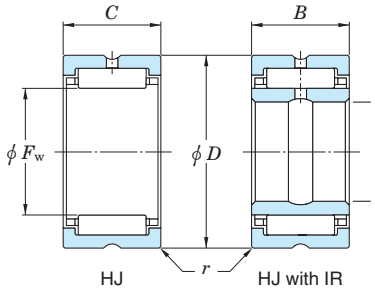
NK ($\phi F_w \geq 12$), NKS, RNA48, RNA49, RNA69 ($\phi F_w \leq 35$)

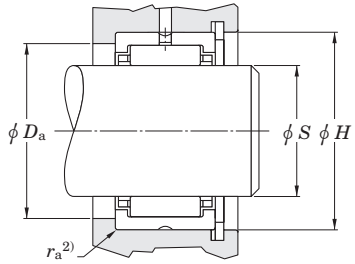


RNA69 ($\phi F_w \geq 40$)

Shaft dia.	Boundary dimensions (mm) F_w	D	C	r min.	Bearing No.	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg)
115	115	140	40	1.1	**RNA4920**	139	296	43.9	2 400	3 700	1.220
120	120	140	30	1	**RNA4822**	90.3	230	33.7	2 300	3 500	0.785
125	125	150	40	1.1	**RNA4922**	147	325	47.0	2 200	3 400	1.320
130	130	150	30	1	**RNA4824**	94.1	249	35.7	2 100	3 200	0.850
135	135	165	45	1.1	**RNA4924**	177	407	58.5	2 000	3 100	1.980
145	145	165	35	1	**RNA4826**	112	323	44.8	1 900	2 900	1.100
150	150	180	50	1.5	**RNA4926**	201	495	68.7	1 800	2 800	2.420
155	155	175	35	1.1	**RNA4828**	116	346	47.1	1 700	2 700	1.170
160	160	190	50	1.5	**RNA4928**	214	549	74.8	1 700	2 600	2.560
165	165	190	40	1.1	**RNA4830**	142	402	53.5	1 600	2 500	1.540
175	175	200	40	1.1	**RNA4832**	146	425	55.6	1 500	2 400	1.910

inch series
HJ type
Shaft dia. 5/8 ~ (1 3/4) in (15.875 ~ (44.450) mm)



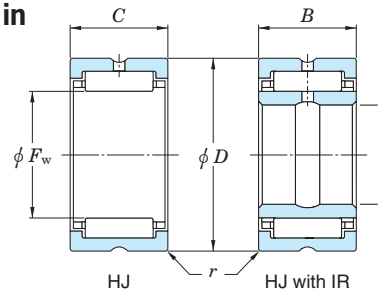


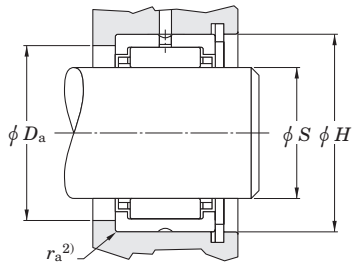
Shaft surface to be 58 HRC or equivalent

Shaft dia. (in)	Boundary dimensions (mm) F_w	D	C (B)	r min.	Bearing No.	Used with inner ring No. 1)	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg)	Recommended dimensions Clearance fit S (h6) max.	min.	H (H7) max.	min.	Tight transition fit S (f6) max.	min.	H (N7) max.	min.	Shoulder dia. D_a ±0.38
5/8	15.875	28.575	19.050	0.64	HJ-101812	IR-061012	19.3	20.7	3.25	20 000	30 000	0.050	15.875	15.865	28.595	28.575	15.860	15.850	28.567	28.547	23.83
3/4	19.050	31.750	19.050	1.02	HJ-122012	IR-081212	20.7	23.3	3.65	16 000	25 000	0.059	19.050	19.037	31.775	31.750	19.030	19.017	31.742	31.717	26.97
	19.050	31.750	25.400	1.02	HJ-122016	IR-081216	27.5	33.7	5.30	16 000	25 000	0.077	19.050	19.037	31.775	31.750	19.030	19.017	31.742	31.717	26.97
7/8	22.225	34.925	19.050	1.02	HJ-142212	IR-101412	23	27.9	4.35	13 000	21 000	0.064	22.225	22.212	34.950	34.925	22.205	22.192	34.917	34.892	30.18
	22.225	34.925	25.400	1.02	HJ-142216	IR-101416	30.7	40.3	6.35	13 000	21 000	0.086	22.225	22.212	34.950	34.925	22.205	22.192	34.917	34.892	30.18
1	25.400	38.100	19.050	1.02	HJ-162412	IR-121612	25.3	32.5	5.10	12 000	18 000	0.073	25.400	25.387	38.125	38.100	25.380	25.367	38.092	38.067	33.32
	25.400	38.100	25.400	1.02	HJ-162416	IR-121616	33.6	47.2	7.40	12 000	18 000	0.095	25.400	25.387	38.125	38.100	25.380	25.367	38.092	38.067	33.32
	25.400	38.100	25.400	1.02	HJ-162416	IR-131616	33.6	47.2	7.40	12 000	18 000	0.095	25.400	25.387	38.125	38.100	25.380	25.367	38.092	38.067	33.32
1 1/8	28.575	41.275	25.400	1.02	HJ-182616	IR-141816	36.3	53.8	8.45	10 000	16 000	0.104	28.575	28.562	41.300	41.275	28.555	28.542	41.267	41.242	36.53
	28.575	41.275	25.400	1.02	HJ-182616	IR-151816	36.3	53.8	8.45	10 000	16 000	0.104	28.575	28.562	41.300	41.275	28.555	28.542	41.267	41.242	36.53
	28.575	41.275	31.750	1.02	HJ-182620	IR-141820	44.9	70.3	10.9	10 000	16 000	0.132	28.575	28.562	41.300	41.275	28.555	28.542	41.267	41.242	36.53
	28.575	41.275	31.750	1.02	HJ-182620	IR-151820	44.9	70.3	10.9	10 000	16 000	0.132	28.575	28.562	41.300	41.275	28.555	28.542	41.267	41.242	36.53
1 1/4	31.750	44.450	25.400	1.02	HJ-202816	IR-162016	37.4	57.4	9.00	9 100	14 000	0.113	31.750	31.735	44.475	44.450	31.725	31.709	44.442	44.417	39.67
	31.750	44.450	31.750	1.02	HJ-202820	IR-162020	46.3	75.2	11.7	9 100	14 000	0.145	31.750	31.735	44.475	44.450	31.725	31.709	44.442	44.417	39.67
1 3/8	34.925	47.625	25.400	1.02	HJ-223016	IR-182216	39.8	64.1	10.1	8 200	13 000	0.127	34.925	34.910	47.650	47.625	34.900	34.884	47.617	47.592	42.88
	34.925	47.625	31.750	1.02	HJ-223020	IR-182220	49.4	84.1	13.0	8 200	13 000	0.159	34.925	34.910	47.650	47.625	34.900	34.884	47.617	47.592	42.88
1 1/2	38.100	52.388	25.400	1.52	HJ-243316	IR-202416	47.6	72.5	11.4	7 600	12 000	0.154	38.100	38.085	52.418	52.388	38.075	38.059	52.380	52.349	47.63
	38.100	52.388	31.750	1.52	HJ-243320	IR-192420	58.7	95.2	14.9	7 600	12 000	0.195	38.100	38.085	52.418	52.388	38.075	38.059	52.380	52.349	47.63
	38.100	52.388	31.750	1.52	HJ-243320	IR-202420	58.7	95.2	14.9	7 600	12 000	0.195	38.100	38.085	52.418	52.388	38.075	38.059	52.380	52.349	47.63
1 5/8	41.275	55.563	25.400	1.52	HJ-263516	IR-212616	48.5	76.5	12.1	7 000	11 000	0.163	41.275	41.260	55.593	55.563	41.250	41.234	55.555	55.524	50.80
	41.275	55.563	31.750	1.52	HJ-263520	IR-212620	60.1	100.5	15.7	7 000	11 000	0.209	41.275	41.260	55.593	55.563	41.250	41.234	55.555	55.524	50.80
	41.275	55.563	31.750	1.52	HJ-263520	IR-222620	60.1	100.5	15.7	7 000	11 000	0.209	41.275	41.260	55.593	55.563	41.250	41.234	55.555	55.524	50.80
1 3/4	44.450	58.738	25.400	1.52	HJ-283716	IR-232816	49.8	81.0	12.8	6 400	9 900	0.177	44.450	44.435	58.768	58.738	44.425	44.409	58.730	58.699	53.98
	44.450	58.738	25.400	1.52	HJ-283716	IR-242816	49.8	81.0	12.8	6 400	9 900	0.177	44.450	44.435	58.768	58.738	44.425	44.409	58.730	58.699	53.98
	44.450	58.738	31.750	1.52	HJ-283720	IR-222820	61.8	106	16.6	6 400	9 900	0.222	44.450	44.435	58.768	58.738	44.425	44.409	58.730	58.699	53.98

[Notes] 1) For inch series inner rings, please refer to JTEKT special catalog "Needle Roller Bearings" (CAT. NO. BS007EN).
2) $r_{a\,max}$ is equal to the minimum bearing chamfer (r_{min}) at unmarked end.

inch series
HJ type
Shaft dia. (1 3/4) ~ 3 1/2 in ((44.450) ~ 88.900 mm)



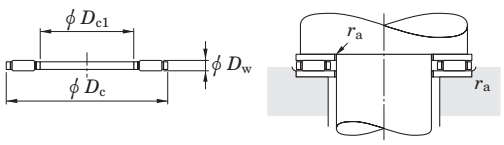


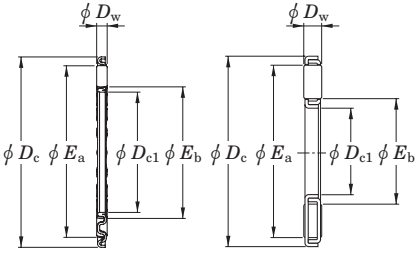
Shaft surface to be 58 HRC or equivalent

Shaft dia. (in)	Boundary dimensions (mm) F_w	D	C (B)	r min.	Bearing No.	Used with inner ring No. 1)	Basic load ratings (kN) C_r	C_{0r}	Fatigue load limit (kN) C_u	Limiting speeds (min^{-1}) Grease lub.	Oil lub.	(Refer.) Mass (kg)	Recommended dimensions Clearance fit S (h6) max.	S (h6) min.	H (H7) max.	H (H7) min.	Tight transition fit S (f6) max.	S (f6) min.	H (N7) max.	H (N7) min.	Shoulder dia. D_a ±0.38
1 3/4	44.450	58.738	31.750	1.52	HJ-283720	IR-232820	61.8	106	16.6	6 400	9 900	0.222	44.450	44.435	58.768	58.738	44.425	44.409	58.730	58.699	53.98
	44.450	58.738	31.750	1.52	HJ-283720	IR-242820	61.8	106	16.6	6 400	9 900	0.222	44.450	44.435	58.768	58.738	44.425	44.409	58.730	58.699	53.98
1 7/8	47.625	61.913	31.750	1.52	HJ-303920	IR-253020	65.4	117	18.1	6 000	9 200	0.236	47.625	47.610	61.943	61.913	47.600	47.584	61.905	61.874	57.15
2	50.800	65.088	25.400	1.52	HJ-324116	IR-273216	53.8	93.0	14.7	5 600	8 600	0.200	50.800	50.782	65.118	65.088	50.770	50.752	65.080	65.049	60.33
	50.800	65.088	31.750	1.52	HJ-324120	IR-243220	66.7	122	19.1	5 600	8 600	0.249	50.800	50.782	65.118	65.088	50.770	50.752	65.080	65.049	60.33
	50.800	65.088	31.750	1.52	HJ-324120	IR-253220	66.7	122	19.1	5 600	8 600	0.249	50.800	50.782	65.118	65.088	50.770	50.752	65.080	65.049	60.33
	50.800	65.088	31.750	1.52	HJ-324120	IR-263220	66.7	122	19.1	5 600	8 600	0.249	50.800	50.782	65.118	65.088	50.770	50.752	65.080	65.049	60.33
	50.800	65.088	31.750	1.52	HJ-324120	IR-273220	66.7	122	19.1	5 600	8 600	0.249	50.800	50.782	65.118	65.088	50.770	50.752	65.080	65.049	60.33
2 1/4	57.150	76.200	38.100	1.52	HJ-364824	IR-283624	89.9	164	25.7	5 000	7 600	0.458	57.150	57.132	76.230	76.200	57.120	57.102	76.192	76.162	68.28
	57.150	76.200	44.450	1.52	HJ-364828	IR-283628	104	198	30.8	5 000	7 600	0.531	57.150	57.132	76.230	76.200	57.120	57.102	76.192	76.162	68.28
2 1/2	63.500	82.550	38.100	2.03	HJ-405224	IR-314024	97.0	187	29.4	4 400	6 800	0.499	63.500	63.482	82.586	82.550	63.470	63.452	82.537	82.502	74.63
	63.500	82.550	38.100	2.03	HJ-405224	IR-324024	97.0	187	29.4	4 400	6 800	0.499	63.500	63.482	82.586	82.550	63.470	63.452	82.537	82.502	74.63
	63.500	82.550	44.450	2.03	HJ-405228	IR-314028	97.0	187	35.2	4 400	6 800	0.499	63.500	63.482	82.586	82.550	63.470	63.452	82.537	82.502	74.63
	63.500	82.550	44.450	2.03	HJ-405228	IR-324028	97.0	187	35.2	4 400	6 800	0.499	63.500	63.482	82.586	82.550	63.470	63.452	82.537	82.502	74.63
2 3/4	69.850	88.900	25.400	2.03	HJ-445616	—	67.2	120	19.1	4 000	6 200	0.363	69.850	69.832	88.936	88.900	69.820	69.802	88.887	88.852	80.98
	69.850	88.900	38.100	2.03	HJ-445624	IR-364424	101	203	31.9	4 000	6 200	0.544	69.850	69.832	88.936	88.900	69.820	69.802	88.887	88.852	80.98
	69.850	88.900	44.450	2.03	HJ-445628	IR-354428	117	245	38.2	4 000	6 200	0.635	69.850	69.832	88.936	88.900	69.820	69.802	88.887	88.852	80.98
	69.850	88.900	44.450	2.03	HJ-445628	IR-364428	117	245	38.2	4 000	6 200	0.635	69.850	69.832	88.936	88.900	69.820	69.802	88.887	88.852	80.98
3	76.200	95.250	38.100	2.03	HJ-486024	IR-404824	107	226	35.5	3 700	5 600	0.585	76.200	76.182	95.286	95.250	76.170	76.152	95.237	95.202	87.33
	76.200	95.250	44.450	2.03	HJ-486028	IR-384828	124	273	42.5	3 700	5 600	0.685	76.200	76.182	95.286	95.250	76.170	76.152	95.237	95.202	87.33
	76.200	95.250	44.450	2.03	HJ-486028	IR-404828	124	273	42.5	3 700	5 600	0.685	76.200	76.182	95.286	95.250	76.170	76.152	95.237	95.202	87.33
3 1/4	82.550	107.950	44.450	2.03	HJ-526828	IR-445228	162	305	48.3	3 400	5 300	1.016	82.550	82.527	107.986	107.950	82.514	82.492	107.937	107.902	98.43
	82.550	107.950	50.800	2.03	HJ-526832	IR-445232	184	358	56.2	3 400	5 300	1.161	82.550	82.527	107.986	107.950	82.514	82.492	107.937	107.902	98.43
3 1/2	88.900	114.300	50.800	2.03	HJ-567232	IR-475632	187	375	58.9	3 200	4 900	1.238	88.900	88.877	114.336	114.300	88.864	88.842	114.287	114.252	104.78
	88.900	114.300	50.800	2.03	HJ-567232	IR-485632	187	375	58.9	3 200	4 900	1.238	88.900	88.877	114.336	114.300	88.864	88.842	114.287	114.252	104.78

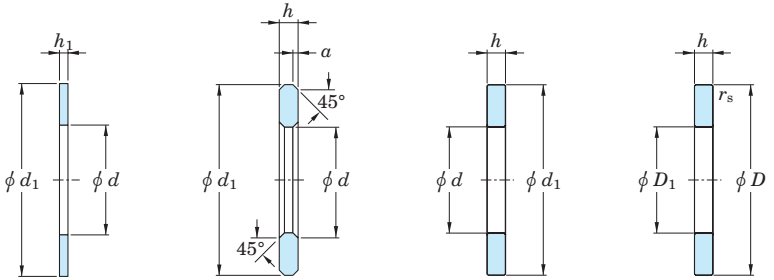
[Notes] 1) For inch series inner rings, please refer to JTEKT special catalog "Needle Roller Bearings" (CAT. NO. BS007EN).
2) $r_{a\,max}$ is equal to the minimum bearing chamfer (r_{min}) at unmarked end.

thrust needle roller and cage assemblies, thrust washers
metric series
AXK, FNT series
Shaft dia. 6 ~ 45 mm





AXK FNT

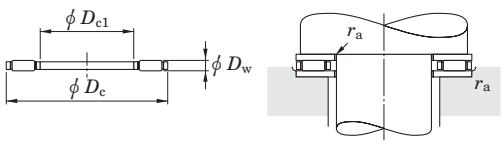


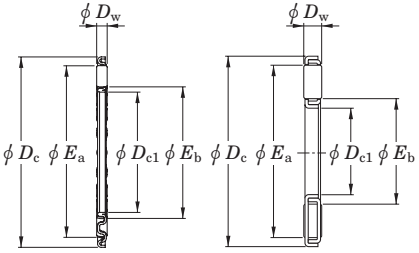
AS (h_1 = 1.0) LS WS.811 GS.811

Shaft dia.	Boundary dimensions (mm) D_{c1}	D_c	D_w	E_a	E_b	r_a max.	Bearing No.	Basic load ratings (kN) C_a	C_{0a}	Fatigue load limit (kN) C_u	Limiting speed (min^{-1}) Oil lub.	(Refer.) Mass (kg)
6	6	19	2	16.9	7.8	0.3	**AXK0619TN**	6.37	14.3	1.40	23 000	0.001
				18.0	8.0	0.3	**FNT-619**	6.82	15.6	1.50	21 000	0.002
8	8	21	2	18.6	9.6	0.3	**AXK0821TN**	8.34	21.1	2.00	20 000	0.001
				20.0	10.0	0.3	**FNT-821**	7.67	19.1	1.85	20 000	0.002
10	10	24	2	22.5	11.0	0.3	**AXK1024**	9.32	25.9	2.90	17 000	0.003
				23.0	12.0	0.3	**FNT-1024**	9.14	25.2	2.40	17 000	0.002
12	12	26	2	24.5	13.0	0.3	**AXK1226**	10.8	32.3	3.40	15 000	0.004
				25.0	14.0	0.3	**FNT-1226**	9.92	29.0	2.75	15 000	0.004
15	15	28	2	27.0	17.0	0.3	**AXK1528**	11.1	35.2	3.35	15 000	0.004
				27.0	17.0	0.3	**FNT-1528**	10.2	31.3	3.00	15 000	0.004
17	17	30	2	28.7	18.3	0.3	**AXK1730TN**	11.7	38.7	3.70	14 000	0.004
				29.0	19.0	0.3	**FNT-1730**	10.8	34.8	3.35	14 000	0.004
20	20	35	2	34.0	22.0	0.3	**AXK2035**	12.8	45.4	4.40	12 000	0.006
				34.0	22.0	0.3	**FNTA-2035**	13.8	50.7	4.80	12 000	0.005
25	25	42	2	41.0	29.0	0.6	**AXK2542**	14.3	56.8	5.50	10 000	0.007
				41.0	27.0	0.6	**FNT-2542**	18.0	75.3	8.05	9 700	0.008
30	30	47	2	46.0	35.0	0.6	**AXK3047**	16.0	68.1	6.60	9 000	0.009
				46.0	32.0	0.6	**FNTA-3047**	18.6	82.4	8.65	8 900	0.009
35	35	52	2	51.0	40.0	0.6	**AXK3552**	17.4	79.5	7.70	8 100	0.010
				51.0	37.0	0.6	**FNT-3552**	21.7	104.0	11.1	7 900	0.010
40	40	60	3	58.0	45.0	0.6	**AXK4060**	27.1	110.0	11.9	7 000	0.016
				57.0	43.0	0.6	**FNT-4060**	31.5	132.0	14.6	7 100	0.020
45	45	65	3	63.0	50.0	0.6	**AXK4565**	29.0	124.0	13.4	6 500	0.020
				63.0	47.0	0.6	**FNT-4565**	37.6	172.0	18.5	6 400	0.024

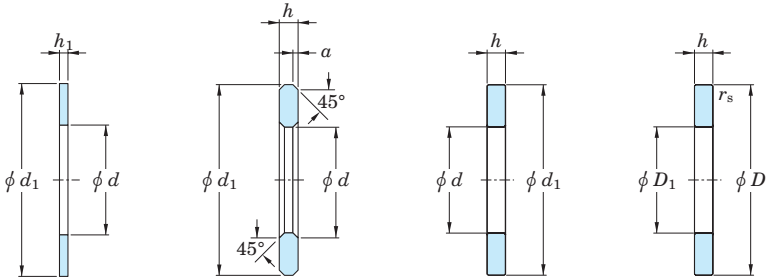
Shaft dia.	Washer dimensions (mm) d	D, d_1	D_1	Thin h_1 (mm)	Thin Washer No.	(Refer.) Mass (kg)	Heavy (LS) h (h11) (mm)	a (mm)	Heavy (LS) Washer No.	(Refer.) Mass (kg)	Heavy h (mm)	Heavy r min. (mm)	Heavy Washer No. Shaft piloted	Heavy Washer No. Housing piloted	(Refer.) Mass (kg)
6	6	19		1.00	AS0619	0.001									
8	8	21		1.00	AS0821	0.002	2.75	0.30	LS0821	0.004					
10	10	24		1.00	AS1024	0.003	2.75	0.50	LS1024	0.008					
12	12	26		1.00	AS1226	0.003	2.75	0.50	LS1226	0.009					
15	15	28	16	1.00	AS1528	0.003	2.75	0.50	LS1528	0.010	2.75	0.30	WS.81102	GS.81102	0.0100
17	17	30	18	1.00	AS1730	0.003	2.75	0.50	LS1730	0.011	2.75	0.30	WS.81103	GS.81103	0.011
20	20	35	21	1.00	AS2035	0.005	2.75	0.50	LS2035	0.014	2.75	0.30	WS.81104	GS.81104	0.014
25	25	42	26	1.00	AS2542	0.007	3.00	1.00	LS2542	0.021	3.00	0.60	WS.81105	GS.81105	0.021
30	30	47	32	1.00	AS3047	0.008	3.00	1.00	LS3047	0.023	3.00	0.60	WS.81106	GS.81106	0.023
35	35	52	37	1.00	AS3552	0.009	3.50	1.00	LS3552	0.030	3.50	0.60	WS.81107	GS.81107	0.032
40	40	60	42	1.00	AS4060	0.012	3.50	1.00	LS4060	0.041	3.50	0.60	WS.81108	GS.81108	0.043
45	45	65	47	1.00	AS4565	0.013	4.00	1.00	LS4565	0.052	4.00	0.60	WS.81109	GS.81109	0.054

thrust needle roller and cage assemblies, thrust washers
metric series
AXK, FNT series
Shaft dia. 50 ~ 160 mm





AXK FNT

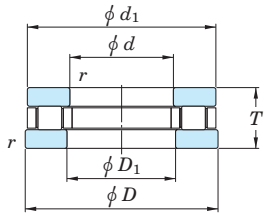


AS (h_1 = 1.0) LS WS.811 GS.811

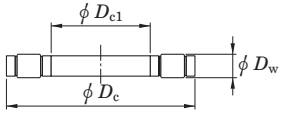
Shaft dia.	Boundary dimensions (mm) D_{c1}	D_c	D_w	E_a	E_b	r_a max.	Bearing No.	Basic load ratings (kN) C_a	C_{0a}	Fatigue load limit (kN) C_u	Limiting speed (min^{-1}) Oil lub.	(Refer.) Mass (kg)	Washer dimensions (mm) d	D, d_1	D_1	Thin h_1 (mm)	Thin Washer No.	Thin (Refer.) Mass (kg)	Heavy (LS) h (h11) (mm)	Heavy (LS) a (mm)	Heavy (LS) Washer No.	Heavy (LS) (Refer.) Mass (kg)	Heavy h (mm)	Heavy r min. (mm)	Heavy Washer No. Shaft piloted	Heavy Washer No. Housing piloted	(Refer.) Mass (kg)
50	50	70	3	68.0	55.0	0.6	**AXK5070**	30.8	137.0	14.9	6 000	0.020	50	70	52	1.00	AS5070	0.014	4.00	1.00	LS5070	0.0560	4.00	0.60	WS.81110	GS.81110	0.059
				68.0	52.0	0.6	**FNT-5070**	37.9	179.0	19.1	5 900	0.026															
55	55	78	3	76.0	60.0	0.6	**AXK5578**	39.4	195.0	20.5	5 300	0.026	55	78	57	1.00	AS5578	0.018	5.00	1.00	LS5578	0.0910	5.00	0.60	WS.81111	GS.81111	0.094
				76.0	57.0	0.6	**FNT-5578**	48.5	254.0	26.3	5 300	0.033															
60	60	85	3	83.0	65.0	0.6	**AXK6085**	44.5	234.0	24.7	4 900	0.035	60	85	62	1.00	AS6085	0.022	4.75	1.50	LS6085	0.102	4.75	1.00	WS.81112	GS.81112	0.106
65	65	90	3	88.0	70.0	0.6	**AXK6590**	46.7	254	26.8	4 600	0.036	65	90	67	1.00	AS6590	0.023	5.25	1.50	LS6590	0.121	5.25	1.00	WS.81113	GS.81113	0.125
70	70	95	4	93.0	74.0	0.6	**AXK7095**	53.8	253	28.0	4 400	0.055	70	95	72	1.00	AS7095	0.025	5.25	1.50	LS7095	0.1280	5.25	1.00	WS.81114	GS.81114	0.133
				93.0	73.0	0.6	**FNTA-7095**	66.6	333	35.3	4 400	0.057															
75	75	100	4	98.0	79.0	0.6	**AXK75100**	55.1	266	29.4	4 200	0.058	75	100	77	1.00	AS75100	0.027	5.75	1.50	LS75100	0.1500	5.75	1.00	WS.81115	GS.81115	0.155
				98.0	78.0	0.6	**FNT-75100**	71.6	374	39.7	4 100	0.064															
80	80	105	4	103.0	84.0	0.6	**AXK80105**	56.4	279	30.8	4 000	0.092	80	105	82	1.00	AS80105	0.028	5.75	1.50	LS80105	0.1580	5.75	1.00	WS.81116	GS.81116	0.165
				103.0	83.0	0.6	**FNTA-80105**	71.3	379	40.1	3 900	0.062															
85	85	110	4	108.0	89.0	0.6	**AXK85110**	57.6	291	32.2	3 800	0.063	85	110	87	1.00	AS85110	0.028	5.75	1.50	LS85110	0.166	5.75	1.00	WS.81117	GS.81117	0.173
90	90	120	4	118.0	94.0	0.6	**AXK90120**	72.9	405	43.0	3 500	0.081	90	120	92	1.00	AS90120	0.038	6.50	1.50	LS90120	0.245	6.50	1.00	WS.81118	GS.81118	0.253
100	100	135	4	133.0	105.0	0.6	**AXK100135**	90.2	552	56.4	3 100	0.106	100	135		1.00	AS100135	0.050									
110	110	145	4	143.0	115.0	0.6	**AXK110145**	93.2	591	59.0	2 800	0.117	110	145		1.00	AS110145	0.055	7.00	1.50	LS110145	0.373	7.00				
120	120	155	4	153.0	125.0	0.6	**AXK120155**	98.5	650	63.5	2 700	0.126	120	155		1.00	AS120155	0.059									
130	130	170	5	167.0	136.0	0.6	**AXK130170**	132	829	78.7	2 400	0.198	130	170		1.00	AS130170	0.074	9.00	1.50	LS130170	0.065					
140	140	180	5	177.0	146.0	0.6	**AXK140180**	136	887	82.5	2 300	0.221	140	180		1.00	AS140180	0.078									
150	150	190	5	187.0	156.0	0.6	**AXK150190**	141	944	86.2	2 200	0.225	150	190		1.00	AS150190	0.083									
160	160	200	5	197.0	166.0	0.6	**AXK160200**	146	1 000	89.9	2 100	0.249	160	200		1.00	AS160200	0.089									

thrust cylindrical roller and cage assemblies, thrust washers
metric series

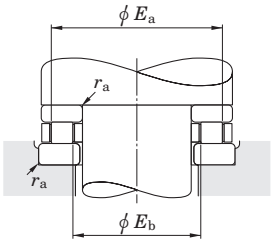
Shaft dia. 15 ~ 55 mm

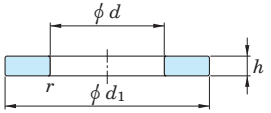


811, 812

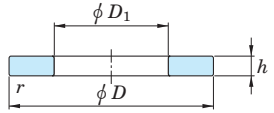


K.811, K.812





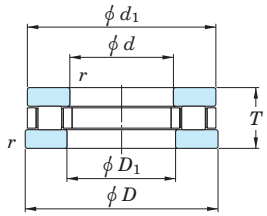
WS.811, WS.812



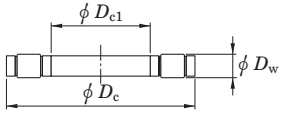
GS.811, GS.812

Shaft dia.	Boundary dimensions (mm) D_{c1} (E11)	D_c (a13)	D_w	T	E_b max.	E_a min.	r_a max.	Bearing No.	Basic load ratings (kN) C_a	C_{0a}	Fatigue load limit (kN) C_u	Limiting speed (min^{-1}) Oil lub.		(Refer.) Mass (kg)	Washer dimensions (mm) d	D_1	D, d_1	h max.	h min.	r min.	Washer No. Shaft piloted	Housing piloted	(Refer.) Mass (kg)
15	15	28	3.5	9	18	25	0.3	**K.81102LPB**	12.1	26.3	3.70	12 000		0.006	15	16	28	2.75	2.64	0.3	WS.81102	GS.81102	0.010
	15	28	3.5	9	18	25	0.3	**K.81102TVP**	12.8	28.6	4.05	12 000		0.006	15	16	28	2.75	2.64	0.3	WS.81102	GS.81102	0.010
17	17	30	3.5	—	20	27	0.3	**K.81103LPB**	12.6	28.6	4.05	11 000		0.008	17	18	30	2.75	2.64	0.3	WS.81103	GS.81103	0.011
	17	30	3.5	9	20	27	0.3	**K.81103TVP**	14.2	33.4	4.70	11 000		0.008	17	18	30	2.75	2.64	0.3	WS.81103	GS.81103	0.011
20	20	35	4.5	10	23	32	0.3	**K.81104TVP**	23.6	56.8	6.85	9 500		0.009	20	21	35	2.75	2.62	0.3	WS.81104	GS.81104	0.014
25	25	42	5.0	11	28	39	0.6	**K.81105TVP**	31.2	81.0	11.4	8 000		0.014	25	26	42	3.00	2.87	0.6	WS.81105	GS.81105	0.021
30	30	47	5.0	—	33	44	0.6	**K.81106LPB**	28.5	69.5	10.7	6 700		0.026	30	32	47	3.00	2.87	0.6	WS.81106	GS.81106	0.023
	30	47	5.0	11	33	44	0.6	**K.81106TVP**	33.0	91.1	12.8	6 700		0.016	30	32	47	3.00	2.87	0.6	WS.81106	GS.81106	0.023
	30	52	7.5	—	33	49	0.6	**K.81206LPB**	53.4	129	13.9	6 300		0.052	30	32	52	4.25	4.12	0.6	WS.81206	GS.81206	0.047
	30	52	7.5	16	33	49	0.6	**K.81206TVP**	56.9	141	15.2	6 300		0.034	30	32	52	4.25	4.12	0.6	WS.81206	GS.81206	0.047
35	35	52	5.0	—	38	49	0.6	**K.81107LPB**	30.8	86.0	12.1	6 000		0.025	35	37	52	3.50	3.34	0.6	WS.81107	GS.81107	0.032
	35	52	5.0	12	38	49	0.6	**K.81107TVP**	34.8	101	14.2	6 000		0.020	35	37	52	3.50	3.34	0.6	WS.81107	GS.81107	0.032
	35	62	7.5	—	41	56	1.0	**K.81207LPB**	58.3	152	16.5	5 300		0.073	35	37	62	5.25	5.09	1.0	WS.81207	GS.81207	0.085
	35	62	7.5	18	41	56	1.0	**K.81207TVP**	61.6	164	17.7	5 300		0.055	35	37	62	5.25	5.09	1.0	WS.81207	GS.81207	0.085
40	40	60	6.0	—	44	56	0.6	**K.81108LPB**	44.2	126	12.0	5 300		0.044	40	42	60	3.50	3.34	0.6	WS.81108	GS.81108	0.043
	40	60	6.0	13	44	56	0.6	**K.81108TVP**	49.8	148	14.1	5 300		0.031	40	42	60	3.50	3.34	0.6	WS.81108	GS.81108	0.043
	40	68	9.0	19	45	63	1.0	**K.81208TVP**	86.8	233	26.9	4 800		0.076	40	42	68	5.00	4.84	1.0	WS.81208	GS.81208	0.093
45	45	65	6.0	—	49	61	0.6	**K.81109LPB**	47.0	140	13.4	4 800		0.035	45	47	65	4.00	3.84	0.6	WS.81109	GS.81109	0.054
	45	65	6.0	14	49	61	0.6	**K.81109TVP**	52.3	163	15.5	4 800		0.035	45	47	65	4.00	3.84	0.6	WS.81109	GS.81109	0.054
	45	73	9.0	—	50	68	1.0	**K.81209TVP**	94.2	266	30.8	4 500		0.083	45	47	73	5.50	5.34	1.0	WS.81209	GS.81209	0.112
50	50	70	6.0	14	54	66	0.6	**K.81110LPB**	49.7	155	14.8	4 300		0.052	50	52	70	4.00	3.84	0.6	WS.81110	GS.81110	0.059
	50	70	6.0	14	54	66	0.6	**K.81110TVP**	54.8	177	17.0	4 300		0.042	50	52	70	4.00	3.84	0.6	WS.81110	GS.81110	0.059
	50	78	9.0	22	55	73	1.0	**K.81210TVP**	101	299	34.6	4 000		0.089	50	52	78	6.5	6.34	1.0	WS.81210	GS.81210	0.144
55	55	78	6.0	16	60	73	0.6	**K.81111TVP**	60.3	207	19.8	4 000		0.066	55	57	78	5.00	4.81	0.6	WS.81111	GS.81111	0.094
	55	90	11.0	—	61	84	1.0	**K.81211LPB**	127	359	39.6	3 600		0.156	55	57	90	7.00	6.81	1.0	WS.81211	GS.81211	0.219
	55	90	11.0	25	61	84	1.0	**K.81211TVP**	138	403	45.2	3 600		0.140	55	57	90	7.00	6.81	1.0	WS.81211	GS.81211	0.219

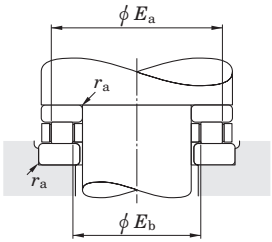
thrust cylindrical roller and cage assemblies, thrust washers
metric series
Shaft dia. 60 ~ 90 mm

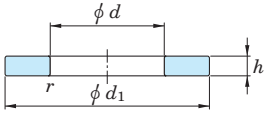


811, 812

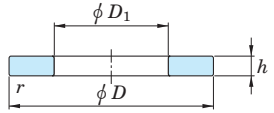


K.811, K.812





WS.811, WS.812

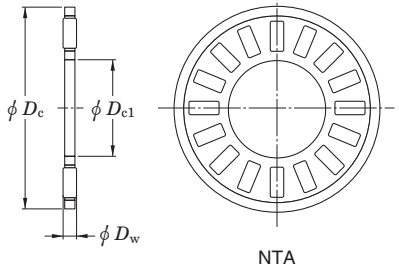


GS.811, GS.812

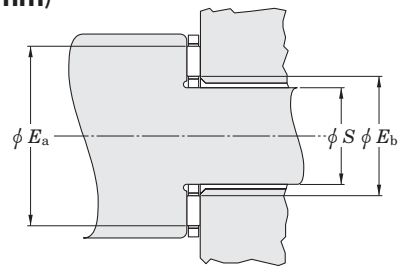
Shaft dia.	Boundary dimensions (mm) D_{c1} (E11)	D_c (a13)	D_w	T	E_b max.	E_a min.	r_a max.	Bearing No.	Basic load ratings (kN) C_a	C_{0a}	Fatigue load limit (kN) C_u	Limiting speed (min^{-1}) Oil lub.		(Refer.) Mass (kg)	Washer dimensions (mm) d	D_1	D, d_1	h max.	h min.	r min.	Washer No. Shaft piloted	Housing piloted	(Refer.) Mass (kg)
60	60	85	7.5	17	65	80	1.0	**K.81112TVP**	84.4	281	30.4	3 600		0.103	60	62	85	4.75	4.56	1.0	WS.81112	GS.81112	0.106
	60	95	11.0	26	66	89	1.0	**K.81212LPB**	129	378	42.4	3 400		0.166	60	62	95	7.50	7.31	1.0	WS.81212	GS.81212	0.251
65	65	90	7.5	18	70	85	1.0	**K.81113TVP**	88.3	305	33.0	3 400		0.109	65	67	90	5.25	5.06	1.0	WS.81113	GS.81113	0.125
	65	100	11.0	27	71	94	1.0	**K.81213LPB**	134	403	45.2	3 200		0.176	65	67	100	8.00	7.81	1.0	WS.81213	GS.81213	0.285
70	70	95	7.5	18	75	90	1.0	**K.81114TVP**	92.1	328	35.5	3 200		0.056	70	72	95	5.25	5.06	1.0	WS.81114	GS.81114	0.133
	70	105	11.0	27	76	99	1.0	**K.81214LPB**	138	428	48.0	3 000		0.186	70	72	105	8.00	7.81	1.0	WS.81214	GS.81214	0.302
75	75	100	7.5	19	80	95	1.0	**K.81115LPB**	86.1	305	33.0	3 000		0.091	75	77	100	5.75	5.56	1.0	WS.81115	GS.81115	0.155
	75	110	11.0	27	81	104	1.0	**K.81215LPB**	143	453	50.9	2 800		0.197	75	77	110	8.00	7.81	1.0	WS.81215	GS.81215	0.319
80	80	105	7.5	19	85	100	1.0	**K.81116LPB**	87.5	316	34.2	2 800		0.103	80	82	105	5.75	5.56	1.0	WS.81116	GS.81116	0.165
	80	115	11.0	28	86	109	1.0	**K.81216LPB**	147	478	53.7	2 600		0.208	80	82	115	8.50	8.31	1.0	WS.81216	GS.81216	0.357
85	85	110	7.5	19	90	105	1.0	**K.81117LPB**	88.9	328	35.5	2 600		0.108	85	87	110	5.75	5.53	1.0	WS.81117	GS.81117	0.173
	85	125	12.0	31	93	117	1.0	**K.81217LPB**	174	572	65.5	2 400		0.376	85	88	125	9.50	9.28	1.0	WS.81217	GS.81217	0.492
90	90	120	9.0	22	96	114	1.0	**K.81118LPB**	119	432	49.3	2 400		0.156	90	92	120	6.50	6.28	1.0	WS.81118	GS.81118	0.253
	90	135	14.0	35	98	127	1.0	**K.81218LPB**	215	691	81.5	2 400		0.540	90	93	135	10.50	10.28	1.1	WS.81218	GS.81218	0.655

thrust needle roller and cage assemblies, thrust washers inch series

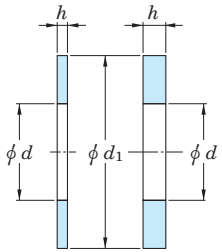
Shaft dia. 1/4 ~ (7/8) in (6.35 ~ (22.23) mm)

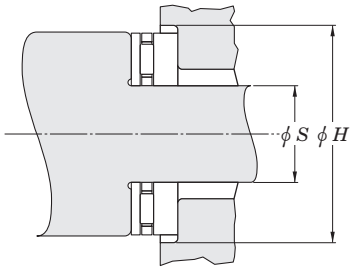


NTA



Raceway hardness to be 58 HRC or equivalent





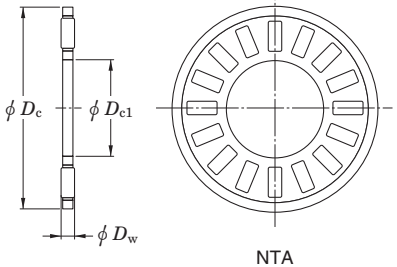
Shaft dia. (in)	Boundary dimensions (mm) D_{c1}	D_c	D_w	E_b	E_a	Bearing No.	Basic load ratings (kN) C_a	C_{0a}	Fatigue load limit (kN) C_u	Limiting speed [1] (min^{-1})	(Refer.) Mass (kg)	Washer No.	Washer dimensions (mm) d	d_1	h max.	h min.	Piloting dimensions (mm) S max.	S min.	Dia. to clear O.D. (mm) H [2]	(Refer.) Washer mass (kg)
1/4	6.35	17.45	1.984	8.636	14.732	**NTA-411**	5.12	10.76	1.05	26 000	0.001	TRA-411	6.35	17.45	0.81	0.76	6.35	6.27	18.26	0.001
												TRB-411	6.35	17.45	1.60	1.52	6.35	6.27	18.26	0.002
												TRC-411	6.35	17.45	2.41	2.34	6.35	6.27	18.26	0.004
5/16	7.92	19.05	1.984	10.16	16.256	**NTA-512**	5.83	13.17	1.30	24 000	0.002	TRA-512	7.92	19.05	0.81	0.76	7.92	7.85	19.84	0.001
												TRB-512	7.92	19.05	1.60	1.52	7.92	7.85	19.84	0.003
3/8	9.53	20.625	1.984	11.68	18.034	**NTA-613**	6.05	14.32	1.40	22 000	0.002	TRA-613	9.53	20.62	0.81	0.76	9.53	9.45	21.44	0.001
												TRB-613	9.53	20.62	1.60	1.52	9.53	9.45	21.44	0.003
												TRC-613	9.53	20.62	2.41	2.34	9.53	9.45	21.44	0.004
1/2	12.70	23.80	1.984	14.99	21.08	**NTA-815**	7.16	19.13	1.85	19 000	0.002	TRA-815	12.70	23.80	0.81	0.76	12.70	12.62	24.61	0.002
												TRB-815	12.70	23.80	1.60	1.52	12.70	12.62	24.61	0.004
												TRC-815	12.70	23.80	2.41	2.34	12.70	12.62	24.61	0.005
9/16	14.275	25.40	1.9837	16.51	22.606	**NTA-916**	7.70	21.53	2.10	18 000	0.003	TRA-916	14.27	25.40	0.81	0.76	14.27	14.20	26.19	0.002
												TRB-916	14.27	25.40	1.60	1.52	14.27	14.20	26.19	0.004
												TRC-916	14.27	25.40	2.41	2.34	14.27	14.20	26.19	0.006
5/8	15.88	28.575	1.9837	18.03	25.908	**NTA-1018**	9.79	30.38	2.85	15 000	0.003	TRA-1018	15.88	28.58	0.81	0.76	15.88	15.80	29.36	0.003
												TRB-1018	15.88	28.58	1.60	1.52	15.88	15.80	29.36	0.005
												TRC-1018	15.88	28.58	2.41	2.34	15.88	15.80	29.36	0.008
												TRD-1018	15.88	28.58	3.20	3.12	15.88	15.80	29.36	0.011
												TRE-1018	15.88	28.58	3.99	3.91	15.88	15.80	29.36	0.013
3/4	19.05	31.75	1.9837	21.34	28.956	**NTA-1220**	10.90	36.48	3.40	14 000	0.004	TRA-1220	19.05	31.75	0.81	0.76	19.05	18.97	32.54	0.003
												TRB-1220	19.05	31.75	1.60	1.52	19.05	18.97	32.54	0.006
												TRC-1220	19.05	31.75	2.41	2.34	19.05	18.97	32.54	0.010
												TRD-1220	19.05	31.75	3.20	3.12	19.05	18.97	32.54	0.012
												TRE-1220	19.05	31.75	3.99	3.91	19.05	18.97	32.54	0.015
7/8	22.23	36.50	1.984	24.38	33.782	**NTA-1423**	13.43	49.82	4.65	12 000	0.005	TRA-1423	22.23	36.50	0.81	0.76	22.23	22.15	37.31	0.004
												TRB-1423	22.23	36.50	1.60	1.52	22.23	22.15	37.31	0.008

[Notes] 1) Limiting speeds listed are based on adequate oil lubrication.
Suggestions for an application requiring O.D. piloting should be determined in consultation with JTEKT.

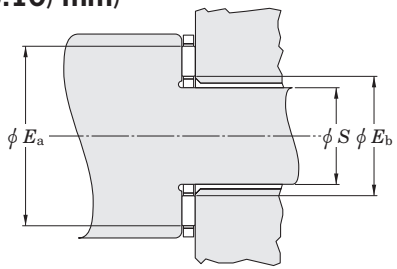
2) If the shaft and the housing adjacent to the bearing O.D. are not concentric, the T.I.R. between the shaft and housing should be added to this dimension.

thrust needle roller and cage assemblies, thrust washers
inch series

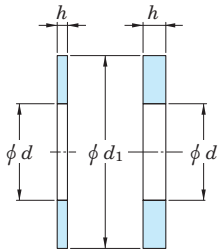
Shaft dia. ($^7/_8$) ~ (1 $^1/_2$) in ((22.23) ~ (38.10) mm)

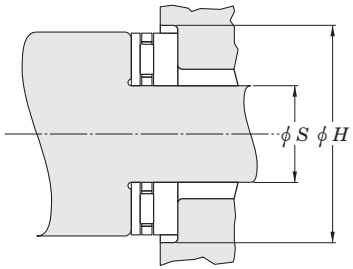


NTA



Raceway hardness to be 58 HRC or equivalent





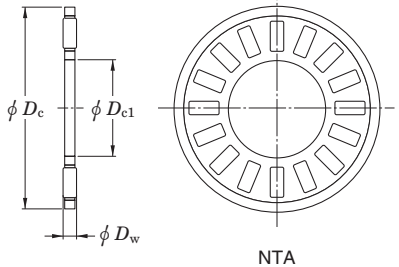
Shaft dia. (in)	Boundary dimensions (mm) D_{c1}	D_c	D_w	E_b	E_a	Bearing No.	Basic load ratings (kN) C_a	C_{0a}	Fatigue load limit (kN) C_u	Limiting speed [1] (min^{-1})		(Refer.) Mass (kg)	Washer No.	Washer dimensions (mm) d	d_1	h max.	h min.	Piloting dimensions (mm) S max.	S min.	Dia. to clear O.D. (mm) H [2]	(Refer.) Washer mass (kg)
$^7/_8$													TRC-1423	22.23	36.50	2.41	2.34	22.23	22.15	37.31	0.012
													TRD-1423	22.23	36.50	3.20	3.12	22.23	22.15	37.31	0.015
	22.23	42.85	1.984	25.91	39.878	**NTC-1427**	18.46	78.29	8.05	9 800		0.008	TRB-1427	22.23	42.86	1.60	1.52	22.23	22.15	43.66	0.013
													TRC-1427	22.23	42.86	2.41	2.34	22.23	22.15	43.66	0.020
													TRD-1427	22.23	42.86	3.20	3.12	22.23	22.15	43.66	0.026
1	25.40	39.675	1.984	27.69	36.83	**NTA-1625**	13.83	53.82	5.00	11 000		0.006	TRA-1625	25.40	39.67	0.81	0.76	25.40	25.32	40.49	0.005
													TRB-1625	25.40	39.67	1.60	1.52	25.40	25.32	40.49	0.009
													TRD-1625	25.40	39.67	3.20	3.12	25.40	25.32	40.49	0.017
													TRE-1625	25.40	39.67	3.99	3.91	25.40	25.32	40.49	0.021
$1\ ^1/_8$	28.58	44.45	1.9837	30.73	41.656	**NTA-1828**	16.68	71.17	7.30	9 600		0.009	TRA-1828	28.58	44.45	0.81	0.76	28.58	28.50	45.24	0.006
													TRB-1828	28.58	44.45	1.60	1.52	28.58	28.50	45.24	0.011
													TRC-1828	28.58	44.45	2.41	2.34	28.58	28.50	45.24	0.017
													TRD-1828	28.58	44.45	3.20	3.12	28.58	28.50	45.24	0.022
$1\ ^1/_4$	31.75	49.20	1.9837	34.04	46.228	**NTA-2031**	20.15	93.41	9.55	8 600		0.010	TRA-2031	31.75	49.20	0.81	0.76	31.75	31.67	50.01	0.007
													TRB-2031	31.75	49.20	1.60	1.52	31.75	31.67	50.01	0.014
													TRC-2031	31.75	49.20	2.41	2.34	31.75	31.67	50.01	0.020
													TRD-2031	31.75	49.20	3.20	3.12	31.75	31.67	50.01	0.026
													TRF-2031	31.75	49.20	4.78	4.70	31.75	31.67	50.01	0.041
$1\ ^3/_8$	34.93	52.375	1.9837	37.08	49.53	**NTA-2233**	21.35	103.20	10.5	8 000		0.010	TRA-2233	34.93	52.37	0.81	0.76	34.93	34.85	53.19	0.007
													TRB-2233	34.93	52.37	1.60	1.52	34.93	34.85	53.19	0.015
													TRC-2233	34.93	52.37	2.41	2.34	34.93	34.85	53.19	0.018
													TRD-2233	34.93	52.37	3.20	3.12	34.93	34.85	53.19	0.029
													TRE-2233	34.93	52.37	3.99	3.91	34.93	34.85	53.19	0.037
													TRF-2233	34.93	52.37	4.78	4.70	34.93	34.85	53.19	0.044
$1\ ^1/_2$	38.10	55.55	1.9837	40.39	52.578	**NTA-2435**	23.22	117.88	12.0	7 600		0.011	TRA-2435	38.10	55.55	0.81	0.76	38.10	38.02	56.36	0.008
													TRB-2435	38.10	55.55	1.60	1.52	38.10	38.02	56.36	0.015

[Notes] 1) Limiting speeds listed are based on adequate oil lubrication.
Suggestions for an application requiring O.D. piloting should be determined in consultation with JTEKT.

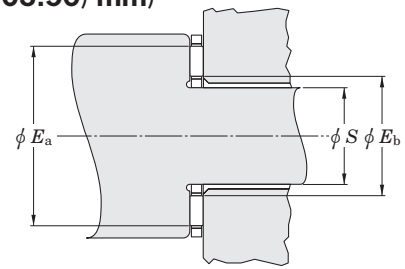
2) If the shaft and the housing adjacent to the bearing O.D. are not concentric, the T.I.R. between the shaft and housing should be added to this dimension.

thrust needle roller and cage assemblies, thrust washers inch series

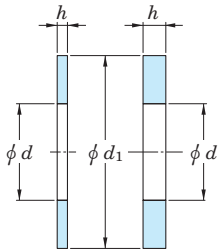
Shaft dia. (1 1/2) ~ (2 1/2) in ((38.10) ~ (63.50) mm)

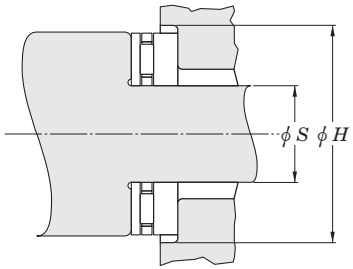


NTA



Raceway hardness to be 58 HRC or equivalent





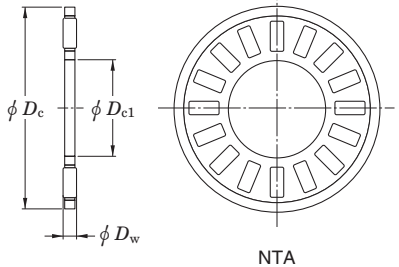
Shaft dia. (in)	Boundary dimensions (mm) D_{c1}	D_c	D_w	E_b	E_a	Bearing No.	Basic load ratings (kN) C_a	C_{0a}	Fatigue load limit (kN) C_u	Limiting speed [1] (min^{-1})	(Refer.) Mass (kg)	Washer No.	Washer dimensions (mm) d	d_1	h max.	h min.	Piloting dimensions (mm) S max.	S min.	Dia. to clear O.D. (mm) H [2]	(Refer.) Washer mass (kg)
1 1/2	38.10	55.55	1.9837	40.39	52.578	**NTA-2435**	23.22	117.88	12.0	7 600	0.011	TRC-2435	38.10	55.55	2.41	2.34	38.10	38.02	56.36	0.023
												TRD-2435	38.10	55.55	3.20	3.12	38.10	38.02	56.36	0.030
												TRF-2435	38.10	55.55	4.78	4.70	38.10	38.02	56.36	0.045
1 3/4	44.45	63.50	1.984	46.74	58.928	**NTA-2840**	25.31	137.45	14.0	6 800	0.014	TRA-2840	44.45	63.50	0.81	0.76	44.45	44.37	64.29	0.010
												TRB-2840	44.45	63.50	1.60	1.52	44.45	44.37	64.29	0.020
												TRC-2840	44.45	63.50	2.41	2.34	44.45	44.37	64.29	0.029
												TRD-2840	44.45	63.50	3.20	3.12	44.45	44.37	64.29	0.038
												TRF-2840	44.45	63.50	4.78	4.70	44.45	44.37	64.29	0.057
2	50.80	69.85	1.9837	53.09	65.278	**NTA-3244**	24.02	132.56	13.5	6 100	0.015	TRA-3244	50.80	69.85	0.81	0.76	50.80	50.72	70.64	0.011
												TRB-3244	50.80	69.85	1.60	1.52	50.80	50.72	70.64	0.022
												TRC-3244	50.80	69.85	2.41	2.34	50.80	50.72	70.64	0.033
												TRD-3244	50.80	69.85	3.20	3.12	50.80	50.72	70.64	0.044
												TRF-3244	50.80	69.85	4.78	4.70	50.80	50.72	70.64	0.066
2 1/8	53.98	73.025	1.984	56.39	68.58	**NTA-3446**	24.42	137.45	14.0	5 800	0.016	TRA-3446	53.98	73.03	0.81	0.76	53.98	53.90	73.81	0.012
												TRB-3446	53.98	73.03	1.60	1.52	53.98	53.90	73.81	0.024
												TRC-3446	53.98	73.03	2.41	2.34	53.98	53.90	73.81	0.035
												TRD-3446	53.98	73.03	3.20	3.12	53.98	53.90	73.81	0.047
2 1/4	57.15	76.20	1.984	59.44	71.628	**NTA-3648**	24.78	142.34	14.6	5 600	0.017	TRA-3648	57.15	76.20	0.81	0.76	57.15	57.07	76.99	0.012
												TRB-3648	57.15	76.20	1.60	1.52	57.15	57.07	76.99	0.022
												TRC-3648	57.15	76.20	2.41	2.34	57.15	57.07	76.99	0.037
												TRD-3648	57.15	76.20	3.20	3.12	57.15	57.07	76.99	0.048
												TRF-3648	57.15	76.20	4.78	4.70	57.15	57.07	76.99	0.071
	57.15	79.375	3.175	59.94	75.184	**NTA-3650**	37.68	177.04	18.6	5 300	0.029	TRC-3650	57.15	79.38	2.41	2.34	57.15	57.07	80.16	0.043
2 1/2	63.50	82.55	1.9837	65.79	77.978	**NTA-4052**	25.53	152.13	15.6	5 100	0.019	TRA-4052	63.50	82.55	0.81	0.76	63.50	63.42	83.34	0.013
												TRB-4052	63.50	82.55	1.60	1.52	63.50	63.42	83.34	0.027
												TRC-4052	63.50	82.55	2.41	2.34	63.50	63.42	83.34	0.041

[Notes] 1) Limiting speeds listed are based on adequate oil lubrication.
Suggestions for an application requiring O.D. piloting should be determined in consultation with JTEKT.

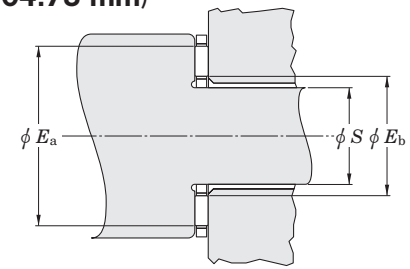
2) If the shaft and the housing adjacent to the bearing O.D. are not concentric, the T.I.R. between the shaft and housing should be added to this dimension.

thrust needle roller and cage assemblies, thrust washers inch series

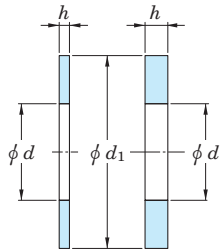
Shaft dia. (2 1/2) ~ 4 1/8 in ((63.50) ~ 104.78 mm)

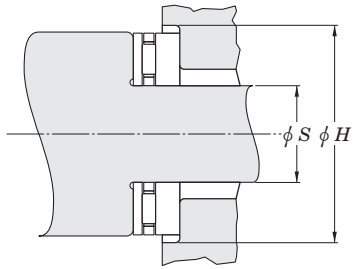


NTA



Raceway hardness to be 58 HRC or equivalent





Shaft dia. (in)	Boundary dimensions (mm) D_{c1}	D_c	D_w	E_b	E_a	Bearing No.	Basic load ratings (kN) C_a	C_{0a}	Fatigue load limit (kN) C_u	Limiting speed [1] (min^{-1})	(Refer.) Mass (kg)	Washer No.	Washer dimensions (mm) d	d_1	h max.	h min.	Piloting dimensions (mm) S max.	S min.	Dia. to clear O.D. (mm) H [2]	(Refer.) Washer mass (kg)
2 1/2	63.50	82.55	1.9837	65.79	77.978	**NTA-4052**	25.53	152.13	15.6	5 100	0.019	TRC-4052	63.50	82.55	3.20	3.12	63.50	63.42	83.34	0.054
2 3/4	69.85	92.075	3.175	72.64	87.884	**NTA-4458**	47.60	255.8	26.8	4 600	0.037	TRA-4458	69.85	92.08	0.81	0.76	69.85	69.77	92.86	0.018
												TRB-4458	69.85	92.08	1.60	1.52	69.85	69.77	92.86	0.035
												TRC-4458	69.85	92.08	2.41	2.34	69.85	69.77	92.86	0.051
												TRD-4458	69.85	92.08	3.20	3.12	69.85	69.77	92.86	0.069
												TRF-4458	69.85	92.08	4.78	4.70	69.85	69.77	92.86	0.104
3	76.20	95.25	1.9837	78.49	90.678	**NTA-4860**	26.96	172.1	17.6	4 400	0.022	TRA-4860	76.20	95.25	0.81	0.76	76.20	76.12	96.04	0.015
												TRB-4860	76.20	95.25	1.60	1.52	76.20	76.12	96.04	0.032
												TRD-4860	76.20	95.25	3.20	3.12	76.20	76.12	96.04	0.061
3 1/4	82.55	104.78	3.175	85.34	100.58	**NTA-5266**	51.60	294.9	30.9	4 000	0.042	TRA-5266	82.55	104.78	0.81	0.76	82.55	82.47	105.56	0.020
												TRD-5266	82.55	104.78	3.20	3.12	82.55	82.47	105.56	0.080
3 3/4	95.25	117.48	3.175	98.04	113.28	**NTA-6074**	56.05	344.3	35.5	3 500	0.050	TRA-6074	95.25	117.48	0.81	0.76	95.25	95.17	118.26	0.023
												TRB-6074	95.25	117.48	1.60	1.52	95.25	95.17	118.26	0.046
												TRC-6074	95.25	117.48	2.41	2.34	95.25	95.17	118.26	0.069
												TRD-6074	95.25	117.48	3.20	3.12	95.25	95.17	118.26	0.092
4 1/8	104.78	128.57	3.175	107.44	124.46	**NTA-6681**	63.61	414.6	41.3	3 200	0.062	TRA-6681	104.78	128.57	0.81	0.76	104.78	104.70	129.39	0.027
												TRC-6681	104.78	128.57	2.41	2.34	104.78	104.70	129.39	0.081
												TRD-6681	104.78	128.57	3.20	3.12	104.78	104.70	129.39	0.109
												TRF-6681	104.78	128.57	4.78	4.70	104.78	104.70	129.39	0.161

[Notes] 1) Limiting speeds listed are based on adequate oil lubrication.
Suggestions for an application requiring O.D. piloting should be determined in consultation with JTEKT.

2) If the shaft and the housing adjacent to the bearing O.D. are not concentric, the T.I.R. between the shaft and housing should be added to this dimension.

Miniature one-way clutches

Miniature one-way clutches consist of a case carburizing steel drawn cup, metal or synthetic resin spring, synthetic resin cage and needle rollers.

They are used in clutch mechanisms of various machines. Use in office automation equipment such as copying and facsimile machines is especially common.

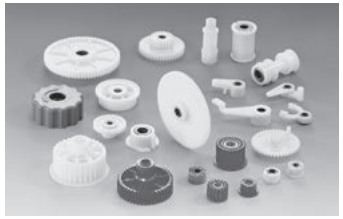
- Useful for making equipment smaller and lighter, due to a drawn cup made of thin sheet steel.
- Locking protrusions are provided around the drawn cup, so that creeping can be prevented without having to hold the surface dimensional accuracy precisely.
- Pre-lubricated with optimum grease, so that no lubrication is necessary under normal operating conditions.
- Unit products with a synthetic resin housing are also available.
  They are compatible with components of various types, such as gears, timing pulleys, cams and rubber rollers. Consult with JTEKT for further information.



1WC series

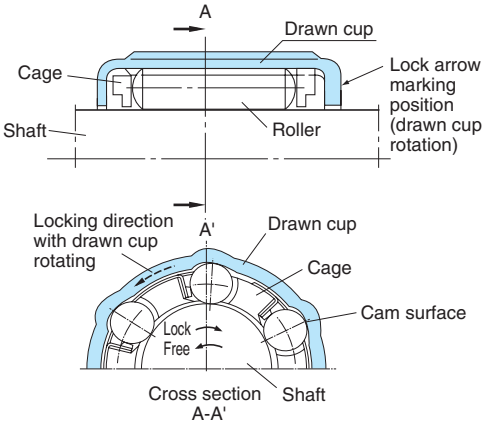


EWC series



Various housings and unit products

Structure and principles



[When the clutch system works]

When the shaft rotates clockwise as in cross section A-A', rollers are locked while engaged with the drawn cup cam surfaces by the effect of springs (wedging of the shaft by the cam surfaces). The drawn cup is driven as a consequence.

[Clutch idle running]

When the shaft rotates counter-clockwise as in cross section A-A', rollers move away from the drawn cup cam surfaces and rotate freely.

Miniature one-way clutch types and characteristics

	1WC series (with metal springs)	EWC series (with synthetic resin springs)	
	Heavy load type	Heavy load type	Light load type
	1WC...	EWC...C	EWC...A
Torque capacity	Heavy load	Heavy load	Light load
Operating temperature range	– 10 to + 90°C	– 10 to + 70°C	
Locking life	Locking system can function more than one million. (Note : this estimation is valid as long as torque magnitude does not exceed the torque capacity shown in the specification table.)		
Insert molding	Possible	Impossible	
Delivery of clutch only	Possible		
Unit delivery	Possible		

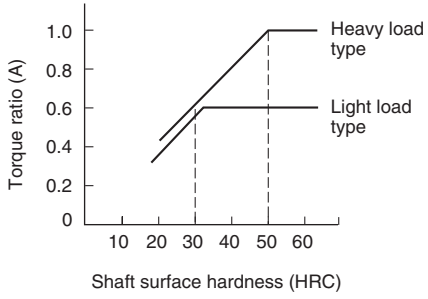
Shaft tolerance

	Heavy load type (1WC... , EWC...C)	Light load type (EWC...A)
Shaft tolerance class	h 8	
Surface hardness	50 HRC or harder	30 HRC or harder
Roughness (Ra)	0.3 a or less	0.8 a or less
Roundness and cylindricity	0.005 mm or less	

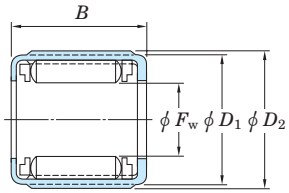
[Remarks] In some operating conditions, shafts need not be as accurate as shown here.
For example :

1. When clutch engaging accuracy is considered unimportant, or when a radial load or moment is not generated, the shaft diameter tolerance can be :
   - shaft diameter 6 mm or less, and EWC0809 (C, A) : 0 to – 0.040 mm
   - shaft diameter 8 mm or more : h 10
2. When the loaded torque is smaller than the torque capacity, shaft surface hardness can be determined as follows :
   * The diagram on the right shows approximate shaft surface hardness relative to torque ratio A.

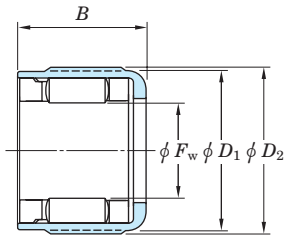
$$\text{Torque ratio (A)} = \frac{\text{Loaded torque}}{\text{Heavy load type torque capacity}}$$



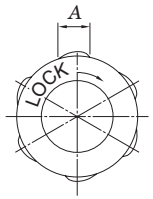
d 4 ~ 12 mm

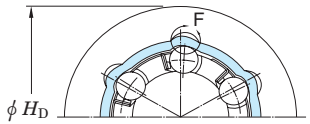


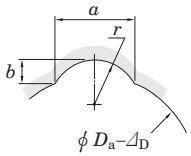
1WC series



EWC series







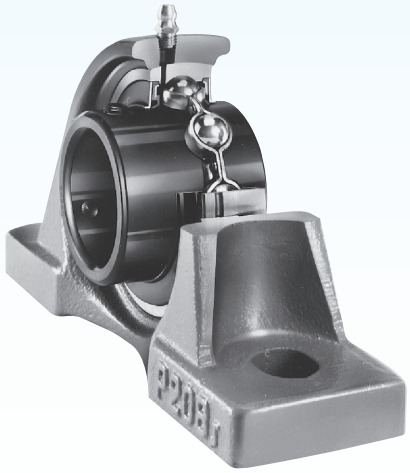
Details of section F

Shaft dia. (mm)	Boundary dimensions (mm)					Torque capacity	Designations		No. of outer ring protrusion		Recommended housing dimensions (Polyacetal resin) (mm)						Recommended housing dimensions (Steel) (mm)						(Refer.) Mass (g)	
							1WC series (With metal springs)	EWC series (With resin springs)																
d	F_w	D_1	D_2	B	A	(N·m)					H_D	a	b	r	D_a	Δ_D	H_D	a	b	r	D_a	Δ_D	1WC	EWC
4	4	8	8.4	6	2.6	0.08	—	**EWC0406A**	4		12	2.65	0.50	2	8	0.06	10	2.9	0.65	2	8	0.03	—	1.0
	4	8	8.4	6	2.6	0.15	—	**EWC0406C**	4		12	2.65	0.50	2	8	0.06	10	2.9	0.65	2	8	0.03	—	1.0
6	6	10	10.4	6	2.8	0.19	—	**EWC0606XC**	6		14	2.8	0.57	2	10	0.08	12.5	3.1	0.85	2	10	0.03	—	1.4
	6	10	10.4	7.5	2.8	0.25	—	**EWC0608XA**	6		14	2.8	0.57	2	10	0.08	12.5	3.1	0.85	2	10	0.03	—	1.7
	6	10	10.4	7.5	2.8	0.44	—	**EWC0608XC**	6		14	2.8	0.57	2	10	0.08	12.5	3.1	0.85	2	10	0.03	—	1.7
	6	10	10.4	8	2.8	0.25	—	**EWC0608A**	6		14	2.8	0.57	2	10	0.08	12.5	3.1	0.85	2	10	0.03	—	1.7
	6	10	10.4	8	2.8	0.44	—	**EWC0608C**	6		14	2.8	0.57	2	10	0.08	12.5	3.1	0.85	2	10	0.03	—	1.7
	6	10	10.4	8	2.8	0.44	**1WC0608**	—	6		14	2.8	0.57	2	10	0.08	12.5	3.1	0.85	2	10	0.03	2.0	—
	6	10	10.4	12	2.8	0.88	**1WC0612**	—	6		14	2.8	0.57	2	10	0.08	12.5	3.1	0.85	2	10	0.03	3.0	—
8	8	12	12.4	9	2.6	0.49	—	**EWC0809A**	6		16	2.6	0.48	2	12	0.10	14	3	0.7	2	12	0.035	—	2.4
	8	12	12.4	9	2.6	0.88	—	**EWC0809C**	6		16	2.6	0.48	2	12	0.10	14	3	0.7	2	12	0.035	—	2.4
	8	14.2	15	12	3.6	1.18	—	**EWC0812A**	6		18.5	3.6	0.87	2.3	14.2	0.11	17.5	3.9	1.15	2.3	14.2	0.035	—	5.8
	8	14.2	15	12	3.6	1.96	—	**EWC0812C**	6		18.5	3.6	0.87	2.3	14.2	0.11	17.5	3.9	1.15	2.3	14.2	0.035	—	5.8
	8	14.2	15	12	3.6	1.96	**1WC0812**	—	6		18.5	3.6	0.87	2.3	14.2	0.11	17.5	3.9	1.15	2.3	14.2	0.035	7.0	—
	8	14.2	15	14.5	3.6	2.65	**1WC0815**	—	6		18.5	3.6	0.87	2.3	14.2	0.11	17.5	3.9	1.15	2.3	14.2	0.035	8.0	—
10	10	16	17	10	5	1.18	—	**EWC1010A**	6		21	5.0	1.20	3.2	16	0.13	19	5.3	1.4	3.2	16	0.035	—	6.0
	10	16	17	10	5	1.96	—	**EWC1010C**	6		21	5.0	1.20	3.2	16	0.13	19	5.3	1.4	3.2	16	0.035	—	6.0
	10	16	17	12	5	1.37	—	**EWC1012A**	6		21	5.0	1.20	3.2	16	0.13	19	5.3	1.4	3.2	16	0.035	—	6.8
	10	16	17	12	5	2.35	—	**EWC1012C**	6		21	5.0	1.20	3.2	16	0.13	19	5.3	1.4	3.2	16	0.035	—	6.8
	10	16	17	12	5	2.35	**1WC1012**	—	6		21	5.0	1.20	3.2	16	0.13	19	5.3	1.4	3.2	16	0.035	8.0	—
12	12	18	19	16	5.1	6.28	**1WC1216**	—	8		23	5.1	1.20	3.3	18	0.14	21	5.4	1.4	3.3	18	0.035	12	—

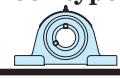
Insert bearing units

Insert bearing units consist of pre-lubricated sealed ball bearings and a housing which varies in shape.

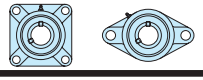
They are capable of aligning themselves efficiently using the spherical fitting surface between the bearing and housing, effectively preventing overloads due to misalignment.



Pillow block type



Flanged type



Flanged type with spigot joint



Take-up type



Cartridge type



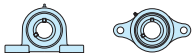
Light duty units



"Compact" series (made from light alloy)



Stainless-series



Pressed steel units



Ball bearings for units



For details, refer to JTEKT separate catalog “Insert bearing units” (CAT. NO. BS006EN).